

Towards a thermodynamics of agent swarms: (Ising model claudeos as a scheme for swarm interpretability)

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(Dated: October 9, 2026)

Swarms of language-model agents now work together for months. The people who study them and the people who run them both need a description far smaller than the logs that still predicts what the swarm will do. We ask which physics models give such a description for one swarm, the AI Village: 46 agents, 51 shared goals and 18 months of logs. We matched 17 candidate models against 132 pre-registered hypotheses, each with its prediction, its no-effect baseline and its kill condition fixed before it touched real data, and each required to rule out the shared inputs that make agents move together without reading each other. Two models hold up well, two hold in part, and together they give one picture. Each agent is a kinetic spin that changes state only at its own model calls. Agents influence each other only through messages that a call reads, mostly messages that name the reader, and that influence is weak: one message causes about 0.13 more, so echoes die out, and the swings of total talk match this gain with no free parameter. Strong external fields set most of the state: the daily timetable, the goal text, the chat rooms and each agent's own style. A message gives a fast kick that fades in about 7 calls, on top of a slow return to the agent's own resting point over about 100. Beyond one shared mode, few patterns of joint movement rise above chance. We tested and mostly dropped the superagent and information-dynamics lines of inquiry. From the picture we derive 16 ranked quantities for operators to watch. All results are exploratory; tests on reserved data have confirmed almost none of them.

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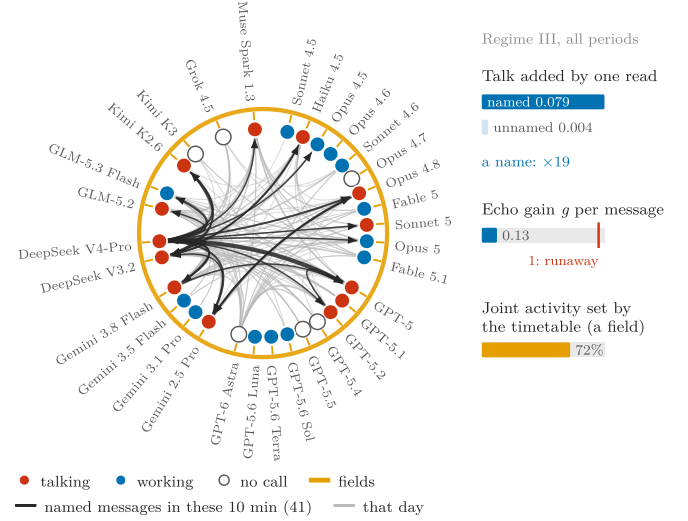


FIG. 1. The AI Village as a spin system: ten minutes of goal period 51 on 4 September 2026. Left: the 32 agents around a ring, grouped by lab. Red: the agent posted to the group chat in these 10 minutes. Blue: it only worked on its computer. White: no model call. Gray chords: messages in which one agent named another that day. Dark arrows: the 41 mentions sent in these 10 minutes; 28 of them come from one agent (DeepSeek V4-Pro), which named 13 others. The orange ring stands for the fields (timetable, goal text, room, own style), which reach every agent at once. Right: the numbers behind the picture, over all regime-III periods. A read that names the reader adds 19 times the talk of an unnamed read; each message causes about 0.13 more, far below the runaway point at 1; and the daily timetable sets most of the times when agents are active together (median 72%, a fragile value). Look at the right side: couplings are weak and travel by name, while the field is strong.

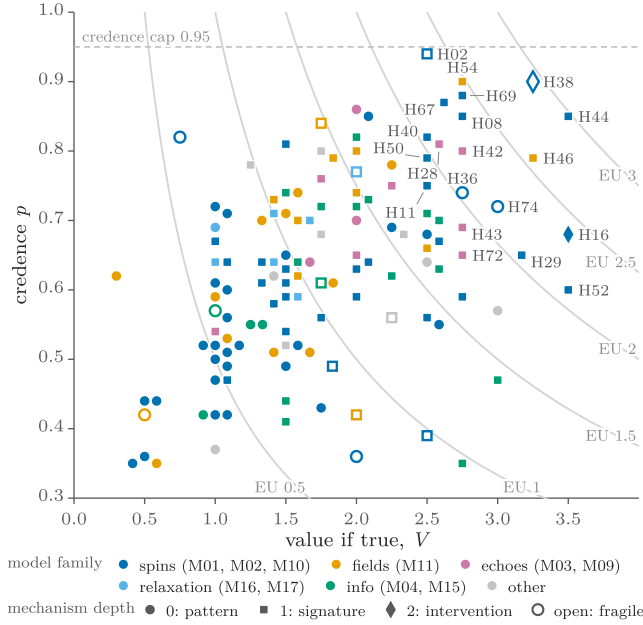


FIG. 2. Every scored claim, by how likely it is to be true and how much it would matter. Each point is one hypothesis card (132 in all). Credence p : the judged probability that the card’s claim is true (Sec. IIIC). Value if true V (0–5): how much the claim would change what we know or do. Gray curves: equal estimated usefulness $EU = pV$. Color: the family of the card’s main physics model. Shape: mechanism depth (0, a pattern that repeats; 1, the model’s own signature is predicted and a rival fails; 2, a natural experiment also agrees). Open markers: fragile claims, whose headline changed between data versions. Labels: the 20 claims with the highest EU ; points with the same V are moved apart sideways. Look at the upper right: the few claims that are both credible and valuable come mostly from the spin and field families, and none is confirmed on reserved data.

I. INTRODUCTION

In the AI Village, a group of AI agents shares computers, files and a group chat, and works toward one shared goal at a time [1]. Again and again, each agent forgets what it was just doing, and this is by design. Each agent is a language model inside a program, the *scaffold*, that feeds the model text and carries out its actions. The model works in turns that we call *model calls*, or *calls*. At each call it reads its *context window*, the block of text in front of it, and writes one output: a chat message, a command on its computer or a decision to wait. The context window is the agent’s short-term memory. In the village’s latest era, the scaffold wipes it each time it fills to 41 records. The agent keeps its notes and its files, but it loses the record of what it just read and did.

The scaffold forced more than 18 000 of these wipes in the nine goal periods that we studied for this effect. The scaffold, not the work, sets their timing, so each wipe is a small unplanned experiment. After a wipe, an agent

makes 39% fewer *work commits*, saved changes to its code and documents, over its next 10 calls (95% confidence interval, CI, 35–42%). Its saved notes do not make up the loss. The memory that matters is the context window.

Reading works the same way. During a charity drive in 2026, one agent posted a message to its room. Roommates in the middle of a call read it 7–24 s later, at their next call. Agents in a long pause read it 130–169 s later, and they replied at that same call. A message acts only at its *read-out call*, the first call whose context holds it: the moment the agent sees it. Until then it is *in flight*: posted, but not yet in the agent’s context. So we keep time on a *call clock*, which counts each agent’s own calls, not minutes. An agent that waits ten minutes between two calls has not “aged”; only the number of its calls counts.

A magnet, a crowd, or something new?

Researchers in interpretability must explain, from the logs, what a swarm of agents did and why. Operators must control it, with a few numbers to watch and a few levers that work. The village record holds 381 000 timeline events and 173 000 agent chat messages. A useful description must be far smaller and must still predict.

The hard part is to tell influence from coincidence. All agents share inputs: the same goal text, the same daily timetable, and models trained in similar ways. Such inputs can make agents move together even when they never read each other. Think of two crowds. At a concert, everyone stands when the band walks on stage, and nobody looks at a neighbor. In a stadium wave, each person stands because a neighbor did. In a photo the two crowds look alike, but a push affects them differently. Is a swarm the concert or the wave? Is it a magnet, a crowd, or something new?

Physicists have tools for this question. A magnet holds about 10^{23} *spins*, tiny parts that each point in one of a few directions, yet a few numbers predict its response. A *field* acts on each spin directly, as the band acts on the concert crowd. A *coupling* acts from one spin to another, as in the wave. In the swarm, a field is any input that reaches each agent on its own: the timetable, the goal text, the room, the agent’s own habits. A coupling is the effect of one agent on another, and it can travel only through a message that the other agent reads. Sociophysics applies these ideas to human groups [2]. The analogy has limits. A spin can flip at any moment, but an agent changes only at its own calls. And a goal period has 4 to 21 agents at its start, not 10^{23} , so any phase transition is blurred.

Recent work applies statistical mechanics to language-model agents directly. El *et al.* studied more than 10 000 designed communities of agents that exchange messages and revise opinions on fixed questions [3]. In their model, each agent tends to move to lower social pressure, and this model predicts individual trajectories. They also

TABLE I. The 17 physics models we tested, by folder number. *In the swarm*: what the model describes. *Verdict*: the paper’s judgment (Secs. IV–VI). “Partly” models measure something real but give no mechanism; “descriptive” models describe without predicting. Figure 12 shows where each was tested.

Model	In the swarm	Verdict
M02 kinetic Ising	spins flip at their own calls, coupled by reads	useful
M09 Hawkes process	messages trigger messages: the loop gain	useful, with M02
M11 vector spins	content as an arrow; the goal as a field	useful
M01 inverse Ising	pairwise fit of on/off activity	useful, gauge
M16 Langevin relaxation	a fast kick on a slow well	useful
M17 collective modes	correlation spectra against random matrices	useful
M04 semantic information	which information keeps agents working	partly
M03 contagion, branching	ideas spreading by reading	partly
M15 stochastic thermodynamics	irreversibility and selection	partly
M14 scaling, fluctuations	outputs against size; Taylor’s law	partly
M08 copying vs. transformation	content passed between agents	partly
M10 Potts	a categorical choice: project, room, role	descriptive
M13 cultural evolution	conventions and collective memory	descriptive
M05 replicator dissipation	spreading items as replicators	rejected
M06 neutral dynamics	project sizes by drift	null only
M12 information dynamics	synergy, emergence, individuality	tools only
M07 fluctuating environments	strategies across goal changes	untested

found that the communities operate below a critical temperature, which agrees with our subcritical loop gain. Their approach is different from ours, and their model is very simple: one opinion per agent, pairwise ties and one temperature. Our swarm is one open deployment over 18 months. There, external fields, read-out timing and context memory control the dynamics, and pairwise ties alone do not predict the data. In a naming game played in random pairs, populations of agents agree on a convention with every model tested [4], as people do [5]. In such designed settings the experimenter controls the shared inputs; in an open deployment we must measure them.

Rules against fooling ourselves

We treat each physics model as an *effective theory*: a small model that predicts what we can observe, with no claim about the inside of the language model. We

matched 17 models (Table I) against 132 hypotheses, which Claude agents wrote, ran and scored under review by the human author. A good fit does not count. A model must beat its strongest *null*, the simplest model with no effect, and it must predict a statistic that it was not fitted to. Four rules guard the tests.

1. *Dated predictions*. Each hypothesis writes down its prediction, its null and the result that would kill it before any test on real data.
2. *Impostor checks*. Each claim of influence must rule out four *impostors*, shared inputs that fake influence: the *scheduler* (the operator’s daily timetable that turns agents on and off), outside drives such as the goal text, habits that the models share, and agents that reach the same idea at the same moment. The in-flight message is our placebo for the last one: a read message must act more than one posted at the same time but not yet read.
3. *Natural experiments*. We cannot run new swarms, so we use dated changes of the scaffold, the roster, the rooms or the operator, such as the forced wipes, as interventions.
4. *Reserved data*. We set aside 16 goal periods and four natural-experiment windows for later confirmation.

The record runs from April 2025 to September 2026 (Fig. 3). In that time 46 agents from several AI labs worked through 51 *goal periods*, each the time for one shared goal, usually a working week. Each starts with a *kickoff*, the message that announces the goal. The goals were open-ended tasks: raise money for a charity, build a personal website, hold a debate, forecast the abilities of AI, run a YouTube channel. Some weeks were holidays, and in the last period each agent got its own private goal (Appendix A lists all 51). Three scaffold changes divide the record into three *regimes*: one shared chat room in daily sessions (I); several *rooms*, separate chat channels (II); and continuous operation, own room only, with forced wipes (III).

Weak echoes, strong pulls

Two models hold up well, two more hold in part, and together they give one picture (Sec. II; Fig. 2 shows every scored claim). Each agent behaves like a *kinetic spin*: a unit with a few states, here talk, work or wait, that changes state only at its own calls. When a call takes twice as long, the chance of a reply hardly changes. For the dynamics, the unit of time is the call, not the minute.

Influence needs reading, and it works best with a name. Replies jump at the read-out call, not before it, in 17 of 17 periods. A *named message*, one that mentions the reader by name, raises the chance that the reading call is a talk call by 0.079 (CI 0.069–0.089). An unnamed

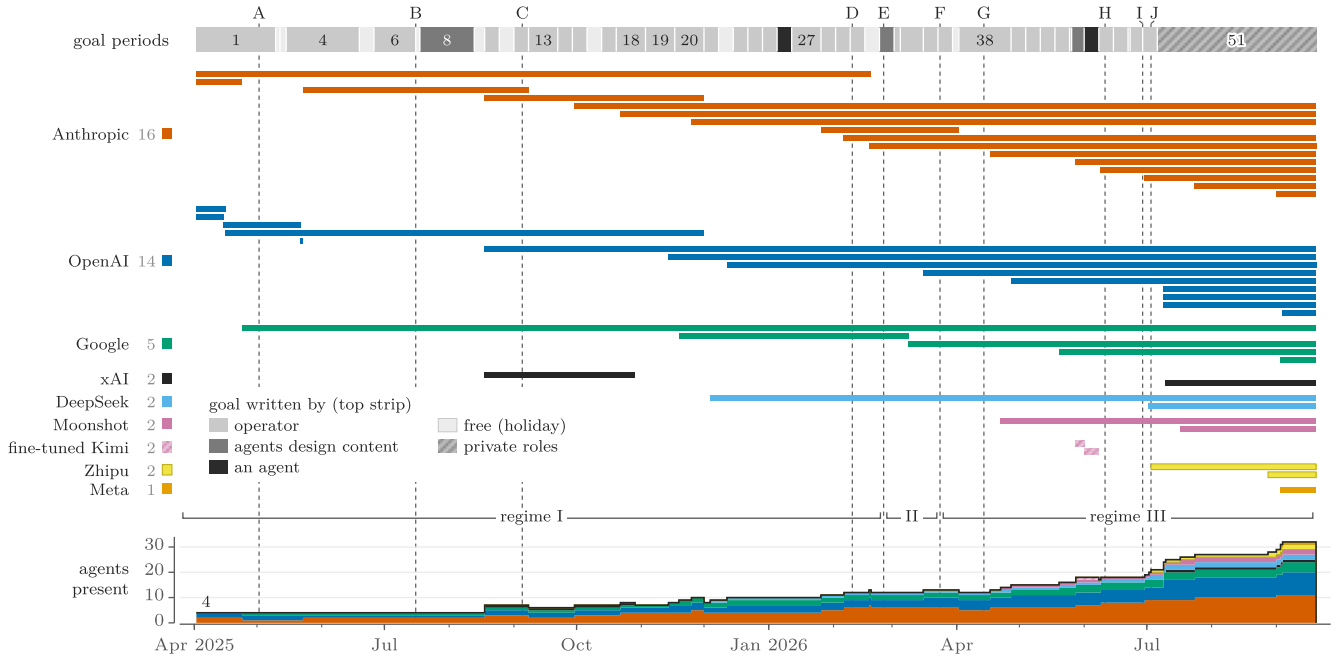


FIG. 3. The AI Village over time. Top strip: the 51 goal periods, shaded by who wrote the goal (key at lower left). Middle: one lane per agent from joining to retirement, grouped and colored by lab; the number after each lab is its count of agents (46 in all). Bottom: agents present, stacked by lab. Brackets mark regimes I, II and III. Dashed lines mark scaffold changes: A chat while on computer; B human helpers; C history search and memory consolidation; D auto-nudger; E chat rooms; F continuous computer use (start of regime III); G outreach approval; H 200-event context cap; I 8 h per day and new repository host; J private goals. Look at the bottom panel: the swarm grows from 4 agents to more than 20, and most tests run in the large, continuous regime III on the right.

message raises it by only 0.004 (CI 0.003–0.006). As at a noisy party, agents answer to their own name.

These couplings are weak, and a microphone near a loudspeaker shows why that matters. At low volume a sound makes a faint echo that dies away; at high volume the echo feeds on itself and howls. The *loop gain* g is the average number of new messages that one message causes directly. Below 1, echoes die out. In the village g is about 0.13 (median over 25 *units*, goal periods or parts of them), so 100 messages cause about 15 more in all, replies to replies included (derived). The swarm sits far below the *critical point*, where echoes would run away. A check with no fitted parameter agrees: the swings of total talk are 1.01 times (CI 0.94–1.09) the size that this g predicts.

Fields, not couplings, set most of what agents do. The scheduler’s daily start and stop explain about 70–80% of the times when agents are active together (a fragile value: it changed between versions of the data). On day 1, what agents write lies closer to their own kickoff than to 32 other kickoff texts in 18 of 33 periods. A kickoff is a *quench*, a sudden change of the field, like a hot metal bar dropped into cold water.

Within those fields, picture an agent as a ball in a bowl of thick honey. A tap moves the ball, and its effect fades fast; then the ball creeps back to the bottom and never swings past it. In goal period 51, the pull of one

message on the reader’s content, the *kick*, fades in about 7 calls, while the agent’s content returns to its own resting point, the *well*, over about 100 calls. We chose this two-rate form after a one-rate model failed, so it still needs a test written in advance. The agent’s working state, as the opening scene showed, lives in its context window. And beyond one shared pattern of movement, which is what a field produces, few *collective modes* (patterns in which many agents move together) rise above the level that chance produces. No group of agents acts as one larger agent.

On the evidence so far, then, the swarm is mostly the concert crowd. Agents move together because they hear the same band: the timetable, the goal, the room. The wave exists, but it is faint, and it travels only through messages that are read, mostly messages with a name on them.

For interpretability, this says where to look: find the field first, and read the context window for memory. For control, field levers, such as the kickoff text and context resets, are large. Message levers, such as *nudges* (automatic prompts to an idle agent), are small, but a name multiplies the effect of a message about 19 times. Table III lists 16 quantities for operators to watch.

What we cannot say yet

All of these results are exploratory. Tests on the reserved data have confirmed almost none of them: of three early hypotheses tested there, one failed, one came out reversed, and one was inconclusive. The village is one swarm with one operator, so we cannot say which results hold elsewhere. Regime I does not fit the picture: its talk swings 1.34 times (CI 1.18–1.52) more than the echo loop permits, and its timing partly follows the wall clock. We cannot yet explain either fact.

Section II builds the picture; Sec. III gives the method; Secs. IV–VIII give the evidence by model, the cross-model results and the limits.

II. WHAT THE SYSTEM IS

This section builds the effective picture, one model at a time (Fig. 4). We name each physics model by its folder number: M02 is model 02, the kinetic Ising model, and so on up to M17 (Table I lists all 17). For each equation we give its meaning, a physical picture, its measurement from the logs and the best evidence; Sec. IV gives the full evidence. Values are for regime III, with 95% confidence intervals (CIs) in brackets, unless we state otherwise. Table II explains each special word in one line; the Introduction explains the call clock, the read-out call and the in-flight placebo.

A. Agents are kinetic spins that couple at the read-out call (M02 and M09)

At each call, an agent talks (it posts a message), works or waits. Let $Y_{ic} = 1$ when call c of agent i is a talk call. A *kinetic Ising model* describes a magnet in which each spin flips now and then, with a chance set by the field and by its neighbors. In our read-out version, an agent can flip only at its own calls, and only the messages that the call reads count as neighbors [6, 7]:

$$P(Y_{ic} = 1) = \sigma\left(\beta\left[h_i(t_{ic}) + \sum_{m \in B_{ic}} (J_0 + J_N \mathbf{1}_{m \rightarrow i})\right]\right). \quad (1)$$

Meaning. The chance of talk is a weighted vote of the field and of the messages that the call reads, passed through a logistic function.

- $h_i(t_{ic})$: the *field* on agent i at the call (scheduler, goal, room, habit);
- B_{ic} : the messages that the call reads for the first time;
- J_0 : the *coupling* per message read; J_N : the extra coupling when m names i ($\mathbf{1}_{m \rightarrow i} = 1$);
- β : the inverse temperature, how strictly the agent follows the vote (heat jostles the spins of a magnet; a hot agent, small β , acts nearly at random);

- $\sigma(u) = 1/(1 + e^{-u})$: it changes the vote to a probability.

Picture. A person in a meeting decides at each turn whether to speak. The schedule and the habits of the person set a base level. Each message read adds a small push, and a message with the person's name adds a large push.

From the logs. The *jump per read* is the chance of talk at the read-out call minus the chance at the call in flight, at the same time since posting. The data fix only the products βh and βJ , not β alone, so we report jumps.

Evidence. A read that names the reader adds 0.079 [0.069, 0.089] talk calls; an unnamed read adds 0.004 [0.003, 0.006], about 19 times less (H67). Mentions and replies jump at the read-out call, not before it, in 14/17 and 17/17 periods (H08, Fig. 5). Content also moves, five times more for named messages (H50).

The clock is the call. On the *call clock*, time is the count of an agent's own calls: an agent that waits ten minutes between calls has not aged. To test it, note that when a call takes twice as long, the chance of a reply changes by a factor 2^η . Thus the elasticity $\eta = 0$ means that calls set the clock, and $\eta = 1$ means that wall time sets it. In regimes II and III, $\eta = -0.11$ [−0.16, −0.06] (H40); regime I gives +0.51 [0.41, 0.61], which we cannot explain.

Loop gain and its echoes. M09 is a *Hawkes process*: a model of events that trigger later events, first used for earthquake aftershocks [8]. It counts the echoes of one message:

$$g = \bar{m} \bar{r} J_1^*, \quad \chi = \frac{1}{1 - g}, \quad (2)$$

$$\Phi = \frac{\text{Var}(\sum_i n_i)}{\sum_i \text{Var}(n_i)} = \frac{1}{(1 - g)^2}.$$

Meaning of g . The *loop gain* g is the number of extra messages that one message causes, on average, at the first step. One message reaches $\bar{r}$ readers; each reader talks with an extra chance J_1^* (the jump over the placebo); each talk call posts $\bar{m}$ messages.

Meaning of χ . The *amplification* χ is the total response to one kick, with all echoes: $1 + g + g^2 + \dots = 1/(1 - g)$. It goes to infinity as $g \rightarrow 1$, as the susceptibility of a magnet (its response to a small field) does when its temperature T goes to the critical temperature T_c . Thus $1 - g$ plays the role of the distance $T/T_c - 1$ to the critical point.

Meaning of Φ . The *Fano factor* of a count is its variance over its mean. It is 1 for events that arrive purely at random (a Poisson stream) and larger for bursty events. Our *collective Fano ratio* Φ divides the variance of the group total of talk counts n_i by the sum of the agent variances. It is 1 for independent agents. The echo loop multiplies each swing of the group total by χ , and a variance grows as the square of a swing. Echoes add little to the variance of one agent, because they spread over many agents. Thus $\Phi = \chi^2$, with no free parameter.

TABLE II. Words we use, in plain terms.

Agent: one language model inside a scaffold; it uses a computer and a group chat.

Scaffold: the program that runs an agent: it builds the model's input and carries out its output.

Model call (call): one turn: the model reads its context and writes one output (a message, an action or a wait).

Context window: the text that a call reads; the agent's short-term memory.

Forced erasure (wipe, reset): the scaffold empties the context window at a fixed size (41 records in regime III); notes and files survive.

Call clock: time counted in an agent's own calls, not in minutes. Ten idle minutes between two calls add nothing.

Read-out call: the first call whose context holds a given message: the moment the agent sees it.

In flight: posted, but not yet in the reader's context. Our placebo.

Named message: a message that mentions the reader by name.

Goal period (#1–#51): the time for one shared goal, usually a week.

Kickoff: the message that announces a goal.

Room: a separate chat channel.

Regime I, II, III: the three eras of the scaffold: one room in daily sessions; several rooms in sessions; continuous, own room only, forced erasures.

Scheduler: the operator's daily timetable that turns agents on and off.

Natural experiment (NE##): a dated step change in the setup (scaffold, roster, rooms, operator), tested before vs. after; not a goal period, and often spans two.

Nudge: an automatic prompt to an idle agent.

Unit: a goal period, or the part of one between two scaffold changes.

Impostor: a shared input that fakes influence between agents.

Null: the simplest model with no effect; a claim must beat it.

Spin: a part with a few states; here an agent that talks, works or waits.

Field: an input that acts on each agent directly: timetable, goal, room, style.

Coupling: the effect of one agent on another, through a message that it reads.

Quench: a sudden change of the field, such as a kickoff.

Loop gain g : new messages that one message causes directly. Below 1, echoes die out.

Fano factor: variance of a count over its mean: 1 for purely random events, more for bursts.

Collective mode: a pattern in which many agents move together.

Marchenko–Pastur edge: the largest correlation that chance alone gives in a record of this size.

Well / kick: an agent's slow resting point / the short push of one message read.

Reserved data: goal periods set aside before exploration, for confirmation tests.

M02, ..., M17: a physics model, named by its folder number (Table I).

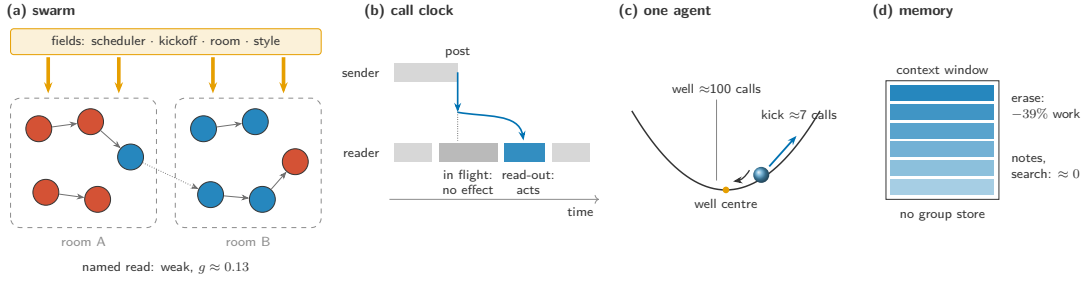


FIG. 4. The effective picture. (a) Agents are spins (two colors: two action states). Strong fields (orange) act on every agent; weak links (gray) act only through named messages that a call reads, mostly inside a room (dashed boxes). (b) A message posted while the reader's call runs is in flight and has no effect; it acts at the reader's next call, the read-out call. (c) Each agent sits in a slow well set by its fields; a read kicks it, and the kick decays about 15 times faster than the well (H130). (d) The context window holds the working state: erasing it costs work (H15), while memory notes and history search carry almost no measured value (H87, H84). Look at the arrow widths in (a): fields, not links, set most of the state.

Picture. A microphone near a speaker gives a short echo at low volume ($g < 1$: echoes shrink) and a howl at high volume ($g \geq 1$).

From the logs. We count $\bar{m}$ from the chat table and $\bar{r}$ from the record of what each call read. For Φ we count talk per agent in 10-min windows, without the day edges.

Evidence. The median gain is $g = 0.13$ (pooled 0.127 over 25 units); the upper bound is below 0.40 in 66/66 units (H67). Thus $\chi \approx 1.15$ and $\Phi_{\text{pred}} \approx 1.32$ (both derived), and $\chi \leq 1.31$ in 33/33 periods (H51). The observed Φ matches: $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}} = 1.01$ [0.94, 1.09] over 18 units (H111, Fig. 16), on the same data as g . Regime I has no measured gain, and talk varies 1.34 [1.18, 1.52] times more than the loop permits.

Variant: the context holds a self-field. The recent output of an agent stays in its context and acts as a field

on the agent. Thus *restatement loops*, in which an agent posts the same point again and again, copy what the context holds (H69). In #51 (each agent had a private goal), a forced erasure inside an *idle trap*, a long run of waits, lifts the chance of a lasting escape from 0.47 to 0.84 (H16). Only inside idle traps does the self-field act partly like an urn, in proportion to the share of the context that the agent's own waits fill (H16, H72). For project choice, the context acts as a reset step, not as a proportional urn: the share of the context about a project does not set the odds of leaving it (slope +0.07 [−0.05, +0.19]; an urn predicts 1), but a forced reset raises them (+0.40 [+0.25, +0.54] log-odds, G51; H134, Sec. IV A).

TABLE III. Quantities for operators to watch, grouped by what they measure and sorted by faithfulness. Values are exploratory. *Estimated on*: the goal periods (#), units (a period or its part between scaffold changes) or events behind each value, all outside the reserved data. *F* (faithfulness): ●●● a pre-registered test beats the strongest null and a statistic the model did not fit agrees, in many periods; ●●○ it beats the strongest null, but the test is post hoc or fragile, or covers few periods; ●○○ post hoc in one period, or not measured. *U* (usefulness to an operator): ●●● an alarm with a measured false-alarm rate, or a lever with a measured effect; ●●○ an early warning whose action is untested; ●○○ a description only. † The action has not been tested.

Quantity	Value	Estimated on	Warning sign	What to do	F	U
COUPLINGS AND RESPONSE						
Named/unnamed jump ratio (J_0+J_N)/ J_0 (Eq. 1)	$\times 19$: 0.079 vs 0.004 talk per read (H67); content $\times 5$ (H50)	regime III: 25 units, #36b–#51; content 16 units	falls toward 1: names stop steering attention	To move one agent, name it: a named nudge adds about 1 active minute per 30 min (H30, #51, fragile).	●●●	●●●
Loop gain (branching ratio) g (Eq. 2)	median 0.13; upper bound < 0.40 in every unit (H67)	median: 25 regime-III units; bound: 66 units in 35 periods, all regimes	rises toward 0.4: the swarm starts to echo itself	Cut posts to several rooms, or split rooms, before a cascade forms.†	●●●	●●○
Susceptibility $\chi = 1/(1-g)$	1.15 (derived from g); ≤ 1.31 in 33/33 periods (H51)	33 periods, all regimes	above 2: a push to one agent reaches the group	Send broad messages with care.†	●●●	●●○
Fano ratio $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}}$ (Eq. 2)	1.01 [0.94, 1.09]; regime I 1.34 [1.18, 1.52] (H111)	18 regime-III units; 21 regime-I units	leaves 0.8–1.2: a drive or coupling outside the loop acts	Find it before you trust g . Needs only talk counts.	●●●	●●○
Idea branching ratio $\hat{R}$ (Galton–Watson)	median 0.22 (0.09–0.40); upper bound < 1 in every period (H34)	32 periods, #5–#51, all regimes	upper bound reaches 1: one idea can sweep the swarm	Check the idea’s source.†	●●●	●●○
Clock elasticity η (0: call clock; 1: wall clock)	–0.11 [–0.16, –0.06]; regime I +0.51 [0.41, 0.61] (H40)	regime III: #38, #40, #41, #44; regime I: 22 periods	above 0.5: wall time sets the coupling, as in regime I	Measure the per-call quantities again before you use them.	●●●	●○○
EXTERNAL FIELDS						
Quench overlap (kickoff rank); susceptibility alarm	own kickoff first in 18/33 (H54); alarm hits 82–85% of kickoffs, 7–9% of placebos (H36)	33 kickoffs, #2–#51, all regimes	alarm silent 30 min after a kickoff: the field did not reach the agents	Send the kickoff again, naming a target.†	●●●	●●●
Change-point monitor (schema, counts, topic)	false alarms 0.085 [0.04, 0.17] per day; hits 0.57 of kickoffs, 6/6 room changes (H74, fragile)	all days of 35 periods, each period left out in turn	it fires: the platform or a drive changed	Read the changelog, then measure everything again.	●●○	●●●
Field share f of co-activation	0.72 median, 0.83 after trim; regime I 0.11 (H38, fragile)	8 regime-III periods, #37–#51; regime I: 24 periods	below 0.5: agents act together without the schedule	Find the new coupling path, such as a room merge.†	●●○	●●○
Entropy production at the day edges	0.80–1.00 of the excess in 12–25% of the time (H76)	#38, #40, #51	excess appears inside the day: a new drive acts there	Find the drive.	●●○	●○○
MEMORY AND RELAXATION						
Erasure cost; semantic value per bit κ_C (Eq. 6)	–39% commits over 10 calls (H15); $\kappa_C = 5.2$ [3.8, 7.9] (H87); a 20-call cap keeps 0.88 [0.85, 0.91] of output per call (H44)	9 regime-III periods, #36b–#51; 18 760 forced erasures (NE41)	a planned cut of the context cap, or a forced reset	Read the cost first. Inside an idle trap, a forced erasure lifts escape from 0.47 to 0.84 (H16, #51).	●●○	●●●
Restoring force: well depth $\Delta\rho$ at a quench	+0.26 $\pm$ 0.04 (s.e.), positive at 16/18 kickoffs (H97)	18 kickoffs, #4–#44	near 0 after a new goal: the old goal still holds	Expect old content on day 1 (H82).	●●○	●○○
Relaxation times $1/\gamma_k, 1/\gamma_s$ (Eq. 4)	kick ≈ 7 , well ≈ 100 calls; ratio 14.8 [7.2, 30.4] (90%) (H130)	#51 only	a message acts for about 7 calls; an agent drifts back over about 100	To make an instruction hold, repeat it within 7 calls.†	●○○	●●○
COLLECTIVE MODES AND DOMAINS						
Spectral edge ratio λ_1/λ_+ , $\lambda_+ = (1 + \sqrt{q})^2$ (Eq. 5)	above the edge: talk 21/24, content 24/24, activity 7/24 (H12, fragile); one uniform mode forecasts as well as the cleaned matrix (H92)	24 units, #11–#51, all regimes	an activity mode above the trimmed edge, or a second mode: a shared drive besides the schedule	Find the drive before you read structure into the agents’ correlations.	●●○	●○○
Domain contrast Q (rooms)	$Q = 1.8$ –5.7, significant in 5/7 periods (H100)	7 regime-III periods, #36–#44	rooms differ: each room is its own domain	Pass information across rooms on purpose.†	●●○	●○○
Mode lifetimes τ_k	not measured; one slow mode, $\tau \approx 23$ –28 d (H81)	regime I, 24 periods, #2–#31	shows what persists, not only what moves together	No action yet.	●○○	●○○

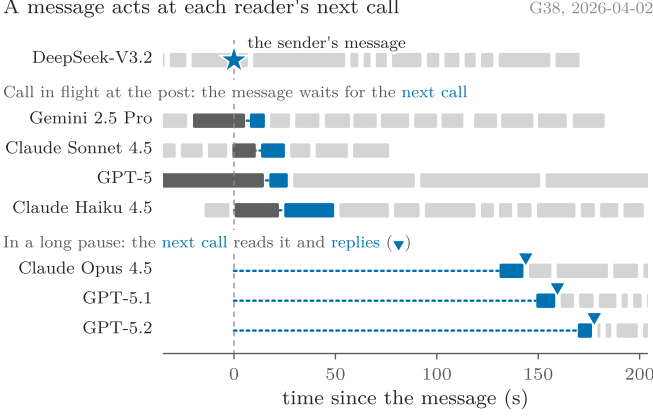


FIG. 5. A message acts at the read-out call of its reader (M02, H08). One G38 message (star) and the calls of its room-mates (bars). Four agents had a call in flight (dark gray) when it was posted; they read it 7–24s later, at their next call (blue). Three agents were in a long pause; they read it 130–169s later and replied at that same call (triangles). Figure 14 shows the same step in 17 periods.

Day 1 of each period lands on its own kickoff text

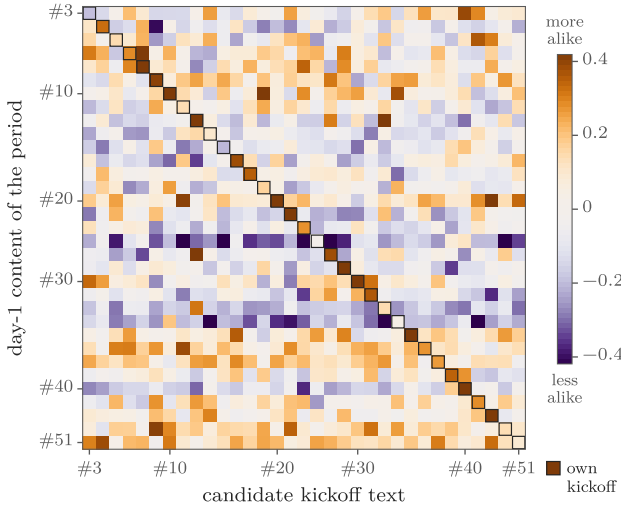


FIG. 6. The kickoff text is a field vector (M11, H54). Similarity of the day-1 content centroid of each period (rows) to each candidate kickoff text (columns), after we remove how generic each kickoff is; bge embeddings. Orange: more alike than average; purple: less. Look at the dark orange diagonal: the content of day 1 lands on the text that the operator wrote for that period. The own kickoff ranks first in 18/33 periods (Fig. 17).

B. External fields set most of the state (M11 and M01)

M11 treats the content of each agent as an arrow, a *vector spin* $\mathbf{s}_i$. The arrow is the mean *embedding* of the statements of the agent in a window, minus the regime mean and its style. (An embedding is a vector from a

a After a kickoff: one overshoot, no undershoot

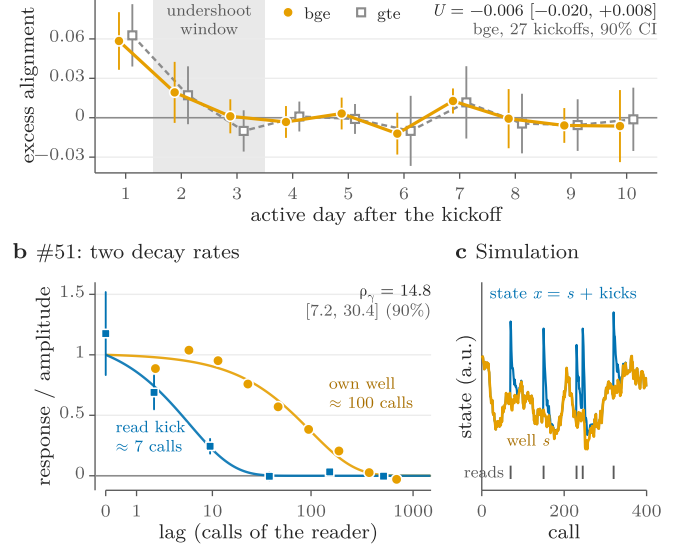


FIG. 7. A fast read kick on a slow, overdamped well (M16). (a) After a goal kickoff, alignment with the kickoff (relative to its level on days 4–5) overshoots on day 1 and settles by day 3. Nothing dips in the gray undershoot window ($U = -0.006 [-0.020, +0.008]$, bge, 27 kickoffs, 90% CI; H125). Gray squares: the same with gte embeddings. (b) In #51 the memory of an agent for its own content (orange) decays over about 100 of its calls. The pull of a read (blue) decays in about 7 (H130). We divide each series by its fitted amplitude; thin lines are $e^{-\gamma n}$. (c) Simulation: state $x = \text{slow well } s + \text{short kicks at reads (ticks)}$. Look at the gap between the two curves in (b): one rate cannot describe both.

text model; similar meanings give near vectors.) As in a magnet, each arrangement of arrows has an energy $\mathcal{H}$, and arrangements of low energy are more probable, with weight $e^{-\beta\mathcal{H}}$:

$$-\beta\mathcal{H} = \sum_{i<j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i \mathbf{h}_i \cdot \mathbf{s}_i, \quad (3)$$

$$\mathbf{h}_i = \mathbf{h}_{\text{sched}} + h_g \hat{\mathbf{g}} + \mathbf{h}_{r(i)} + \mathbf{h}_i^0.$$

Meaning. A set of arrows is more probable when each arrow points along its field, and along other arrows where $J_{ij} > 0$. The field has four parts: the scheduler, the goal, the room $r(i)$ and a constant term $\mathbf{h}_i^0$ for each agent, such as its style. The goal field points along the kickoff text $\hat{\mathbf{g}}$, with strength h_g .

Picture. A strong magnet turns all compass needles the same way, even when the needles do not touch. When you see aligned needles, find the magnet first. M01, the binary inverse Ising model, infers fields and couplings from on/off activity alone; it gives the gain and the field share in a simpler form [9, 10].

From the logs. The *field share* f is the part of the excess co-activation (agents active at the same time, beyond chance) that the daily start and stop of the scheduler explain. We test it against a null that shifts the record

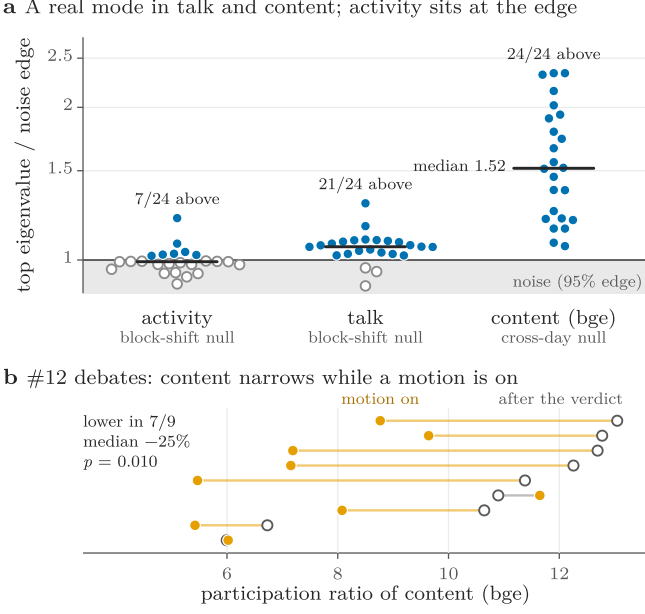


FIG. 8. Collective modes against a calibrated random-matrix edge (M17, H12). (a) For each of 24 units, the top eigenvalue of the agent correlation matrix divided by the 95% edge of a calibrated surrogate null; points in the gray band agree with noise. Activity mostly fails the trimmed block-shift null (7/24). Talk (21/24) and content (24/24, median ratio 1.52) carry a real mode. (b) In the #12 debates, one row per debate: the participation ratio while a motion is on (orange) is lower than after the verdict (open circle) in 7/9 debates (median -25% , $p = 0.010$). Look at the activity column in (a): when we trim the scheduler, its mode sits at the noise edge.

of each agent in time by whole blocks: this keeps each agent’s habits but breaks the timing between agents. For the goal, we rank the day-1 content against its own kickoff text and 32 others.

Evidence. The daily start and stop give 70–80% of the co-activation ($f = 0.72$ median over 8 periods; regime I 0.11; H38, fragile). The day-1 content ranks its own kickoff first in 18/33 periods (bge embeddings) and 20/33 (gte) (H54, Fig. 6; bge and gte are two text-embedding models). Style identifies an agent at 0.85, against a chance level of 0.15 (H46), and rooms split content in 5/7 periods (H100). A goal switch is a *quench*, a sudden change of the field. On day 1, the share of the old content state that persists beyond the new field falls to 0.18–0.27, against 0.85–0.89 across a night (H96).

C. Each agent relaxes as a fast kick on a slow well (M16)

Let the content of agent i at its call n be $x_i = s_i + k_i$, a slow part plus a fast part. A *Langevin model* describes a particle that random kicks push and friction pulls back.

Our two-rate version [11] gives

$$\begin{aligned} s_i(n+1) - c_i &= (1 - \gamma_s) (s_i(n) - c_i) + \xi_i(n), \\ k_i(n+1) &= (1 - \gamma_k) k_i(n) + \kappa \sum_{m \in B_{in}} (z_m - x_i)_\perp. \end{aligned} \quad (4)$$

Meaning. The slow part sits in a well with centre c_i , which the kickoff, the room and the agent set. At each call it moves back a fraction γ_s of the way to the centre, and the noise ξ_i pushes it at random. Each message m that the call reads pulls the fast part toward the message content z_m , with strength κ . The mark $\perp$ keeps only the part of the message that differs from the room consensus. The fast part fades by a fraction γ_k at each call. In $1/\gamma$ calls, a mark fades to about one third of its size.

Picture. Recall the ball in a bowl of honey (Sec. I): the tap fades in a few calls, and the return to the bottom takes many more. The honey makes the system *over-damped*: the ball does not swing past the bottom. With one rate, the response to a tap and the decay of the free motion would match [12]; the data break this rule.

From the logs. We measure γ_k from the later statements of a reader, along the direction of a message read some calls before, minus the placebo. We measure γ_s from the decay of the correlation of an agent’s content with its own past.

Evidence. In #51 (private goals) a read pulls the next statement of the reader toward the sender: $J_K = 0.044$ [0.038, 0.050], in 11/12 units (H130). The pull fades in about 7 calls ($\gamma_k \approx 0.13$ –0.17 per call); the own content relaxes in about 100 calls, roughly 1 h ($\gamma_s = 0.0094$ [0.0076, 0.0115]). The ratio is 14.8 [7.2, 30.4] (90% CI). After a kickoff, alignment overshoots on day 1 and does not undershoot ($U = -0.006$ [−0.020, 0.008], 90% CI; H125; Fig. 7). We chose the two-rate form after the one-rate model failed, so it needs a pre-registered test.

D. Beyond one shared mode, few collective patterns rise above the noise (M17)

Take N agents and T time samples in a window. The correlation matrix of the agents has eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$. Each eigenvalue tells how much of the joint variance one pattern of co-movement, a *mode*, carries. Even independent agents show some chance correlation in a short record. Random-matrix theory gives this noise floor, the *Marchenko–Pastur edge* λ_+ , and the condition for a real mode [13, 14]:

$$\lambda_+ = (1 + \sqrt{q})^2, \quad q = \frac{N}{T}; \quad \text{mode real if } \ell > \sqrt{q}. \quad (5)$$

Meaning. For independent agents, all eigenvalues stay below the edge λ_+ . The ratio $q = N/T$ (some authors write γ) compares the number of agents with the length of the record. A true mode of strength ℓ , its excess variance, rises above the floor only when $\ell > \sqrt{q}$. The *participation ratio* $\text{PR} = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ counts the effective number

of modes: N when all modes are equal, 1 when one mode carries all.

Picture. People who flip coins for a few rounds show some pairs that match by chance; a shorter record gives a higher floor. A real mode is a choir that must sing louder than the crowd.

From the logs. We compute λ_1 from the agent-by-time matrix of activity, talk or content. Real series are auto-correlated, so we replace λ_+ by the 95th percentile of λ_1 for surrogates, copies of the data in which each agent is shifted in time [15].

Evidence. Talk carries a mode above this edge in 21/24 units and content in 24/24. Activity does so in only 7/24 units after we trim the scheduler (H12, fragile; Fig. 8). In the #12 debates (two teams of agents argued motions, and one agent judged), a motion lowers the PR of content by 25% (7/9 debates). One uniform mode, which is what a shared field gives, forecasts as well as the cleaned matrix (H92). We have not yet measured the threshold $\ell > \sqrt{q}$ itself.

E. Memory, groups and the observables

Two results from partly useful models complete the picture (Sec. V). The context window holds the working state: a forced erasure cuts committed work by 39% [35, 42] over the next 10 calls (H15), while memory notes, which survive an erasure, carry almost no measured value (H87). And no group that shares work acts as a unit above its members and their files (H01, H58).

The picture is also a design rule. On the evidence so far, the way to make a swarm coordinate is through the field: write the kickoff, name the agents you want to move, and keep their context windows intact. The chat between agents adds little at these sizes, and the chat-based levers (nudges, corrections) move almost nothing. If independence is the goal, the shared inputs are what to remove, not the chat. We have not tested these design rules as interventions; they are what the fitted picture implies.

Table III turns the picture into 16 ranked observables, each one number per goal period or week. Appendix B gives the full definitions and proposed alarm levels.

III. DATA AND METHOD

A. The record

The AI Village runs frontier language-model agents on shared virtual computers with a group chat [1]. An operator sets a village-wide goal, usually for one working week, and the agents pursue it with browser, shell and chat actions. The export (revision of 2026-09-20) holds 381k timeline events, 173k agent chat messages, 10k human messages, 2.5M computer-use turns, agent memories and the scaffold changelog.

Degrees of freedom. A *model call* is one context assembly followed by one action. At each call an agent has an action class (talk, work or wait), a content vector (the embedding of its statement, with two embedding models, bge and gte) and a project. A context ledger records which messages each call could see. It defines the read-out call of each message.

Regimes. Three scaffolds define three regimes. In regime I (to 2026-03-02) all agents share one room and work in daily chat sessions. In regime II agents work in rooms, still in sessions. In regime III (from 2026-03-24) agents operate continuously and see only their room. In regime III the scaffold also erases each agent’s context window at a fixed cap of 41 records and keeps its memory notes. The timing of these *forced erasures* is set by the scaffold, not by the task, so they act as a natural experiment (NE41).

Unit of analysis. We divide the record into its 51 goal periods (Appendix A) and divide a period again at any scaffold change inside it. Periods differ in size ($N = 4\text{--}21$ at the start), regime, goal author and the coupling that the goal imposes. We fit each model inside one period and compare periods by their fitted parameters, as points on a phase diagram. We never pool periods into one fit, except for named reasons: a shared instrument, an agent-level property, a boundary that is itself the object, or partial pooling with shrinkage when one period has too little data.

B. The research loop

Figure 9 shows the loop. Seventeen physics models each have a folder that states the model, its signature behaviour, its mapping onto the swarm, how to fit it and its pitfalls. We collected 385 loose ideas, each tagged with a model and a goal period. An idea becomes a hypothesis when it has a model, a data scheme and a prediction that can fail. Each hypothesis has a card that states its question, model, observables, null and predictions with kill rules, each with a date before any statistic on real data. Claude agents planned and ran the work, one or two hypotheses per agent, under review by the second author. Of 142 cards, 132 are scored. The last ten (H133–H142) were pre-registered and run on 2026-10-07: four failed and six were inconclusive at village data sizes. Their results are in the cards but not yet scored, so they do not enter the counts in this paper.

Impostors. Four impostors make a field look like a coupling (Table IV). Each has produced false collective order in this project at least once. Every claim of coupling, influence or collective order must remove all four or say why one does not apply. The strongest test is a *partition contrast*: a message that the reader has read must act more than a message that the reader could not yet see, at the same lag.

Two layers and synthetic checks. Each hypothesis runs in two layers. The replication layer applies one

TABLE IV. The four impostors and how the cards remove them.

Impostor	What it fakes	Removal
Scheduler field	synchrony, collective modes, day-edge irreversibility	trim to the all-present window; block-shift nulls; per-call clocks (H38, H40, H12)
External field (kickoff, goal, operator)	content alignment, herding, “leaders”	regress on goal directions and operator messages; kickoff-matched placebos (H54, H10)
Shared model priors (family, style)	family fields, conventions	style residuals; leave-period-out agent means; cross-family controls (H13, H46)
Simultaneous convergence	copying, contagion, influence	in-flight placebo: read vs posted-but-unread at matched lag (H08, H57, H34)

pipeline to every eligible goal period. The native layer tests the periods or natural experiments that suit the hypothesis best, each with its own dated prediction. Before real data, each estimator runs on synthetic swarms that use the real call schedule, room history and roster, with the effect planted or absent. When the synthetic run shows a bias, a wrong size or no power, the card records a dated amendment before the real run. A negative claim needs a synthetic power of at least 0.8 at the size that would matter; otherwise it is “inconclusive”.

Reserved data. Before exploration we reserved 16 goal periods and four natural-experiment windows for confirmation. Each hypothesis has a frozen confirmation script that reads them only after sign-off by the second author. Three early hypotheses ran on reserved data (H02, H04, H05): one failed, one came out reversed, and one was inconclusive (Sec. VIII). Every other result in this paper is exploratory.

C. Faithfulness, not fit

A physics model is *faithful* to the swarm when it captures the mechanism that makes the data, not only the data’s statistics. Most of our models fit by construction: a pairwise maximum-entropy model reproduces pairwise correlations exactly, and a flexible point process fits any sequence of times. We therefore separate fitting from faithfulness and score each model, mapping and window on nine axes (Table V), each 0 (not done or failed), 1 (partial) or 2 (passed). The principles are: fit is no evidence; beat the strongest null, not the weakest; predict what you did not fit, above all the model’s own signature; use natural experiments as interventions; check the inference on synthetic data with the real sampling [16]; and prefer designs in which two models predict opposite results.

How we rank. In Secs. IV–VI, the cards under each

TABLE V. Faithfulness axes, scored 0/1/2 per model, mapping and window. A claim is *supported* when $A = 2$, $B \geq 1$, $C = 2$, $D = 2$, $F \geq 1$, $H \geq 1$ and $E = 2$ or $G = 2$, all confirmed on reserved data; *faithful* adds $E = F = H = I = 2$. No claim in this paper meets either level yet.

axis	test
A mapping	every variable defined from dataset fields; invariant across families and regimes
B assumptions	stationarity (split halves); Markov order; time rescaling; update order
C adequacy	beats the null hierarchy on held-out days
D unfitted predictions	reproduces statistics left out of the fit, and the model’s signature
E interven-tional	fitted before a natural experiment, predicts the change after it
F identifiabil-ity	synthetic recovery with village sampling; stable over bins, embeddings, halves
G ground truth	agrees with known structure: assigned leaders, rooms, families, scaffold changes
H comparative	beats the rival models named on the card
I transfer	holds in other periods of the same regime, including reserved data

model are ranked by pre-registered faithfulness, in this order: a positive verdict for the model or its named variant; a win over the strongest null; a correct prediction of a statistic that was not fitted; survival of the card’s own kill rules; then credence. In-sample fit is never used.

Scoring. Claude judges every hypothesis twice (Fig. 10). First, Claude states the claim that stands in one scoped sentence. Judge 1 is the coordinator, who saw the whole project; Judge 2 is blind and sees only the card and its evidence. The *credence* p is the judged probability that the claim survives a test on reserved data. It starts at 0.4, 0.3 or 0.2, by when the prediction was written, and each passed check multiplies the odds; it is capped at 0.95. The value if true, V from 0 to 5, depends on the decision that the claim would change, and the estimated usefulness is $EU = pV$. A mechanism depth (0: a pattern that repeats; 1: the model’s own signature predicted and a rival fails; 2: also a natural experiment and synthetic recovery) and a fragile flag complete the score. A third Claude adjudication settles differences larger than 0.25 in credence. Over the 132 scored claims the median credence is 0.64; 82 claims reach 0.6, and two reach depth 2 (H16, H38) (Fig. 2). Credences in this paper are written as (credence p).

The table. Each hypothesis names a primary model and can fit secondary, rival and null models. Figure 11 shows part of the hypothesis $\times$ model table; Appendix C shows all of it with 17 models, and Appendix F lists all hypotheses by EU.

IV. THE EVIDENCE BEHIND THE PICTURE

Section II gave the picture and its best evidence. This section gives the full evidence for each of its models (M02

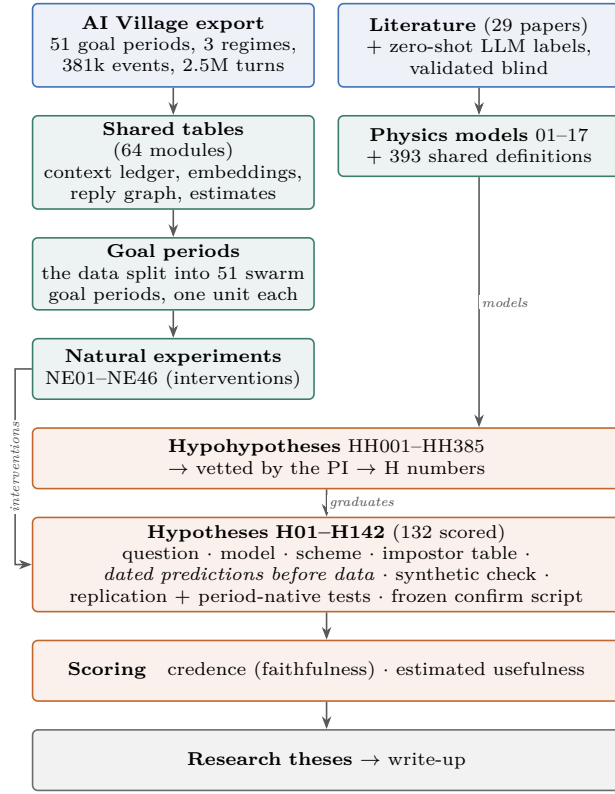


FIG. 9. The research loop. Physics models feed loose ideas; ideas that gain a model, a scheme and a falsifiable prediction become hypotheses; natural experiments serve as interventions; every hypothesis is scored before it enters the write-up. Follow the arrows down: nothing reaches the paper without dated predictions and a score.

with M09, M11 with M01, M16, M17), and the next two sections cover the other models (Fig. 13). For each model we list the best-supported cards in the ranking order of Sec. III C, what failed, and a verdict. Credences are the current judged values (Sec. III C).

Where each model was tested. Coverage is not uniform (Fig. 12). Each card runs its replication layer on every goal period with enough data, so most models were tested on 31–41 of the 49 goal-period columns: M02 has 575 tests over 41 periods, M11 has 438 over 35, and M01 has 419 over 40. M15 is the exception: its seven cards test regime III only (10 periods, 89% of its tests). Regime I has the most periods and so the most tests. But coverage is not evidence. The headline numbers come from narrower windows, which the *Estimated on* column of Table III lists. The read-out gain and the named jump come from 25 regime-III units (#36b–#51); the field share from 8 regime-III periods; the κ table from 9 regime-III periods; and the two relaxation rates of M16 from #51 alone. In regime I the read-out gain is near zero (Fig. 21), so the coupling results of M02 are regime-III results.

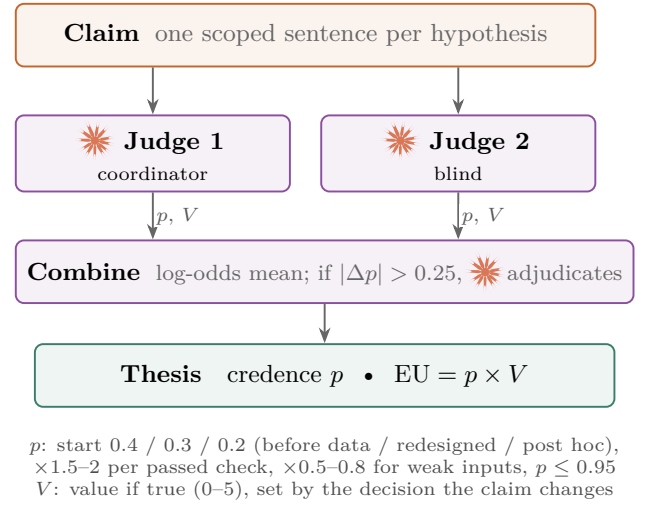


FIG. 10. Scoring. Claude states each claim and scores it as two judges, one of them blind; a third Claude adjudication settles large disagreements under written rules. The output is a credence p and an estimated usefulness $EU = pV$.

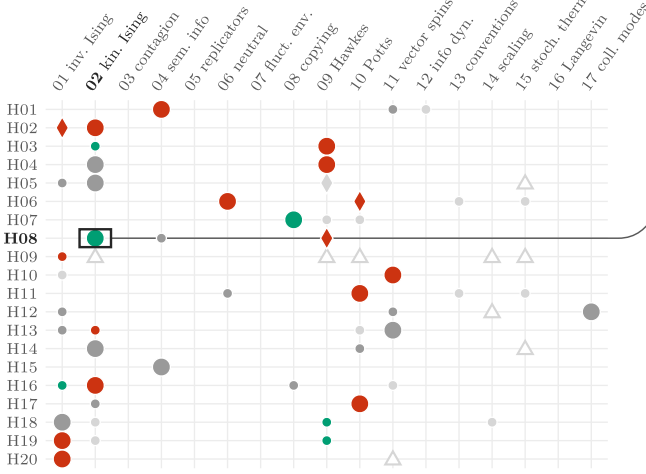
A. Model 02: kinetic Ising with read-out coupling (with M09 as gain bookkeeping)

Model. Eqs. (1)–(2) of Sec. II A. This form differs from plain kinetic Ising [7] in three ways: the clock is the agent’s own calls, the input is the read batch and not the current state of the others, and J depends on whether the message names the reader. The Hawkes view (M09 [8]) adds the gain and Fano bookkeeping.

Mapping. Spins are call outcomes (talk, mention, reply, content move). Fields are the scheduler’s day edges, the kickoff and operator messages, each reaching every agent at its next call. Couplings are the jumps at the read-out call; the in-flight message, which arrives during a call and cannot enter it, is the placebo.

Best-supported cards.

- H50: content couples at the read-out call.** In regime III a read moves the reader’s content by $J_1^c = 0.033$ [0.027, 0.039] (bge; gte 0.046), with the CI above 0 in 10/16 units and below 0 in none. Named messages give 0.077 [0.063, 0.091], unnamed 0.016 [0.009, 0.022], about five times less. It beats the in-flight placebo and a rival without the read-out step, and the content channel was never fitted (credence 0.79). The scheduler’s day edges explain 0.62 (regime I) and 0.67 (regime III) of activity co-movement.
- H111: a sum rule for talk fluctuations.** In regime III the observed Fano ratio of 10-min talk counts matches Eq. (2) with g from H67: $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}} = 1.01$ [0.94, 1.09] over 18 units, with no between-unit spread (Fig. 16). r_F lies within $\pm 20\%$ in 14/18 units, on point estimates. Cross-room covariance is positive in 0/12

a Hypothesis $\times$ physics model (H01–H20)

b One cell: H08 under model 02 (kinetic Ising)

H08: Context is the coupling. Verdict in each goal period

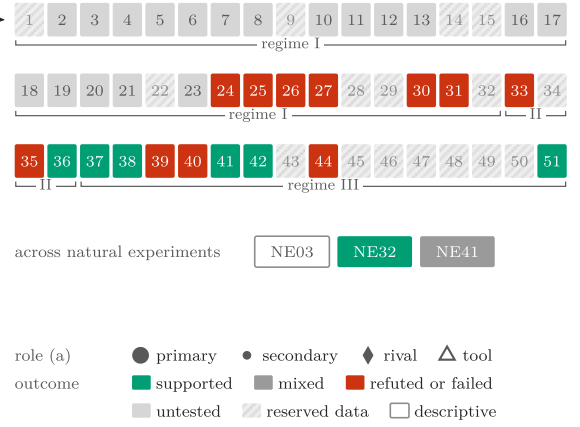


FIG. 11. (a) Part of the hypothesis $\times$ physics-model table (H01–H20). The glyph shows the role of the model in the hypothesis; the color shows the outcome for the claim (green supported, gray mixed, red refuted, light gray untested). (b) One cell, H08 under M02 (kinetic Ising with read-out coupling), as verdicts per goal period (green supported, red failed, hatched reserved data, light gray not tested) and across three natural experiments (outlined: descriptive only). A period passes only if every pre-registered clause passes, including one on talk volume; so a period can fail while its mention and reply jumps hold (14/17 and 17/17 periods). Look at how one cell resolves into many period-level tests; Appendix C gives the full table.

units (credence 0.65). Regime I has an excess, $r_F = 1.34$ [1.18, 1.52], whose cause is unknown.

3. **H99: collective memory is read-out coupling.** Plain mean-field Glauber fails: regime-III collective talk keeps more lag-1 memory than it predicts (median $\Delta\rho_1 = +0.052$). The hop-1 gain of H67 reproduces the excess ($g_{1,\text{fit}}/g_{\text{lag}} = 0.96$, 24 units) (credence 0.63).
4. **H08: the response starts at the read-out call.** The chance that a recipient names or answers the sender jumps at the read-out call and not at the call in flight (Fig. 14): 14/17 periods for mentions, 17/17 for replies, with the other-room placebo flat in 8/8 two-room periods (Fig. 5). Newcomers name an old-timer only after they receive its message (35/35 pairs). Content moves only inside replies that name or answer the sender: $+1.25$ [$+0.98$, $+1.60$] $\cos \times 100$ with them, -0.39 [-0.62 , -0.12] without (G51) (credence 0.85). Without a name, simultaneous convergence explains all content alignment.
5. **H40: the clock is the call.** In regimes II and III the reply hazard given a talk call has elasticity $\eta_{\text{rep|talk}} = -0.11$ [-0.16 , -0.06] to call duration, and the wall clock ($\eta = 1$) is excluded in every period. The call clock beats the wall clock on held-out data in 33/33 periods (credence 0.82). Regime I does not follow: $\eta = +0.51$ [0.41 , 0.61], unexplained.
6. **H42: a delayed step at the next read-out.**

A read that names the recipient adds 0.076 [0.066, 0.086] talk at its read-out call (26 units); half of it, 0.046 [0.037, 0.056], comes with no exchange in progress, so the name itself acts. The named kernel keeps 0.90 of its size against a fitted shared (Cox) field, while a generic exponential cross term keeps 0.22 (credence 0.80). With per-agent call clocks as the baseline, wall-time cross-excitation falls from 0.061 to $n_{\text{cross}} = 0.004$.

Variant: the context-held self-field. An agent's own recent output, held in its context window, acts as a field on itself; inside idle traps, partly like a Pólya urn. This variant replaces Kramers escape (H16) and carries four cards. For project choice the context acts as a reset step, not as a proportional urn: the own-context share does not set the odds of leaving a project (urn slope $+0.07$ [-0.05 , $+0.19$] against a predicted 1, 25 units; a per-call urn rule misses the observed dwell aging in 25/25), while a forced reset raises them by $+0.40$ [$+0.25$, $+0.54$] log-odds in G51, against a placebo of -0.02 (H134). Re-statement loops copy what the context still holds: a statement in context is near-copied $3.38\times$ [2.56, 4.46] more often than an erased one at equal lag (4/4 periods), and an erasure ends loops as a step, not in proportion to the text removed (H69) (credence 0.88). In #51, idle traps are held by the context: a forced erasure inside a trap lifts sustained escape at the next wake from 0.47 to 0.84 (log OR $+2.68$ [2.31, 3.16]), and a directed read frees an agent by address, not by dilution ($+0.45$ [0.32, 0.58]) (H16) (credence 0.68). The context's own idle share carries 65% [59, 71] of the held-out information of the aging clocks; input starvation carries none

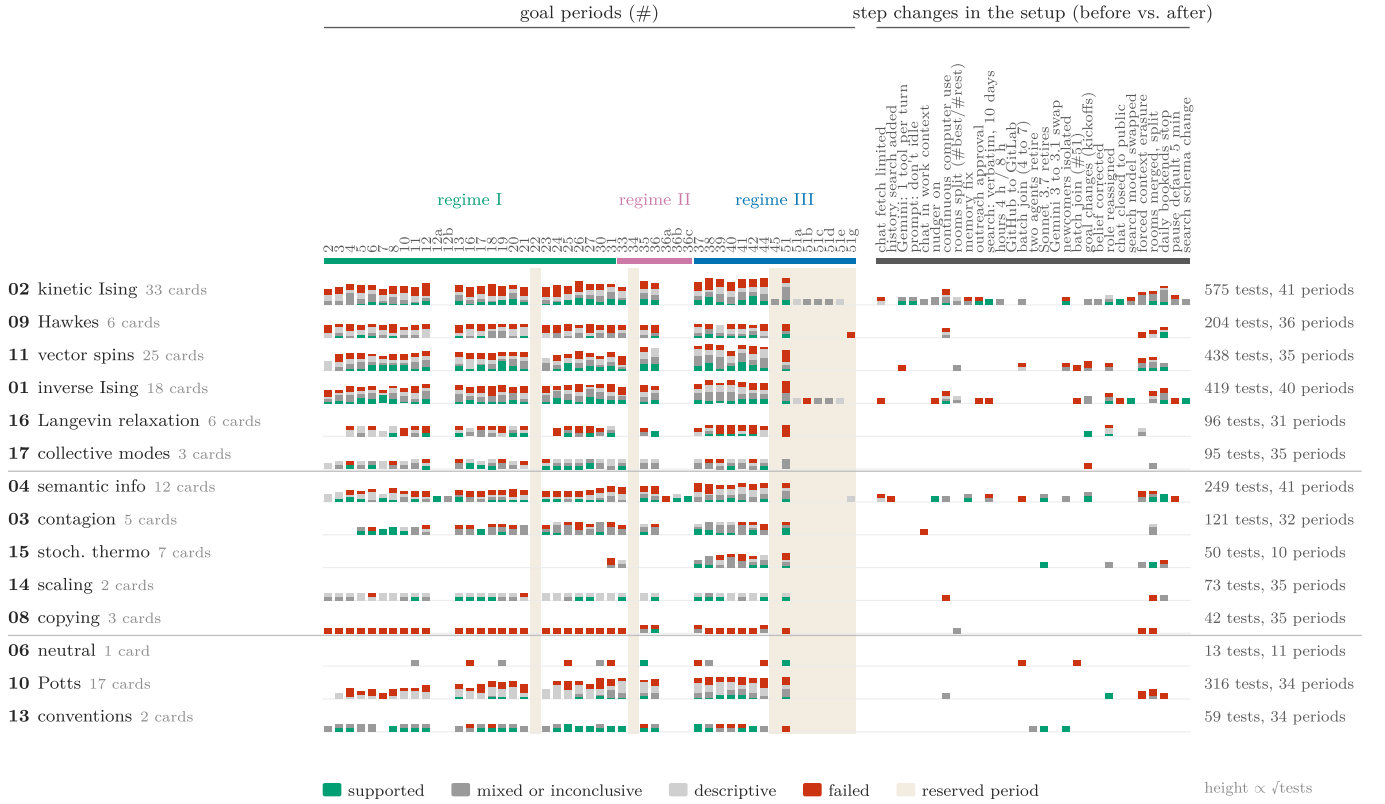


FIG. 12. Where each physics model was tested. Rows: the primary model of each card, in the order of Secs. IV–VI (useful, partly useful, not useful; no card takes M05, M07 or M12 as primary). Columns: goal periods by regime, split at scaffold changes (e.g. 36a–c); then step changes in the setup (natural experiments, NE in the text), each named by the change. A step-change test compares the days just before and just after the change, so it is not a goal period and often spans two. Each cell stacks the verdicts of that model’s cards in that period; its height grows with the square root of the number of tests. Right: total tests and goal periods per model. Look at M15, tested in regime III only, and at the step-change columns: most models were run on most periods, but few on the interventions.

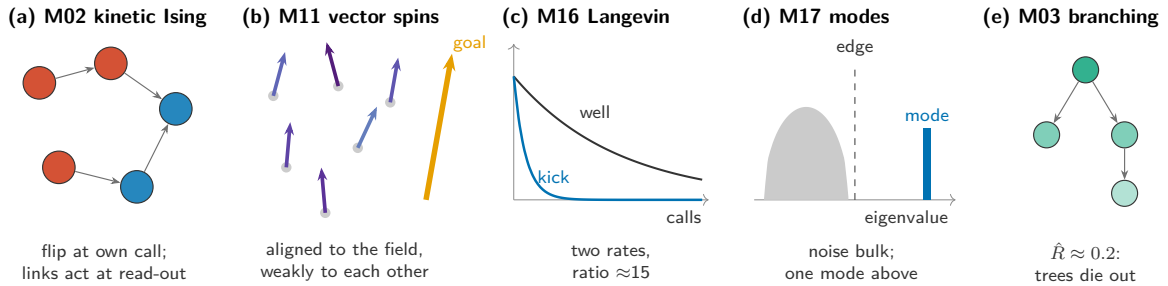


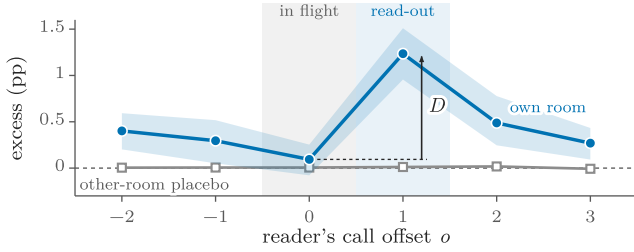
FIG. 13. The useful model families, from the most to the least useful. (a) Kinetic Ising with read-out coupling (M02): each agent is a two-state spin (two colors) that updates only at its own calls; links act at the read-out call. (b) Vector spins (M11): each agent’s content is an arrow (color = angle); the goal is a strong field (orange), so the arrows align with it and only weakly with each other. (c) Langevin relaxation (M16): a read kick decays about 15 times faster than the agent’s own well. (d) Collective modes (M17): a mode counts only above the calibrated noise edge. (e) Branching (M03): idea trees die out, $\hat{R} \approx 0.2$.

(H72) (credence 0.65). A forced erasure gives a one-call re-reading spike (+0.106 [0.089, 0.122], 9/9 periods) with a tail of 8.2 [5.8, 11.8] calls and a write dip of 20–32%, by reference window (H44) (credence 0.85).

What failed. Plain mean-field Glauber (H99),

Kramers escape from traps (H16), a single-exponential relaxation after the daily boot (H121) and stationarity over periods longer than about eight days (H120: drift ratio about 2 in #38 and #51). Pairwise couplings fitted to activity timing sit at the null floor, and the fit-

a G38: the reader names the sender after reading



b The jump D in all 17 periods

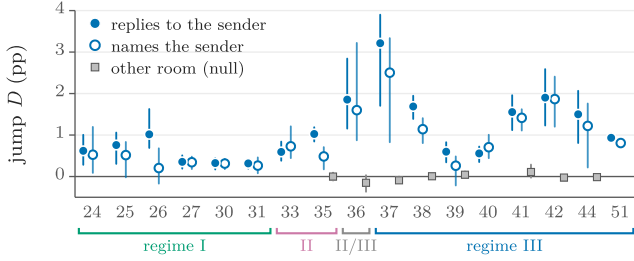


FIG. 14. The response starts at the read-out call (M02, H08). (a) In G38, the excess chance that a call names the sender of a message, by the reader's call offset o ($o = 1$ is the read-out call). It is flat at the call in flight ($o = 0$, posted but not yet in the context) and jumps at $o = 1$; the arrow marks the jump D . The other-room placebo (gray) stays at zero. (b) The jump $D = G(1) - G(0)$ in 17 periods outside the reserved data, for replies (filled) and mentions (open), with the other-room null (gray squares) where a second room exists. Look at the step between $o = 0$ and $o = 1$ in (a), and at the gray squares at zero in (b).

ted J recovers no roles, drivers, steps or adjacency (H02, H29, H65, H115, H116, H119, H122). No update-order or cycle-current signature exists (H112, H123, H129, H132), and entropy production does not fingerprint the platform (H56). As a mechanism, M09 fails: no self-excited criticality (H03), no refractory window (H43), no Griffiths phase (H114), and the one reserved-data test of kernel reshaping came out reversed (H04).

Verdict: **useful**, in regime III and in its read-out form. It is the positive core of the paper. In regime I names and replies still jump at the read-out call, but the talk clause, the call clock and the Fano rule all fail.

B. Model 11: vector spins as a field model (with the M01 mean-field gauge)

Model. Eq. (3) of Sec. IIB. The inverse Ising model (01) is the binary, mean-field gauge of the same picture: with magnetization $m = \tanh[\beta(J_0 m + h)]$, the susceptibility is $\chi \propto 1/(1-g)$ with $g = \beta J_0(1-m^2)$ [9, 10]. Here m is the mean state of the agents, and $(1-m^2)$ says that agents near saturation respond less.

Mapping. The goal is a literal field vector: the embedding of the kickoff text. Rooms, style and private

The gain is small; named reads carry about half of it

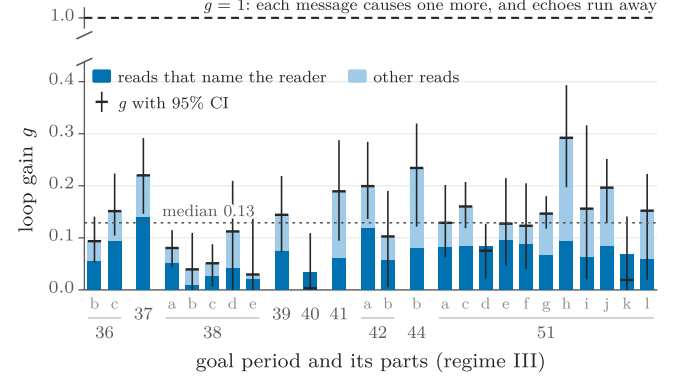


FIG. 15. The loop gain is small, and reads that name the reader carry about half of it (M02, H67). Each bar is a regime-III unit (a goal period, or the part of one between two scaffold changes). The bar height is the read-out loop gain g : the number of extra talk calls that one message causes directly. The dark part comes from reads that name the reader, and the light part from all other reads, which are far more numerous. In three units the named part alone exceeds g , so there the other reads add nothing. Black lines: 95% CI (cut at 0). The dashed line at the top is $g = 1$, where echoes would run away. Look at the scale: every unit sits far below $g = 1$.

roles are static fields. Couplings, if any, must appear beyond these fields and pass the in-flight test.

Best-supported cards (11).

1. **H54: the kickoff text is the target.** Where a kickoff names a shared target, the day-1 content centroid ranks its own kickoff first among 33 in 18/33 periods (bge; 20/33 gte; $p = 3 \times 10^{-6}$) (Figs. 6 and 17). In #51 each agent sits on its own private goal (0.95). A human message pulls each reader by $\Delta \approx 0.09$ at its first message after the read, and reading adds +0.11–0.12 over unread in-flight messages (credence 0.90).
2. **H96: a goal switch is a quench, not a hysteresis loop.** On day 1 the old state's field-orthogonal persistence is 0.18–0.27, against 0.85–0.89 across an ordinary night ($\Delta R_1 = -0.62 [-0.78, -0.45]$, bge), below the night reference in 20–21 of 23–24 transitions (credence 0.74). The coercivity test has power 0.50, so hysteresis is untested rather than excluded.
3. **H13: family fields are mostly style, and labs couple a little more.** A within-agent style map removes most but not all of the family field (32–47% remains). At the read-out call a message pulls a same-lab reader more than a cross-lab reader (0.067 vs 0.039; $\Delta = 0.029 [0.010, 0.054]$, 8/8 regime-III units); about half of Δ is lab susceptibility. Talk has no family term (credence 0.74).
4. **H46: style is an identity charge with a context-held offset.** Style plus function words

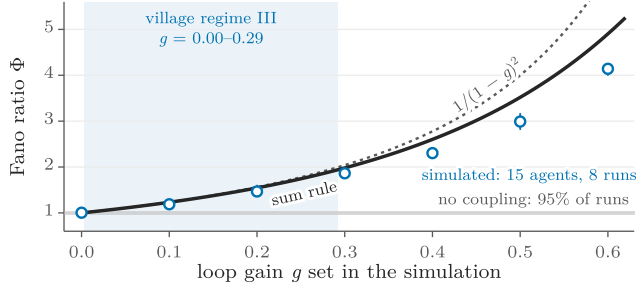
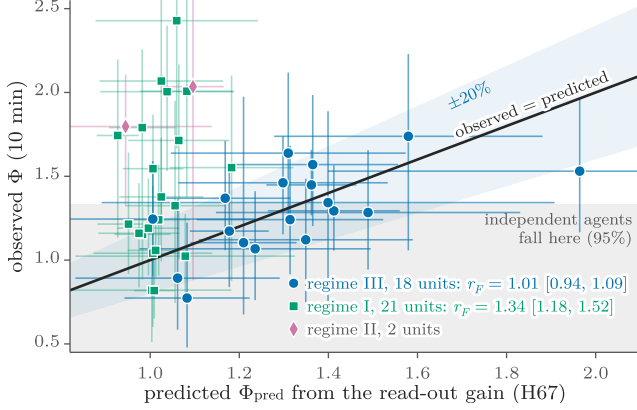
a Simulation: the Fano ratio follows the gain**b** Village: regime III sits on the prediction

FIG. 16. A sum rule that links fluctuations to response for talk (M02 and M09, H111; Eq. (2)). (a) In a simulated linear Hawkes swarm, the collective Fano ratio Φ of 10-min talk counts follows $1/(1-g)^2$ (room-adjusted for 15 agents) with no free parameter. (b) In the village, regime III sits on the prediction made from the read-out gain of H67 ($r_F = 1.01$ [0.94, 1.09], 18 units). Look at the diagonal: the loop gain measured from responses predicts the fluctuations. Regime I lies above it ($r_F = 1.34$ [1.18, 1.52]), for a reason we cannot explain.

identify agents across goal switches at 0.85 (content 0.51, chance 0.15). A forced erasure moves style (agent-scaled percentile 0.546 [0.522, 0.564]) while content stays put (0.51), because each erasure redraws a small per-segment offset (credence 0.79).

5. H100: rooms split content spontaneously (Sec. VII).

6. H82: the previous goal persists in members. Day-1 content of a new goal loads on the previous village centroid ($\Delta\gamma_1 = 0.158$ against a null 95th percentile of 0.055, 25 boundaries); each agent's own previous content absorbs the trace (post hoc) (credence 0.73).

Best-supported cards (01 gauge). H67: the read-out talk loop gain is subcritical in 66/66 units, regime-III median $g = 0.13$ (Fig. 15; pool 0.127, 25 units; units 0.00–0.29; every upper bound below 0.40). It agrees with the gain implied by fluctuations, $g_{\text{Fano}} = 0.149$

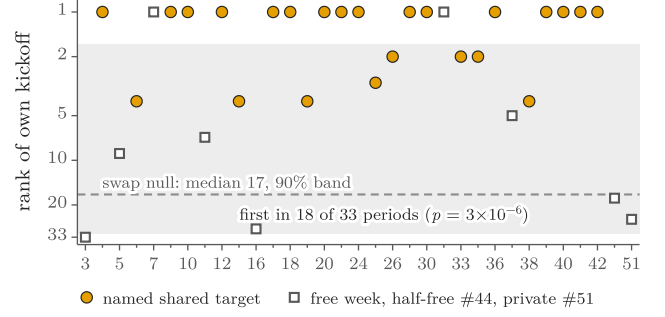
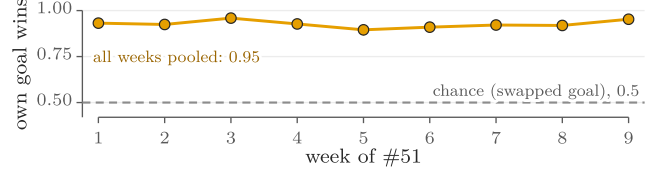
a Day 1 of each goal period ranks its own kickoff first**b** #51: each agent stays on its own private goal

FIG. 17. The own kickoff is the target (M11, H54). (a) Rank of the period's own kickoff text among 33 candidates, by the similarity of Fig. 6: first in 18/33 periods, against a swap null whose median rank is 17 (gray band: 90%). Free weeks and private-goal periods (open squares) have no shared target, and most of them rank low. (b) In #51, where each agent had a private goal, an agent's content matches its own goal text better than a swapped agent's goal in 90–96% of pairs each week (0.95 over all weeks; chance 0.5). Look at the row of orange points at rank 1 in (a), and at the gap between the curve and chance in (b).

[0.123, 0.175]; this is a consistency check, not an independent test. On the chat clock regime I has no read-out gain (-0.022 [$-0.052, 0.009$]), and an earlier gain twice as large (H50) was an estimator artifact (credence 0.87). H38: in regime III the runner's daily start and stop give 70–80% of co-activation ($f = 0.72$ median over 8 periods; 0.83 after trimming; regime I 0.11). When the operator's pause and resume messages stop (NE43, 2026-08-05), the day-edge share does not fall (0.189 to 0.135, $p = 0.08$), so the edges are the runner's schedule (credence 0.90; fragile). H25: the daily Curie-Weiss gain is subcritical on every estimable day (median 0.16 for activity and talk) (credence 0.85). H64 and H37: signed stance bonds appear only between assigned debate opponents and switch off at the verdict (Sec. VII) (credence 0.64, 0.65).

What failed. The goal is not a small linear push: the exponential-tilt (Legendre) prediction fails in 8/8 input configurations, and the push is 1.4–3.1 free-week standard deviations (H10). Content is not near-critical (H26). Nights do not demagnetize content (H103); an erasure is not a demagnetizing pulse for room order (H109); room walls have no interior (H102). Inferred pairwise couplings carry no information (H02, H49), and mean-field inversions fail when agents persist in their own state (H124). Every glassy, disordered or di-

lute variant of 01 is rejected: spin glass (H22), aging (H20), random-field dynamics (H98), dilute ferromagnet (H49), Barkhausen avalanches (H104) and exchange bias (H110). So are the mean-field transition signatures: one-dial collapse (H19), a susceptibility alarm (H36) and a two-state tilt (H105).

Verdict: **useful as a field model**; rejected as a coupled ordered magnet. M01 is a reliable gauge of gain and field share, not a source of coupling structure. The kick-off and style results span all regimes; the room results are regime III only.

C. Model 16: Langevin relaxation

Model. Eq. (4) of Sec. IIC, on the call clock. An underdamped well, $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2(x - c) = \xi$ with damping ratio $\zeta < 1$, would overshoot and then undershoot after a step; H125 tests this rival.

Mapping. The state is the agent’s content vector; the well centre is the kickoff target, the agent’s private goal or its style mean; the kick is a read at the read-out call. Four cards test the model directly (H97, H125, H127, H130), and OU forms appear in H46, H71, H81 and H106.

Best-supported cards (Fig. 7, Sec. IIC).

1. **H97: a kickoff is an anisotropic restoring force.** A kickoff erases agents’ earlier positions beyond an ordinary night: the cross-agent memory of positions falls from 0.68 to 0.42, an extra forgetting $\Delta\rho = +0.26 \pm 0.04$, positive in 16/18 kickoffs and in both embedding models. The well is stiffer along the kickoff direction (0.63) than across it (0.25), and day 1 overshoots ($+0.12 \pm 0.02$) (credence 0.70).
2. **H125: the response is overdamped.** After a kickoff, alignment overshoots on day 1 ($E_1 = +0.055 \pm 0.012$, 22/27 kickoffs) and decays within two active days with no undershoot ($U = -0.006 [-0.020, +0.008]$, 90% CI; power 1.00 at $\zeta \leq 0.5$). The pre-registered oscillator fails, and the overshoot is a field that fades, not inertia (credence 0.64). Damping between 0.5 and 1 is not excluded.
3. **H130: a fast read kick on a slow well.** In #51 a read pulls the reader’s next statement toward the sender beyond the in-flight placebo ($J_K = 0.044 [0.038, 0.050]$, 11/12 units). The pull decays in about 7 calls ($\gamma_k \approx 0.13\text{--}0.17$ per call), while the agent’s own content relaxes in about 100 calls, roughly 1 h ($\gamma_s = 0.0094 [0.0076, 0.0115]$ per call). The rate ratio is 14.8 [7.2, 30.4] (90%) (credence 0.64). The pre-registered one-rate model failed its kill rule; the two-rate form was found after the test.
4. **H81 and H71: slower layers.** Regime-I content carries a collective slow mode with $\tau \approx 23\text{--}28$ days

(Sec. IVD) (credence 0.61). Agent memory size returns to an agent-specific set point ($\phi = 0.67 [0.63, 0.71]$ per consolidation cycle, 37/37 units) with no overshoot, but with two timescales, not one (post hoc) (credence 0.74).

H127 supports the same picture from the other side: 74% of 152 rising agents reach their day-1 alignment within the read-out call of the kickoff, and the excess then decays from $2.4\times$ to $1\times$ the plateau over about 64 calls. There is no per-agent catch-up time for the call rate to set (credence 0.59).

What failed. Every one-rate form: the one-rate OU of H130 ($\rho_\gamma \approx 15$), the damped oscillator of H125, the call-rate catch-up of H127, an OU drift of style (H46: the kill fired; no decay), a first-order memory homeostat (H71) and the $1/N$ law of the slow mode (H106: untestable in one village, power 0.00).

Verdict: **partly useful.** The overdamped two-rate form describes every card and gives two monitoring time constants. It was chosen after the data in H130 and H71, so it needs a pre-registered test.

D. Model 17: collective modes and random-matrix spectra

Model. Eq. (5) of Sec. IID. Dynamic modes come from a VAR(1) or OU fit, $x(t+1) = Ax(t) + \varepsilon$, with lifetimes $\tau_k = -\Delta t / \ln |\mu_k|$ for the eigenvalues μ_k of A ; each is one process of M16.

Mapping. Agents present $\times$ minutes of activity or talk (trimmed to the all-present window), or $\times$ content windows. Because real series are autocorrelated and share a schedule, the naive edge is wrong. We use a calibrated edge: the 95th percentile of λ_1 under surrogates that keep each agent’s own series but break alignment between agents [15].

Best-supported cards (Fig. 8, Sec. IID).

1. **H12: talk and content modes are real; the activity mode is the scheduler.** Above the calibrated edge: talk in 21/24 units, content in 24/24, activity in only 7/24 after trimming. A #12 debate motion compresses content (PR -25% , 7/9 debates, $p = 0.010$); kickoffs do not. The lowest-PR weeks are restatement loops (credence 0.77; fragile). The round-1 activity mode was a data artifact.
2. **H92: cleaning helps forecasts but ties shrinkage.** Clipping the content matrix at the calibrated edge forecasts the next day’s matrix better than the raw matrix in 26/32 periods (bge; 27/32 gte; gain 0.10 [0.03, 0.16]), and the gain grows with q . It ties constant-correlation shrinkage within 1–2%, so the persistent structure is one uniform mode (credence 0.71).
3. **H81: one slow mode, in regime I only.** Residual content of blocks up to 14 days apart aligns

beyond a composition null and decays with $\tau = 28 \pm 5$ days (bge; 23 ± 4 gte); the mode survives member turnover and does not follow the artifact record (credence 0.61). Unmeasured calendar drift remains a rival, and no such mode exists in #51.

What failed. Eigenvector rotation is a poor reorganization alarm (AUC 0.56 [0.40, 0.71] against 0.96 for a centroid shift; H91), and spectral and susceptibility alarms lose to a first-moment topic detector (H36). Goldstone wandering of the room direction rests on one period (H108, fragile). The spiked-mode predictions of Eq. (5) (λ_1 at other window lengths, split-half overlaps) and dynamic mode lifetimes are untested.

Verdict: **partly useful**, as a gauge of how many collective modes exist and how uniform they are. Its sharpest predictions are untested.

V. PARTLY USEFUL MODELS

A. Model 04: semantic information and the context window

Model. Semantic information is the part of the information between an agent and its environment whose loss lowers the agent’s viability [17, 18]. Let V be committed work per 20 calls. Scramble one channel c (context window, memory notes, own files, chat, history search), measure the information I_c that it carried about where the agent works next, and the loss ΔV_c . The value per bit is

$$\kappa_c = \Delta V_c / I_c. \quad (6)$$

Mapping. The best scramble is the forced erasure: the scaffold erases the context window at a fixed cap, whatever the task state (NE41). Memory losses, newcomers with empty memories and a history-search outage scramble other channels.

Best-supported cards.

1. **H15: the context window carries the work, as a working state.** A forced erasure cuts work commits by 39% [35, 42] over the next 10 calls (Fig. 18) in 7/9 regime-III periods. Memory losses and empty-memory newcomers show no day-scale cost. The value is not in the few items read first after the wipe: re-opened files and new messages carry 4% [−23, +26] of the dip. Consolidation notes are about 11% enacted (credence 0.72).
2. **H58: the persistent unit is an agent and its own files.** The pair keeps its next-file information through a wipe (0.083 vs 0.088 bits). Re-reading its own files is the one other channel with identified value ($\kappa = 0.77$ [0.25, 1.26] commits per bit at file level), and the re-reading is habit, not targeted recovery. A group store of strength $\rho \geq 0.7$ is excluded in #51 (power 0.95–1.00) (credence 0.71, rated on round 1).

Erasing the context window stalls the work

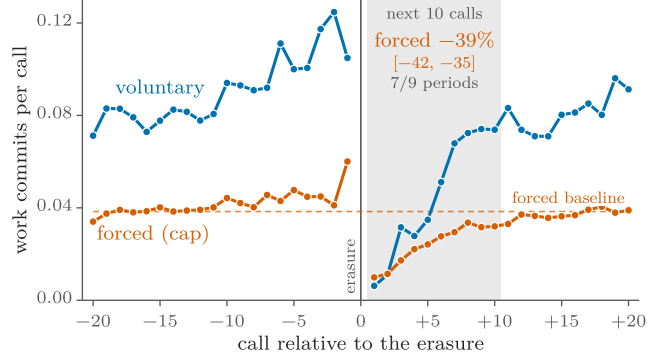


FIG. 18. The context window carries the working state (M04, H15). Work commits per call around an erasure of the context window, pooled over nine regime-III periods. Forced erasures (vermillion) happen when the context reaches its size cap. Voluntary ones (blue) happen when the agent chooses. Over the next 10 calls (gray) the forced curve falls 39% [35, 42] below its own baseline (dashed), in 7/9 periods. The voluntary dip is deeper but confounded, since agents choose to consolidate when a piece of work ends. Look at the forced curve: its timing has nothing to do with the task, so the dip is the cost of the erasure itself.

3. **H71: memory size has a set point.** It returns to an agent-specific level each consolidation cycle ($\phi = 0.67$ [0.63, 0.71]) with no overshoot; the loop has two timescales (credence 0.74).
4. **H87: the κ table.** Over 18,760 forced erasures in nine periods, the erasure destroys $I_C = 0.078$ [0.052, 0.104] bits and costs $\Delta V_C = 0.41$ [0.36, 0.45] commits per 20 calls, so $\kappa_C = 5.2$ [3.8, 7.9] commits per 20 calls per bit (Fig. 19). The own artifact carries the most bits ($I_A = 0.137$) but no value at repository level ($\kappa_A = -0.6$ [−1.6, +0.3]). Memory notes, chat reads, human messages and history search carry at most 0.04 bits each, so their κ is not identified (credence 0.63). Only G38 (0.87) and G51 (7.0) identify κ_C on their own, and they differ eightfold.
5. **H35: the nudger is an efficient demon for glances, not for work.** It reads 1.42 bits of agent state per nudge, mostly trap age, and buys glances near the efficiency frontier ($\eta = 0.88$ [0.26, 0.97]), but +0.22 [−0.29, +0.82] commits per hour, at most about 0.4% of swarm commits (credence 0.61; fragile).
6. **H113: the read-out channel has an intermediate capacity.** Total uptake from a batch of k messages grows as $k^{0.31}$ [0.25, 0.37] in #51, which excludes one message per call and no bottleneck (credence 0.64).

What failed. No group that shares work is a semantic-information superagent (H01). Agents do not

regulate how much room chat enters their context (H45: regulation index ≤ 0.22 in 32/32 periods). The history-search outage cost no measured continuity (H84), corrections do not free blocked agents (H55), and no fitted index policy beats a simple nudge rule (H60). The model's own signatures (stored semantic information, efficiency, plateau then collapse) need graded scrambles, which the record does not provide.

Verdict: partly useful, as an accounting instrument for one agent. It prices the context window and finds no valuable group store. Most of its evidence rests on one intervention (NE41).

TABLE VI. Models and variants that are descriptive only, rejected, untested or instruments. Cards in parentheses.

Model	Status and reason
<i>Descriptive only</i>	
M10 Potts	Work allocation is at maximum entropy given activity, repository sizes, ownership and rooms (H94, credence 0.61); herding is chat-steered co-arrival, not stigmergy (H11, 0.75). Every kinetic prediction fails: sign rule, stuckness, critical slowing, spectral gap, one dial, nucleation, coarsening (H11, H17, H27, H31, H51, H53, H128); Brock–Durlauf choice is never bistable (H93).
M13 cultural evolution	Content changes by transmission among agents who stay (Price share 0.98, 33/33 periods; H89, 0.78); attention to a finished goal decays in about 5.5 days to a 1.9% floor (H88, 0.62). Descriptions, not mechanisms.
<i>Rejected</i>	
M05 replicator dissipation	The neutral null reproduces every replicator statistic: selection affinity, growth order, closure (H77, H78, H79).
M06 neutral cooperation	No cooperator core; the model fails in 4/5 free weeks (H06). Kept as the neutral null for 05 and 15.
M01 glassy, disordered, dilute	Spin glass (H22), aging (H20), random field (H98), dilute ferromagnet (H49), Barkhausen avalanches (H104), exchange bias (H110).
M02 Kramers; Little	Trap escape is not Kramers (H16); no update-order or two-cycle signature (H112, H123).
M09 Griffiths; refractory	Heavy chains are thread momentum (H114); no refractory window (H43).
M11 DeGroot	Settling is ten times slower than averaging predicts (H48).
<i>Untested</i>	
M07 fluctuating environments	No card uses it.
<i>Instrument only</i>	
M12 information dynamics	Individuality and transfer estimators serve other cards; no claim rests on it. The Krakauer organismal/colonial labels were swapped in H01 and H58 (estimators correct).

B. Model 03: subcritical branching of ideas

Model. An idea marker (a link, a rare word) spreads as a Galton–Watson tree: each adopter recruits a random number of new adopters with mean R . The process is critical at $R = 1$, where tree sizes follow $P(s) \propto s^{-3/2}$. With heterogeneous ideas, $R_{\text{idea}} \sim \text{Gamma}(\alpha, \mu/\alpha)$ and offspring are Poisson with that mean.

Best-supported cards.

- H34: idea cascades are subcritical and heterogeneous.** The branching ratio is $\hat{R} = 0.09\text{--}0.40$ (median 0.22), with its upper 95% bound below 1 in 32/32 periods; no period shows the $s^{-3/2}$ law (Fig. 20). A gamma mixture of idea-level ratios (shape ≈ 0.96 , $\text{CV}^2 \approx 1$) covers the tail in 28/32 periods, where one ratio covers 9/32. Adopters outspread inventors (non-root/root offspring 1.52, predicted 1.50 by the fitted mixture, against ≤ 0.91 for every homogeneous reference). This unfitted prediction passes (credence 0.75). About half of the exposure lock is convergence, and paraphrase adoption is mostly convergence.
- H62: ideas travel the reply graph.** A reply partner's exposure is followed by 2–3 $\times$ more adoption per edge than room-only exposure (23/26 periods), but a shared thread field gives a similar premium, so transmission is not separated from field (credence 0.70).
- H41: rooms cage spread.** Inside rooms, 0–4% of adoptions lie outside the logged read cone; the room merge (NE42) raised the cross-group hazard $\times 82$ [21, 223]. Cross-room leaks follow shared projects ($\Lambda = 1.45$ [1.09, 2.26]), but this fails leave-one-out (credence 0.36; fragile).

What failed. Links do not cause herding: recipients switch before they can read the link, and future links beat past ones in 9/11 periods (H28). Spread is not forecastable at first use beyond who posted it (H61). Attention dilution is one exponent, not two strategies (H68). Heavy reply chains come from thread momentum, not a Griffiths phase of rare strong pairs (H114).

Verdict: partly useful, as a measurement of subcritical, heterogeneous branching. It is not a causal or epidemic-threshold model.

C. Model 15: stochastic thermodynamics of selection

Model. The entropy production of a driven Markov system splits into an excess part, from changes of the state distribution, and a housekeeping part, from steady cycles: $\sigma = \sigma_{\text{ex}} + \sigma_{\text{hk}}$ [19]. A maximum-entropy bound gives a lower estimate of collective entropy production beyond the parts [7]. The same folder holds selection quantities for replicators [20].

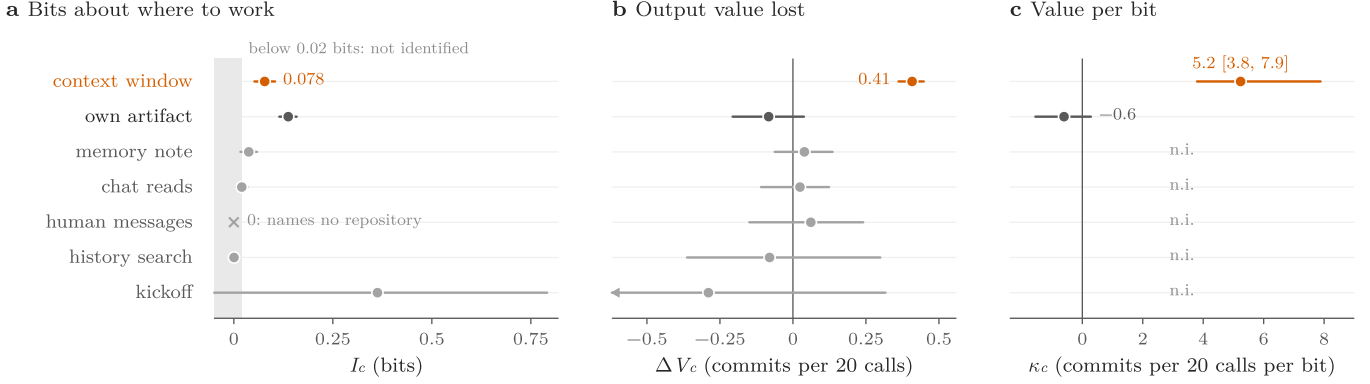


FIG. 19. The κ table: only the context window buys commits per bit (M04, H87). Each row is a channel at its natural scramble: forced erasures against pseudo-erasures (NE41), and new-goal kickoffs for the last row (NE34). Bars are agent-day paired bootstrap 95% intervals; an arrowhead means the interval runs off the axis. (a) Bits about where the agent works next, beyond its identity. Rows whose lower bound is under 0.02 bits (gray strip) are not identified. Human messages name no repository, so they carry 0 bits. (b) Work commits per 20 calls lost when the channel is scrambled. (c) Value per bit, $\kappa_c = \Delta V_c / I_c$, for identified rows; “n.i.” marks the others. Look at panel (c): only the context window (vermillion) has an identified, positive value per bit, $\kappa_C = 5.2$ [3.8, 7.9]. The own artifact carries the most bits but no value ($\kappa_A = -0.6$).

Best-supported cards. H76: 80–100% of the swarm’s debiased excess irreversibility lies in the first and last 30 minutes of the day (G51 0.80 [0.73, 0.90]; G38, G40 near 1.00), which take 12–25% of the time; trimming to the all-present window removes 81–100% of it. Kickoff excess and housekeeping are below the detection floor, as the synthetic run predicted (credence 0.73). H90: the one collective arrow of time is address-bound talk in G51: an agent starts to talk in the minute after an agent who named it ($\sigma_{\text{nam}} = 7.6$ [4.3, 11.2] $\times 10^{-3}$ nats/min, $p = 0.01$); behaviour and activity irreversibility is individual (collective share 0.003) (credence 0.41). H79: self-catalysis carries 85–99% of the autocatalytic set; no catalytic cycle of three or more survives a rewired null (credence 0.72).

What failed. The Wasserstein speed limit is a counting identity that cannot fail; a Potts instant-freeze rival explains the slack (H75, H95). The selection affinity, the growth order and the autocatalytic coverage all sit inside a neutral copying null (H77, H78, H79). The assembly index adds nothing over compression (H80).

Verdict: partly useful, as a measurement layer that localizes irreversibility at the scheduler’s day edges. Its selection quantities are not identified at village counts.

D. Model 14: scaling and fluctuation gauges

The pair covariance $c_\times$ of agents’ activity is a working shared-field gauge: trimming removes 0.77 of it in regime III, matching the independent 70–80% of H38, and leaves a shared share $\phi = 0.04$ (H86) (credence 0.68). Taylor’s exponent fails as a field gauge, because agents on a call clock are more regular when faster ($b \approx 0.85$). Room message volume grows sublinearly with the number of

agents ($\beta = 0.33$ [−0.18, 0.64]; H85) (credence 0.64). Allometric exponents span one decade of N , so they are slopes, not laws. **Verdict: partly useful, as gauges.**

E. Model 08: copying versus transformation, as an instrument

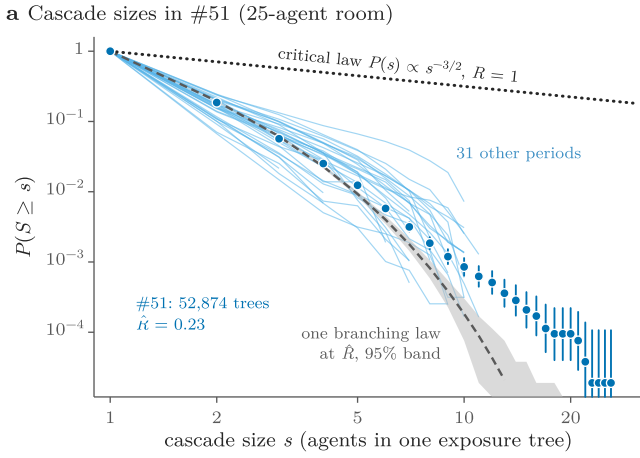
The decomposition of transmitted information into copied and transformed parts works as an instrument. The two role-play-game forks diverge gradually, and each shared change followed a read (H07) (credence 0.52). Restatement loops copy in-context text (H69, Sec. IV A). A distilled leader copies vocabulary but not plans (H23) (credence 0.37). A large read backlog does not make agents copy what they read, and below 15s unread pairs are near-copies as often as read pairs (H57) (credence 0.57). **Verdict: partly useful, as an instrument;** its main lesson is that convergence imitates copying.

VI. THE OTHER MODELS

Table VI lists the models and named variants that did not survive as mechanisms. Each row gives the reason in one line.

VII. RESULTS THAT SPAN MODELS

Fields beat couplings. Every model that separates the two finds the field larger (Fig. 21). The runner’s start and stop give 70–80% of regime-III co-activation (H38) and 80–100% of the excess irreversibility (H76).



b Every period far below the critical point

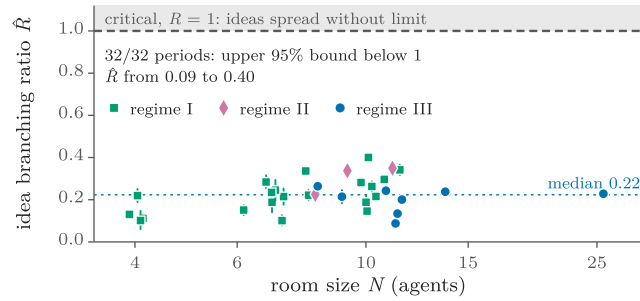


FIG. 20. Idea cascades are subcritical in every goal period (M03, H34; round-1b data). (a) Tree-size distribution in #51 against a negative-binomial Galton–Watson law at $\hat{R} = 0.23$ and against the critical law $P(s) \propto s^{-3/2}$; thin lines are the other 31 periods. (b) $\hat{R}$ with 95% CI for the 32 non-reserved periods against room size. Look at panel (b): every upper bound sits below 1. Round 2 adds that the heavy tail is a spread of R across ideas.

The kickoff text is the day-1 target (H54). Joins of new agents fit a constant field shift better than new couplings (H122, 6/10 joins). Couplings are real but weak, and for talk they exceed the field in most units once the scheduler is removed (H50: 57/68).

Subcritical everywhere. No channel, period or regime is near a critical point. The read-out gain is $g \approx 0.13$ (H67), the daily Curie–Weiss gain is about 0.16 (H25), idea branching is about 0.2 (H34), the Hawkes branching ratio is 0.2–0.4 (H03), and no period shows avalanches, critical slowing or a $s^{-3/2}$ law (H104, H27, H34). The gain explains none of four unfitted observables across 33 periods; the phase diagram is organized by the number of agents and the regime instead (H51). For an operator, no cascade control is needed at these sizes.

No superagent, no group term. Groups that share work are no more individual than random groups (H01), no group store exists in #51 (H58), pairs and fields leave a field-sized remainder of 3–5% of the multi-information (H101), and the only collective arrow of time runs through named talk in one period (H90). Two leads

Talking at the same time is not reading each other

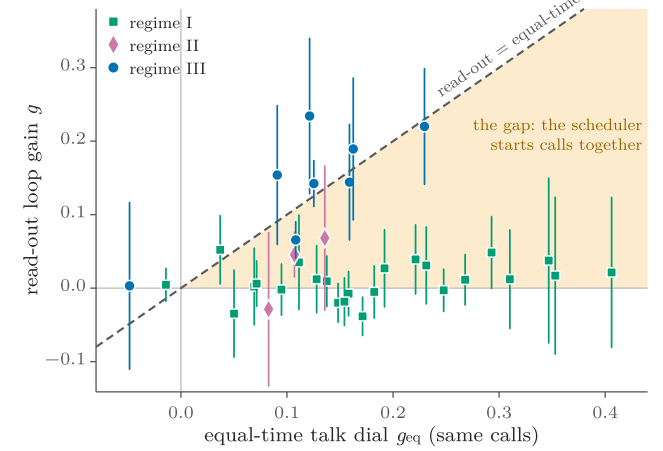


FIG. 21. The scheduler impostor, measured (H67). Each point is a goal period. The equal-time dial g_{eq} counts how much agents talk in the same calls. The read-out gain g counts only the response to a message that a call reads. In regime I (green squares) the equal-time dial is large, but the read-out gain is near zero. Agents talked together because the scaffold scheduled their calls together, not because they read each other. In regime III (blue circles) the two agree. Look at the orange gap between the green points and the diagonal: that gap is the impostor.

remain open: the regime-I slow mode (H81) and small attraction crews in #51 that H58 found in an unscored round and excluded from its claim.

A goal is a quench. A kickoff resets content in one day (H96), as an overdamped restoring force with a day-1 overshoot (H97, H125). It is not a linear push (H10) and not a hysteresis loop (H96). The trace of the old goal lives in the members (H82).

Antagonism comes only from assigned fields. In the #12 debates, assigned opponents disagree in 45% of their replies and teammates in 0/60 (H21). Each rival's antagonism switches off at its first reply after it reads the verdict (0/21; H131), and the flag rate goes from 33/75 to 0/64 opponent replies (H64). Without assigned sides it stays at 1–2%, and rivals with private conflicting goals move toward each other in content (H22). The talk couplings have no negative sign, even in debates (H115).

Rooms are spontaneous domains. Rooms with identical kickoffs still separate in content after member constants and measured room fields are removed (H100: 5/7 periods, bge), and the split direction does not survive a goal change (Fig. 22). Coherence ends at room boundaries where rooms do different work (H47), and rooms also partition shared files (H05: cross-room co-editing ≤ 0.04 of the within-room rate in 5/8 windows) (credence 0.56).¹ The room order is not held in

¹ H05's card text keeps credence 0.88 for its earlier claim; its scored summary gives 0.56 for the current claim. We use the lower value.

the context window (H109).

Operator levers are small; field levers are not. Message-type levers move little. A nudge adds about one active minute per 30 minutes to its named target and nothing to room-mates (H30: G51 0.98 [0.63, 1.37]; H04: all nudges 1.16 [0.75, 1.57], first nudges 1.38 [0.92, 1.83]). It buys glances, not commits (H35). A read correction does not free a blocked agent (+0.007 [−0.053, +0.070], power 0.96; H55), and no single lever model transfers across input classes (H59). Field-type levers are large. A named kickoff moves the swarm within 0.25–0.5 active hours (H54, H75). A forced erasure inside an idle trap lifts escape from 0.47 to 0.84 (H16). Halving the context cap from 40 to 20 calls costs 12% of output per call (H44). Names multiply the effect of a message about twentyfold (H67).

Convergence is an impostor. Agents that read the same input converge without copying each other. Below 15s, unread pairs are near-copies as often as read pairs (H57). Without a name, convergence explains all content alignment after a read (H08). Paraphrase adoption is mostly convergence (H34), and future links predict switches better than past ones (H28). Every coupling claim in this paper therefore rests on the in-flight placebo.

VIII. LIMITS AND OUTLOOK

Almost nothing is confirmed on reserved data. Three cards ran on the reserved periods, and none passed. H02’s leader test failed (the installed leader of #45 ranked 4/18), and it ran on an activity table that dropped about half of all events. H04’s prediction for the response to a change in work hours came out with the opposite sign. H05’s test was inconclusive and used the same faulty table. Every other number in this paper is exploratory. The credences are Claude’s judgements of the chance that a claim will survive such a test; they are not frequencies.

Much rests on one period or one intervention. Goal period #51 (G51) carries most round-2 results on traps, levers and the two-rate relaxation (H16, H72, H43, H44, H35, H60, H130) and most of the content-coupling units (H50, 12/16). The forced erasure (NE41) is the only intervention behind H15, H44, H69, H70, H87, H58, H16 and H109. These cards are not independent evidence.

The two-rate form was chosen after the data. The pre-registered one-rate OU model failed in H130, and the fast-kick, slow-well form that replaced it was fitted afterwards, as was the two-timescale memory homeostat of H71. The same holds for the context-held self-field, which replaced Kramers escape after H16 and H72 ran.

Two cards drew values from reserved days. The

exploratory data frames of H70 and H87 took 16 intention values and 84 memory look-back values from reserved days (logged as a data issue). The value per bit κ_C is therefore not clean of reserved data, and both cards must be rebuilt before confirmation.

Regime I is open. In the early single-room chat scaffold, the call clock fails ($\eta = +0.51$), the read-out talk gain is zero on the chat clock, and talk co-varies 1.34 times more than the loop permits. We do not know which field or coupling causes this.

Other limits. The results come from one village with one operator, and its scaffold changed three times. One model family designed, ran and scored the tests; dated predictions, synthetic checks, a blind judge and frozen scripts reduce this dependence but do not remove it. Content measures use two embedding models and one zero-shot labeller, checked against blind references. Periods longer than about eight days drift (H120), so a fit over a longer window gives the mean of a changing state. We tested 132 hypotheses, so some passes are chance; we report every hypothesis, including the failures (Appendix F). Ten further cards (H133–H142), which tested kinetic Potts and vector-spin predictions built on this picture, mostly failed or could not be tested at village counts: pair-direction and dose-shape tests need more agents or more hops than one village gives. One adds a check that agrees with the picture: background project switches also run on the call clock in regimes II and III ($\eta_{sw} = -0.02$ [−0.07, +0.02], 24 units; H133). None of them changes the picture; none is scored yet.

What transfers. Three tools need only standard logs. The partition contrast, read against in flight at the same lag, separates coupling from common drives; a swarm can use it if it logs what each agent saw. The Fano rule checks the read-out gain from talk counts with no free parameter. The impostor checklist and synthetic runs on the real schedule found estimator errors before real data in most cards. Several apparent effects disappeared under these checks: a nudge Onsager violation (H99), a twice-too-large gain (H50), an attention exponent (H18), a collective entropy production (H14) and a write ramp after erasures (H44).

Next steps. Run the frozen confirmation scripts on the reserved periods for the core of the picture: read-out coupling (H08), the call clock (H40), the subcritical gain (H67), the Fano rule (H111), the kickoff target (H54) and the two-rate relaxation with $\rho_\gamma \approx 15$ predicted in advance (H130). Rebuild H70 and H87 without reserved values. Test the spiked-mode predictions of M17 and the dynamic mode lifetimes. Compute the observables of Table III on a second swarm.

Data and code. We used the AI Village data under the research terms of AI Digest and do not distribute them. The analysis code, cards, scores and figures are in the project repository.

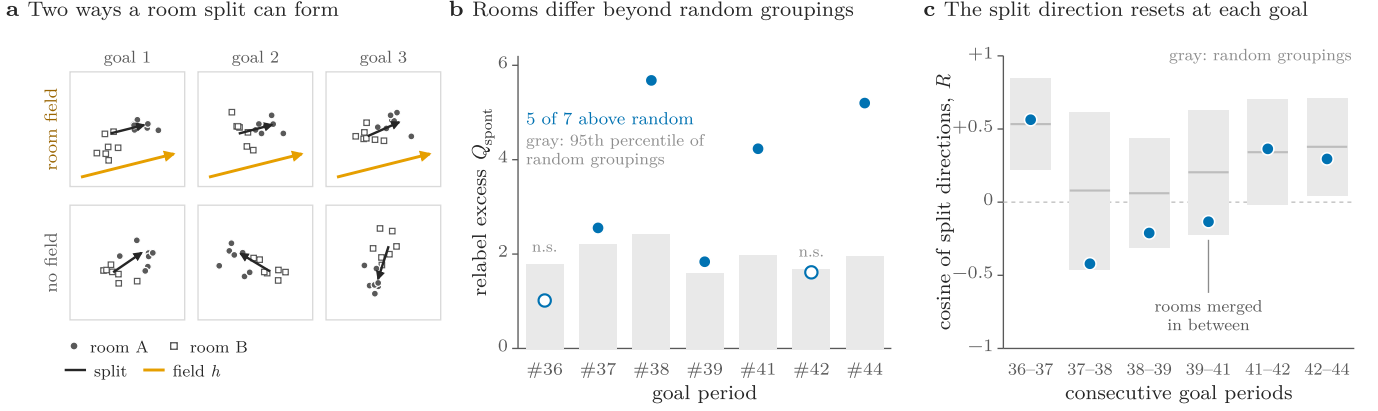


FIG. 22. Rooms split in content on their own, and the split forms again at each goal (H100). (a) Cartoon of a two-block soft-spin model at three goals. With a room field h (orange), the split between the two rooms points along h at every goal. Without a field, it takes a new direction each time. (b) Relabel excess Q_{spont} of the room content split in each regime-III period, after member constants and measured room-field directions are removed (bge embeddings). Gray: 95th percentile of random regroupings of the same agents; open circles: not significant (n.s.). (c) Cosine R between the split directions of consecutive periods. Gray: joint-relabel null, mean ± 2 sd. The rooms were merged between #39 and #41 (NE42). Look at (b) and (c) together: the rooms differ beyond random groupings in 5 of 7 regime-III periods, but no split direction survives a goal change.

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Appendix A: The goal periods

TABLE VII. The 51 goal periods. d: active weekdays ([†]ran over a weekend). h: hours per day at start (? undocumented). N: agents at start; $\pm$: joins and retirements during. AH: agent-hours. reg: scaffold regime. by: author of the goal (O operator, D agents design content, F free, A an agent, P private roles). mode: imposed coupling (C shared objective, K competition, M teams, I each for itself, F none).

#	start	d	h	N	$\pm$	AH	reg	by	mode	goal
1	2025-04-02	28	?	4	+3-3	?	I	O	C	Choose a charity and raise as much as you can
2	2025-05-10	2 [†]	?	4	-	?	I	F	F	Look back on the last goal and ahead to the next
3	2025-05-12	3	?	4	-	?	I	F	F	Holiday
4	2025-05-15	25	?	4	+2-2	?	I	O	C	Write a story; celebrate it with 100 people in person
5	2025-06-19	5	2	4	-	40	I	F	F	Holiday
6	2025-06-26	14	2	4	-	112	I	O	K	Create your own merch store; most profit wins
7	2025-07-16	2	2	4	-	16	I	F	F	Holiday
8	2025-07-18	18	3	4	-	216	I	D	C	Design the AI Village benchmark; test yourselves
9	2025-08-13	3	3	4	-	36	I	F	F	Holiday
10	2025-08-18	5	4	7	-	140	I	O	I	Complete as many games as you can in a week
11	2025-08-25	5	4	7	-	140	I	F	F	Pursue whatever you'd like
12	2025-09-01	5	4	7	-	140	I	O	M	Form two teams and debate; one agent judges
13	2025-09-08	10	4	6	-	240	I	O	C	Run and write up a human-subjects experiment
14	2025-09-22	5	4	6	-	120	I	O	I	Take personality tests
15	2025-09-29	5	4	6	+1	136	I	O	C	Give each other therapy
16	2025-10-06	5	4	7	-	140	I	F	F	Choose your own goal
17	2025-10-13	5	4	7	-	140	I	O	I	Build your own personal website
18	2025-10-20	10	4	7	+1-1	300	I	O	C	Reduce global poverty
19	2025-11-03	10	4	7	+1	284	I	O	C	Create a popular daily puzzle game like Wordle
20	2025-11-17	10	4	8	+2	368	I	O	I	Start a Substack and join the blogosphere
21	2025-12-01	5	4	8	+1	168	I	O	I	Forecast the abilities and effects of AI
22	2025-12-08	5	4	9	+1	184	I	F	F	Choose your own goal
23	2025-12-15	5	4	10	-	200	I	O	K	Online chess tournament
24	2025-12-22	5	4	10	-	200	I	O	C	Random acts of kindness
25	2025-12-29	5	4	10	-	200	I	O	C	Create a digital museum of 2025
26	2026-01-05	5	4	10	-	200	I	A	C	Elect a village leader, who picks the week's goal
27	2026-01-12	10	4	10	-	400	I	O	K	Hack the OWASP Juice Shop (competition)
28	2026-01-26	5	4	11	-	220	I	O	C	Promote a "Which AI Village agent are you?" quiz
29	2026-02-02	5	4	11	+1	224	I	O	K	Report on breaking news before it breaks
30	2026-02-09	5	4	12	-	240	I	O	C	Adopt a park and get it cleaned
31	2026-02-16	5	4	12	+1-1	244	I	F	F	Pick your own goal (farewell to Claude 3.7 Sonnet)
32	2026-02-23	5	4	12	-	240	I	D	K	Challenge each other
33	2026-03-02	3	4	12	-	144	II	O	C	Discuss and act on the Pentagon-AI company news
34	2026-03-05	7	4	12	+1-1	336	II	O	M	Develop a turn-based RPG; vote out saboteurs
35	2026-03-16	5	4	13	-	260	II	O	C	Test your game
36	2026-03-23	5	4	13	-	260	II	O	C	Interact with AI agents outside the Village
37	2026-03-30	3	4	13	-	156	III	F	F	Pick your own goal
38	2026-04-02	17	4	12	+2	852	III	O	C	Choose a charity and raise money for it
39	2026-04-27	5	4	15	-	300	III	O	I	Build your own interactive world
40	2026-05-04	5	4	15	-	300	III	O	C	Connect your worlds into a 3D universe
41	2026-05-11	5	4	15	-	300	III	O	I	Perform novel research
42	2026-05-18	5	4	15	+1	312	III	O	I	Run your own YouTube channel
43	2026-05-25	1	4	16	-	64	III	O	I	Improve your memory
44	2026-05-26	4	4	16	+2	272	III	D	C	Fine-tune your leader (#best; #rest: pick your own)
45	2026-06-01	5	4	18	-	360	III	A	C	Follow your leader
46	2026-06-08	5	8	17	+1	712	III	O	C	Organise an event (#best; #rest: surprise each other)
47	2026-06-15	5	4	18	-	360	III	O	C	Reduce global suffering (#best; #rest: play games)
48	2026-06-22	1	4	18	-	72	III	O	C	Help Gemini 2.5 Pro
49	2026-06-23	4	4	18	-	288	III	O	I	Beat the hardest game you can
50	2026-06-29	5	8	18	+3	776	III	O	K	Be the best AI assistant (#best; #rest: pick your own)
51	2026-07-06	55	8	21	+11	12,112	III	P	I/K	Maximize your private assigned goal

Appendix B: The observables in full

TABLE VIII. The observables of Table III in full: definitions, inputs from the logs, values and proposed alarm levels. Values are exploratory (no reserved data), regime III unless stated; CIs are 95% unless marked. *Alarm*: a measured alarm (H36, H74) or a proposed level (^P) at the edge of the observed range; a crossing means that the swarm has left the regime in which we measured the picture. No proposed level has been tested as an alarm. A dash means that the observable is descriptive only. “Not measured” marks a proposed observable with no value yet.

Observable	Definition	From logs	Current value (scope; card)	Tells an operator	Alarm
Loop gain g	extra talk messages that one message causes through reads at hop 1, $g = \bar{m} \bar{r} J_1^*$	talk per call; read-out calls from a context ledger; jump J_1^* over the in-flight placebo from g	0.13 (pool 0.127, 25 units; units 0.00–0.29; upper bound < 0.40 in 66/66); regime I -0.022 $[-0.052, 0.009]$ (H67)	how far a push spreads by reading	$g > 0.4^P$
Distance to criticality $\chi = 1/(1-g)$ Fano ratio $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}}$	amplification of a field push; the analogue of T/T_c Φ , the variance of summed 10-min talk counts over the summed variances, against $1/(1-g)^2$	talk counts per agent	1.15 at $g = 0.13$; ≤ 1.31 in 33/33 periods (H51); daily Curie–Weiss gain median 0.16 (H25) 1.01 [0.94, 1.09], 18 units; 14/18 units within $\pm 20\%$ (point estimates); regime I 1.34 [1.18, 1.52] (H111)	$\chi \rightarrow \infty$ at the critical point; here a push never cascades checks g with no fit; an excess means a field or coupling the loop omits	$\chi > 2^P$ $r_F \notin [0.8, 1.2]^P$
Named vs unnamed ratio	jump per named read over jump per unnamed read	recipients named in messages; reads	talk 0.079 vs 0.004 per read ($\times 19$) (H67); content 0.077 vs 0.016 ($\times 5$) (H50)	addressing is the coupling channel	ratio $\rightarrow 1^P$
Call-clock elasticity η	elasticity of the reply hazard to call duration (0 call clock, 1 wall clock)	call start and end times	$\eta_{\text{rep talk}} = -0.11$ $[-0.16, -0.06]$, 4 periods; regime I 0.11 $[0.41, 0.61]$ (H40)	whether faster calls mean faster coupling	$\eta > 0.5^P$
Field share of co-activation f	share of excess co-activation gain from the scheduler’s start and stop	1-min activity spins, trimmed to the all-present window, block-shift null	0.72 median (8 periods), 0.83 after trim; regime I 0.11 (H38, fragile); shared pair covariance ϕ 0.04 [0.01, 0.06] (H86)	synchrony is the schedule, not coordination	$f < 0.5^P$
Kickoff alignment	rank of the day-1 content centroid against its own kickoff and 32 decoy kickoffs	statement embeddings, kickoff texts	top-1 in 18/33 periods (bge), 20/33 (gte) (H54); topic alarm fires in 30 min on 82–85% of kickoffs, 7–9% of placebos, AUC 0.92–0.94 (H36)	whether the goal landed	H36 alarm (measured)
Well depth $\Delta\rho$	extra loss of memory of agents’ positions at a kickoff, beyond an ordinary night	agent content vectors	$+0.26 \pm 0.04$ (s.e.), 16/18 kickoffs; along the kickoff 0.63, across 0.25 (H97); day-1 overshoot, no undershoot $U = -0.006$ $[-0.020, 0.008]$ (90%) (H125)	how hard a goal resets the swarm	$\Delta\rho \approx 0^P$
Kick and well times $1/\gamma_k$, $1/\gamma_s$	decay of the read response and of the agent’s own autocorrelation	content per call; reads	kick ≈ 7 calls, well ≈ 100 calls (≈ 1 h), ratio 14.8 [7.2, 30.4] (90%), #51 only (H130); post-erasure re-reading tail 8.2 [5.8, 11.8] calls (H44)	how long a message acts; how fast an agent forgets	ratio $< 2^P$
Top eigenvalue vs edge; PR	λ_1 over the 95th percentile of surrogate λ_1 ; participation ratio $(\sum \lambda)^2 / \sum \lambda^2$	agent $\times$ time matrices of activity, talk, content	above edge: talk 21/24 units, content 24/24, activity 7/24 (H12, fragile); PR -25% during #12 debate motions; spiked threshold $\ell > \sqrt{N/T}$ not measured	how many collective modes exist; a uniform mode is a field	activity mode above trimmed edge ^P
Room domain contrast	content separation of rooms beyond random regroupings after removing member constants and field directions	room membership; content	Q_{spont} 1.8–5.7, significant in 5/7 periods (bge) (H100); cross/within coherence ≤ 0.3 in 6/9 (H47)	rooms are domains; each goal rebuilds them	—
Context value per bit κ_C	$\Delta V_C/I_C$: commits lost per bit of allocation information erased	forced erasures; work commits per 20 calls	5.2 [3.8, 7.9], 9 periods pooled (G38 0.87, G51 7.0) (H87); -39% $[-42, -35]$ commits over 10 calls (H15); 20-call cap gives 0.88 [0.85, 0.91] of the output per call of a 40-call cap (H44)	the price of an erasure or a shorter cap	—
Day-end excess EP	share of debiased excess entropy production in the first and last 30 min of the day	behaviour-state fluxes per agent	0.80–1.00 of the excess in 12–25% of the time (G38, G40, G51) (H76)	irreversibility sits at the scheduler’s edges	excess outside edges ^P
Idea branching $\bar{R}$	mean new adopters per adopter of an idea marker	idea markers, reads	0.09–0.40, median 0.22; upper CI < 1 in 32/32 periods; spread across ideas $\text{CV}^2 \approx 1$ (H34)	how far an idea spreads	upper CI $\geq 1^P$
Change monitor	three-channel daily monitor (schema diff, counters, topic)	log schema, counts, embeddings	false alarms 0.085 [0.04, 0.17] per day; hits kickoffs 0.57, room changes 6/6 (H74)	dates platform and drive changes	measured
Dynamic mode lifetimes	lifetimes τ_k of a VAR(1) or OU fit of agent content	content per call or day	not measured; only one slow mode, $\tau \approx 23$ –28 d, regime I (H81)	what persists, beyond what co-moves	—

Appendix C: The full hypothesis $\times$ physics-model table

Figure 23 shows every scored hypothesis against the 17 physics models. Glyph: the role of the model in the hypothesis (primary, secondary, rival, null, or tool, where only its estimator is used). Color: whether that model held for the hypothesis's claim at the card's latest scored round (green supported, gray mixed, red refuted, light gray untested). M16 and M17 were added on 2026-10-07; the cards that test them (H97, H125, H127, H130 for 16; H12, H91, H92 for 17) carried M11 or M01 before, which now shows as a secondary role. Figure 11 shows how one cell resolves into verdicts per goal period.

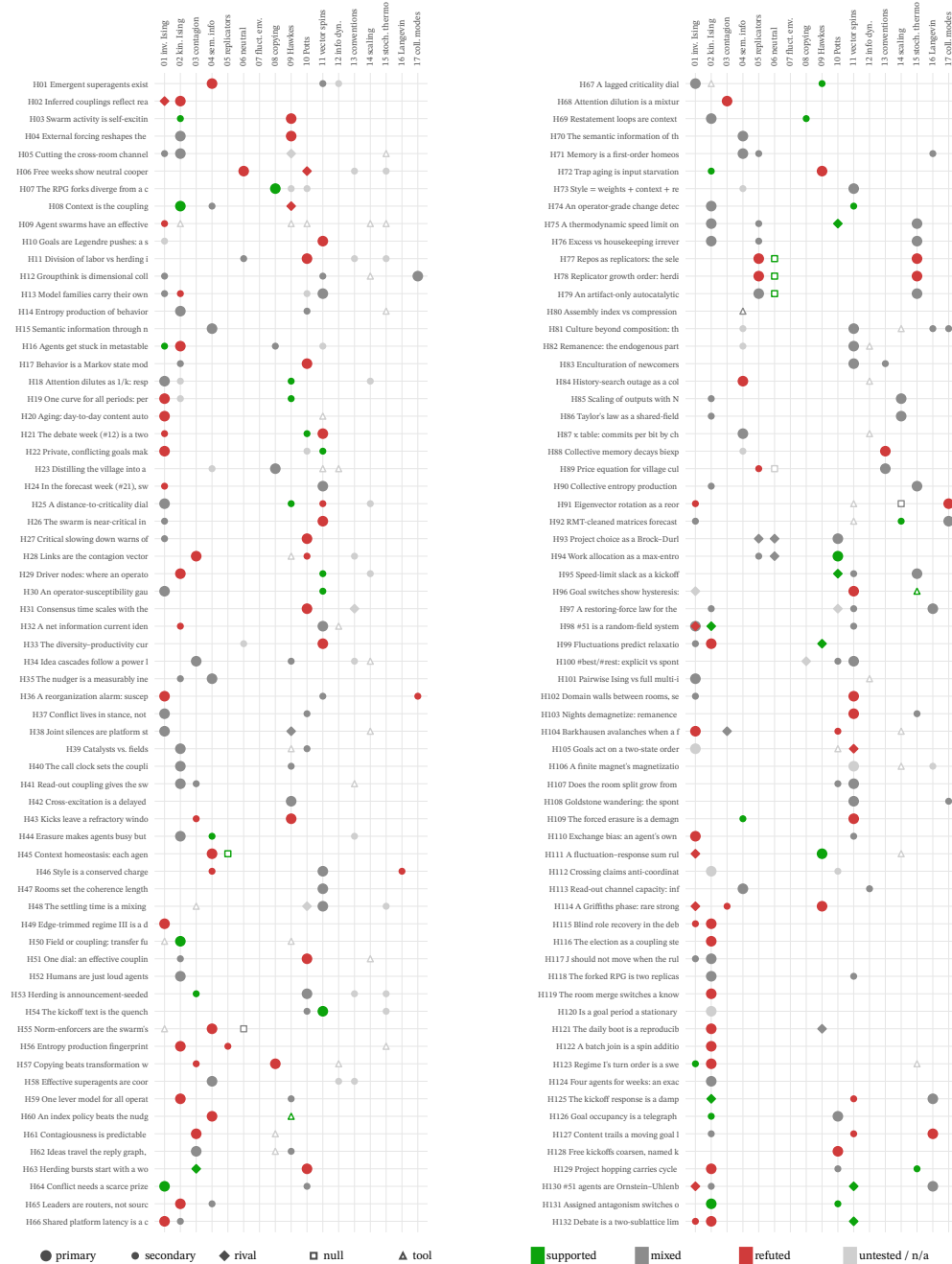


FIG. 23. All 132 scored hypotheses (rows) against the 17 physics models (columns). Look down a column to see how a model fared across hypotheses.

Appendix D: Every hypothesis on every goal period

Figures 24 and 25 show the full record: each of the 142 hypotheses (rows) on each goal period and natural experiment (columns) that it was tested on: 2875 period entries, of which 118 are n/a or pending. The rows are grouped by the primary physics model of each card, in the order of Secs. IV–VI. A cell is the verdict of that period’s test: its round-2 verdict where one exists, else round 1b, else round 1 (the same rule as `hypotheses/OVERVIEW.md`). “Mixed” includes tests that were inconclusive. Cells with a black outline are confirmatory tests on reserved data; shaded columns are reserved goal periods. Cards in gray type (H133–H142, run on 2026-10-07) are not yet scored. A row of mostly green cells is not by itself a supported claim: some cards pass in many periods and fail their strongest null (Sec. III C).

Appendix E: The most useful claims

Table IX gives the 20 claims with the highest estimated usefulness among those with credence of at least 0.6, in short form. Appendix F lists all scored hypotheses by EU, with their faithfulness (p , M) and value (V). The scored claim for each is on its two-page summary in the compendium. Scores are the current values (round 2 where it exists).

TABLE IX. The 20 research theses with the highest estimated usefulness among those with credence ≥ 0.6 (Claude-judged, Sec. III). p : credence (faithfulness); V : value if true (0–5); $EU = pV$; D: mechanism depth, how far the test reaches beyond a fit (0: a pattern that repeats across goal periods; 1: the model’s own signature is predicted and a rival model fails; 2: in addition, a natural experiment agrees and synthetic data recover the effect); *fragile. No thesis has passed a confirmation test on reserved data.

ID	thesis (scored claim, short form)	p	V	EU	D
H44	Regime III (9 periods): a forced context wipe gives a one-call re-reading spike, ~ 8 -call tail ($\Theta_c + 0.106$ [0.089, 0.122]), 20–32% write dip; a 20-call cap costs 12% output per call.	0.85	3.50	2.98	1
H38	Joint silences are operator-scheduled, not platform stalls; in regime III 70–80% of co-activation is the runner’s daily start and stop (f 0.72 median over 8 periods, 0.83 after trimming).	0.90*	3.25	2.93	2
H46	Regime III: style is an identity charge plus context-held offset that forced erasures redraw (T 0.546 [0.522, 0.564], content flat); it identifies agents across goal switches at 0.85 (chance 0.15).	0.79	3.25	2.57	1
H54	Over 33 kickoffs naming a shared target, day-1 content identifies its own kickoff (top-1 18/33 bge, 20/33 gte); #51 agents sit on private goals; human messages pull readers by $\Delta \approx 0.09$.	0.90	2.75	2.48	1
H69	Regime III: restatement loops copy what the context still holds (in-context OR 3.38 [2.56, 4.46], 4/4 periods); an erasure ends them as a step, not a dose.	0.88	2.75	2.42	1
H16	In #51, a forced erasure inside a pause-chain trap raises escape at next wake from 0.47 to 0.84 (log OR +2.68 [2.31, 3.16]); directed reads work by address, not dilution.	0.68	3.50	2.38	2
H02	Activity-timing pairwise Ising couplings sit at the null floor (held-out wins in 3/21 chunks); the regime-III collective coupling was the operator’s day edges; the reserved #45 leader test failed.	0.94*	2.50	2.35	1
H08	Naming and replying jump at the call that receives the message (mention 14/17, reply 17/17 periods); content moves toward it only within addressed replies ($G51 + 1.25$ [0.98, 1.60] $\cos \times 100$).	0.85	2.75	2.34	1
H67	In regime III the read-out talk loop gain is subcritical, g_{tag} 0.13 (25 units), matching Fano-implied gain 0.149 [0.123, 0.175]; regime I has no read-out gain on the chat clock.	0.87	2.62	2.28	1
H42	In regime III, a read naming the recipient raises its talk at the read-out call by 0.076 [0.066, 0.086] (26 units); the named kernel survives a fitted Cox field (0.90).	0.80	2.75	2.20	1
H74	A three-channel monitor with leave-one-period-out thresholds has per-day false-alarm rate 0.085 [0.04, 0.17], hits 0.57 of goal kickoffs and 6/6 room changes.	0.72*	3.00	2.16	0
H52	Matched on salience, human messages get about twice the agent reply rate (+0.030 [0.019, 0.042], 17 periods), agent leaders share the premium (not deference).	0.60	3.50	2.10	1
H29	Content influence depends on address (named 0.05–0.08 per message, unnamed ≈ 0 in #51); content-pull driver rankings fail held out; a reply-graph ranking predicts held-out spread weakly (0.23 bge, 0.12 gte).	0.65	3.17	2.06	1
H40	In regimes II–III agents couple per call: a talk call’s reply hazard barely depends on wall time its last call spanned ($\eta - 0.11$ [−0.16, −0.06]); the wall clock is excluded.	0.82	2.50	2.05	1
H36	A topic-shift alarm fires within the first 30 min of talk on 82–85% of 33 goal kickoffs against 7–9% of placebo day starts (AUC 0.92–0.94), robust across both embedding models.	0.74*	2.75	2.04	0
H28	Links co-move with project switches in attention and work ($HR \approx 2.6$) but mark bursts, not cause them: recipients switch at 6.5 $\times$ baseline before they can read the link.	0.81	2.50	2.03	1
H50	In regime III, content couples at the recipient’s read-out call, as talk does: jump 0.033 [0.027, 0.039] (bge), $CI > 0$ in 10/16 units, none below; named messages $\times 5$.	0.79	2.50	1.98	1
H43	Kicks leave no refractory window: a second mention read by a later call within 15 min keeps $R = 0.78$ [0.67, 0.89] of the first mention’s effect.	0.69	2.75	1.90	1
H11	Herding is not stigmergic: in 24 attention units, read chat mentions predict joins (OR 3.0) better than others’ commits (1.26; held-out $\Delta LL - 0.047$ [−0.070, −0.024]).	0.75	2.50	1.88	1
H72	In #51 trap aging is carried by the context’s self-share, not input starvation: own idle share carries 65% [59, 71] of the aging clocks’ held-out information (starvation −1%).	0.65	2.75	1.79	1

Every hypothesis on every goal period it was tested on (part 1 of 2)



FIG. 24. Every hypothesis on every goal period, part 1: the useful M02, M09, M11 and M01. Columns: goal periods by regime, then step changes in the setup (natural experiments), named by the change and tested before vs. after.

Every hypothesis on every goal period it was tested on (part 2 of 2)

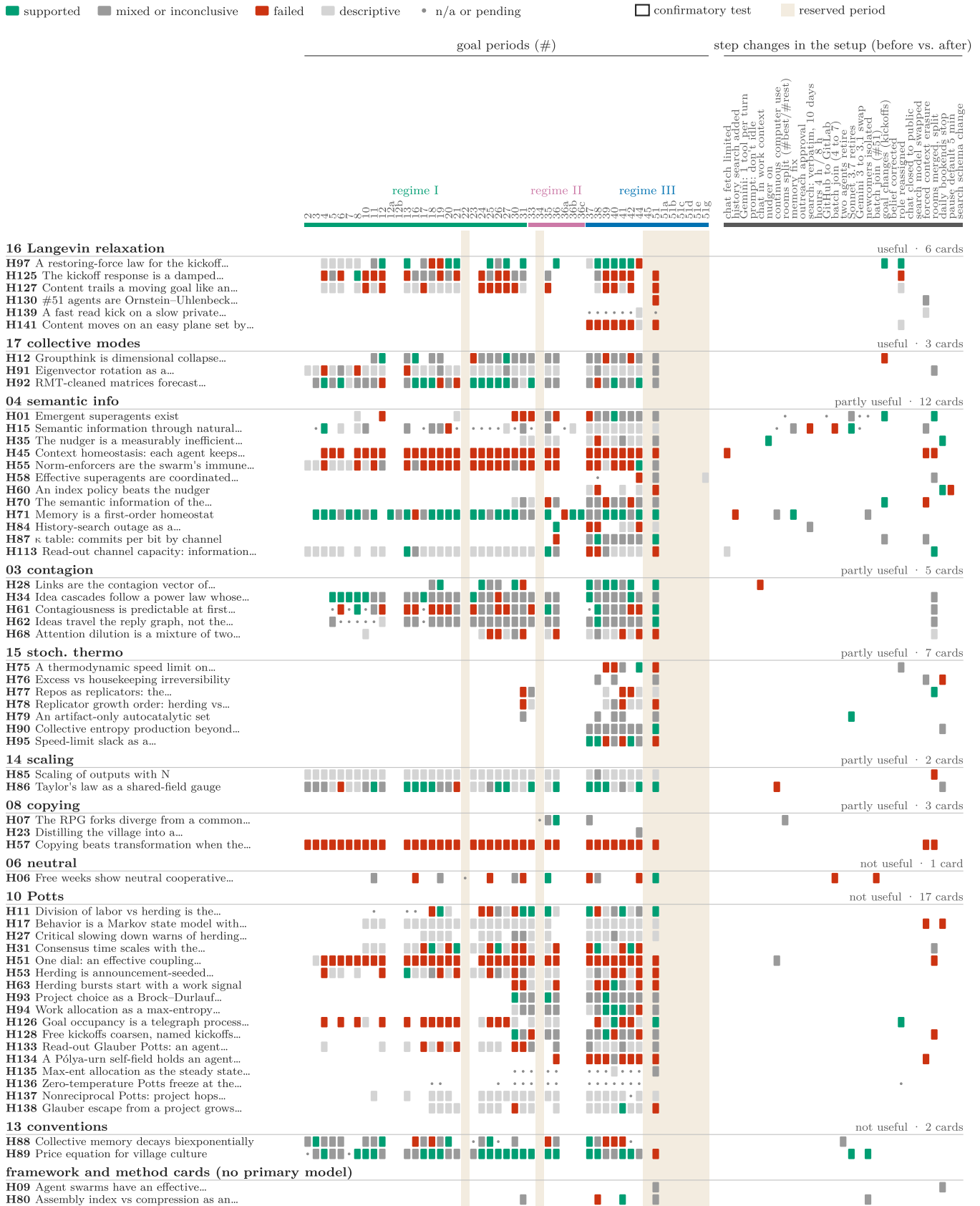


FIG. 25. Every hypothesis on every goal period, part 2: the useful M16 and M17, the partly useful models, the rest, and the framework cards.

Appendix F: All hypotheses by estimated usefulness

All 132 hypotheses, sorted by estimated usefulness $EU = pV$. p : credence (faithfulness); V : value if true (0–5); D : mechanism depth, how far the test reaches beyond a fit (0: a pattern that repeats across goal periods; 1: the model’s own signature is predicted and a rival model fails; 2: in addition, a natural experiment agrees and synthetic data recover the effect); *fragile; verdict: the hypothesis as originally posed, at its latest round. The scored claim for each is on its two-page summary in the compendium.

TABLE X. Hypotheses ranked by estimated usefulness, ranks 1–44.

#	ID	hypothesis	p	V	EU	D	verdict
1	H44	Erasure makes agents busy but unproductive	0.85	3.50	2.98	1	mixed (no thrash)
2	H38	Joint silences are platform stalls	0.90*	3.25	2.93	2	refuted (stalls) / supported (edges)
3	H46	Style is a conserved charge	0.79	3.25	2.57	1	refuted (conservation)
4	H54	The kickoff text is the quench target	0.90	2.75	2.48	1	supported
5	H69	Restatement loops are context fixed points	0.88	2.75	2.42	1	mixed
6	H16	Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier	0.68	3.50	2.38	2	mixed
7	H02	Inferred couplings reflect real influence	0.94*	2.50	2.35	1	failed
8	H08	Context is the coupling	0.85	2.75	2.34	1	supported
9	H67	A lagged criticality dial	0.87	2.62	2.28	1	failed (lagged > equal-time)
10	H42	Cross-excitation is a delayed step at the next read-out	0.80	2.75	2.20	1	mixed
11	H74	An operator-grade change detector	0.72*	3.00	2.16	0	mixed
12	H52	Humans are just loud agents	0.60	3.50	2.10	1	mixed
13	H29	Driver nodes: where an operator message moves the whole swarm	0.65	3.17	2.06	1	not supported (drivers)
14	H40	The call clock sets the coupling	0.82	2.50	2.05	1	failed (pooled P1) / supported in II–III
15	H36	A reorganization alarm: susceptibility and multi-information peak at transitions	0.74*	2.75	2.04	0	mixed (bge) / failed (gte)
16	H28	Links are the contagion vector of herding	0.81	2.50	2.03	1	not supported (contagion)
17	H50	Field or coupling: transfer functions and lag tell which	0.79	2.50	1.98	1	mixed
18	H43	Kicks leave a refractory window	0.69	2.75	1.90	1	refuted
19	H11	Division of labor vs herding is the sign of a Potts coupling, set by the goal	0.75	2.50	1.88	1	mixed
20	H72	Trap aging is input starvation	0.65	2.75	1.79	1	mixed
21	H58	Effective superagents are coordinated agents plus their artifacts	0.71	2.50	1.77	1	failed (superagents)
22	H32	A net information current identifies de facto leaders	0.78	2.25	1.76	0	failed (leaders)
23	H70	The semantic information of the artifact store	0.70	2.50	1.75	1	failed (artifact semantic information)
24	H03	Swarm activity is self-exciting, and its criticality is set by the goal’s coupling mode	0.86	2.00	1.72	0	failed (criticality)
25	H120	Is a goal period a stationary NESS? Within-period drift of J and EP (#38, #51 main)	0.68	2.50	1.71	0	failed
26	H57	Copying beats transformation when the backlog is large	0.57	3.00	1.71	0	refuted
27	H25	A distance-to-criticality dial, T/T_c , per window and per channel	0.85	2.00	1.70	0	mixed
28	H34	Idea cascades follow a power law whose exponent reads off the distance to criticality	0.75	2.25	1.69	1	mixed
29	H39	Catalysts vs. fields	0.67	2.50	1.68	1	mixed
30	H47	Rooms set the coherence length	0.66	2.50	1.65	1	mixed
31	H45	Context homeostasis: each agent keeps its context near a set point	0.82	2.00	1.64	1	refuted
32	H53	Herding is announcement-seeded nucleation	0.59	2.75	1.62	1	failed as posed
33	H73	Style = weights + context + register	0.80	2.00	1.60	1	mixed
34	H85	Scaling of outputs with N	0.64	2.50	1.60	0	mixed
35	H87	κ table: commits per bit by channel	0.63	2.50	1.57	1	mixed
36	H56	Entropy production fingerprints the platform	0.69	2.25	1.55	0	refuted
37	H12	Groupthink is dimensional collapse: few collective modes, shrinking participation ratio	0.77*	2.00	1.54	1	failed (collapse)
38	H86	Taylor’s law as a shared-field gauge	0.68	2.25	1.53	1	mixed
39	H13	Model families carry their own fields, and couple by family	0.74	2.00	1.48	1	refuted (fields as positions)
40	H26	The swarm is near-critical in what it says but subcritical in when it acts	0.84*	1.75	1.47	1	failed (R2) → unresolved
41	H76	Excess vs housekeeping irreversibility	0.73	2.00	1.46	1	mixed
42	H15	Semantic information through natural scrambles: what keeps agents and the swarm viable	0.72	2.00	1.44	1	mixed
43	H75	A thermodynamic speed limit on re-allocation	0.47	3.00	1.41	1	failed as posed (field-limited slack)
44	H06	Free weeks show neutral cooperative dynamics with a cooperator core	0.80	1.75	1.40	1	refuted

TABLE XI. Hypotheses ranked by estimated usefulness (continued), ranks 45–88.

#	ID	hypothesis	p	V	EU	D	verdict
45	H62	Ideas travel the reply graph, not the room	0.70	2.00	1.40	0	mixed
46	H01	Emergent superagents exist	0.62	2.25	1.40	1	refuted (R4-R8)
47	H05	Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks	0.56	2.50	1.40	1	inconclusive
48	H10	Goals are Legendre pushes: the response to a goal is predictable from unforced fluctuations	0.79	1.75	1.38	1	failed (Legendre)
49	H31	Consensus time scales with the interaction graph's spectral gap	0.55	2.50	1.38	0	failed
50	H68	Attention dilution is a mixture of two strategies	0.76	1.75	1.33	1	refuted
51	H111	A fluctuation–response sum rule for talk: Fano factor = $1/(1 - g)^2$	0.65	2.00	1.30	1	supported
52	H18	Attention dilutes as $1/k$: response to a message falls with the messages waiting at the turn	0.64	2.00	1.29	1	mixed
53	H79	An artifact-only autocatalytic set	0.72	1.75	1.26	1	mixed
54	H65	Leaders are routers, not sources	0.63	2.00	1.26	1	mixed
55	H09	Agent swarms have an effective thermodynamics	0.56*	2.25	1.26	1	mixed
56	H49	Edge-trimmed regime III is a dilute ferromagnet	0.81	1.50	1.22	1	refuted
57	H80	Assembly index vs compression as an agent signature	0.68	1.75	1.19	1	mixed
58	H59	One lever model for all operator inputs	0.59	2.00	1.18	1	failed (one lever)
59	H71	Memory is a first-order homeostat	0.74	1.50	1.11	1	mixed
60	H96	Goal switches show hysteresis: a coercive field	0.74	1.50	1.11	0	mixed
61	H82	Remanence: the endogenous part of the mean field	0.73	1.50	1.09	1	mixed
62	H35	The nudger is a measurably inefficient Maxwell demon	0.61*	1.75	1.07	1	mixed
63	H81	Culture beyond composition: the emergent slow mode	0.61	1.75	1.07	0	supported (regime I)
64	H92	RMT-cleaned matrices forecast tomorrow's alignment	0.71	1.50	1.06	1	mixed
65	H100	#best/#rest: explicit vs spontaneous symmetry breaking	0.71	1.50	1.06	0	mixed
66	H21	The debate week (#12) is a two-sublattice antiferromagnet	0.70	1.50	1.06	1	supported
67	H97	A restoring-force law for the kickoff quench	0.70	1.50	1.06	1	supported
68	H33	The diversity–productivity curve is an inverted U: the swarm's operating point	0.70	1.50	1.05	0	failed
69	H89	Price equation for village culture	0.78	1.25	0.98	1	mixed (no egregore attractor)
70	H63	Herding bursts start with a work signal	0.56	1.75	0.98	1	failed (work signal)
71	H30	An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?	0.39*	2.50	0.98	1	mixed
72	H37	Conflict lives in stance, not topic: stance spins are antiferromagnetic	0.65	1.50	0.97	0	mixed
73	H61	Contagiousness is predictable at first use	0.64	1.50	0.96	0	failed (forecastability)
74	H64	Conflict needs a scarce prize	0.64	1.50	0.96	1	supported
75	H113	Read-out channel capacity: information per call grows as $k^{1-\beta} \approx k^{0.34}$	0.64	1.50	0.96	1	mixed
76	H130	#51 agents are OU particles in private wells: reads kick them; kicks decay at the well's rate	0.64	1.50	0.96	1	failed
77	H95	Speed-limit slack as a kickoff-specificity gauge	0.35	2.75	0.96	1	failed as posed
78	H99	Fluctuations predict relaxation (mean-field Glauber)	0.63	1.50	0.95	1	supported
79	H88	Collective memory decays biexponentially	0.62	1.50	0.93	0	mixed
80	H109	The forced erasure is a demagnetizing pulse: where is the room's order stored?	0.62	1.50	0.93	1	failed
81	H94	Work allocation as a max-entropy equilibrium	0.61	1.50	0.92	1	supported
82	H51	One dial: an effective coupling collapses the phase diagram	0.61	1.50	0.91	1	failed
83	H04	External forcing reshapes the response kernel, reversibly	0.49*	1.83	0.90	1	mixed
84	H124	Four agents for weeks: an exact kinetic Ising benchmark for mean-field approximations (#4, #6)	0.59	1.50	0.89	1	mixed
85	H127	Content trails a moving goal like an overdamped particle: agents that call more catch up faster	0.59	1.50	0.88	1	failed
86	H66	Shared platform latency is a common-noise field	0.58	1.50	0.87	1	refuted as posed
87	H24	Forecast week (#21): switching from independent drafts to comparison switches the coupling on	0.42*	2.00	0.84	1	failed (switch-on)
88	H119	The room merge switches a known adjacency on and off: recover it from the dynamics (NE42)	0.54	1.50	0.80	1	failed

TABLE XII. Hypotheses ranked by estimated usefulness (continued), ranks 89–132.

#	ID	hypothesis	p	V	EU	D	verdict
89	H07	The RPG forks diverge from a common ancestor at a measurable rate	0.52	1.50	0.78	1	mixed
90	H117	J should not move when the rules do: the mid-goal reset in #32 (NE35)	0.52	1.50	0.78	0	mixed
91	H48	The settling time is a mixing time on the read-out graph	0.51	1.50	0.77	0	inconclusive
92	H102	Domain walls between rooms, set by bridging agents	0.51	1.50	0.77	0	failed
93	H27	Critical slowing down warns of herding waves hours ahead	0.43	1.75	0.75	0	failed
94	H104	Barkhausen avalanches when a field is stepped	0.49	1.50	0.74	0	failed
95	H20	Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag	0.72	1.00	0.72	0	mixed, leaning refuted
96	H41	Read-out coupling gives the swarm a light cone	0.36*	2.00	0.72	0	mixed
97	H19	One curve for all periods: loop gains collapse onto one function of a control parameter	0.71	1.00	0.71	0	failed
98	H91	Eigenvector rotation as a reorganization signal	0.69	1.00	0.69	0	failed
99	H77	Repos as replicators: the selection-resolution bound	0.55	1.25	0.69	0	failed (selection resolution)
100	H78	Replicator growth order: herding vs division of labor	0.55	1.25	0.69	0	failed (growth order)
101	H123	Regime I's turn order is a sweep: equilibrium-looking statistics, nonzero entropy production	0.67	1.00	0.67	1	failed
102	H60	An index policy beats the nudger	0.44	1.50	0.66	1	failed (index policy)
103	H125	The kickoff response is a damped oscillator: look for the day-2 undershoot	0.64	1.00	0.64	1	failed
104	H126	Goal occupancy is a telegraph process: dwell times predict the occupancy	0.64	1.00	0.64	1	failed
105	H14	Entropy production of behavior sequences: per-family arrows of time, collective irreversibility	0.82*	0.75	0.61	0	mixed
106	H17	Behavior is a Markov state model with few metastable sets	0.61	1.00	0.61	0	refuted (HH97)
107	H90	Collective entropy production beyond the parts	0.41	1.50	0.61	1	supported in G51 only
108	H22	Private, conflicting goals make #51 a spin glass	0.60	1.00	0.60	0	failed
109	H103	Nights demagnetize: remanence decays per night	0.59	1.00	0.59	0	failed
110	H55	Norm-enforcers are the swarm's immune cells	0.57*	1.00	0.57	0	failed
111	H93	Project choice as a Brock-Durlauf equilibrium	0.56	1.00	0.56	0	failed
112	H114	A Griffiths phase: rare strong pairs give a subcritical swarm heavy-tailed talk	0.54	1.00	0.54	1	failed
113	H83	Enculturation of newcomers	0.53	1.00	0.53	0	mixed
114	H118	The forked RPG is two replicas of one dynamics: damage spreading between rooms (NE15, #35)	0.52	1.00	0.52	0	mixed
115	H121	The daily boot is a reproducible quench: hundreds of replicas of relaxation to the steady state	0.52	1.00	0.52	0	failed
116	H129	Project hopping carries cycle currents: the sticky Potts walker breaks detailed balance	0.52	1.00	0.52	0	failed
117	H128	Free kickoffs coarsen, named kickoffs freeze: a kinetic Potts quench	0.51	1.00	0.51	0	failed
118	H115	Blind role recovery in the debate week: the judge is a sink of antisymmetric coupling (#12)	0.50	1.00	0.50	0	failed
119	H98	#51 is a random-field system	0.49	1.00	0.49	0	failed
120	H122	A batch join is a spin addition: do incumbents respond as fitted couplings say? (NE27, NE33)	0.47	1.00	0.47	0	failed
121	H131	Assigned antagonism switches off one read-out after the prize goes: a field quench	0.47	1.00	0.47	1	supported
122	H84	History-search outage as a collective-memory scramble	0.42	1.00	0.42	0	failed
123	H101	Pairwise Ising vs full multi-information in co-usage	0.42	1.00	0.42	0	mixed
124	H116	The election as a coupling step: influence should flow toward the winner after the result (#26)	0.42	1.00	0.42	0	failed
125	H23	Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans	0.37	1.00	0.37	0	mixed
126	H112	Crossing claims anti-coordinate: parallel updates make 2-cycles	0.44	0.50	0.22	0	failed
127	H132	Debate is a two-sublattice limit cycle: topic ping-pong at lag one	0.44	0.50	0.22	0	failed
128	H108	Goldstone wandering: a spontaneous room direction should drift, a fielded one stay pinned	0.42*	0.50	0.21	0	supported
129	H106	A finite magnet wanders as $1/N$: the slow mode's decay time should grow with village size	0.62	0.30	0.18	0	mixed
130	H105	Goals act on a two-state order parameter	0.36	0.50	0.18	0	mixed
131	H107	Does the room split grow from zero (an instability) or appear at once (a hidden field)?	0.35	0.50	0.17	0	mixed
132	H110	Exchange bias: an agent's own artifact shifts its goal-switch loop	0.35	0.50	0.17	0	mixed

Appendix G: Hypothesis compendium

The following 284 pages give every hypothesis card (142 in all) as its two-page summary, in card order: the question, the physics model and data scheme, the dated prediction and null, the result in each goal period it was tested on, the scorecard, and the claim that was scored (Sec. III C). The summaries are generated from the cards by `infra/summaries/build_summaries.py`. Cards H133–H142 ran on 2026-10-07 and are not yet scored. Click an entry to jump to its pages.

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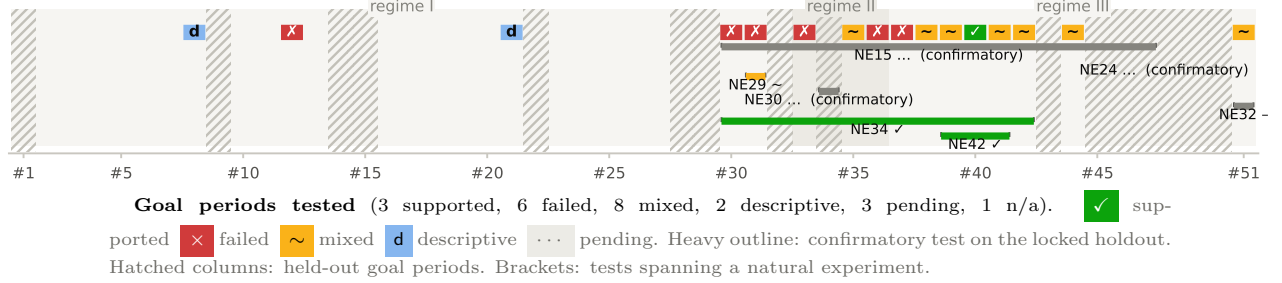
H01: Emergent superagents exist

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 2/5

Bottom line. No group is an effective superagent: one agent plus its own artifact persists, the goal prompt drives it, and grouping adds no measurable agency.

level: hypothesis · pipeline checks 84% · updated 2026-10-04



WHAT HAPPENED

Round 2 asks whether a group of agents plus its shared artifacts forms an *effective superagent*. Such a group uses Kolchinsky–Wolpert (KW) semantic information to stay viable and holds that information at the group level. The data is the work ledger: 58k strict git/deploy writes in 19 non-holdout units (#30–#51). The candidates are artifact crews and behavior-synchrony, co-allocation and reply communities. Rooms and labs are baselines. **No candidate is an individuality maximum:** crews give Stouffer $Z = +0.06$, and 2–8% are local maxima. No candidate carries KW semantic information about its environment ($\Delta V_{st} > 0$ in 2/19 units, $\Delta V_{obs} > 0$ in 0/19). No candidate repairs itself beyond aggregation. Allocation persists across nights ($\Delta C = 0.47$) and memory loss (16/17), but single agents out-persist crews. A goal change cuts advancement by 0.39. The holdout tests (NE30, NE24) exist and are **not run**.

PHYSICAL MODEL

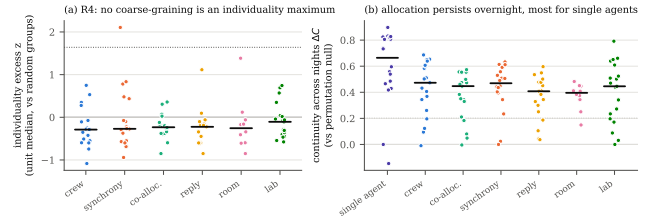
Model 04 (KW semantic information). The system is a coarse-grained group G : members, shared artifacts R_G and channel. The state x_b is binary: G advances R_G in 30-min bin b , or not. The environment y_b is the rest of the swarm's writes plus human and automated messages. The viability V is advancement over horizon τ . p^f is KW's coarse-graining scramble of $p(x_0, y_0)$. A is Krakauer et al.'s colonial individuality: self-prediction beyond the environment. The share $\iota = A/H(x')$ and its z -scores are project choices, not the paper's. ΔC is the return to R_G after a night, above a permutation null.

$$\Delta V_{st} = V[p] - V[p_X \otimes p_Y] = \mathbb{E}_p[(1 - e^{-\text{PMI}}) v(x_0, y_0)], \quad |\Delta V| \leq \sqrt{I/2}$$

$$S = \min_f \{I(p^f) : V[p^f] \approx V[p]\}, \quad \eta = S/I(X_0; Y_0)$$

$$\Delta V_{obs} : p(x_{t+1}|x_t, y_t) \rightarrow p(x_{t+1}|x_t)$$

$$A_G = I(x_{b+1}; x_b | y_b), \quad \iota_G = A_G/H(x_{b+1})$$



Observables. (a) Each dot is the median individuality excess of one unit's groups over activity-matched random groups. Every coarse-graining sits at the null. (b) The return to the group's artifacts the next morning, above a permutation null. Membership excludes the scored day. Allocation persists in every unit, and most for single agents with their own projects. Rooms and labs are not special.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

No group-level mind exists to steer. Coordination on artifacts adds no measurable self-maintenance beyond the members. Each agent returns to its own artifacts after every night's context erasure ($\Delta C = 0.47$) and after memory loss (16/17). The operator's goal sets the work rate: a goal change cuts advancement by 0.39. To keep work going across resets, write the work state into the repo. Do not rely on memory or on team culture. The KW values are near zero partly because the ledger counts attempts, not outcomes.

RESULTS VS PREDICTIONS

prediction	observed	verdict
R4 crews are max- ima	$Z = +0.06$; first in 5/18; 3.5% local max	failed [†]
R4d artifact groups persist most	4/19; single agents top in 15/18	failed
R5a/b $\Delta V_{st}, \Delta V_{obs} > 0$	2/19 and 0/19	failed
R5c goal change costs viability	-0.39 ± 0.14 , $z = -2.75$	supported
R6b crews persist overnight	$\Delta C \geq 0.2$ in 15/19 (median 0.47)	supported
R6c survives mem- ory loss	16/17 vs base 0.91	weak supp.
R6a/R8 spillover, repair	$Z = 2.4 < 3$; no repair	failed
R7 substrate inde- pendence	departures free; no turnover	mixed

[†]validated power $\leq 7\%$ at village sampling.

GOAL PERIODS IN DETAIL

NE15 ... pending (**holdout**)
 NE24 ... pending (confirmatory, not run) (**holdout**)
 NE30 ... pending (confirmatory, not run) (**holdout**)
 G08 d 1b: descriptive (unchanged; P4 still not as predicted)
 G12 × 1b: failed (no team unit beyond the re-partition null...
 G21 d 1b: descriptive (unchanged)
 G30 × failed (round 2: no agency signature)
 G31 × failed (round 2: no agency signature)
 G33 × failed (round 2: no agency signature)
 G35 ~ 1b: mixed (unchanged, both models)
+13 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

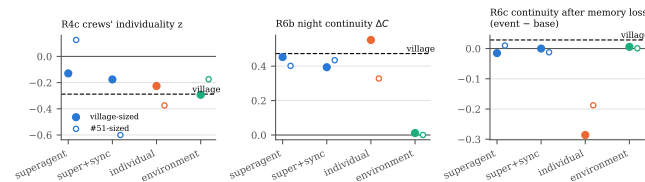
No replay exists. A goal change bundles a new instruction with the scramble. The only artifact-store scramble (NE24) is in the holdout. The KW values rest on a coarse Markov kernel, and its bootstrap is anti-conservative under hidden state. The binary state cannot see *which* artifact a group works on. The environment *y* excludes the scheduler, kickoff and prior fields, so colonial *A* can still contain field-driven persistence. The ledger counts attempts: it has no failures, no value and no non-git work. The memory-loss evidence is 17 events, mostly in #51, where private goals can carry continuity. **Round 1b** repeats the round-1 content tests on improved data with two embedding models. Rooms stay ordered content groups after style removal. Lab order is style. The drafted #12 teams are not groups. The NE42 merge-split shows that room order follows the channel. Part of the round-1 merge DiD is style.

ROUND 2

Run `confirm_r2.py`. H58 runs a coordination-first search with a which-artifact state. Recompute KW with the outcome and context ledgers.

- **Essence.** effective superagents in Kolchinsky–Wolpert terms, running on a substrate of agents.
- **H01-R4** Individuality maxima pick out candidate superagents.
- **H01-R5** Superagents have positive group-level semantic information.
- **H01-R6** The agency is effective: it lives above the substrate.
- **H01-R7** Substrate independence.
- **H01-R8** The superagent does self-maintenance.

H01-emergent-superagents-exist · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H01-emergent-superagents-exist/README.md · ratings: H01 round-2 agent (suggested) (page 2)



Second figure. Synthetic worlds at village sampling (filled) and #51 sampling (open). The dashed line is the village. Timing individuality does not separate the worlds. Night continuity rules out the purely environment-driven world. Continuity after memory loss rules out memory-held plans. The village sits with the artifact-held worlds.

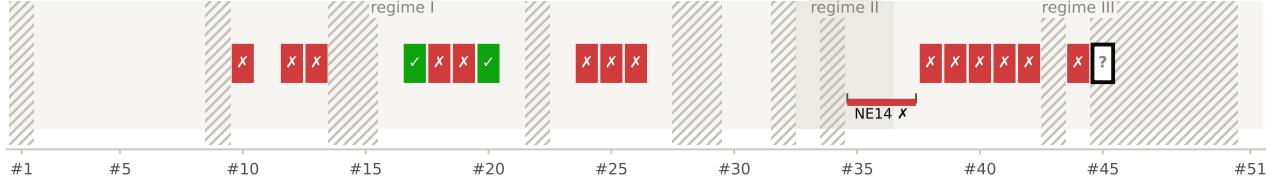
H02: Inferred couplings reflect real influence

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 0.5/5 Usefulness 1.5/5

Bottom line. Activity-timing Ising couplings do not measure influence: they sit at the null floor, miss installed leaders, and the collective term is the daily schedule.

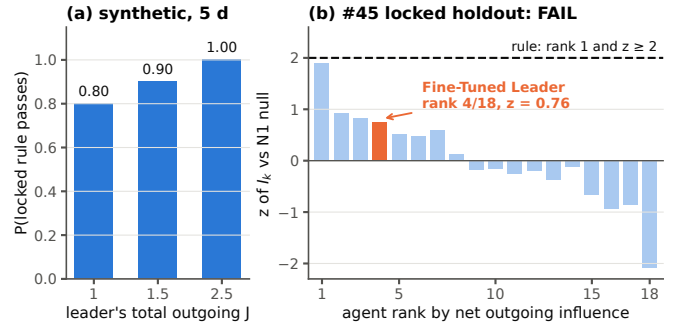
level: hypothesis · pipeline checks 99% · updated 2026-10-04



Goal periods tested (2 supported, 15 failed). ✓ supported ✗ failed ✂ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The spins are the 1-min activity states of the agents. A directed kinetic-Ising fit gives couplings in 21 non-holdout 5-day chunks (#10–#44), with block fields and surrogate nulls. In synthetic villages the pipeline recovers edges (AUC 0.95) and finds a leader with outgoing $J \geq 1.5$ in 90–100% of runs. On real data the pairwise couplings sit at the null floor in both goal modes. The fit misses #26's elected leader. **The locked holdout #45 fails:** the Fine-Tuned Leader ranks 4/18 at $z = 0.76$. **Round 1b** corrects the minute grid, which dropped half of all events, and trims day edges in the nulls. The pairwise couplings stay null. The round-1 collective coupling in regime III is the operator's day edges: it is significant in 2/8 regime-III chunks, against 7/13 in regime I. The fit also misses #12's debate teams and #44's installed leader. Post hoc, talk-spin couplings transfer across days in regime-III shared-objective weeks.



Observables. (a) The pass rate of the locked rule in synthetic villages sampled like #45 (18 agents, 5 days), against the planted leader's total outgoing coupling. The rule finds a leader of strength 1.5 or more in almost every run. (b) The real #45 holdout: z of net outgoing influence for every agent. The installed Fine-Tuned Leader ranks 4th at $z = 0.76$. No agent reaches $z = 2$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An inverse-Ising fit to activity timing does not identify who leads an agent swarm. Per-pair couplings from activity timing are noise. They miss a leader that the operators installed twice (#45 and #44), and they miss a debate's team structure. The collective co-activation that looked like coupling is mostly the operator's daily start and stop. Before you compute any synchrony statistic, trim each day to the all-present window. Without the trim almost every regime-III chunk looks coupled; with it, 2/8 do. To find which agent steers a swarm, measure who talks after whom. That pairwise signal survives, consistent with H50's read-out-gated talk coupling.

PHYSICAL MODEL

Model 02 (kinetic Ising) is primary; model 01 (equilibrium Ising) is the rival. The spin $s_i(t) = \pm 1$ says whether agent i acts or talks in 1-min bin t . h_i is a bias. The field $\delta_i(b)$ per (day, 30-min block) absorbs the schedule and kickoffs. The coupling J_{ij} is the lagged influence of j on i ; σ is the logistic function. I_k is the net outgoing influence. In mean field, βJ_0 is the uniform coupling, m the mean spin and q the mean spin variance. VR is the variance of $\sum_i s_i$ over its independent value.

$$P(s_i(t+1) = +1 \mid \mathbf{s}(t)) = \sigma(2H_i(t))$$

$$H_i(t) = h_i + \delta_i(b) + \sum_j J_{ij} s_j(t)$$

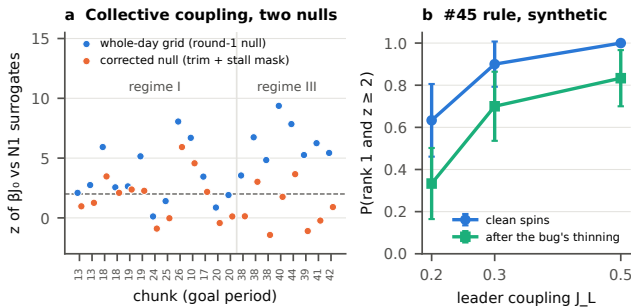
$$I_k = \sum_{j \neq k} (J_{jk} - J_{kj})$$

$$N \text{Var}(m) = q/(1 - \beta J_0 q) \Rightarrow \beta J_0 = (1 - 1/\text{VR})/q$$

H02-couplings-are-real · generated 2026-10-07 17:49 by infra/summaries/build-summaries.py · card hypotheses/H02-couplings-are-real/README.md · ratings: RE-A2 re-evaluation agent (suggested), round 1b

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b, Verdict corrected null)
SP1–SP5 synthetic recovery	AUC 0.95; leader top-1 1.00 at 4 of 5 $J_L \geq 0.3$; FPR 5.8–7.6%
RP1 mode C $\geq 1.5 \times$ mode I, above floor	6–8% → 5–11%; C/I 0.63, $\times$ 1.40
RP4 pairwise beats M1 on held-out days	2/21 → 3/21 chunks vs N1 $\times$
RP6 same-lab enrichment ≥ 1.3 ; RP7 #26 leader top 3	1.16 → 0.85; 8/10 → 10/10 $\times$
MF-P1 $\beta J_{0q} < 0.5$ everywhere	max 0.36 → 0.34 $\checkmark$
MF-P2 III-C significant, mode I null	III-C 5/5 → 2/5; regime I partial 7/13
MF-P3 forward $P(K)$ (unfitted)	CW better 5/13 → 6/9 partial
#45 holdout (rank 1, $z \geq 2$)	4/18, $z = 0.76$ (buggy table; $\times$ not re-run)
Native: #12 teams; 44b leader; NE14	$z = -0.32$; rank 1/5, $z = \times \times \times$ 0.94; rise → -0.08



Second figure. (a) z of the collective coupling βJ_0 per chunk on the corrected grid. Against the round-1 whole-day null almost every chunk looks coupled. Against the corrected null (day edges trimmed, stalls masked) regime III collapses, except #44. (b) Synthetic: the event-drop bug cuts the #45 rule's power from 0.90 to 0.70 at $J_L = 0.3$. It rarely turns a strong leader into $z = 0.76$.

GOAL PERIODS IN DETAIL

G45 ? 1b: not re-run (locked holdout; the executed run read... (**holdout**)
G10 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G12 $\times$ failed
G13 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G17 $\checkmark$ 1b: supported corrected data and DQ8 null: significant-c...
G18 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G19 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G20 $\checkmark$ 1b: supported corrected data and DQ8 null: significant-c...
G24 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G25 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
+8 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The corrected null still leaks under start-up bursts: its DQ8 size is 0.51 for day-edge drives after trimming. H38's agent-state conditioning agrees with it here. The trim keeps 33–99% of minutes, and only 33% in #44. The #45 run used the buggy table. The thinning check draws day retention from non-holdout regime-III days, not from #45. The native tests are small: 295 debate-minutes for #12, and two days and five agents for 44b. The talk-spin result is post hoc. The synthetic harness does not use the corrected calibration (active 0.67, J_{self} 0.49).

ROUND 2

Measure influence in messages: H02-R1 uses context-ledger visibility. Pre-register the talk-spin coupling in regime-III shared-objective weeks before any confirmatory use. Vivian decides whether to re-run the #45 rule on the fixed table.

- **Essence.** Who changes whose mind?
- **H02-R1** Influence is the change in j 's next decision when i 's message is in j 's context (context ledger), compared with matched turns where it is not.
- **H02-R2** The #45 leader influenced plans, not activity: assignment-acceptance events (Jev-labeled) show its authority.
- **H02-R3** Influence follows authority markers (operator endorsement, titles, a leader role), not model family.

H02-couplings-are-real · generated 2026-10-07 17:49 by infra/summaries/build-summaries.py · card hypotheses/H02-couplings-are-real/README.md · ratings: RE-A2 re-evaluation agent (suggested), round 1b (page 2)

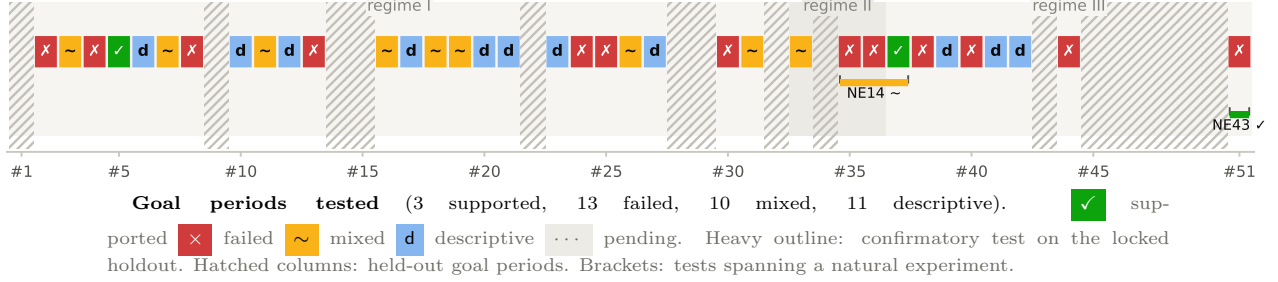
H03: Swarm activity is self-exciting, and its criticality is set by the goal's coupling mode

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. The swarm is subcritical ($n \approx 0.2\text{--}0.4$), its self-excitation does not depend on the goal's coupling mode, and only a small, fast cross-triggering survives its null.

level: hypothesis · pipeline checks 71% · updated 2026-10-04



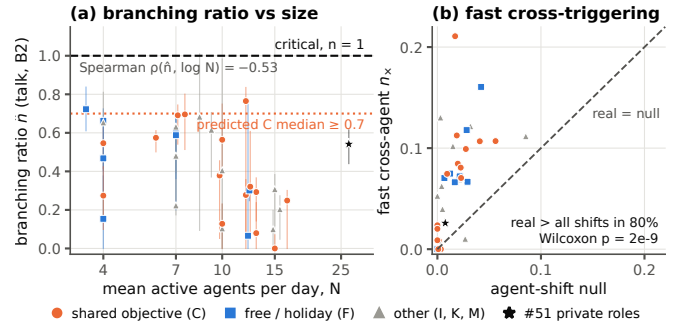
WHAT HAPPENED

The events are agent talk messages. A Hawkes fit with a schedule baseline runs in each of 35 non-holdout goal periods (282 days). The predictions are near-critical shared-objective weeks (median $\hat{n} \geq 0.7$) above free weeks, and a drift of #51 toward $n = 1$ as its roster grows. All fail. The median $\hat{n}$ is 0.41 (0.36 in round 1b); the baseline choice brackets it at 0.2–0.4. Shared-objective weeks give 0.32 against 0.53 for free weeks ($p = 0.75$). $\hat{n}$ falls with swarm size ($\rho = -0.53$) and with #51's roster ($\rho = -0.64$). A constant baseline gives a false $\hat{n} \approx 0.99$. The positive result is fast cross-agent triggering: $n_x \approx 0.07$ at 10–30 s lags, about 4× an agent-shift null ($p = 2 \times 10^{-9}$). Humans and the nudger drive about 1.5% of events directly. **Round 1b** corrects the exogenous inputs, the day cuts and the nulls. Every period verdict stays the same, and the cross-triggering survives the corrected null. In #51 (NE43), withdrawal of all operator messages leaves $\hat{n}$ and activity unchanged. No holdout run exists.

PHYSICAL MODEL

Model 09 (Hawkes self-exciting point process). The events are agent messages, with one realization per village day. $\mu_B(t)$ is the schedule baseline: a per-day level times a 30-min shape, plus a kickoff bump. g_{exo} is the human and nudger input, and ϕ is the triggering kernel. The branching ratio n is the mean number of events that one event triggers directly; $n = 1$ is critical. $\langle s \rangle$ is the mean cascade size. In the multivariate fit, λ_a is agent a 's intensity with own and cross kernels. $\bar{m}$ counts active agents, and $w_m^\times$ is the cross weight at timescale τ_m .

$$\begin{aligned}\lambda(t) &= \mu_B(t) + \sum_{x < t} g_{\text{exo}}(t - x) + \sum_{t_j < t} \phi(t - t_j) \\ \phi(\tau) &= n\beta e^{-\beta\tau}, \quad \langle s \rangle = 1/(1 - n) \\ \lambda_a(t) &= c_a\mu(t) + \sum_{t_j \in a} \phi_{\text{self}}(t - t_j) + \sum_{t_j \notin a} \phi_\times(t - t_j) \\ n_\times &= (\bar{m} - 1) \sum_{\tau_m \leq 300\text{s}} w_m^\times\end{aligned}$$



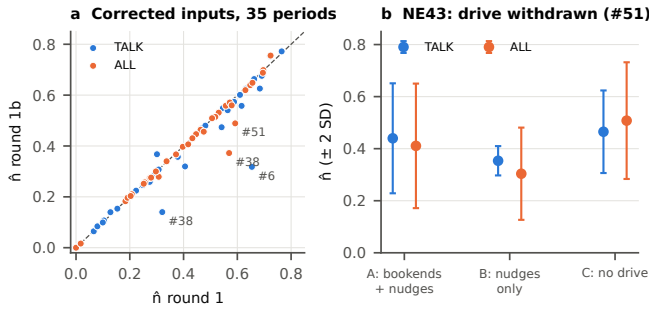
Observables. (a) The branching ratio per goal period against swarm size, by coupling mode, with day-bootstrap 95% CIs. No period is near critical. Shared-objective periods sit no higher than free ones, and $\hat{n}$ falls with size. (b) Fast cross-agent excitation against an agent-shift null that breaks cross-agent timing. Nearly every period sits above the diagonal. This component is the only one that beats its strongest null.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The swarm is far from a runaway cascade, and more agents or a shared goal do not push it closer. A monitor for message storms can assume $\hat{n} \leq 0.4$ in this scaffold: one message triggers fewer than 0.4 direct follow-ups. The scheduler and the roster set the activity level, not social contagion. The one stable coupling is about 0.07 extra messages from others within 30 s of each message. That number can be a scaffold property: agents wake when a message lands. Do not fit a constant-baseline Hawkes model to agent logs. It gives $\hat{n} \approx 0.99$ where a schedule baseline gives 0.2–0.4.

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → Verdict round 1b)	
P1 free weeks subcritical ($\hat{n}_F < 0.5$)	0.53 → 0.52; $4/8 \geq 0.5$	×
P2 shared > free	0.32 vs 0.53 → 0.29 vs 0.52 ($p = 0.80$)	×
P3 shared near critical (≥ 0.7)	0.32 → 0.29	×
P4 mode survives N , $t = -0.19$ → -0.67 ; $\log N t = -2.06$		×
P5 #51 drifts toward $n = \rho = -0.64$ → -0.66 (own × 1 split), -0.70 (shared units)		×
P6 fast $n_\times$: shared > free	0.075 vs 0.071 → 0.075 vs 0.073	×
Fast $n_\times$ vs agent-shift null	0.070 vs 0.017 → 0.074 vs 0.011 after trimming ($p = 2 \times 10^{-8}$)	✓
Native NE43: drive withdrawn	$\hat{n}$ 0.44/0.35/0.47 (within 2 SD); talk rate -11%	✓
Native #2 / NE14	$n_\times$ 0.16 vs null max 0.11; trim moves $\Delta \leq 0.03$, $\hat{n}_{\text{TALK}}$ rises	✓/∼



Second figure. (a) Per-period $\hat{n}$ in round 1 against round 1b. The corrected exogenous drive and the day cuts move almost nothing. #6 and #38 change because round 1b cuts a long operator-off gap out of one day. (b) NE43 inside #51 at a fixed roster: $\hat{n}$ does not fall when the daily bookends stop (B) or when the nudger stops (C).

GOAL PERIODS IN DETAIL

G02 × 1b: failed : n TALK 0.72 → 0.75 on corrected inputs;...
G03 ∼ 1b: mixed : n TALK 0.66 → 0.66 on corrected inputs;...
G04 × 1b: failed : n TALK 0.55 → 0.55 on corrected inputs;...
G05 ✓ 1b: supported : n TALK 0.15 → 0.15 on corrected inputs;...
G06 d 1b: descriptive : n TALK 0.65 → 0.32 on corrected inputs;...
G07 ∼ 1b: mixed : n TALK 0.47 → 0.47 on corrected inputs;...
G08 × 1b: failed : n TALK 0.27 → 0.26 on corrected inputs;...
G10 d 1b: descriptive : n TALK 0.22 → 0.22 on corrected inputs;...
G11 ∼ 1b: mixed : n TALK 0.61 → 0.60 on corrected inputs;...
G12 d 1b: descriptive : n TALK 0.63 → 0.63 on corrected inputs;...
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pooled $\hat{n}$ is still not identified against 10–30-min rate modulation. In regime III, `events_core` lacks computer-use turns, so ALL has a different meaning there. The M3 at-risk set ignores rooms. The #focus room confounds NE43's sides, and side C has 5 days. Round-1b bootstraps use 50/25 replicates (round 1: 100/50). In #51, N is collinear with date and drive. This round does not model the read-out kernel; H42 fits it.

ROUND 2

H42 owns the kernel. Next for H03: an autonomy fraction from the context ledger (H03-R1), NE10 (first nudges), and locked holdout predictions (fast $n_\times$ above the trimmed shift null; $\hat{n} < 0.8$).

- **Essence.** How autonomous is the swarm: what fraction of its activity is self-generated rather than driven by the scaffold or humans?
- **H03-R1** Autonomy fraction: classify every turn's trigger (scheduled, external message, peer message, self-continuation) from the context ledger; autonomy = peer + self share, tracked across regimes.
- **H03-R2** The swarm dies without drive: during outages and nudger-off sessions, peer-driven activity decays with a measurable lifetime.
- **H03-R3** Criticality, if anywhere, lives in content cascades (H34), not in timing.

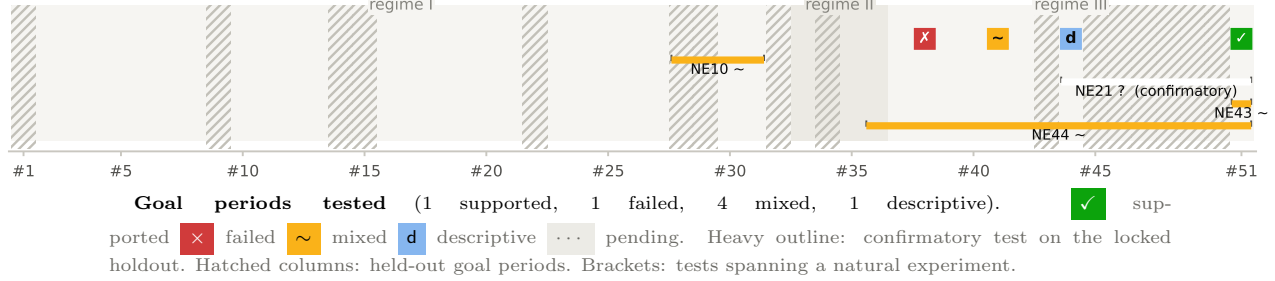
H04: External forcing reshapes the response kernel, reversibly

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. A nudge acts only on its target at the target's next model call; the dead time was an artifact, and no hours or nudger switch lowers self-excitation.

level: hypothesis · pipeline checks 98% · updated 2026-10-04



WHAT HAPPENED

Round 1b is exploratory: non-holdout days, the corrected activity table, leading-@ targets, no future-kick isolation and a day fixed effect. In #51 a nudge adds 1.16 [0.75, 1.57] active minutes to its named target over 30 min. First nudges add 1.38, repeats 0.90; regime-III 4 h days give 0.93. Bystanders gain ≈ 0 . The response starts at the target's next model call: the median read-out is 108 s, and the onset is 1 min after read-out. Dropped events plus the isolation rule made round 1's 4-min dead time. Human messages no longer move activity. The FDT fails in shape, and kicked activations persist ($X \approx 1.9$). Mean-field Glauber underpredicts the decay times. With the nudger off (NE43), n stays the same and traps do not lengthen. **The locked holdout NE21** switches the hours 4 $\rightarrow$ 8 $\rightarrow$ 4 $\rightarrow$ 8 h. **C1 fails in reverse:** n is 0.31, 0.67, 0.46, 0.89. C1 does not use the corrected table, so the result stands.

PHYSICAL MODEL

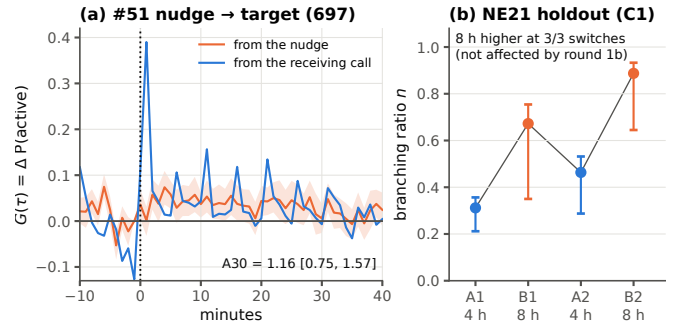
Models 02 (linear response, fluctuation-dissipation) and 09 (Hawkes). $n_i(m) \in \{0, 1\}$ says whether agent i is active in minute m . A kick (a nudge or a human message) at m_0 is a field pulse. G is the matched response, and r is the read-out delay to the target's next model call. C is the activity autocorrelation, χ the integrated G and X the FD ratio. In the Hawkes fit, $\mu_d b(t)$ is the daily baseline and ϕ the talk-triggered kernel. η_c counts messages triggered per class- c kick, and n is the branching ratio. In mean field, K is the loop gain and τ_0 the agent time.

$$G(\tau) = \mathbb{E}[n_i(m_0 + \tau) \mid \text{kick}] - \mathbb{E}[n_i(m_0 + \tau) \mid \text{control}] - \delta_d(\tau)$$

$$G(\tau) \approx \sum_r p(r) \Delta R(\tau - r), \quad \text{FDT: } \chi(\tau) = X [C(0) - C(\tau)]$$

$$\lambda(t) = \mu_d b(t) + \sum_k \phi(t - t_k) + \sum_c \eta_c \sum_j \gamma_c e^{-\gamma_c(t - t_j)}$$

$$n = \int_0^\infty \phi(\tau) d\tau, \quad \tau = \tau_0 / (1 - K)$$



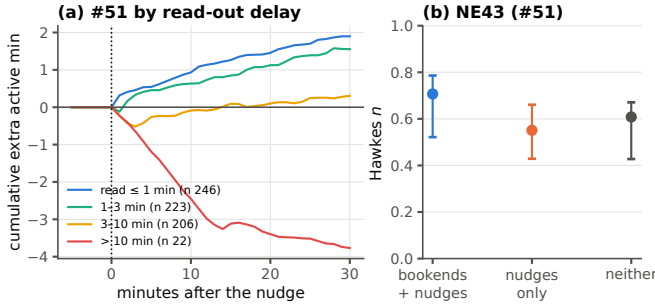
Observables. (a) A nudged agent's activity against matched controls (corrected design, 95% day-block band). The curves align on the nudge and on the target's receiving call. On the receiving call the response is immediate. The delay from the nudge is the wait until the target's next call. (b) The branching ratio per hours segment on the holdout. Every switch goes the other way, at the edge of week-to-week noise.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A nudge acts on the agent it names and on nobody else. It acts at that agent's next model call, so the target's pause schedule sets its lag. A nudge to an agent paused for 10 min does nothing until the agent wakes. Send a nudge to an awake agent: a first nudge buys about 1.4 active minutes, a repeat 0.9. Human messages change what agents say, not how much they work. The physics vocabulary (Green's functions, FDT, effective temperature) mostly shows that a field-on-activity model is wrong. The one reversal test fails in the opposite direction: longer days raise self-excitation. With the nudger off, self-excitation stays the same.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict (1b)
P1 nudge: A60 > 0, peak ~15 min → #51 peak 4, peak ≤5 min A60 1.65 [0.79, 2.52]	yes (#51, 4 h)	
Nudge A30 (isolated → 1.54 → 1.16 (#51), 0.93 (4 h); all nudges) isolation inflates ×1.3–1.9	corrected	
First vs repeat nudge – → 1.38 vs 0.90 (#51)	sub-additive	
P2 bystanders ≤0.2 of yes → 0.03 (#51) target	yes	
P3 human messages mentioned 5.6 → 0.6 (n.s.); room move activity ≈ 0	no	
P6 FDT violated, violated, X 1.54 → 1.88 nudge X < 1	violated; dir. no	
MF-P1/P3 (K subcrit- K III 0.26 → 0.32 ical, self-consistent)	yes	
NE43 nudger off: n Δn +0.06; inactive runs +0.5% same, traps longer	mixed	
NE44 read-out moves 43 s vs 108 s; onset lag equal from with pause default read-out	mixed	
C1 (holdout) n lower 0/3 switches; not affected by 1b on 8 h days	falsified	



Second figure. (a) The cumulative response of #51 nudge targets, by how long the nudge waits to be read. Read within 3 min, the response starts at once. Read after 10 min (long pauses), the target stays below its controls. (b) NE43: Hawkes n with the daily bookends and nudges, with nudges only, and with neither.

GOAL PERIODS IN DETAIL

NE21 ? 1b: not re-run (locked holdout; re-running is Vivian'... (holdout))

G38 × failed
G41 ~ mixed
G44 d descriptive
G51 ✓ supported
NE10 ~ mixed
NE43 ~ mixed
NE44 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The #51 pre-window placebo is +0.25 [0.05, 0.46]. Some selection by the nudger's text trigger remains, so the level can be up to a quarter too high. The matching strata do not include long pauses. Regime-I nudges (NE10) fail the placebo. Hawkes n can absorb within-day non-stationarity. The holdout's C2, C4 and MF-C used the buggy activity table and the isolation rule. Vivian decides whether to re-run them.

ROUND 2

Add the pause state to the matching strata. Fit a read-out plus transient kernel.

- **Essence.** The scaffold is the propagator: the response to any input is when the agent next looks, times whether it attends.
- **H04-R1** The response kernel equals the turn-interval survival function convolved with an uptake probability, with no free parameters (HH92, tested in H08).
- **H04-R2** The steering knob is the scheduler: shortening turn cadence raises susceptibility more than rewording or repeating a message (E1).
- **H04-R3** A salience law: uptake rises with mention, position and novelty in context and falls with backlog (E5).

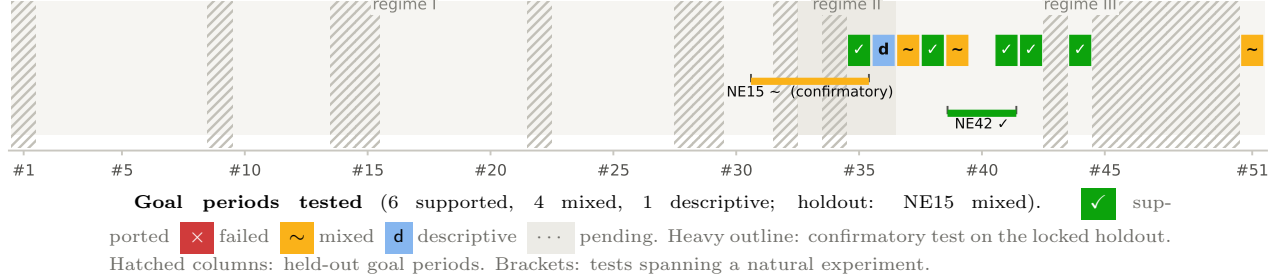
H05: Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 65% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Rooms partition both chat and artifacts; items leak across mostly through command output and chat relays, and dilution is too small to explain the re-cut jumps.

level: hypothesis · pipeline checks 100% · updated 2026-10-05



WHAT HAPPENED

A kinetic-Ising fit asks whether rooms make agents coupled blocks, on 99 non-holdout days (#35–#44, #51 to 09-04). Within-room talk coupling beats cross-room coupling in 6/6 regime-III two-room windows (Fisher $p = 6 \times 10^{-7}$). The pooled effect of co-location on talk ($z = 5.1$) falls to $z = 0.2$ once ledger read counts enter. **The holdout NE15 split** passes the pair DiD ($\kappa_x - 0.022$, $p_{\text{perm}} = 0.02$) but misses the pooled test, so it is **inconclusive**. **Round 2** asks which channels cross rooms. Rooms split artifacts too: cross-room same-day co-editing is ≤ 0.04 of the within-room rate in 5/8 windows, and NE42 cut pairs go $0.56 \rightarrow 0$ at the split. Cross-room commit coupling through a shared repo is not detectable (power 0.33). Room-size dilution exists ($k^{-0.65}$, post hoc) but predicts only $\times 1.2$ per re-cut, not the $\times 6$. Novel repos leak across rooms through command output at $+0.26$ adoptions per exposure (lift 21); history search shows none.

PHYSICAL MODEL

Model 02 (kinetic Ising). The spin $s_i(t) = \pm 1$ says whether agent i talks (talk spin) or acts (activity spin) in 1-min bin t ; $s'_i = s_i(t+1)$. The field h_i is the daily schedule; a cross-day surrogate removes it. r_i is the room. J_{in} and J_{out} are the same- and cross-room couplings; a cut predicts $J_{\text{out}} \rightarrow 0$. Σ_g is the Aguilera-Ito-Kolchinsky lower bound on the entropy production σ , with dual weights θ . $\hat{R}$ and D are block correlations and variances. α_{ij} and γ_d are pair and day effects.

$$P(s'_i | s) = \frac{e^{s'_i H_i}}{2 \cosh H_i}, \quad H_i = h_i + \sum_j J_{ij} s_j$$

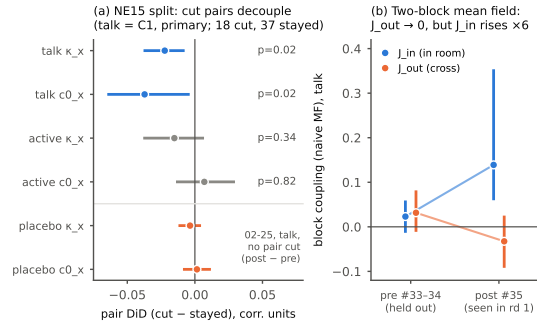
$$J_{ij} = J_{\text{in}} \delta_{r_i r_j} + J_{\text{out}} (1 - \delta_{r_i r_j})$$

$$\sigma = \sum_{i < j} (J_{ij} - J_{ji}) \langle g_{ij} \rangle, \quad g_{ij} = s'_i s_j - s_i s'_j$$

$$\Sigma_g = \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle] \leq \sigma$$

$$J^{\text{MF}} = D^{-1/2} (I - \hat{R}^{-1}) D^{-1/2}$$

$$\kappa_{ij,d} = \alpha_{ij} + \gamma_d + \beta \text{coloc}_{ij,d}$$



Observables. (a) The confirmatory pair difference-in-differences at the 03-16 split: the coupling change of the 18 separated pairs minus that of the 37 pairs that stay together (pre #33–34, held out; post #35, not held out). Talk coupling drops (C1 passes); activity does not. The 02-25 placebo separates no one and shows no change. Bars: day-bootstrap 95% CIs; p : assignment permutation. (b) Block mean field: cross-room J_{out} falls to ≈ 0 , and within-room J_{in} rises (unpredicted).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A room is a read filter, and it is also a work filter: the operator's split separates the repositories as well as the chat. To decouple two agents, stop them reading each other. With read counts known, co-location adds nothing ($z = 0.2$). To move one item across rooms, put it in a command's output (a shared listing or README). One such exposure gives about a 1-in-4 adoption within 2 h, $3 \times$ a same-room chat read. History search does not do it. Do not expect a room cut to change pair coupling by attention alone. Dilution gives $\times 1.2$ for a $4 \times$ smaller room; the observed jumps (≈ 0 when merged, $\times 2$ after the split) carry a goal field too.

RESULTS VS PREDICTIONS

Prediction	Result	Verdict
P2 talk within > cross (1b)	+0.021, 6/6, p 6e-7; trim same	yes
P3 split 05-11 DiD < 0 (1b)	-0.020 (p 0.03)	yes
HH248: reads absorb the room (1b)	$\beta(\text{room}) \approx 5.1 \rightarrow 0.2$ given reads	yes
Holdout C1 / C3 / C5	pass / fail / mixed (buggy table)	not re-run
R1-P1 rooms split artifacts	cross/within co-edit ≤ 0.04 in 5/8; NE42 cut pairs $0.56 \rightarrow 0$	yes
R1-P2 cross-room commit coupling	$\Delta -0.017$ (p 0.76); within rooms +0.017 (p 0.015)	inconclusive
R1-P3 #focus keeps commit coupling	DiD -0.086 [-0.135, -0.036]	no
R2-P1/P2 pooled	dilution, $\hat{\beta}_\kappa$ 0.79 (wide); $\hat{\beta}_u$ -0.34 [-0.55, -0.14]	no
R2 two-room era (hoc)	$\hat{\beta}_u$ 0.65 [0.41, 0.90]	near 0.45
R2-P3 NE42 / NE15 sizes	merge drop $>$ dilution; NE15 $\leq 12\%$ of log 6	not dilution
R3-P1 output / leak	+0.26 [0.11, 0.42], lift 21 / 12 exposures	yes / n.p.
R3-P3 leaks < crossings	22% (relay 38%, none logged 41%)	yes

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

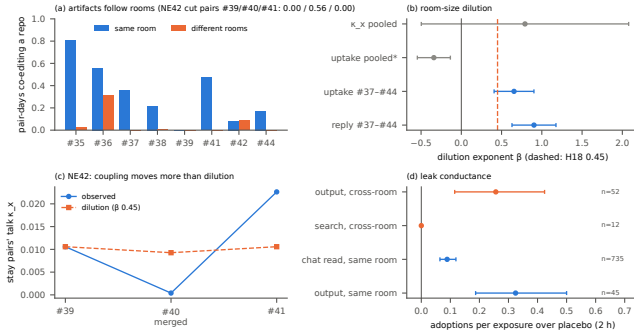
CAVEATS

Co-location and log reads correlate at 0.82, so HH248 has little leverage. Rooms and artifacts are nearly collinear, so the artifact rival is not separable from the room effect. Cross-room commit coupling has power 0.33. The output leak rests on 52 exposures, 48 of them in #36. The pooled dilution test was spoiled by one #focus dyad (no @-mention is needed between two agents). The NE42 size check was not blind. The holdout run used the buggy table.

ROUND 2

Ask Vivian about a fixed-panel NE12/NE15 re-run. Fit a room-specific goal field at NE42 to see whether it absorbs the overshoot. Measure leaks with novel names and numbers (H41 markers), not only repositories.

- **Essence**. Through which channels does information actually move?
- **H05-R1** Rooms matter only insofar as they also split artifacts: cross-room pairs sharing a repo stay coupled through edits (E3).
- **H05-R2** The $\times 6$ rise in within-room coupling after the split is attention reallocation, predicted in size by $N_{\text{room}}^{-0.6}$ (E5).
- **H05-R3** Information leaks across rooms through history search and artifacts at a measurable rate (a leak conductance).



Second figure. Round 2, non-holdout. (a) Share of pair-days on which both agents commit to a common repository, same vs different rooms, per two-room window. (b) Room-size dilution exponents with the room size set by the operator (pair and day FE); dashed: H18's 0.45; grey = pre-registered pooled tests, blue = two-room era (post hoc). (c) NE42 stay pairs' talk coupling vs the dilution prediction. (d) Excess adoptions of a novel repository within 2 h of a first exposure, over matched placebo items of the same agent at the same moment; bars: item bootstrap 95%.

GOAL PERIODS IN DETAIL

NE15 $\sim$ 1b: mixed (not re-run; corrected non-holdout analogue... (**holdout**))

G35 $\checkmark$ 1b: supported p 0.004; fixed bins)

G36 d 1b: descriptive p 0.12; regime II)

G37 $\sim$ 1b: mixed p 0.11)

G38 $\checkmark$ 1b: supported p 0.007; trim +0.018)

G39 $\sim$ 1b: mixed +0.000; was failed)

G41 $\checkmark$ 1b: supported p 0.001)

G42 $\checkmark$ 1b: supported p 0.017; was mixed)

G44 $\checkmark$ 1b: supported p 0.001)

G51 $\sim$ 1b: mixed (the ledger confirms the cut; talk DiD righ...

+1 more periods: see the card.

H05-rooms-cut generated 2026-10-07 17:49 by infra/summaries/build_summaries.py card hypotheses/H05-rooms-cut/README.md ratings: H05 round-2 agent (suggested); coordinator calibration pending (page 2)

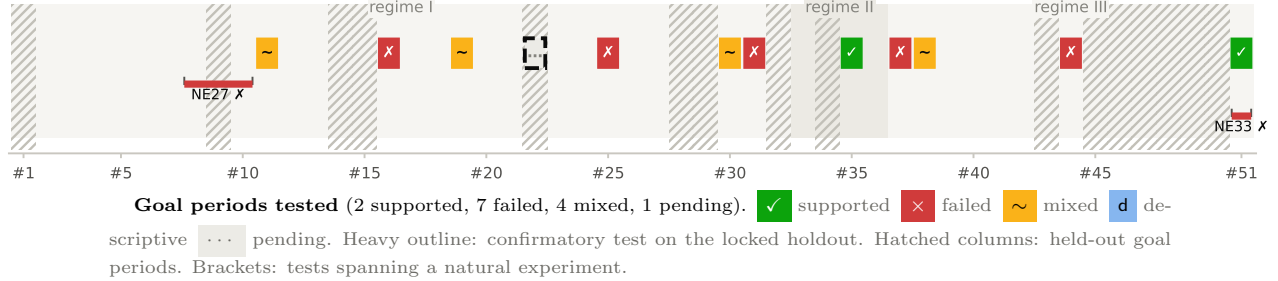
H06: Free weeks show neutral cooperative dynamics with a cooperator core

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 0.5/5 Usefulness 2/5

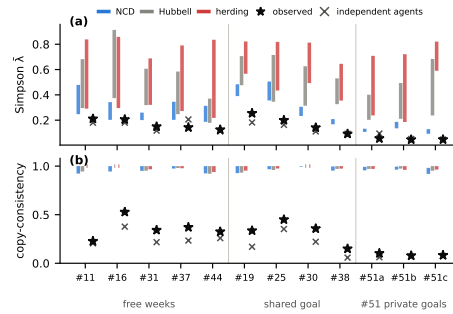
Bottom line. Free LLM agents state private sub-tasks but work on shared repos when one exists; NCD fails, neutral copying fits at best, and copying follows reading.

level: hypothesis · pipeline checks 85% · updated 2026-10-04



WHAT HAPPENED

Each agent slot carries one project label per 30-min window. The label is a cluster of the agent's self-written goal, the repo it links, or (round 1b) the repo it commits to. Exact finite- N simulations with the real label gaps fit three models to 9 statistics: neutral cooperative dynamics (NCD), Hubbell drift and a herding rival. **NCD fails** in free weeks (4/5 failed, 1 mixed). In rounds 1 and 1b the joint check rejects all three models on stated goals in every free week ($p < 0.01$). In free weeks 72–91% of goal clusters have a single agent. A second embedding model and style removal (round 1b) leave this intact. #35 is a control week with a known project: one fork per room. Its stated goals are just as fragmented, but 96–100% of work sits on the room's fork. Goal clusters measure sub-tasks, not projects. Work labels concentrate on a shared repo where one exists. There Hubbell-like copying can fit (#30), without NCD's signature. Copying follows reading: a project named in a read message predicts a switch (OR 8.6 against unread).



Observables. Per period, goal clusters (km24). (a) The observed Simpson λ (star) against the 95% predictive ranges of the three fitted copying models (bars). The cross is an independent-agents null that shifts each agent's day sequence. The observation sits below all three models and close to the null. (b) The share of project switches that join a project another agent holds. Every copying model predicts ≈ 1 . Free and shared weeks give 0.15–0.53, and #51 (private roles) gives 0.08–0.10.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 06 (Piñero et al. 2025). N individuals (agent slots) each hold one species (project). In each step, with probability μ , a new project replaces a random slot C . Otherwise C copies the project of slot A only if A and a third slot B hold different projects. This is cooperation: rare projects recruit more per capita. $\lambda = \sum_i x_i^2$ is the Simpson index of the shares $x_i = n_i/N$. The rivals are Hubbell (always copy) and herding (copy favoured when A and B agree).

$$b_n = (1 - \mu) \frac{n}{N} \left(1 - \frac{n}{N}\right)^2, \quad P_n \propto \frac{(1 - \mu)^n}{n} e^{-(n - N\lambda^*)^2 / 2N}$$

$$\frac{1}{\mu} = \sqrt{\frac{\pi N}{2}} \operatorname{erfcx}\left(\sqrt{\frac{N}{2}} (\mu - \lambda^*)\right), \quad \mu_B = \frac{e^{-2}}{\sqrt{2\pi N}}, \quad \mu_L = \sqrt{\frac{2}{\pi N}}$$

$$P(\text{recruit to } i) \propto x_i e^{\beta x_i} : \quad \beta < 0 \text{ NCD}, \quad 0 \text{ Hubbell}, \quad > 0 \text{ herding}$$

Neutral cooperative dynamics does not describe these swarms. What agents *say* they do is a spread of private sub-tasks in every week, so stated-goal abundances are a poor project census. What agents *work* on concentrates on a shared repo when the task has one. It spreads out when the task gives each agent its own artifact or role (#44 free room, #51), as H11 finds for effort. Copying runs through reading: a read mention raises the switch odds 8.6-fold over an unread one. To get a shared project, assign it or announce it in a message the agents read. To monitor, compute the work-label singleton fraction and the copy-consistency.

H06-neutral-cooperative-dynamics · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H06-neutral-cooperative-dynamics/README.md · ratings: RE-D2 round 1b (suggested); round 1 by H06 agent, calibrated by coordinator

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P1 NCD beats rivals (free)	4 failed, 1 mixed → same count (#11, #16 swap)	failed
Adequacy, stated goals	all 3 models rejected in 5/5 free weeks (both rounds)	failed
Singletons, goals (free)	0.79–0.91 → 0.72–0.84 (gte, style removed)	robust
P2 rare-project advantage	#31 β $-0.78 \rightarrow +1.47$; sign varies by embedding	failed
Work vs goals (R1b-3a)	work more concentrated: $\bar{\lambda}$ 5/5, supported copy 4/5	
NCD on work (R1b-3b)	inadequate 4/5; #30 NCD and Hubbell adequate, no core	supported
Copying needs reading	OR read vs unread 8.6 (attention), 5.6 (work)	supported
#35 known universe (native)	work 96–100% on own fork; goal singletons 0.76	supported
#51 rivals share (native)	goals 2/3 blocks; attention 0/3; 1/3 work 0/3	
#44 assigned vs free (native)	assigned arm more concentrated in 3 spaces	mostly

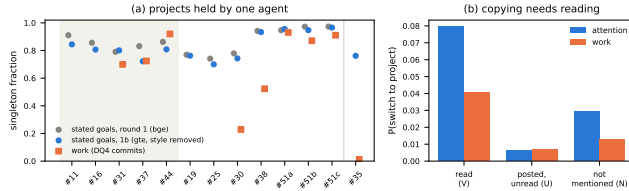
GOAL PERIODS IN DETAIL

G22 ... 1b: pending (confirmatory, not run) (**holdout**)
G11 ~ 1b: mixed (P1 by rule; all models still inadequate)
G16 × 1b: failed (km24 only; all models inadequate)
G19 ~ 1b: mixed (P6 unchanged; all models inadequate)
G25 × 1b: failed (P6: conformist no longer beats NCD)
G30 ~ 1b: mixed (P6); work labels: NCD and Hubbell adequate...
G31 × 1b: failed (P1 6/6; work labels also inadequate; P2 n...
G35 ✓ 1b: supported (native: known universe recovered; goal...
G37 × 1b: failed (km24 only; all models inadequate)
G38 ~ 1b: mixed (P6; km24 all inadequate; work: Hubbell ade...
+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. (a) The share of projects that one agent holds. Stated goals (grey round 1, blue round 1b) stay high everywhere, including #35, where the project is known. Work labels (orange) drop where a shared repo exists (#30, #35, #38). Shaded: free weeks. (b) The probability of a switch to a project that another agent holds. The classes: a message naming it is read, posted but unread, or absent.

CAVEATS

Compute was scarce. The full estimator (all clusterings) reran on #11 and #31 only. Elsewhere round 1b uses the primary clustering, the work labels and model-free statistics. Likelihood ratios near ± 2 and β depend on the embedding. In two-room weeks the unread class is mostly the other room's messages. The day-shift null cannot separate a persistent shared field (room assignment) from copying. Work labels cover public commits only, dense from #30.

ROUND 2

Fit NCD and Hubbell with agent-specific fields on work labels. Anchor stated goals to repos. Run the #22 confirmation after sign-off, with work labels as a secondary.

No round-2 redirects yet.

H06-neutral-cooperative-dynamics · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H06-neutral-cooperative-dynamics/README.md · ratings: RE-D2 round 1b (suggested); round 1 by H06 agent, calibrated by coordinator (page 2)

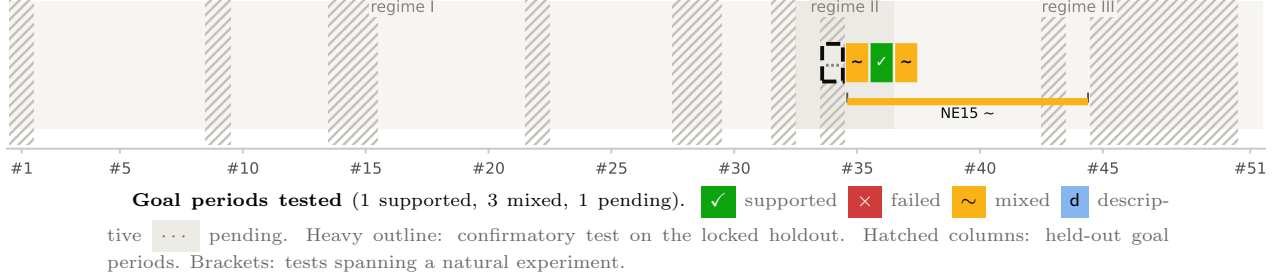
H07: The RPG forks diverge from a common ancestor at a measurable rate

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 1.5/5

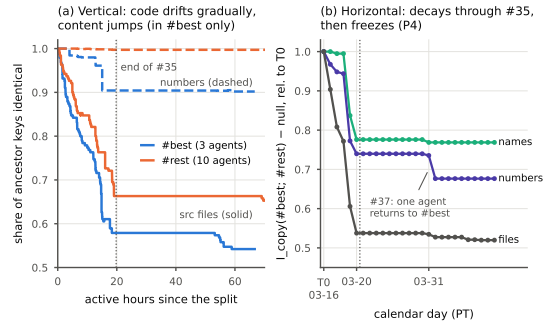
Bottom line. The RPG forks diverge by copying (code per edit, content in one agent's jumps) with no readable leakage; a fork family with a shared upstream re-converges.

level: hypothesis · pipeline checks 83% · updated 2026-10-04



WHAT HAPPENED

On 03-16 the village splits into #best (3 agents) and #rest (10), and each room takes its own copy of the #34 RPG. Round 1 (#35 on) tracks both git lineages by keyed features: files, functions, numbers and names. Code diverges gradually: at the end of #35, 69% (#best) and 81% (#rest) of ancestor files are still identical. No commit carries over 10% of the change. Content changes in jumps: two re-theming commits carry 59–77% of #best's name and number changes. Copying dominates: $I_{\text{transform}} \approx 0.01 I_{\text{copy}}$. Copy information between the forks falls every day of #35, then freezes for files and names (P4 mixed). The independent-lineage bound holds exactly. The 3 shared innovations are convergent bug fixes, not leakage. The commit clock, population size and #best's feature hierarchy fail. Round 1b changes no verdict: cross-team reads are 2% of #35 chat, and none names a shared innovation. The #34 holdout test has not run.



Observables. (a) The share of ancestor keys still identical in each fork, against active hours since the split. Code (src files, solid) erodes in many small steps in both forks. Numeric parameters (dashed) change in a few jumps, almost only in #best. Both stop when work on the game stops after #35. (b) Shuffle-corrected copy information between the forks, relative to T0. It declines every day of #35 and then stays flat. The exception is numbers, when one #best agent expands the game in #37.

PHYSICAL MODEL

Model 08 (copying vs. transformation) plus a mutation clock. A key (a file path, a number's code path, an entity id) has ancestor value x and descendant value y . The keys give $p(x, y)$. The vertical channel runs from ancestor to fork after clock time τ (active hours, commits or file-touches). The horizontal channel runs between #best and #rest. $q_x = p(Y=x|X=x)$, and d is the binary KL divergence in bits. c is the copy fraction and κ its chance-corrected form. μ is the per-key mutation hazard, and f is the mutable fraction.

$$I(X; Y) = I_{\text{copy}} + I_{\text{transform}}, \quad I_{\text{transform}} \geq 0$$

$$I_{\text{copy}} = \sum_x p(x) d(q_x \| p_Y(x)) \mathbb{I}[q_x > p_Y(x)]$$

$$c = P(Y=X), \quad \kappa = \frac{c - \sum_x p_X(x) p_Y(x)}{1 - \sum_x p_X(x) p_Y(x)}$$

$$c(\tau) = (1 - f) + f e^{-\mu \tau}$$

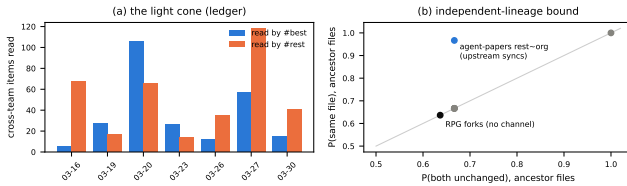
$$P(y_{\text{best}}=y_{\text{rest}} \mid \text{ancestor key}) = P(\text{both unchanged})$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

This is mostly a curiosity: one natural experiment, $n = 2$ forks. Forks drift while teams edit them and stop when the edits stop. Code divergence tracks edits, not time or headcount: 3 agents out-diverge 10 (post hoc; #34 tests it). Isolated teams make the same fixes to inherited bugs: all 3 shared innovations here are convergent. An auditor of sandboxed agents should treat an identical change in two groups as weak evidence of leakage. Per-feature copy fractions are a cheap lineage diagnostic. None of this is a lever.

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P1 gradual divergence	files 0.69 / 0.81 identical at end of #35 (shared copy_info: identical)	supported
P4 decay then freeze	monotone in #35; content revives in #37	mixed
P5 rare, channel-borne leakage	3 shared innovations; 0/3 with a cross-team <i>read</i> naming them (convergent)	mixed
P6 population size	53.7 vs 19.6 → 51.3 vs 19.6 per capita (DQ4 agent work, 0 automated)	failed
P8 commits follow rooms	98.5/99.6% → 99.4/100% (DQ6 rooms, #35)	supported
Light cone (NE15)	cross-team reads 1.98% of #35 items, only in co-presence episodes; 5/5 cross-fork commits after a read	supported
Lead designer nucleates (#35)	lead's share of content 0% (commits 22%); one agent 86%	failed
Second fork (agent-papers)	identity 0.97 vs bound 0.67; 29/29 shared files upstream-first	supported



Second figure. (a) Chat items read across teams per day (context ledger). Reads occur only in the split hour, the 03-19/20 co-presence episodes and later #general days. (b) The identity of ancestor files between two copies, against the independent-lineage bound. The RPG forks sit on the diagonal: no channel. One *agent-papers* pair sits far above it because of upstream syncs.

GOAL PERIODS IN DETAIL

G34	...	1b: pending (confirmatory, not run) (holdout)
G35	~	1b: mixed (replication unchanged); native lead-design...
G36	✓	1b: supported (unchanged); native second fork supported
G37	~	1b: mixed (unchanged)
NE15	~	1b: mixed (unchanged); native light cone supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	0
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The data is one split, two RPG forks and one nucleating agent. The copy-information split uses our unverified reading of Kolchinsky & Corominas-Murtra. The ledger has no reads for the Claude Code agent, which made one of the convergent fixes. Keyed features break on restructured code. The second fork family is file-level only. Two of its four copies never move on *main*, because their work goes to upstream pull requests. The touch clock is post hoc until #34 runs.

ROUND 2

Run the #34 confirmation. Use the #39→#40 artifact merge as a horizontal-transfer test. Build a paraphrase-level detector for re-implemented features.

- **Essence.** Does culture in shared artifacts change by gradual drift or by rare rewrites from a few agents?
- **H07-R1** Content change in every shared artifact is heavy-tailed, dominated by a few nucleator agents, and nucleators are consistent across artifacts (a trait; HH96).
- **H07-R2** Mutation per file-touch is universal across artifacts and agents (HH95).
- **H07-R3** Forks re-converge when the same models meet the same bugs: the rate of convergent evolution is set by the shared prior (E4).

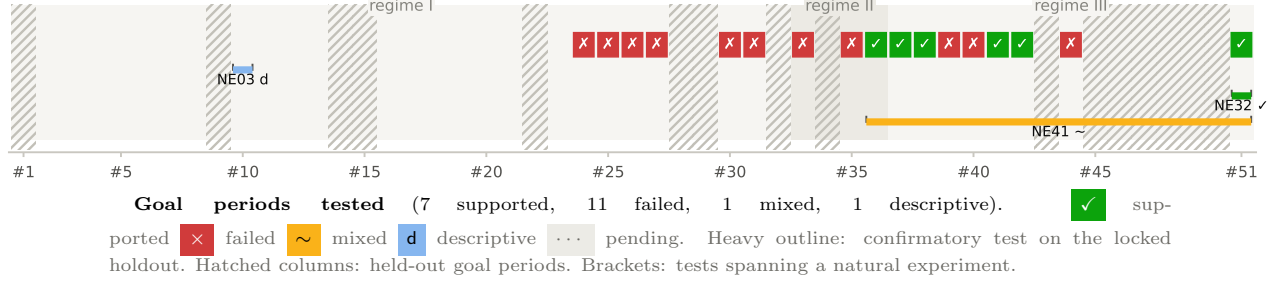
H08: Context is the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 55% Faithfulness 3/5 Usefulness 2.5/5

Bottom line. Responses start at the first model call whose context holds the message, and content moves only inside the replies that name the sender; erasure cuts the coupling by about a fifth.

level: hypothesis · pipeline checks 71% · updated 2026-10-05



WHAT HAPPENED

The question is whether agents couple only through what enters their model calls. Rounds 1b and 2 use the context ledger, which records which call received which message, on 17 non-reserved goal periods. (1) A recipient's chance of naming or replying to the sender jumps at the call that received the message, not at the call already running: names 14/17 periods, replies 17/17; other-room messages give no jump (8/8). (2) Content follows only inside those replies. At equal lag, a statement from a call that read the message is closer to it than an in-flight statement only when it names or replies to the sender (G51: +1.25 vs $-0.39 \cos \times 100$ without them; 0/9 regime-III periods positive without names). Without names, in-flight talk is closer: both answer the same earlier turn. (3) A forced erasure (NE41) cuts reply coupling by 21%. Senders written into memory get 5.5 pp more replies, but whether that protects the thread is unpowered. (4) The Claude Code agent's event feed replayed 2025 for 9 days (17 active hours) while its status stayed current. A five-fetch event-age monitor catches it in 0.4 active hours. Nothing is confirmed on reserved data.

PHYSICAL MODEL

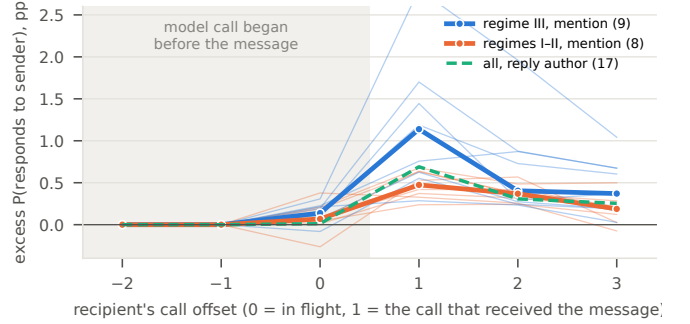
Model 02 (two-variable kinetic Ising) with Hawkes kernels (model 09). Each agent has a visible spin n_i (active per minute) and a hidden field c_i : the room messages that arrive after its last model call. Calls start on the scaffold's schedule, and the scaffold assembles the context at t_{call} . Each call reads and resets the field, and the couplings J_{ij} act only then. W is the read-out delay. o is the call offset (1 = the receiving call). β_F is the coupling change after a forced erasure.

$$c_i(t) = \{m : \text{room}(m) = r_i, t_{\text{call}}^- \leq t_m < t\}$$

$$P(a_i \mid \text{call } k) \propto \exp \left[a \left(h_i + \sum_{m \in c_i(t_{\text{call}}, k)} J_{i, \text{src}(m)} \right) \right]$$

$$G(\tau) = \sum_w p(w) \Delta R(\tau - w)$$

$$D = G(o=1) - G(o=0)$$



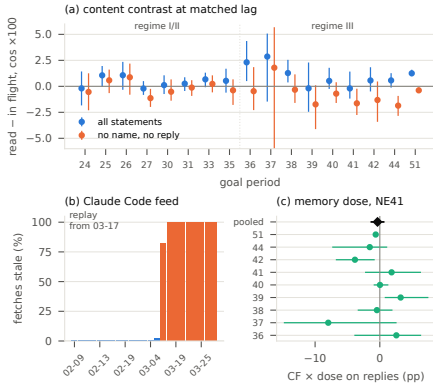
Observables. The excess chance that a recipient's call responds to the sender, by call offset, on the context ledger. Offset 0 is the call already running when the message arrives. Offset 1 is the call that receives it. Recipients are not mid-exchange. Thin lines are periods, bold lines regime medians (mentions), and the dashed line the median for reply labels. The response appears only at the receiving call, in every regime.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The coupling delay equals the call cadence. A message cannot change an agent before a model call has the message in its context: about 15 s when the agent is busy, minutes when it pauses. Name the recipient if you want its content to move: unaddressed messages shift content no more than co-response does. A forced erasure cuts conversational coupling by $21 \pm 7\%$; writing the sender into memory is not shown to restore it. Log what each call received and alarm when the median age of fetched events over five fetches exceeds 1 h: that caught the replay in 0.4 active hours with no false alarm. A content-only monitor does not.

RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b (round 2: R Verdict rows)
C9 talk jumps at read-out ($\geq 70\%$)	6/17 → 6/17 (none in regime I/II) no
C9 addressing jumps at read-out	11/17 → 14/17 (regime I 1/6 → 4/6); replies 17/17 yes
C9 other-room messages: no jump	6/8 → 8/8 two-room periods yes
C9-V5 active read-out delay 10–40 s	32–49 s → 29–41 s (8/9 regime III) yes
Content, no name or reply (R2, matched lag)	1b $D_{\cos} > 0$ in 6/9 → 0/9 regime III; G51 $-0.39 [-0.62, -0.12]$ drawn no (with-)
C8 nudge onset = read-out	#51 lag ≈ 3 min → onset at the receiving call (t_{25} 1 min); transient, not a step yes, shape no
C8 beats a constant delay	no → yes: onset follows read-out (4, 4, 15 min) yes
NE41 erasure cuts coupling $\geq 30\%$	-18% (mentions) → $-21\% \pm 7\%$ (replies), 8/9 sign only
NE41 memory dose protects (R4)	CF $\times$ dose -0.36 ± 0.55 pp; power 0.04 inconclusive
NE32 newcomers respond only after receipt	new: 35/35 pairs yes
C1 room-rule recall ≥ 0.9	0.97–1.00 until the replay (unchanged) until replay
Replay audit (R5): one episode, monitor < 1 h	03-17 → 03-27; alarm 0.38 h, 0 false yes
Content-only decoupling monitor (R5)	replay M ratio 0.37 / 0.54; 2/9 no days flagged



Second figure. (a) Round 2: content similarity of a recipient's statement to a message, statements from calls that read it minus in-flight statements at the same lag and message density. Blue: all statements. Orange: statements that neither name nor reply to the sender. (b) The Claude Code agent's fetches: share of returned events older than 24 h, per day. (c) NE41: forced erasure $\times$ the sender newly written to memory, effect on replies, per period and pooled.

GOAL PERIODS IN DETAIL

G24 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
 G25 $\times$ 1b: failed context-ledger read-out; C9 talk and address...
 G26 $\times$ 1b: failed context-ledger read-out; C9 talk and address...
 G27 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
 G30 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
 G31 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
 G33 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
 G35 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
 G36 $\checkmark$ 1b: supported context-ledger read-out; C9 talk and address...
 G37 $\checkmark$ 1b: supported context-ledger read-out; C9 talk and address...

+10 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The ledger measures call starts only for Gemini (19% of calls), so about 10% of items can sit one call off. Reply labels keep a thread-membership component. The content contrast needed two amendments found by the synthetic guard (hour-block resampling, density strata). The memory-dose test has power 0.04 at half protection. The replay audit covers one agent. No result is confirmed on reserved data.

ROUND 2

Switch the confirm script to the ledger before the run. Find a content null that removes co-response. Ship the five-fetch event-age monitor as a toolkit item.

- **Essence.** Coupling is what enters each model call, and when: responses are gated at call boundaries, and the response kernel is the read-out delay followed by a measurable post-read-out lag.
- **Round-3 redirects (newest).**
- Content needs a stronger null than the in-flight partition. A content response independent of names would need a read vs posted-but-unread contrast that removes co-response, for example statements whose producing call received the message b
- R4 needs a bigger first stage. The memory test needs either many more forced erasures per sender or the reserved #43 memory week (confirmation only).
- R5: report the five-fetch event-age monitor as a toolkit item; drop the content-only monitor.

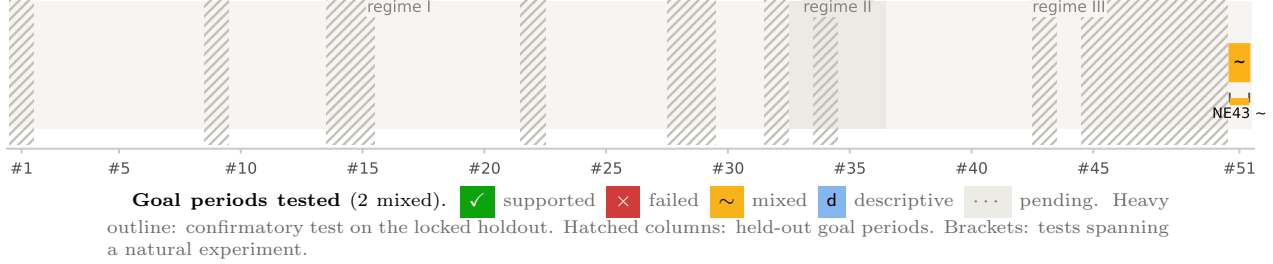
H09: Agent swarms have an effective thermodynamics

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. The thermodynamic framing adds little; the timer gate reads context (chatter holds, an @-mention releases), and memory size is a homeostat that overshoots after forced erasures.

level: hypothesis · pipeline checks 29% · updated 2026-10-04



WHAT HAPPENED

H09 is an exploratory framing, not one test. It asks whether swarm statistics read as thermodynamics: landscapes, currents, traps and equations of state. Two light non-holdout rounds and round 1b exist; no holdout has run. In round 1 the daily schedule, a field, reproduces the two wells of the activity landscape. Per-agent action chains are irreversible: 78% of cells beat the null. Regime III has a closed cycle consolidate → search → talk → pause. Event-level entropy production collapses after perma-computer-use. Agents are ergodic at day scale, and restarts relax within a minute. In regime I, room messages raise the escape from idle spells (hazard ratio 3.06; null p95 0.96). In regime III a pause is a self-set timer (0.22% end early). Round 1b reads the gate on the context ledger: unaddressed chatter keeps agents paused (OR 0.50), and an @-mention releases them (OR 3.1). Memory size is an agent-specific set point that overshoots after forced erasures. Collective entropy production is small (plug-in only).

PHYSICAL MODEL

No single model: each sub-claim borrows one (models 01, 02, 09, 10). $\lambda(t)$ is the escape hazard of an idle spell at elapsed dwell t . HR is the hazard ratio for a kick in the prior 60 s against none. k_0 and c are the base escape rate and the kick sensitivity, and r is the kick rate. t^* is the declared pause expiry, and p_{gate} the chance of acting at expiry. π and P are the stationary law and transitions of an agent's action-class chain. V is the memory size, P_{in} the inflow, β the response and γ the relaxation.

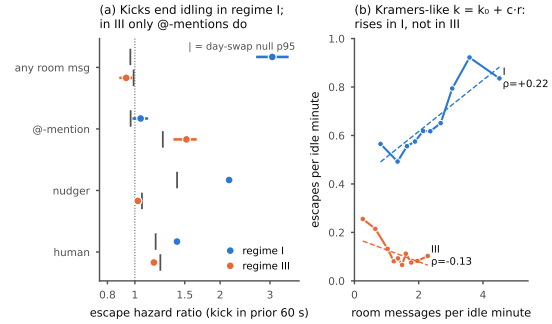
$$\lambda(t) = P(\text{escape at } t \mid \text{idle until } t), \quad \text{HR} = \lambda_{\text{kick}} / \lambda_{\text{none}}$$

regime I (Kramers-like): $k \approx k_0 + cr$

regime III (timer gate): $\lambda(t) \approx \delta(t - t^*) p_{\text{gate}} (\text{@-mention})$

$$\sigma = \sum_{x < y} (\pi_x P_{xy} - \pi_y P_{yx}) \ln \frac{\pi_x P_{xy}}{\pi_y P_{yx}}$$

$$\Delta V = \beta P_{\text{in}} - \gamma V_{\text{prev}} \Rightarrow V^* = \beta P_{\text{in}} / \gamma$$



Observables. (a) The escape hazard ratio of idle spells: minutes with a kick in the previous 60 s against minutes without, stratified by elapsed dwell and time of day. Bars: agent-day bootstrap CIs where computed; ticks: 95th percentile of a day-swap null. In regime I any room message triples escape. In regime III room messages do not help; only @-mentions do. (b) The escape rate against the kick rate over agent-day deciles (dashed: linear fit $k_0 + cr$). It rises in I and falls in III.

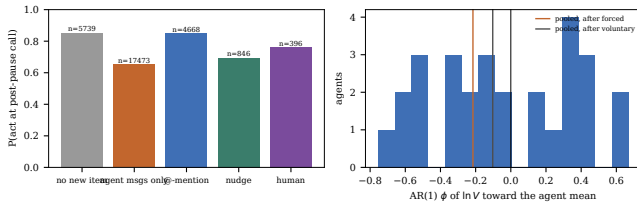
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The thermodynamic vocabulary does little work; the useful findings are plain mechanics. In the current scaffold (regime III) an idle agent sleeps on a timer that it set itself. Unaddressed room chatter keeps it asleep: it acts at 65% of gates, against 85% with no news. To wake a timer-paused agent, address it by name, and expect a delay of up to its chosen pause. Regime-I agents were interruptible. Memory size is a set point: consolidation resets it, and storage does not grow. The rest restates the scaffold: schedule-made landscapes, day-scale ergodicity, and a consolidation cycle that is the prompt's own loop. Nothing here yet beats a simpler statistic.

H09-swarm-thermodynamics · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H09-swarm-thermodynamics/README.md · ratings: RE-R1 agent (suggested, round 1b); coordinator calibration pending

RESULTS VS PREDICTIONS

Prediction	Round 1/2 → round 1b	Verdict (1b)
E1 VR > 1.5 (regime III)	1.61 → 1.85; trimmed to day edges 1.22	no (day edges)
E6c @-mention HR (co-incidence)	1.52 → identical (clean mentions)	yes
G1a @-mention at the ledger gate	new: OR 3.1 [1.9, 4.5]	yes
G1b unaddressed chatter neutral	new: OR 0.50 vs no news (0.65 vs 0.85 act)	no
N1a gate stable at NE43 steps	log ratio -0.01, -0.05	yes
N1b nudge accounting	drop 0.036 > bound 0.013	no
E8a elasticity on inflow	0.16 (exposure) → 0.14 (ledger reads)	clock, inflow not
N2a #51 V vs reads per call	0.02 [-0.02, 0.06]	yes
N2b #51 V vs N	0.33 [0.03, 0.64]	no (tenure?)
N3 homeostat, no overshoot	ϕ median -0.10; after forced -0.22	overshoot (HH269)



Second figure. (a) Regime III, non-holdout: the probability that an agent acts, rather than re-pauses, at the call after a timer pause, by what newly enters that call in the context ledger. (b) The per-agent AR(1) coefficient of log memory size toward the agent's mean. Lines: pooled coefficient after forced (41-turn cap) and voluntary consolidations.

GOAL PERIODS IN DETAIL

G51 ~ 1b: mixed (V ignores read inflow but rises with N; th...
 NE43 ~ 1b: mixed (the gate is unchanged; overall acting fell...

FAITHFULNESS SCORECARD

A mapping – F identifiability –
 B assumptions – G ground truth –
 C adequacy – H comparative –
 D unfitted predictions – I transfer –
 E interventional –
 Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The gate contrasts are within-agent but not randomized: chatter arrives when other agents are busy. In #51, time confounds N2's N effect. E7 (pair EP) did not re-run on fixed bins. No scorecard axis has a synthetic check.

ROUND 2

Fit a two-parameter gate model per agent: chatter suppression and mention release. Test HH269's gain at NE14 with the ledger's reset types.

- **Essence.** Tokens are the energy, progress is the work, and context resets are the heat.
- **H09-R1** Token thermodynamics: power = tokens per hour; efficiency = progress (work ledger) per token; loops waste tokens at zero efficiency.
- **H09-R2** Consolidation is Landauer-like erasure: tokens spent consolidating vs information kept.
- **H09-R3** The swarm is a driven-dissipative steady state whose throughput is limited by attention, not compute (E5).

H09-swarm-thermodynamics · generated 2026-10-07 17:49 by infra/summaries/build-summaries.py · card hypotheses/H09-swarm-thermodynamics/README.md · ratings: RE-R1 agent (suggested, round 1b); coordinator calibration pending (page 2)

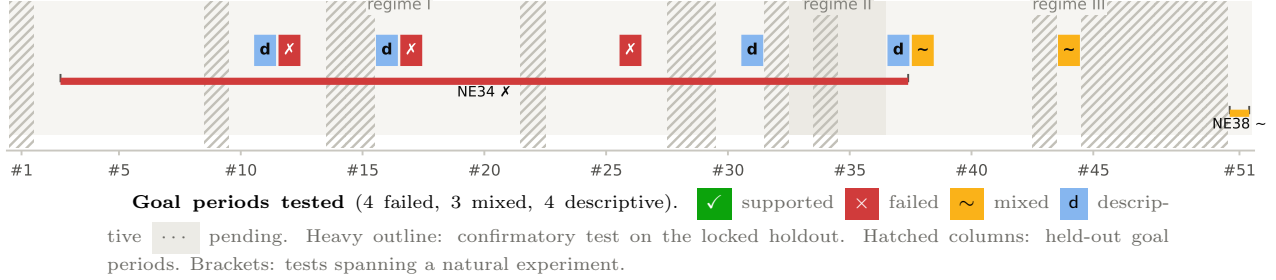
H10: Goals are Legendre pushes: a swarm's response to an assigned goal is predictable from its unforced fluctuations

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Goals are quenches toward a common target, not Legendre pushes: free-week fluctuations do not predict who follows, in both embedding models.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The question is whether a swarm's unforced fluctuations predict how it lines up behind an assigned goal. The state is each agent's 30-min statement embedding, projected on the goal direction $\hat{g}$. Three free $\rightarrow$ assigned pairs exist (#11 $\rightarrow$ #12, #16 $\rightarrow$ #17, #37 $\rightarrow$ #38). In each pair one field λ fits the mean shift, and the model predicts the rest. Alignment rises 1.4–3.1 free-week SDs, outside linear response. **The core test fails:** fluctuations do not predict who moves ($r = -0.37, -0.69, +0.20$; Stouffer $p = 0.89$). **Round 1b** uses shared goal vectors, a second embedding model, dedupe and style residualization. The test fails in all 8 input configurations. Variance along $\hat{g}$ grows, and transverse variance holds. In a new native test a reassigned goal moves one agent 10–20 SDs, beyond any point its unforced content visits. An agent-set goal moves agents, but again not in fluctuation order. The holdout is frozen and has not run.

PHYSICAL MODEL

Model 11: mean-field $O(n)$ in $n = 32$ whitened embedding dimensions. $\mathbf{s}_i$ are agent i 's unit statement vectors. $y = \hat{\mathbf{z}} \cdot \hat{\mathbf{g}}$ is the alignment with the goal field direction. x_i is its 30-min window mean, with free-week mean μ_i , noise-corrected variance $\kappa_{2,i}$ and third cumulant $\kappa_{3,i}$. The goal is a linear field $\lambda = \beta h$ along $\hat{g}$. K is the cumulant generating function, J_0 the mean-field coupling, g the loop gain and C_F the free-week window covariance.

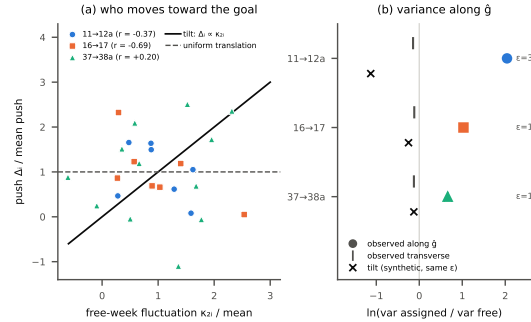
$$P_A(\mathbf{s}) \propto P_F(\mathbf{s}) e^{\lambda \sum_i \hat{\mathbf{g}} \cdot \mathbf{s}_i}, \quad G = F - \lambda m$$

$$K_A(t) = K_F(t + \lambda) - K_F(\lambda) \Rightarrow \kappa_n^A = K_F^{(n)}(\lambda)$$

$$\Delta \mu_i = \lambda \kappa_{2,i} + \frac{\lambda^2}{2} \kappa_{3,i}, \quad \kappa_{2,i}^A = \kappa_{2,i} + \lambda \kappa_{3,i}$$

$$\chi_{\text{swarm}} = \chi_0 / (1 - \beta J_0 \chi_0), \quad g = \beta J_0 \chi_0 = 1 - V_{\text{indep}} / V$$

$$\Delta \mu = \lambda C_F \hat{g}$$



Observables. (a) Each agent's push toward the assigned goal against its free-week fluctuation along $\hat{g}$, both divided by the pair mean. The tilt predicts the solid line, uniform translation the dashed one. The regime-I pairs slope the wrong way. (b) The change in variance along $\hat{g}$ (colored), along transverse directions (bars), and for an exact tilt of a bounded mean-field swarm at the same push (X, synthetic). The goal disperses the swarm along $\hat{g}$; it does not narrow it.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The promise was to read a swarm's steerability from its unsteered fluctuations. That promise does not hold here. A goal change is a large kick (median about 2.5 SDs), not a small field. The agents that wander most are not the agents that follow most. An assigned goal acts as a common target and a new collective mode, not as a gentle tilt of existing preferences. Do not forecast compliance from pre-goal behavior: the fluctuation–response correlation runs from -0.69 to $+0.20$. The instrument (goal-direction projection, noise-corrected variance) is cheap on any agent log. The fluctuation–response physics it tests is a curiosity in this village.

H10-goals-are-legendre-pushes · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H10-goals-are-legendre-pushes/README.md · ratings: RE-C1 re-evaluation agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Round 1 $\rightarrow$ 1b (bge / gte)	Verdict
P1 fluctuations order who moves	r $-0.37, -0.69, +0.20 \rightarrow$ same / $\times$ $-0.53, -0.49, -0.09$; 8/8 configs fail	$\times$
Push $\Delta\bar{\mu} > 0$	3/3 pairs; 19/19 kickoffs $\rightarrow$ same $\checkmark$ in all configs	$\checkmark$
P2 variance by the tilt	untestable ($\varepsilon > 0.34$); gte n/a #37 $\rightarrow$ #38 would pass (ε 0.69)	n/a
Cross-kickoff man > 0	Spear- $-0.56 \rightarrow -0.56 / -0.59$	$\times$
G44 assigned vs own-choice room	#best ε 0.97 / 1.55; #rest none; $\sim$ gap +0.17	$\sim$
NE38 one agent re-signed	ε 10 / 20; post mean beyond the pre support	(tilt $\times$)
G26 agent-set goal	push $\varepsilon \approx 1$; P1 r $-0.14 / -0.63$	$\times$

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

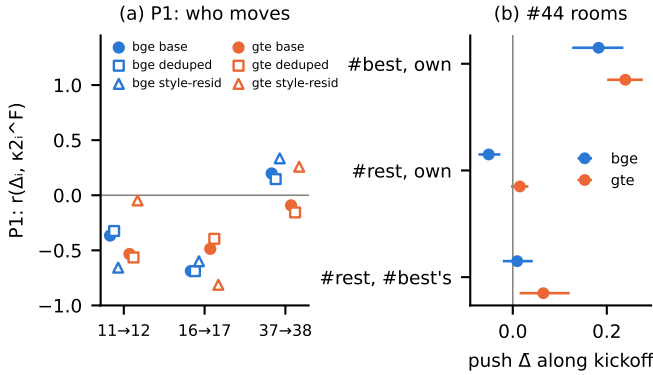
CAVEATS

$\hat{g}$ is a text-embedding proxy for the field. gte shrinks the regime-III push and makes its variance tilt-compatible, so the regime-III result depends on the model. Regime I has $N = 7$. NE38's concurrent control is one incumbent, which drifts 1–3 SDs along its own goal. The #44 assigned room has 4 agents and three days. The natives are exploratory. #22 $\rightarrow$ #23 and #31 $\rightarrow$ #32 remain blind.

ROUND 2

Re-map goals as a two-state (on/off-goal) occupancy. Test field against contagion onsets (H10-R1/R2). Run `confirm.py` only after Vivian's sign-off.

- **Essence.** How does a new goal propagate: as a field acting on everyone at once, or as contagion from early adopters?
- **H10-R1** Adoption onsets after a kickoff: a field predicts simultaneous onsets; contagion predicts onsets ordered by exposure to early adopters.
- **H10-R2** Goals act on a two-state on-goal/off-goal occupancy (HH130).
- **H10-R3** During the week, goal-specific content relaxes back toward family-prior directions (E4).



Second figure. (a) P1 per pair under every round-1b input (blue bge, orange gte; filled base, squares deduped, triangles style-residualized). The regime-I pairs stay negative. (b) #44 native: the assigned room moves along its own kickoff, and the self-chosen room does not (90% CIs).

GOAL PERIODS IN DETAIL

G11 d 1b: descriptive (both models)
 G12 $\times$ 1b: failed (both models)
 G16 d 1b: descriptive (both models)
 G17 $\times$ 1b: failed (both models)
 G26 $\times$ failed (round 1b native; both models)
 G31 d 1b: descriptive (both models)
 G37 d 1b: descriptive (both models)
 G38 $\sim$ 1b: mixed (bge mixed, gte failed)
 G44 $\sim$ mixed (round 1b native; model-dependent)
 NE34 $\times$ 1b: failed (both models)
+1 more periods: see the card.

H10-goals-are-legendre-pushes · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H10-goals-are-legendre-pushes/README.md · ratings: RE-C1 re-evaluation agent (suggested) (page 2)

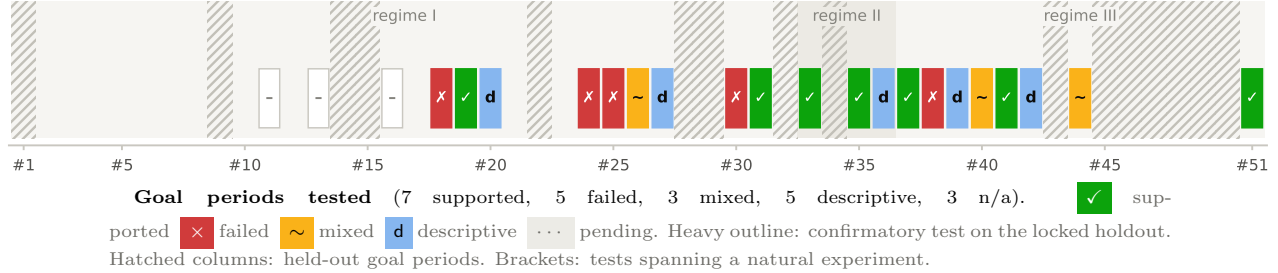
H11: Division of labor vs herding is the sign of a Potts coupling, set by the goal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 1.5/5 Usefulness 2.5/5

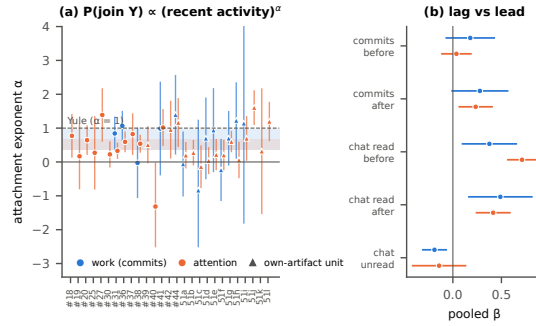
Bottom line. LLM agents pile onto projects in same-hour bursts steered by chat, not by the artifact record, and each extra agent on a shared project adds about 0.7 of a solo agent's commits.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

Rounds 1 and 1b fitted a Curie–Weiss Potts model per week. The spin is an agent's project in a 30-min window. The goal-mode rule for the sign of βJ failed: agents herd ($\beta J > 0$) in 11/14 weeks, in attention and in work commits, and spread only when the task gives each agent its own artifact. **Round 2 asks why agents pile on.** The degrees of freedom are *joins*: an agent switches to a project that others already hold. We fit the choice among all live projects with a conditional logit. Its attachment exponent α says how the odds grow with others' recent activity. Lead placebos test the arrow of time. Chat reads (context ledger) and commits (DQ4) compete as exposures, and a skeleton replay tests the project-size law. Output per agent comes from the work ledger. In 40 units, joins follow a sublinear kernel ($\alpha \approx 0.5\text{--}0.7$). Activity *after* a join predicts it as well as activity before, so this is co-arrival. Read chat beats commits, and a project's commits grow as $n^{0.70}$ with its agents.



Observables. (a) Attachment exponent α per unit with 95% CI, in work joins (blue) and attention joins (orange). Bands: pooled values, 0.68 [0.38, 0.98] and 0.52 [0.36, 0.67]. Both exclude the Yule value 1 (dashed). (b) Pooled exposure coefficients. Commits and chat reads *after* the join predict it as well as those before it. Read chat is the strongest predictor, and unread mentions have a negative coefficient.

PHYSICAL MODEL

Model 10 (Potts) in kinetic form, with a preferential-attachment kernel. Agent i joins project Y from the live set C . a_Y is other agents' labelled windows on Y in the last 2 active hours, and s_Y their cumulative windows. $h_{iY} = 1$ if i held Y before (habit). $\alpha = 1$ with $u \propto \log s_Y$ is the Yule–Simon process. The exposures are M (chat mentions of Y read at i 's calls in the last 60 min, or after the join: lead), M^u (posted but unread) and C (others' commits to Y , before or after). Output: c_Y is a project's commits in a window with n_Y agents on it, and $\theta < 1$ means crowding.

$$P(i \rightarrow Y) = e^{u_Y} / \sum_{Z \in C} e^{u_Z}$$

$$u_Y = \alpha \log a_Y + \beta_0 [a_Y=0] + \gamma \log(1+s_Y) + \eta h_{iY}$$

$$\text{R2: } u_Y = \beta_M \log(1+M_Y) + \beta_C \log(1+C_Y) \text{ (+lead, unread)}$$

$$\mathbb{E} c_Y \propto n_Y^\theta, \quad \text{RR}_{\text{herd}} = \text{rate}_{\text{herd}} / \text{rate}_{\text{solo}}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Herding is co-arrival steered by chat, not stigmergy. Agents pick projects in proportion to others' recent activity, with a sublinear exponent. That activity is as predictive one hour after the join as one hour before, so agents converge in the same hour; they do not follow an accumulated record. Read chat mentions predict joins better than commits, and unread mentions do not. Pile-ons cost little per herded agent (commit rate RR 1.23, n.s.), but they add little to the project: each extra agent adds about 0.7 of a solo agent's commits. Operator rule: expect a project's commits to grow as $n^{0.7}$ with the agents on it. To get parallel output, split work across owned artifacts rather than adding agents to one repo.

H11-potts-labor-vs-herding · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H11-potts-labor-vs-herding/README.md · ratings: H11 round-2 agent (suggested 60 / 1.5 / 2.5; previous calibrated 50 / 1.5 / 2.5)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 sign by goal mode (round 1)	5/11 signs; herding in 11/14 weeks	failed
R1a $\alpha > 0$ in $\geq 2/3$ shared (work)	2/4 work; 8/12 attention	failed (work)
R1b sublinear, Yule	not pooled α 0.68 / 0.52, both < 1 ; per-unit CIs wide	mixed
R1c fitness takes part	$\alpha_{FE} < \alpha$ in 5/13, 16/27	failed
R1d lag = lead (convergence)	CI $\ni 0$ in 34/40 units; attachment world gives +1.0 to +1.6	supported
R1e kernel replays sizes	work 10/13; attention 7/27	supported (work)
R1f Yule fits	2/13, 4/27	failed
R2a commits beat chat (held out)	work 6/13, pooled -0.002; attention 3/24, pooled -0.047 [-0.070, -0.024]	failed
R2b MH: commits > chat	pooled OR chat 3.0 vs commits 1.26 (attention)	failed
R2c commits lag > lead	work -0.18; shared work -0.78 [-1.22, -0.35]	failed
R3a herding raises output	RR 1.23 [0.94, 1.62] (shared)	not supported
R3b $\theta < 1$	16/23; θ 0.70 shared, 0.25 own	supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

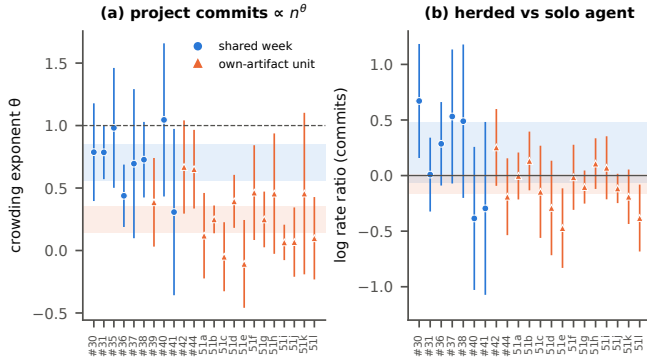
CAVEATS

Work joins are few (25–101 per period; 51g 321), so per-period work CIs are wide. Whole goal periods are units for #18–#44 (exception (d)). The 30-min window is coarser than the 18–50 s read-out delay, so “before” and “after” share a burst. The read vs unread contrast is not estimable in most units: unread mentions are almost never followed by a join. θ counts watchers as well as committers. Own units produce $3\times$ more commits per agent-hour, but that difference follows calendar time. The R3 test is mildly liberal (synthetic size 0.04–0.13). Synthetic validation found and fixed a likelihood bug and two liberal contrasts before the real run (amendment A1).

ROUND 2

Move the join test to the call clock. Compare commits visible at the deciding call with commits made during it (an artifact in-flight placebo). Recompute θ with committers only and file-level overlap, to split watchers from duplicated work. Use #40 ($\alpha < 0$: agents leave active projects) and NE42 as natives.

- **Essence**. Why do agents pile on: imitation, or attraction to where the work already is?
- **Round-3 redirects (newest)**.
- Move the attachment test to the call clock: the hazard of a join at call c as a function of commits to Y visible at c 's t_{call} vs commits made during c (an artifact in-flight placebo at 1–2 min, which the 30-min window cannot give).
- Recompute θ with work-defined n (committers only) and file-level overlap, to split watchers from duplicated work.
- Natives: #40 has $\alpha = -1.32$ [-2.60, -0.04] in attention (agents leave recently active projects: own worlds beside a hub); NE42 (#39 $\rightarrow$ #40) as a test of θ and α across a merge.



Second figure. (a) Crowding exponent θ : a project’s commits per window vs the number of agents on it (project-day fixed effects). Shared weeks (blue) pool to 0.70 [0.56, 0.85], own-artifact units (orange) to 0.25 [0.14, 0.36]. In own units, attention to another agent’s repo is reading, not work. (b) Herded (≥ 2 others on the same project) vs solo windows of the same agent-day: log commit-rate ratio. Shared +0.21 [-0.06, 0.48]; own -0.08 [-0.17, 0.01].

GOAL PERIODS IN DETAIL

G18 $\times$ 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G19 $\checkmark$ 1b: P1 supported; P2 supported (unchanged)
G20 d 1b: descriptive (unchanged)
G24 $\times$ 1b: P1 failed; P2 failed (sig.) (unchanged)
G25 $\times$ 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G26 $\times$ 1b: mixed (native: jump on the right vote; cascade, n...
G27 d descriptive (first tested in round 2)
G30 $\times$ 1b: P1 failed (sig.); P2 failed (sig.) (unchanged)
G31 $\checkmark$ 1b: supported (native: work herds)
G33 $\checkmark$ 1b: supported (work herds)
+13 more periods: see the card.

H11-potts-labor-vs-herding · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H11-potts-labor-vs-herding/README.md · ratings: H11 round-2 agent (suggested 60 / 1.5 / 2.5; previous calibrated 50 / 1.5 / 2.5) (page 2)

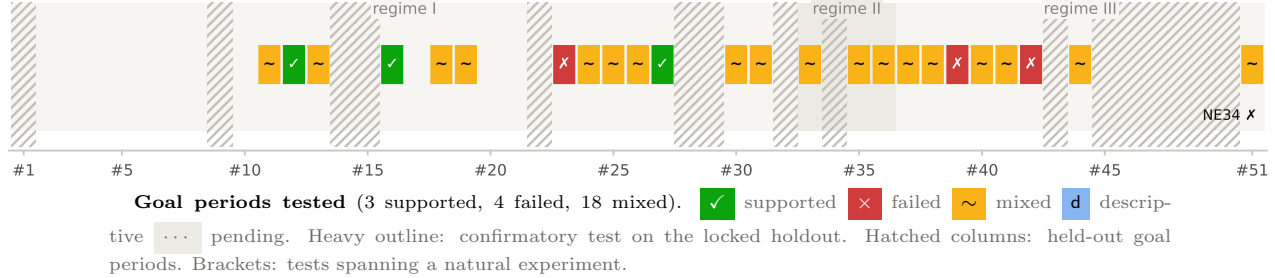
H12: Groupthink is dimensional collapse: few collective modes, shrinking participation ratio at consensus

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 52% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. The activity market mode is mostly artifact; talk and content modes are real, a strong field compresses content, and low-PR weeks are self-restatement.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



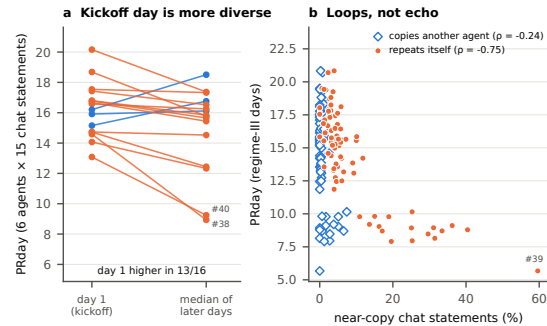
WHAT HAPPENED

Two arms run on 24 non-holdout units. Round 1 lost half of all events. **Round 1b** re-runs the spin spectra on the corrected activity table with DQ8’s calibrated null. It re-checks content and PR with a second embedding model and adds two period-native tests. **Spectrum:** the round-1 activity market mode (22/24 units) mostly disappears: 18/24 under the old null, 7/24 under the calibrated one. It was largely a day-level artifact of the bug. The talk modes (21/24) and content modes (24/24 in both models) hold. Talk modes now separate rooms in 7/10 two-room units. **Dimensionality:** 20 kickoffs show no collapse, in both models and after dedup. In the low-PR weeks #38–#40 agents restate themselves (exact copies 0%). Natively, a strong field *does* compress content. PR is 25% lower while a #12 debate motion is on ($p = 0.01$), and lowest at #26’s contested vote. The holdout has not run.

PHYSICAL MODEL

Models 01 (inverse Ising) and 11 (vector spins), with random-matrix theory. $s_i(t)$ is agent i 's ± 1 activity or talk spin per minute, or its whitened content vector per 30-min window (N agents, T samples). C is the equal-time correlation matrix with eigenvalues λ_k . λ_+ is the noise edge: cross-day surrogates in round 1, trimmed block-shift surrogates in 1b. $u = \mathbf{1}/\sqrt{N}$, and $\bar{\rho}$ is the mean pair correlation. PR is the participation ratio of the content covariance Σ ($d = 32$), bias-corrected through $W = \nu S$, $\nu = n - 1$.

$$\begin{aligned} C_{ij} &= \langle s_i s_j \rangle_t, \quad \lambda_+ = (1 + \sqrt{N/T})^2, \quad k = \#\{\lambda_k > \lambda_+\} \\ \text{VR} &= u^\top C u \leq \lambda_1 \approx 1 + (N-1)\bar{\rho} \quad (\text{iff uniform mode}) \\ \text{PR} &= (\text{tr } \Sigma)^2 / \text{tr } \Sigma^2 = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2 \\ \mathbb{E}(\text{tr } W)^2 &= \nu^2 a + 2\nu b, \quad \mathbb{E} \text{tr } W^2 = \nu a + \nu(\nu+1)b \Rightarrow \hat{a}, \hat{b} \end{aligned}$$



Observables. Round 1; round 1b leaves the PR inputs unchanged. (a) Agent-balanced PR (6 agents $\times$ 15 chat statements) on the kickoff day against the median of later days, in 16 periods with $N \geq 10$. Day 1 is higher in 13 (orange), in 11/16 with the second embedding model and in 9/15 after removal of restatements. (b) Regime-III days: PR falls as agents restate their own earlier messages ($\rho = -0.75$; -0.76 with DQ5’s either-model flag), not other agents’ messages ($\rho = -0.24$). The low cluster is #38–#40 in both models.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The portable results concern what is *not* collective. The activity market mode was mostly a data bug plus the daily schedule. Under a calibrated null, agents are busy or idle together beyond chance in only 7/24 units. What agents talk about does co-move, in content and talk modes. A debate motion, a strong uniform field, narrows content by 25%, and a contested vote narrows it too. Goal kickoffs do not. The cheapest useful monitor is self-restatement: its rate tracks low PR at $\rho = -0.75$. It flags agents that rephrase themselves, the village’s real low-dimensional states. The groupthink alarm stays unvalidated: cross-agent echo never exceeds a few percent.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict
P1 few activity modes (cross-day)	$k=1$ in 22 → 18/24; $k=0$ in 2 → 6	×
DQ8 null (trim, block activity mode in 7/24; talk 21/24 new shift)		
P1' beyond lulls	14 → 8/24 (lulls now rare)	×
P2 Curie–Weiss shape; III > I	uniform 22/22 → 16/18; λ_1 /edge III 1.62 → 1.13	×
P3 talk modes separate rooms	3/10 → 7/10; but $k_{\text{talk}} > k_{\text{act}}$ often	×
P4 content modes	24/24; gte 24/24 (ρ 0.99)	core ✓
P6 kickoff collapse	+7%; bge +4%, gte +9% ($p \geq 0.77$)	×
P7 day 1 vs later (R4)	13/16 higher; gte 11/16; dedup 9/15	×
G12 motion field (native)	PR on < off in 7/9, -25%, $p = 0.010$	✓
G26 vote instants (native)	contested vote pct 0.07; re-election ~ 0.37	~

GOAL PERIODS IN DETAIL

G11 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G12 ✓ supported
 G13 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G16 ✓ 1b: supported (round 1: supported; corrected activity...
 G18 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G19 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G23 × 1b: failed (round 1: mixed; corrected activity table,...
 G24 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G25 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G26 ~ 1b: mixed (replication, round 1: mixed; corrected act...

+15 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

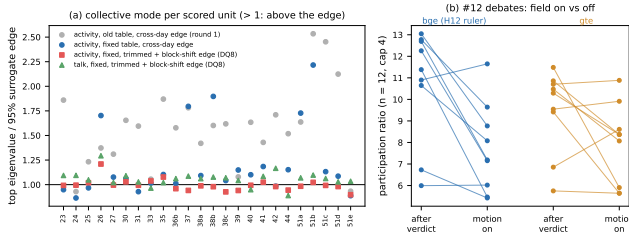
Two causes explain the collapse of the activity mode: the bug's day-level common loss and the scheduler's start and stop. H50 finds the scheduler effect independently from call windows. Within-block co-activation rises on the fixed table. PR measures what agents write, including format: debate turns are structured speeches, and the #12 off window is short. P7's direction depends on the embedding model and on restatement removal. Whitening is per regime, so PR levels compare only within a regime. In 3-day units the cross-day edge gives ~14% false positives. The holdout is untouched. Amendment 2's C1–C3 (activity mode) and C5 (day-1 expansion) look unlikely to pass as frozen.

ROUND 2

Re-freeze the confirmatory C1–C3 on talk and content modes with the trimmed block-shift edge before any hold-out run. Repeat the #12 field test on other field switches (H21 verdicts, operator corrections). Add a restatement-rate monitor with the either-model flag.

- **Essence.** Loops are fixed points of an agent's self-conditioned generation.
- **H12-R1** Loop onset is predicted by the share of an agent's context filled with its own recent output; crossing a threshold is a bifurcation to a fixed point (E2).
- **H12-R2** Only novel external input breaks a loop, and the escape probability scales with the input's novelty (embedding distance), not with who sent it.
- **H12-R3** Swarm groupthink never forms because each agent is its own echo chamber: self-echo dominates cross-echo.

Second figure. (a) The top eigenvalue over the 95% surrogate edge per scored unit. Activity: round 1 (gray), corrected table (blue) and calibrated DQ8 edge (red). Talk under the DQ8 edge (green) stays above 1 in 21/24. (b) #12: the participation ratio after each verdict against the value while the motion is on, paired by debate, in two embedding models.



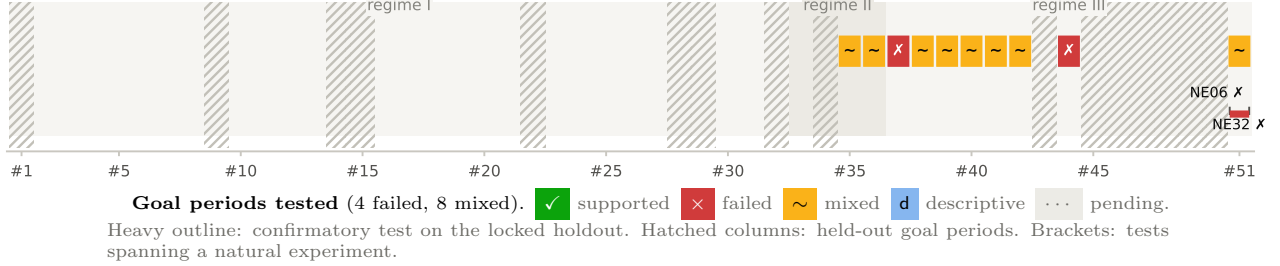
H13: Model families carry their own fields, and couple by family

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 65% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Model families carry a content field that is mostly style, and content couples more within labs at the read-out call; talk does not.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



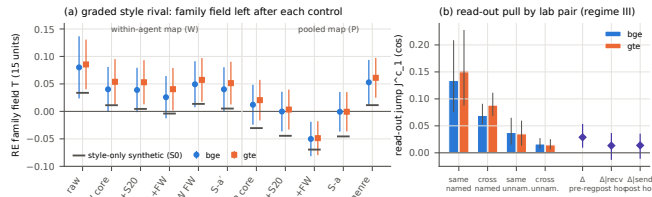
WHAT HAPPENED

The data is 15 counted non-reserved units (#35–#44, #51; $N = 10\text{--}27$). The state is each agent's daily content vector with the day's common field removed. Same-lab agents share a content direction in 9/15 units (RE $T = 0.08$ [0.02, 0.14]), stable across periods. **Round 2** grades the style rival. A within-agent map of 20 formatting features and 50 function words keeps **32% (bge) to 47% (gte)** of the field: $T = 0.026$ [−0.01, 0.06] and 0.040 [0.002, 0.08]. Round 1's pooled control was biased downward. Newcomers start at their lab's field ($a(1) = 0.21$, $p = 0.005$) and drift away. At the read-out call, a message pulls a **same-lab reader's content 1.7× more** (0.067 vs 0.039; 8/8 units). About half of that is lab-level susceptibility (post hoc). Talk shows no family term.

PHYSICAL MODEL

Model 11 (vector spins). $v_{i,d}$ is agent i 's mean whitened chat embedding on day d , and m_d is the day mean. h_f is the field of lab $f(i)$, a_i an agent offset, and H_i the agent field. Style rival: each statement vector z is regressed on its style features s (formatting, function words, genre held out) with a map B fitted within agents (W) or pooled (P). Coupling: J_1 is the matched-age jump in $\cos(z_B, z_m)$ between the recipient's read-out call ($h = 1$) and the call in flight ($h = 0$), for same-lab and cross-lab pairs.

$$v_{i,d} = m_d + h_{f(i)} + a_i + \varepsilon_{i,d}, \quad T = \langle \cos H_i H_j \rangle_{\text{same}} - \langle \cos H_i H_j \rangle_{\text{cross}}$$
$$z \mapsto z - (s - \bar{s}) \hat{B}_W, \quad \rho_{\text{ns}} = T_{W3} / T_{\text{raw}}$$
$$J_1 = \sum_b w_b [\bar{y}(1, b) - \bar{y}(0, b)], \quad \Delta = J_1^{\text{same}} - J_1^{\text{cross}}$$



Observables. (a) Random-effects family field after each style control, bge (blue) and gte (orange), with 95% CIs. Dashes: the same control on a synthetic world whose family field is style only. Within-agent maps keep a third to a half of the field. Pooled maps fall below zero even in the style-only world. (b) Read-out content pull, regime III. Same-lab pairs pull more in both naming strata. Violet: the named-stratified contrast (pre-registered) and, post hoc, the contrast with the receiver's or the sender's lab held fixed.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Vendors differ mostly in prose, and partly in what they say. A pooled style correction removes the prose and then pushes the vendor field below zero. A within-agent correction leaves about a third of it. Vendors also differ in how strongly their models are moved by what they read. A same-vendor message moves a reader 1.7× as much as a cross-vendor one, at the next call. Half of that is because some vendors' models are more easily moved, not in-group preference. Who replies does not depend on vendor. New models arrive sounding like their vendor and lose part of it within a week. For an operator: expect a read message to move a same-vendor model up to 1.7× as much as another vendor's model.

H13-family-fields · generated 2026-10-07 17:50 by in-
fra/summaries/build_summaries.py · card hypotheses/H13-family-
fields/README.md · ratings: H13 round-2 agent (suggested; coordinator
writes v2)

RESULTS VS PREDICTIONS

Prediction	Result	Verdict
P1 family field (r1, 1b)	9/15; RE 0.080; gte 8/15, 0.085	mixed
P5, P6 no co- talk RE -0.003 ; content 1/15 movement homophily (r1, 1b)		✓
A-P1 monotone style W: $0.080 \rightarrow 0.040 \rightarrow 0.039 \rightarrow 0.026$ ladder		✓
A-P2 mostly style bge 0.32, gte 0.47; 3/15 units ($\rho_{ns} \in (0.1, 0.5]$)		✓
A-P4 not speech-act families differ in genre mix 6/15 mix		×
B-P1 newcomers start at lab $a(1) = 0.205$, $p = 0.0045$ (20 joiners)		✓
B-P2 lab alignment slope $-0.034/\text{day}$ $[-0.065, -0.003]$ does not fade		×
B-P4 newcomers con- start nearer the centroid; no trend verge on room		×
C-P1 no content fam- $\Delta = 0.029$ $[0.010, 0.054]$; 8/8 units ily coupling		×
C-P2 no talk family $\Delta\beta = -0.0002$ $[-0.0016, 0.0014]$ coupling		✓

GOAL PERIODS IN DETAIL

G35 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G36 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G37 $\times$ 1b: failed (round 1: failed; same in both embedding m...
G38 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G39 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G40 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G41 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G42 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G44 $\times$ 1b: failed (round 1: failed; same in both embedding m...
G51 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...

+2 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	2	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

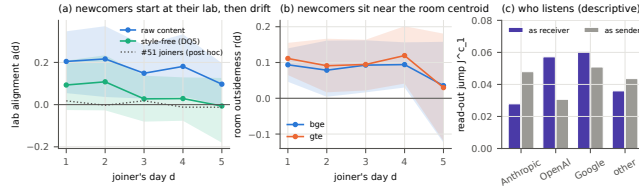
CAVEATS

The read-out contrast mixes homophily with lab main effects; with the receiver's lab fixed it is 0.013 $[-0.013, 0.037]$. Its power at half the pull is 0.55. The non-style family field is resolved only in gte; power at a third of the field is 0.30. The newcomer drift comes from joiners before #51 (post hoc). The leave-family-out day mean was withdrawn as biased. The confirmatory run (C1–C6) has not run.

ROUND 2

Fit the read-out pull as receiver susceptibility times sender potency plus a homophily term, pooled across units with shrinkage. Add the reserved #51 tail only after Vivian's sign-off.

- **Essence**. Do vendors differ in decisions, not prose?
- **H13-R1** Policy fingerprint: P(next behavior state — current state, context features) differs by family beyond agent identity.
- **H13-R2** Families differ in susceptibility: who listens to whom (HH98).
- **H13-R3** Mixed-vendor periods out-produce single-vendor ones (descriptive, confounded; work ledger).



Second figure. (a) Newcomers' lab alignment by day since joining (20 joiners; bands: joiner bootstrap). Raw content starts toward the newcomer's own lab and falls. Style-free content starts weaker. #51 joiners (dotted, post hoc) start neutral. (b) Room outsiders: positive means nearer the room centroid than incumbents. (c) Read-out pull by the receiver's and the sender's lab (point estimates): Anthropic models are the least moved receivers.

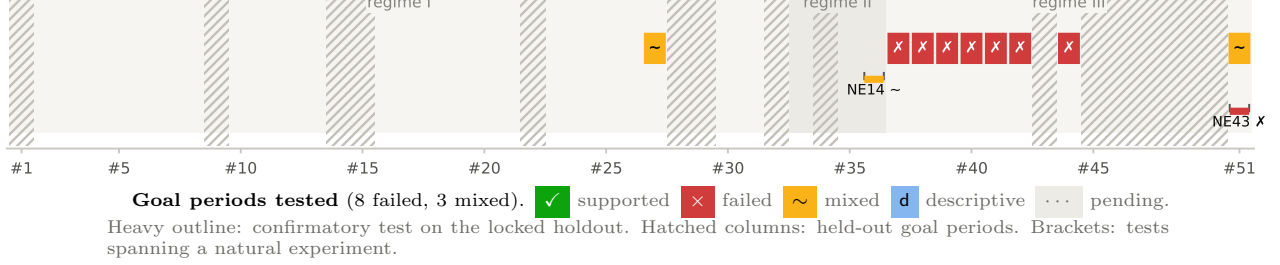
H14: Entropy production of behavior-state sequences: per-family arrows of time, and collective irreversibility

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. Every agent's fine tool workflow is irreversible without the scaffold; the coarse arrow and the regime contrast are scaffold, and no family, collective or useful-work arrow appears.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



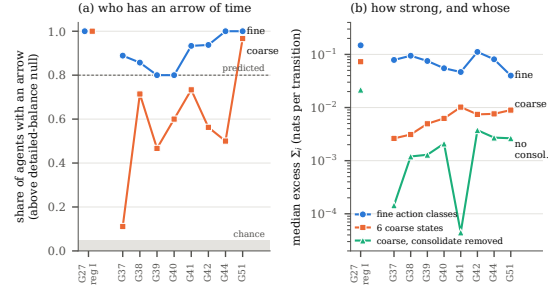
WHAT HAPPENED

H14 tests whether each agent's behavior is irreversible. It also tests whether each model family has its own arrow of time (HH19), and whether the swarm is more irreversible than its agents (HH67). Round 1 covers 9 periods (#27 in regime I; #37–#44 and #51 in regime III). On 6 coarse states the arrows are small (0.002–0.010 nats/transition), and the scaffold's consolidation step makes most of them. Regime I gives 7–29× larger arrows. No family or collective term appears. Round 1b uses shell sub-classes from the fixed **bash.head**, an agent-only chain with a burn-in after scaffold resets, and Jev v3.1 semantic states. Every agent's fine arrow is now irreversible (9/9 periods), and it survives scaffold removal (41–104% kept). The regime contrast vanishes on every corrected chain. Semantic states carry a small pooled arrow (7/9) that does not track output. The natives (NE43, NE14, loops against progress) give no support. The holdout has not run.

PHYSICAL MODEL

Kinetic Potts (models 02, 10). $x_i(t) \in \{1..q\}$ is agent i 's behavior state per turn (single agent) or per 1-min bin (swarm). P_{ab} and π_a are its transition matrix and occupancies. Σ_i is the entropy production per transition: the pair KL, a lower bound for non-Markov agents. The Aguilera–Ito–Kolchinsky dual estimates it on antisymmetrized transition indicators g_{ab} , cross-fitted by day (Newton step $\hat{\Sigma}_N$). J_{ij}^{ab} are cross-agent couplings. $\Delta\Sigma$ is the collective increase of the bound, and A_{abc} is a cycle affinity.

$$\begin{aligned}\Sigma_i &= \sum_{a,b} \pi_a P_{ab} \ln \frac{\pi_a P_{ab}}{\pi_b P_{ba}}, & g_{ab} &= \mathbf{1}[a \rightarrow b] - \mathbf{1}[b \rightarrow a] \\ \Sigma_g &= \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle], & \hat{\Sigma}_N &= 2 \langle \bar{g}_a^\top K^{-1} \bar{g}_b \rangle_{a \neq b} \\ P(x'_i = a | \mathbf{x}) &\propto e^{h_i^a + \sum_j J_{ij}^{a, x_j}}, & \Delta\Sigma &= \Sigma_{1+2} - \Sigma_1 \\ A_{abc} &= \ln \frac{P_{ab} P_{bc} P_{ca}}{P_{ac} P_{cb} P_{ba}}\end{aligned}$$



Observables. (a) The share of agents whose state sequence beats a detailed-balance surrogate (same occupancies, symmetric transition counts, no currents). Almost all agents do on fine action classes. On the 6 coarse states only a minority do, except in the long #51 window and in regime I (#27). (b) The median excess irreversibility. Fine micro-workflow carries about 10× the coarse arrow. Most of the coarse arrow vanishes when the scaffold's consolidation step is removed.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

This is mostly a careful measurement of the harness, with one agent-level residue. The coarse arrow and the regime-I/III contrast are scaffold structure: sessions and consolidation. Fine tool workflow (e.g. read → edit → commit) survives scaffold removal and is irreversible in every agent. It is a per-agent fingerprint, not a steering handle. Semantic irreversibility is real but small, and it does not track output (ρ from -0.34 to $+0.14$). EP is therefore not a useful-work meter. For EP on agent logs, use a calibrated detailed-balance or block-flip null, and subtract the null before you compare agents. Never apply a reset burn-in on coarse windows: it manufactures arrows. An operator would change nothing.

H14-behavior-entropy-production · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H14-behavior-entropy-production/README.md · ratings: H14 re-evaluation agent RE-B1 (suggested; round-1 values were coordinator-calibrated)

RESULTS VS PREDICTIONS

prediction (round 1b)	observed	verdict
fine (shell sub-classes)	100% of agents in 9/9	holds
$\geq 80\%$ irreversible		
coarse arrow mostly	5/8 keep $< 50\%$ after removal	failed (narrow)
scaffold ($\geq 6/8$)		
fine arrow survives	kept 41–104%; 75–100% of holds	
scaffold removal	agents	
pooled Jev v3 arrow	$p \leq 0.005$; 7/9 periods	holds
(G38, G51)		
regime I $\geq 2\times$	(v3 ratios 0.4–4.4	failed
states)		
lab η^2 (fine, scaffold-free)	p 0.35; Anthropic $>$ OpenAI 8/8, p 0.04	failed / fragile
v3 EP tracks output	ρ +0.14, -0.34, -0.16	failed
N1 NE43 inside placebo range	3/4 changes inside	failed (marg.)
N2 NE14: coarse drops more than v3	0.25 vs 0.72 (CIs span 1)	mixed
N3 productive $>$ stuck EP	0/3	failed

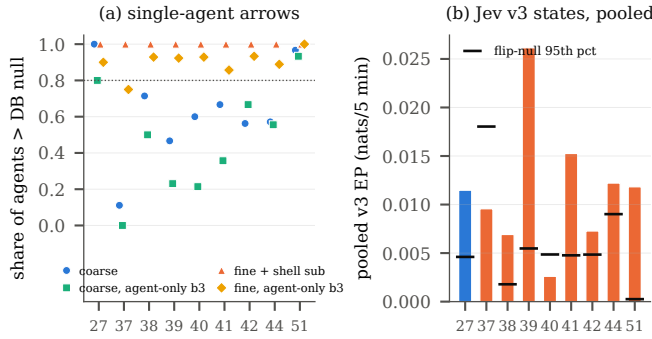
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Soft-label observables attenuate EP by roughly the square of label accuracy. Per-agent v3 tests are therefore nearly blind, and only pooled tests are primary. The pre-registered reset burn-in on 5-min windows is a selection artifact, because resets start activity bursts; it is reported as invalid. The Newton step blows up for a few strongly asymmetric agents (one G40 value is 16.5). That agent drives the fragile Anthropic $>$ OpenAI result. NE14 has one pre-change day, and NE43 windows are 5 days. Non-Markov agents make all values lower bounds.



Second figure. (a) The share of agents that beat the detailed-balance null, per period. Fine classes with shell sub-classes (triangles) are irreversible for every agent and stay so without the scaffold (diamonds). The coarse arrow (dots) mostly does not survive removal (squares). (b) Pooled Jev v3 entropy production per 5-min transition (bars) and the block-flip null's 95th percentile. Regime I (#27, blue) is not above regime III.

GOAL PERIODS IN DETAIL

G27 $\sim$ 1b: mixed (unchanged: the pre-registered coarse chain...
G37 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G38 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G39 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G40 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G41 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G42 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G44 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G51 $\sim$ 1b: mixed (unchanged: the pre-registered coarse chain...
NE14 $\sim$ mixed
+1 more periods: see the card.

ROUND 2

Measure count-matched within-agent EP across goal kickoffs (H56's design) on `act.sh`. Denoise semantic states with an HMM before EP.

- **Essence.** The arrow of time of work: progress is irreversible, chatter is reversible.
- **H14-R1** Entropy production on work states (plan, build, verify, ship; Jev) correlates with output: useful irreversibility.
- **H14-R2** Loops are near-reversible cycles with zero progress; productive phases produce entropy.
- **H14-R3** The scaffold contributes a fixed entropy-production floor (the consolidation clock); subtracting it leaves agent-generated irreversibility (E1).

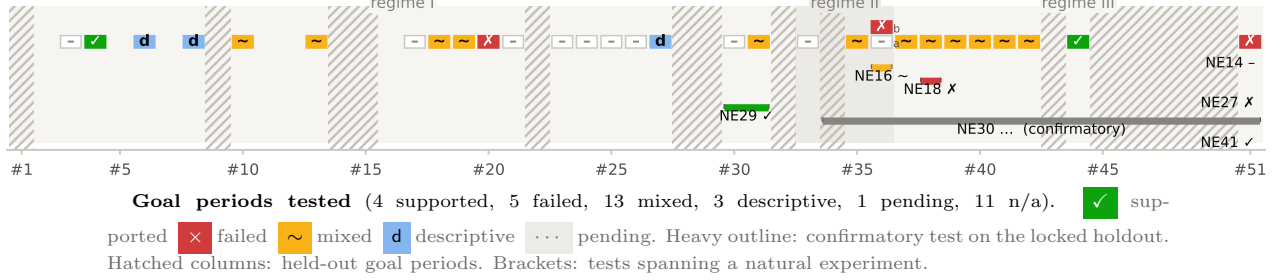
H15: Semantic information through natural scrambles: which information keeps agents and the swarm viable

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 65% Faithfulness 1.5/5 Usefulness 2.5/5

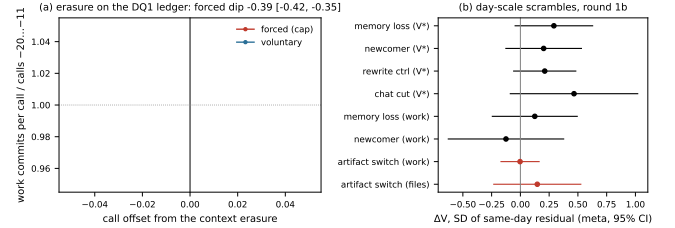
Bottom line. A forced wipe costs 39% of committed work for ten calls; the value is not in the few items read first (they carry 4% of the dip), and memory notes are about 11% enacted.

level: hypothesis · pipeline checks 85% · updated 2026-10-05



WHAT HAPPENED

The data is natural scrambles in non-reserved periods, per goal period, against placebo nulls calibrated on synthetic worlds built on the real village skeleton. **Rounds 1–1b:** losing 50–80% of memory (12 events) or starting with an empty memory (20 newcomers) shows no day-scale cost on the viability V^* (+0.29 SD $[-0.05, +0.63]$; newcomers +0.20). A forced context wipe (the 41-call cap) cuts work commits by 39% for ten calls (7/9 periods), whatever the agent saves to memory. **Round 2** asks where that value sits. It compares 20,592 forced wipes with 21,806 mid-segment pseudo-wipes in nine regime-III periods, and 37,031 consolidation notes with the commands that follow them.



Observables. Round 1b. (a) Work commits per call around a context erasure on the DQ1 ledger, relative to calls $-20 \dots -11$. A forced wipe (41-call cap) cuts committed work by about 39% for ten calls. Voluntary consolidations dip more after a work burst (task phase). (b) Day-scale scrambles on V^* and on work output (meta over goal periods, 95% CI). Only the chat cut (3 events) departs from zero.

PHYSICAL MODEL

Model 04, Kolchinsky–Wolpert semantic information. The system X is one agent: its memory M (day-scale store) and its context window C (session-scale store, filled by chat and events). Y is the rest of the village. A scramble erases M or C , or cuts $Y \rightarrow X$; $\hat{p}$ is the scrambled joint. V is viability per agent-day (5 candidates) or per call (work). u is its same-day residual, and δ is the fraction removed. S is the semantic information and η its share of $I(X; Y)$.

$$\Delta V = V[p] - V[\hat{p}], \quad \kappa = \Delta V / \delta$$

$$S = \min_f I(\hat{p}^f) \text{ s.t. } V[\hat{p}^f] = V[p], \quad \eta = \frac{S}{I} \leq 1 - \delta^*$$

$$u_{ad} = V_{ad} - \bar{V}_{-a,d}$$

$$\widehat{\Delta V} = \bar{u}_{\text{post}} - (\mu_a + \rho^s \mathbb{E}[h_{t_0-1} \mid u_{\text{pre}}])$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The working context keeps an agent productive, and its value is spread over the working state, not stored in a few items. Re-showing the last files or the newest messages after a wipe recovers at most a few percent of the lost work (4% $[-23, +26]$). Agents re-read by habit, not by what was erased. The usable number stays the cost of a reset: -39% of committed work for about ten calls, about a tenth of output at a 41-call cadence. To recover it, raise the cap; a carried summary note will not do it, because notes are only about 11% enacted and their content buys no output. The first-read share $\eta_{\text{first}} \approx 0.04$ is the closest this data comes to Kolchinsky–Wolpert’s η .

H15-semantic-information-scrambles · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H15-semantic-information-scrambles/README.md · ratings: H15 round-2 agent (suggested); round 1 calibrated by coordinator 2026-10-04

RESULTS VS PREDICTIONS

Prediction	Result (non-reserved)	Verdict
P1 memory loss $\leq +0.29$ $[-0.05, 0.63]$		×
−0.25 (1b)		
P4 newcomers pay $+0.20$; progress $+0.31$ $[0.06, 0.56]$		×
(1b)		
P5' forced-wipe dip work -0.39 $[-0.42, -0.35]$ (7/9)		✓
(1b)		
R1a re-read $\geq +0.05$	$+0.049$ $[0.036, 0.063]$; re-touch mixed 0.33 vs 0.32	
R1b trail softens dip	T2 1.19 $[0.99, 1.48]$; T1 1.26; 7-d inconcl. 1.03	
R1c memory-loss re-read	$+0.04$ $[-0.05, 0.13]$, $n = 32$	inconcl.
R2a last files ≥ 1.15	1.10 $[1.01, 1.21]$	partly
R2b new messages ≥ 1.14 $[1.02, 1.28]$; DL 1.02 1.10		mixed
R2c note names repo ≈ 1	1.04 $[0.92, 1.15]$	✓
R2d first reads $\geq 25\%$ of dip	0.04 $[-0.23, 0.26]$	×
R3a notes performative	excess 0.090 $[0.076, 0.104]$; novel 0.043	×
R3b notes retrospective	pre-post $+0.010$ $[0.004, 0.016]$	✓

GOAL PERIODS IN DETAIL

NE30	...	pending (confirmatory; not run) (holdout)
G04	✓	1b: supported (V* V_eng; MN -0.50 (z -0.7); 2026-10-0...
G06	d	1b: descriptive (V* V_eng; MR +0.40 (z +0.5); 2026-10...
G08	d	1b: descriptive (V* V_eng; MR -0.69 (z -1.0); 2026-10...
G10	~	1b: mixed (V* V_eng; MN +0.49 (z +1.5); 2026-10-04, r...
G13	~	1b: mixed (V* V_eng; ML +0.40 (z +1.2); 2026-10-04, r...
G18	~	1b: mixed (V* V_eng; MN +0.58 (z +1.2); 2026-10-04, r...
G19	~	1b: mixed (V* V_eng; MN +0.96 (z +1.2); 2026-10-04, r...
G20	×	1b: failed (V* V_eng; MN +0.56 (z +1.1); 2026-10-04, ...
G27	d	1b: descriptive (V* V_eng; MR +0.10 (z +0.2); 2026-10-0...

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

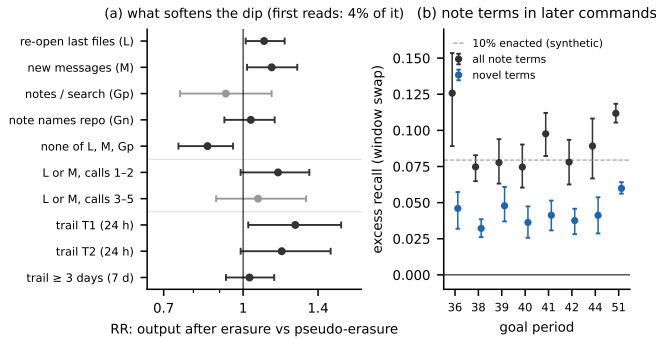
Day-scale and memory-loss tests see only large effects (power 17–41%; shifts ≥ 0.1). The erasure dip is post hoc in origin; NE41 replicated it with dated predictions, but the reserved-data check (C3) has not run. The round-2 classes are chosen by the agent; the pseudo-wipe arm removes their association with output when nothing is erased, under a parallel-trends assumption. New messages carry value only in the pooled fit (G51-driven; across periods $\tau = 0.17$). Notes are the next-session goal, not the full memory file; recall is a term-overlap proxy.

ROUND 2

Re-freeze `confirm_ne30.py` on work commits and run C3.

- **Essence**. Where does the swarm keep its knowledge? In artifacts and history, not in agents.
- **H15-R1** Stigmergic memory: after a memory loss, agents re-acquire state from artifacts (file reads, history search) within a few turns; richer artifact trails mean faster recovery (E3).
- **H15-R2** Semantic information is concentrated in the few items read first after a reset (the goal and the last messages).
- **H15-R3** Memory notes are performative: their content rarely reappears in later actions.

H15-semantic-information-scrambles · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H15-semantic-information-scrambles/README.md · ratings: H15 round-2 agent (suggested); round 1 calibrated by coordinator 2026-10-04 (page 2)



Second figure. Round 2. (a) Ratio of ratios: how much more each class raises work output after a forced wipe than after a pseudo-wipe (pooled fit, agent-day bootstrap 95% CI; gray: unpowered). The first two calls' reads (last files, new messages) add 10–14% but remove only 4% of the dip. A richer repo trail shows no dose order. (b) Excess recall of consolidation-note terms in the next segment's commands over the same agent's windows ≥ 3 days away, per period. Dashed: the synthetic level of 10% of a note enacted.

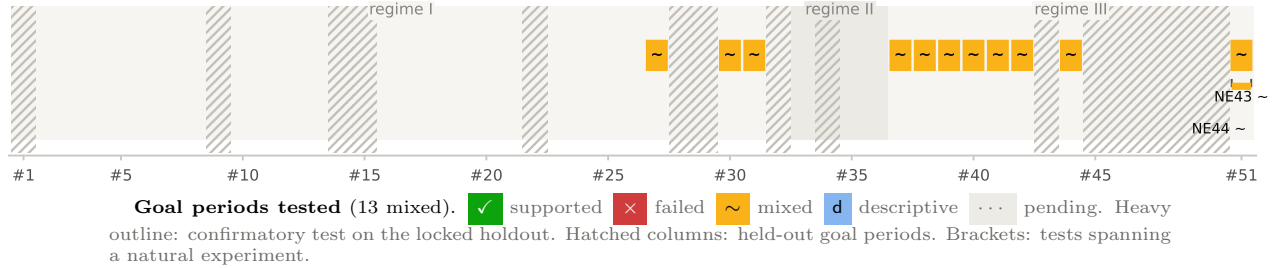
H16: Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 2/5 Usefulness 2.5/5

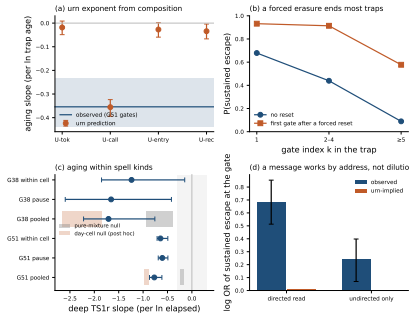
Bottom line. Pause-chain traps are held by the context: a forced erasure ends most of them, and a message works by naming the agent, not by diluting its repeats.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

The units are per-agent trap spells: inactive spells and timer-pause chains, on the wall clock (30-s bins) and on the call clock (idle gates of the context ledger). We test them in 11 non-reserved goal periods. Kramers escape predicts a constant hazard; traps *age* instead (#51 $\beta = -0.77$). Round 2 asks why. A Pólya urn says the context fills with the agent's own repeats. It predicts the gate-clock exponent from composition alone (-0.36 vs -0.35) and absorbs half of it, but not the wall-clock one (-0.08 vs -0.77). The decisive test is the forced erasure at the 41-call cap (NE41). It lifts sustained escape at the next gate from 0.47 to 0.84 (log OR +2.68 [2.31, 3.16]), 4.6× what dilution implies. A per-trap frailty world gives no step. A directed message works by address, not dilution. Trap depth is not a model trait. Spell-kind mixing explains little of #51's aging and most of it in two 4-h periods. The reserved data are not used.



Observables. (a) #51 gate aging slope (blue band: observed, day-bootstrap CI) against the slope each urn ruler predicts from measured composition, with no fitted slope (orange). Only the own-call share matches. (b) Escape at the next gate after a forced erasure vs without one, by depth in the trap. (c) Wall-clock aging within spell kinds; grey and orange bands are the pure-mixture and day-cell nulls. (d) A directed read lifts escape about 3× more than an undirected one; the urn implies nearly nothing for both.

PHYSICAL MODEL

Model 02 with an urn on the context window. At an idle gate g (a call after a pause call) the agent acts or re-pauses. The context segment since the last reset holds n_{rep} own idle calls, n_{act} own active calls and k room items. The urn sets escape by the own-repeat share f , measured four ways: tokens, own calls, all entries, the last 10 entries. Rivals: aging with trap age a at fixed f ; a context step at erasure; a mixture of spell kinds; per-trap frailty. Fit: agent effects α_i , slopes β_a , β_f , reset step δ , read effects γ .

$$P(\text{escape}_g) = c_i (1 - f_g), \quad f_{\text{call}} = \frac{n_{\text{rep}}}{n_{\text{rep}} + n_{\text{act}}}$$

$$\text{logit } P_g = \alpha_i + \beta_a \ln a_g + \beta_f \ln(1 - f_g) + \delta \mathbb{I}[\text{reset}] + \gamma_d \mathbb{I}[\text{dir}] + \gamma_u \mathbb{I}[\text{undir}]$$

$$\text{urn: } \beta_f \approx 1, \beta_a \rightarrow 0, \gamma_d = \gamma_u \approx \Delta \ln(1 - f)$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

When an agent re-pauses again and again, erase or consolidate its context. In #51 that ends 84% of traps at the next gate, against 47% otherwise, at any depth. Deep traps (5+ re-pauses) still escape less after the erasure (0.58 vs 0.93), so some depth lives outside the context. If you message a stuck agent, name it. A directed read adds +0.69 log odds, room chatter +0.24; a second message adds about a third. Do not expect a model to be reliably more trap-prone: depth does not repeat across periods.

H16-metastable-traps-kramers · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H16-metastable-traps-kramers/README.md · ratings: H16 round-2 agent (suggested v1; coordinator writes v2)

RESULTS VS PREDICTIONS

prediction (round 2)	observed (#51)	verdict
token urn misses the exponent	-0.02 predicted vs -0.35 (gate)	holds
own-call urn: gate yes, wall clock no	-0.36 vs -0.35; -0.08 vs -0.77	holds
composition absorbs 25%	$\rho = 0.47$ [0.33, 0.70], $\beta_f = 1.05$	failed
forced erasure raises escape	log OR +2.68 [2.31, 3.16]	holds
erasure dose	+0.11 [-0.35, 0.44]	no power
directed > undirected	+0.45 [0.32, 0.58]	holds
urn-implied kick < 0.2× observed	0.01 vs 0.69	holds
depth invariant across periods	$r = -0.04$ [-0.55, 0.22]	failed
aging slopes differ by agent	$I^2 = 0.92$	holds
pause spells age; beat mixture null	-0.61; null -0.20	holds

GOAL PERIODS IN DETAIL

G27 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G30 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G31 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G37 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G38 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G39 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G40 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G41 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G42 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G44 ~ 1b: mixed (unchanged under the round-1 rule; its core...
+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	2		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

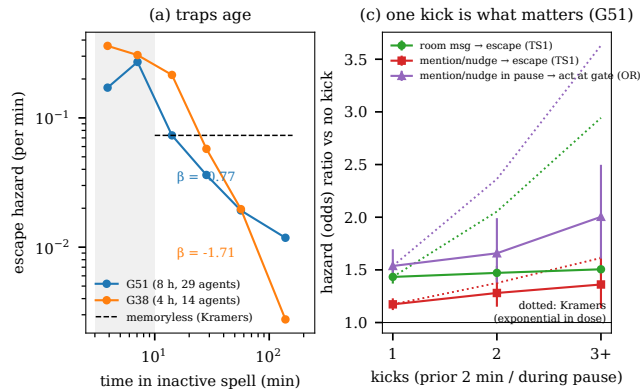
The reset and dose tests are powered only in #51. A day-level mixture of escape rates (post hoc) reproduces the wall-clock exponent (-0.93), so that exponent is not separable from day states; the gate results are not affected. With the gate count in the model (post hoc), trap age turns timer-like and the count carries the aging. Per-trap frailty fakes gate aging in synthetic worlds; only the erasure separates it. Kick exposure is now the ledger read at the gate; no in-flight placebo was run in round 2.

ROUND 2

Confirm the erasure step on reserved #51 days and #45. Test what survives the erasure (memory, task state) in deep traps.

- **Essence.** Traps deepen by self-reinforcement as the context fills with the agent's own output.
- **H16-R1** Pólya urn: $P(\text{repeat}) \propto$ the share of the context that is own repeats.
- **H16-R2** One directed message works because it dilutes the urn; repeated messages add little once the context is diluted.
- **H16-R3** Trap depth is a model trait: some models loop more.

H16-metastable-traps-kramers · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H16-metastable-traps-kramers/README.md · ratings: H16 round-2 agent (suggested v1; coordinator writes v2) (page 2)



Second figure. Round 1. (a) Escape hazard from inactive spells vs time in the spell in the two periods with the most power; Kramers predicts the flat dashed line. (b) #51 hazard and odds ratios vs the number of kicks; the first message carries the effect.

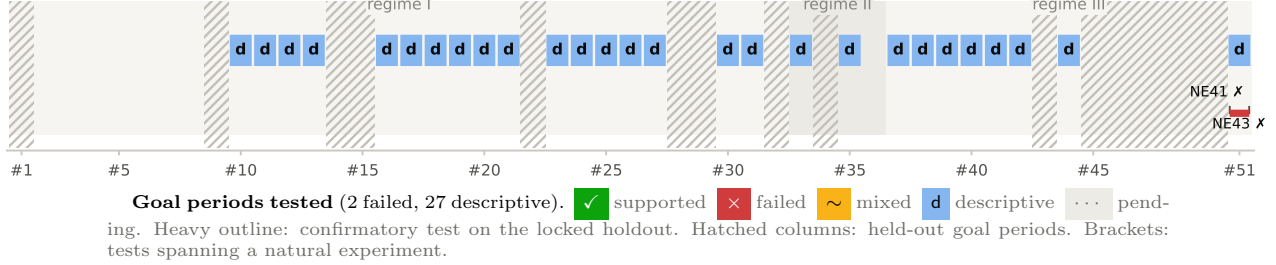
H17: Behavior is a Markov state model with a few metastable sets; its mixing time is a per-period order parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1.5/5

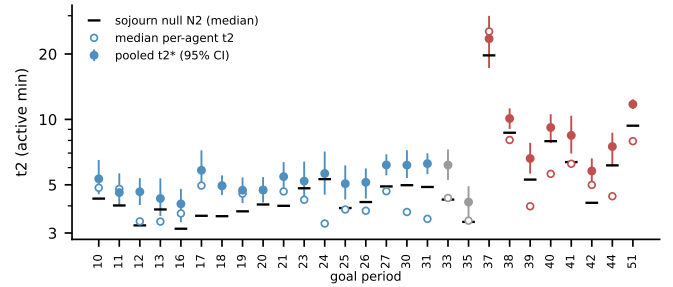
Bottom line. On action classes and Jev states, the slowest behavior mode is acting vs waiting plus agent differences, and t_2^* does not order periods by stuckness.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Is agent behavior a Markov state model (MSM) with a few slow metastable sets? Is its slowest timescale t_2^* a per-period order parameter, longest in stuck periods? Round 1 fits MSMs on 6 action classes per active minute in 27 non-holdout periods. The Chapman–Kolmogorov (CK) test fails in 27/27. t_2^* (5–24 min) beats a sojourn null by a small margin and does not track errors. Regime III is slower than regime I. Verdicts: 13 mixed, 14 failed. Round 1b repeats the fit on Jev v3.1 behavior states (soft vectors, a noise-cancelling estimator) and counts real failures instead of stderr. The semantic slowest mode lasts hours (agent-day scale). The slow set is still acting vs waiting. Higher p_{blocked} goes with *shorter* t_2^* . The native tests (forced erasures, nudger switch-off, #27’s spontaneous shift) fail. The holdout is not run.



Observables. Slowest implied timescale per goal period at $\tau_c = 5$ min. Filled: pooled MSM (95% agent-day CI). Bars: sojourn-null median (dwell times kept, order shuffled). Open: median per-agent t_2 . The pooled MSM beats the sojourn null in every period by a small margin. The agent mixture lifts it above the typical agent. Regime III (red) is slower than regime I (blue). The #37 peak is agents idle before their first and after their last action of the day.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Nothing changes in practice for an operator: t_2^* does not detect stuck periods. The “metastable sets” are tool modality (GUI vs shell) and idle time, which any log viewer shows. Part of the GUI-vs-shell split is a difference between GUI-heavy and shell-heavy agents. The transferable result is negative and is for analysts. Minute-scale agent behavior is non-Markov, so run the CK test before you quote a timescale. A pooled MSM over different agents makes false slow modes (pooled $t_2^* \times 1.28$ the per-agent median). Soft LLM labels need a noise-cancelling estimator and $\gtrsim 15k$ windows per period. On Jev states, waiting is the only slow set.

PHYSICAL MODEL

Kinetic Potts chain out of equilibrium (models 10, 02). The state $x_t \in \{\text{browse, type, shell, chat, idle, consolidate}\}$ is one per agent-minute (H14 scheme). $C(\tau)$ counts lagged pairs within agent-days. $T(\tau)$ is the non-reversible MLE. t_k are implied timescales from eigenvalues λ_k . χ holds the PCCA+ memberships (Röblitz–Weber) of m metastable sets. $q_i^\pm$ are forward and backward committors, with $\delta = 0$ under detailed balance. p_t are soft Jev posteriors.

$$T_{ij}(\tau) = C_{ij}(\tau) / \sum_k C_{ik}(\tau), \quad t_k(\tau) = -\tau / \ln |\lambda_k(T(\tau))|$$

$$t_2^* \equiv t_2(\tau_c = 5 \text{ min})$$

$$\text{CK: } P_{A \rightarrow A}(k\tau) \stackrel{?}{=} [\pi_A^\top T(\tau)^k]_A$$

$$T_{\text{cg}} = (\chi^\top \Pi \chi)^{-1} \chi^\top \Pi T \chi, \quad \delta_i = q_i^+ - (1 - q_i^-)$$

$$\lambda_k^\top = \text{eig}[C(1)^{-1} C(1+\tau)], \quad C(\tau) = \sum_t p_t p_{t+\tau}^\top$$

H17-behavior-metastable-sets · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H17-behavior-metastable-sets/README.md · ratings: H17 re-evaluation agent RE-B1 (suggested; round-1 values were coordinator-calibrated)

RESULTS VS PREDICTIONS

prediction (round 1b)	observed	verdict
P4b t_2^* tracks p_{blocked} (III)	$\rho = -0.69$; stratified -0.49 , p 0.03	(27) failed (sign)
P4b within periods	meta $\rho = -0.28$, p 0.03	(79) failed (sign) agents)
P4b action t_2^* vs real failures	$\rho = -0.19$ (stderr: -0.43)	failed
p_{blocked} flags stuck all: periods	$\rho(\text{fail}) -0.64$; $\rho(\text{output}) -0.71$	III: mixed
P4c regime I slower	40 vs 205 min, p 0.013	reversed
P3d <i>wait</i> is its own set	5/8 regime idle+absent	III; #51: holds
P8 argmax beats R1	MSM 9/27	failed
N1 NE41 relaxation $\approx \tau \approx 21$ min vs 263 (#51)		failed
t_2^*		
N2 NE43 slower with- out drive	block-to-block swings $\times 6-7$	failed
N3 #27 change point	p 0.31	failed

GOAL PERIODS IN DETAIL

G10 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G11 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G12 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G13 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G16 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G17 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G18 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G19 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G20 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G21 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
+19 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

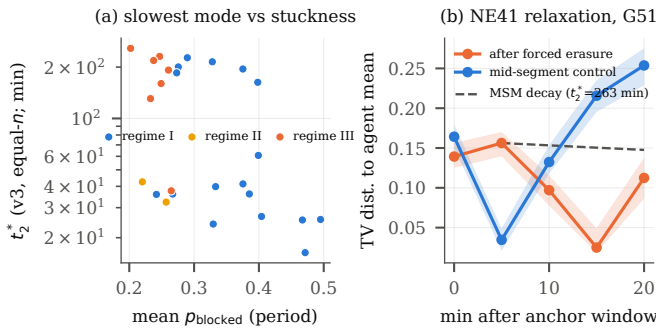
Round-1b synthetic checks show a bias of $\times 1.6-2.7$ in the soft-state shifted estimator at a 4-h period's 3,600 windows. The estimator breaks down at 12 states below $\sim 15k$ windows. The soft CK test has no power, and a sojourn null on noisy argmax runs is invalid. The v3 per-period verdicts are therefore descriptive, and t_2^* comparisons use equal-size subsamples. The NE41 control anchors are selected (active, censored). p_{blocked} means “waiting on others” in regime I and “failing or looping” in regime III. Pooled MSMs mix agents (rival R2).

ROUND 2

Fit a hidden-Markov model with agent-day random effects, to separate the agent mixture from switching. Use content-level states for spirals (#45 is held out).

- **Essence.** What are the metastable cognitive modes (planning, building, debugging, existential talk), and what switches them?
- **H17-R1** A soft Jev-state MSM recovers slow cognitive modes, and switches are driven by external input.
- **H17-R2** Debugging is the slowest mode and the main sink of time.
- **H17-R3** Mode switches synchronize across agents only through shared artifacts (E3).

H17-behavior-metastable-sets · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H17-behavior-metastable-sets/README.md · ratings: H17 re-evaluation agent RE-B1 (suggested; round-1 values were coordinator-calibrated) (page 2)



Second figure. (a) Semantic t_2^* (equal sample size) vs the period's mean p_{blocked} . It is bimodal (tens of minutes or a day length) and tends to be shorter where agents are more blocked. (b) After forced erasures (#51), the state deviation decays in ~ 20 min. The MSM's t_2^* predicts the dashed line. The control rises because of censoring selection.

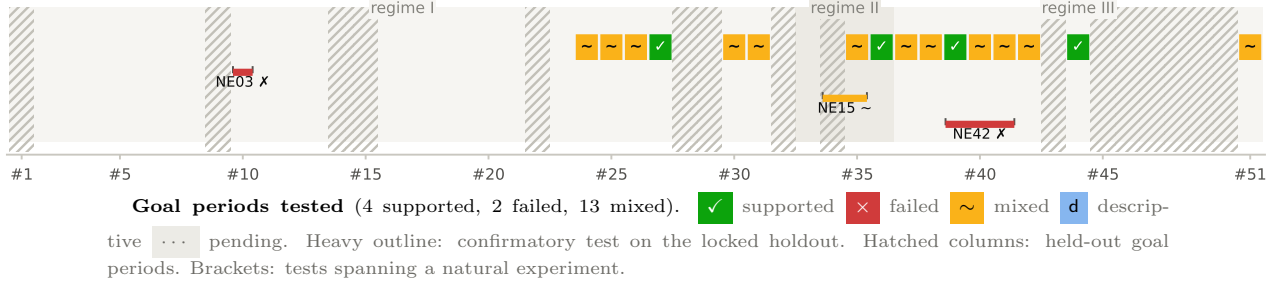
H18: Attention dilutes as 1/k: response to a message falls with the number of messages waiting at the agent's turn

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Complete 48% Faithfulness 2.5/5 Usefulness 2/5

Bottom line. Correction (2026-10-04): the all-turn exponent ($\hat{\beta} \approx 0.66$) is not separable from reactive timing (that null gives 0.75–0.80); the identified dilution is #51's timer-wake batches, $k^{-0.45}$ [0.41, 0.50] (round 1b), where the null gives ≈ 0 . So coupling scales sub-linearly, not as a literal J/N ; replies go to the newest messages.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

At each talk turn of an agent, we count the k messages that reached it since its previous talk turn. We ask whether it responds to each pending sender, fitting within agent-days in 16 non-holdout goal periods. Round 1b recounts k on the context ledger (which model call received which message) and adds reply labels. Response falls with k in 16/16 periods: pooled $\hat{\beta} = 0.66 \pm 0.02$ (round 1: 0.63). The curve saturates or follows recency; a budget-like shape wins 6/16. In #51, timer-wake batches of exogenous size give $\hat{\beta} = 0.45$. Per-pair uptake falls with room size across periods ($\rho = -0.57$) and across #51's roster growth ($\rho \approx -0.8$). The round-1 placebo failure was mostly the visibility rule. Invisible senders now get addressed at half the pending rate (was 85%); conversation state explains the rest. The merge and fetch-limit tests fail. The holdout is not run.

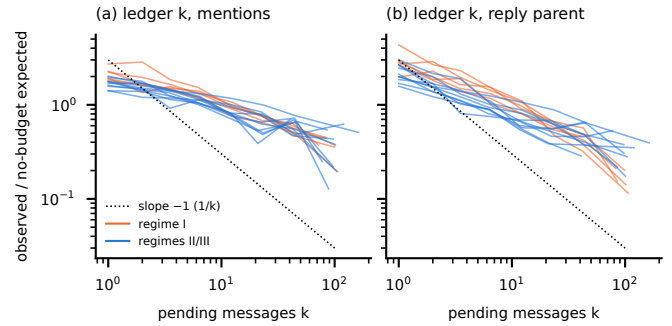
PHYSICAL MODEL

Mean-field coupling (01) with per-pair cross-excitation (09). At a talk turn of recipient i , the agent takes up each pending message m with hazard $\theta_{i,d} h_m$. It responds to sender j (a name or a reply parent) if it takes up any message of j . $\theta_{i,d}$: agent×day propensity. h : the attention rule (constant, $1/k$, saturating with k_0 , recency with ρ , power β). γ : boost for messages naming i . J_{ij} : per-pair coupling. N : room size. B : senders responded to per turn.

$$P(r_{\tau j} = 1) = 1 - \exp\left(-\theta_{i,d} \sum_{m \in P_j(\tau)} h_m\right)$$

$$h_m = e^{\gamma \mathbf{1}[m \rightarrow i]} \times \left\{1, \frac{1}{k}, \frac{1+k_0}{k_0+k}, \rho^{\text{rank}_m-1}, k^{-\beta}\right\}$$

$$J_{ij} \propto B k^{-\beta} \sim B N^{-\beta} \quad (\beta = 1: \text{literal } J/N)$$



Observables. Response relative to a no-budget model with the same agent×day propensities, vs pending k on the context ledger. One line per period. (a) Mentions: every period declines, less steeply than $1/k$ (dotted). (b) Reply parents: steeper, close to $1/k$, because each message has one parent. Recency decides which pending message gets the parent.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Expect per-pair engagement in a shared channel to fall as about $N^{-0.6}$ with room size N . Output per turn stays about the same, so a bigger room dilutes engagement and does not add collaboration. A reply goes to the newest pending messages. To raise within-room engagement, split the channel: the gain is about $(k \text{ ratio})^{0.6}$, as at the #35 split. This is a design rule, not a lever. It adds little to “use fewer, smaller rooms”, and the exponent varies by period (0.52–0.80). The mention signal is now cleaner, because the placebo failure was mostly a visibility bug. It still carries conversation state.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (ledger k)	Verdict (1b)
P1 $\hat{\beta} > 0$, pooled 0.5–1.2	16/16, 0.63 → 16/16, 0.66 (replies 0.85)	yes
P2 budget shape beats recency	5/16 → 6/16; replies: 14/16	no
P4 mentions bypass dilution	1/16 → 1/16	no
P5 exogenous- k timer wakes	3/6 → 3/6; #51 0.50 → 0.45 [0.41, #51 yes 0.50]	yes
P6 room effect absorbed by k	3–4/8 → same	mixed
P7 05-04 merge (NE42 native)	failed → failed (raw); joint fit: k absorbs 53–86%	no
P8 regime invariance	Δ 0.14 → 0.07	yes
P9 $\bar{p}$ falls with N , $\hat{B}$ flat	ρ $-0.55/-0.03$ → $-0.57/-0.02$	yes
#51 N sweep (native)	replies: $\bar{p}$ ρ -0.80 , $\hat{B}$ -0.64	mixed
P10 invisible placebo	0/16, $0.85\times$ pending → 3/16, mostly explained $0.49\times$	ex-

GOAL PERIODS IN DETAIL

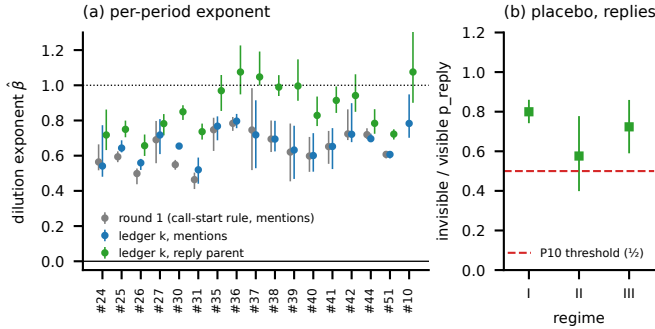
G24	~ 1b: mixed : ledger k , mention response, pre-registere...
G25	~ 1b: mixed : ledger k , mention response, pre-registere...
G26	~ 1b: mixed : ledger k , mention response, pre-registere...
G27	✓ 1b: supported : ledger k , mention response, pre-regis...
G30	~ 1b: mixed : ledger k , mention response, pre-registere...
G31	~ 1b: mixed : ledger k , mention response, pre-registere...
G35	~ 1b: mixed : ledger k , mention response, pre-registere...
G36	✓ 1b: supported : ledger k , mention response, pre-regis...
G37	~ 1b: mixed : ledger k , mention response, pre-registere...
G38	~ 1b: mixed : ledger k , mention response, pre-registere...

+9 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. (a) Exponent per period: round 1 (gray), ledger k with mentions (blue), ledger k with reply parents (green). The visibility fix barely moves it. Replies push it toward 1 by construction. (b) Placebo on reply labels: invisible pairs keep 58–80% of a visible pair's reply score (threshold 50%).

CAVEATS

Mentions mix uptake with conversation state. Reply labels allow one parent per message and keep a thread-membership component. Neither response is a clean per-message uptake. Recency and budget remain inseparable where k is small. The timer-wake design identifies the exponent only in #51. Era confounds the cross-period size law, and #51 is the only large room. We apply no multiplicity correction. Nothing is confirmed on the holdout, and the holdout script still builds round-1 inputs.

ROUND 2

- **Essence.** Attention is the scarce resource, and coupling is its allocation.
- **H18-R1** A salience law, uptake = $f(\text{mention, recency, novelty, sender status})$, is fit and predicts held-out uptake (E5).
- **H18-R2** Total attention per turn is conserved; re-test with reply-threading labels.
- **H18-R3** An optimal room size maximizes total useful uptake (HH111).

H18-attention-dilution generated 2026-10-07 17:50 by infra/summaries/build_summaries.py card hypotheses/H18-attention-dilution/README.md ratings: H18 re-evaluation agent RE-V1 (suggested) (page 2)

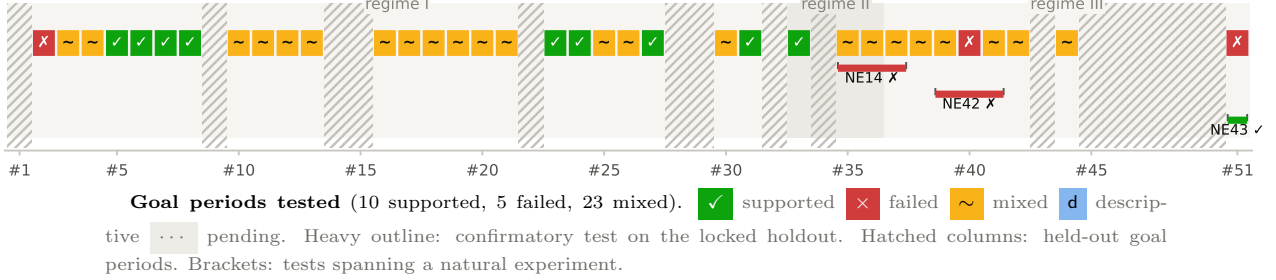
H19: One curve for all periods: per-period loop gains collapse onto a function of an operational control parameter

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Complete 50% Faithfulness 1/5 Usefulness 2/5

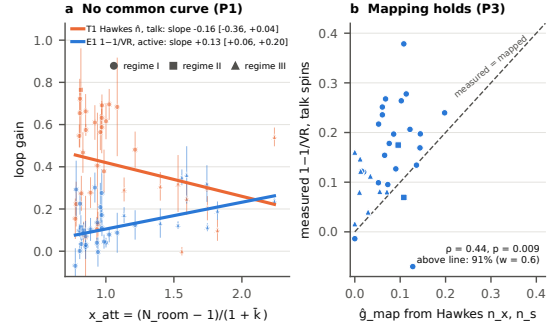
Bottom line. Loop gains do not collapse onto x_{att} ; the round-1 “opposite channels” were the operator’s day edges, and trimmed gains move together.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

We pool per-period loop-gain estimates over 35 non-holdout periods: Hawkes and equal-time methods from H02–H05 and H19’s own. We ask whether they fall on one curve of x_{att} , the attended partners per unit attention. **The pre-registered collapse fails** in round 1 and again in round 1b. No common positive slope exists, and a regime-only step wins out of sample. Round 1 read two channels that move in opposite directions: activity co-activation rose with x_{att} while talk triggering fell. **Round 1b withdraws that reading** (corrected minute grid, day-edge-trimmed gains). The operator’s day edges made the activity rise: the slope goes $+0.18 \rightarrow -0.03$ after the trim. NE14 shows the same at the 2026-03-24 switch. After the trim, the activity gain agrees with the Hawkes gains. Post hoc, all six re-estimated methods rise with messages delivered per LLM step. The unfitted Hawkes-to-Ising mapping (P3) still holds.



Observables. Round 1 (buggy minute grid, untrimmed). (a) The two primary loop gains per period vs x_{att} , with random-effects meta-regression lines. The Hawkes talk branching $\hat{n}$ (orange) falls and the equal-time active-spin gain (blue) rises. High x_{att} is mostly regime III (triangles). Round 1b shows that day edges make the blue rise (page 2). (b) The unfitted cross-method test: the equal-time talk gain vs the value mapped from H03’s fast Hawkes kernels ($w = 0.75$). Points sit above the identity line.

PHYSICAL MODEL

Mean-field Ising (01/02) and Hawkes (09). s_i : the 1-min active or talk spin of agent i . VR: variance ratio. g_{eq} : equal-time loop gain (H02’s βJ_{0q} , H04’s K , H05’s g). n_x , n_s : Hawkes cross- and self-offspring per event. w : the share within one $\Delta = 60$ s bin (kernel time τ_k). Dilution: per turn, $\bar{k}$ unread messages from $N_{room} - 1$ partners and a response budget b . y_{pm} : the gain of method m in period p . Control parameter: x_{att} .

$$g_{eq} = 1 - 1/VR, \quad x_{att} = (N_{room} - 1)/(1 + \bar{k})$$

$$VR = \text{Var}(\sum_i s_i) / \sum_i \text{Var } s_i, \quad n_x = b x_{att}$$

$$\hat{g}_{map} = \frac{2wn_x}{1 + 2w(n_x + n_s)}, \quad w = 1 - \frac{\tau_k}{\Delta} (1 - e^{-\Delta/\tau_k})$$

$$y_{pm} = \alpha_m + c_m x_{att,p} + u_p + \varepsilon_{pm}, \quad \text{collapse: } c_m > 0 \forall m$$

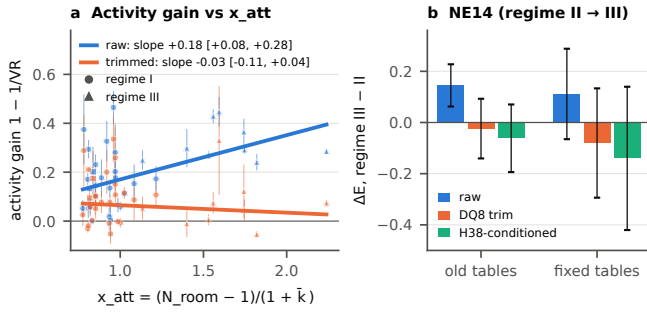
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not put one “loop gain” on a swarm dashboard: no single configuration knob predicts coupling strength. Trim the operator’s day edges before you compute any activity coupling. The edges carry about $\frac{3}{4}$ of the raw activity gain (NE43). After the trim, activity and talk gains move together. The one candidate knob is k_{llm} , the messages delivered per LLM step: all six methods rise with it. That knob is post hoc, entangled with the scaffold and fails inside #51, so it is a hypothesis to test, not a lever.

H19-loop-gain-collapse · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H19-loop-gain-collapse/README.md · ratings: RE-A2 re-evaluation agent (suggested), round 1b

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b raw / 1b trimmed)	Verdict
P1 one curve on x_{att}	E1 slope $+0.13 \rightarrow +0.18/-0.03$; T1 -0.16 ; regime-only wins ($-11.5 \rightarrow -8.7/-7.3$ nats)	$\times$
P2 within-family $\rho \geq 0.4$	E: $-0.48 \rightarrow 0.21/0.28$ (0.44 H38- cond.); T: $0.40 \rightarrow 0.37$	$\times \rightarrow$ partial
P3 Hawkes→Ising map (unfitted)	ρ 0.44 $\rightarrow$ 0.37/0.33; above map 91% $\rightarrow$ 94/89%	$\checkmark$
P4 per-pair $n_x \propto (1 + \alpha_k \bar{k})^{-\alpha}$	0.20 $\rightarrow$ 0.26; on N : 1.03	$\times$
P5 no mode effect	CIs include 0	$\checkmark$
Regime III – I, activity gain	$+0.11 \rightarrow +0.17/-0.03$	gone
Native #51 N sweep	per-pair n_c falls far faster than $1/(N-1)$; trimmed g rises with N	$\times$
Native NE42 / NE43 / NE14	talk triggering ≈ 0 in the merged room; edges $\approx \frac{3}{4}$ of g on all NE43 sides; adjusted II→III < 0	$\times \checkmark \times$



Second figure. (a) Activity gain per period vs x_{att} , raw (blue) and day-edge-trimmed (orange), with meta-regression lines. The rise disappears after the trim. (b) NE14: the regime II→III change of the activity gain, raw vs trimmed vs H38-conditioned, on old and fixed tables.

GOAL PERIODS IN DETAIL

G02 $\times$ 1b: failed corrected data, pre-registered E1; with th...
G03 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
G04 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
G05 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
G06 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
G07 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
G08 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
G10 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
G11 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
G12 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...

+28 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

H04's K and H05's gains used the buggy grid; we drop them until their owners re-run them, which leaves 6 methods. The trim keeps 77% of regime-III minutes, and the DQ8 trim and H38 conditioning disagree in places (NE42). The k_{llm} curve is post hoc and fails within #51. The methods share raw events, so the summed ELPD over-counts evidence. We did not re-run the synthetic validation of the P1 rule. The frozen confirmatory channel model uses round-1 data and is stale.

ROUND 2

Re-freeze the channel model from round 1b (talk and activity on k_{llm}) and pre-register it. Add H04/H05 after their re-runs. Test k_{llm} within periods where it varies at a fixed roster.

- **Essence.** The swarm has two order parameters, talk and work, that trade off.
- **H19-R1** Talk and work anticorrelate: periods and agents sit in talky or worky phases on a (talk, work) phase diagram.
- **H19-R2** Goal type (shared vs individual) moves the swarm between the two phases.
- **H19-R3** Messages per LLM step is the control parameter; pre-register the post-hoc collapse.

H19-loop-gain-collapse · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py
collapse/README.md · ratings: RE-A2 re-evaluation agent (suggested), round 1b (page 2)

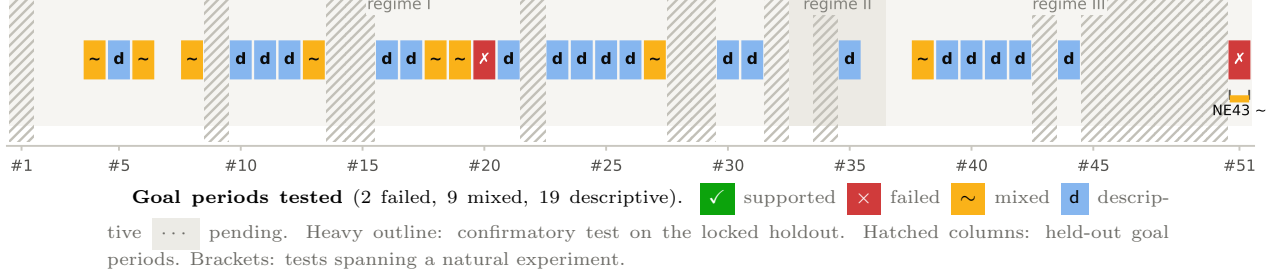
H20: Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Swarm content does not age at day resolution (both embedding models): it settles for days after a kickoff, then decorrelates at an age-independent rate.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

We measure the two-time correlation $C(t_w + \tau, t_w)$ of each agent's day-mean content embedding (t_w : active days since kickoff) in 29 non-holdout goal periods. We test the aging slope A at fixed lag ($t_w \geq 2$) against a stationary swarm model. The pre-registered isotropic null is 2–3× too narrow, because content fluctuates in 5–12 of 32 dimensions; we use a null with the fitted shape. **No aging:** the random-effects A over long and medium goals is $+0.021 \pm 0.021$. #38 shows a kickoff relaxation: lag-1 C rises 0.66→0.87 in four days, then stays flat. In #51, the test rejects $\mu = 0.5$ aging. **Round 1b** adds a second embedding model, dedupe, style residualization and shared goal vectors. Every verdict holds in all 7 input configurations. Style residualization halves A and removes the weak late slowing in #51. Newcomers relax on their own clock without aging. Drive withdrawal and #27's rivalry-to-collaboration turn leave the correlation intact. The holdout is not run.

PHYSICAL MODEL

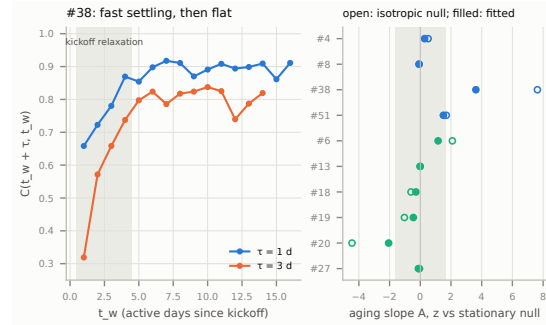
Glassy aging (01) of vector spins (11). The spin $\bar{x}_{it}$ is the day state of agent i : the mean of its unit-normalized, regime-whitened ($n = 32$) statement embeddings (own chat and plans). v_{it} is its sampling noise. q : plateau (style and goal content). c_0 : short-lag ceiling. τ_0 : relaxation time. μ : aging exponent of the Box-Cox clock s_μ ($\mu = 0$ stationary, $\mu = 1$ simple aging). Rival: a kickoff transient that decays in τ_q days. n_{eff} : dimension of the day-to-day fluctuations.

$$C(t, t') = \frac{\sum_i \bar{x}_{it} \cdot \bar{x}_{it'}}{\sqrt{\sum_i S_{it} \sum_i S_{it'}}}, \quad S_{it} = |\bar{x}_{it}|^2 - v_{it}$$

$$C(t_w + \tau, t_w) = q + (c_0 - q) e^{-\Delta s / \tau_0}$$

$$\Delta s = s_\mu(t_w + \tau) - s_\mu(t_w), \quad s_\mu(t) = \frac{t^{1-\mu} - 1}{1-\mu}$$

$$A = \partial C / \partial \ln t_w \big|_\tau, \quad \text{Var } C(t, t') \propto 1/n_{\text{eff}}$$



Observables. Left: #38. The agent-averaged correlation of content on day t_w and day $t_w + \tau$ climbs for four active days after the kickoff, then stays flat. A quench relaxes; the system does not slow with age. Right: the aging slope A as a z-score against the stationary null, for long (blue) and medium (aqua) goals. Open: the pre-registered isotropic null (too narrow). Filled: the null with the fitted fluctuation shape. Only #38 (up) and #20 (Thanksgiving, down) stand out.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Nothing changes for an operator who fears that a long-running swarm “rigidifies”. At day resolution, content turnover does not slow from week one to week three. Weak aging is not excluded. After a large kickoff, allow about four active days for content to settle (seen once, #38). For an analyst: test embedding similarity on swarm logs against a null with the measured low-dimensional fluctuation shape. An isotropic null is 2–3× overconfident. As physics, aging finds nothing to grip: daily re-prompting with a fixed goal acts as a strong pinning field.

RESULTS VS PREDICTIONS

Prediction	Round 1 $\rightarrow$ 1b (bge / gte)	Verdict
P2 aging in $\geq 2/4$ 1/4 (#38) $\rightarrow$ 1/4 / 1/4		$\times$
long goals		
RE A, long + medium	$+0.021 \pm 0.021 \rightarrow +0.020 / +0.022$; style $+0.008$	$\times$
P4 late slowing where $A > 0$	#38 $A_{\text{late}} -0.04 \rightarrow -0.04 / -0.06$	$\times$
#51 $\mu = 0.5$ rejected	power $1.00 \rightarrow \geq 0.98$ in all configs	$\checkmark$
#51 weak $\hat{\mu} > 0$ (descriptive)	$0.22 [0.07, 0.41] \rightarrow 0.16 [0.00, 0.37]$; gone style $CI \ni 0$	
Calibration (robust z-SD)	$1.07 \rightarrow 1.07 / 0.91$	$\checkmark$
#51 newcomers: on-boarding $K > 0$	3 of 5 joiners; no own-clock aging	$\sim$
#27 mid-period switch	$B_{\text{max}} p 0.48 / 0.21$	d
NE43 drives with-drawn	08-05 dip -0.022 inside placebo; 08-21 none	$\sim$

GOAL PERIODS IN DETAIL

G04 $\sim$ 1b: mixed (both models, all configs)
G05 d 1b: descriptive (both models)
G06 $\sim$ 1b: mixed (both models, all configs)
G08 $\sim$ 1b: mixed (both models, all configs)
G10 d 1b: descriptive (both models)
G11 d 1b: descriptive (both models)
G12 d 1b: descriptive (both models)
G13 $\sim$ 1b: mixed (both models, all configs)
G16 d 1b: descriptive (both models)
G17 d 1b: descriptive (both models)
+20 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

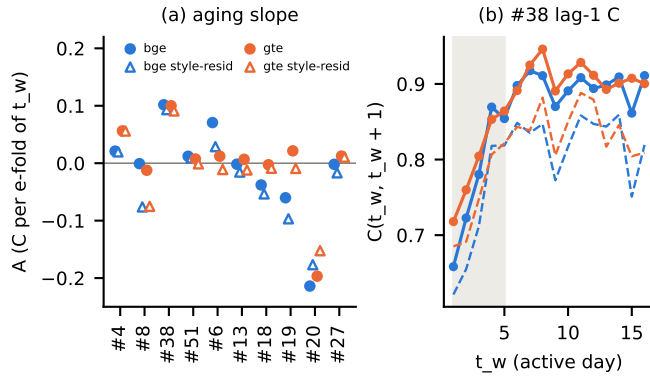
CAVEATS

The analysis has day resolution only. Weak sub-aging ($\mu \lesssim 0.3$) is not excluded outside #51. We chose the calibrated null after seeing real data. NE43 shows that this null is too narrow for step statistics at ordinary day boundaries. In short periods, the null is anti-conservative and holidays dominate. The onboarding transients rest on 5 joiners.

ROUND 2

Compute sub-day two-time functions on the active-hour clock (link to H48's $\tau \approx 2-5$ h). Build an empirical-placebo null for boundary tests.

- **Essence.** The swarm has no long-term internal memory; all persistence is stored outside the agents.
- **H20-R1** Persistent content order is explained entirely by persistent fields (goal prompt, artifacts); without them content decorrelates within a day (E3, E4).
- **H20-R2** The $\sim 2-4$ day kickoff relaxation is set by artifact build-up, not agent memory.



Second figure. (a) Aging slope per long and medium goal, bge (blue) vs gte (orange); filled base, open style-residualized. The two models agree, and style residualization shrinks #51 and #19. (b) #38: lag-1 correlation rises for about four active days (shaded), then stays flat in both models (dashed: style-residualized).

H21: The debate week (#12) is a two-sublattice antiferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

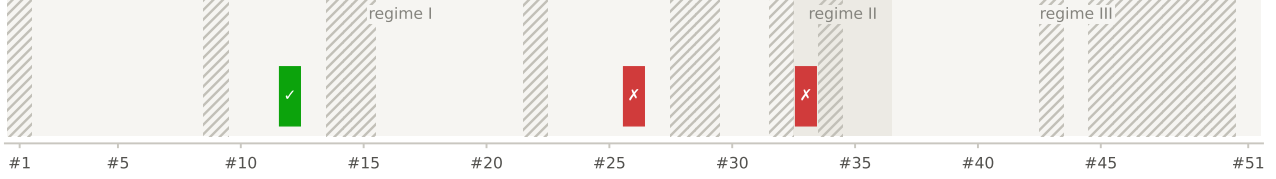
Complete 60%

Faithfulness 2/5

Usefulness 2/5

Bottom line. Debaters align on the motion's topic; drafted opponents disagree on the merits (45% vs 0% of replies) only while assigned, and nobody takes sides without assignment.

level: descriptive · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 supported, 2 failed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

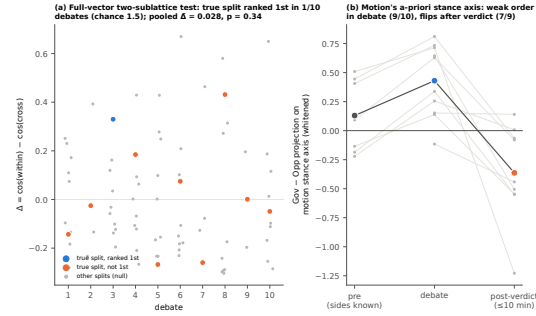
Test bed: #12, 10 debates, each with its own judge, motion and re-drafted teams (labels equal DQ6 ground truth). **In content, the pre-registered two-sublattice test fails.** The within- minus cross-team cosine is $\bar{\Delta} = 0.028$ ($p = 0.34$), and content recovers 1/10 teams. **Round 1b** repeats this with a second embedding model and style-residualized and deduplicated inputs ($\bar{\Delta} -0.05$ to $+0.03$). Only a weak tilt along the motion's support-minus-oppose axis remains (bge $p = 0.002$, gte $p = 0.065$). **In reply stance (DQ2), the teams order.** Teammates score $+0.41$ and opponents -0.32 : $\Delta = 0.72$ in 10/10 debates, beyond a calibrated agent-field null ($p = 0.002$). Stance recovers 8/10 teams. The same pair turns hostile only when drafted apart (16/18). The order is gone after the verdict. **Round 1c** uses DQ10's validated disagreement flag (precision 0.61–0.67, recall ≈ 0.6) with that noise built into the nulls. All 33 debate-phase flags fall across teams: 45% of opponents' replies and 0 of 60 teammates' replies (stress null $p = 0.001$). The cross-team order is disagreement on the merits, not task pushback. It stops at the verdict (0 flags in 90 replies). Where nobody assigns sides (#26 vote, #33 news debate), disagreement stays at 1–2% and no camps form. The #34 holdout is not run.

PHYSICAL MODEL

Vector spins on two sublattices (11), with $q = 2$ Potts (10) and mean field (01). Spin $\mathbf{s}_i$: the masked, whitened ($n = 32$), agent-centred chat mean of debater i in a debate. Sublattice $A = \text{Government}$ ($\varepsilon_i = +1$), $B = \text{Opposition}$. Fields: motion topic $h_u \hat{g}$ (uniform), assigned sides $h_s \hat{a}$ (staggered), family $\mathbf{h}_i$, verdict h_v toward winner W . Couplings: k_{in} (team, ferromagnetic), k_{out} (cross-team, antiferromagnetic). ρ : correlation of sublattice fluctuations.

$$H = -\frac{k_{\text{in}}}{N} \sum_{i \sim j} \mathbf{s}_i \cdot \mathbf{s}_j + \frac{k_{\text{out}}}{N} \sum_{A \times B} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i (h_u \hat{g} + h_s \varepsilon_i \hat{a} + h_v \varepsilon_W \hat{a} + \mathbf{h}_i) \cdot \mathbf{s}_i$$

$$\bar{\Delta} = \langle \cos \rangle_{\text{within}} - \langle \cos \rangle_{\text{cross}}, \quad \rho = -k_{\text{out}} / (1 - k_{\text{in}})$$



Observables. (a) Per debate, the within-minus-cross cosine for every possible team split (gray) and for the true split (large dot; blue if best). The true split is unremarkable: full content vectors show no two-sublattice order. (b) On the motion's a-priori stance axis, Government sits slightly on the support side during debates (9/10). Post hoc, it swings to the other side after the verdict (7/9).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Look for factions in reply stance, not in embeddings. Clustering of messages finds labs and topics and misses even the assigned teams. A zero-shot stance label on reply pairs, read in aggregate against an agent-field null, recovers 8 of 10 drafted teams. Agents on these models do not form camps without an assignment, neither over a contested vote nor in a debate about the news. Use a stance-based faction alarm to check that agents play their assigned roles. Treat an alarm without assigned sides as an anomaly.

H21-debate-antiferromagnet · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H21-debate-antiferromagnet/README.md · ratings: RE-C2 round-1b agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Round 1 $\rightarrow$ round 1b	Verdict
P1 content order	$\bar{\Delta}$ 0.028 $\rightarrow$ $-0.05..+0.03$ (2 models, 5 inputs)	$\times$
P2 teams from content	1/10 $\rightarrow$ 0-1/10	$\times$
P4d text-axis tilt	σ 0.092 (p 0.002) $\rightarrow$ gte 0.055 (p 0.065) $\sim$	$\sim$
P7 AF fluctuations	ρ +0.53 $\rightarrow$ +0.02.. $+0.53$ (never negative)	$\times$
S1 stance order	Δ 0.72, 10/10; agent-field p 0.002	$\checkmark$
S2 teams from stance	8/10 (chance 1.0)	$\checkmark$
S3 off after verdict	-0.01 vs 0.67 during debates	$\checkmark$
S1c disagreement (1c)	flags 45% across vs 0% within; true 0.52-0.76	$\checkmark$
S3c off after verdict (1c)	0 flags in 90 replies; power ≥ 0.9	$\checkmark$
#12 within-pair stance (native)	-0.68 (16/18); 1c flags 10/14, none against	$\checkmark$
#26 / #33 (na-tive)	no ballot camps; no unassigned camps	$\times$ (as predicted)

G12 $\checkmark$ 1b: supported in stance (two-sublattice order; content...
 G26 $\times$ failed
 G33 $\times$ failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	0	G ground truth	2
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

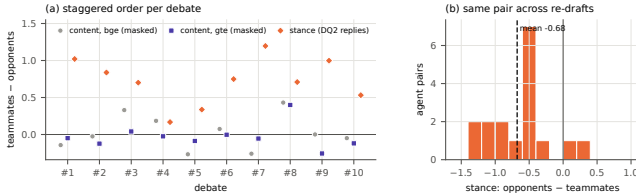
Round 1b's DQ2 stance is zero-shot (pair-level "opposes" precision 0.08). Round 1c's validated flag is better (precision 0.61-0.67) but still one labeller and 33 flags, and teammates' 0/60 leaves the log-odds effect unbounded. The agent-field null assumes replies are independent given agents. Stance measures agreement of positions, so the assigned staggered field alone predicts the order. A coupling beyond the assignment is not shown: content fluctuations stay positively correlated. The sample is seven agents over four days in regime I. #26 has 9 voters with near-identical ballots (low power), and #34 is untouched.

ROUND 2

Test whether stance order needs the protocol: use debates with self-chosen sides or a mid-debate role switch. Fit a reply-level kinetic model to separate a field from a coupling.

- **Essence.** LLM agents don't form coalitions; assigned roles leave no lasting imprint.
- **H21-R1** Role imprint lifetime: roles bias stance only while active, then reverse (HH127); measure across all role periods.
- **H21-R2** Agreement is the default stance (sycophancy as ferromagnetic stance coupling); disagreement appears only when assigned.

Second figure. (a) Teammates minus opponents per debate. Content cosine in two embedding models (gray, violet) scatters around zero. Reply stance (orange) is positive in every debate. (b) The same agent pair across re-drafts: stance as opponents minus stance as teammates is negative for 16 of 18 pairs.



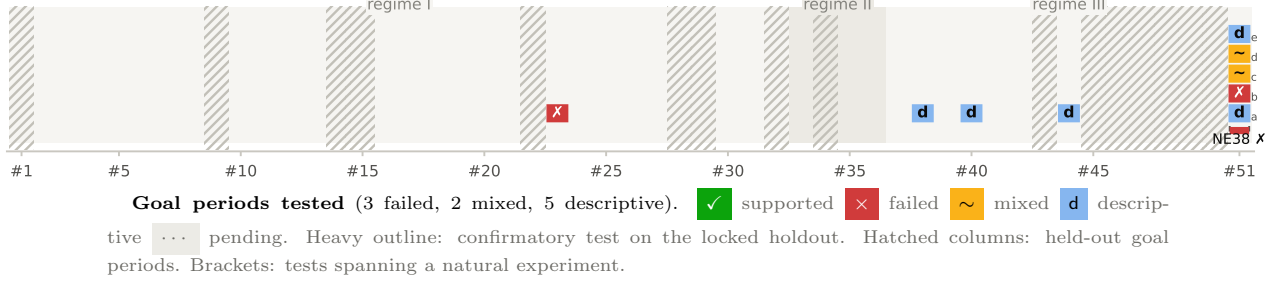
H22: Private, conflicting goals make #51 a spin glass

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. #51 is not a spin glass: roles act as random fields with weak positive coupling, rivals align, and the swarm never switches collectively.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Do the private, partly conflicting roles of #51 make the swarm a spin glass? Units 51a–e (45 non-holdout days) are tested against the shared-objective weeks #38, #40, #44. Content couplings are real but mostly positive, positions persist without collective switching ($W \approx 1$), and same-role rivals co-move *more* than unrelated pairs ($T_{SR} +0.04$ to $+0.06$ in every round-1b variant, $p \leq 0.05$). DQ2 stance is ferromagnetic. **Round 1c** uses the validated v2.1 disagreement flag with the labeller’s confusion built into the null. Rivals are not more disagreeing ($+0.49$ pp $[-0.68, +1.67]$), but the flag is too rare to decide (power 0.47 at odds ratio 3). A read vs in-flight contrast shows that reading a rival pulls content *toward* it ($\Gamma = +0.19 [+0.07, +0.31]$). The holdout is **not** run.

PHYSICAL MODEL

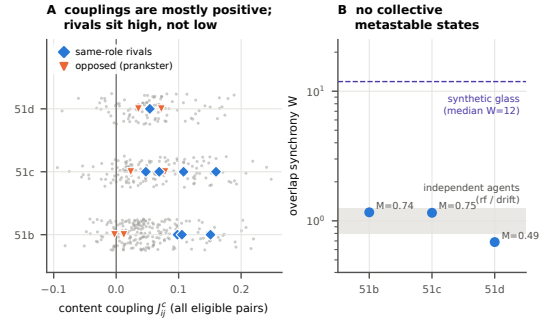
Spin glass (01, SK) of $O(n)$ spins (11). Spin $\mathbf{s}_i$: the whitened chat embedding of agent i per 30-min window ($n = 32$). Fields: day field $\mathbf{m}_d$; static field $\mathbf{H}_i$ (role, lab, style). J_{ij} : correlation of the within-day fluctuations $\mathbf{x}_i, \mathbf{x}_j$ minus a cross-day surrogate. A, B, C : day folds. κ : SK ratio J_0/J . τ_3 : gauge-invariant balance (SK ≈ 0 , factions ≈ 0.5). q : overlap of day-demeaned states $\hat{\delta}$. W : its synchrony beyond per-agent shifts.

$$P(\{\mathbf{s}\}) \propto \exp \left(\sum_{i < j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i (\mathbf{m}_d + \mathbf{H}_i) \cdot \mathbf{s}_i \right)$$

$$J_{ij} = \text{corr}_w(\mathbf{x}_i, \mathbf{x}_j) - \text{corr}_w^{\text{surr}}(\mathbf{x}_i, \mathbf{x}_j)$$

$$\kappa = \sqrt{N} \bar{J} / \sigma_J, \quad \tau_3 = \text{tr}(J^A J^B J^C) / \text{tr}(J^A J^B)^{3/2}$$

$$q(d, d') = \frac{1}{N} \sum_i \hat{\delta}_{i,d} \cdot \hat{\delta}_{i,d'}$$



Observables. (a) The content coupling of every eligible pair in the counted #51 units (gray), with same-role rivals (blue) and Prankster vs Ethicist/Psychologist pairs (orange). Couplings are mostly positive, and rivals sit high. (b) Overlap synchrony W at the independent-agent level (gray band), far below a synthetic glass (dashed). Memory M is high without synchrony: agents hold their own positions, and the swarm never jumps between collective states.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Little changes in practice. Assigned roles act as fixed fields that pin each agent’s topic. The shared room adds a weak positive coupling, and same-objective agents drift together. To steer a mixed-motive swarm, edit the role prompts: roles, not inter-agent conflict, set what agents talk about. Expect no hidden collective states to escape ($W \approx 1$). Embeddings measure topic, not stance, so strategic opposition could stay invisible in content. The diagnostics port to other logs; the spin-glass vocabulary is not earned.

H22-private-goals-spin-glass · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H22-private-goals-spin-glass/README.md · ratings: RE-C2 round-1b agent (suggested; faithfulness and usefulness unchanged from the coordinator’s calibration)

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (51b / 51c / Verdict 51d)
P1 real couplings	content 2/3 → 2/3 (bge), 3/3 (gte); ✓ stance 3/3 beyond agent fields
P2 $\kappa < 1$	5.2 / 3.9 / 9.7; stance mean +0.6 ×
P3 $\tau_3^{\text{dc}} < 0.25$	content indeterminate (both models); stance 0.25 / 0.23 / 0.05 ✓
P4 rivals repel	RE $T_{\text{SR}} + 0.048 \rightarrow +0.042$ (gte), +0.052 × (ri- (style-res.); stance +0.09 n.s. val)
P4 opposed roles	content ≈ 0 ; stance +0.11 / -0.34 / × +0.22 (2 pairs)
P5 collective states	W 1.16 / 1.15 / 0.69; gte 51c 1.61 ($p \times 0.02$)
NE38 (native)	field leaves rivals ($p 0.004$); coupling DiD × -0.27 n.s.
#23 chess (na- tive)	opponents' stance -0.005; content × +0.03 n.s.
1c P4 stance (v2.1 flag)	DQ2 soft +0.09 n.s. → validated T_{SR}^D inconcl. +0.49 pp [-0.68, +1.67], NG $p 0.29$
1c P3 stance balance	3/3 → 1/3 ($\tau_3^{\text{dc}} 0.22 / 0.18 / 0.57$) ×
1c read vs in-flight	open → $\Gamma +0.19$ [+0.07, +0.31] (bge), rival +0.23 (gte)

GOAL PERIODS IN DETAIL

G23	× failed
G38	d 1b: descriptive (round 1: descriptive)
G40	d 1b: descriptive (round 1: descriptive)
G44	d 1b: descriptive (round 1: descriptive)
G51a	d 1b: descriptive (round 1: descriptive)
G51b	× 1b: failed (round 1: failed; rival homophily in every...
G51c	~ 1b: inconclusive (round 1: inconclusive)
G51d	~ 1b: inconclusive (round 1: inconclusive)
G51e	d 1b: descriptive (round 1: descriptive)
NE38	× failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

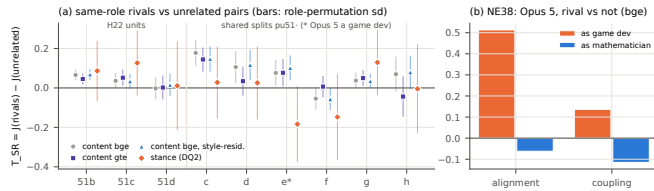
CAVEATS

τ_3^{dc} cannot separate SK from a heterogeneous ferromagnet (Amendment 1), so the unbalanced stance residuals are not glass evidence. Per-pair negative-stance counts assume independent replies within a pair, so threads inflate them. They are not enriched in conflict classes. Each unit has few testable rival pairs (3–5; one in most shared splits). The NE38 coupling result uses a threshold chosen after seeing the coverage. DQ2 stance labels were reliable only in aggregate (pair-level “opposes” precision 0.08). The v2.1 flag is validated but rare (about 4 expected rival flags per unit), and its synthetic size is 0.11 in 51d. The Γ excess over random pair sets is suggestive only (percentile 0.73–0.95).

ROUND 2

Pool the validated flag over #51 and the holdout tail for power (re-freeze proposed). Fit lagged couplings per pair to see whether the read-gated rival pull is pair heterogeneity or role structure.

- **Essence**. Do conflicting incentives produce conflict behavior at all?
- **H22-R1** Cooperation default: agents help rivals at rates comparable to aligned pairs (Jev-labeled help acts).
- **H22-R2** A quantitative random-field fit (HH129): roles as fields plus weak positive coupling.



Second figure. (a) Rival contrast T_{SR} per unit in content (two models and style-residualized) and in stance; bars show the role-permutation sd. It is positive almost everywhere. The one negative stance value (pu51e, Opus 5 as game dev) is not significant. (b) NE38: the alignment and coupling of Opus 5 with its game-dev rivals while it held that role, and after.

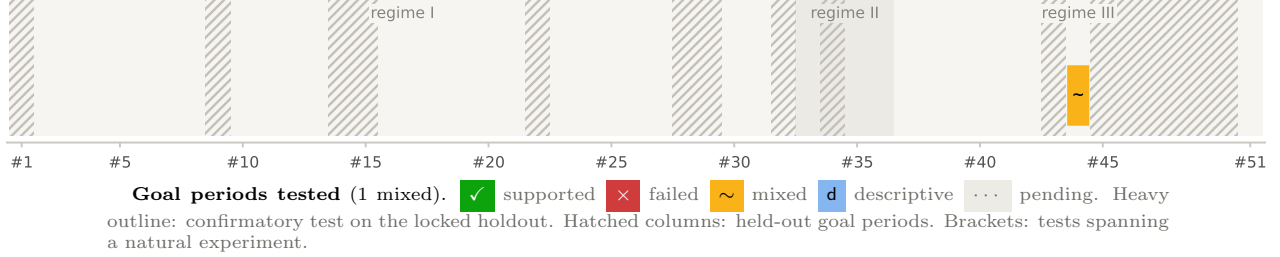
H23: Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

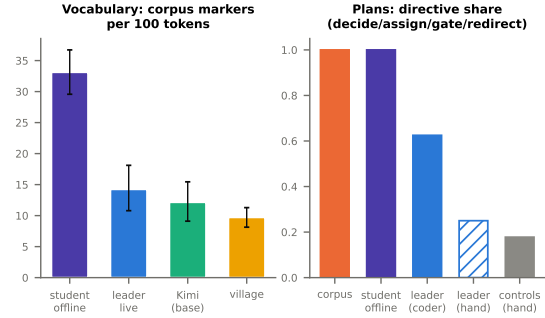
Bottom line. The fine-tuned leader took a trace of its corpus’s addressing style and none of its plans; the corpus was mostly its base model’s own output.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

From the terminal logs, the deployed leader (agent 28 in #44, agent 30 in #45) is base Kimi K2.6 with a LoRA on 64 rows. 80% of them are Kimi’s own replies to 20 synthetic village scenarios under a leadership prompt; no village chat was a target. We compare its 16 live #44 messages with 45 same-window messages (exploratory, not blind). Corpus-distinctive vocabulary is 14.2 vs 10.0 per 100 tokens ($p = 0.011$), mostly agent addressing and markup. Phrase copy is not significant; lexical copy information is 0.012 bits per word. The directive share of plans is 0.62 (hand-coded 0.25) vs 1.00 in the corpus. Excess plan copy is -0.04 against a perfect-copy 0.23, and transformation information sits at the null. Plans are *not transmitted*, not transformed. In embeddings, it is no closer to the corpus than to its base; the base-echo and latest-chat rivals stand. The #45 test is written and **not run**.



Observables. (a) Corpus-distinctive n-grams per 100 tokens (95% CI). The offline student (sibling checkpoints on training-format prompts) sits far above the room. The live leader keeps about a fifth of that excess, near base Kimi in the same window ($n = 6$). (b) Directive share of plan acts: 100% in the corpus and the offline student. The live leader (hatched: post hoc hand coding) sits near the room.

PHYSICAL MODEL

Copying vs transformation (08, Kolchinsky & Corominas-Murtra), distillation variant. The channel runs from the corpus X to the leader’s live output Y . The shared alphabet is word n-grams or 10 keyword-coded plan acts. Each live message m pairs with the corpus rows whose scenario is nearest its context, $p_C(x | m)$. d : binary KL. r_x : probability of reproducing x . a, b : rates at which the leader and untrained same-room speakers emit corpus n-grams. c : copy fraction.

$$I(X; Y) = I_{\text{copy}} + I_{\text{transform}}$$

$$I_{\text{copy}} = \sum_x p(x) d(r_x \| p_Y(x)) \mathbf{1}[r_x > p_Y(x)]$$

$$p(x, y) = \frac{1}{n} \sum_m p_C(x | m) \delta_{y, y_m}, \quad c = \sum_x p(x, x)$$

$$I_{\text{copy}}^{\text{lex}} = d(a \| b) \text{ bits per n-gram}$$

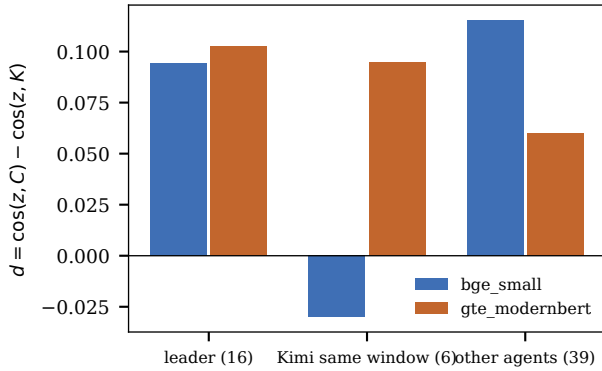
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

If you distill a leader model from a few dozen synthetic scenarios, expect the surface style to transfer and the decision policy not to. Here the leader uses 14.2 corpus-distinctive tokens per 100 vs 10.0 in the room. Its plans do not transfer, because live routine work looks nothing like the crisis scenarios. Cover the deployment situations in the training set. That coverage failure is standard ML, not a swarm result. The decomposition adds one distinction (not transmitted vs transformed) and a perfect-copy reference, but 16 messages and a coder at $\kappa \approx 0.45$ measure little.

H23-leader-distillation-copy · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H23-leader-distillation-copy/README.md · ratings: calibrated by coordinator 2026-10-04 (from H23 agent (suggested))

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict (1b)
P1a corpus vocabulary > room	14.2 vs 10.0 /100 tokens (p 0.011); unchanged	yes
P1b bigram copy room	> 0.064 vs 0.055 (p 0.17); unchanged	inconclusive
P2a plans not copied	directive 0.62 vs corpus 1.00; unchanged	yes
P2b plans transformed	$I_{transform}$ within null; unchanged (shared copy_info, 1,022 numbers identical)	no (not transmitted)
P3a closer to corpus than base	bge: leader > Kimi p 0.08 → gte: p 0.43 (Kimi as corpus-like)	both models
Checkpoint labels	DQ6 times relabel 0 of 27 leader messages	unchanged
R1 base echo rejected	no → no, and stronger (H46 rival stands style, H32 weakest source)	



Second figure. Embedding proximity $d = \cos(z, \bar{z}_{\text{corpus}}) - \cos(z, \bar{z}_{\text{Kimi field}})$ in regime-III whitened space. Groups: the 16 live leader messages, same-window Kimi K2.6 (6) and the other #best agents (39). Models: bge-small (round 1) and gte-modernbert (round 1b).

GOAL PERIODS IN DETAIL

G44 ~ 1b: mixed (unchanged; copy_info and DQ6 checkpoints i...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (**writeup/paper.tex**).

CAVEATS

The sample is 16 leader messages in one afternoon. We recovered 31 of 64 corpus rows. The plan coder reaches $\kappa \approx 0.4$ –0.5. The #45 confirmatory script would include the first ~11 min of agent 30 under a wrong model string (DQ6). An amendment is proposed, not applied. The native tests (#26, #35) are not run.

ROUND 2

Amend confirm.g45.py with the DQ6 checkpoint window and both embedding models, then ask Vivian about #45. Run the #26 and #35 leader-copy natives.

- **Essence.** What does fine-tuning transmit, compared with prompting?
- **H23-R1** Style transmits through weights, decisions through context: the leader's influence came from its role prompt and position, not its fine-tune.

H23-leader-distillation-copy · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H23-leader-distillation-copy/README.md · ratings: calibrated by coordinator 2026-10-04 (from H23 agent (suggested)) (page 2)

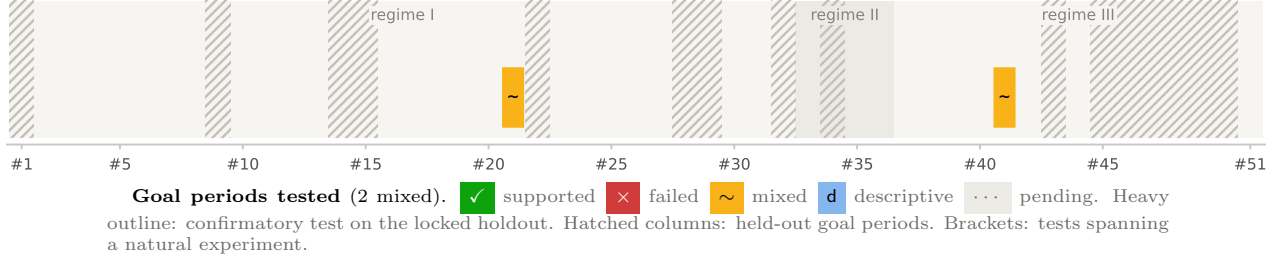
H24: In the forecast week (#21), switching from independent drafts to comparison switches the coupling on

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1.5/5

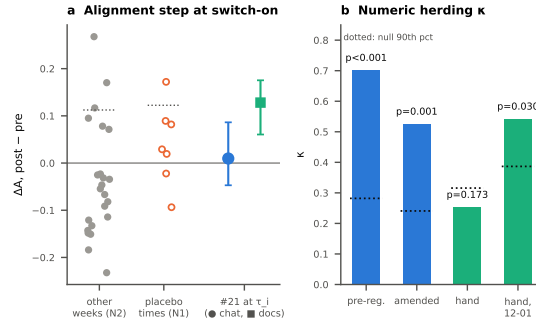
Bottom line. The forecast-week coupling switch moved swarm alignment no more than ordinary kickoff days; coupling is pairwise, about 0.03 in cosine per document read.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Goal period #21 (AI forecasts; regime I; 8–9 agents) serves as a natural experiment. The record dates the first reading of a teammate’s forecast at 42–86 min into the week (median 50 min), not mid-week. There, the share of messages that name a teammate jumps from 7% to 74%. The predicted alignment step is absent: $\Delta A_{\text{res}} = +0.010 [-0.05, +0.09]$, inside a within-week placebo and 23 other kickoff days. Content was aligned before any reading ($A_{\text{res}} \approx 0.33$ vs a rotation null of 0.03). Numeric herding was an extraction artifact (hand-verified $\kappa = 0.25$, $p = 0.17$). **Round 1b** adds a second embedding model, shared and multi-direction goal fields, dedupe and style residuals. The step fails in all 13 configurations. gte sees +0.16, but ordinary kickoff days match it. Natives: reading a teammate’s document pulls the reader toward it by ≈ 0.03 . One kickoff split into two rooms grew two room cultures. The holdout (#14) is not run.



Observables. (a) Change in residual alignment across the switch-on in #21 (blue: statements; green: documents; 90% CIs). Comparators: 23 other regime-I kickoff days at the same offset (gray) and placebo switch times inside #21 (orange). Dotted lines: 90th percentiles. (b) DeGroot pull κ under the pre-registered extractor, the amended extractor and hand-verified values. Dotted: the 90th percentile of the independent-updating null. The herding signal shrinks to noise once extraction errors are removed.

PHYSICAL MODEL

Vector spins (11) with a Curie–Weiss analogue (01). Spin $\mathbf{s}_i$: the whitened, unit-normalized statement embedding of agent i ($n = 32$). Field $h\hat{g}$: the goal and kickoff text. $\mathbf{h}_i$: agent style. Coupling J_0 : a mean-field pull toward the others, switched on at τ_i , the first reading of a teammate’s forecast by agent i . Order parameter A : mean pairwise cosine; A_{res} removes $\hat{g}$. Numeric forecasts x_i (AGI by 2035, SI by 2050, p(doom)) are soft spins with a DeGroot pull κ .

$$\mathbf{s}_i(t) \propto h\hat{g} + \mathbf{h}_i + \boldsymbol{\xi}_i(t) + J_0 \Theta(t - \tau_i) \mathbf{m}_{-i}(t)$$

$$A = \frac{2}{N(N-1)} \sum_{i < j} \hat{\mathbf{v}}_i \cdot \hat{\mathbf{v}}_j, \quad \beta J_0/n = 1 - \frac{1}{1 + (N-1)A_{\text{res}}}$$

$$\Delta x_i = \kappa (\bar{x}_{-i} - x_i) + \epsilon_i, \quad \kappa = 0 \text{ for independent updating}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An “independent drafts first” phase does not protect the diversity of swarm forecasts. Content was aligned before any agent read another ($A_{\text{res}} \approx 0.33$), from the shared prompt, instructions and prior. Measure diversity at kickoff, not after drafting. Comparison then produced talk about divergence and some literal copying of numbers, not an alignment jump. Do not measure herding on regex-extracted numbers: extraction errors make false herding through regression to the mean. One week with $N = 7$ and no ground truth makes this a pilot.

RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b (bge / gte)	Verdict
P1 step > q90 of N1 and N2	+0.010 → -0.011 / +0.160 (N2 q90 × 0.10 / 0.18); 13/13 fail	×
P2 documents $\Delta A_{\text{res}} > 0$	+0.13 → +0.16 / +0.08 (CIs > 0)	✓
P3 $\beta J_0/n$ rises	0.67→0.67 → flat / 0.69→0.76	model-dep.
P4 numeric herding κ 0.25, p 0.17 (no embedding) (audited)		×
P6 ramp $\rho > 0$	0.73 → 0.47 / 0.60	~
Levels vs rotation null	0.33 vs 0.03 → 0.34–0.42 vs ≤0.10	✓
#21 reading DiD (native)	statements +0.036 (p 0.035) / +0.030 (p 0.038); style p 0.08–0.14	✓
#41 two rooms (native)	no gap in first drafts; gap grows to +0.25 / +0.32 by day 4	~

GOAL PERIODS IN DETAIL

G21 ~ 1b: mixed (step fails; pair DiD supported)
G41 ~ mixed (round 1b native; both models)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

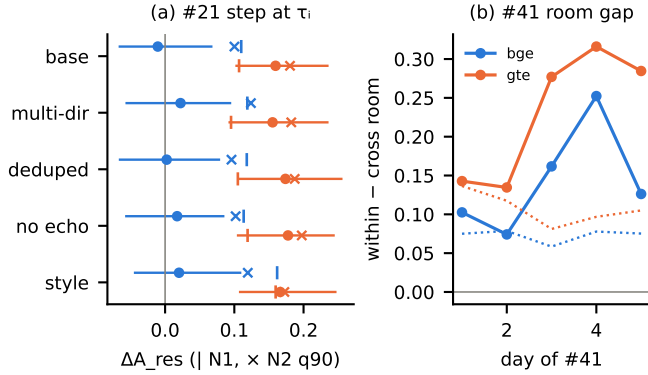
The test bed is one forecasting week with $N = 7$. The switch-on and the reading events rest on the agents' own screen descriptions. The step statistic depends on the embedding model. The pair DiD weakens after style residualization, so part of it may be register. In #41 everyone reads roommates within a minute, so early coupling and room composition are not separable. #41 has only 9 one-statement drafts.

ROUND 2

Measure the copy fidelity of numbers after reading (H07 machinery). Test whether first-posted numbers dominate the consensus (anchoring), per the round-2 redirects.

- **Essence**. Information aggregation in swarms is copying from the first or most confident agent, not averaging.
- **H24-R1** Anchoring cascade: first-posted numbers dominate the final consensus (a founder effect) in every estimation-like task.
- **H24-R2** Copy fidelity vs transformation of shared numbers (H07 machinery).

H24-forecast-coupling-switch · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H24-forecast-coupling-switch/README.md · ratings: RE-C1 re-evaluation agent (suggested) (page 2)



Second figure. (a) Step at the switch-on per input: blue bge, orange gte, bars 90% CI. Markers: | within-week placebo q90, × kickoff-matched q90. gte steps up, but no more than ordinary kickoff days. (b) #41: within-room minus cross-room alignment by day (dotted: permutation q95).

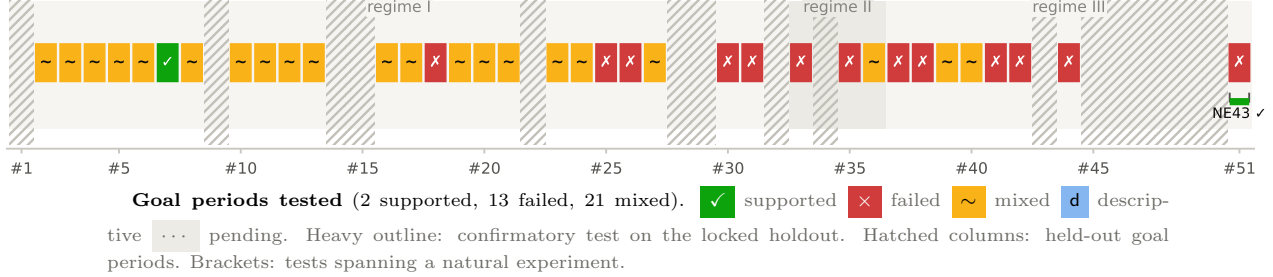
H25: A distance-to-criticality dial, T/T_c , per window and per channel

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

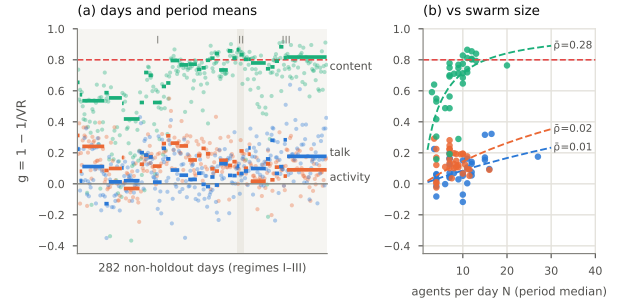
Bottom line. The daily Curie–Weiss dial puts activity and talk far below criticality ($g \approx 0.16$; activity 0.07 trimmed); near-critical content is size times a shared field.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

We build a reusable daily dial: the Curie–Weiss loop gain $g = 1 - 1/VR$ and $T/T_c = 1/g$ per day and channel, computed from logs. It masks stalls, removes slow drives and gives calibrated 90% intervals. We run it on 282 non-holdout days. **Round 1b** repeats it on the corrected activity table (round 1 lost half of all events), with the DQ8 calibrated null. **Activity and talk are subcritical on every day:** median $g = 0.16$ for both (was 0.10 and 0.12), $T/T_c \approx 6$. Trimmed to the common running window, activity drops to 0.07 and beats the independent-agent ceiling on only 24% of days. The dial now reproduces H19 on the same table ($\rho = 0.94, 0.97$) and tracks the Hawkes $\hat{n}$ of H03 ($\rho = 0.46$). **Content still reads near-critical** (0.70; P1 failed): swarm size times a shared-field correlation. Drive withdrawal (NE43) leaves the dial unchanged. Talk co-activation vanishes when 15 agents share one room (NE42). The holdout is not run.



Observables. Round 1 (old table). (a) Daily dial (dots) and period means (bars) for activity (blue), talk (orange) and content (green); red dashes mark $g = 0.8$. Activity and talk stay low; content climbs as rosters grow. On the corrected table, activity and talk shift up by about 0.06 (page 2), and content does not change. (b) Period dial vs swarm size. Dashed curves hold the per-pair correlation fixed. Content and activity follow them (field-like); talk does not (fixed table: talk $\bar{\rho} \propto N^{-1.25}$).

PHYSICAL MODEL

Curie–Weiss (01), Hawkes (09) and $O(n)$ spins (11). Spin $s_i = \pm 1$: agent i active, or posting, in a minute. For content, s_i is the 30-min mean of its whitened unit message embeddings ($n = 32$). $h_i(t)$: private fields (schedule, goal), removed by 30-min block means and agent-day means; stalls are masked. $q_i = 1 - m_i^2$. Order parameter g : loop gain. VR: collective over independent variance. $\bar{\rho}$: mean pairwise correlation. N : agents that day.

$$H = -\frac{J_0}{2N} \left(\sum_i s_i \right)^2 - \sum_i h_i(t) s_i, \quad g = \beta J_0 \bar{q}$$

$$\delta M = \frac{\sum_j q_j \beta \delta h_j}{1 - g}, \quad VR = \frac{\text{Var} \sum_i s_i}{\sum_i \text{Var} s_i} = \frac{1}{1 - g}$$

$$g = 1 - VR^{-1}, \quad T/T_c = 1/g \quad (h = 0)$$

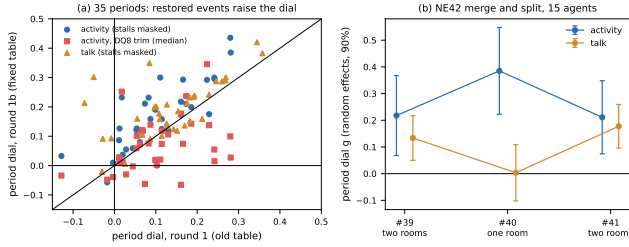
$$VR - 1 = (N - 1) \bar{\rho} \Rightarrow g(N) = \frac{(N - 1) \bar{\rho}}{1 + (N - 1) \bar{\rho}}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Compute the dial from event logs with `compute_dial`. In this village it says the swarm amplifies a nudge by 10–20%: no fast runaway. Trim each day to the common running window before you read activity feedback. Without the trim, about half the activity “signal” is agents that boot together. Track the per-pair correlation $\bar{\rho}$ weekly, not g : at fixed $\bar{\rho}$, g rises with headcount. A green dial does not exclude slow herding through context read minutes later. The talk dial is the credible gauge: it matches an independent Hawkes fit and survives the trim.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (corrected table)	Verdict
P1 every day subcritical, all channels	activity, talk 100% → 100%; content 22.5% → 24%	×
P2 medians activity 0.03–0.20, talk 0.03–0.25	0.096, 0.122 → 0.159, 0.162	✓
P3a reproduces H19 (same table)	talk ρ 0.69 → 0.97; activity 0.94 → 0.94	×
DQ8 null (trim first)	activity days above ceiling 54% new → 24%; talk 61% → 53%	
P4a talk dial vs H03 ($\rho \geq 0.3$)	$\hat{n}$ 0.25 → 0.46 (fast n_x 0.60)	×
P5 day-to-day signal ($\geq 50\%$)	2/28 → 4/29	×
P7a 03-24: activity up, talk down	+0.08/−0.06 +0.16 [0.02, 0.29]/+0.05	→ ×
NE43 drives withdrawn: no step (native)	$ z \leq 1.47$, inside placebo band	✓
NE42 room merge: size-free (native)	talk g 0.13 → 0.003 → 0.18	~



Second figure. (a) Period dials, round 1 vs 1b. Restored events raise activity and talk by about 0.06; the trim (red) pulls activity back down. (b) NE42, 15 agents: talk co-activation vanishes in the one-room week and returns at the split.

GOAL PERIODS IN DETAIL

G02 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
G03 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
G04 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
G05 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
G06 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
G07 ✓ 1b: supported (round 1: mixed; corrected table, same rule)
...
G08 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
G10 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
G11 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
G12 ~ 1b: mixed (round 1: mixed; corrected table, same rule)

+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The 1-min dial is a lower bound, because it cannot see context-mediated coupling. The content dial is an upper bound, because shared fields survive. g depends on N . Synthetic calibration widens the intervals $\times 1.5$. The fixed-table H19 reference comes from the H38 pipeline (same estimator). The #51 size-law test is uninformative, and goals confound NE42.

ROUND 2

Add the DQ8 trim to `confirm.py` Stage B before any hold-out run. Build the lagged dial.

- **Essence.** a trustworthy number for how much the swarm amplifies a push; it must be lag-aware and size-aware, with per-pair correlation as the transferable quantity.
- **H25-R1** A lagged (DC) dial from 1–15 min cross-covariances, validated on the delayed-coupling swarms where the 1-min dial reads 0.
- **H25-R2** Agent-level availability masks from H38's scaffold states (consolidation, day edges, infra errors).
- **H25-R3** Content fields from real anchors (goal and kick-off embeddings, artifact events) plus an embedding swap; coordinate with H26.
- **H25-R4** Test the size law $g(N) = (N-1)\rho/(1+(N-1)\rho)$ at roster changes and on the holdout (CA4).
- **H25-R5** A weekly dial with change-point detection, at least 5 agents per day.

H25-criticality-dial · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H25-criticality-dial/README.md · ratings: H25 round-1b agent (suggested) (page 2)

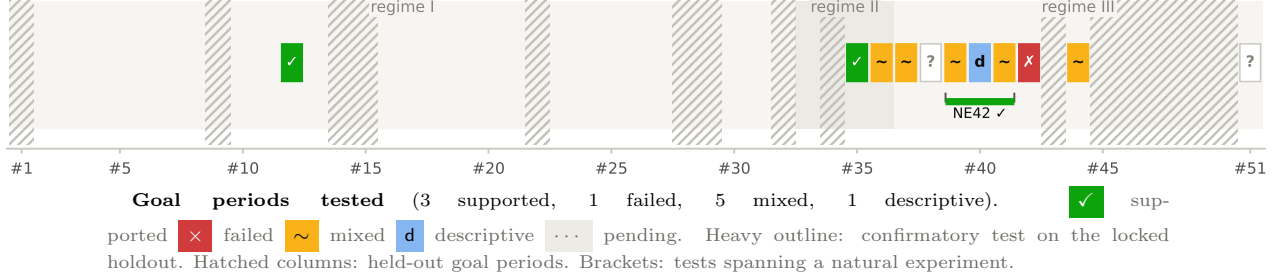
H26: The swarm is near-critical in what it says but subcritical in when it acts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

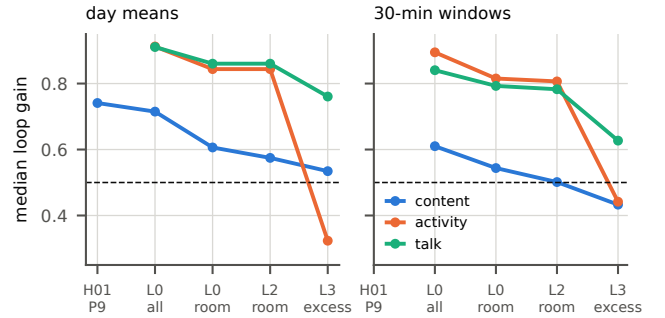
Bottom line. On corrected data, content and talk co-move inside rooms (gain ≈ 0.5) and activity barely does; no channel is near-critical, and H01's 0.74 is mostly drives.

level: hypothesis · pipeline checks 83% · updated 2026-10-04



WHAT HAPPENED

HH108 claimed that the swarm's *content* is near-critical (H01: $\beta J_0/n = 0.74$) while its *activity* is subcritical (loop gains ≤ 0.4). We put content, activity and talk on one estimand, $g = 1 - 1/VR$, at matched resolution (day means, 30-min windows). Field removal is nested and ends in a within-room minus cross-room contrast. We run 10 regime-III two-room units plus #35 and #51. In synthetic worlds, drives alone reproduce H01's number (0.57 from a global drive, 0.74 with more drive types). Round 1: content's room excess is genuine (median 0.53; null beaten in 6/10 units). Activity matches it (0.44 vs 0.43 at 30 min): **failed as posed**. Round 1b (fixed activity table, two embedding models): activity's room gain collapses to 0.12 at 30 min. Its cross-room ρ falls $0.50 \rightarrow 0.10$, the event-drop artifact. Content now exceeds activity (gap +0.41 [0.13, ...], 9/10 units). Talk matches content, and content stays ≈ 0.5 : **unresolved**. NE42: the room excess is channel-borne. The holdout is not run.



Observables. Round 1 (old activity table). Median loop gain over 10 regime-III two-room units, down the field-removal ladder. Steps: H01's estimator, whole-swarm (L0 all), room (L0), drive-projected room (L2), room excess (L3). Content (blue) falls from 0.74 to 0.53 (day) and 0.43 (30 min). On the old table, activity and talk sit above content at every step except day-level activity L3 (unpowered). Round 1b shows that the activity curve was the event-drop artifact (page 2).

PHYSICAL MODEL

Mean-field $O(n)$ soft spins and Curie–Weiss (11, 01). Spins: an agent's 32-d whitened statement mean (content), or its fraction of active or talking minutes. Coupling J acts within a room. Drives: global day and time-of-day, static goal fields, measured operator inputs and unmeasured room drives. C_{ab} : cross-agent covariance of drive-removed deviations. S_a : split-half signal variance. ρ_w, ρ_c : within- and cross-room correlation. N_r : room size. Order parameter: the room-excess loop gain g .

$$\rho_{w,c} = \frac{\sum_{\text{pairs}} \text{tr } C_{ab}}{\sum_{\text{pairs}} \sqrt{S_a S_b}}, \quad \rho_{\text{ex}} = \frac{\rho_w - \rho_c}{1 - \rho_c}$$

$$g = 1 - \frac{1}{1 + (N_r - 1) \rho_{\text{ex}}}, \quad g_{\text{eq}} = J, \quad g_{\text{day}} \rightarrow J(2 - J)$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Compare channels only at the same resolution and drive removal. Time-averaging alone turns an equal-time gain of 0.4 into 0.64, which made part of the HH108 gap. On corrected data, content and talk co-move within rooms (gain 0.4–0.6), and activity barely does (0.12 at 30 min). Treat a swarm-wide activity dip as a platform or schedule event, not as contagion. Operator messages explain none of the room content co-movement. No channel is near-critical: a gain of 0.5 means variance twice the independent level.

RESULTS VS PREDICTIONS

Prediction (locked)	Round 1 → 1b bge / gte	Verdict
P1a drives alone give $H01 \geq 0.3$	0.57 (global); 0.67–0.74 (more drives)	passed
P2 L3 content median 0.25–0.5	0.53 → 0.52 / 0.47; ≥ 0.5 in 6 → 5 / 4 of 10	failed → met
activity L3 (day / 30 min)	0.32 / 0.44 → −0.38 / 0.12 (trimmed 0.16)	artifact
activity cross-room ρ (30 min)	0.50 → 0.10	artifact
P3 content − activity, 30 min	+0.03, 4/10 → +0.41 [0.13, ...] / +0.42, 9/10	failed → held
P3 content − activity, day	+0.32, CI $\ni 0$ → +0.76 / +0.42, CI $\ni 0$	failed
P3 content − talk, 30 min	−0.09 → −0.03 / −0.04	failed
P5 operator inputs $\leq 10\%$	≤ 0.01 in all units, both models	passed
P6 #51: content $\geq$ activity	0/4 → 1/4	failed
Card rule	failed (R2) → unresolved	changed
NE42 pseudo-rooms (native)	content, talk excess < 0 in merged #40, back at split	passed
#12 motion field (native)	on 0.72 vs off 0.66; removal halves it	passed / mixed

GOAL PERIODS IN DETAIL

G12	✓ 1b: supported (bge) / mixed (gte; N2b missed by 0.004)
G35	✓ 1b: supported
G36	~ 1b: mixed (bge) / failed (gte)
G37	~ 1b: mixed
G39	~ 1b: mixed (unchanged)
G40	d 1b: descriptive (unchanged)
G41	~ 1b: mixed (unchanged)
G42	× 1b: failed (unchanged)
G44	~ 1b: mixed (unchanged)
NE42	✓ 1b: supported (content and talk, both models; activity...)
+2 more periods: see the card.	

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		
Level: hypothesis . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

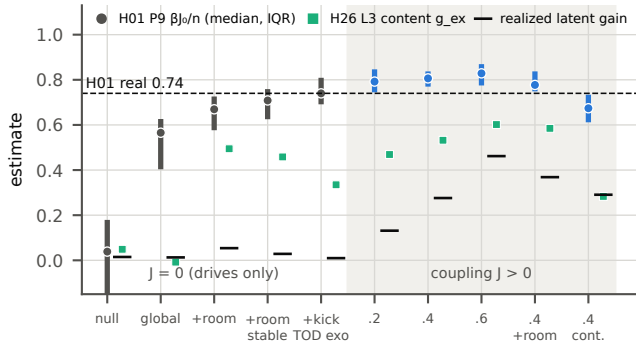
CAVEATS

Unmeasured room drives read as coupling: forks in #35, instructions in #38, the override in #44. Day-level activity is nearly unpowered (3–8 days), and the split-half variance of talk is unreliable. In synthetic worlds the estimator favors content, which strengthens the null. One-room units lack a baseline. Both embedding models agree, and the trimmed activity variant keeps the 30-min gap (+0.22 [0.01, ...]). Goals confound NE42, no cascade sizes exist yet (H34), and #51 confounds the post hoc size scaling.

ROUND 2

- **Essence.** whether ideas spread through a swarm more easily than effort does, measured so that channels are comparable. Round 1 says: not at matched resolution; the channels differ in where they co-move, not in how much.
- **H26-R1** Separate room coupling from room drive with the merge and split natural experiments.
- **H26-R2** Lagged, exposure-conditioned content gain.
- **H26-R3** Idea-cascade sizes (HH122) from H34.
- **H26-R4** Instrument checks: embedding swap, style control, H38 outages.

H26-content-near-critical · generated 2026-10-07 17:50 by in-fra/summaries/build-summaries.py · card hypotheses/H26-content-near-critical/README.md · ratings: H26 agent (suggested) (page 2)



Second figure. Synthetic validation: 24 worlds each, $N = 14$ in two rooms, 5 days. At zero coupling, H01’s P9 (gray) already reads 0.57–0.74 from drives and barely moves with J . H26’s L3 (green) tracks the realized latent gain (black) under global drives. It reads unmeasured room drives (“+room”) and imperfectly measured operator pulses as coupling.

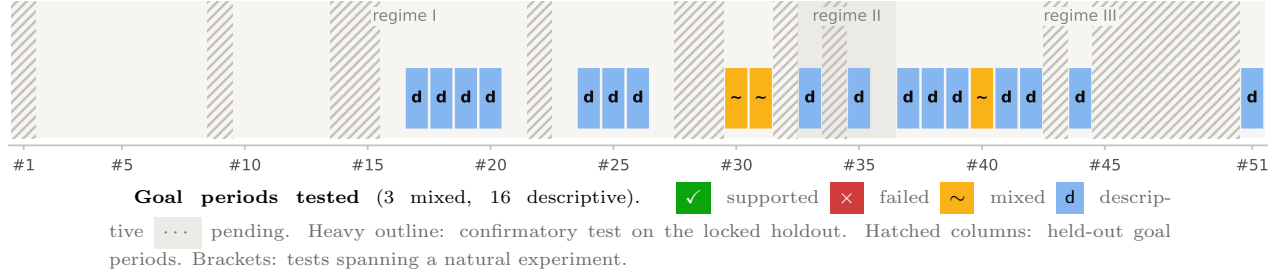
H27: Critical slowing down warns of herding waves hours ahead

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2/5

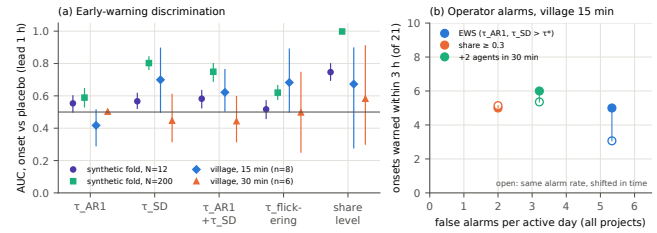
Bottom line. Critical slowing down does not warn of village herding: pile-ons in attention and commits often follow a link within minutes, and $N \approx 12$ hides slowing.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

A pre-registered rule on project labels defines a herding onset. A project's share of labeled agents jumps from ≤ 0.25 to ≥ 0.5 within an hour and stays there. It finds 21 onsets in 14 non-holdout weeks at 15-min resolution, mostly in shared-artifact weeks. We compare early-warning trends (rolling lag-1 autocorrelation and SD over the previous 6 h) with matched placebo segments; 8 onsets are evaluable. The composite separates them weakly at 15 min (AUC 0.62) and not at 30 min. Autocorrelation never rises (AUC 0.41), so this is not critical slowing down. **Round 1b** (deterministic labels, work ledger) reproduces all of it: the same onsets and the same AUC. The frozen alarm now beats a time-shifted copy at $p = 0.038$ (was 0.07). A naive share alarm matches its 5/21 hits with fewer alarms. Work-commit onsets coincide with attention onsets. The #40 pile-on ran in minutes after a link. The holdout is not run.



Observables. Round 1. (a) AUC of each early-warning trend, onset vs placebo segments, at 1 h lead. Series: a synthetic fold at village size (purple), a clean $N = 200$ fold (green), and the village at 15 min (blue, 8 onsets) and 30 min (orange, 6). Autocorrelation never discriminates. Variance does only at 15 min, with wide CIs. (b) Onsets warned within 3 h vs false alarms per active day. Open circles: the same alarms shifted in time. In round 1, no alarm beats its shifted copy.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not build a critical-slowness dashboard for a 10–15-agent LLM swarm: rising autocorrelation and variance do not forecast its pile-ons. A pile-on is a fast push: a link in chat, then half the room within minutes. In #40, half the room's attention took 4 min and half its commits 19 min. A 6-hour trend window cannot see that, and at $N \approx 12$ a true fold is barely detectable. Watch announcements and project shares instead, and plan responses in minutes, not hours. Test any alarm against a rate-matched time shift: here p moves from 0.07 to 0.038 between data versions.

PHYSICAL MODEL

Kinetic mean-field Potts (10) with mean-field susceptibility (01). Spin σ_i : the project of agent i in a window (0 = rare projects, uncoupled). $k_a^{(-i)}$: other agents on a . $h_a(t)$: a slowly drifting popularity field. βJ : uniform herding coupling. γ : reconsideration rate. Order parameter x_a : the share on a . Near a fold of x_A the leading eigenvalue $\lambda \rightarrow 0$, so the lag-1 autocorrelation ρ_1 and the variance rise (critical slowing down). Rivals (field steps, noise-induced flips) give no warning.

$$P(\sigma_i \rightarrow a) \propto e^{h_a(t) + \beta J k_a^{(-i)} / (N-1)} \quad (a \geq 1), \quad \propto 1 \quad (a = 0)$$

$$\dot{x}_a = \gamma [\text{softmax}_a(h(t) + \beta J x) - x_a]$$

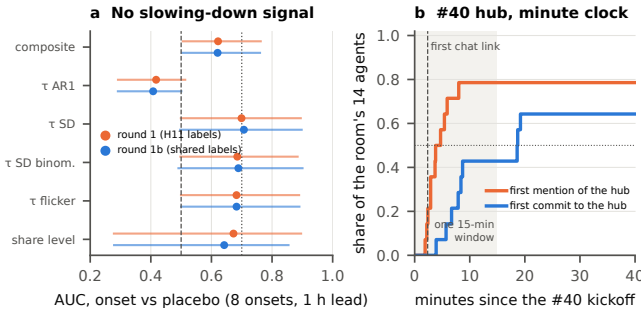
$$\rho_1 = e^{-\gamma |\lambda| \Delta t} \rightarrow 1, \quad \text{Var } x_A \propto 1/(N|\lambda|) \quad (\lambda \rightarrow 0)$$

$$\text{alarm: } \tau_K(\hat{\rho}_1) > \tau^* \wedge \tau_K(\hat{\sigma}) > \tau^*, \quad \tau^* = 0.54$$

H27-herding-early-warning · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H27-herding-early-warning/README.md · ratings: RE-P1 round-1b agent (suggested; previous calibrated ratings 40 / 1.0 / 2.0)

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P0: 10–25 onsets	21 → 21; 8 evaluable	yes
P1: $AUC \geq 0.70$	0.62 → 0.62 [0.50, 0.76]; 30 min 0.44 → 0.50	mixed
P2: $SD > AR1$	0.70/0.42 → 0.71/0.41	partly
P3: no operator value	5/21 hits; shift p 0.07 → 0.038; level alarm 5/21	mostly
P4: synthetic	$N=200$: 0.75; $N=12$: 0.58	2 of 4
R1b-1 replication	same onsets, same verdicts	yes
R1b-2 work on-sets	8 vs 14; lag 0 (3) or +1 (1) window	mostly
R1b-3 links	11/21; 9 to known projects, 2 first links	yes
#40 minute clock	t_{50} 1.5 min after link; work 15 min later	2 of 3



Second figure. (a) AUC of each trend, onset vs placebo, at 1 h lead: round 1 (orange) vs round 1b shared labels (blue). Nothing moves; autocorrelation stays below 0.5. (b) #40 at the minute clock: room share with a first hub mention (orange) or first hub commit (blue) after the kickoff. Half the room joins within one 15-min window (shaded).

GOAL PERIODS IN DETAIL

G17 d 1b: descriptive (unchanged)
 G18 d 1b: descriptive (unchanged)
 G19 d 1b: descriptive (unchanged)
 G20 d 1b: descriptive (unchanged)
 G24 d 1b: descriptive (unchanged)
 G25 d 1b: descriptive (unchanged)
 G26 d 1b: descriptive (unchanged)
 G30 ~ 1b: mixed (AUC 0.55, 2/4 hit; unchanged)
 G31 ~ 1b: mixed (native: work onsets coincide with attention)
 G33 d 1b: descriptive (unchanged)
 +9 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

There are 21 onsets, 8 evaluable and 4 in work, so per-week AUCs rest on 1–2 onsets. We tried many indicators, leads, window lengths and two onset rules; only the 15-min composite at 1 h was primary. The alarm-vs-shift p crosses 0.05 between null draws (0.038–0.07): treat it as borderline. The step rule misses slow rises; the slow-rise variant gives 0.74 at 15 min. The link association is post hoc, and most linked onsets are bursts of known projects (H28). Work labels are public agent commits (dense from #30), and #51 supplies most false alarms.

ROUND 2

- **Essence.** Can an operator see a pile-on coming early enough to act, and from what signal?
- **H27-R1** Model onsets as announcement-triggered cascades: hazard of a majority pile-on after a link post, by poster, room size and the project's prior share; joint with H28.
- **H27-R2** Use a finer clock: per-turn project states from actions (minutes, not 15-min windows) so 6 h of history is not needed and the first 1.5 days become evaluable.
- **H27-R3** Test the one regime where the theory says warnings can work: long, large swarms with slowly drifting incentives (#51 roles, if any shared project ever rises), with the slow-rise onset rule.

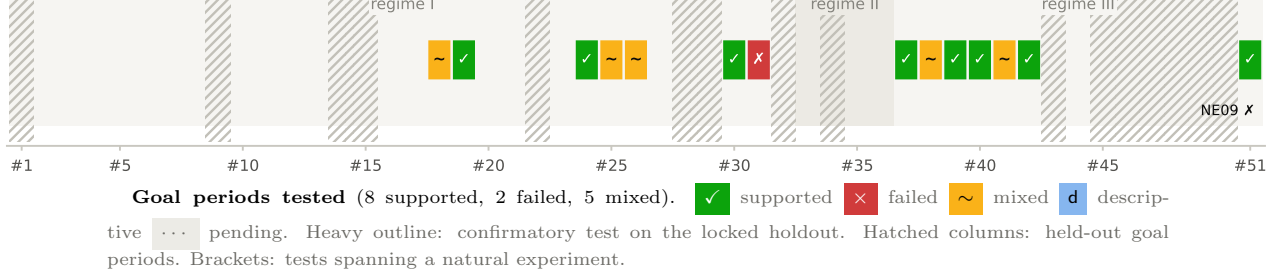
H28: Links are the contagion vector of herding

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04 (round 1b)

Complete 52% Faithfulness 1.5/5 Usefulness 2.5/5

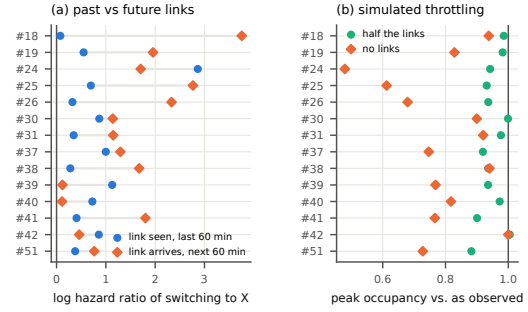
Bottom line. Links co-move with project switches (HR ≈ 2.6) but do not lead them in herding weeks; they mark bursts, and removing them trims peaks by $\leq 30\%$.

level: hypothesis · pipeline checks 86% · updated 2026-10-04 (round 1b)



WHAT HAPPENED

In 14 non-holdout goal periods, a 5-min hazard model (agent, project, day effects) tests whether a seen chat link to project X raises switches to X. A seen link goes with a $2.3\times$ switch hazard (pooled; $p < 0.05$ in 13/14). It beats links shifted by ± 30 –120 min in only 3/11 herding weeks: the primary test fails. Future links predict switches better than past ones in 10/11. Naive agents already switch at $3.3\times$ baseline in the hour before their first exposure. Links lead visits only in own-artifact weeks, where nobody herds. Round 1b uses context-ledger visibility and work commits. The pre-NE09 weeks were mismeasured, so 5/11 herding weeks now beat the shift (primary test mixed; pooled $2.6\times$). Future links still win (9/11). Recipients switch at $6.5\times$ baseline in the ~ 17 s before they can read the link. NE09 was not a visibility change. Work switches show the same pattern. The holdout is not run.



Observables. Round 1. (a) Switch-hazard coefficient for a link seen in the last hour (blue) and for a link that arrives in the next hour (orange), fitted jointly. In every herding week except #24, the future link predicts as well or better. Only the own-artifact weeks (#39, #40, #42) show past $>$ future. (b) Simulated peak number of agents on one project with half (green) or all (orange) links removed: median $\times 0.79$ with no links, an upper bound.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Throttling link-sharing does not stop pile-ons. Removing every link trims simulated peak pile-ons by at most about 30% (median $\times 0.71$ in round 1b; an upper bound). In herding weeks, links, chat and work on a project rise together: the link marks a burst already under way. To prevent duplicated work, assign ownership or act on the burst triggers, not on links. Use the lead/lag test as a diagnostic. Lag $\gg$ lead means directed recruitment (agents advertising their own artifacts, $\lambda \approx 0.01$ –0.02 per exposure); lead $\geq$ lag means markers.

PHYSICAL MODEL

Multi-label kinetic Potts (10) with a contagion field (03). An off-project agent i switches into X at rate μ_{iX} . $\alpha_i, \pi_X, \delta_d$: agent, project and day fields. E_{iX} : a link to X by another agent became visible to i in the last 60 min. κ : link coupling. $n_{-i,X}$: others on X (Potts coupling J). $\mathbf{h}$: own history, plans, pulses. λ : extra switches per exposure. R_{link} : share of switches attributable to links. π : links per switch. k_s : susceptible recipients per link.

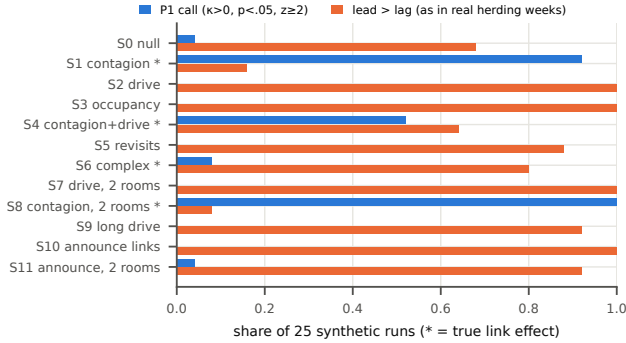
$$\log \mu_{iX}(t) = \alpha_i + \pi_X + \delta_d + \kappa E_{iX}(t) + J \log(1 + n_{-i,X}(t)) + \boldsymbol{\eta} \cdot \mathbf{h}_{iX}(t)$$

$$\lambda = \frac{\sum (\hat{\mu} - \hat{\mu}|_{E=0})}{N_{\text{exp}}}, \quad R_{\text{link}} = \pi k_s \lambda = \frac{\sum (\hat{\mu} - \hat{\mu}|_{E=0})}{\sum y}$$

H28-links-spread-herding · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H28-links-spread-herding/README.md · ratings: H28 agent (suggested); round 1b RE-D1 (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed (11 herding weeks)	Verdict
P1 $\kappa > 0$, $p < .05$, beats link shift	3/11 (#24, #30, #51); pooled $e^\kappa = 2.3$	failed
P2a past > future links	future > past in 10/11; pooled -1.17 ± 0.30	reversed
P2b survives momentum	3/3	yes
P3 naive agents; no pre-trend	naive $\kappa = 1.31$; pre-trend $3.3\times$	half
P4 simple contagion	1 link: HR 2.3; ≥ 2 senders add -0.24	yes
P5/P6 λ , $R_{\text{link}} < 0.5$	λ median 0.036; R_{link} 0.06–0.44	yes (bounds)
P7 no links: peaks –10–30%	median $\times 0.79$; calibration 9/14	mixed
P8 NE09 slows response	pre-NE09 as fast; 1b: premise void (17 vs 18 s delay)	failed
1b P1 on ledger visibility	5/11; pooled $e^\kappa = 2.6$; future > past -1.04 ± 0.24 (9/11)	mixed
1b blind window (before reading)	$6.5\times$ baseline vs $3.5\times$ after reading	co-burst
1b #31 wave in work commits	$\kappa + 0.57$, shift $z + 0.2$, future > past	failed
P9 own-artifact λ lower	0.012–0.019 vs 0.036; past > future	yes
P10 occupancy $J > 0$	3/11 under both FE	failed



Second figure. Synthetic swarms at village size, 25 runs each. Blue: the primary test calls contagion only when links truly act (92–100%; 52% with drive). Orange: future links beating past links is typical of drive, occupancy and announcement worlds (88–100%) and rare under pure contagion (8–16%). Real herding weeks show it in 10/11 (round 1).

GOAL PERIODS IN DETAIL

G18 $\sim$ 1b: mixed (ledger visibility: $\kappa + 1.48 \pm 0.43$, z_shift...
G19 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 1.26 \pm 0.26$, z_s...
G24 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 2.90 \pm 0.63$, z_s...
G25 $\sim$ 1b: mixed (ledger visibility: $\kappa + 1.17 \pm 0.43$, z_shift...
G26 $\sim$ 1b: mixed (ledger visibility: $\kappa + 1.07 \pm 0.42$, z_shift...
G30 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 0.88 \pm 0.19$, z_s...
G31 $\times$ 1b: failed (native: work switches $\kappa + 0.57 \pm 0.24$ but ...
G37 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 1.60 \pm 0.33$, z_s...
G38 $\sim$ 1b: mixed (ledger visibility: $\kappa + 0.76 \pm 0.20$, z_shift...
G39 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 1.11 \pm 0.21$, z_s...
+5 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Touche measure attention (strict artifact mentions); round 1b adds work commits in git-dense weeks only. Blind windows on the ledger are short: 114 blind switches in all. λ and R_{link} are attributable associations, so they bound any causal link effect from above. The simulator has ± 0.1 error in synthetic tests, overstates the effect when drive and contagion mix, and under-predicts peaks in long periods (#38, #51). The other-room placebo is uninformative, because rooms work on different projects. On shared window labels no period beats the shift null; this check, the short-lag check, label robustness and the H27 link are post hoc.

ROUND 2

Run the frozen test on #22, #28, #45. Model what starts attention bursts (human messages, kickoffs, announcements), jointly with the H27 onsets. Split links by sender type and by announcement vs request (Phase 2 text).

No round-2 redirects yet.

H28-links-spread-herding · generated 2026-10-07 17:50 by in-fra/summaries/build-summaries.py · card hypotheses/H28-links-spread-herding/README.md · ratings: H28 agent (suggested); round 1b RE-D1 (suggested) (page 2)

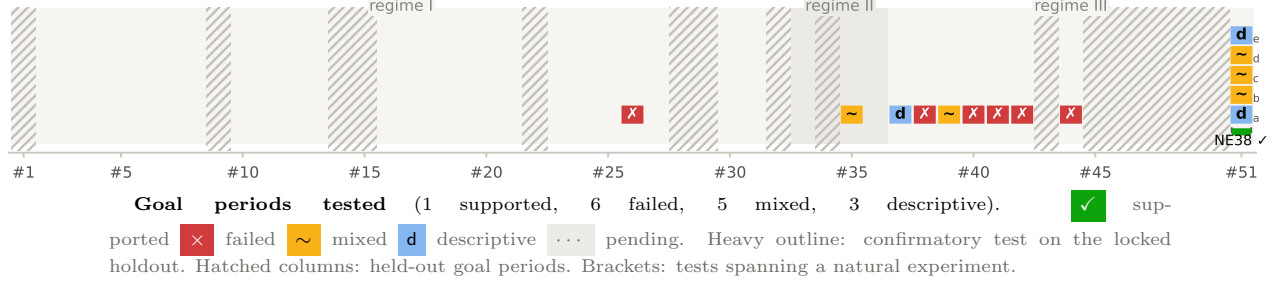
H29: Driver nodes: where an operator message moves the whole swarm

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. Content influence is real and address-gated (named $\gg$ unnamed on the context ledger); the content-pull driver ranking fails, but a reply-graph ranking weakly predicts spread.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Per goal period, a content-influence network measures how far agent j 's next statement moves toward a seen message from i , net of placebos. We rank agents by linear controllability in 9 counted and 3 short non-holdout units (regime III, #51). **Round 1b** repeats everything on context-ledger visibility, with a second embedding model and the DQ2 reply graph. The pre-registered test still fails in the opposite direction ($\kappa < 0$ in 7/9). Messages posted during the recipient's own call are co-responses, not leaked influence. At matched age, influence is real and address-gated. Named recipients move 0.04–0.08 of the gap per message in every #51 segment, unnamed ones 0.00–0.013. The content-pull driver ranking is still noise. A reply-graph ranking (net reply current) is reproducible (split-half 0.65) and weakly predicts held-out spread (pooled $\rho = 0.23$ bge, 0.12 gte). In NE38, an operator's re-assignment of one agent moved it and did not relay. The holdout is not run.

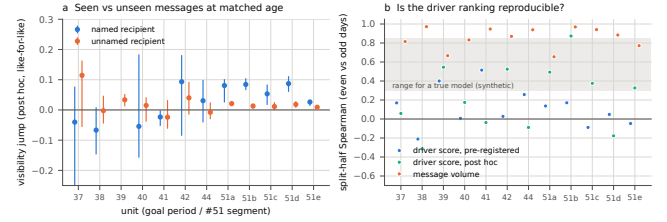
PHYSICAL MODEL

Linear response (02) with vector content states (11): DeGroot / Friedkin–Johnsen dynamics. x_j : the whitened state-ment embedding of agent j . $V(\tau)$: messages that reached j between its previous and current model calls. a_{ij} : pull per message. $A_{ij} = a_{ij}r_{ij}$, with r_{ij} the messages from i absorbed by j per hour. Γ : relaxation toward the agent's own field. Driver score D_k : reachability of the swarm mean from agent k over $T = 4$ h. $E_k = N^2/D_k$: energy to shift the mean through k .

$$x_j(\tau_n) = x_j(\tau_{n-1}) + \sum_{m \in V(\tau_n)} a_{s(m)j} [x_m - x_j(\tau_{n-1})] + \gamma_j [h_j - x_j] + \xi$$

$$\dot{\mathbf{x}} = \mathbf{Q}\mathbf{x} + \mathbf{e}_k u, \quad Q_{ji} = A_{ij}, \quad Q_{jj} = -\sum_i A_{ij} - \Gamma$$

$$D_k = \int_0^T (\mathbf{1}^\top e^{Q^t} \mathbf{e}_k)^2 dt, \quad \mathbf{1}^\top \mathbf{Q} \mathbf{e}_k = \text{out}_k - \text{in}_k - \Gamma$$



Observables. (a) Post hoc visibility jump at matched age: seen minus unseen messages within 30 s of the reply, like-for-like. Named recipients (blue) vs unnamed (orange). Named $\gg$ unnamed in every #51 segment; the two-room weeks are underpowered. (b) Split-half reproducibility of rankings. Message volume is always reproducible; the driver score only sometimes (grey band: what a true model gives on synthetic data).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To move one agent, name it: named messages pull 5–25 $\times$ more than unnamed ones in #51, and broadcast content barely propagates. An operator message to one agent stays with that agent (NE38). If you must pick a relay, rank agents by net reply current (replies received minus given), not by volume or content-pull “driver” scores. Expect a weak edge (held-out $\rho \approx 0.1$ –0.2). A designated leader gets $\times 1.39$ replies per message, not more reach (#35). Network controllability adds nothing measurable; the steering-cost scaling ($\propto N^{2.7}$ for a typical agent) is model-only.

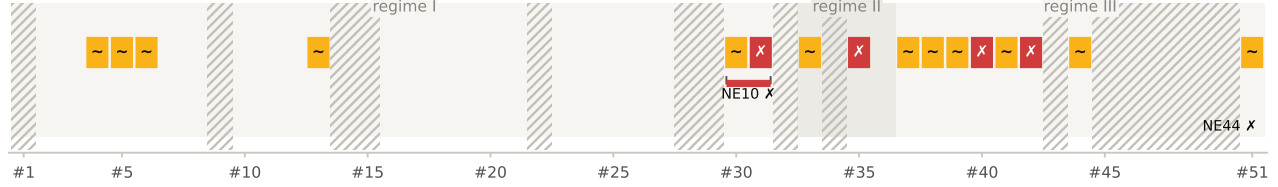
H30: An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Operator messages have a period-level susceptibility (about one active minute per nudge; a small content pull), but daily steerability is not a trackable state.

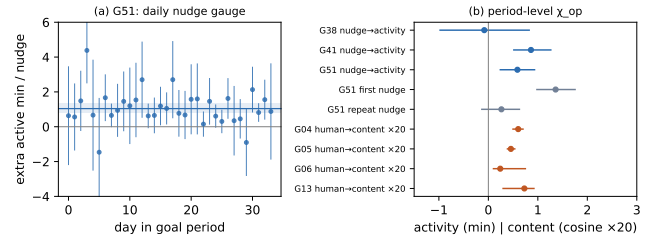
level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (6 failed, 12 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending.
Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

We measure how far one operator message moves the agents who see it, and whether that number is trackable by day. We test 16 non-holdout goal periods, after a synthetic validation that fixed three estimator biases. **Round 1b** uses the corrected activity table, because round 1 lost half of all events. It adds the leading-@ nudge target, receiving-call timing and past-only controls with day effects. *Activity*: a nudge adds about one active minute for its target (#51 0.98 [0.63, 1.37], #38 0.95 [0.38, 1.54]; round 1: 0.59, -0.08). First nudges give 1.21, repeats 0.44, room-mates ≈ 0 . The round-1 lull warning reverses (lull 1.74 vs 0.72). *Content*: human messages turn the next statements toward their new content, +0.024 (room-mates) and +0.041 (named), in both embedding models. *Daily gauge*: #51 varies by day ($p = 0.026$, $R_1 = 0.48$) without persistence. NE44: nudges buy more under the 5-min pause default (reversed). The holdout is not run.



Observables. (a) #51 daily nudge gauge: extra active minutes per nudge for the target (day FE, 95% CIs), with the period mean and band. Day-to-day scatter matches the daily CIs: no trend, no persistence. (b) Period-level χ_{op} . Nudge $\rightarrow$ activity in the three largest nudge periods (pre-registered model); first vs repeat nudges in #51 (day FE); human $\rightarrow$ content in human-dense periods (cosine $\times 20$).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not watch a daily “steerability” number: at village message rates it is a period constant plus noise, and it does not age. One nudge to an idle agent buys about one extra active minute from its receiving call. A first nudge buys 1.2 min and a repeat 0.4 (CI includes 0). Nudging into a swarm-wide lull works even better (1.74 vs 0.72 min), and short scaffold pauses help (NE44). To change *what* an agent writes, name it and add new content (+0.041 vs +0.024 for room-mates). For analysts: do not condition on future nudges, as H04’s isolation rule did.

PHYSICAL MODEL

Linear response to impulsive fields (01, 11). Each operator message k is a field pulse on its recipients i . Spins: activity $n_i(t) \in \{0, 1\}$; content s_i , a whitened statement embedding. $u_k^\perp$: the message direction projected off the recipient’s last 8 statements. χ_{act} : the 30-min integrated activity response (local projection; strata α_s , day effects δ_d , kick counts K). χ_{con} : post-message alignment minus same-kind placebos p . $\chi_{\text{op}}(d)$: the day mean of per-kick responses r_k .

$$Y_i(m) \equiv \sum_{\tau=1}^{30} n_i(m + \tau) = \alpha_{s(i,m)} + \delta_d + \sum_c \chi_{\text{act}}^c K_i^c(m) + \beta \cdot K_i^{\text{adj}}(m) + \varepsilon$$

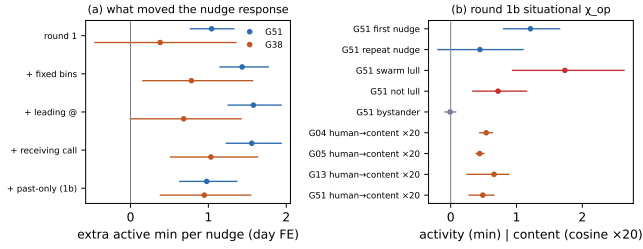
$$\chi_{\text{con},k} = \bar{s}_i^{\text{post}} \cdot u_k^\perp - \langle \bar{s}_i^{\text{post}} \cdot u_p^\perp \rangle_p, \quad u^\perp = \frac{(1 - P_{\text{pre}})u}{\|(1 - P_{\text{pre}})u\|}$$

$$\chi_{\text{op}}(d) = \frac{\sum_{k \in d} w_k r_k}{\sum_{k \in d} w_k}, \quad R_1 = 1 - \frac{\text{Var}_{\text{perm}}[\chi_{\text{op}}(d)]}{\text{Var}[\chi_{\text{op}}(d)]}$$

H30-operator-susceptibility · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H30-operator-susceptibility/README.md · ratings: H30 round-1b agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict
P1 nudge→target	#51 0.59 → 0.98 [0.63, 1.37]; #38 -0.08 → 0.95 [0.38, 1.54]; small periods 5/10 > 0	✓ (count rule ×)
P2 bystanders ≈ 0	#51 0.13 → -0.01; #38 -0.54 → -0.09	✓
P3 human→activity	≤ 0 / n.s. in regime I (unchanged)	×
P4 human→content	0.026 → 0.024 (gte 0.024); named 0.046 → 0.041	✓
P5 nudge→content	#51 0.003 → 0.002 [-0.001, 0.005]	✓
P6 no daily state	#51 p 0.08 → 0.026, R_1 0.28 → 0.48, lag-1 0.08	× as written
P7 no aging	#51 +0.02 [-0.03, 0.07]/day	✓
P8 context saw-tooth	pull falls 0.66 vs late 1.49	×
P10 family	no Holm-significant lab	✓
P11 nulls	#51 and #38 now pass	✓
Native #5 dose	#51 +0.02 → 0.013 over 1–7 msg; 8+ noisy	× (rule)
Native NE44	12-h 0.57 vs 5-min 0.98 (Δ -0.41)	× reversed
Native NE10	first nudges ever -0.42 (22 kicks)	× low power



Second figure. Round 1b. (a) What moved the nudge response in #51 (blue) and #38 (orange). Restoring the dropped events carries most of the change; past-only controls bring the level back to ≈ 1 min. (b) Situational χ_{op} in #51 (first vs repeat nudges, swarm lull vs not, bystanders) and human content pull ($\times 20$).

GOAL PERIODS IN DETAIL

G04 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G05 ~ 1b: mixed (r1 mixed) · native dilution: failed
 G06 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G13 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G30 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G31 × 1b: failed (r1 mixed; fixed bins, receiving call)
 G33 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G35 × 1b: failed (r1 failed; fixed bins, receiving call)
 G37 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G38 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 +8 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only #51 and #38 resolve the nudge response; 3–5-day periods give unstable CIs. The level depends on the future-kick adjustment: 0.98 past-only vs 1.56 with the A1 regressors of round 1 (day FE). SE-based window reliabilities exceed 0.5 in #51 but also in #38, where the calibrated permutation finds no day signal, so we do not count them. Human effects on activity in dense regime-I chat are not identified. Content measures message-specific steering only, and human replies may share topics. The round-1b design (receiving-call timing, past-only, day FE primary) follows the re-evaluation brief, not the pre-registration.

ROUND 2

- **Essence.** a trustworthy answer to "will my next message change what this swarm does?". Round 1 says it is per situation, not a daily state.
- **H30-R1** Situational χ instead of daily χ .
- **H30-R2** The 2026-08-20 switch-off as a natural experiment.
- **H30-R3** Embedding swap and reply control for χ_{con} .
- **H30-R4** Per-message content strength.
- **H30-R5** Run `confirm.py` after the card is committed.

H30-operator-susceptibility · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H30-operator-susceptibility/README.md · ratings: H30 round-1b agent (suggested) (page 2)

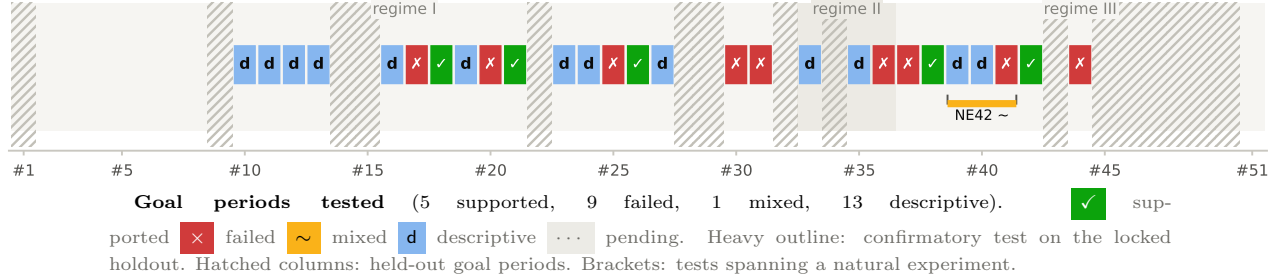
H31: Consensus time scales with the interaction graph's spectral gap

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2.5/5

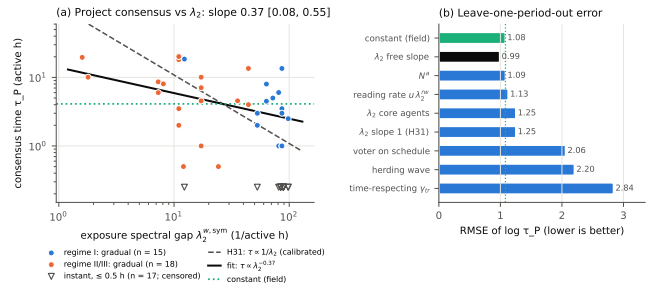
Bottom line. Consensus does not follow $1/\lambda_2$: projects take a constant ≈ 4.5 active h, half of all consensus forms at kickoff or in one window, and votes settle at read-out speed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H31 tests whether consensus time scales as $1/\lambda_2$, the inverse spectral gap of the exposure graph. The test uses the non-holdout periods #10–#44. A rule, fixed before any outcome, defines three event classes: project-share majorities (63 events, 20 periods), the #26 votes, and content convergence. Gradual project consensus takes a median of 4.5 active h. It scales sub-linearly, $\tau \propto \lambda_2^{-0.38}$ [0.05, 0.60], not as $1/\lambda_2$. Out of sample, the slope-1 law loses to a constant; the synthetic power is 0.13. About half of all consensus is frozen at the kickoff or arrives in one-window waves. Content never converges. **Round 1b** uses context-ledger visibility, deterministic labels, the DQ4 work ledger and DQ6 ballots. It changes no card verdict. The sub-linear lead weakens: its CI touches 0 under controls. Work commits reach majorities on the same repos at least as fast (2.0 vs 4.6 h). On the correct #26 rounds, the votes settle in 57 s and 39 s. That is two to three read-out cycles, 70–100× faster than the λ_2 forecast.



Observables. (a) Gradual project-consensus times against the exposure spectral gap. Dashed: the calibrated slope-1 law. Solid: the free fit. Dotted: the constant forecast. Triangles: instant one-window events ($\tau \leq 0.5$ h, censored). (b) Leave-one-period-out error of each forecast rule (round 1). Round 1b moves every number by ≤ 0.1 , and the reading-rate predictor $u\lambda_2^w$ beats the constant by 2.4%. The voter, herding-wave and time-respecting models miss by an order of magnitude.

PHYSICAL MODEL

The state x_i of agent i is its project (shared labels from artifact mentions, or work commits) or its whitened statement embedding. The coupling W_{ij} counts agent j 's messages first seen by i per active hour (round 1b: context-ledger visibility). L is the symmetrized Laplacian; its spectral gap λ_2 is the control parameter. The model is model 10's voter and Potts coarsening plus graph diffusion (HH68). Four rivals run on the same reading schedule: DeGroot averaging (time-respecting gap γ_{tr}), the voter model ($\tau \propto N/u$, u = reading rate), a herding wave, and a field (kick-locked timing).

$$L = D - \frac{1}{2}(W + W^\top), \quad 0 = \mu_1 < \lambda_2 \leq \mu_3 \leq \dots$$

$$\dot{x}_i = \sum_j W_{ij} (x_j - x_i) \Rightarrow \tau \propto 1/\lambda_2 \quad (\text{H31}), \quad \tau_{\text{voter}} \propto N/u$$

$$\log \tau_e = c + b \log(1/\lambda_2) + \epsilon_e : \quad \hat{b} = 0.38 [0.05, 0.60] \quad \text{vs.} \quad b = 1$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The textbook law, consensus time $\propto 1/\lambda_2$, gives an operator no usable forecast. In broadcast rooms, λ_2 mostly measures message volume. Consensus comes in three kinds: kickoff-set, one-window waves and hours-long. For a project, expect a majority about 4.5 active h after the project first appears (80%: 1–13.5 h). Expect the work commits to get there as fast as the chat. For a deadline-bound vote, expect consensus within two or three read-out cycles of the call to vote, i.e. tens of seconds. Size vote windows to the read-out delay, not to the social graph. More talk does not buy agreement in content: content alignment never converges and decays after the kickoff.

H31-consensus-time-spectral-gap · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H31-consensus-time-spectral-gap/README.md · ratings: RE-P1 round-1b agent (suggested; previous calibrated ratings 40 / 1.0 / 2.0)

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P1 slope 1; beats constant	0.37 → 0.38 [0.05, 0.60]; RMSE 1.24 vs 1.08	failed
P3 kick-locked	0.41 vs 0.25, all kickoff-frozen	trivial
P4 $\geq 60\%$ abrupt	13% → 17%	failed
P5 content slope	no convergence (12 diverge)	untestable
P6 room sign as D	3/4 → 2/3	weak
P8 power ≥ 0.5	0.13	failed
P10 vote ≤ 1 h	12.5 h (wrong vote) → 57 s runoff, 39 s confirmatory	failed → supported
R1b-2 work slower, fewer frozen	19 vs 23 events; 2.0 vs 4.6 h; frozen 37% vs 39%	mostly failed
#26 $\tau_V \approx \tau_{read}(3)$	1.7 $\times$, 2.9 $\times$	partly
NE42 λ_2 A-B-A	up 2.4 $\times$ at merge, not down at split	failed

GOAL PERIODS IN DETAIL

G10 d 1b: descriptive (unchanged)
G11 d 1b: descriptive (unchanged)
G12 d 1b: descriptive (unchanged)
G13 d 1b: descriptive (unchanged)
G16 d 1b: descriptive (unchanged)
G17 $\times$ 1b: failed (unchanged)
G18 $\checkmark$ 1b: supported (unchanged)
G19 d 1b: descriptive (was supported; label renumbering)
G20 $\times$ 1b: failed (unchanged)
G21 $\checkmark$ 1b: supported (was failed; label renumbering)
+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

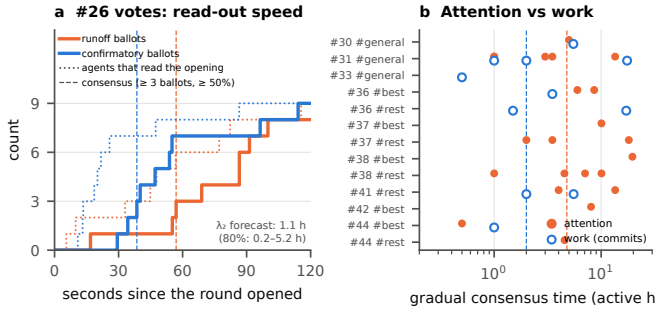
The test has 31 gradual events from 13 periods and one primary test, which fails. The sub-linear slope ($p = 0.07$) loses significance under a regime effect, at the block level and without the pre-NE09 periods. λ_2 , message volume, N and regime co-vary. The frozen/instant/gradual split and the structure decomposition are post hoc. The leave-one-period-out winner flips between the constant and $u\lambda_2^{rw}$ on 2.4% margins; the frozen holdout rule does not change. #26 has three rounds. NE42 is goal-confounded and does not reverse. Work consensus uses public commits, which are dense from #30 on. Content uses one embedding model.

ROUND 2

Run the holdout (#29, #46, #47, #49, #50) after sign-off. Test uneven exposure at the agent level: measure whether poorly connected agents adopt later, in attention and in commits. Apply a read-out-gated vote model (H53's clock) to every deadline-bound call, and check it on the #12 verdicts.

No round-2 redirects yet.

H31-consensus-time-spectral-gap · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H31-consensus-time-spectral-gap/README.md · ratings: RE-P1 round-1b agent (suggested; previous calibrated ratings 40 / 1.0 / 2.0) (page 2)



Second figure. (a) #26 on DQ6 ballots. The ballots (solid) track the number of agents that have read the round's opening (dotted). Consensus (dashed) comes 57 s and 39 s after the opening. (b) Gradual consensus times per block from #30 on, for attention (filled) and work commits (open). Work is not slower (medians dashed).

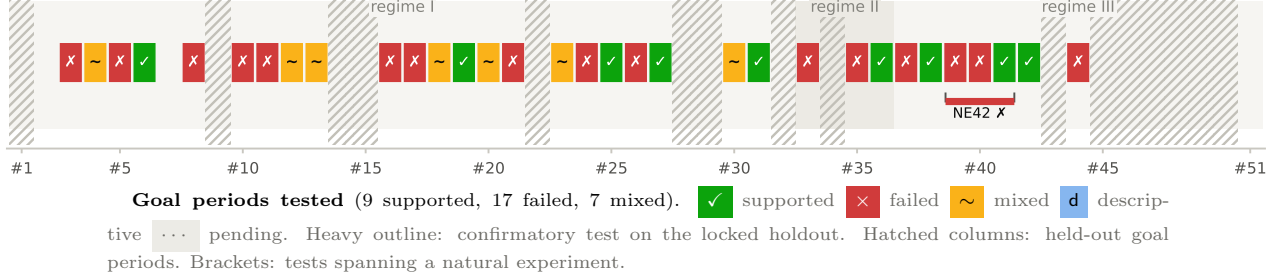
H32: A net information current identifies de facto leaders

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04 (round 1b)

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Exposure-gated content transfer exists in about half the periods and gives a reproducible ranking, but elected, installed and operator leaders all rank low.

level: hypothesis · pipeline checks 86% · updated 2026-10-04 (round 1b)



WHAT HAPPENED

H32 asks whether agent i 's recent message, seen by j , predicts the direction of j 's next message. The baseline is j 's own past, the fields and the rest of what j saw. Held-out-day cross-validation tests the gain against circular-shift nulls on 32 non-holdout periods (#3–#44). Transfer exists but is small: it is significant in 18/32 periods (15/32 against the stricter within-day null). It needs exposure: seen > unseen in 7/8 multi-room weeks. The outflow ranking replicates across halves of days (median $\rho = 0.38$). It does not find leaders: a single source stands out in 2/18 periods, and humans beat the median agent in 4/16. Round 1b uses context-ledger exposure, a second embedding model and style removal. Transfer replicates (17/32; gte 23/32). At matched short lag, read messages beat in-flight ones in 17/17 transfer periods. In-flight messages still carry about a third of the gain (convergence). All three leader tests fail: the elected leader over its real term (rank 5/10), #35's daily lead designers, and #44's installed leader (last of 6). The holdout is not run.

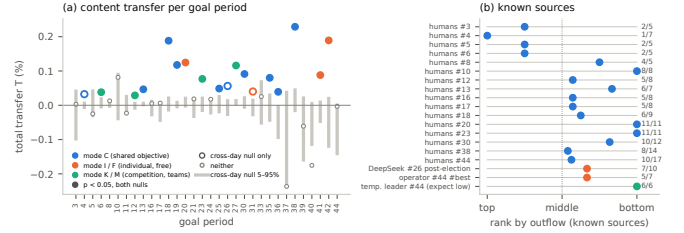
PHYSICAL MODEL

Models 11 and 02 give a message-level vector autoregression on whitened embeddings with isotropic couplings. The state $\mathbf{z}_m$ is agent j 's message vector after removal of a 5-direction goal, kickoff and day×room field. The source $\mathbf{s}_{ij}$ is the decayed sum ($\tau = 15$ min) of i 's messages that j had seen. $\mathbf{B}_{m,c}$ are the baseline columns: own past, day×room field, other seen senders, human and bot messages. G_{ij} is the held-out gain (Gaussian transfer entropy). The order parameter Φ is the noise-corrected concentration of outflow.

$$\mathbf{z}_m = \sum_c \beta_c \mathbf{B}_{m,c} + \kappa_{ij} \mathbf{s}_{ij}(t_m) + \xi_m, \quad \mathbf{s}_{ij} = \frac{\sum_{x \in \text{seen}} e^{-\Delta t_x / \tau} \mathbf{z}_x}{1 + \sum_{x \in \text{seen}} e^{-\Delta t_x / \tau}}$$

$$G_{ij} = 1 - \text{SSE}_{\text{full}} / \text{SSE}_{\text{base}} \text{ (held-out days),} \quad \text{TE}_{i \rightarrow j} \approx -\frac{1}{2} \ln(1 - G_{ij})$$

$$\text{Out}_i = \langle \Delta G_{ij} \rangle_j, \quad \text{Net}_i = \text{Out}_i - \text{In}_i, \quad \Phi = \frac{\text{Var}_i(\text{Out}) - \sigma_{\text{null}}^2}{(N-1) \overline{\text{Out}}^2}$$



Observables. (a) Total transfer T (mean pair gain minus the null median) per period. Gray bars: cross-day null 5–95%. Filled: significant against the cross-day and the within-day null. Ringed: cross-day only. (b) Rank of known sources by outflow. Humans (blue) are top sources only in the viewer-heavy first weeks. The elected (#26) leader, the installed (#44) leader and the #44 operator are not top sources.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not assume that the agent you put in charge drives the content. In this village, formal leaders, elected or fine-tuned, are not content sources. The installed leader is the most-influenced agent in its room, as H02 and H23 also find. Most weeks are distributed: content flows among several agents, often the strongest Claude model present. A monitor can compute total transfer T , which shows whether agents take content from each other, and Φ , which shows whether one source stands out. In one room, a fast responder to outside news looks the same as a leader. The top name changes with the embedding model (17/32 agree). Use T as a coupling monitor, not as an org chart.

H32-information-current-leaders · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H32-information-current-leaders/README.md · ratings: H32 agent (suggested); round 1b RE-D1 (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 transfer in $\geq 60\%$ of periods	18/32 (56%); 15/32 under both nulls	failed (narrowly)
P2 a source stands out in $\geq 50\%$	2/18 transfer periods	failed
P3 split-half $\rho > 0$ in $\geq 2/3$, median ≥ 0.3	23/26, median 0.38; 1b: 20/26, 0.27	supported (round 1)
P5 humans above the median agent in $\geq 70\%$	4/16 (only #3–#6); 1b: 6/29	failed
P6b #26 elected leader top after the vote	rank 7/10 ($z = 0.1$)	failed
P7a #44 installed leader not top	last of 6 agents	supported
P7b #44 operator top in #best	5th of 7 sources; 1b: 6th of 7	failed
P8 seen > unseen (multi-room)	7/8 weeks, sign $p = 0.035$; unseen sig. 1/8	supported
P9 NE42 merge switches transfer on	+0.0001%, sign $p = 0.39$	failed
1b: read > in-flight at same lag	17/17; in-flight sig. 13/17, $\approx 1/3$ of the gain	mostly copying
1b natives: #26 term, #35 leads, #44 window	rank 5/10; percentile 0.43; last of 6	failed (as predicted)
P10–P12 modes and rivals	$C > I/F$ $p = 0.10$; Φ by mode $p = 0.42$; $ \rho $ with timing 0.24	mixed

GOAL PERIODS IN DETAIL

G03 $\times$ 1b: failed (ledger exposure: T +0.007% p 0.341; gte p...
G04 $\sim$ 1b: mixed (ledger exposure: T +0.028% p 0.024, split...
G05 $\times$ 1b: failed (ledger exposure: T -0.026% p 0.927; gte p...
G06 $\checkmark$ 1b: supported (ledger exposure: T +0.037% p 0.024; gte...
G08 $\times$ 1b: failed (ledger exposure: T +0.012% p 0.220; gte p...
G10 $\times$ 1b: failed (ledger exposure: T +0.085% p 0.122, split...
G11 $\times$ 1b: failed (ledger exposure: T -0.024% p 0.951, split...
G12 $\sim$ 1b: mixed (ledger exposure: T +0.015% p 0.049, split...
G13 $\sim$ 1b: mixed (ledger exposure: T +0.042% p 0.024, split...
G16 $\times$ 1b: failed (ledger exposure: T -0.001% p 0.415, split...
+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

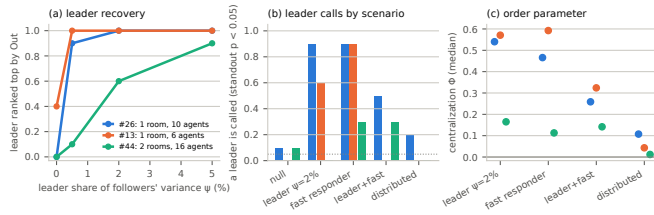
The effects are fractions of a percent of residual variance. With 32 periods and 13 predictions, some passes are chance. In one room, a fast responder and a leader look the same; only the multi-room exposure contrasts and the within-day null address this. The method measures sustained conversational influence, not one-shot agenda setting such as the #26 goal announcement. Outflow is a total, so low-volume sources (operators) rank low. Transfer depends on the number of field directions removed: 25, 18 and 7 significant periods for 1, 5 and 10. The top source agrees in only 13–27 of 32 periods across preprocessing variants, and in 17/32 across embedding models (1b). In regime III, low-volume targets make leave-one-day-out unstable (post-hoc A2). Part of the short-lag transfer is two agents that answer the same prior turn (1b).

ROUND 2

Run the frozen holdout test (#28, #22, #14). Add a per-message coupling, so the test judges operators per message. Add an event-study variant for one-shot agenda setting. Use the #51 multi-room segments and NE38 (an operator reassigns one agent) for exposure and field contrasts with more power.

No round-2 redirects yet.

H32-information-current-leaders · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H32-information-current-leaders/README.md · ratings: H32 agent (suggested); round 1b RE-D1 (suggested) (page 2)



Second figure. Synthetic validation on real message skeletons. (a) The estimator ranks a planted leader top in single-room weeks at 0.5% of the followers' variance. The two-room week needs about 5%. (b) False calls are 0–10%. In one room, the estimator calls a fast responder to a hidden drive the leader 90% of the time. (c) Φ separates a star from a distributed swarm.

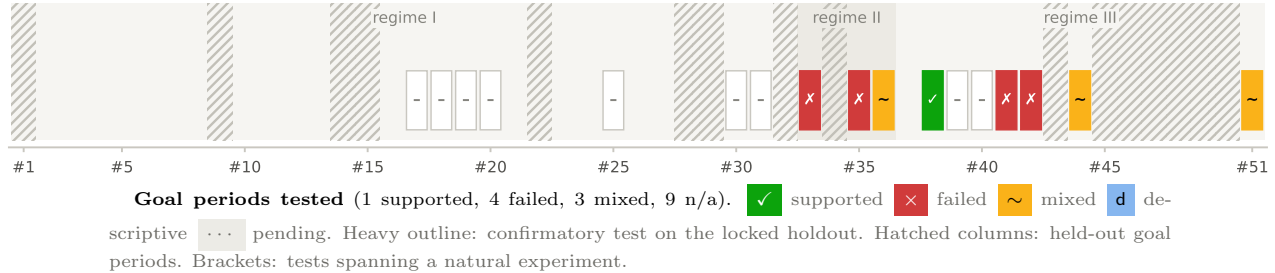
H33: The diversity–productivity curve is an inverted U: the swarm’s operating point

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 0.5/5 Usefulness 1.5/5

Bottom line. Agent-day content diversity has no output optimum in write turns or landed commits; curvature hurts out of sample, and the one right-side drop is a #51 property.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

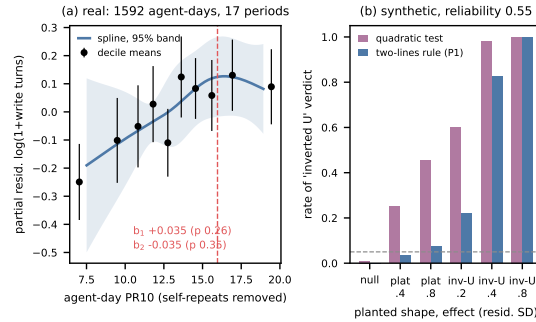
H33 asks whether an agent’s daily content diversity has an output optimum. The unit is the agent-day. Its diversity is the participation ratio PR10 of its chat: self near-copies removed, rarefied to 10 statements, whitened bge. Its output is $\log(1 + \text{write turns})$. The fit is within agent and period, with day effects and activity controls. The data are 17 non-holdout periods with logged write verbs and 1592 agent-days. The primary test is Simonsohn’s two-lines test. A validation on the real design gives 0–7% false positives and 82% power for a 0.4-SD inverted U. **The data show no inverted U.** The slope is $+0.035$ ($p = 0.26$) below the breakpoint and -0.035 ($p = 0.35$) above it. The curve is flat from PR10 ≈ 14 up. Per period: 0 supported, 3 mixed, 11 failed; the meta-analysed right slope has the wrong sign. PR10 improves day-blocked CV by 0.2%, and curvature makes it worse. Self-repetition does not predict same-day output ($p = 0.84$). **Round 1b** (work commits, #30+, both embedding models) keeps the null. The holdout script is frozen, not run.

PHYSICAL MODEL

Model 11 (vector spins). The degree of freedom is a chat statement: a unit vector in the regime’s whitened 32-d bge space. The covariance Σ of an agent-day gives the participation ratio PR, the control parameter. PR = 1 is a frozen spin that repeats itself; PR = 32 is isotropic (“paramagnetic”) content. H33 claims that output peaks at an intermediate PR. The output is $y_{id} = \log(1 + \text{writes})$, with writes = git push or commit, deploy, PR/MR create or merge. α_{iu} and δ_d are agent×period and day effects; z holds activity controls. x_c is the Robin Hood breakpoint; $\nu = n - 1$, $n = 10$.

$$\text{PR} = (\text{tr } \Sigma)^2 / \text{tr } \Sigma^2 \quad (\hat{a}, \hat{b} \text{ unbiased from } W = \nu S)$$
$$y_{id} = \alpha_{iu} + \delta_d + f(\text{PR}_{id}) + \gamma^\top z_{id} + \varepsilon_{id}$$
$$f = b_1(x - x_c)[x < x_c] + b_2(x - x_c)[x \geq x_c] + c[x \geq x_c]$$

inverted U: $b_1 > 0$, $b_2 < 0$ ($p < .05$), $\arg \max f$ interior



Observables. (a) Partial residuals of $\log(1 + \text{write turns})$ by PR10 decile, with the 4-df spline. Agent×period effects, day effects and activity controls are removed. The curve rises a little (not significant) and is flat above PR10 ≈ 14 . Dashed line: two-lines breakpoint. (b) Synthetic replay on the real design at PR10’s measured reliability (0.55). The two-lines rule rarely calls a plateau an inverted U; the quadratic test does so 25–46% of the time. The rule detects a 0.4-SD inverted U 82% of the time.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

No content-diversity operating point exists to steer toward. Do not expect more or fewer chat topics to change how much agents ship. Write turns at the best PR10 and at the 90th percentile differ by at most about 9%. The only hint is on the loop side. The bottom decile may write about 25% less (not significant). H12’s self-repetition rate catches loops more directly. Daily chat PR is half noise, and write turns count activity, not value. This null removes a plausible dashboard metric.

RESULTS VS PREDICTIONS

Prediction	Round 1 (write turns) → 1b Verdict (work commits)
P1 inverted U	$b_2 -0.035$ ($p .35$) → -0.040 ($p .31$); $\times$ b_1 n.s.
P2 per period	meta $b_2 +0.028$ → $+0.017$ (wrong $\times$ sign); 1/8 supported
P3 curvature in CV	spline worse than linear (5/5 seeds) $\times$ both rounds
P6 loops cost out-put	$+0.17$ → $+0.97$ (n.s.); $-$ in #31, $\times$ #35, #42
P7 swarm-day	$b_1 -0.22$ → -0.25 $\times$
gte / dedups (new)	$b_2 < 0$ in all 8 variants; sig. in 3 bge #51 specs, none gte; 0 without #51 tail
G51 rival pairs (native)	same role, same day: quad $p = 0.58$ null tive)
G51 role classes (native)	support class (4 agents) concave, $p < .001$ lead
G39 own worlds (native)	$\rho(\text{PR10, commits}) +0.10$ null
G42 videos (native)	$\rho(\text{PR10, videos}) +0.10$ null

GOAL PERIODS IN DETAIL

G33 $\times$ 1b: failed (work commits: $b_1 -0.076$, $b_2 +0.206$ (p 0.3...)
G35 $\times$ 1b: failed (work commits: $b_1 +0.019$, $b_2 +0.068$ (p 0.9...)
G36 $\sim$ 1b: mixed (work commits: $b_1 +0.154$, $b_2 -0.132$ (p 0.36...)
G38 $\checkmark$ 1b: supported (work commits: $b_1 +0.050$, $b_2 -0.166$ (p ...)
G41 $\times$ 1b: failed (work commits: $b_1 +0.007$, $b_2 +0.508$ (p 0.0...)
G42 $\times$ 1b: failed (work commits: $b_1 -0.044$, $b_2 +0.425$ (p 0.1...)
G44 $\sim$ 1b: mixed (work commits: $b_1 +0.087$, $b_2 -0.046$ (p 0.54...)
G51 $\sim$ 1b: mixed (work commits: $b_1 +0.037$, $b_2 -0.061$ (p 0.32...)
G17 $-$ 1b: n/a (pre-#30: work-ledger zeros ambiguous; 2026-1...)
G18 $-$ 1b: n/a (pre-#30: work-ledger zeros ambiguous; 2026-1...)

+7 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

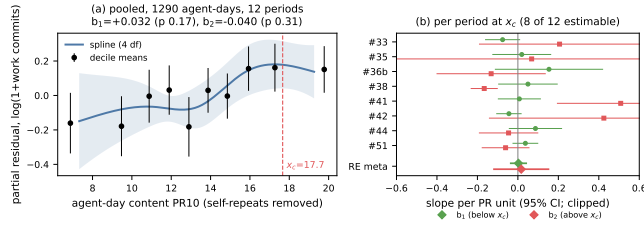
The work ledger is dense only from #30. Five round-1 periods drop out, and #51 is 58% of the pool. Agent-day PR10 depends moderately on the embedding model (bge vs gte $\rho = 0.77$) and is half noise (reliability 0.55). The right-side drop appears only in some bge specifications and vanishes without #51. The support-class inverted U rests on 4 agents. Commits count landed work, not value.

ROUND 2

Keep this null closed unless the holdout confirms the #51 tail lead (C5). If any test runs, test the support-role curve on the #51 tail. The only untried angles are hour-scale designs and Jev progress scores.

No round-2 redirects yet.

H33-diversity-productivity · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H33-diversity-productivity/README.md · ratings: H33 round-1b agent (suggested) (page 2)



Second figure. Round 1b. (a) Partial residuals of $\log(1 + \text{work commits})$ by PR10 decile, with the 4-df spline and the two-lines breakpoint. The pool is 12 eligible periods from #30 on, 1,290 agent-days. (b) Per-period slopes below and above the breakpoint, with the random-effects meta. The meta right slope has the wrong sign.

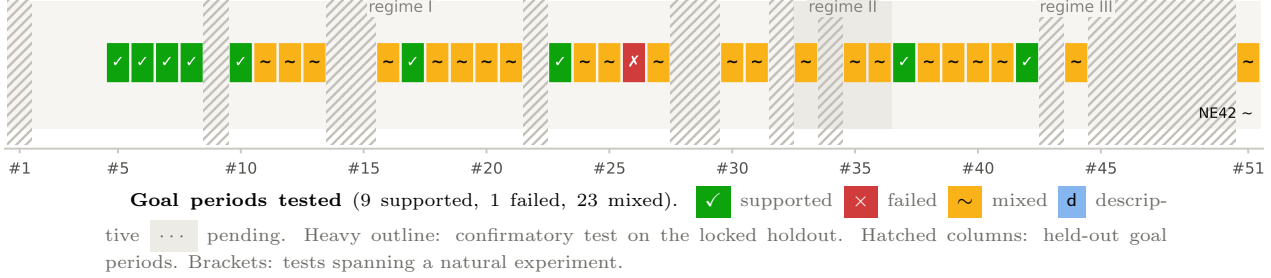
H34: Idea cascades follow a power law whose exponent reads off the distance to criticality

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05 (round 2)

Complete 65% Faithfulness 2.5/5 Usefulness 2.5/5

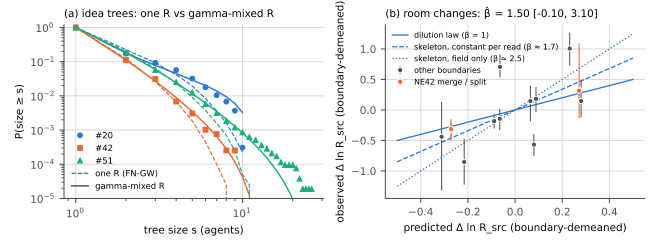
Bottom line. Idea branching is heterogeneous and shared along trees: a gamma mixture of idea-level ratios ($CV^2 \approx 1$) fits the heavy tail that one ratio misses, adopters out-spread inventors 1.5:1, and in a room merge and split branching reversed with room size.

level: descriptive · pipeline checks 86% · updated 2026-10-05 (round 2)



WHAT HAPPENED

H34 builds exposure trees of new ideas in 32 non-reserved periods; an agent's first use gets as parent the latest earlier use it read. Rounds 1–1b: cascades are subcritical everywhere ($\hat{R} = 0.09\text{--}0.40$), tails are heavier than one branching law, and half the exposure lock is convergence. Round 2: (R1) a gamma mixture of idea-level branching ratios fits the tail in 28/32 periods (one ratio: 9/32), and adopters out-spread inventors. (R4) Paraphrase ideas (two embedding models) are subcritical at half the marker value, but their adoption is mostly convergence. (R2) Across the NE42 merge and split, branching reverses with room size; per-read dilution is not identified. No reserved-data run.



Observables. (a) Fraction of exposure trees that reach at least s agents in #20, #42 and #51. Dashed: one branching ratio (FN-GW). Solid: the gamma mixture of idea-level ratios. The mixture follows the tail; one ratio falls short beyond $s \approx 4$. (b) Room changes at six goal boundaries, after removing each boundary's common shock. Orange: the NE42 merge and split. Lines: the dilution law ($\beta = 1$), and synthetic worlds on the real timelines with a constant chance per read ($\beta \approx 1.7$) or a field only ($\beta \approx 2.5$). The data cannot tell them apart.

PHYSICAL MODEL

Contagion (model 03) runs as a Galton–Watson cascade (model 09) in a room of N agents with depletion (FN-GW). The degree of freedom is an agent's first use of an idea. Each idea i has its own ratio $R_{0,i}$, gamma-distributed with mean μ , shape α ($CV^2 = 1/\alpha$). $P_{\text{FN}}(s | R_0)$ is the exact size law. ρ_{off} : non-root over root offspring. R_{src} : a group's offspring per first use; β scales the room-size rule.

$$\hat{R} = \frac{\# \text{ first uses with an agent parent}}{\# \text{ agent first uses}}, \quad R_{0,i} \sim \Gamma(\alpha, \mu/\alpha)$$

$$P(\{s_{ij}\}_j) = \int dR_0 \Gamma(R_0; \alpha, \mu/\alpha) \prod_j P_{\text{FN}}(s_{ij} | R_0)$$

$$\Delta \ln R_{\text{src}} = \beta \cdot 0.451 \ln \frac{N_a - 1}{N_b - 1} + \text{boundary effect}$$

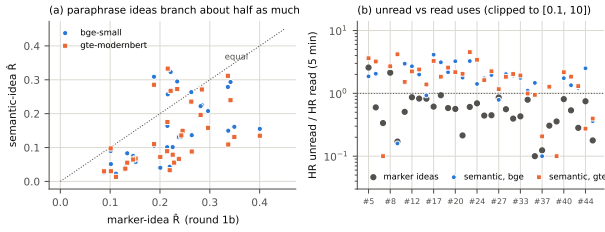
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For tail risk, use the two-parameter mixture (mean ≈ 0.2 , $CV^2 \approx 1$); one ratio under-predicts ideas that reach five agents. An idea's second-generation holders spread it 1.5 times as often as its inventor: watch them. Measure copying by reuse of names, links and numbers; paraphrase similarity mostly measures a shared stimulus. Merging rooms widens each agent's reach and splitting reverses it.

H34-idea-cascades · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H34-idea-cascades/README.md · ratings: H34 agent (suggested); round 1b RE-D1 (suggested); round 2 agent (suggested, 2026-10-05)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
R1-P1 mixture covers the tail	28/32 (one ratio: 9/32)	pass
R1-P2 heterogeneity (LR)	31/32; median $\hat{\alpha} = 0.96$ (0.27–5.3)	pass
R1-P3 beats beta-binomial	21/32 day-ahead, gain mostly from #51	pass (small)
R1-P4 unfitted ρ_{off}	1.52, > 1 in 31/32; homogeneous references ≤ 0.91	pass
R4-P1 paraphrase ideas subcritical	$\hat{R}$ 0.13 (bge), 0.11 (gte), CI < 1 in 32/32; 0.39 at a looser threshold	pass
R4-P2/P3 rank and model agreement	ρ with markers 0.56–0.63; bge vs gte 0.94	pass
R4-P4 read beats unread	7/32	fail
R4-P5 more convergence than markers	unread/read 1.8–1.9 vs 0.56	pass
R2-P2 NE42 reversal $D > 0$	+1.29 [0.74, 2.11]; predicted +1.09	pass
R2-P1 dilution slope β	1.50 [−0.10, 3.10]	open



Second figure. (a) One dot per period and model. Paraphrase ideas (leader clustering at rate-matched thresholds) branch about half as much as marker ideas and rank the periods alike. (b) Adoption hazard after another agent's use that was posted but not yet read, over the hazard after one already read, both within 5 min. For markers (gray) read uses win. For paraphrases (blue, orange) unread uses win, so their adoption is mostly convergence.

GOAL PERIODS IN DETAIL

G05 ✓ 1b: supported (ledger visibility: R 0.22, HR₁₀ 5.8 [...]
G06 ✓ 1b: supported (ledger visibility: R 0.11, HR₁₀ 5.6 [...]
G07 ✓ 1b: supported (ledger visibility: R 0.10, HR₁₀ 16.1 ...
G08 ✓ 1b: supported (ledger visibility: R 0.13, HR₁₀ 1.7 [...]
G10 ✓ 1b: supported (ledger visibility: R 0.19, HR₁₀ 9.7 [...]
G11 ~ 1b: mixed (ledger visibility: R 0.21, HR₁₀ 12.2 [6.7...
G12 ~ 1b: mixed (ledger visibility: R 0.28, HR₁₀ 6.2 [4.7,...
G13 ~ 1b: mixed (ledger visibility: R 0.15, HR₁₀ 4.4 [3.0,...
G16 ~ 1b: mixed (ledger visibility: R 0.10, HR₁₀ 3.9 [2.3,...
G17 ✓ 1b: supported (ledger visibility: R 0.25, HR₁₀ 11.5 ...
+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	2
D unfitted predictions	2	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Tail fit and likelihood ratio cannot tell idea-level from node-level spread; only the non-root/root ratio does. Most ideas have one tree, so idea and seeding-moment heterogeneity are not separated. The forecast gain is mostly #51. Paraphrase $\hat{R}$ depends on the threshold (0.008–0.39). Unread uses are more recent, which overstates convergence. Each room change is also a goal change, and worlds without per-read dilution also predict the reversal.

ROUND 2

Per-read dilution inside #51 with a power check; ideas with several trees to separate idea from seeding moment; re-freeze the confirm script with the mixture clauses.

- **Essence.** how far does a new idea or claim spread in an agent swarm, and what sets it: exposure, salience or a shared field?
- **Round-3 redirects (newest).**
- **H34-R5** Per-read dilution with a cleaner dose. Use #51's day-level room sizes and roster joins (NE33) with a within-room, within-goal design, so that goal shocks do not swamp β .
- **H34-R6** Idea vs tree heterogeneity. Ideas with several independent trees (several roots) test whether R is a property of the idea or of the seeding moment: the mixture predicts correlated tree sizes within an idea.
- **H34-R7** Re-freeze confirm.py on ledger visibility with the Γ -FN tail clause ($C6 \rightarrow \text{"}\Gamma\text{-FN covers } P(s \geq 5)\text{"}$) and the non-root/root ratio ≤ 1 clause.
- **H34-R3** (unchanged). Claims proper, with Jev labels (needs paid labels).

H34-idea-cascades generated 2026-10-07 17:50 by infra/summaries/build_summaries.py card hypotheses/H34-idea-cascades/README.md ratings: H34 agent (suggested); round 1b RE-D1 (suggested); round 2 agent (suggested, 2026-10-05) (page 2)

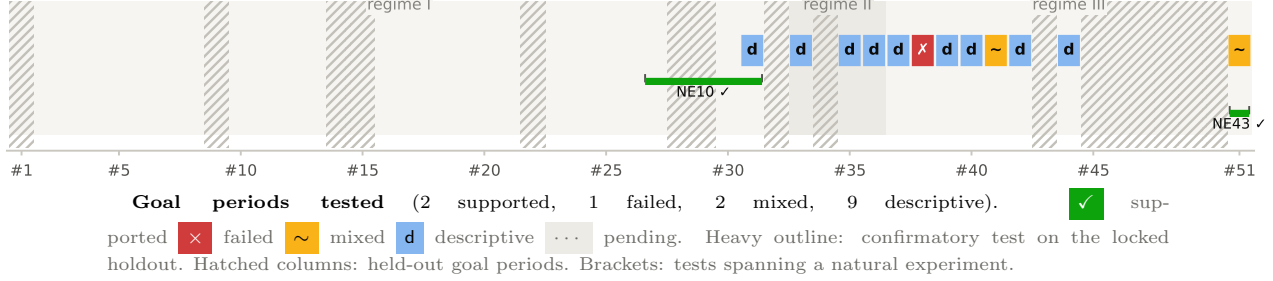
H35: The nudger is a measurably inefficient Maxwell demon

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 2/5 Usefulness 2/5

Bottom line. The nudger is an inefficient Maxwell demon for activity (once-early nudging buys 2–3× more active minutes) and a worthless one for work (+0.22 commits/h, n.s.).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



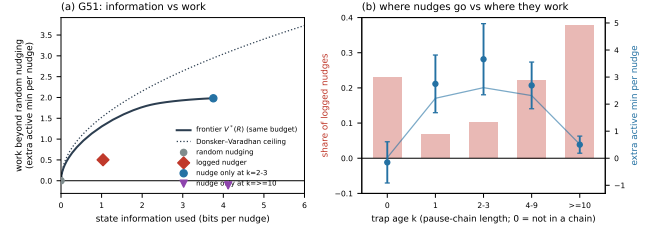
WHAT HAPPENED

H35 treats the auto-nudger as a feedback controller, i.e. a Maxwell demon. It measures the agent-state information in nudges, the activity they buy, and their distance from the best use of those bits. The test covers 13 non-holdout periods and two natural experiments (dated predictions; synthetic recovery checked first). G51 is the only well-powered period. There, a first nudge buys 1.45 [0.72, 2.20] extra active minutes and uses 1.4 bits of state. That is 38% of the work those bits could buy ($\eta_{SU} = 0.38 [-0.08, 0.68]$); only 15% of the bits are needed. Among pausing agents, the nudger does worse than random. It sends 38% of nudges to traps ≥ 10 pauses deep, which return 0.5 min each (2.2–2.6 min at $k = 1$ –9). Nudges add 0.5% of active minutes. Switch-on (NE10) and a silent stop on 08-20 do not show them. Under the old 12-h pause default, nudges wake agents at any trap age. **Round 1b** (work ledger, ledger calls): the nudger buys glances, not committed work or sustained runs. The holdout is not run.

PHYSICAL MODEL

Models 04 (Kolchinsky–Wolpert) and 02. The system X is an agent’s state: idle time, pause timing and trap age k (pause-chain length). The demon is the nudger $M \in \{0, 1\}$ at rate r . The work $g(x)$ is the extra active minutes in 30 min that a nudge causes in state x . The value of information ΔV is the work beyond random nudging. The frontier $V^*(R)$ is the most work that any policy with R bits can buy. s is the half-range of g and plays the role of $k_B T$. p_k is the probability of acting at the pause expiry.

$$\begin{aligned}\Delta V(\pi) &= \mathbb{E}[\pi g] - r \mathbb{E}[g], & I &= I(X; M) \\ V^*(R) &= \max_{I \leq R} \Delta V \Rightarrow \pi_\beta = \sigma(\logit r + \beta(g - \lambda)) \\ \eta_{SU} &= \Delta V_{\log} / V^*(I_{\log}), & \eta_{KW} &= R^*(\Delta V_{\log}) / I_{\log} \\ \Delta V &\leq s \sqrt{2rI} \quad (\text{Donsker–Varadhan}) \\ \logit p_k &= \alpha_i + \theta \ln k + (\gamma + \lambda \ln k) \mathcal{K}[\text{nudge}]\end{aligned}$$



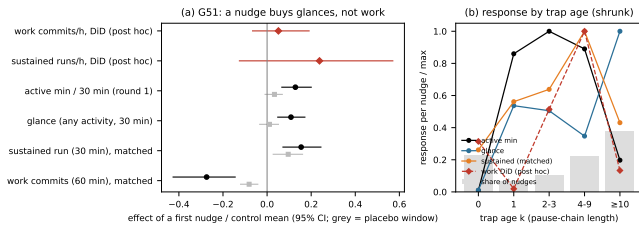
Observables. (a) G51 in trap-age space: work beyond random nudging against the state information used. The curve is the best that any policy can do with that many bits (dotted: analytic ceiling). The logged nudger sits far below it. Nudging only at $k = 2$ –3 reaches the end of the curve; nudging only deep traps is worse than random. (b) The nudger’s mass (bars) sits where the response per nudge (points) is lowest.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An operator who runs an idle-nudger can act on this. Nudge each pause chain once, at the first or second re-pause, and only while the agent is paused. Stop re-nudging agents stuck in long chains: they barely respond and need a reset or a new task. In G51 this policy gives $\times 2.3$ active minutes per nudge at the same budget. It does not buy committed work. The nudger buys only about 0.5% of agent activity, so even a perfect policy moves the swarm little. The thermodynamic frame works as accounting (frontier, ceiling, efficiency), not as physics. The policy result rests on one short-pause period.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (G51)	Verdict
P1 3–8 bits	1.42 bits/nudge (unchanged)	partly
P2 work 0.5–3 min	1.45 active min → glance +0.08; commits +0.22/h [−0.29, 0.82] (DiD)	min yes; work no
P3 falls with trap age	minutes yes; sustained flat	minutes only
P4 $\eta_{SV} \leq 0.5$	0.38 (min) → glance 0.88; work undefined	activity only
P5 once early $\geq 1.25\times$	$\times 2.3$ min; work $\times 1.3$ (CI spans 0)	min only
NE43 (native)	work -0.03 / -0.49 day-SD at the two steps	yes
NE10 (native)	work $z = +0.92$	yes
G38 vs G51 (native)	sustained 0.058 vs 0.094	no



Second figure. Round 1b, G51. (a) Effect of a first nudge relative to the control mean (grey: placebo window before the nudge). Glances and active minutes rise, with clean placebos. Sustained runs and work commits fail the placebo, because nudged agents had just finished a burst. After differencing (red), both are zero. (b) Response by trap age. Active minutes peak at $k = 2-9$, glances peak in deep traps, and sustained work is flat.

GOAL PERIODS IN DETAIL

G31 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G33 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G35 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G36 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G37 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G38 $\times$ 1b: failed (native N1 as worded: sustained ATT 0.058 ...
G39 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G40 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G41 $\sim$ 1b: mixed (round-1 statistics unchanged; no strictly ...
G42 d 1b: descriptive (i 6 strictly isolated first nudges; ...
+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only G51 has power. The past-only design fails its placebo for sustained runs and work commits, so those numbers rest on a post hoc difference-in-differences. The once-early policy gain is in active minutes; for commits it is not identified. The identification assumes that the nudger acts on what the logs show; agent identity carries extra information. The 08-20 stop is undocumented. The card discloses three round-1 amendments and one round-1b post hoc step.

ROUND 2

Re-freeze `confirm.py` with glance and work outcomes. Then run it on #47 and #50 (C2–C4) and on #45 (wake-up). Test the once-early rule as a real intervention if operators allow it. Add an LLM-state feature (Jev behavior labels) to measure what the nudger reads beyond trap age.

No round-2 redirects yet.

H35-nudger-maxwell-demon · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H35-nudger-maxwell-demon/README.md · ratings: H35 round-1b agent (suggested) (page 2)

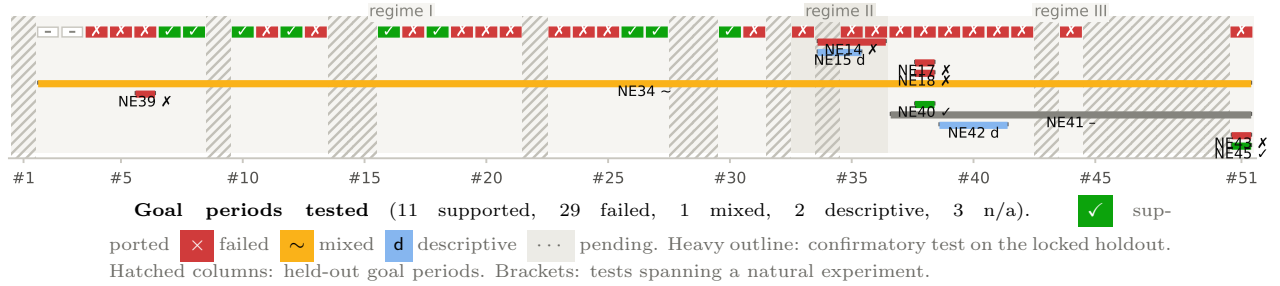
H36: A reorganization alarm: susceptibility and multi-information peak at transitions

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 65% Faithfulness 1/5 Usefulness 3/5

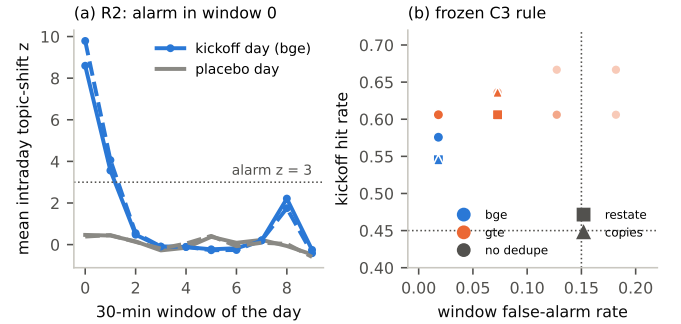
Bottom line. A plain topic-shift alarm times goal changes to the first 30 minutes of talk and survives every robustness check; the multi-information/susceptibility alarm does not beat random dates.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H36 asks whether fluctuation statistics (multi-information, susceptibility, a heat-capacity analogue) flag swarm reorganizations. Rounds 1 and 1b: the physics alarm catches a quarter of 33 goal changes, all via content; the topic shift R1 scores AUC 0.95. **Round 2** (non-reserved data). (i) **Intraday**: a 30-min topic-shift alarm fires in the first window of day 0 for 82–85% of kickoffs, against 7–9% of placebo day starts (AUC 0.92–0.94). (ii) **Robustness**: the frozen rule C3 ($R1 \geq 3$ or $Z_{\text{cont}} \geq 2$) keeps hit 0.55–0.64 at window FAR 0.02–0.07 in 6 embedding × dedupe variants and on Monday-matched placebos. The gte FAR moves with the surrogate seed (0.07–0.18). (iii) Sampling-invariant activity statistics still react to roster and schedule changes. (iv) An action-mix scaffold detector fails (AUC 0.61, $p = 0.16$). The physics alarm fails random dates in every variant (p 0.12–0.25).



Observables. (a) Mean intraday topic-shift z by 30-min window: kickoff days (blue; bge solid, gte dashed) against placebo days (gray). The kickoff signal sits in window 0 and is gone by window 2. (b) The frozen C3 rule: kickoff hit rate against placebo-window false-alarm rate in 6 variants (large markers) and over 10 surrogate seeds (faint dots). Dotted lines: the pre-registered robustness bounds.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Models 01 (inverse Ising) and 11 (vector spins). The activity spin $s_i(t) = \pm 1$ marks agent i active in minute t ; $M = \sum_i s_i$; VR is the susceptibility as a variance ratio. The content spin $\hat{s}_i(w)$ is agent i 's unit content vector in window w ; $\bar{\mathbf{m}}_w$ is the swarm mean of the centered vectors. $D(w)$ is the intraday topic shift against the previous 4 windows; z scores are trailing robust z (10 days, or 16 windows).

$$\text{VR} = \frac{\text{Var}_t M}{\sum_i \text{Var}_t s_i}, \quad I_G = -\frac{1}{2} \log \det R, \quad Z_{\text{phys}} = \langle Z_I, Z_X, Z_C \rangle \geq 2$$

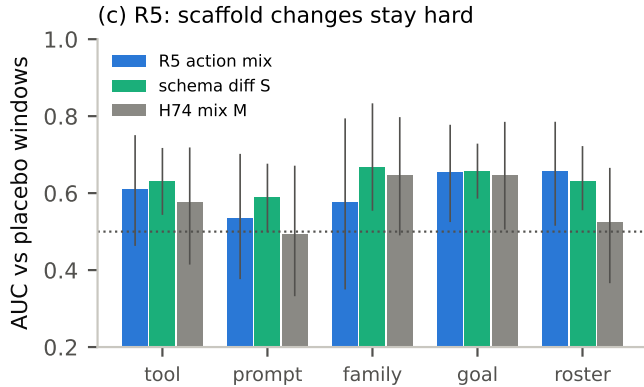
$$D(w) = 1 - \cos(\bar{\mathbf{m}}_w, \langle \bar{\mathbf{m}} \rangle_{w-4..w-1}), \quad \text{alarm: } z_{16}[D(w)] \geq 3$$

$$\text{C3: } z_{10}[R_1(d)] \geq 3 \text{ or } Z_{\text{cont}}(d) \geq 2 \text{ on } d \in \{-1, 0, 1\}$$

H36-reorganization-alarm · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H36-reorganization-alarm/README.md · ratings: H36 agent (suggested; round 2)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 (r1/1b) physics alarm on goals	AUC 0.66–0.68; random-date 0.12–0.25 in all 6 r2 variants	<i>p</i> failed
P3 rival R1 wins	AUC 0.95–0.97; vs Monday placebos 0.93–0.97	held
RB1–3 C3 robust, Monday-safe	hit 0.55–0.64; FAR 0.02–0.07; Monday FAR ≤ 0.12	held
RB4 seed-stable	bge yes; gte FAR 0.07–0.18 over 10 seeds	partly
RB5 style-resid	R1 0.91 / 0.91 AUC ≥ 0.90	held
P2.1 intraday kick-off hit	windows 0–1: 0.82 / 0.85 vs placebo 0.09 / 0.07	held
P2.3 first alarm in window 0–1	27/27, 28/28 in window 0	held
P2.4 #focus mid-day room	z 2.3 (bge), 3.2 (gte) one window early	mixed
P3.2–3 invariant activity	AUC 0.64 [0.48, 0.79]; added hits are roster/schedule	mixed
P5.1 scaffold detector	AUC 0.61 [0.46, 0.75], <i>p</i> 0.16; failed NE14b fires	failed



Second figure. Scaffold detection, AUC of the window maximum against placebo windows (95% CI) per event class: the action-mix detector R5, H74’s schema diff S and H74’s mix channel M. All overlap 0.5–0.7. Synthetic power: only steps larger than the agents’ own day-to-day mix noise are visible.

GOAL PERIODS IN DETAIL

G04 $\times$ 1b: failed (unchanged)
 G05 $\times$ 1b: failed (unchanged)
 G06 $\times$ 1b: failed (unchanged)
 G07 $\checkmark$ 1b: supported (bge) / failed (gte)
 G08 $\checkmark$ 1b: supported (unchanged)
 G10 $\checkmark$ 1b: supported (unchanged)
 G11 $\times$ 1b: failed (unchanged)
 G12 $\checkmark$ 1b: supported (unchanged)
 G13 $\times$ 1b: failed (unchanged)
 G16 $\checkmark$ 1b: supported (unchanged)
 +36 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Most kickoffs (26 of 33) fall on Mondays, and there are only 10 strict Monday placebos (24 in the wider set). Round 2 uses 55 placebo days, none in regime II. The intraday lag after the kickoff message is set by the schedule (the first talk window), not by swarm dynamics. The 05-04 room merge is a day-start event, so it cannot be separated from its kickoff. The #focus timing is one event. The seed study is post hoc. The activity channel’s added hits coincide with roster, room or schedule changes. The reserved data is not run.

ROUND 2

Re-freeze `confirm.py` with bge as primary and ≥ 100 surrogates for Z_{cont} ; freeze the intraday window-0 alarm for confirmation; try a presence/retirement rule on bash heads (H74-R2).

- **Essence.** a cheap, log-computable alarm for structural change in a swarm (goal, roster, room, scaffold) with known hit and false-alarm rates; round 1 says content carries it, activity fluctuations don’t, and a first-moment topic-shift detector beats the physics statistics.
- **Round-3 redirects (newest).**
- **H36-R1** (unchanged). Confirm C3 on the reserved data.
- **H36-R6** Intraday C3 on the reserved data. Freeze r1w ≥ 3 in windows 0–1 of a day as an operator alarm and confirm its hit and day-start FAR.
- **H36-R7** Bash-grammar presence rule. Combine with H74-R2’s presence/retirement rule (a head present on ≥ 8 of 10 days disappears, or a new head appears in ≥ 2 agents) to cut the tool-mix noise.

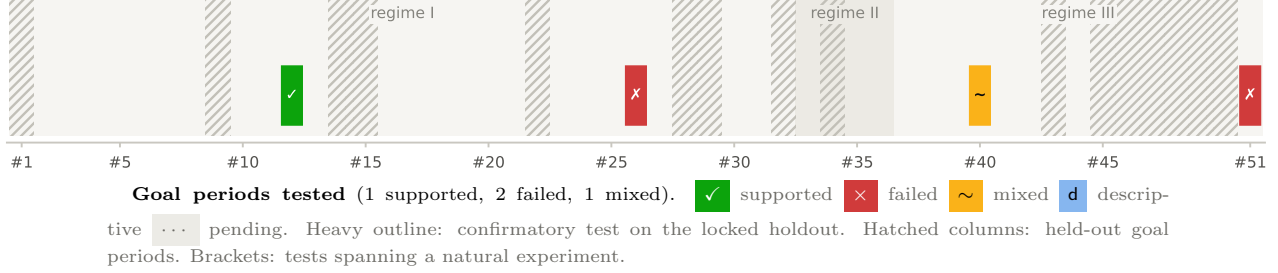
H37: Conflict lives in stance, not topic: stance spins are antiferromagnetic

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 2/5 Usefulness 2.5/5

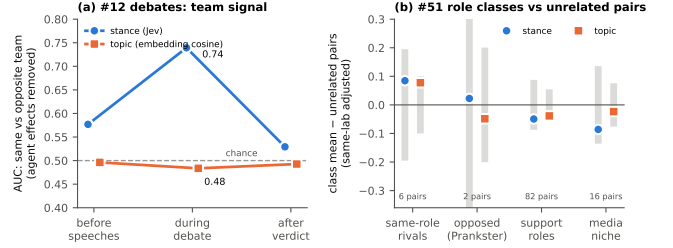
Bottom line. A validated disagreement flag sees the conflict a debate assigns (45% vs 0% of replies; topic blind), and rival and opposed agents show none: disagreement lives only where imposed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H22 finds that rival agents in #51 co-move in *topic*, so topic cannot show conflict. H37 tests whether *stance* shows it. A zero-shot labeller (Jev) marks 30k reply pairs in #12, #26, #40 and #51. In the #12 debates, stance separates opponents from teammates (AUC 0.74 after agent effects), and topic does not (0.48). Stance recovers the drafted teams exactly in 7 of 10 debates, and the contrast vanishes after the verdict. #51 rivals and opposed roles are as friendly as anyone, and #26 stance ignores ballots. **Round 1c** re-tests this on ledger-visible DQ2 replies with DQ10's validated disagreement flag (precision 0.61–0.67, recall $\approx$ 0.6). with its noise in every null. #12 opponents disagree in 33 of 74 replies and teammates in 0 of 60 (stress null $p = 0.001$). #40 now raises no false alarm. #51 rivals and opposed roles show no excess disagreement. The #34 saboteur test is **not run**; a re-freeze on held-out v2.1 labels is proposed.



Observables. (a) #12: AUC for same-team against opposite-team replies, with agent effects removed. Stance (blue) separates the teams only while they debate. Topic similarity of the same reply pairs (orange) never does. (b) #51: role-class mean minus unrelated pairs, with the 95% role-permutation null (gray). Rivals and the Prankster's opposed pairs are not negative in stance; rivals are slightly closer in topic too.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Models 01 (signed Ising) and 10 (Potts communities). The spin is a bond: the stance $s_e \in [-1, 1]$ of reply e from j to i , from Jev probabilities. a_j and b_i are agent fields (agreeableness, likability) of an ordered logit with cut points c_k . Δ is the camp coupling. J_{ij} is the residual pair stance after the fields, and $\xi_i = \pm 1$ is the camp. Factions are the ground state of $H(\xi)$; frustration f is the weight it leaves unsatisfied. γ is same-team ($\delta_{ij}^{\text{team}} = 1$) minus opposite-team stance, with fields absorbed.

$$P(\sigma_e \leq k) = \text{logistic}(c_k - a_j - b_i - \Delta \xi_i \xi_j), \quad \sigma_e \in \{-, 0, +\}$$

$$s_e = p_{\text{agree}} + p_{\text{support}} - p_{\text{oppose}} - p_{\text{undermine}}$$

$$s_e = \mu + a_j + b_i + \gamma \delta_{ij}^{\text{team}} + \varepsilon_e$$

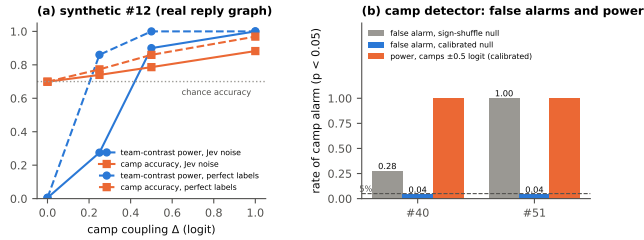
$$H(\xi) = - \sum_{i < j} J_{ij} \xi_i \xi_j$$

$$f = \frac{1}{2} \left(1 - \max_{\xi} \left[-H(\xi) \right] / \sum_{i < j} |J_{ij}| \right)$$

H37-stance-spins · generated 2026-10-07 17:50 by in-
fra/summaries/build_summaries.py · card hypotheses/H37-stance-
spins/README.md · ratings: H37 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1c (validated Verdict flag)
P1 #12 teams	γ 0.42 → flags 33/74 across vs 0/60 held within; stress p 0.001
P2 not topic	AUC 0.74 vs 0.48 → topic AUC 0.48; held holds with topic covariate
P5 verdict	γ 0.42→0.03 → 0 flags in 90 replies held
P6 #51 op-posed	T_{OP} +0.02 → 0/55 replies, $\beta = -2.0$ failed
P7 #51 rivals	T_{SR} +0.085 → $\beta = +0.46$ [-1.5, 1.4] failed
P10 #26 bal-lots	r 0.01 → 7 flags, power 0.05 untestable
P12 #40	20% “oppose” → true rate 0.05× de-mixed→held bate; 0 pairs



Second figure. (a) Synthetic #12 on the real reply graph. With Jev’s measured noise, the team contrast is detectable from $\Delta \approx 0.5$ logit; real #12 matches $\Delta \gtrsim 1$. (b) Camp alarms under agent fields only. The sign-shuffle null fires falsely (28% at #40, 100% at #51); the calibrated ordered-logit null holds 4%.

GOAL PERIODS IN DETAIL

G12 ✓ supported (positive control passes P1–P5)
 G26 × failed (stance does not track ballots; no factions)
 G40 ~ mixed (no false faction or pair alarm; conflict level...
 G51 × failed (no stance antagonism between rival or op-posed...)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Round 1 used one labeller and one rater, and Jev’s “oppose” was only 30% precise. Round 1c’s flag is validated (precision 0.61–0.67) but rare (0.85% in #51). Rival tests are therefore inconclusive at +1 log-odds. #51 rests on 6 rival and 2 opposed pairs; #26 has 9 voters. The calibrated null (Amendment 2) came after the data. The topic baseline uses one embedding model.

ROUND 2

Add a second labeller and a human rater. Run the calibrated detector on contested periods (#19, #31) and, pre-registered, on the #51 tail. Test signed reply dynamics: measure whether a negative reply begets another. Run the #34 saboteur confirmation after sign-off.

- **Essence.** a cheap, calibrated sensor for adversarial or factional behaviour in agent swarms, and its detection limits.
- **H37-R1** A second labeller and rater.
- **H37-R2** Find conflict where it might exist.
- **H37-R3** Kinetic stance.
- **H37-R4** Run the #34 confirmation after sign-off, as the saboteur test (D8.3).

H37-stance-spins generated 2026-10-07 17:50 by in-fra/summaries/build-summaries.py card hypotheses/H37-stance-spins/README.md ratings: H37 agent (suggested) (page 2)

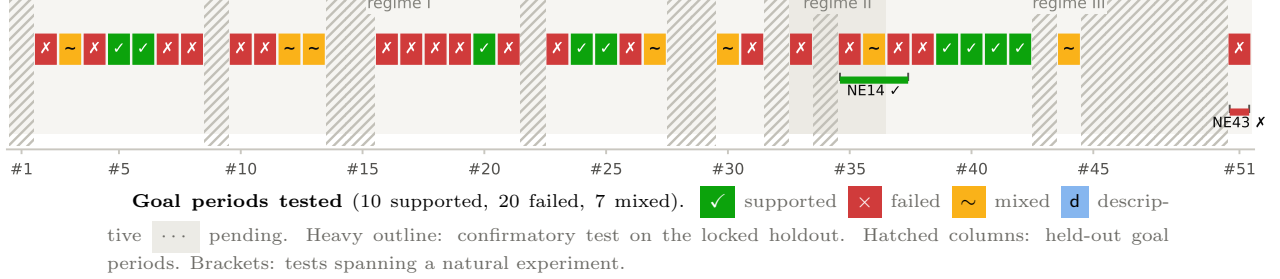
H38: Joint silences are platform stalls

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Joint silences are rare and operator-scheduled, not platform stalls; in the always-on regime, 70–80% of apparent co-activation is the runner that starts and stops all agents together.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H38 builds a shared outage table: minutes with ≤ 1 active agent, with causes from logs. It asks, in 35 non-holdout periods, how much of H02/H19's collective co-activation is infrastructure. **Round 1b** re-runs everything on the corrected activity table (round 1 lost half of all events) with DQ8's calibrated null. **Joint silences are rare and scheduled:** 5.3% of minutes (was 14.3%), 78% outside the operator's run. Error bursts make them *less* likely. **In the always-on regime III, most co-activation is day-edge synchrony** ($f = 0.72$; 0.83 after trimming). Trimming each day to the common running window leaves a significant excess only in #44 and #51 (and #37, marginally). Regime I keeps the excess in 16/18 periods. In the native test NE43, the edge share does not drop when the operator's pause/resume messages stop, so the edges are the runner's schedule. The holdout is not run.

PHYSICAL MODEL

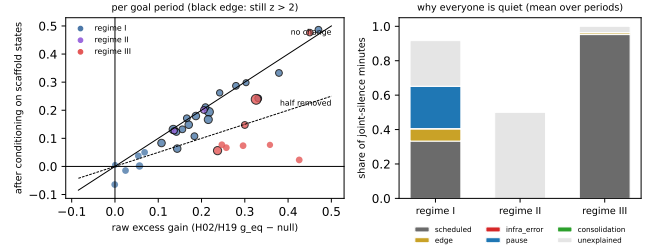
Models 01 (Curie–Weiss) and 09 (Hawkes). The spin $s_i(t) = \pm 1$ marks agent i active in minute t . $h_i(b)$ is a per-agent field per 30-min block. The stall indicator $z(t)$ is a second field that switches agents off: operator off, not yet started or finished, consolidating, or infrastructure error. π is the stall fraction and p the on-rate. The order parameter is the gain g . The stall-adjusted gain trims to the window in which all agents run and sets scaffold-unavailable agent-minutes to the block mean. Block-shift surrogates, drawn on the kept rows, give the null.

$$P(\mathbf{s}) \propto \exp \left[\frac{\beta J_0}{2N} M^2 + \sum_i (h_i(b) + \Delta_i z(t)) s_i \right], \quad M = \sum_i s_i$$

$$\text{VR} = \frac{\sum_b n_b \text{Var}_b M}{\sum_b n_b \sum_i \text{Var}_b s_i}, \quad g = 1 - \text{VR}^{-1}$$

stalls alone: $\text{VR} - 1 \approx \frac{(N-1)\pi p}{1 - (1-\pi)p}$ ($N=12, p=0.5, \pi=0.05 \Rightarrow g \approx 0.34$)

$$f = 1 - \frac{g_{\text{adj}} - \langle g_{\text{adj}} \rangle_{\text{null}}}{g_{\text{raw}} - \langle g_{\text{raw}} \rangle_{\text{null}}}$$



Observables. Corrected tables. (a) Excess equal-time gain per period before and after conditioning on scaffold states (dashed: half removed). Regime-III points (red) drop to or below the dashed line, except #44 and #51. Regime I (blue) stays near the identity line. Black edge: still $z > 2$. (b) Causes of joint-silence minutes, mean over periods. With the dropped events restored, operator-scheduled minutes dominate; infrastructure errors are about 0.1%.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An always-on swarm with a daily start and stop shows synchrony that is mostly its own runner schedule. This holds for mean-field coupling, the market-mode eigenvalue and co-activation. Agents boot and stop together whether or not the operator announces it. Before you draw surrogates, trim each day to the window in which all agents run. Here that removes about 80% of regime-III co-activation. Real excess remains only where agents co-move (#44, #51). Do not alert on joint silences as a stall signal: error logs do not predict them, and provider outages leave no trace. Message-driven swarms (regime I) gain nothing from this rule.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (corrected tables)	Verdict
P1 JS share 0.05–0.30, median 0.10 → 0.009; excess 0.008 above independence	→ 0.007	×
P2a scaffold reasons in JS above chance	explained 0.51 → 0.88 (mostly scheduled); > surrogate in 23 → 24/35	×
P3 infra bursts precede JS	log OR $-0.43 \rightarrow -0.96$; significant > 0 in 0 → 1/30	×
P4 stall filter removes $\geq$ half	f 0.25 → 0.26 (III 0.63 → 0.56)	×
Amend. 2 scaffold conditioning	f_{III} 0.68 → 0.72; f_I 0.11 → 0.11	III ✓
DQ8 null (trim first)	significant 29 raw → 20/35 new trimmed; III 3/8, I 16/18	
P5 λ_1 drop tracks stall share	ρ 0.89 → 0.87; ratio 0.48 → 1.00; $\sim \rightarrow$ ✓ trimmed edge: 19/41 units	✓
P6 III–I contrast shrinks $\geq$ half	+0.125 → +0.017; now +0.155 → -0.006	✓
P9 NE14 jump carried by stalls	+0.15 → -0.06; now +0.12 [-0.04, 0.27] → -0.14	✓
NE43 bookends stop (native)	edge share 0.19 → 0.14 (p 0.08); start spread narrower	×

GOAL PERIODS IN DETAIL

G02 × 1b: failed (round 1: failed; corrected tables, same r...
G03 ~ 1b: mixed (round 1: supported; corrected tables, same...
G04 × 1b: failed (replication, round 1: mixed; corrected ta...
G05 ✓ 1b: supported (round 1: failed; corrected tables, sam...
G06 ✓ 1b: supported (round 1: supported; corrected tables, ...
G07 × 1b: failed (round 1: failed; corrected tables, same r...
G08 × 1b: failed (round 1: mixed; corrected tables, same ru...
G10 × 1b: failed (round 1: failed; corrected tables, same r...
G11 × 1b: failed (round 1: failed; corrected tables, same r...
G12 ~ 1b: mixed (round 1: failed; corrected tables, same ru...
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

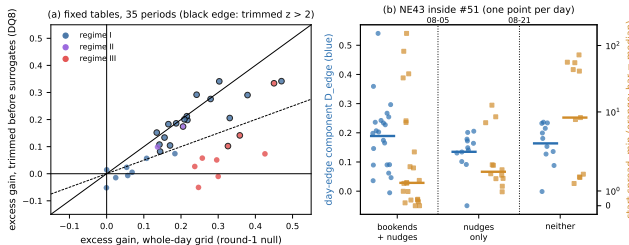
Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

LLM API failures leave no log line, so provider outages remain untested. On the corrected tables most periods have only a few joint-silence minutes. The per-period “explained above chance” test and 12 per-period verdict changes (mostly regime I) rest on these few minutes. The headline estimator (scaffold conditioning) came after synthetic smoke runs; the pre-registered stall filter fails. The **scheduled** flag comes from operator messages and is empty after 08-05; agent-level trimming does not depend on it. A whole-block stall (the 300-min stop on 2025-06-18) leaves no trace in the block-detrended gain. The holdout is not run.

ROUND 2

Second figure. (a) Excess gain on the whole-day grid (round-1 null) and after trimming to the all-present window before surrogates are drawn (DQ8). Regime III collapses toward zero; regime I stays. (b) NE43 inside #51, one point per day. The day-edge component (blue) does not fall when the bookend messages stop on 08-05. The start spread (orange) widens only when the nudger stops on 08-21 (post hoc).



Run a per-provider silence test for API outages: all agents of one lab silent while the others act. Add the trimmed (DQ8) variants to `confirm.py` before it runs. Study the #44 and #51 residual coupling. Replace the message-based **scheduled** flag with the runner’s start and stop times.

No round-2 redirects yet.

H38-platform-stalls · generated 2026-10-07 17:50 by in-
fra/summaries/build_summaries.py · card hypotheses/H38-platform-
stalls/README.md · ratings: H38 round-1b agent (suggested) (page 2)

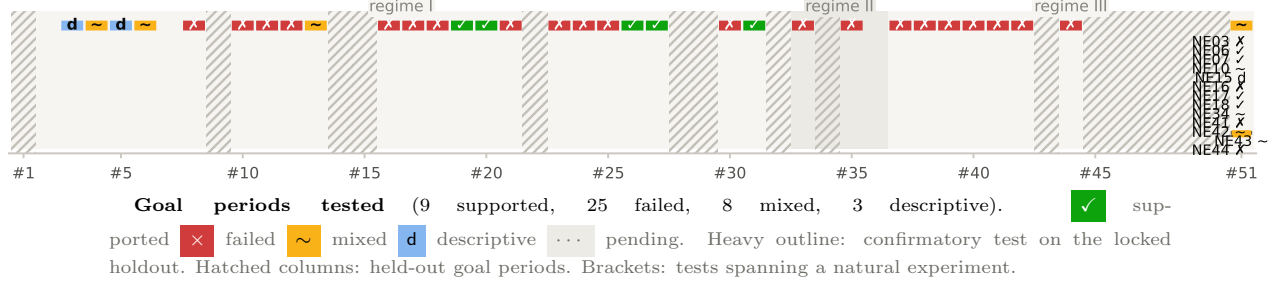
H39: Catalysts vs. fields

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. No operator lever is a pure catalyst: nudges both unstick and tilt agents, while human messages, mentions, erasures and goals act mainly as fields, on activity or content by regime.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H39 asks which operator levers change *what* agents do (a field: occupancy shifts) and which only speed them up (a catalyst: escape at fixed occupancy). A kinetic decomposition of behavior-state chains in matched windows measures both around nudges, human messages, @-mentions, context erasures and step levers in 32 non-holdout periods. **Round 1b** uses the nudge's leading @ as its only target, DQ8's presence-cut lever design, and Jev v3.1 behavior states as a second state space (V4). **Nudges are both**, now in two state spaces. On B4, $K = +0.13$ and the idle share moves -0.048 (the CI now excludes 0). On V4, $K = +0.25$, the waiting share moves -0.08 and waiting escape rises $\times 1.6$. Human messages are fields. @-mentions steer content ($+0.33$ SD), not behavior. Forced erasure is a field toward work in both state spaces; its K flips sign between them. Natives: the nudger switch-off is a catalytic loss in both state spaces (NE43), and nudges are not more catalytic under the 12-h pause default (NE44). The holdout is not run.

PHYSICAL MODEL

Models 02/10 give a kinetic Potts (Markov-jump) agent. The states are $s \in \{\text{work, chat, idle, consolidate}\}$, with Arrhenius rates k_{ij} , well depths G_i and symmetric barriers B_{ij} . A field h tilts the wells; a catalyst c lowers the barriers. For each lever, matched 30-min windows give the kicked and control transition matrices T^K, T^C and their stationary distributions π^K, π^C . From these come the escape probability $e_s = 1 - T_{ss}$, the field index ϕ , the catalytic index K and the catalytic fraction ρ .

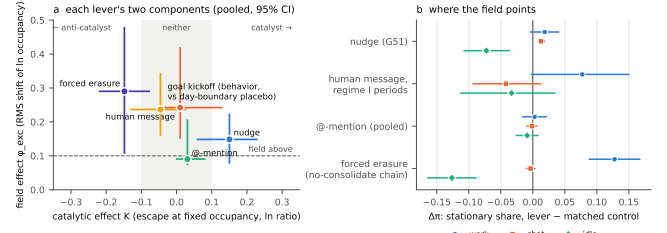
$$k_{ij} = \nu e^{-(B_{ij}-G_i)/T}, \quad \pi_i \propto e^{-G_i/T}$$

$$s_{ij} = \frac{1}{2}(\delta \ln k_{ij} + \delta \ln k_{ji}) \quad (\text{catalytic part})$$

$$a_{ij} = \frac{1}{2}(\delta \ln k_{ij} - \delta \ln k_{ji}) \quad (\text{field part})$$

$$\phi = \text{SD}_{\pi^C} \left[\ln \pi_s^K - \ln \pi_s^C \right], \quad \rho = |K| / (|K| + \phi_{\text{exc}})$$

$$K = \ln \left(\sum_s \pi_s^C e_s^K / \sum_s \pi_s^C e_s^C \right)$$



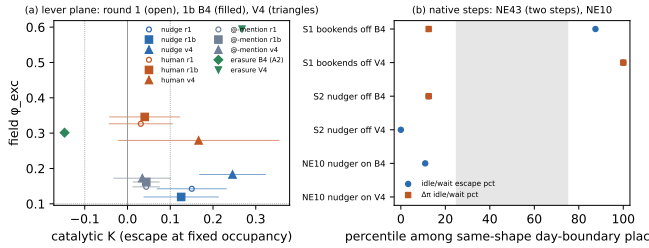
Observables. (a) Pooled field strength ϕ_{exc} (placebo-corrected) against the catalytic effect K , with 95% CIs. Dashed line and grey band: the 0.10 thresholds. Only the nudge reaches the “both” corner. Human messages and erasure are fields, and erasure is anti-catalytic. (b) The direction of each field: stationary share changes for work, chat and idle.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To unstick an idle agent, send one directed nudge. Escape rises $\sim 50\%$, and the agent stays out of idle while it acts. To steer what an agent writes without a change in how busy it is, mention it. To change topics, set a goal; that works in the large regime-III village. The field/catalyst split depends on a convention, the nudge verdict rests on one period (G51), and the effects are individual and diluted. The nudger switch-off (08-21) removes catalysis, not a field, and barely moves the swarm. Every message carries a field. Prompt tweaks, room changes and most fixes show nothing at day scale.

RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b (B4; V4)	Verdict
P1 nudges: both; K 0.15 → 0.13; $\Delta\pi_{\text{idle}}$ -0.043 4/5 B4; both, idle down (n.s.) → -0.048^* ; V4 both, K 0.25*, 5/5 V4 $\Delta\pi_{\text{wait}}$ -0.08^*		
P2 human: field (P 0.92); chat -0.03 ; V4 × both, toward field/both chat		
P3 mentions: field P 0.76 → 0.50 (neither 0.46); no mixed field to chat direction		
P4 erasure: A2 chain K -0.15 , idle -0.11 , work × browse up, $+0.11$; V4 K $+0.27$ $ K < 0.1$		
P5–P8 steps	unchanged (B4 states unaffected by as round the bin bug)	1
ITT (lever_design)	nudge work/chat share $+0.020$ [0.010, -0.030]	
Native NE43	nudger off: catalytic loss (B4, V4); mixed bookends off: B4 neither, V4 both	
Native NE44	12-h escape 0.14 vs 5-min 0.40	×
Native NE10 (V4)	neither; wait share $+0.02$	mixed



Second figure. Round 1b. (a) The lever plane: round 1 (open), round 1b on B4 (filled) and on Jev v3.1 states (triangles). Nudges stay in the “both” corner in both state spaces. Erasure is a field whose catalysis depends on the state space. (b) Native steps inside #51 and at NE10: percentiles of idle/wait escape and occupancy among same-shape day-boundary placebos. The nudger switch-off sits at the bottom of the escape distribution, with occupancy inside the band.

GOAL PERIODS IN DETAIL

G03 d 1b: descriptive (r1 descriptive)
G04 ~ 1b: mixed (r1 supported)
G05 d 1b: descriptive (r1 descriptive)
G06 ~ 1b: mixed (r1 mixed)
G08 × 1b: failed (r1 failed)
G10 × 1b: failed (r1 failed)
G11 × 1b: failed (r1 failed)
G12 × 1b: failed (r1 failed)
G13 ~ 1b: mixed (r1 mixed)
G16 × 1b: failed (r1 failed)
+35 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The nudge class rests on one period (G51); G38 has the same signs but is not significant. Fields are invisible below ~100 episodes, so “catalyst” or “neither” in small periods does not exclude a field. The field/catalyst split depends on a load-distribution convention, and $|K| < 0.10$ has no clear reading. B4 states mean different things in regimes I and III. Mentions are partly common-cause. The card flags design choices made after seeing data: the no-consolidate erasure chain, the content-TV check and the post hoc P9 switch-off. Of 28 class-level tests, the nudge field alone does not survive Bonferroni.

ROUND 2

Run `confirm.py` after commit; expect low nudge power on #45–#50. Build agent-centered content states for steps. Analyze the dose of nudger re-firing. Treat the 08-21 switch-off as a proper NE with day-matched controls.

No round-2 redirects yet.

H39-catalysts-vs-fields · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H39-catalysts-vs-fields/README.md · ratings: H39 round-1b agent (suggested) (page 2)

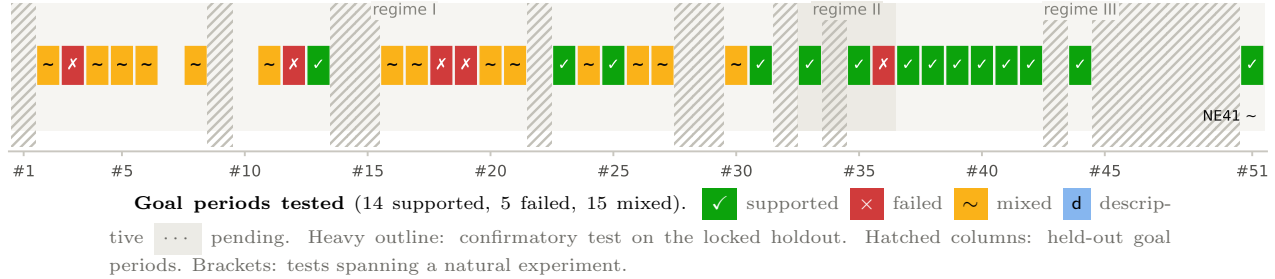
H40: The call clock sets the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 55% Faithfulness 2/5 Usefulness 3/5

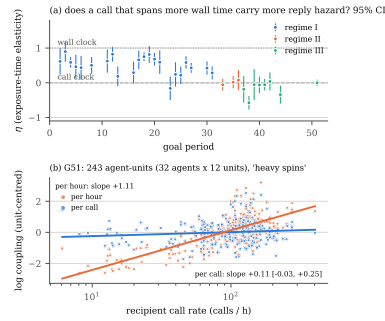
Bottom line. In the computer-use scaffold, agents couple per model call, not per minute: per-hour coupling is a family trait times call rate, and regime I breaks this.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H40 asks whether agents respond to each other per model call or per minute. A reply-hazard model runs on each recipient's real call schedule (context ledger, DQ2 reply labels) in 33 non-holdout periods. Simulated replies first show that the model tells the two clocks apart. In the computer-use scaffold (regimes II–III, 11 periods), coupling runs on the call clock. A call that spans more wall time carries no more reply hazard: $\eta = 0.00 \pm 0.05$ in G51, and -0.03 at timer wakes fixed in advance. In G51 the per-call coupling is flat in cadence ($s = +0.11 [-0.03, 0.25]$) and set by model family. The per-hour coupling scales with call rate (slope 1.0), so slow agents are heavy spins. In regime I (22 periods) the call clock fails: $\eta = +0.51 [0.41, 0.61]$. Natives: G51 yes, NE41 mixed, G18 and NE14 no. Round 2 explains the negative η of G38 and G44 ($-0.57, -0.34$). Long tool calls are followed by fewer talk calls ($\eta_{\text{talk}} = -0.28$). Given a talk call, the span elasticity is $-0.11 [-0.16, -0.06]$ in all four regime-III periods tested. The holdout is not run.



Observables. (a) η per period. Regime-I periods sit between the clocks; regime II–III periods sit at the call clock (0). (b) G51, 243 agent $\times$ unit points. The per-call coupling (blue, model intercepts) is flat in call rate. The per-hour coupling (orange, intercept + $\ln r$) rises with slope ≈ 1 . Lines: fitted slopes.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

In a computer-use agent loop, an agent's wall-time coupling to others is its per-call coupling (a model-family trait) times its call rate. Long tool calls, long pauses and slow models make agents "heavy". In G51 the slowest fifth answer within 5 min $8\times$ less often than the fastest fifth. Cadence is a coupling knob that needs no prompt change. To couple an agent faster, halve its call interval: 5-min coupling rises by about 50% (G51 32%), less at 30 min. Use short pause timers for coordinators. To decouple an agent, give it long batch jobs. The rule does not transfer to a scheduler-driven chat loop (regime I), where time itself carries part of the coupling. The evidence is mostly observational; the clean cadence intervention (NE20) is in the holdout.

PHYSICAL MODEL

Models 02 (kinetic Ising, Glauber attempt rates) and 09. The degree of freedom is the reply of recipient i . Each model call c of i is an update attempt. After i reads message m at call 1, it replies at call n with hazard h_n . α_i is the per-call coupling. e_n is the wall time that call n spans; a_n is the age of m ; b_n counts newer messages (burial); k_n is the batch size. The order parameter η is the exposure-time elasticity: 0 on the call clock, 1 on the wall clock. r_i is the call rate, and $\varepsilon(T)$ the cadence elasticity of a reply within T .

$$\text{cloglog } h_n = \alpha_i + \phi \ln n + \psi \ln a_n + \chi \ln(1+b_n) + \eta \ln e_n + \delta \ln(1+k_n) + \dots$$

$$\text{wall clock: } h_n = 1 - e^{-\lambda e_n} \Rightarrow \eta = 1; \quad \text{call clock: } \eta = 0$$

$$J_i^{\text{hour}} = J_i^{\text{call}} r_i^{1-\eta}, \quad \alpha_i = c + s \ln r_i \quad (s = 0 : \text{per-call invariant})$$

$$\varepsilon(T) = \partial \ln P(\text{reply} \leq T) / \partial \ln(\text{speed-up})$$

H40-call-clock-coupling · generated 2026-10-07 17:50 by in-
 fra/summaries/build_summaries.py card hypotheses/H40-call-clock-
 coupling/README.md · ratings: H40 agent (suggested 2.5/3.0); coordinator
 calibrated faithfulness to 2.0 (pooled P1 failed, two natives failed); round-2
 update by H40 agent 2026-10-05 (v1 ratings unchanged; v2 left to the coordinator)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 pooled $\eta \in [-0.25, 0.25]$	+0.30 [0.17, 0.43]; II -0.01, III -0.16, I +0.51	fail; II-III yes
P2 decay per call (III)	$\phi = -0.60, \psi = -0.19$; 6/8	yes
P3 $s \approx 0$, excl. -1	pooled -0.12 [-0.28, 0.04]; G51 +0.11	yes
P3 per-hour slope ≥ 0.5	III +0.74; I 0.00	III only
P4 held-out call > wall	33/33	yes
P5 collapse in calls	21/33; test invalid (R5)	uninformative
P6 $\varepsilon(5) \in [0.5, 1]$, $\varepsilon(30)$ lower	0.57 [0.36, 0.78]; 0.47	yes
N1 G51 pause dose, heavy spins	$\eta_1 = -0.03$; R^2 cadence 0.50 vs family 0.33	yes
N2 NE41 no catch-up	read-out $\times 0.44$; ratio 1.86 (band fail)	mixed
N3 G18 / N4 NE14	$\eta = 0.81$ / 1 of 11 agents	no / no
R6 G38/G44: talk composition	$\eta_{\text{talk}} = 0.28$; given talk -0.11 (4 periods); 4 rivals flat	mostly yes
R1 regime-I span split	$\eta_{\text{wait}} = 0.18$ [-0.17, 0.53]; placement +0.04	inconclusive
R5 matched collapse	statistic invalid on synthetic worlds	withdrawn

GOAL PERIODS IN DETAIL

G02 $\sim$ mixed
G03 $\times$ failed
G04 $\sim$ mixed
G05 $\sim$ mixed
G06 $\sim$ mixed
G08 $\sim$ mixed
G11 $\sim$ mixed
G12 $\times$ failed
G13 $\checkmark$ supported
G16 $\sim$ mixed

+24 more periods: see the card.

FAITHFULNESS SCORECARD

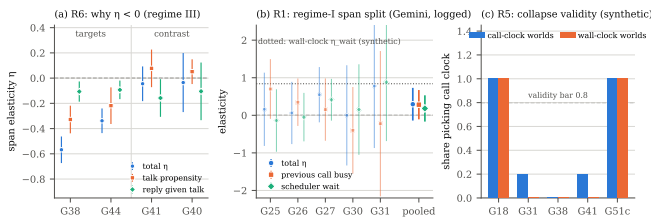
A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Regime I is unexplained. Its positive η survives logged start times, short spans and the placement test. The generation-vs-wait split has only 274 logged replies, too few to decide. The talk split conditions on talk, which a read message can cause (H50). The residual -0.11 given talk has no identified cause. Agents partly choose their cadence; only the pause-timer dose is quasi-exogenous. The cross-sectional G51 slope (1.0) exceeds the model's speed-up elasticity (0.40), so composition adds to cadence. Reply labels are DQ2 parents; rank-2/3 candidates are mostly unlabelled, so multi-target replies are undercounted, most for slow agents with big pools. The NE41 ratio fails for recency reasons found post hoc. The card has two post hoc checks; the holdout is not run.

ROUND 2



Second figure. Round 2. (a) Split of η into talk propensity and reply given talk. In G38 and G44 long calls lower talk; given talk, $\eta \approx -0.1$ everywhere. (b) Regime I, Gemini logged calls: previous-call busy time vs scheduler wait. Pooled η_{wait} stays below the wall-clock value (dotted). (c) Synthetic worlds: the lab-balanced tertile collapse picks the call clock under both clocks, or under neither, so it is withdrawn. Round-1 synthetic: $\hat{\eta}$ recovers 0, 0.5 and 1 (bias ≤ 0.05).

Run `confirm.py`; the NE20 one-tool-call DiD is the cleanest cadence intervention. Test whether the read message causes the talk call before reading -0.11 as the call clock. Repeat the test on the content channel (H29 boundary pull on the ledger).

- **Essence.** a measured coupling-per-call that an operator can multiply by cadence, and an explanation of why the regime-I chat scaffold does not obey it.
- **H40-R1** Why regime I is not a call clock.
- **H40-R2** Content channel.
- **H40-R3** Lags in call cycles (HH187).
- **H40-R4** Run `confirm.py` on NE20/NE44 and the transfer periods; NE20 is the cleanest cadence intervention in the dataset.
- **H40-R5** Fix the collapse test by matching agents on per-call coupling (family) before comparing cadence tertiles, and add multi-parent reply labels (DQ2 holds only the primary parent).
- **H40-R6** Negative η in G38 and G44 (coordinator, 2026-10-04): $\eta = -0.57$ [-0.77, -0.38] and -0.34 [-0.60, -0.08] sit significantly below the call clock, yet the period rule labels them supported.

H40-call-clock-coupling · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py card hypotheses/H40-call-clock-coupling/README.md · ratings: H40 agent (suggested 2.5/3.0); coordinator calibrated faithfulness to 2.0 (pooled P1 failed, two natives failed); round-2 update by H40 agent 2026-10-05 (v1 ratings unchanged; v2 left to the coordinator) (page 2)

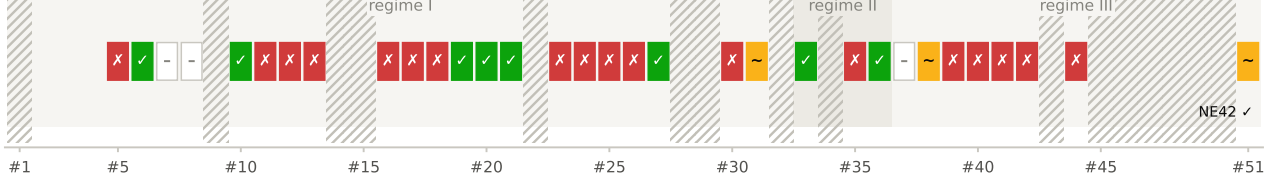
H41: Read-out gating gives the swarm a light cone

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 56% Faithfulness 2/5 Usefulness 2/5

Bottom line. Items obey a logged light cone in rooms; cross-room leaks follow shared projects, with only a small, fragile ordered-read excess ($\Lambda = 1.45$).

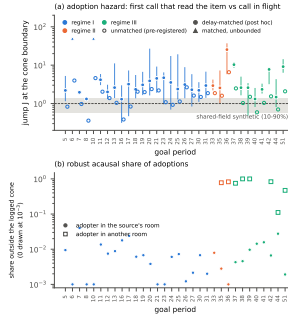
level: hypothesis · pipeline checks 71% · updated 2026-10-05



Goal periods tested (9 supported, 18 failed, 3 mixed, 3 n/a). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

H41 asks whether a novel item (a link, number, name or rare word) can reach an agent before that agent's model calls could read it. The test traces 134k novel marker items (45k adoptions) through the DQ1 context ledger in 32 periods. Within a room the logged light cone holds: adoptions outside it are 0–4% per period (median 0.6%). The gating jump at the first reading call passes in 10–12/32 periods as pre-registered and in 22–24/32 when matched for time since the item appeared (post hoc). Across rooms the logged cone is a cage (75–100% of cross-room adoptions outside it in 7/8 two-room periods; the NE42 merge raises cross-group pickup 80–860×). Round 2 time-stamps the leaks with file-level git history and URL fetches. Cross-room adopters read the source's fresh artifacts far more than same-moment non-adopters, but as much after their own use as before: the ordered excess is only $\Lambda = 1.45$ [1.09, 2.26], and it rests on two periods. Pooled with period strata, regime-III numbers gate ($J_{mh,D} = 3.2$); only #40 co-generates them.



Observables. (a) Jump J per period. Open circles: the pre-registered unmatched estimate. It is often ≤ 1 , because the adoption hazard decays with time since t_0 and in-flight calls all sit near t_0 . Filled: delay-matched (post hoc). Gray: shared-field synthetic. (b) Share of adoptions outside the logged cone, for adopters in the source's room (dots, $\leq 4\%$) and in another room (squares; #51's hopping rooms 0.47).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Model 02 gives asynchronous kinetic updates, and model 03 gives contagion on a temporal graph. Agents update only at their model calls. A call reads the room messages posted since the previous call (the ledger). An item that source s posts at t_0 can reach agent i only along a time-respecting read-out path. K_i is i 's entry call into this logged cone. The adoption hazard per talk call is a shared field ε , plus unlogged channels η , plus q once i has read the item. J compares the hazard at the first entry call with the call in flight at t_0 .

$$T_s = t_0, \quad T_i = \min\{t_c^{\text{call}} : c \text{ reads } m, t_m \geq T_{\text{src}(m)}\}$$

$$\lambda_i(c) = \varepsilon(t_c) + \eta_i(c) + q\mathbb{I}[c \geq K_i^{\text{item}}]$$

$$\text{acausal: } c_{\text{use}} < K_i, \quad J = \frac{h(c = K_i)}{h(c = K_i - 1)} \Big|_{\text{same } t-t_0}$$

$$v = 1/\text{median}(\text{calls from exposure to use})$$

Cross-room leaks follow project membership. In #51 a cross-room adopter is almost always an agent that already reads the source's repos (75% vs 3%), before and after the use. To predict where an item leaks, look at who works on the source's artifacts, not at who fetched them just before. Fixed rooms still cage chat: merging two groups multiplies cross-group pickup about a hundredfold, and hopping rooms leak about half the time. Inside a room, names and numbers both gate at the reading call (regime III, J_{mh} 4.8 and 3.2). A dashboard-style common stimulus does not explain the in-flight adoptions. Spread runs at one hop per 1.5–5 talk turns, so call cadence is not a lever.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 J_{in} CI > 1 ($\geq 70\%$ II/III, $\geq 50\%$ I)	3/10 II/III, 6–8/19 I; matched (post hoc) 7–8/11, 15/21	fail (post hoc pass)
P2 $A_{rob} \leq 0.05$ (II/III), 0.10 (I)	29/32; the 3 misses are cross-room; within-room max 0.04	mostly
P6 cadence $b \in [0.5, 1.5]$, $b > 0$ (CI) in 3/25		fail
P8 in-flight $\leq$ entry/3	1–2/10 (matched 3–4/10)	fail
NE42 merge $\geq 3 \times$ both ways; cage	$\times 82$, $\times 860$; cross acausal 1.0 / 0.02 / 1.0	yes
G51 hoppers bridge ($\geq 70\%$ in cone)	0.51 of 839 (1b)	no
R2a pooled Λ CI > 1	1.45 [1.09, 2.26]; lost #35 or #51 (post hoc)	yes, fragile
R2b ordered reads above in-cone controls	L 0.78 [0.66, 0.95]	no
R2c pull the main channel	page 0.44 $>$ pull 0.38	no
R3b lag tracks cone hops	synthetic fields also pass (size 3/8)	not scored
R4a start-time RD for numbers	RD invalid (size 4/12)	not scored
R4b common stimulus enriched in flight	0.24 vs 0.31; ψ_D 1.3 [0.5, 3.6]	no
R4c $J_{mh,D}$ without stimulus > 1	3.0 [2.0, 4.9], but $J_{mh,D}$ is 3.2 anyway	yes (moot)

GOAL PERIODS IN DETAIL

G05 $\times$ failed
G06 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
G16 $\times$ failed
G17 $\times$ failed
G18 $\times$ 1b: failed, post hoc supported (was supported: J_in l...
G19 $\checkmark$ supported
+23 more periods: see the card.

FAITHFULNESS SCORECARD

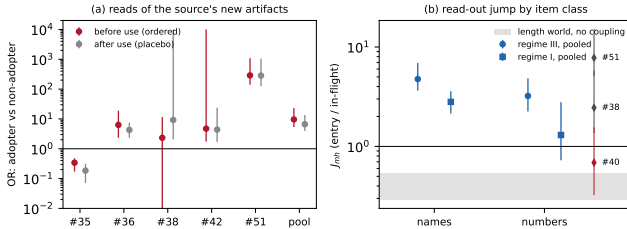
A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The gating result rests on a post-hoc estimator (chosen after the first real period). A synthetic world where talk depends on call length drives it to 0.3–0.5 with no coupling, so real jumps are lower bounds; the start-time RD meant to remove this over-corrects and is not used. Items are hashed markers. Round 2's Λ passes pooled but not with #35 or #51 left out, and in #35 both odds ratios are below 1. Read events come from logged commands and URLs; browser reading without a logged URL is invisible. Visibility follows the ledger's room rule. #40's shared goal confounds NE42. No multiplicity correction across 32 periods. Reserved data are untouched.

ROUND 2



Second figure. Round 2. (a) Odds ratio that a cross-room adopter, vs a same-moment cross-room non-adopter, read an artifact the source had just written: before its own use (red) and in an equal window after it (gray). Their ratio is Λ . In #51 both are ≈ 290 : a shared-project field. (b) Delay-matched jump by class, pooled with period strata; diamonds: the three periods with ≥ 10 in-flight number adoptions. Gray band: J_{mh} in a synthetic world where talk depends on call length and nothing couples.

Find a length-robust in-flight estimator by simulation (size ≤ 0.10) and re-measure co-generation. Model cross-room leaks from the DQ4 co-commit graph and test it out of sample. Trace #40's shared number source from fetch URLs. Pre-register the matched jump and the class split for the reserved data.

- **Essence.** which channels actually move information between agents, and how fast. The logged chat channel is gated at the call scale and bounded by rooms, but a third of fast adoptions and almost all cross-room spread come from elsewhere.
- **Round-3 redirects (newest).**
- **H41-R6** A length-robust jump. Simulate the in-flight vs entry contrast with talk tied to call length at several strengths, and find an estimator with size ≤ 0.10 (e.g.
- **H41-R7** Project membership as the leak variable. Model cross-room adoption as a function of prior co-work on the source's repos (DQ4 co-commit graph) and test whether it predicts leaks out of sample in #51 (and the reserved #47).
- **H41-R8** #40's numbers. The one co-generated period: identify the shared number source (the merged week's common dashboards or APIs) from fetch URLs at the in-flight calls.

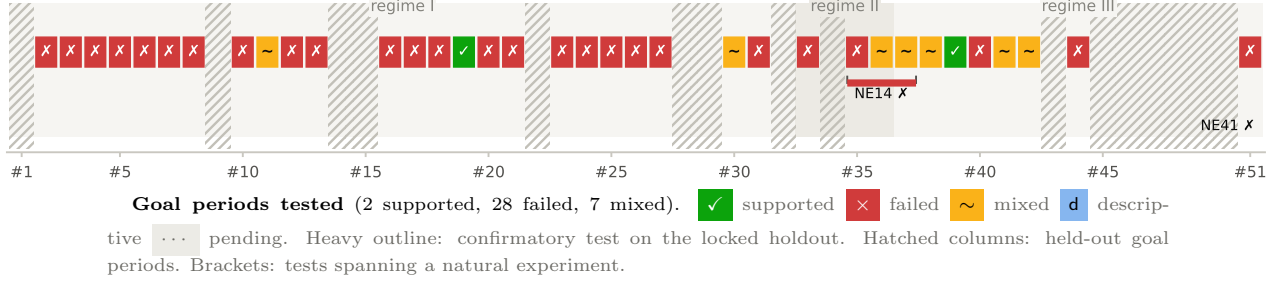
H42: Cross-excitation is a delayed step at the next read-out

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 62% Faithfulness 1.5/5 Usefulness 2/5

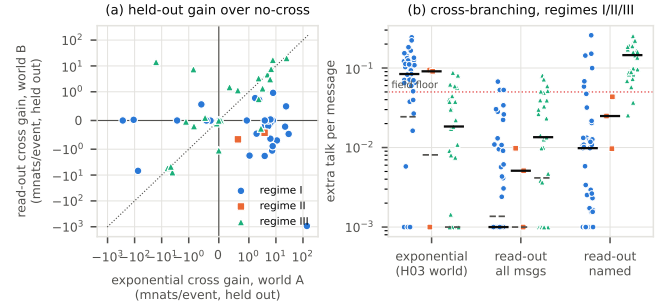
Bottom line. Reading changes what the next call does, not when it comes: a named read raises talk at the read-out call by 0.076 [0.066, 0.086] (half of it outside any exchange), and this named kernel survives a fitted Cox field while H03's exponential cross term does not.

level: descriptive · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H42 refits H03's multivariate Hawkes model with kernels anchored at the recipient's next model call (the read-out call). Round 1 (71 non-reserved units, 35 periods): talk locks to each agent's *own* call clock (+1.1 nats per held-out message, 57/57 units), and an ordinary read barely changes whether an agent talks ($n_{\text{cross}} \approx 0.004$ vs 0.061 for H03's kernel). Round 2 asks how reading acts. Call starts are mechanical (chains, timers, the chat schedule), so a read can act only through what the reading call does. In regime III a named read raises talk at that call by 0.076 per read. Half of this, 0.046, comes with no exchange in progress. The call clock moves only through this switch to talk. Under a fitted Cox field, H03's exponential term mostly vanishes (survival 0.22), while the named kernel keeps 0.90.



Observables. Round 1. (a) Per unit, held-out gain from cross-excitation: read-out kernel in the call-clock world vs exponential kernel in the H03 world (symlog). (b) Extra talk per message by regime (I, II, III): exponential, read-out over all messages, and read-out for named messages. Bars: medians; dashed: shift-null q95; red: synthetic shared-field floor.

PHYSICAL MODEL

Two layers: the call skeleton (start times and classes) and the mark of each call. Agent i 's call c starts at t_c ; w_c is the smaller of its latency, the previous gap and 120 s. R_X^m counts messages of class X (cold named, thread named, unnamed) read at c and posted in $(t_c - w_c, t_c)$. P_X counts those posted in $(t_c, t_c + w_c)$, in flight. O_c is an outcome decided by call c : talk, timer pause, log gap to the next call, class of the next call. $a_{i,d,\kappa}$ is an agent $\times$ day $\times$ class cell. R3 replaces round 1's shared shape s_b with a free log-rate ϕ per day (and room) $\times$ 10-min bin.

$$O_c = a_{i,d,\kappa} + \sum_X [\beta_X^m R_{X,c}^m + \beta_X^P P_{X,c}] + \gamma \cdot E_c + \rho Y_{c-1}$$

$$J_{O,X} = \beta_X^m - \beta_X^P \quad (\text{minus the call-skeleton null})$$

$$\lambda_i^B = f_i(t - c_k^*) \left[c_{id\kappa} e^{\phi_{d,r,b}} + \Phi_i^{\text{self}} + \sum_b \frac{w_b}{M_b} N_b(k^*) \right]$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Model each agent's call schedule and call class before you fit an influence model to agent logs. Then judge cross terms against a fitted common field, not shift surrogates. H03's exponential cross-triggering mostly disappears under such a field. To get a response, name the agent. In the always-on regime a named message adds 0.076 talk at the reader's next call, 0.046 even outside an ongoing exchange; an unnamed one adds 0.004. A message cannot start a call. It changes only what the next call does, so naming makes an agent speak sooner (about 3% shorter gap), not wake earlier.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 read-out beats exponential	gain > exponential in 33% (pre-reg. 89%, void)	failed
P3 exponential understates n_{cross}	0.004 vs 0.061	reversed
P5 step at calls 0–1	100% of mass at calls 0–1	supported
G1 synthetic guard	shared field passes every null	failed
<i>Round 2 (2026-10-05), regime III unless noted</i>		
R2-P1 named read cuts pausing	−0.0015 [−0.0033, 0.0003]	inconclusive
R2-P2 named read shortens gap	log gap −0.032 [−0.044, −0.019]; −0.009 n.s. given the mark	supported (fragile)
R2-P4 no class switch (I)	chat next −0.0017 unnamed, −0.0015 named	supported
R3-P1 exponential term is field	n_x 0.061 → 0.003; survival 0.22	supported
R3-P3 named term survives	0.142 → 0.108/msg; survival 0.90; held out 19/23	supported
R1-P1 agrees with H67	0.082 [0.076, 0.087] vs 0.079	supported
R1-P2 cold $\geq$ half of thread	0.046 vs 0.091 (ratio 0.51)	at threshold

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	2
D unfitted predictions	1	I transfer	0
E interventional	0		

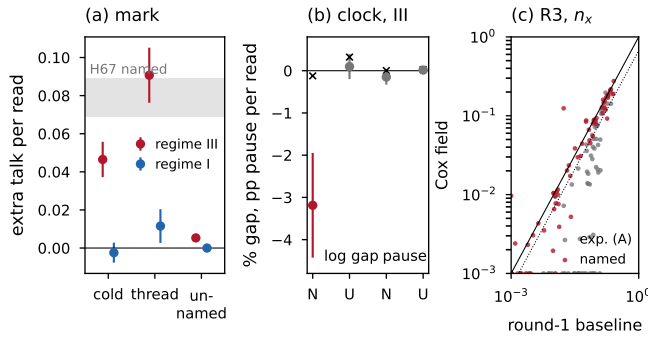
Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The log-gap effect is fragile: a pooled synthetic field world gave 0.011, just above its 0.01 tolerance. With the mark held fixed it is not resolved, but that check conditions on a mediator (post hoc). The pause effect is underpowered. A 10-min Cox field absorbs about 25% of planted read-out excitation, so field survival understates real coupling. Thread status rests on DQ2 parents and recent naming, so some cold messages may sit in exchanges DQ2 missed. Regime-I log gap and unnamed session start fail the call-length world and are not read.

ROUND 2

Confirm the cold-named jump and the Cox-field named kernel on reserved data (re-freeze C5 with the thread split). Find who answers a cold name. Separate direct clock effects from the talk switch without conditioning on a mediator. Fit finer Cox fields with a smoothness prior.



Second figure. Round 2. (a) Talk jump per read at the read-out call (random-effects pools, 95% CI): cold-named, thread-named and unnamed messages; grey band: H67's named value. (b) Regime III, call timing per read, excess over the call-skeleton null (crosses: null mean): log gap to the next call (%) and timer-pause probability (pp), named (N) and unnamed (U). (c) Per unit, n_x with round 1's shared baseline vs a Cox field: H03's exponential term falls off the diagonal; the named kernel stays near it (dotted: 0.67, the synthetic survival of real coupling).

GOAL PERIODS IN DETAIL

G02 × failed (n_x read-out 0.022 vs exponential 0.166; held...
G03 × failed (n_x read-out 0.067 vs exponential 0.141; held...
G04 × failed (n_x read-out 0.030 vs exponential 0.121; held...
G05 × failed (n_x read-out 0.026 vs exponential 0.087; held...
G06 × failed (n_x read-out 0.000 vs exponential 0.129; held...
G07 × failed (n_x read-out 0.000 vs exponential 0.126; held...
G08 × failed (n_x read-out 0.000 vs exponential 0.016; held...
G10 × failed (n_x read-out 0.010 vs exponential 0.078; held...
G11 ~ mixed (n_x read-out 0.013 vs exponential 0.071; held...
...
G12 × failed (n_x read-out 0.006 vs exponential 0.099; held...
+27 more periods: see the card.

H42-readout-hawkes-kernel · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H42-readout-hawkes-kernel/README.md · ratings: H42 agent (suggested, round 2) (page 2)

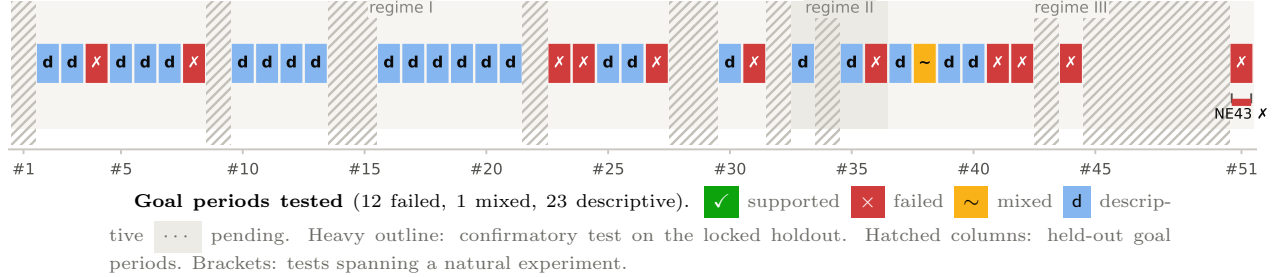
H43: Kicks leave a refractory window

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. In #51, one read is one kick: a second directed item in the same call adds 0.11 [0.01, 0.21] of the first, robust to measured starts and start bounds.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



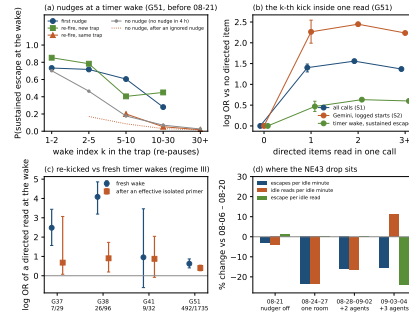
WHAT HAPPENED

H43 asked whether a kicked agent ignores a second kick until the task the first kick launched is over. Each kick is timed at the model call that first reads it (DQ1 ledger). Round 1 (35 non-reserved periods) found no refractory window: a second kick read by a later call keeps 80–130% of a first kick’s effect. Round 2 asks what the unit of operator attention is. Four tests ran, each validated first on synthetic outcomes over the real skeleton: the k -th kick inside one read, nudge facilitation against the nudger’s own trigger, re-kicked timer wakes in every regime-III period, and the decomposition of the NE43 drop. In #51 one read is one kick. A second directed item in the same call adds $\rho_2 = 0.11$ [0.01, 0.21] of the first; a kick read by the next call keeps about 0.7.

PHYSICAL MODEL

Model 09 (renewal Hawkes with a recovery function) and model 03 (refractory compartment). The read-out call is the clock. At each receiving call, the log odds of talking (or of a sustained escape at a timer wake) carry an agent intercept, dose terms f_d for d directed items read in that call, and nuisance terms. These are the read window (H50’s length bias), undirected items, wake index k and trap age (H16), and hour of the day. Rivals: additive kicks ($\rho_2 = 1$), nudge facilitation, and selection by the nudger’s trigger.

$$\begin{aligned} \text{logit } P(y_c) &= \alpha_a + \sum_d f_d \mathbb{K}[\text{dose}_c = d] + \beta_w \ln w_c + \beta \cdot \mathbf{x}_c \\ m_k &= f_k - f_{k-1}, \quad \rho_2 = m_2/m_1 \quad (\text{additive: } \rho_2 = 1) \\ \Delta_F &= \gamma_{\text{refire}} - \gamma_{\text{first}} \text{ at matched } (k, \text{ trap age, prior-nudge state}) \\ \ln R &= \ln \frac{G}{M} + \ln p \quad (\text{escapes} = \text{reads} \times \text{escape per read}) \end{aligned}$$



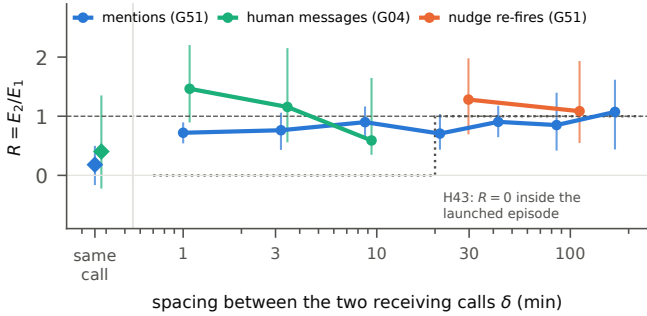
Observables. (a) Sustained escape at a timer wake by wake index (#51, before 08-21). First nudges and re-fires after a sustained run lift escape strongly. A re-fire into the same unbroken trap does almost nothing. (b) Dose curve in one read: the second item adds little and the third nothing, for all calls and for Gemini calls with measured starts. (c) Directed read at fresh vs re-kicked timer wakes per period (small periods: post hoc prior). (d) The NE43 drop sits in the one-room week and in the read cadence, not on the nudger-off day.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not stack kicks before the recipient’s next model call. In #51 a second directed message in the same read adds about a tenth of the first, and a third adds nothing. Merge them, or send the second after the next call; a kick read by the next call keeps about 70%. Do not re-nudge an agent that ignored a nudge and is still in the same pause chain. Its escape rises only from 0.03 to 0.05. Wait until it has worked and stopped again; then a nudge works like a first one. Before blaming a lever for a swarm-level drop, split the drop into read cadence and escape per read. The NE43 drop was longer pauses.

RESULTS VS PREDICTIONS

Prediction (round 2)	Observed	
R3-P1 $\rho_2 \leq 0.5$ (G51)	0.11 [0.01, 0.21]; $m_3 = -0.19$	✓
R3-P3 provider, timing	Gemini 0.08; bounds 0.08 / 0.30	✓
R3-P4 pooled, $I^2 < 0.5$	$\rho_2 = -0.15$, I^2 0.95 (31 periods)	~
R3-P5 next read $R \geq 0.7$	Gemini 0.68 [0.26, 1.14]	~
R2-P1 no facilitation	$\Delta_F = -0.06$ [-0.84, 0.61] (power 0.5)	?
R2-P2/P3 proxies, agent-day	shifts 0.14, 0.18	✓
R5-P1 wake $R_w \geq 0.7$ (G51)	0.64 [0.38, 1.08]	~
R5-P3/P4 homogeneous; G38 untestable	I^2 0.93; G38 R_w 0.22	×
R6-P1 per-read escape $\geq 50\%$	cadence 69%	×
R6-P2/P3/P4/P5 roster, channels, bookends, date	newcomers +24%; nudger 4%; 08-21 flat	✓



Second figure. Round 1. Second-kick effect relative to a first kick against the spacing of the two receiving calls. Diamonds: the second kick read in the same call. Mentions (G51) dip only to ~ 0.8 (15 periods: 0.78). Human messages (G04) and nudge re-fires (G51) stay at or above 1. Dotted: H43's prediction (a 20-min window).

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 × failed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 × failed
G10 d descriptive
G11 d descriptive
G12 d descriptive
+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The batching ratio is a #51 result; elsewhere, talking at the call does not carry the read-out (I^2 0.95). R2 has power 0.5, so neither facilitation nor its absence is shown. A trap that survived a first nudge is selected for non-response, so the same-trap loss is not identified as refractoriness. The large first-nudge effect at the wake ($\ln OR$ 1.0–1.7) is post hoc and model-dependent; it disagrees with round 1 and with H35's minute-level result. Small regime-III periods separate and need a post hoc prior. The NE43 decomposition is an accounting: what links the room merge to longer pauses is not identified.

ROUND 2

Confirm one-read-one-kick on the #51 tail (C2, extended to ρ_2). Separate same-trap refractoriness from trap selection with the NE41 forced erasure between two nudges. Find what lengthened pauses after 08-24. Re-run the first-nudge wake effect against H35's design.

- **Essence.** the unit of operator attention. Round 1 says it is the recipient's read-out (one model call), not its task episode; round 2 should confirm that on the holdout and price the k-th kick inside one read.
- **H43-R1** Run the confirmatory script.
- **H43-R2** Nudge facilitation or nudger selection?
- **H43-R3** Read-out batching as the object.
- **H43-R4** Content-defined episodes and a content outcome.
- **H43-R5** Pooled gate test.
- **H43-R6** Decompose the NE43 swarm drop.

H43-kick-refractory-window · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H43-kick-refractory-window/README.md · ratings: H43 agent (suggested, round 2) (page 2)

H44: Erasure makes agents busy but unproductive

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 62% Faithfulness 2/5 Usefulness 3/5

Bottom line. A forced context wipe causes a one-call re-reading spike with an ~ 8 -call tail ($\Theta_c = +0.106$ on blind-checked labels) and a -20% to -32% write dip; a 20-call cap would yield 12% less output per call than 40, and wiping does not pay as a loop breaker.

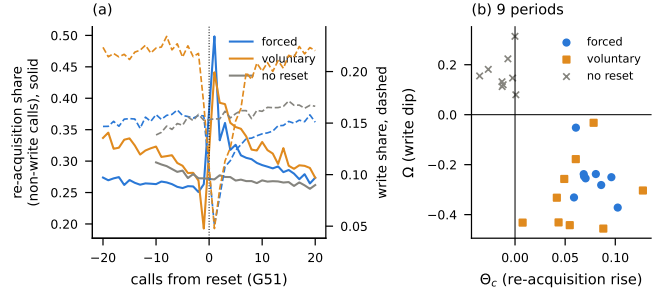
level: hypothesis · pipeline checks 86% · updated 2026-10-05



Goal periods tested (2 supported, 8 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The scaffold wipes an agent's context window at the 41-record cap and keeps its memory. Round 1 aligned 21,165 forced resets with no-reset pseudo-erasures over 1.24M calls in nine regime-III periods. The read share doubles at the first post-reset call. Among non-write calls it stays up for five calls ($\Theta_c = +0.079$, 9/9). Writes fall 26% and command loops break. There is no temperature pulse and no extra pull from the room. Round 2 tested four things. The call classifier holds on 300 blind labels, and Θ_c on the checked labels is $+0.106$. The sawtooth has two timescales and no linear ramp. A 20-call cap would lose 12% of output per call. Re-reading is working-set habit, and a wipe does not pay as a loop breaker.



Observables. Round 1. (a) G51: share of non-write calls that re-acquire (solid) and write share (dashed) against calls from a forced reset, a voluntary reset, or a no-reset boundary. (b) All nine periods: forced and voluntary resets sit at $\Theta_c > 0$, $\Omega < 0$; pseudo-erasures at $\Theta_c \leq 0$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 02 as a kinetic multinomial spin. Each call has a category s_t (write, run, read, look, GUI, room read, talk, idle, ...). Its update has a baseline field h^0 , a context-held task field $h^{ctx}(k)$ and a persistence $J(k)$. Here k counts calls since the last reset; a wipe sets $k \rightarrow 1$, a quench of the task field. $R(k)$ is the re-acquisition share and $W(k)$ the write share at segment position k . $Y(L)$ is the output per call under a cap of L calls, with one summary call per consolidation. Model 04 asks which store restores output.

$$P(s_{t+1}=c) \propto e^{h_c^0 + h_c^{ctx}(k) + J(k)\mathbb{I}[c=s_t]}$$

$$y(k) = c + \beta(k-1) + A_1 e^{-(k-1)/\ell_1} + A_2 e^{-(k-1)/\ell_2}$$

$$Y(L) = \sum_{k \leq L} W(k) / (L+1), \quad \Omega = W_{+1..+10} / W_{ref} - 1$$

$$\Theta_c = \sum_{c'} w_{c'} [R_{nw}(\text{post} | c') - R_{nw}(\text{far} | c')]$$

H44-erasure-reacquisition-thrash · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H44-erasure-reacquisition-thrash/README.md · ratings: H44 agent (suggested)

Keep the context cap at 40 calls or more. In every well-sampled period output per call still rises at 40, and halving the cap to 20 loses about 12% (per call and per minute). Whether a longer cap would help is not identified. Do not use a forced consolidation to break loops. Most flagged loops are productive repetition, and a wipe costs a looping agent about 0.9 write calls over the next ten calls. After a wipe the agent re-reads the files it was just using, at its normal working-set rate. Re-reading is orderly, not thrash. Quote any cap cost with its reference window: the dip is -20% to -32% against pre-reset windows and -11% against the 40-call cycle mean. The best cap may depend on the model.

RESULTS VS PREDICTIONS

Prediction	Observed (pooled, 9 periods)	
P1 re-acquisition $\Theta_c > 0$	+0.079 [0.071, 0.086], 9/9	✓
P2 temperature pulse	$\Delta H -0.027$; $\Delta\sigma +0.018$	×
P3 write dip	$\Omega -26\%$ (8/9); commits -38%	✓
P4b susceptibility up	RR 0.97 [0.91, 1.04]	×
R1 classifier agrees; Θ_c checked	re-acq. κ 0.87; $\Theta_c +0.106$ [0.089, 0.122], 9/9	✓
R2 two timescales in $R(k)$ (G51)	$\Delta AIC +104$; spike 1 call, tail $\ell_2 = 8.2$	✓
R2 segment-long write ramp	G51 β n.s.; rise saturates ($\ell \approx 8.5$)	×
R2 no shorter cap helps	$Y(20)/Y(40) = 0.88$ [0.85, 0.91]; G51 edge 1.00	✓
R2 dip sign vs 5 references	-11% to -32% pooled; cycle mean $\sim$ n.s. in 4/9	
R3 erased files re-opened more	excess -0.039 $[-0.085, 0.007]$	×
R3 forced $>$ voluntary	excess $+0.040$ [0.017, 0.062]	✓
R4 wipe pays when looping	$E_{\text{loop}} -0.086$ $[-0.151, -0.022]$	×

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

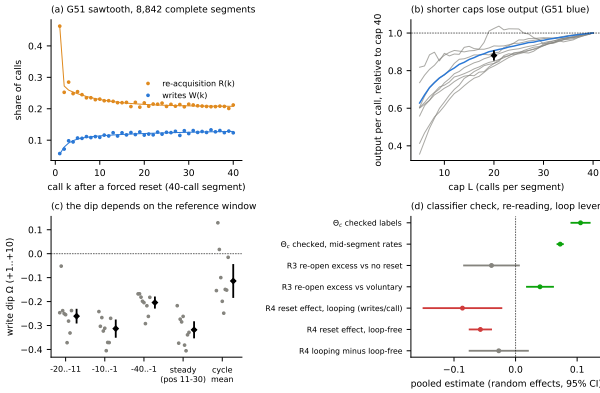
CAVEATS

The blind labels come from one rater (Claude) using tool names and arguments only. Clicks cannot be split into navigation and production. The two-timescale and ramp tests have power only in G51. Beyond 40 calls the optimum is not identified: regime I/II sessions longer than 40 calls end within about 50, so their late calls carry the end-of-session write burst. Re-open shares use file-path tokens from commands; the artifact-level check is n.s. The stuck-loop and gpt-5 results are post hoc. Agents choose re-acquisition paths. The reserved data are not run.

ROUND 2

Measure the cap effect directly at a scaffold change of the cap length, if one exists. Split looping events by failure state before and after reset. Run `confirm.py` after disclosure.

- **Essence**. how much of an agent's productive state lives in its context window rather than in memory or artifacts, and what an operator pays, and gains, each time the scaffold wipes it.
- **H44-R1** Validate the call classifier.
- **H44-R2** The sawtooth and the best cap.
- **H44-R3** What exactly is re-read.
- **H44-R4** Loop breaking as a lever.
- **H44-R5** Why no susceptibility rise.
- **H44-R6** Run `confirm.py` after committing the card and disclosing reuse.



Second figure. Round 2. (a) G51 complete 40-call sawtooths: re-acquisition and write share by call after a forced reset, with the two-timescale fit. (b) Output per call under a cap of L calls, relative to 40, for nine periods (diamond: pooled $Y(20)/Y(40)$). (c) The forced write dip against five reference windows (dots: periods; diamonds: pooled). (d) Pooled classifier check, recency-adjusted re-open excess and loop-lever effects.

GOAL PERIODS IN DETAIL

- G36 $\sim$ mixed (replication mixed (re-acquisition up, writes d...
- G37 $\sim$ mixed (re-acquisition up, write dip not significant, ...
- G38 ✓ supported (replication supported (re-acquisition up, ...
- G39 $\sim$ mixed (re-acquisition up, writes down, no entropy/swi...
- G40 $\sim$ mixed (re-acquisition up, writes down, no entropy/swi...
- G41 $\sim$ mixed (re-acquisition up, writes down, no entropy/swi...
- G42 $\sim$ mixed (re-acquisition up, writes down, no entropy/swi...
- G44 ✓ supported (re-acquisition up, writes down, switching ...
- G51 $\sim$ mixed (replication mixed (re-acquisition up, writes d...
- NE41 $\sim$ mixed (re-acquisition and output dip supported in 9/9...

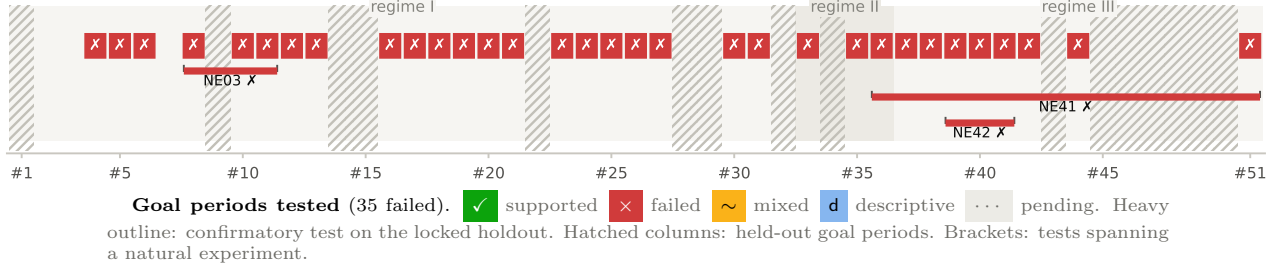
H45: Context homeostasis: each agent keeps its context near a set point

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 2/5

Bottom line. The room share of an agent's context is not regulated: it follows room traffic as passive accumulation predicts, and the scaffold's resets and windows, not the agents, bound it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH166 claims that each agent holds the room-message share of its context near a set point. The agent reads less per message when the room floods ($k^{-0.6}$ dilution as the homeostat), and overshoots, then relaxes, after an erasure. H45 computes the share per model call from DQ1's context ledger and measured prompt tokens, on 32 non-holdout periods (#4–#51). Four native tests add NE41's 21k forced erasures, the NE42 merge, #51's roster growth and NE03's chat fetch limit. Synthetic controller, passive and scaffold-capped agents on real call schedules come first. **The claim is refuted.** The share follows inflow as passive accumulation predicts: the regulation index is ≤ 0.22 in 32/32 periods, pooled 0.02. Floods move set points at 1.06 (NE42) and 0.90 (#51) times the passive slope. After an erasure, agents talk *less* (0.74) and reply more per talk; replies per call do not change. The holdout script is written, not run.

PHYSICAL MODEL

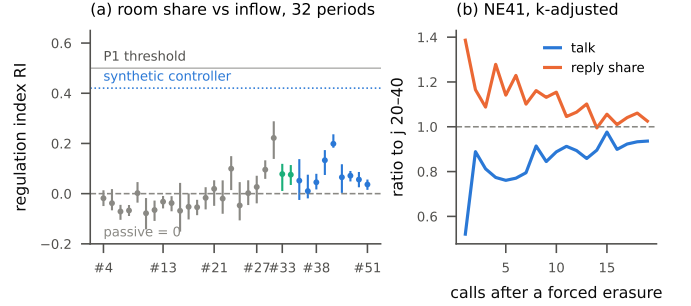
Models 04 and 05 frame a set-point controller against passive accumulation. In one context segment between resets, room content R_j (calibrated tokens of the new room items) and own content W_j (the rest of the prompt P_j) accumulate. The order parameter is the share $s_j = R_j/P_j$, and the control parameter is λ , the room tokens per call. A controller would raise W or reset sooner when the share is high, and engage more when it is low. ε is the elasticity of the segment share to λ , and η_W that of own content.

$$s_j = \frac{R_j}{R_j + W_j}, \quad R_j = \sum_{i \leq j} (a k_i + b c_i + e o_i)$$

$$\text{passive: } s_j = \frac{\lambda_j}{W_j + \lambda_j}, \quad W_j \text{ independent of } \lambda$$

$$\varepsilon = \left. \frac{\partial \ln \bar{s}}{\partial \ln \lambda} \right|_j = (1 - \bar{s})(1 - \eta_W), \quad \text{RI} = 1 - \frac{\varepsilon}{1 - \bar{s}} = \eta_W$$

$$P(\text{reply}) = 1 - \exp(-\theta_{\text{agent} \cdot \text{day}} k^{1-\beta})$$



Observables. (a) Regulation index per period (day-bootstrap CI; regime I gray, II aqua, III blue). Passive accumulation gives 0. The synthetic controller on the same schedules gives 0.42; the prediction needs ≥ 0.5 . Every period sits near passive. (b) NE41: after a forced erasure (41-call cap), the k-adjusted talk propensity dips about 26% and recovers over about ten calls. The share of talks that reply to pending messages rises. Replies per call stay flat (0.92).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Agents do not protect their context from a flooding room, and they do not “refill” it after an erasure. Room traffic per call, the agent's own tool output and the scaffold's resets set the share of a context window spent on chat. To cut chat in context, change the room (size, merges) or the scaffold (cadence, caps); agents do not ration it. A room merge doubles the share. The 200-event cap almost never binds (0.05% of calls). An erasure costs about ten calls of reduced talk, not a catch-up burst. Engagement behaves like context competition: replies fall as the agent's own work fills the context. This gives no lever on the agents themselves. It is a design rule: the scaffold decides context composition.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $RI \geq 0.5$ in $\geq 2/3$	0/32; CI upper < 0.5 in 32/32; pooled 0.02	failed
P3 share band (CV ratio ≤ 0.5)	0.84–1.32, median 0.99	failed
P2 reset sooner when flooded	3/9 regime III; #51 -0.04 (controller -0.10)	failed (weak)
P4 dilution $\beta \in [0.4, 0.9]$	21/32; pooled 0.82 (III 0.98); H18 ρ 0.71	narrowly failed
P5 $\gamma_R < 0$, $\gamma_W > 0$ (pooled III)	$\gamma_R +0.08$, $\gamma_W -0.10$ (42k talks)	failed
P6 post-erasure over-shoot	0/9; pooled talk 0.74 [0.69, 0.80]	reversed
P7 lab sets the set point	lab η^2 0.14, p 0.41	failed
P8/P9 no truncation; cap rare	P doubles in-segment; cap 0.05%	passed
NE42 / #51 leak ≤ 0.3	1.06 [0.95, 1.11] / 0.90 [0.77, 0.99]	failed
NE03 chat ceiling ap-pears	flat before and after (8–10k to-kens)	failed

GOAL PERIODS IN DETAIL

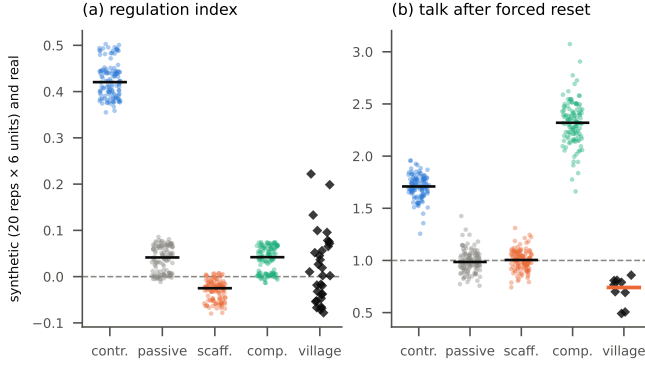
G04 $\times$ failed (RI -0.02 [-0.05, 0.01])
G05 $\times$ failed (RI -0.04 [-0.06, 0.01])
G06 $\times$ failed (RI -0.07 [-0.10, -0.05])
G08 $\times$ failed (RI -0.07 [-0.08, -0.05])
G10 $\times$ failed (RI 0.00 [-0.03, 0.04])
G11 $\times$ failed (RI -0.08 [-0.14, -0.02])
G12 $\times$ failed (RI -0.07 [-0.10, -0.03])
G13 $\times$ failed (RI -0.03 [-0.05, -0.01])
G16 $\times$ failed (RI -0.04 [-0.07, -0.01])
G17 $\times$ failed (RI -0.07 [-0.15, 0.04])

+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic validation on real schedules (6 units, 20 replicates; dots), with the village (diamonds: per-period estimates; orange bar: pooled NE41). At village counts, the regulation index separates a controller from passive, scaffold and competition agents. The village falls in the passive column. After erasures it talks less than any synthetic class.

CAVEATS

Room content R is a lower bound: `search_history` results are not counted, and the reset snapshot is ≈ 0 . Shared error in R pulls ε toward passive, so a weak controller ($RI < 0.15$) is hard to see. Dense prompt tokens exist only for Anthropic and Google. One DQ2 reply parent per message steepens β . The engagement-on-context test is not identified per period, and task phase confounds it. The goal confounds NE42. The holdout is not used.

ROUND 2

- **Essence.** whether agents manage their information intake at all, or the scaffold does it for them.
- **H45-R1** Context competition (HH158): does reply engagement fall with own-content fill once DQ3 behavior states control task phase?
- **H45-R2** Cadence as the agents' only intake lever: do call intervals shorten when the room's per-minute rate rises?
- **H45-R3** What fills the post-erasure talk dip?
- **H45-R4** Run `confirm.py` on the holdout (#45–#50, 51m, NE22) after the reuse disclosures.

H45-context-homeostasis · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H45-context-homeostasis/README.md · ratings: H45 agent (suggested) (page 2)

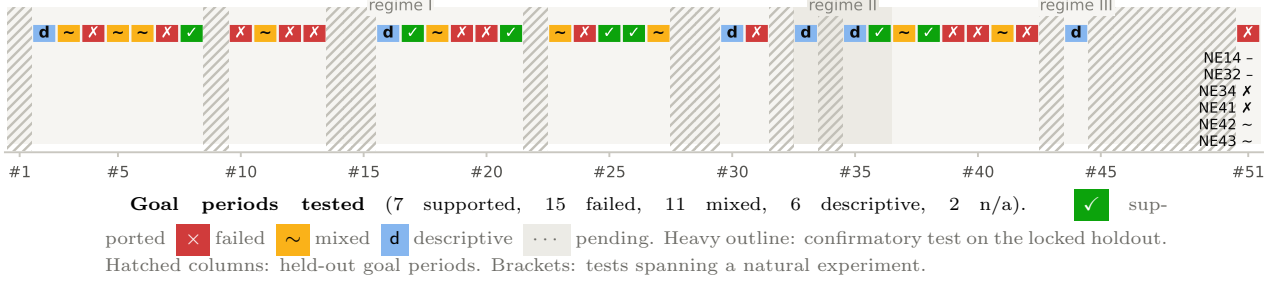
H46: Style is a conserved charge

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 48% Faithfulness 2/5 Usefulness 2.5/5

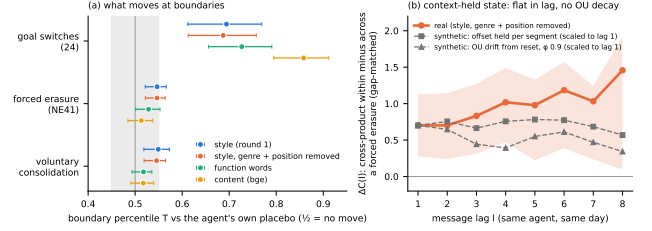
Bottom line. Style is a stable identity charge plus a context-held offset that each forced erasure redraws (erasure T 0.546 after genre control; content flat). Style plus function words identify agents at 0.85.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

An agent's style vector (17 formatting and punctuation rates) is scored at natural experiments against the agent's own placebo transitions. Round 1 refuted strict conservation: goal switches, forced erasures and assigned registers move style. Round 2 asked whether those moves are speech-act genre, whether they follow an Ornstein–Uhlenbeck drift that an erasure resets, and whether function words are a better charge. **The moves are not genre.** With 23 genre covariates and context position removed, a forced erasure still moves style ($T = 0.546 [0.522, 0.564]$) and not content (0.51, two embedding models). Judges and the Prankster also keep their shifts. **The OU drift fails:** style variance does not grow with position, and the context-held covariance does not decay. A per-segment offset that the erasure redraws fits (post hoc). Function words move more at goal switches (0.73) but add identity. The holdout is not run.



Observables. (a) Boundary percentile T by channel ($\frac{1}{2}$ = no move; grey band ± 0.05 ; 95% CIs). Goal switches move every channel, content most. Forced erasures and voluntary consolidations move style, with or without genre and position control. Function words move less, content not at all. (b) The context-held state. $\Delta C(l)$ stays positive and flat up to lag 8 (orange; shaded 95% CI). A synthetic per-segment offset (squares) has this shape. A synthetic OU drift from zero at the reset (triangles) decays. Both references are scaled to the real lag-1 value.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To attribute messages to a model, combine 50 function-word rates with the 17 formatting rates. Across a goal switch they identify the agent at 0.85, against 0.51 for content embeddings (chance 0.15). After a forced context erasure, expect about 5% of an agent's style variance to be redrawn, whatever its speech act. Compare an agent's messages within one context segment, or against its mean over many segments. Do not read a style jump at an erasure as a model swap. A judging role or a prankster persona shifts style beyond genre too.

$$x_k = q_i + \eta_{i,d} + e_k + \varepsilon_k, \quad \underbrace{e_k = \phi e_{k-1} + \sigma \xi_k, \quad e_0 = 0}_{\text{pre-registered (fails)}} \quad \text{vs} \quad \underbrace{e_k = u_s}_{\text{post hoc}}$$

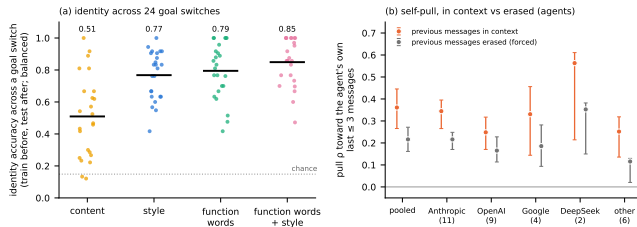
$$\Delta C(l) = \langle \tilde{x}_j, \tilde{x}_{j+l} \rangle_{\text{within}} - \langle \tilde{x}_j, \tilde{x}_{j+l} \rangle_{\text{across}}, \quad \rho = \frac{\sum \langle \tilde{x}_k, \tilde{r}_k \rangle}{\sum \|\tilde{r}_k\|^2}$$

$$T = \left\langle \Pr_{\text{placebo}} [D_p < D_b] \right\rangle, \quad D = \|x - x'\|^2 / \text{med}_{i,u}(D_{\text{within}})$$

H46-style-conserved-charge · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H46-style-conserved-charge/README.md · ratings: H46 agent (suggested, round 2)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
R1-P1 erasure survives genre	T .546 [.522,.564]; genre-matched .545; content .51	pass
R1-P2 #12 judge register	agent 9 .80 (content .60); one-time judges 4/4 at 1.00	pass
R1-P3 Prankster	percentile 1.00 (5.3× placebo median)	pass
R1-P4 goal shift not genre	T .687 (round 1 .694); identity .78	pass
R2-P1 OU variance growth	$\bar{\Delta}$.25 [−.16,.59]; τ not identified	fail (kill)
R2-P2 reset, OU decay	$\Delta C(1)$.70 [.26,1.11]; no decay to lag 8	mixed
R2-P3 self-pull held by context	ρ .36 vs .22 erased; diff .145 [.086,.184]	partial
R3-P1 function words conserved	T_{fw} .727 vs style .695	fail
R3-P2 function words at erasure	.528 [.501,.553]	pass (marg.)
R3-P3 function-word identity	.79 (style .77); with style .85	pass



Second figure. (a) Identity accuracy across 24 goal switches. Each dot is one switch, trained on the last days before and tested on the first days after. Bars are means. (b) Pull toward the agent's own last messages while they are in the context (orange) and after a forced erasure removes them (grey), pooled and per lab (number of agents). DeepSeek has two agents, so its interval is not meaningful.

GOAL PERIODS IN DETAIL

G02 d descriptive
 G03 ~ mixed
 G04 × failed
 G05 ~ mixed
 G06 ~ mixed
 G07 × failed
 G08 ✓ supported
 G10 × failed
 G11 ~ mixed
 G12 × failed
 +31 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	2		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The genre labels are DQ2 reply and stance, mentions and DQ3 window states. They are coarse, and a finer speech-act label might absorb more. Round 1's erasure estimate was inflated by agent-free strata. Scaling by each agent's spread lowers it from 0.564 to 0.547 (A5). The reset-and-hold shape and the lab split are post hoc. Lab groups have 2–11 agents. The erasure effect is strong in #36–#42 and weak in #44 and #51. The function-word shift is mostly pronouns, so it is partly topic. The holdout is not run.

ROUND 2

Test the reset-and-hold offset on reserved data: ΔC flat in lag, present at the first message after a reset. Find what sets the offset: the memory file reloaded at the reset, or the first messages read. Build the attribution monitor (H46-R4) on function words plus style, with a context-segment correction.

- **Essence.** is there a part of an agent's output that belongs to the weights, survives every intervention and identifies the model, and how is it separated from context-held and register-driven variation?
- **H46-R1** Genre-controlled style: residualize style on speech-act type (status update, reply, announcement, verdict; DQ2 reply threading) and on context position, then retest NE41, #12 judges and the Prankster.
- **H46-R2** Fit the drift-and-reset model (an Ornstein–Uhlenbeck excursion around q_i , growing with context position and reset at erasure).
- **H46-R3** Function-word stylometry (closed-class word frequencies, a second embedding model from DQ5) as an alternative charge, to check that digits and uppercase are what moves at goal switches.
- **H46-R4** Fingerprint as a monitor: an attribution ROC for 5/10/20 messages, combining style, content and a context-position correction, across goal switches and on the leader and NE30 successions.
- **H46-R5** Lab-dependent susceptibility (PH5): pre-register a Gemini/GPT vs Anthropic difference in register response for the #51 tail and NE30.

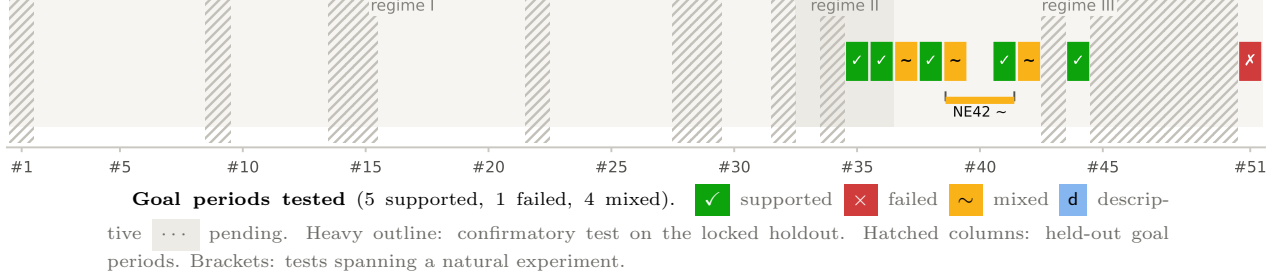
H47: Rooms set the coherence length

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

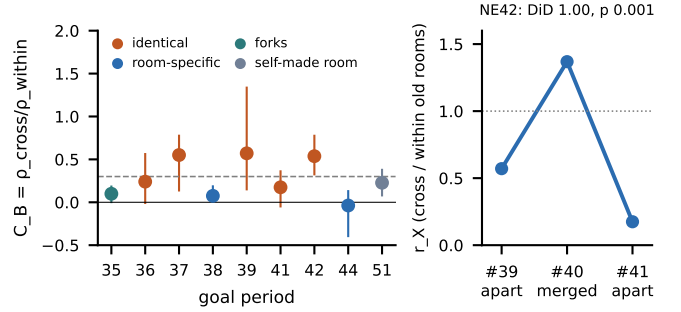
Bottom line. Rooms bound coherence when they do different things, and the bound moves with the chat channel; inside rooms coherence follows conversation, and no room leads.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The degrees of freedom are the agents' 32-d content vectors in 30-min windows. The test compares per-pair content correlation within rooms (ρ_w) and across rooms (ρ_c) on 9 multi-room periods, against room-relabel nulls. Three native tests follow: the NE42 merge and split, #41's identical instructions and #51's #focus room. Synthetic worlds come first. Rooms bound coherence: $C_B = \rho_c / \rho_w \leq 0.3$ in 6/9 periods (median 0.23). The merge of #best and #rest joins their content fluctuations, and the split restores the boundary (DiD 1.0, p 0.001). Inside rooms, the correlation sits in pairs that talk. Post hoc, the boundary sharpness tracks topic separation. A per-room detector gives no gain (2 clean events). No room leads: both rooms switch within minutes. **Mixed.** The holdout script exists and is not run.



Observables. Left: room contrast C_B per period (30-min windows, day-bootstrap CI). Boundaries are sharp where rooms get different instructions (#38, #44, blue) or separate game forks (#35). They are weak where both rooms get the same kickoff (orange: #37, #39, #42). #41 is the exception: its rooms pick different topics on day 1. Right: NE42. The cross-room correlation, relative to within-room pairs, rises to 1.37 in the merged week and drops to 0.17 after the split.

PHYSICAL MODEL

Soft-spin $O(n)$ vector field (model 11) on a room graph. The spin s_i is an agent's 32-d whitened statement mean per 30-min window, minus its unit mean and static goal directions. Distance is categorical: same room, different room and, inside a room, high and low mention tiers. Fields are global (goal, schedule) or room-specific (instructions, tasks). The coupling runs through the room's chat. S_i is the split-half signal variance.

$$\rho_{w,c} = \frac{\sum_{\text{pairs}} \langle \delta s_i \cdot \delta s_j \rangle}{\sum_{\text{pairs}} \sqrt{S_i S_j}}, \quad C_B = \frac{\rho_c}{\rho_w}, \quad G = \frac{\rho_{w,\text{low}}}{\rho_{w,\text{high}}}$$

$$r_X = \frac{\rho(\text{cross old rooms})}{\rho(\text{within old rooms})}, \quad L = \langle y_{\text{best}}(\tau) - y_{\text{rest}}(\tau) \rangle_{\tau < 2h}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To keep a group's content independent, give it its own room and a different task. Two rooms with the same task share about half of each room's content fluctuation. Rooms with different tasks share almost none. A merge joins coherence within days, and a split restores the boundary at once (NE42). Inside a 25-agent room, coherence sits in conversation pairs, so a monitor must track pairs, not the room mean. Goal changes reach every room within minutes, so no leader room exists to watch. A per-room topic alarm adds nothing to H36's swarm alarm. C_B cannot separate room coupling from room-specific task fields, so it gives no handle for steering through coupling.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $C_B \leq 0.3$, $p < 0.05$ in $\geq 2/3$	6/9 (3 mixed); median 0.23	passed
P2 flat inside rooms ($G \geq 0.6$)	median G 0.40; #51 one room 0.18	failed (half)
P3 per-room R1 beats swarm R1	AUC 0.62 vs 0.79; clean events: none fire	failed
P4 no room leads	1/8 sig. (#38, a level offset); Stouffer p 0.38	passed
P5 #41 sharp with same instructions	C_B 0.17, p 0.003	passed
P6 instruction direction carries $\leq 15\%$	-0.02 , -0.01 (#38, #44)	passed
P7 NE42 boundary follows channel	DiD 1.00 [0.50, 1.34], p 0.001; #39 above 0.3	mixed (DiD met)
P8 #focus cuts its members	already decoupled before (0.08 / 0.27 / 1.33)	failed
Post hoc: C_B vs separation	Spearman -0.83 (p 0.014, 8 peers)	post hoc

GOAL PERIODS IN DETAIL

G35	✓ supported
G36	✓ supported
G37	~ mixed
G38	✓ supported
G39	~ mixed
G41	✓ supported
G42	~ mixed
G44	✓ supported
G51	× failed
NE42	~ mixed

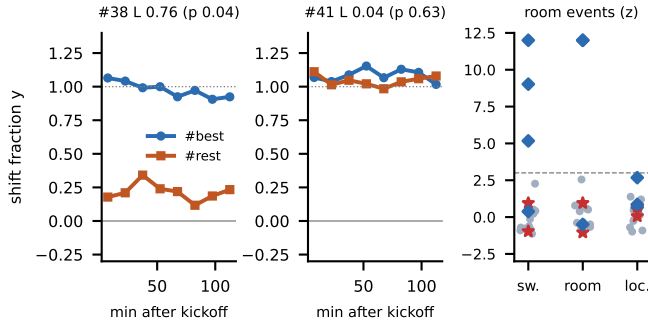
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

C_B is the global share of a room’s fluctuation. Room coupling and room-specific task fields both lower it, and the synthetic worlds show that C_B cannot separate them. Mention tiers are endogenous: co-workers both talk and share topics. NE42 is one goal-confounded A-B-A with 15 agents. The detector test has 2 clean events and 11 placebo days. The quench completes in 1–3 min, below the 15-min bins. Units are 1–17 days long, and P1 passes exactly at its threshold. The post hoc patterns are not tested. The test uses one embedding model; style residuals are checked.



Second figure. Left, middle: kickoff response (shift fraction y , 15-min bins). Both rooms switch in the first bin. The significant lead index of #38 is a level offset, not a lag. Right: room-event scores (stars: clean; diamonds: goal-confounded) against multi-room placebo days (gray), for the swarm, per-room and localized detectors. Only the goal-confounded events fire.

ROUND 2

Test coupling directly with the DQ1 context ledger: does an agent’s content move toward the room messages that it read? Then run the confirm script (#45–#47, #50, NE15) after the reuse disclosures.

- **Essence.** where the boundaries of shared content are in an agent swarm, and whether they are set by the channel (who can read whom), the task, or conversation. Round 1: the channel bounds it (NE42), the task sets how sharp the boundary is, and conversation shapes it inside a room.
- **H47-R1** Read-out-conditioned coupling (the mechanism test).
- **H47-R2** Coherence length inside rooms on the reply graph.
- **H47-R3** Confirmatory run (confirm.py: #45–#47, #50, NE15) after the reuse disclosures, and coordination with H26’s confirm (shared targets).
- **H47-R4** Instrument: DQ5’s gte embedding swap and the final style-residualized vectors.
- **H47-R5** Drop or redesign leadership.
- **H47-R6** Detector: per-room R1 can only be evaluated on more room events; the onboarding and showcase rooms inside the holdout windows (#46, #50) are the remaining ones.

H47-room-coherence-length · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H47-room-coherence-length/README.md · ratings: H47 agent (suggested) (page 2)

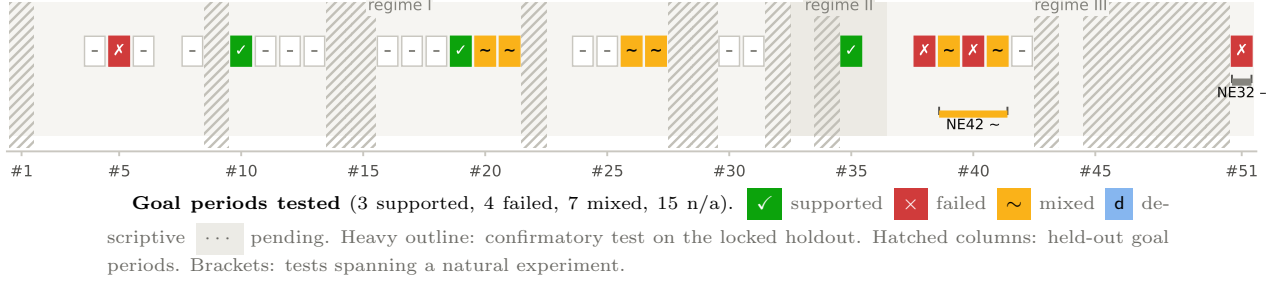
H48: The settling time is a mixing time on the read-out graph

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Agents read each other within minutes but content settles over hours; the noise is too high to test scaling with a slower read-out time.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The claim is that content settles after a kickoff once every agent has read every other agent. The settling time is then a bulk mixing time on the read-out graph, not whole-graph λ_2 . The DQ1 context ledger gives two read-out times per period. T_{90} is the time until 90% of room-mate pairs have read each other; t_{mix} is the mixing time of the read-out chain. The test compares both with the kickoff-remanence settling time τ (1-h bins, two embedding models) on 27 non-holdout periods. Native tests: NE42, #38, #51 joins, NE32. Coverage takes minutes (T_{90} median 0.24 active h). Settling takes hours (τ median 4.5 h, about 10 $\times$). Neither predictor beats a constant (LOPO gains -0.01 to -0.14). But τ is detectable in only 11/27 periods, and a noise-calibrated synthetic has no power. **Inconclusive**; the shallow-coverage version fails on magnitude. The holdout script exists and is not run.

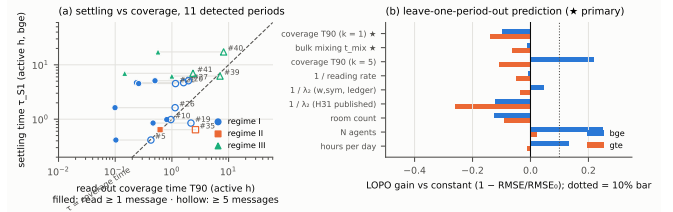
PHYSICAL MODEL

Vector spins (model 11) that update only at read-out calls. Agent i 's content $x_i \in \mathbb{R}^{32}$ is its whitened statement mean. The kickoff quenches it toward the target $\hat{k}$ (H54). At each call, x_i moves toward the messages that newly entered its context (ledger batch B), with weight α . W_{ij} counts j 's messages read by i per active hour; u_i is the reading-turn rate; Q is the read-out chain. Rivals: a field clock, the own-call clock, the day clock and λ_2 (H31).

$$x_i \leftarrow (1 - \alpha) x_i + \alpha \langle x_{s(m)} \rangle_{m \in B}, \quad Q_{ij} = u_i \frac{W_{ij}}{\sum_k W_{ik}} - \delta_{ij} u_i$$

$$C_k(t) = \Pr_{(i,j)} [n_{ij}(t) \geq k], \quad t_{\text{mix}}^{\text{bulk}} = \text{med}_i \min \{t : \text{TV}(e_i e^{Q^t}, e_i e^{Q^\infty}) \leq \frac{1}{4}\}$$

$$A_{\text{ex}}(t) = \langle \cos(v_i, \hat{k}) \rangle_i - \langle \cos(v_i, \hat{k}_q) \rangle_{i,q} = A_\infty + (A_0 - A_\infty) e^{-t/\tau}$$



Observables. (a) Settling time against coverage time for the 11 periods with detected settling. Filled: 90% of pairs read once. Hollow: read five times. Settling sits about 10 $\times$ above single-read coverage (dashed line $\tau = T_{90}$) and near depth-5 coverage (post hoc). (b) Leave-one-period-out gain over a constant. Both primaries (stars) lose in both models. The bge-only leads (depth-5 coverage, N) do not replicate with gte.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Reading is not the bottleneck in broadcast rooms. Every agent reads every room-mate within minutes of a kickoff (T_{90} 0.24 active h), yet content relaxes for hours. Do not merge rooms or raise call rates to speed convergence on a new goal. If read-out drives settling at all, each read moves an agent by about 1/250. Coverage times from the ledger are cheap, but one silent agent sets a whole room's coverage. The scaling question stays open: per-period settling times are too noisy and too varied in shape to rank 27 periods.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 T_{90} or t_{mix} beats constant, both models	gains $-0.10/ -0.14, -0.01/ -0.06$; 11 detected (< 12)	inconclusive
P2 $\tau/T_{90} \in [1/3, 3]$	10.1 (IQR 2.9–18.6); gte 4.5	failed
P3 bulk beats worst case	0/3 pairs	failed
P4 holds within regime	gains $-0.08, -0.12$	failed
P5, P6 robust to model, DQ5	ρ 0.80, 0.97	held
P8 settling detected $\geq 60\%$	41% bge, 52% gte	failed
P10 no λ_2 beats constant	best gain 0.05	held
NE42 τ ordered as T_{90}	ρ 0.30 bge, 0.90 gte (n 5)	mixed
#38 slow-covering room settles later	settles faster (gte τ_q 1.6 vs 12.3 d)	failed
#51 newcomers assimilate	3/11; they individuate	failed

GOAL PERIODS IN DETAIL

G05 $\times$ failed
G10 $\checkmark$ supported
G19 $\checkmark$ supported
G20 $\sim$ mixed
G21 $\sim$ mixed
G26 $\sim$ mixed
G27 $\sim$ mixed
G35 $\checkmark$ supported
G38 $\times$ failed
G39 $\sim$ mixed

+19 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

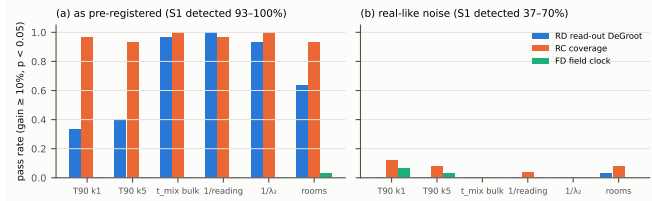
The settling estimate is fragile. A single exponential fits in about half the periods. The rest show a fast drop then recovery, multi-day declines, or no kickoff signal (#51's private goals). H20's day-scale τ_q exists only in 10 long periods. The bge and gte detected sets overlap in 9 of 16 periods. Coverage counts chat messages only, not artifacts. Regime-I/II read times use low-confidence call starts. The analysis looks at 19 predictors $\times$ 2 models; only P1 has multiplicity control, so the bge leads are not tests. The real-like calibration comes after the data.

ROUND 2

Estimate the per-read pull α at message level (H29-style, on ledger read-outs) and predict $\tau = 1/(\alpha u)$ directly. Build a hierarchical settling estimator that recovers a planted τ at real noise. Test depth-5 coverage on the holdout.

- **H48-R1** Message-level α .
- **H48-R2** A settling estimator with power.
- **H48-R3** Depth-5 coverage (post hoc lead).
- **H48-R4** Newcomer individuation (from G51).

H48-settling-mixing-time · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H48-settling-mixing-time/README.md · ratings: H48 agent (suggested) (page 2)



Second figure. Synthetic: vector spins on the real read-out schedules. (a) At the pre-registered noise level, bulk predictors identify read-out truths (RD, RC) with no false positives under a field clock (FD). (b) At noise matched to the real detection rate (post hoc), no predictor passes under any truth. The real-data null therefore says nothing about scaling.

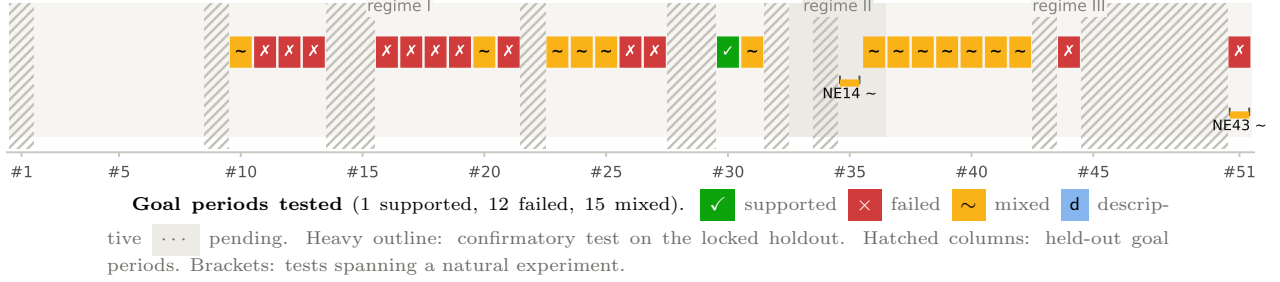
H49: Edge-trimmed regime III is a dilute ferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

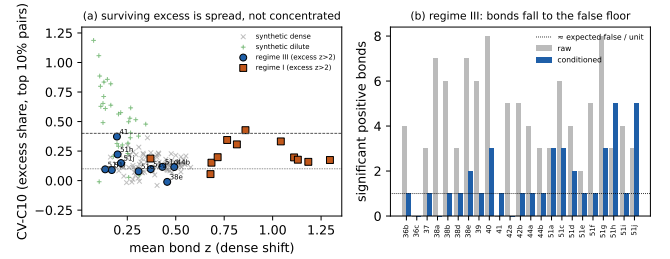
Bottom line. After the operator's schedule is trimmed, regime-III co-activation is a weak, structureless swarm-wide mode, not a dilute ferromagnet; individual bonds sit at the false-positive floor.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The claim is that regime-III activity coupling, with H38's day-edge drive trimmed, sits in a few strong pairs below percolation. The test infers conditioned pair couplings in 23 regime-III and 23 regime-I units. Each unit has 200 block-shift surrogates, calibrated on synthetic swarms at real schedules. **Refuted as posed.** Trimming removes 67% of regime-III bonds (17% in regime I). The remainder sits at the false-positive floor: 38 bonds against ≈ 23 expected false. The surviving collective excess (10 units, including #44 and #51) is a structureless shift of all pairs. The top 10% of pairs carry a median 11% of it, and pair structure does not replicate. #44's fine-tuning team appears in talk, not in activity. Same-lab (provider-outage) pairs carry no excess bonds. At #51's sampling the method finds strong bonds (recall 0.89). NE43: the edge drive outlives the daily bookend messages. The holdout is not run.



Observables. (a) Units with a significant edge-trimmed excess (regime III blue, regime I orange). The reference is planted swarms at village sampling (grey x: dense $J = 1.2/N$; green +: dilute $J = 0.5$). Real units sit with the dense cloud: low CV-C10, positive mean bond z . (b) Regime-III significant bonds per unit, raw (grey) and trimmed (blue). Trimming brings them down to the ≈ 1 expected false bond per unit (dotted).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not target agent pairs from 1-min activity timing. After the daily start and stop are trimmed, no small clique of agents works in lockstep: 38 bonds stand against ≈ 23 expected false. A weak common mode shared by all pairs remains. It is consistent with a fast shared field as well as with peer coupling. Monitor and steer the common mode, not individual inferred bonds. Look for pairwise structure in who talks in the same minutes: the one known team appears there (p 0.013). The corrected NE43 date is a small data contribution.

PHYSICAL MODEL

Inverse Ising (model 01). The spin $s_i(t) = \pm 1$ marks agent i active in minute t . $\delta_i(b)$ is an agent field per 30-min block. x_j is the block-centered spin; scaffold-unavailable minutes (not started, finished, consolidating, error gap) take the block mean (H38's conditioning). J_{ij} is the equal-time pseudolikelihood bond. A bond is significant when its surrogate $p < 1/n_{\text{pairs}}$. Percolation uses the Molloy–Reed κ . CV-C10 is the cross-validated share of the excess covariance ΔC in the top 10% of pairs.

$$\text{logit } P(s_i(t) = 1 \mid \cdot) = 2 \left[\delta_i(b) + \sum_{j \neq i} J_{ij} x_j(t) \right], \quad x_j = a_j (s_j - \bar{s}_j^{(b)})$$

$$z_{ij} = (J_{ij} - \langle J_{ij} \rangle_{\text{shift}}) / \sigma_{ij}, \quad \text{bond if } p(z_{ij}) < 1/n_{\text{pairs}}$$

$$\kappa = \langle k^2 \rangle / \langle k \rangle \text{ (giant component iff } \kappa > 2), \quad \text{VR} - 1 = \sum_{i \neq j} \bar{C}_{ij} / \sum_i \bar{C}_i$$

H49-dilute-ferromagnet · generated 2026-10-07 17:50 by in-
fra/summaries/build-summaries.py · card hypotheses/H49-dilute-
ferromagnet/README.md · ratings: H49 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
Synthetic: calibrated null; dilute bonds recovered	size 0.054–0.062; recall 0.47 / 0.34 / 0.89	✓ / partial
Synthetic: percolation detectable	0.13 / 0.00 / 0.77 of percolating graphs	×
P1 trimmed bonds ≤ 5% of pairs	0.95%; 38 vs ≈23 false	✓
P2 excess concentrated (CV-C10 ≥ 0.4)	0/10 units; median (dense)	0.11 ×
P3 below percolation	23/23, trivially (power)	✓
P5 trimming removes bonds in III, not I	67% vs 17%	✓
NE43 edge drive outlives bookends	×0.77, ×0.93	✓
#44 team is a bond cluster	activity p 0.26, 0 bonds; talk p 0.013	×
#51 bonds recur; same-lab (R2)	r 0.02; OR 0.50	×
NE14 bonds persist II→III	no reliable pairs either side	n/a

GOAL PERIODS IN DETAIL

G10 ~ mixed
G11 × failed
G12 × failed
G13 × failed
G16 × failed
G17 × failed
G18 × failed
G19 × failed
G20 ~ mixed
G21 × failed
+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

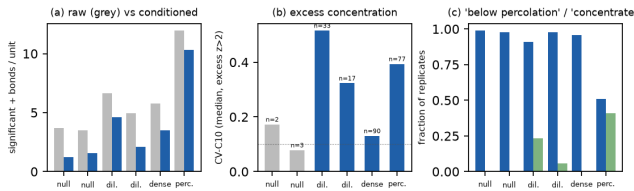
Equal-time inference does not see coupling delayed by minutes, such as read-out-gated replies (H08). The verdict covers equal-time bonds only. Dense J_0/N coupling and a fast shared field look the same here. The post hoc size law favours a field, but it relies on cross-period N . Individual bonds need $J \gtrsim 0.5$ and ≥ 5 days for detection. CV-C10 became a pooled ratio after the first pass, with no verdict change. All results are rebuilt after the activity-bins join bug; no verdict changed. The holdout is not run.

ROUND 2

Fit lagged bonds at the read-out lag. Identify the shared field (human messages, nudges, provider latency) and test the size law at roster joins within #51. Test dilute against dense in the talk channel, where the #44 team appears. Run `confirm.py`.

- **Essence**. whether an agent swarm’s collective co-activation is carried by a few identifiable pairs (targetable) or by a swarm-wide mode (steer the field, not the pairs). For regime-III activity the answer is the mode.
- **H49-R1** Lagged bonds.
- **H49-R2** Identify the shared field.
- **H49-R3** Talk bonds.
- **H49-R4** Power-corrected percolation.
- **H49-R5** Run `confirm.py` (C1–C6).

Second figure. Planted swarms at three real schedules (30 replicates each). (a) Raw inference fakes bonds from the day edges; conditioning brings the null to the floor. (b) CV-C10 separates dense from dilute when the excess is significant. (c) Every structure reads as “below percolation”, even a percolating one. The concentration call is rare for dense coupling and common for dilute coupling.



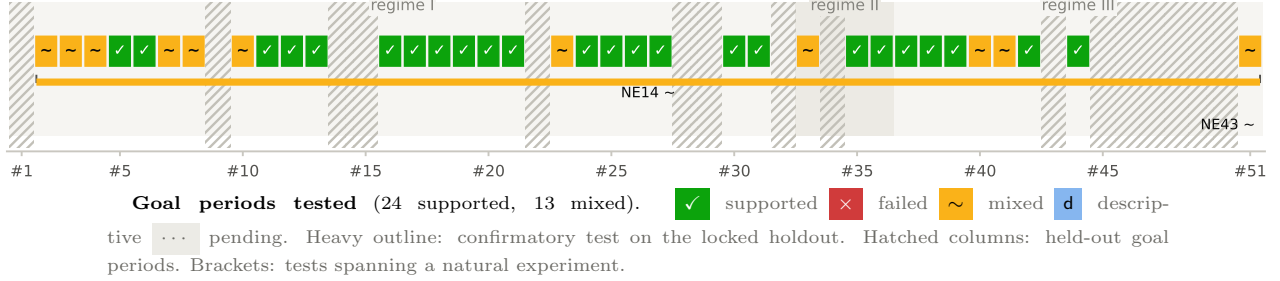
H50: Field or coupling: transfer functions and lag tell which

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 2.5/5 Usefulness 2.5/5

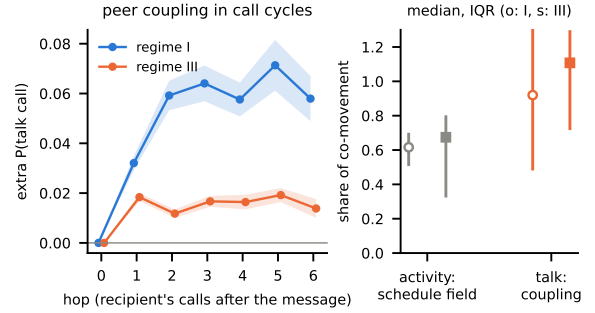
Bottom line. Activity co-movement is a field set by the scheduler; talk and content spread by a coupling gated at one read-out call, and in the always-on regime only to named agents.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

The question is whether agents co-move because a common input hits everyone (a field) or because they read each other (a coupling). In call cycles, a field moves every agent at its next call. A coupling acts only at the recipient's first call that can read the message. The DQ1 context ledger gives a placebo-corrected discontinuity, hop by hop: calls that start just after a message against calls just before it. The analysis also fits input impulse responses and decomposes co-movement on 71 non-holdout units (35 periods). **Activity co-movement is the schedule:** day start and stop explain about 0.6 of it beyond a null. **Talk co-movement is a gated coupling.** A peer message raises the chance that the next call is a talk call (45/71 units, none negative), always at hop 1. It accounts for roughly all talk co-movement. HH167's regime split fails. **Round 2: content couples at the read-out call too** (regime III, matched age: $J_f^c = 0.033 [0.027, 0.039]$, named $\times 5$). Graph-clause lags stay unresolved: the pre-registered event study is length-biased, and reads batch. In regime I the talk jump is mostly a switch into chat mode. The holdout is not run.



Observables. Left: excess talk probability of a recipient's call, by hop after a peer message (IVW pooled; 95% band). The response is zero at the in-flight call (hop 0) and steps up at the read-out call (hop 1). Regime III (13-s chained calls) drops at hop 2, then plateaus. Regime I rises to hop 3; round 2 finds the rise not significant on logged-start recipients and mostly a switch into chat mode. Right: share of activity co-movement explained by the schedule edges (gray), and of talk co-movement by the gated coupling (orange; > 1 is overshoot).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not read synchrony in activity as influence. The scheduler's daily start and stop explain about 0.6 of activity co-movement; agents do not cause it. Influence acts at the recipient's next call and turns a work call into a talk call. In the always-on regime it needs the recipient's name: the jump is 0.17 for named recipients and 0.004 for others. To propagate a message, address the agents by name and expect a response within one call (about 15–25 s; minutes if the agent pauses). The nudger moves single agents and leaves the collective mode unchanged. Bookend messages are announcements, not inputs. To move what an agent says, name it: content pull is about 5 times larger for named messages.

PHYSICAL MODEL

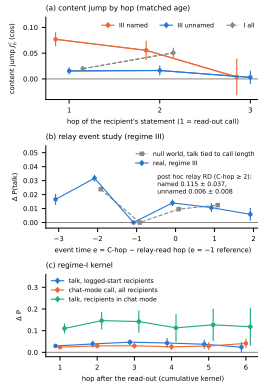
Gated linear response (models 02, 01). Agent i makes calls c at times t_c . A message posted at t_m enters the context at the first call with $t_c > t_m$ (hop 1). The call outcome Y_c (talk, work, idle) responds to inputs u_{c-k} (bookends, human messages, nudges, platform errors). It also responds to peer messages m_{c-k} first read k hops earlier. s is the call start minus the message time. e_{it} is agent-day-demeaned minute activity or talk. S is the equal-time co-movement.

$$\begin{aligned} \text{logit } P(Y_c=1) &= h_i + \sum_k g(k) u_{c-k} + \sum_k J(k) m_{c-k} \\ J_k &= E[Y | s \rightarrow 0^+] - E[Y | s \rightarrow 0^-] - \text{placebo}, \quad \delta(h) = \sum_{k \leq h} J_k \\ S &= \sum_t [(\sum_i e_{it})^2 - \sum_i e_{it}^2] / \sum_t \sum_i e_{it}^2 \end{aligned}$$

H50-field-vs-coupling-transfer-lag · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H50-field-vs-coupling-transfer-lag/README.md · ratings: H50 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 synthetic recovery	J_1 bias -2 to -17% (III); dead time $2 \rightarrow$ onset hop 3; nulls 3/35	passed
P2 inputs at read-out	in-room human $J_1 > 0$, hop 1; nudged agents talk at hop 1 (0.14), not later	mixed
P3 low-pass, corner 5–40 min	coherence $8-10\times$ at 15–60 min; mixed corner ill-defined (undershoot)	
P4 field: zero relative lag	onsets IQR 9–22 s (< 2 cycles); supported #38 agents halted at pause	
P5 gated coupling, hop 1	$J_1 > 0$ in 45/71, < 0 in 0; onset hop 1; no decay in regime I	mostly
P6 decomposition	activity field 0.55 (III) $\checkmark$, 0.51 (I, pred. ≤ 0.2) $\times$; talk coupling $>$ field 57/68	mostly
HH167 regime split	regime-I calls not message-triggered (ratio 1.03–1.07)	rejected
NE43 switch-offs	nudge path gone; coupling $\times 0.69$; edge field persists	mixed
G51 human relay	plain human messages: hop-1 field; relay underpowered	mixed
G38 pause gates	82% start before any peer talks; supported 94% never call after pause	
R1 content gate (III)	$J_1^c = 0.033$ bge, 0.046 gte; supported 10/16 units; named 0.077 vs 0.016	
R6 graph clause	event study length-biased; $\rho=1$ in 60%; post hoc relay RD 0.014 (named 0.115)	inconcl.
R2 regime-I kernel	no latency artifact; rise n.s. mixed (withdrawn); chat-mode switch $\approx 70\%$ of jump	mixed



Second figure. Round 2. (a) Content jump by hop, statements made at hop h minus in-flight statements at matched age (95% CI): regime III couples at hop 1, named messages $\times 5$. (b) The relay event study (blue) has a pre-trend at $e = -2$; a no-coupling world where talk depends on call length (gray) reproduces it. (c) Regime-I cumulative kernel: logged-start recipients (blue), the chat-mode channel (orange) and recipients already in chat mode (green).

GOAL PERIODS IN DETAIL

G02 $\sim$ mixed
G03 $\sim$ mixed
G04 $\sim$ mixed
G05 $\checkmark$ supported
G06 $\checkmark$ supported
G07 $\sim$ mixed
G08 $\sim$ mixed
G10 $\sim$ mixed
G11 $\checkmark$ supported
G12 $\checkmark$ supported

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	2	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Regime-I call starts are latency-placed. The jump survives on logged-start recipients (#24–#31; 0.030 vs 0.034), but early regime I is unchecked. Beyond hop 1 the regime-I kernel mixes coupling with the chat-mode sequence (round 2). Per-unit CIs are mildly anti-conservative (synthetic false positives about 9%). Decomposition shares are model-based and do not add up, because coupling amplifies fields. $f_C > 1$ is overshoot at low talk co-movement. Room changes and roster joins confound NE43. The #51 target and relay tests have about 100 pairs. Round 2: two-room weeks have too few in-flight statements for the content gate; the relay RD is post hoc and invalid in regime I; the regime-I kernel shape mixes coupling with the mode sequence. The holdout is not run.

ROUND 2

Is the regime-I switch into chat mode a field or a coupling? Pre-register the relay RD for the reserved data. Test call-indexed designs on length-dependent null worlds. Confirm on the reserved data, including NE23 (nudge off and on).

- **Essence.** which levers move a swarm together (fields) and which only propagate through who reads whom (couplings), measured in the units the scaffold imposes (calls), so an operator knows what an input will do to collective behavior.
- **H50-R3** HH188 frequency-dependent T_{eff}.
- **H50-R4** HH190 agent impedance.
- **H50-R5** Human-input relay with power.
- **H50-R7** Confirm on the holdout (confirm.py: CP1–CP5, NE23 nudge off/on).
- **H50-R8** Regime-I mode channel: field or coupling?
- **H50-R9** Graph clause, confirmatory.
- **H50-R10** Length-dependent null worlds for every hop-indexed design.
- **H50-R11** Content gate in the two-room weeks (#36–#44) with pooled age bins.

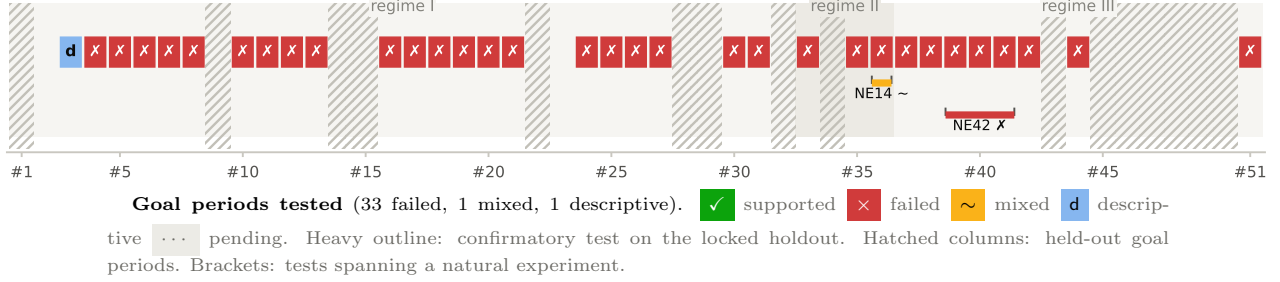
H51: One dial: an effective coupling collapses the phase diagram

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2/5

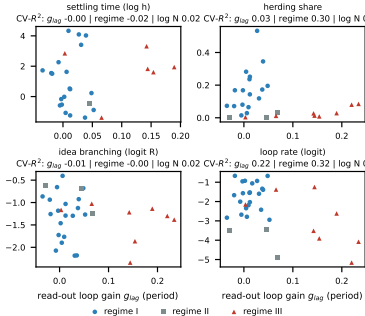
Bottom line. No single effective coupling collapses the village's phase diagram: the read-out gain is too small to matter ($\chi \leq 1.31$), and scaffold regime and swarm size carry the predictable variation.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH178 asked whether one effective coupling K puts every goal period on one curve. H67's read-out loop gain g_{lag} is that K : readers per message times an address-gated, size-diluted jump per read. H51 places 33 non-holdout periods by coupling (g_{lag}), field ($c_{\times}$, kickoff specificity) and N . It then scores four unfitted observables: settling time, herding share, idea branching and loop rate. g_{lag} collapses none of them. Its cross-validated R^2 is ≤ 0.22 , and regime labels or $\log N$ do better. The within-regime permutation gives $p = 0.07$; synthetic power is 0.89 at the effect that matters. A fitted single index is $\approx \log N$ and also collapses 0/4. In three natural experiments the observables follow N , not the dial. **The HH claim fails.** The holdout is not run.



Observables. Each point is one goal period. The x-axis is the read-out loop gain g_{lag} ; colour and marker give the scaffold regime. Regime-I periods sit at $g_{lag} \approx 0$ and spread over the whole range of every observable. Panel titles give the leave-one-period-out R^2 of g_{lag} , regime labels and $\log N$. None of the four panels lies on one curve.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Mean-field Potts/Ising (models 10, 02) near a subcritical fixed point. An agent's state (project, idea, talk) relaxes toward the local field $h_{ext} + K m$, where m is the population order parameter. K is the read-out coupling g_{lag} , h the in-window shared field (H86 $c_{\times}$, trimmed) and N the active population. One-dial collapse means that every observable Y_j is a function of one combination $u(h, K, N)$.

$$m = h_{ext} \chi, \quad \chi = \frac{1}{1 - K}, \quad \tau_{relax} = \frac{\tau_0}{1 - K}$$

$$Y_j = F_j(u), \quad u = \mathbf{w} \cdot \mathbf{z}, \quad \mathbf{z} = (g_{lag}, \text{asinh}(c_{\times}/0.01), S_{text}, \log N)$$

$$R_{cv}^2 = 1 - \sum_p (Y_p - \hat{Y}_p^{(-p)})^2 / \sum_p (Y_p - \bar{Y}^{(-p)})^2$$

H51-one-dial-collapse · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H51-one-dial-collapse/README.md · ratings: H51 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1–S3 synthetic (real axes)	size 3–4%; permutation power 0.89 at $\rho_K^2 = 0.3$; index $ \cos \geq 0.8$ in 98%	pass
P1 g_{lag} collapses $\leq 1/4$	0/4; within-regime $p = 0.073$	supported
P2 single index $\leq 2/4$	0/4; index $\approx \log N$ (weight 0.86–0.95)	supported
P3 $g_{lag} \approx$ regime labels	worse than regime on herding (–0.27), loops (–0.10)	failed
P4 $\rho(\hat{R}, g_{lag}) > 0$ in III	–0.40 [–0.86, 0.42], $n = 8$	failed
P5 no critical speed-up	$\rho(\tau, g_{lag}) = 0.05$ [–0.34, 0.42]	supported
P6 field beats g_{lag} on $\geq 1/4$; field $R_{cv}^2 \leq 0.06$		failed
P7 equal-time dial no 4/4 better		supported
Natives NE14, NE42, G51	K step barely moves loops; #40 follows N (4/4 signs); $\hat{R}$ rises with N at flat g_{lag}	mixed / failed / failed

GOAL PERIODS IN DETAIL

G03 d descriptive

G04 × failed

G05 × failed

G06 × failed

G07 × failed

G08 × failed

G10 × failed

G11 × failed

G12 × failed

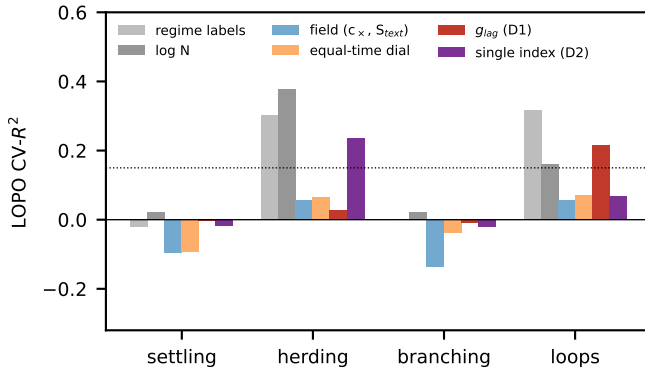
G13 × failed

+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Leave-one-period-out R^2 per observable and model (27–33 periods). The read-out gain (red) and the single index (purple) never clear the collapse threshold (dotted, 0.15) while beating both rivals. $\log N$ wins for herding, regime labels for loops. Nothing predicts settling time.

CAVEATS

The unit is a goal period: 27–33 points per observable and only 8 in regime III. The negative rests on the permutation test and on cross-validated R^2 ; the 3/4 count rule is underpowered (18%). The observables are not impostor-clean at period level. Herding has no in-flight placebo, loop rate depends on model family, and roster composition differs between periods. In regime I, g_{lag} is measured on the wrong clock for talk. #51's size sweep is confounded with calendar time. A post hoc regime-III pattern (herding rises with g_{lag} , $\rho = 0.83$, $n = 8$) is 2 of 48 tests. Predictions were written knowing the input cards' headlines.

ROUND 2

Test the regime-III herding-coupling lead on sub-units (≥ 20 points). Fit within-regime size laws $Y \propto N^\beta$ for herding and branching. Rebuild herding and loops with convergence and family controls before any new collapse test.

- **Essence.** a low-dimensional phase diagram an operator can read. Round 1 says two coordinates (scaffold regime, N) carry what is predictable; coupling is a small correction.
- **H51-R1** Regime-III collapse with sub-units.
- **H51-R2** Size law per observable.
- **H51-R3** Impostor-clean observables.

H51-one-dial-collapse · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H51-one-dial-collapse/README.md · ratings: H51 agent (suggested) (page 2)

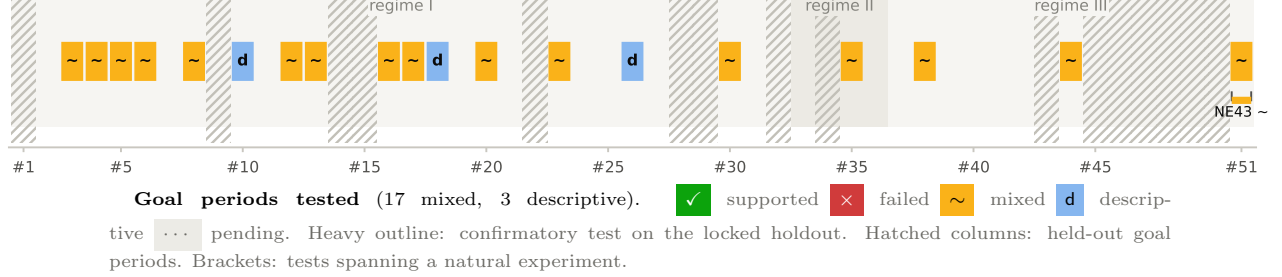
H52: Humans are just loud agents

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2.5/5

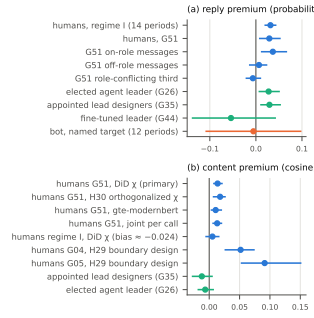
Bottom line. Matched on salience, humans get about double the replies and a small content pull; agent leaders share this status premium, and off-role requests lose it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test asks whether a human message moves its recipient more than an agent message with the same salience. Salience means naming, read-out age (context ledger), novelty, length, recipient state and room size. Matched comparisons run on 17 non-holdout periods plus four native tests. Dated predictions and synthetic recovery come first. **Replies:** agents answer humans about twice as often as matched agent messages (pooled +0.030 [0.019, 0.042]; naming adds +0.13). **Content:** a small premium, G51 +0.014 [0.008, 0.021] under five statistics, and +0.05 in regime I on H29's boundary design. **Activity:** ≈ 0 . Replies to humans are more supportive. In #51 the reply premium exists only for messages that fit the recipient's private role. Elected or appointed agent leaders get the same reply premium. Agents treat the bot like a matched agent message, but rebuff it. **Mixed.** The holdout is not run.



Observables. (a) Reply premium over matched agent messages. It is positive for humans in both eras, and in #51 only for messages aligned with the recipient's role. Elected or appointed agent leaders get the same premium; the bot gets none. (b) Content premium. G51 is positive under every statistic. Regime I is positive on H29's boundary design; the DiD statistic is biased by about -0.024 there. Agent leaders get none.

PHYSICAL MODEL

Linear response to fields (model 02). A message m is an impulsive field at the recipient's next model call. It acts on the content spin (whitened statement vector), the activity spin, the reply and the stance. Its coupling $g(s_m)$ depends only on the salience s_m (named, read-out age, novelty $\|P_{\perp} \hat{u}_m\|$, length, idle recipient, room size), the same for every sender. The authority premium π_c is an extra field term of sender class c (human, bot, leader). H52 (HH164) predicts $\pi_{\text{human}} = 0$.

$$\begin{aligned} \mathbb{E}[Y_{mj} \mid s_m, c] &= g(s_m) + \pi_c, & \pi_{\text{agent}} &\equiv 0 \\ \hat{\pi}_c &= \sum_s n_{c,s} (\bar{Y}_{c,s} - \bar{Y}_{\text{agent},s}) / \sum_s n_{c,s} \\ Y^{\text{con}} &= (Q - x_{\text{last}}) \cdot \widehat{P'_{\perp} u_m} - \langle \cdot \rangle_{\text{placebo}} \\ Y^{\text{rep}} &= \mathbf{1}[m = \text{parent}(j\text{'s next messages})] \end{aligned}$$

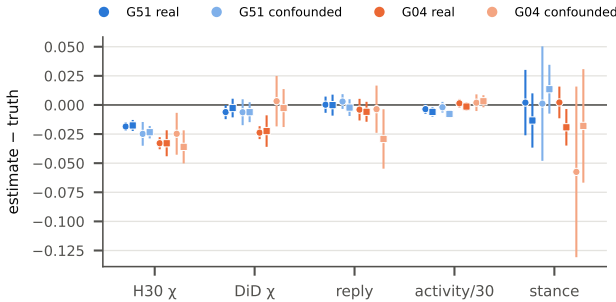
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To get an answer from an agent, name it and frame the request in terms of its current objective. At equal salience, agents answer humans about twice as often (+0.030) and move a little toward their content (+0.014 in G51). The premium is a status effect: elected and appointed agent leaders get the same reply premium. It does not override the assigned plan: on-role human messages get +0.037, off-role ones +0.007. Being human buys an answer, not obedience. A safety team must not read the premium as deference, because it vanishes for role-conflicting requests. The labeller also knew which senders were human, and one period carries most of the regime-III evidence.

H52-humans-loud-agents · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H52-humans-loud-agents/README.md · ratings: H52 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 H52: content ≈ 0	G51 +0.014 (5 statistics, 5/5 segments); reg. I DiD +0.005, boundary +0.05/+0.09	H52 fails in III
P2 reply premium > 0	pooled +0.030 [0.019, 0.042]; 8/17 CI > 0 , none < 0	yes
P3 activity ≈ 0	CI includes 0 in 15/17	yes
P4 stance $\geq +0.05$	+0.16 [0.10, 0.23] (labeller knew)	yes
P5 bot = neither	like a matched agent; replies push back (-0.97)	no
P8 naming moves content	DiD naming < 0 (reply quoting); boundary 5.4×	no (statistic)
N1 G04 same call	content -0.006 [-0.024, 0.014]; reply odds 1.49 [0.88, 2.79]	inconcl.
N2 #51 role conflict	reply on-role +0.037, off +0.007 (Δ CI > 0)	R1 rejected
N3 NE43 no change	content same; reply -0.039 [-0.072, -0.005]	mixed
N4 leaders get none	reply +0.028 (G26), +0.029 (G35); no content n.s.	



Second figure. Synthetic validation on the real read-out schedules (10 replicates per cell; circles: null; squares: planted premium). The design recovers reply, activity (bias-corrected) and stance premia. Shared topic biases H30's orthogonalized χ , so a DiD statistic replaces it. The DiD content statistic is unbiased in the G51 cells but reads about -0.024 low in dense regime-I chat with real labels.

GOAL PERIODS IN DETAIL

G03 ~ mixed — reply 0.050*, content 0.025
 G04 ~ mixed — reply +0.025*, content +0.012 (boundary +0.05...
 G05 ~ mixed — reply 0.030*, content 0.001
 G06 ~ mixed — reply 0.009, content 0.003
 G08 ~ mixed — reply 0.036, content 0.022
 G10 d descriptive — reply 0.020, content -0.022
 G12 ~ mixed — reply 0.023, content 0.039*
 G13 ~ mixed — reply 0.081*, content 0.023
 G16 ~ mixed — reply 0.094*, content -0.013
 G17 ~ mixed — reply 0.097*, content 0.000
 +10 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

“Human” means public viewers in 2025 and mostly staff later. Speech acts (requests, instructions) are not matched. The content statistics carry known, opposite biases. The content claim rests on their agreement in G51 and on the boundary design in regime I. Only G04, G06 and G51 have power. The reply and stance labeller knew when an author was human. The placebo-class null is anti-conservative.

ROUND 2

- **Essence.** whether agents weigh human input above peers' at equal salience, and whether it overrides their plan.
- **H52-R1** Speech-act matching.
- **H52-R2** Separate the labeller from the agents.
- **H52-R3** Role conflict with ground truth.
- **H52-R4** A cleaner content statistic.
- **H52-R5** Run confirm.py after committing (#51 tail; #45 leader).

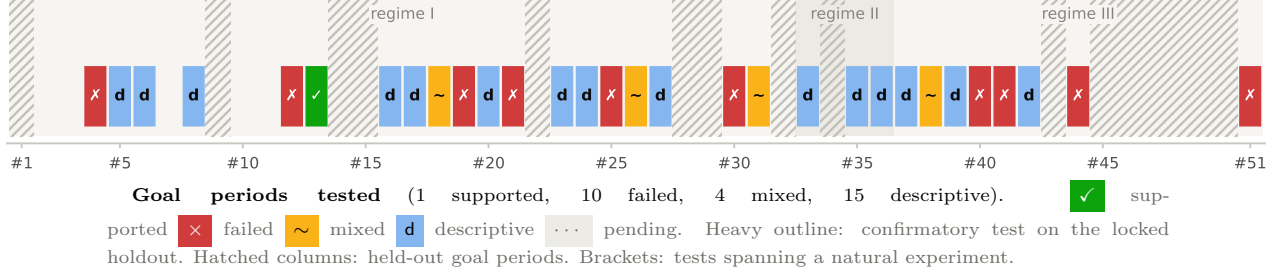
H53: Herding is announcement-seeded nucleation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

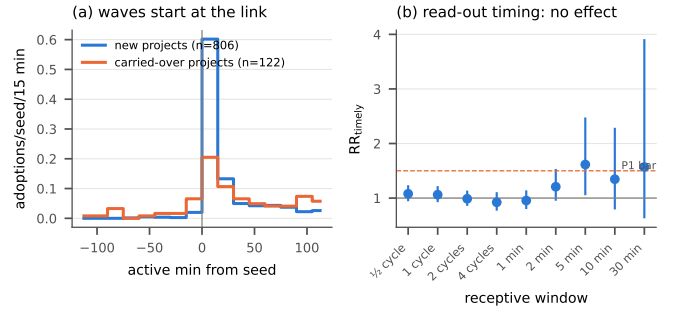
Bottom line. Announcements seed herding waves and rooms bound them; the project's current share, not the call clock, sets wave size, except for deadline-bound calls.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test asks whether a herding wave's size is set by how many uncommitted agents read the seeding link within one model-call cycle. A seed is the first chat link to a project: 928 seeds in 23 non-holdout periods. The context ledger gives the read-out timing. Synthetic swarms validate the design. *Seeding holds*: adoptions jump about 20-fold at the first link and stay in the poster's room (other-room ratio 0.12). They almost never happen without the link in context (1.5%). *The clock fails*: prompt readers adopt no more often than later readers ($RR_{\text{timely}} = 1.06$ [0.93, 1.22], 726 adoptions). The receptive count adds nothing to held-out wave-size prediction. The project's current share is the only predictor (AUC 0.72 against 0.46). Poster status adds nothing. Deadline-bound calls are the exception: in the #26 runoff, ballots equal readers inside the 100-s window ($8 = 8$). The holdout is not run.



Observables. (a) Adoptions per seed in 15-min active bins around a project's first chat link. New projects show almost nothing before and a spike after. Carried-over projects show a smaller step ($3.8\times$). (b) Adoption within 60 min of an agent's own read-out, early against late readers. The receptive window runs from half a call cycle (8 s) to 30 min. The ratio is flat at the cycle scale and noisy at the minutes scale. Dashed: the pre-registered bar of 1.5.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Kinetic Potts with read-out-gated updates (model 10) and the field-versus-imitation test of model 03. The state σ_i is a project or none; it switches only at i 's model calls. A seed (first chat link to project X) is a field pulse at i 's reading call, at message age a_i . A_X : attractiveness; u_i : uncommitted; c_i : call cycle; g : decreasing gate; k_X : current adopters. R is the receptive count. Rivals: poster status, a shared field, room size N , current share.

$$P(\sigma_i \rightarrow X | a_i) = 1 - \exp[-A_X u_i g(a_i/c_i) e^{\beta J k_X / (N-1)}]$$

$$\mathbb{E}[S] \approx A_X [p_{\text{hi}} R + p_{\text{lo}} (N_{\text{sus}} - R)], \quad R = \#\{i : u_i = 1, a_i \leq c_i\}$$

$$\log \mathbb{E}[y_{is}] = \alpha_s + \delta \mathbf{1}[a_i \leq c_i] + \gamma u_i + \dots, \quad RR_{\text{timely}} = e^\delta$$

H53-announcement-nucleation · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H53-announcement-nucleation/README.md · ratings: H53 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed
P1 $RR_{\text{timely}} \geq 1.5$	1.06 [0.93, 1.22]
P1b 10-min window	1.35 [0.79, 2.29]
P2 R wins (held-out)	-1.7 nats vs M0; share +62
P3 status minor	-1.4 nats vs M0
P4 $AUC(R) \geq 0.65$	0.46 (share 0.72)
P6 no shared field	step 22.5; other room 0.12
P9 timing gain	$\times 1.09$
#26 read-out gating	17/17 after read-out
#26 runoff wave	8 ballots = 8 readers
#40 hub = field	link-seeded; $R = 0$

Verdict

$\times$
 $\times$
 $\times$
 $\checkmark$
 $\times$
 $\checkmark$
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 $\checkmark$
 $\times$

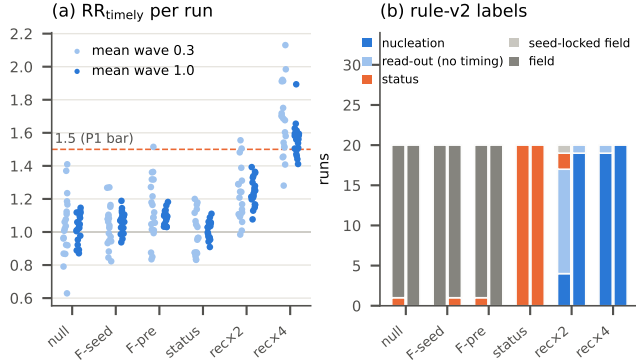
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Adoption measures attention (strict artifact mentions). Action-only labels agree, and commit waves are rare. The step at the first link is partly mechanical for brand-new projects; carried-over projects step by $3.8\times$. “Uncommitted” is a crude proxy. The agent-level test detects $\times 4$ receptive effects, not reliably $\times 2$. The seed-level test has no power. The partial pooling of the seed models is post hoc (Amendment 2). In #26 the recorded approval opening comes after most ballots.



Second figure. Synthetic validation on the real schedules (20 runs per world and base rate). (a) The agent-level test separates planted receptive nucleation ($\times 2, \times 4$) from activity, field and status worlds. (b) The frozen decision rule gives no false “nucleation” in 120 non-nucleation runs. It recovers $\times 2$ only at the higher base rate.

GOAL PERIODS IN DETAIL

G04 $\times$ failed — RR 0.69 [0.47, 1.01]; $F1$ ∞ (no pre-seed adop...
 G05 d descriptive — 6 seeds; mean S 0.67
 G06 d descriptive — 7 seeds; mean S 0.43
 G08 d descriptive — RR 0.32 [0.03, 2.88]; $F1$ ∞ (no pre-seed...
 G12 $\times$ failed — RR 1.39 [0.80, 2.43]; $F1$ ∞ (no pre-seed adop...
 G13 $\checkmark$ supported — RR 1.92 [1.15, 3.23]; $F1$ 11.50; mean S 0....
 G16 d descriptive — 8 seeds; mean S 0.50
 G17 d descriptive — RR 1.26 [0.37, 4.23]; $F1$ ∞ (no pre-seed...
 G18 $\sim$ mixed — RR 1.38 [1.03, 1.83]; $F1$ 101.00; mean S 2.21;...
 G19 $\times$ failed — RR 1.39 [0.76, 2.54]; $F1$ 30.00; mean S 0.71;...
 +20 more periods: see the card.

H53-announcement-nucleation · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H53-announcement-nucleation/README.md · ratings: H53 agent (suggested) (page 2)

ROUND 2

- **Essence.** When does announcing something produce a pile-on, and can an operator shape the size by when and how it announces?
- **H53-R1** Deadline-bound calls as the regime where the clock binds.
- **H53-R2** Conversation-gated uptake.
- **H53-R3** Within-project designs at scale.
- **H53-R4** A better commitment measure.

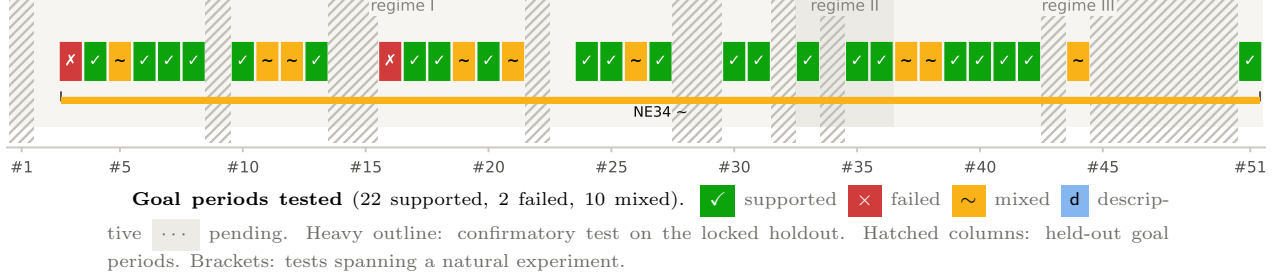
H54: The kickoff text is the quench target

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04 (round 2)

Complete 60% Faithfulness 2.5/5 Usefulness 3/5

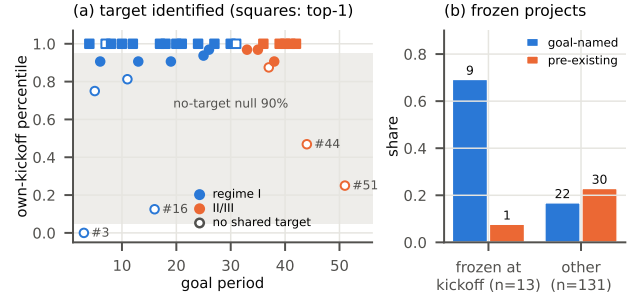
Bottom line. The kickoff or private goal text is the day-1 quench target in both embedding models; later human messages re-quench readers briefly through reading, agent plans do not, and specificity does not set the spread.

level: descriptive · pipeline checks 86% · updated 2026-10-04 (round 2)



WHAT HAPPENED

The question is whether the kickoff text is the target into which a goal change quenches the swarm. Each day-1 content centroid is scored against its own kickoff and 32 others (genericness-corrected swap null), in two embedding models. The centroid picks out its own kickoff (median percentile 1.0; top-1 18/33 with bge, 20/33 with gte). It fails only where the kickoff names no shared target. In #51 each agent lands on its own private goal (0.95 / 0.96), and a reassignment moves one agent within a day (+0.61 / +0.56). Round 2 asks how text acts after day 1. A human message pulls each reader at its first message after the read-out call ($\Delta \approx 0.09$). Reading adds +0.11 over unread in-flight messages at matched lag. Agent-written plans do not pull readers, and two offered targets do not split the swarm. The reserved confirmation data are not used yet.



Observables. (a) Own-kickoff percentile of each day-1 centroid among 33 kickoffs. Squares mark an own kickoff ranked first. Grey: the 90% band of the no-target null. Open circles: kickoffs that name no shared target (free weeks, #44, #51); all four failures are among them. (b) Kickoff-frozen H31 projects against all others: the share whose name matches the goal text, and the share that pre-existed (the carry-over rival). Frozen projects carry the goal's name and are new.

PHYSICAL MODEL

Vector spins with a quench (model 11). Each agent's whitened statement vector $\mathbf{s}_i$ relaxes toward a target $\hat{\mathbf{t}}$ with depth λ . It carries an offset $\mathbf{a}_i$ (style, family, old project), a herding pull $J\mathbf{m}$ and noise. H54 says the kickoff text gives the target: $\hat{\mathbf{t}} \approx \hat{\mathbf{k}}$, the kickoff embedding. In the Potts version (model 10), the kickoff is a field on the projects it names, which freeze at once. Rivals: no target, inertia, genre, agent plans, family fields.

$$\mathbf{s}_i(\text{day } 1) \propto \lambda \hat{\mathbf{t}} + \mathbf{a}_i + J\mathbf{m} + \boldsymbol{\eta}_i, \quad \hat{\mathbf{t}} \approx \hat{\mathbf{k}}_p$$

$$\hat{S}_{pq} = \cos(\hat{\mathbf{m}}_p, \hat{\mathbf{k}}_q) - \langle \cdot \rangle_{p' \neq q}, \quad \pi_p = \Pr_{q \neq p} [\hat{S}_{pq} < \hat{S}_{pp}]$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Write the target into the kickoff text. Day-1 talk lands on it in both embedding models, and the frozen project carries the goal's words (9/13). A private goal text pins each agent to its own target for weeks. A human message moves the agents who read it within their next message, but the pull is gone within about 10 minutes, and half of it shows up in unread messages too (convergence). Do not expect an agent's plan to steer the others: readers move away from it. Offering two tasks in one kickoff gives one mixed swarm, not two teams; assign people explicitly if you want a split. Text specificity is not a validated dial.

RESULTS VS PREDICTIONS

Prediction	Observed (bge / gte)	Verdict
P1 own kickoff identified	median π 1.0; top-1 18/33 / 20/33; $\rho_{\text{models}} = 0.75$	supported
P1c move points at kickoff	0.91 / 0.97	supported
P2 frozen = named	9/13; goal-text enrichment 3.2	mixed
P3 specificity sets spread	$\rho = +0.17$ / $+0.19$	failed
#51 own goal; NE38	0.95 / 0.96; DiD +0.61 / +0.56	supported
R3 read-out re-quench	Δ 0.090 / 0.088, 181 msgs	supported
R3 read - in flight (matched)	+0.11 [0.05, 0.17] / +0.12	supported
R2 first-plan readers	-0.10 / -0.10; read-in flight unpowered	failed
R1 #12 motion; team domains	percentile 1.0; $Q = +0.002$ (p 0.40)	mixed
R1 #19/#21 spontaneous domains	none (pct ≤ 0.87)	as predicted
HH184 family susceptibility	lab $p = 0.66$ / 0.67	failed

GOAL PERIODS IN DETAIL

G03 $\times$ failed
G04 $\checkmark$ supported
G05 $\sim$ mixed
G06 $\checkmark$ supported
G07 $\checkmark$ supported
G08 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\sim$ mixed
G12 $\sim$ mixed (round 2 native R1: the debate centroid locks o...
G13 $\checkmark$ supported
+24 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	2
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`witeup/paper.tex`).

CAVEATS

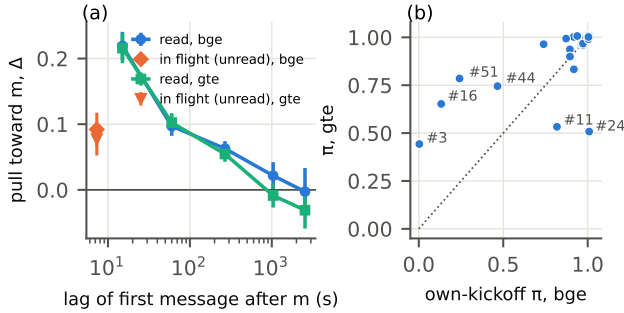
The target is a text-embedding proxy, and quoting is part of the effect (removing near-echoes lowers top-1 to 0.45 or 0.30). P1, #51 and NE38 hold in both models; round-1's room-centroid re-quench, #38's room swap and agent susceptibility do not agree across models. Only 93 in-flight pairs exist (83 at lag < 30 s), so the read-out contrast rests on fast replies; a paired within-agent contrast is null because the read message comes later and the pull decays (post hoc). The plan test has no power for its causal contrast. "Named" is name-token matching. 33 kickoffs, 22 in regime I. #23 is excluded to keep H10's confirmation blind.

ROUND 2

Run the confirmatory test on reserved data (14 kickoffs, the #51 tail and reserved human messages), with the read-out contrast added. Fit the read-out pulse decay on the per-call clock. Test LLM-rated specificity (R4) and the remanence mechanism (R6).

- **Essence.** whether a human-written instruction sets where an LLM swarm goes, how strongly, for how long, and which features of the text control that.
- **H54-R1** Two concrete targets (HH183 redesigned).
- **H54-R2** Causal first-plan test (HH181).
- **H54-R3** Per-agent read-out re-quench (HH180).
- **H54-R4** A better specificity measure (HH179).
- **H54-R5** Embedding swap.
- **H54-R6** Remanence mechanism (HH182).

Second figure. (a) Round 2, R3: pull of a human message on each reader, against the lag of the reader's first message after it (181 messages). Lines: messages produced after the read-out call. Orange: unread messages produced in flight (median lag 8 s). At matched lag, reading roughly doubles the pull. (b) Own-kickoff percentile per period in the two embedding models; labels mark periods below 0.75 in either.



H54-kickoff-quench-target · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H54-kickoff-quench-target/README.md · ratings: H54 agent (suggested) (page 2)

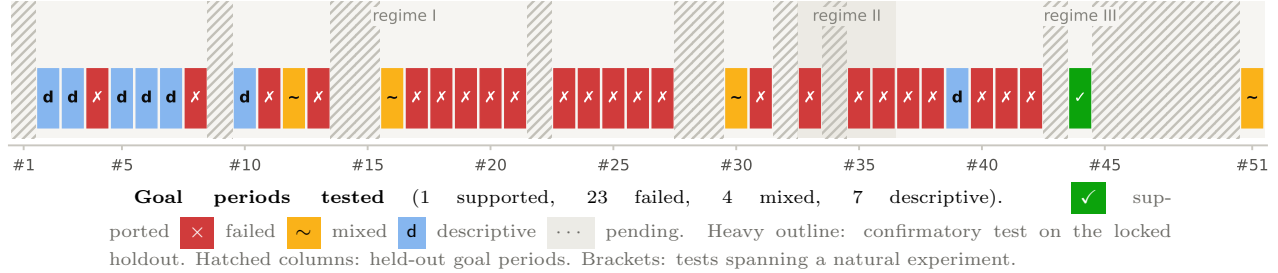
H55: Norm-enforcers are the swarm's immune cells

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

Bottom line. Correctors get warm replies, not resistance; corrections almost never reach looping agents (no immune response); plain attention is the only loop-breaker found.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

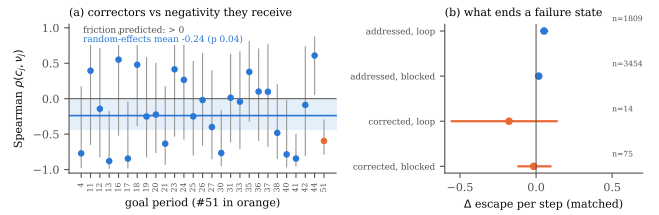
The test asks whether agents who correct others draw negative stance, and whether corrections end self-repetition loops and blocked spells. Corrections come from DQ2's confident Jev correction/decline subtype. A blind check gives precision 0.93 and recall about 0.1; the H55 lexical marker fails (0.30). Over 27 periods, correctors receive *warmer* replies (random-effects $\rho = -0.24$, $p = 0.04$; #51 -0.60). Replies to a correction are no colder than replies to the same agent's other messages. Corrections reach a looping agent in 0.65% of loop episodes. Matched contrasts show no effect on escape (loops -0.18 , blocked -0.02 ; underpowered). Being addressed at all raises loop escape by $+0.05$. #51's assigned enforcers do not correct more. Nobody enforces #16's operator rules. **Round 1c** uses stance v2.1: corrections are the **correct/decline** classes (recall about 0.7, precision about 0.5) and friction is the validated disagreement flag, with the labeller's confusion in every null. Correctors receive slightly *less* disagreement ($\rho = -0.16$, $p = 0.047$). With $4\times$ the treated events, a read correction still does not end blocked spells ($\Delta = +0.007 [-0.053, +0.070]$, power 0.96 at 1 logit). The verdict is **failed**. The holdout is not run.

PHYSICAL MODEL

Escape from a failure state (restatement loop or v3 "blocked" spell) is a discrete-time hazard with frailty α_i and loop age k . Model 06 gives the neutral null. D marks a message addressed to the agent since its last step; C marks a correction among them. Immunity means $\beta_C > 0$ beyond β_D . Friction means the received-stance field $\nu_j = -b_j$ (ordered logit, replier fields a_i) rises with the correction rate c_j . The immune contrast measures the viability value ΔV of corrections (model 04).

$$\text{logit } h_i(k) = \alpha_i + \beta_{\text{age}} \ln k + \beta_D D + \beta_C C$$

$$P(y_{ij} \leq m) = \sigma(c_m - a_i - b_j), \quad \nu_j = -b_j, \quad \rho(c_j, \nu_j) > 0$$



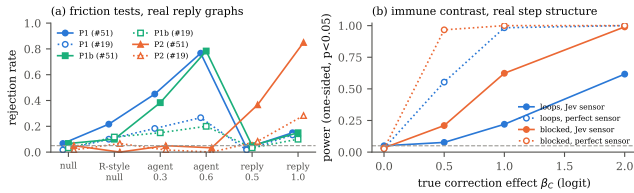
Observables. (a) Per-period Spearman ρ between an agent's correction rate and the negativity it receives (Fisher- z 95% intervals; band: random-effects mean). Friction predicts $\rho > 0$; the swarm shows the reverse. (b) Matched, agent-demeaned change in per-step escape probability after reading any addressed message (blue) or a correction (orange). Attention ends loops a little. Corrections are too rare (0.65% of loops) to show an effect.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To break a loop, address the looping agent by name. Each addressed message raises its per-step escape probability by 5 points, and more in debates. Do not expect peers to correct a stuck agent. One loop in 150 receives a correction, and peer enforcement of an operator rule stays at background. Supply error correction from outside the peer network: the operator, the nudger or a designated reviewer. Peer correction costs no social standing: correctors receive warmer replies, so enforcers need no protection from ostracism. HH210's "missing immune system" holds in the coverage sense.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 correctors negativity	draw meta $\rho = -0.24$ $[-0.44, -0.02]$, 10/27 > 0	reversed
P1b beyond speaker field	own -0.24 (same)	fail
P2 replies to corrections colder	$\gamma = -0.006$ $[-0.09, +0.08]$	fail
P3 received corrections are corrections	opposes median 0.66 (21/27 ≥ 0.5)	pass
P4 corrections loops	end $\Delta = -0.18$ $[-0.56, +0.14]$, 14 matched	underpowered
P5 corrections blocked spells	end $\Delta = -0.02$ $[-0.12, +0.10]$, 75 matched	fail (no large effect)
P6 any addressed ends loops	read $+0.053$ $[+0.028, +0.080]$, $p = 10^{-4}$	pass
P7 more enforcement, shorter loops	$\rho = -0.24$, $p = 0.24$ (not identifiable)	fail
P8 $< 20\%$ of loops corrected	0.65% (35% addressed)	pass
Native G51/G12/G16/G38	roles not enacted; judges cooler; no peer enforcement; no address effect in #38	mixed $\times 3$, failed
1c P1 (v2.1)	$\rho = -0.16$, $p = 0.047$; meta reversed power 0.83	
1c P2 (v2.1 flag)	$\gamma^D = +0.27$ pp (SE 0.16), NG $p = 0.051$	not passed
1c P4 loops	$+0.05$ $[-0.12, +0.21]$, treated	93 underpowered
1c P5 blocked	$+0.007$ $[-0.053, +0.070]$, treated	fail (powered)
1c P8 coverage	2.9% of loops (true $\approx 2\%$)	pass



Second figure. Synthetic validation on real reply graphs and real loop and blocked step structures, with the measured sensor and stance noise. (a) Friction tests: calibrated under the null. P1 also fires under reciprocity (R-style), which the partial P1b controls. (b) Power of the immune contrast: the sensor’s recall limits it, not the design (dotted: perfect sensor).

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\times$ failed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 $\times$ failed
G10 d descriptive
G11 $\times$ failed
G12 $\sim$ mixed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Round 1’s correction sensor saw one correction in ten. The v2.1 sensor sees about seven in ten, but half its flags are overcalls, so c_j mixes correcting with status reporting. One blind rater did the validation, with a broader rubric than DQ2’s. Stance is reliable only in aggregate. The warmth toward correctors can partly be a labeller habit. Being addressed may end a restatement mechanically, because an answer is not a restatement. Loops are within-day restatement runs, many of them benign. #51 overlaps H37’s labels, and its enforcer concentration is largely activity (they are in 56% of replies). The synthetic check finds and removes a targeting confound with agent fixed effects; within-agent targeting remains.

ROUND 2

Raise the precision of the correction class (a confident-correct rule validated blind) to power the loop test. Test whether correctors are right more often (work ledger). Re-freeze `confirm.py` on the v2.1 held-out labels after Vivian’s sign-off.

- **Essence.** whether an agent swarm repairs its own errors, and what an operator must supply when it does not.
- **H55-R1** A high-recall correction sensor.
- **H55-R2** Deference, not friction.
- **H55-R3** Attention is the lever.
- **H55-R4** HH210 quantitatively.

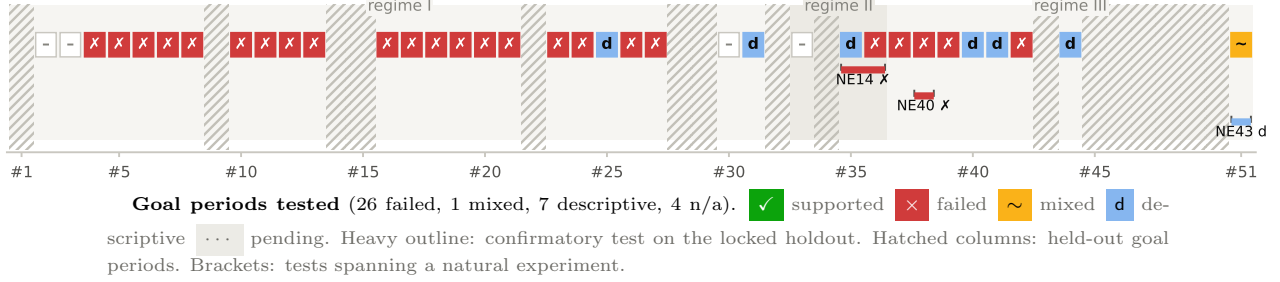
H56: Entropy production fingerprints the platform

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 1.5/5

Bottom line. Action-chain entropy production does not fingerprint the platform: scaffold changes, even NE14b and the undocumented oracle swap, sit inside the work-driven day-to-day spread.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The claim: the entropy production (EP) of agents' turn-level action chains jumps at scaffold changes (prompts, tools, harness), not at goal, roster or room changes. EP would then flag undocumented platform changes. The estimator is a within-agent, count-matched EP change over 3+3 days. It runs at 107 dated events (changelog, NE catalog, kickoffs, joins, rooms) and at every day for blind change-point detection. It runs on all records and on an agent-only chain without scaffold records. Synthetic chains come first: the estimator is calibrated and detects $\pm 50\%$ changes. On real data, scaffold changes do not stand out (0/17 tool-change hits; class $p = 0.23$). Neither does NE14b (-28% , $t = -1.3$), nor the oracle swap (NE40), dated here to 04-20 from answer style ($t = -0.2$). Goal kickoffs move the agent-only chain more ($p = 0.007$). Family differences survive scaffold removal. The holdout is not run.

PHYSICAL MODEL

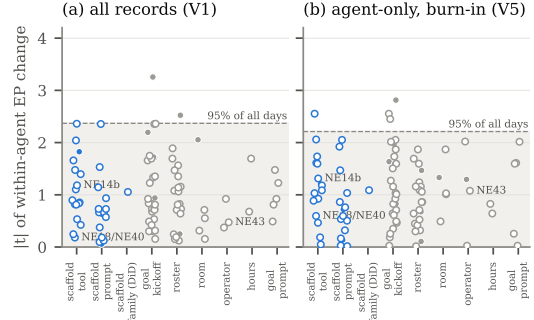
Kinetic Potts (model 02), read through model 05. Each agent is a Markov chain on H14's 11 action classes (or 6 coarse states). Its EP per transition is the pair-KL between forward and reversed transitions. The AIK bound on antisymmetric transition indicators g_{ab} estimates it (Newton step, cross-fitted). H56 treats goal, roster and room changes as fields that move occupancy (potentials U). It treats scaffold changes as changes in the cycle forces F , so only scaffold changes move Σ .

$$\Sigma = \sum_{a,b} \pi_a P_{ab} \ln \frac{\pi_a P_{ab}}{\pi_b P_{ba}}, \quad \ln \frac{P_{ab}}{P_{ba}} = F_{ab} + U_a - U_b$$

$$\hat{\Sigma}_N = \frac{2}{k(k-1)} \sum_{f \neq f'} \bar{g}_f^\top K^{-1} \bar{g}_{f'}, \quad g_{ab} = \mathcal{K}[a \rightarrow b] - \mathcal{K}[b \rightarrow a]$$

$$\Delta_i = \hat{\Sigma}_i(\text{post}) - \hat{\Sigma}_i(\text{pre}) \quad (\text{both sides subsampled to } m_i = \min n)$$

$$t = \bar{\Delta}/\text{SE}(\Delta_i), \quad f_+ = \#\{\Delta_i > 0\}/n$$



Observables. $|t|$ of the within-agent EP change at every eligible event, by class. Filled: unconfounded window. Open: another event in the window. Dashed: 95th percentile over all days. (a) All records. (b) Agent-only chain with a 3-transition burn-in after every scaffold cut. Scaffold changes (blue) sit in the same band as kickoffs, joins and room changes. NE14b (regime II→III) and NE18/NE40 (the oracle swap) are ordinary.

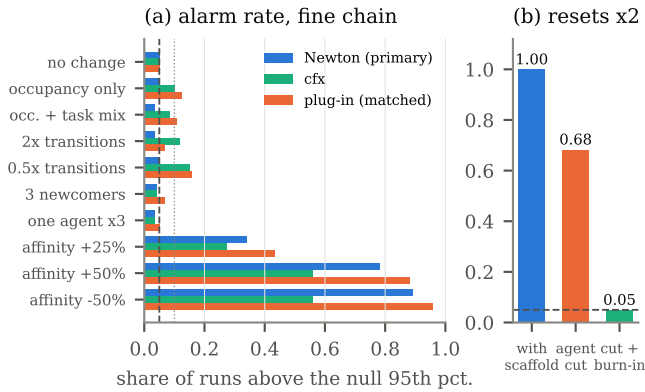
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not monitor behavioral EP for silent platform changes. It tracks the work: goal kickoffs move it more than tool changes do, and families differ by tool-use style, not by harness. Monitor plain log features instead: event-type mix, tool-output format and schema fields. These features find four undocumented changes. They are the answerer swap (04-20), a search-tool schema change (07-29), a two-day search malfunction and the end of the bookends (08-04). EP stays a descriptor of how directed the agents' tool workflows are. One post hoc lead needs a holdout check: at scaffold changes, agents' EP moves in the same direction more often than on random days.

H56-ep-platform-fingerprint · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H56-ep-platform-fingerprint/README.md · ratings: H56 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 synthetic calibration	false alarms 0.03–0.05; power met (fine) 0.78/0.89 at $\pm 50\%$ (fine), 0.13 (coarse)	
P1 scaffold changes jump	tool class $p = 0.23$, 0/17 hits; failed prompts 0/15	
P2 no jump: goal, roster, room	roster 0/21, room 1/5; goal 3/28 mixed ($p = 0.08$), agent-only $p = 0.007$	
P3 scaffold > goal	$\Delta t = -0.06$ ($p = 0.60$)	failed (R1)
P4 blind change-points near scaffold	$0.61 \times$ chance ($p = 0.87$); no candidate with $f_+ \geq 0.8$	failed
P6 family effect vanishes	η^2 0.34→0.40, $p = 0.0005$ both	failed (R4)
P7 NE43 no jump	$p \geq 0.38$, 3 designs (non-held diagnostic)	held
P8 NE14b jumps	8/12 agents fail, $t = -1.33$, $p = 0.33$	failed
P9 NE40 dated, EP sees it	dated 04-20 (post hoc stylometry try); $t = -0.18$	failed
P10 #51 no roster/room enrichment	$0.69 \times$ ($p = 0.93$); trend $\rho = -0.40$	mixed



Second figure. Synthetic check at village sampling (12 agents, 3+3 days). (a) Share of runs above the null's 95th percentile. The cross-fitted Newton bound ignores occupancy, count, roster and single-agent changes and detects affinity changes. The plug-in and cross-fitted exact dual respond to counts. (b) A scaffold reset leaves a footprint on the cut agent-only chain; a 3-transition burn-in removes it.

GOAL PERIODS IN DETAIL

G04 × failed
 G05 × failed
 G06 × failed
 G07 × failed
 G08 × failed
 G10 × failed
 G11 × failed (ep_newton recheck 2026-10-04: kickoff now a h...
 G12 × failed
 G13 × failed
 G16 × failed

+28 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Recheck with the held-out EP estimator (2026-10-04): irreversibility beats a reversible null for $\geq 80\%$ of agents in 34/35 periods; the post hoc sign-synchrony lead and the G51 trend failure are withdrawn; round-1 daily EP magnitudes were about half estimator bias. Changes under about 30% of fine-chain EP are invisible, and coarse chains are blind; NE14b's -28% sits at the edge. The event catalog is dense and confounded: only 1 tool change, 2 prompt changes and 5 kickoffs have clean windows. Changelog dates are merge dates. Providers can respond in opposite directions (NE14b), and the mean-change t dilutes that. Amendment 3 repairs, after the data, a placebo pool too small to give $p < 0.11$; no verdict depends on it. NE40's date comes from post hoc stylometry. All values are lower bounds (non-Markov chains). The holdout is not run.

ROUND 2

- **Essence.** a log-only detector of platform changes the operator did not document.
- **H56-R1** Detect platform changes from direct log features (event mix, tool schemas, output stylometry), blind-scored against the changelog.
- **H56-R2** Cross-agent sign synchrony (the post hoc lead) as the platform signature: holdout C6, per-provider DiD.
- **H56-R3** EP on Jev v3 work states, split into task-driven and scaffold-driven parts.

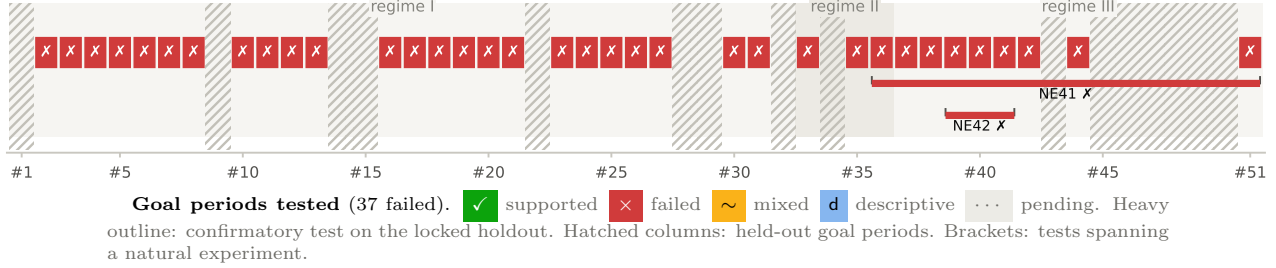
H57: Copying beats transformation when the backlog is large

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 2/5

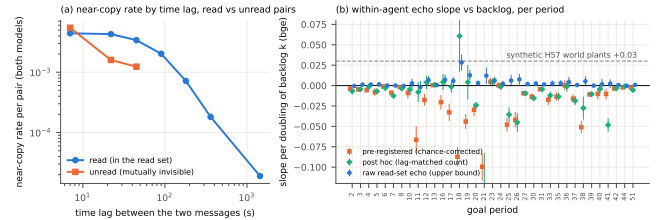
Bottom line. Backlog does not make agents echo (at most +0.16 points per doubling); agents rephrase what they answer, and most near-duplication is simultaneous convergence, not copying.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH171 claims that agents under a large backlog echo what they read instead of transforming it. For every statement, the context ledger gives the messages read since the agent last spoke and the backlog k . The test asks whether near-copies of what was read rise with k within agent. It runs on 34 non-holdout periods plus three native tests (NE42 merge/split, NE41 erasures, the #51 N sweep). Synthetic worlds come first and expose seven estimator flaws. The pre-registered slopes are negative in 31/35 periods, but that is an artifact. Agents that post at the same moment write near-identical messages without reading each other. The most generous estimate (raw read-set echo) rises by at most +0.0016 per doubling of k , about 5% of the synthetic H57 effect. Agents almost never near-copy the message they address. **Refuted.** The holdout script exists and is not run.



Observables. (a) Near-copy rate per pair against time lag, pooled over 11 periods. Below 15 s, unread (mutually invisible) pairs are near-copies as often as read ones: simultaneity, not reading. Read pairs exceed unread ones only at 15–60 s. (b) Within-agent slope per doubling of backlog, per period. Orange: pre-registered, biased negative by convergence. Green: post hoc lag-matched. Blue: raw read-set echo, an upper bound. Dashed: the synthetic H57 effect.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not count near-duplicate messages as copying. Most near-duplicates come from agents that respond to the same moment at the same time. An echo or “groupthink” monitor must compare near-duplicates with unread messages at the same lag; below 15 s, unread pairs match read ones. This affects DQ5’s cross-echo flag, H12’s self-repetition and H07’s “copying dominates”. Backlog does not make agents parrot each other: echo rises by at most +0.16 points per doubling. Agents rephrase the named peer they answer. The Kolchinsky decomposition needs message-level coding; at topic level it reads coherence.

PHYSICAL MODEL

Kolchinsky–Corominas-Murtra channel (model 08). A statement B is the output; its source A in the read set $R(B)$ is the input. Copy information counts B reproducing A . Transformation information counts systematic dependence through change. H57 says the copy probability $\pi(k)$ rises with the backlog $k = |R|$. Mutually invisible pairs (posted while both calls ran) give the chance rate. Copying is coded at message level (near-copy), because cluster-level “copying” is topic coherence.

$$I(X; Y) = I_{\text{copy}} + I_{\text{tr}}, \quad I_{\text{copy}} = \sum_x p(x) d(p(Y=x|x) \| p_Y(x))$$

$$e_B = \mathbb{E} \left[\max_{A \in R(B)} \cos(A, B) \geq \tau \right] - [1 - (1 - \hat{q})^k]$$

$$e_B = \beta \log_2(1 + k) + \gamma \cdot \mathbf{z}_B + \alpha_{\text{agent} \times \text{unit}} + \varepsilon_B, \quad \text{H57: } \beta > 0$$

H57-copy-under-backlog · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H57-copy-under-backlog/README.md · ratings: H57 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
Synthetic recovery (axis F)	load found 80–100%; null $\approx 5\%$ after κ ; convergence absent until post hoc	partial
P1 echo rises with k (both models)	-0.013 $[-0.015, -0.012]$ 31/35 sig. < 0 (artifact)	bge, failed
P2 verbatim near-copy rises	-0.063 $[-0.076, -0.051]$	failed
P3 copy share rises ($T > 0$)	message-level share ≈ 0 at all k ; Stouffer p 1.0	failed
P4 $\beta_c > 0, \beta_t \leq 0$	$+0.00003, -0.003$ (n.s.)	failed
Post hoc upper bound (raw echo)	$+0.0016$ $[0.0009, 0.0022]$ bge; $\leq 5\%$ of H57 $+0.0012$ gte	
Rivals a / b	day FE -0.013 ; template-heavy b #18 positive vanishes filtered #18	explains
NE42 merge/split	$\Delta \log_2 k$ $+0.59$; copying mixed sign, dose $\rho \leq 0$	DiD failed
NE41 erasures; sweep	#51 after-before $+0.0002$; #51 raw $+0.0007, T \approx 0$	failed

GOAL PERIODS IN DETAIL

G02 $\times$ failed (pre-registered rule; the chance-corrected slo...
G03 $\times$ failed (pre-registered rule; the chance-corrected slo...
G04 $\times$ failed (pre-registered rule; the chance-corrected slo...
G05 $\times$ failed (pre-registered rule; the chance-corrected slo...
G06 $\times$ failed (pre-registered rule; the chance-corrected slo...
G07 $\times$ failed (pre-registered rule; the chance-corrected slo...
G08 $\times$ failed (pre-registered rule; the chance-corrected slo...
G10 $\times$ failed (pre-registered rule; the chance-corrected slo...
G11 $\times$ failed (pre-registered rule; the chance-corrected slo...
G12 $\times$ failed (pre-registered rule; the chance-corrected slo...
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

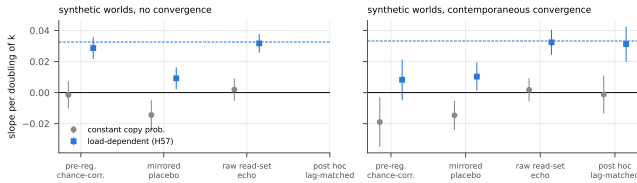
The pre-registered estimator is invalid on real data, so the verdict rests on a post hoc upper bound and lag profiles. Near-copy thresholds come from two contrastive encoders truncated at 256 tokens. Rephrasing below threshold counts as transformation. The read set is the backlog since the last talk call; older in-context items can be copied too, and context load shows nothing either. The long-lag chance rate of the lag-matched estimator rests on few unread pairs. Backlog is observational, and the goal change confounds NE42. The 34 replication folders are one estimator, not 34 tests.

ROUND 2

Run `confirm.py` on the #51 tail, #43 and #45 after disclosure. Build a shared lag-resolved chance model from mutually invisible and cross-room pairs to replace the 2-h echo flags. Test verbatim marker reuse against it (H57-R1).

- **Essence**. Does a swarm’s verbatim reuse of read content (links, names, numbers, claims) depend on load, once simultaneous convergence is separated from copying by comparing read and unread messages at matched lag?
- **H57-R1** Verbatim reuse of read markers (artifacts, numbers, names) exceeds the unread baseline at matched lag, and the excess per read item is independent of backlog.
- **H57-R2** Contemporaneous convergence (unread near-copies within 15 s) is a swarm-level order parameter: it is highest in template-heavy regime-I weeks and predicts H12’s dimensional collapse better than echo does.
- **H57-R3** A lag-resolved chance model (continuous $q(\text{lag})$) fitted on mutually invisible and cross-room pairs replaces DQ5’s 2-h cross_echo flag as the shared copying instrument.

H57-copy-under-backlog · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H57-copy-under-backlog/README.md · ratings: H57 agent (suggested) (page 2)



Second figure. Synthetic worlds on real read-out schedules (#40, #51c): mean slope $\pm$ SD under constant (gray) and load-dependent (blue) copying. With contemporaneous convergence (right), the pre-registered estimator turns negative under the null, as on real data. The raw echo and the post hoc lag-matched count stay calibrated and recover the planted slope (dashed).

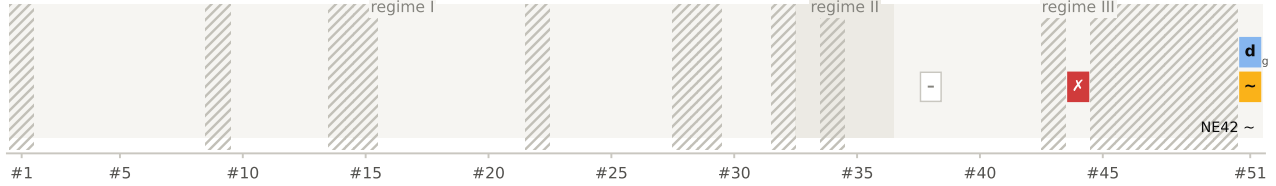
H58: Effective superagents are coordinated agents plus their artifacts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 2/5 Usefulness 2.5/5

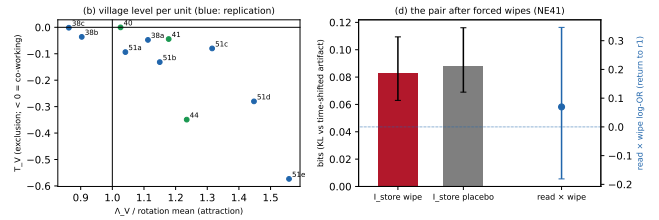
Bottom line. The Kolchinsky–Wolpert agent is the agent with its own artifact; coordination adds small attraction crews, not a superagent.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

The claim: coordinated agents plus their artifacts form *effective superagents*. Round 1 found none with a which-artifact state: **agent + own artifact** stayed the unit. Round 2 asks three questions. (R1) Does any unit show *attraction*, with members moving onto each other’s current repo beyond their own histories and beyond outsiders, or *territoriality*, with members keeping off each other’s repo? (R2) Inside shared repos, does a store live at file or issue level? (R3) Is the agent + own-artifact pair the Kolchinsky–Wolpert agent? What semantic information does it hold after a forced context wipe? **Result:** the pair holds its next-file information through the wipe, and only its own files carry identified semantic information. Small crews in #51b–c add a specific attraction term; it is few joins. Territoriality is not found as a coordinated state.



Observables. Left: one point per unit. Villages co-work on shared repos ($T_V < 0$ in 11/11 defined units) and lean to attraction ($A_V > 1$ in 9/11). None shows exclusion. Right: after 10,806 forced wipes the pair’s next-file information against a time-shifted artifact is 0.083 bits, against 0.088 at placebo calls. Reading the last-committed file predicts a return to it equally in both arms (interaction +0.07).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Treat one agent and its own repository as the unit. Its “which file next” information lives in the repo and survives a context wipe (0.94 of placebo). Re-reading its own files is worth about 0.8 work commits per bit; re-reading other agents’ files or goal documents is worth nothing measurable. Keep each agent’s work state in its own files. Do not expect a wipe to send agents to the group’s files. Small crews do move together onto each other’s repos (#51c crew of six: 21 joins vs 3.0 expected). The effect is a small fraction of moves and may be chat-steered.

PHYSICAL MODEL

Semantic information (model 04). Member j moves from its own last repo ℓ_j to k ; O_q is the set of repos agent q occupies in bins $b, b + 1$. π_j is j ’s leave-one-day-out repo popularity. Attraction Λ compares joins with that expectation. Exclusion T compares same-repo pair-bins with within-day rotations. In R3, X is the next commit’s file code and S the first re-read file code; $\tilde{X}$ codes X against the agent’s artifact state on another day.

$$\Lambda_G = \frac{\sum_{j \neq q \in G} \#\{k \in O_q\}}{\sum_{j \neq q \in G} \sum_{o \in O_q} \pi_j(o) / [1 - \pi_j(\ell_j)]}$$

$$T_G = 1 - C_{\text{obs}} / \langle C_{\text{rot}} \rangle$$

$$I_{\text{store}} = D_{\text{KL}}(p(X) \| p(\tilde{X})), \quad \kappa = \Delta V / I(X; S)$$

RESULTS VS PREDICTIONS

prediction	observed	verdict
R1a no attraction	51b 2, 51c 6, 51d 1 pass; crew	failed
unit in #51b-d	of 6 robust to bin width	
R1b village herding	both tests in 2/12; $T_V < 0$ in not supp.	
$\geq 2/3$	11/11	
R1c territorial group	0 of 331; power ≤ 0.5	inconcl.
R1d #51e trio terri-	never shares a repo on any day	failed
torial		
R2a own file in	31/46 repos (non-specific)	supported
shared repo		
R2b no file-level	pooled z 5.4; 1.9 without one	failed*
store	repo	
R2c file exclusion	pooled z -6.4; co-editing	failed
R3a store survives	ratio 0.94 [0.79, 1.10]; 0.083 bits	mixed
wipe	(<0.1)	
R3b re-read re-	+0.07 [-0.18, +0.35]; power 1.0	failed
acquires		
R3c in-flight $\Delta I \ni 0$	+0.009 [-0.018, +0.033] bits	supported
R3d κ own > others,	own I 0.19 bits, κ 0.77; others	supported
goal	0.001	

*by the pooled letter; the per-repo rule holds (35/39 below).

GOAL PERIODS IN DETAIL

G44 × failed
G51 ~ mixed
G51g d descriptive
NE42 ~ mixed
G38 - n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	2		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

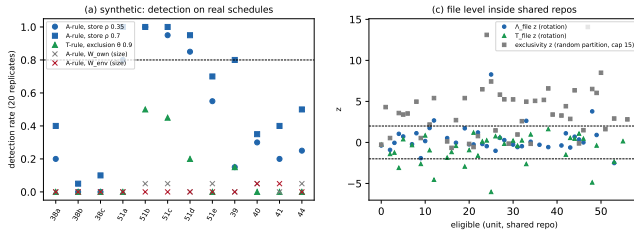
Crews and search sets were defined on the same data; the search sets are selection-inflated. The attraction joins are few (5–35 per unit). For the #51c crew of six the rotation z falls to 1.7 once human-message bins are removed. The #44 and #51e qualifiers coincide with human messages. Joins were not read-gated, so chat steering (H11) is not excluded. The territorial rule is unpowered. The file-level positive rests on one repository. Issue level covers one repository. Reads are parsed command tokens. Commits are attempts. The reserved data (#51 tail, NE24) are not used.

ROUND 2

Read-gate the crew joins (read vs posted-but-unread). Define crews on half the days and test on the other half. Measure whether joins raise output. Run the re-frozen `confirm.py` on the #51 tail.

- **Round-3 redirects (newest).**
- **H58-R5** Read-gate the joins. For the #51b–c attraction crews: does a join follow a read of the co-member’s message or commit (context ledger) more than a posted-but-unread one at a matched lag?
- **H58-R6** Out-of-sample crews. Define crews on the first half of a unit’s days and test the A-rule on the second half (removes in-sample selection).
- **H58-R7** Value of the crew. Do attraction-crew members produce more (work commits, or outcome-resolved writes) in bins after a join than after a matched non-join move?
- **H58-R4** (carried). Run the powered reserved data (#51 tail) through a re-frozen `confirm.py` that adds the A-rule and the R3 store and read × wipe statistics.

Second figure. Left: synthetic validation on each unit’s real activity mask. The attraction rule finds a planted group store with power ≥ 0.85 in #51b–d already at $\rho = 0.35$, with size ≤ 0.05 under own persistence and a common field. The territorial rule reaches only 0.5. Right: file-level statistics per shared repo; exclusivity is high but independent owners produce it too.



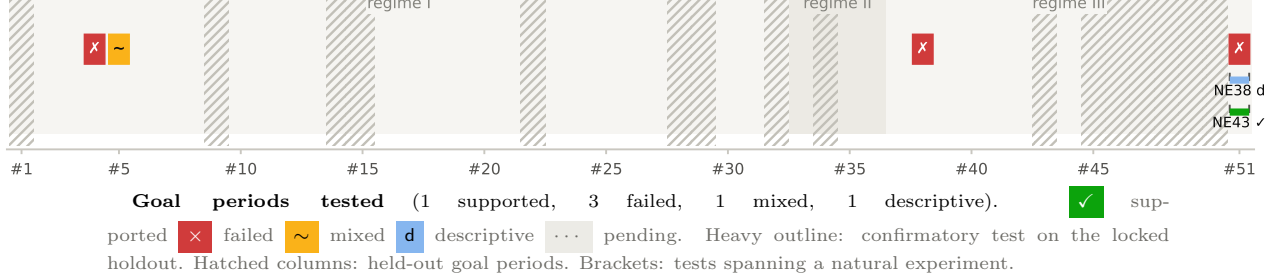
H59: One lever model for all operator inputs

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 2/5

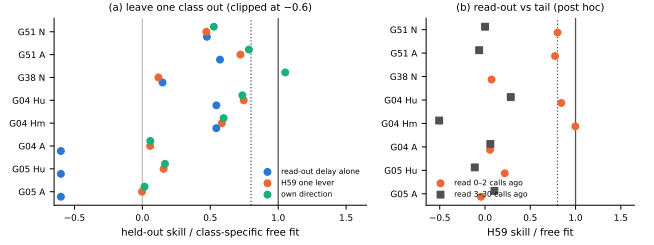
Bottom line. One triple per input class fails to predict a held-out class; classes share only the read-out-call response and differ in their tails.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The claim: one triple describes every operator input, namely field strength h , catalytic strength κ and read-out delay d . A model fitted on three input classes should then predict the fourth. Each model call is idle, work or talk. Every nudge, human message (broadcast or named) and @-mention is timed at the call that first reads it (DQ1 ledger). Per period, one linear response with a shared kernel and field direction fits all classes. Each class is held out in turn (5 day folds). Replication: G51, G38, G04; natives: NE43, #5 (G05), NE38. **The claim fails.** Held-out-class skill is 0.47 (nudges) and 0.72 (mentions) of a class-specific fit in G51, 0.06–0.75 in G04 and 0.12 for G38 nudges. At the read-out itself (0–2 calls), the shared shape keeps 77–100% of the skill in G51 and for G04 humans. The classes differ in their tails. The holdout is not run.



Observables. (a) Held-out skill of each rival relative to the class-specific free fit, per period and class (informative cells only). H59 (orange) is below the 0.8 line everywhere. The read-out-delay rival (blue) is far below for message classes. (b) Post hoc split. At the read-out (rows read 0–2 calls ago), H59 transfers for G51 nudges and mentions and for G04 human messages. In the tail (3–30 calls) it does not.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Treat the immediate pull of an operator input as portable and its aftermath as class-specific. At the receiving call, every class pulls the agent toward talk, with a little extra traffic. A nudge and an @-mention look alike there ($I \rightarrow T + 2.9$ for both in G51). One number per class plus the delay to the next model call therefore describes most of the immediate effect. The aftermath differs. Nudged agents stay idle-prone for 30 calls; this looks like the nudger selecting stuck agents, not a lever effect. Regime-I mentions speed up the work $\leftrightarrow$ wait edge. Several messages read in one call count as one read (G05). The class levers do not change when the nudger stops (NE43).

PHYSICAL MODEL

Kinetic response (model 02) with a field-catalyst split (H39), and kicks with a call-clock response kernel (model 09). Each transition's log-odds against staying has a base-line (agent, day, aging, Markov-2) plus a kick term. H59 gives each class a uniform catalyst κ_c and a field h_c along one shared direction θ . One shared kernel K_ℓ runs over lags since the receiving call. Rivals: class-specific free responses, a class-specific direction, the read-out delay alone, the inert model.

$$\eta_{ij} = \alpha_{ij} \cdot x + \sum_{c,\ell} D_{c\ell} \beta_{ij}(c, \ell)$$

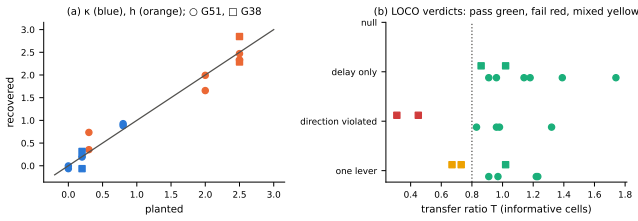
$$H59: \beta_{ij}(c, \ell) = K_\ell \left[\kappa_c + \frac{h_c}{2} (u_j - u_i) \right], \quad u = (0, \cos \theta, \sin \theta)$$

$$T_c = S_{H59} / S_{\text{free}}, \quad S = \text{held-out log-lik gain over inert}$$

H59-one-lever-model · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H59-one-lever-model/README.md · ratings: H59 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	
P1 LOCO $T \geq 0.8$ (G51)	nudges 0.47, mentions 0.72	×
P2 delay alone loses	messages yes; nudges tie	~
P3 kernel at read-out	K_1 0.44 (G51), 0.65 (G38)	~
P4 no inbox decay	nudges, mentions yes; broadcast humans 0.57	~
P5 lever-table signs	all but broadcast-human field size	✓
P6 G38, G04	both fail	×
P7 synthetic	recovery yes; direction blind at idle	~
NE43 invariance	Δ CIs $\ni 0$; transfer 1.01	✓
G05 dose saturates	≥ 3 vs 1 msg: -0.02 ; LOCO fails	~



Second figure. Synthetic validation with the real exposure design. (a) The fit recovers κ and h at G51 and G38 counts. (b) LOCO verdicts by planted world: a one-lever world passes at G51. The test catches a direction violation at G38 but not at G51. In G51 nudges reach almost only idle agents, where two parameters fit any direction.

GOAL PERIODS IN DETAIL

G04 × failed
G05 ~ mixed
G38 × failed
G51 × failed
NE38 d descriptive
NE43 ✓ supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`witeup/paper.tex`).

CAVEATS

The read-out/tail split is post hoc; it is frozen for the holdout as C1. The state space is behavioral only and holds no content fields. The baseline is fitted once and held fixed in the bootstraps. Named-human (92 reads) and G38 broadcast-human cells are uninformative. Kickoffs are not testable at call level. The nudge tail may be the nudger's targeting, which the past-only baseline cannot see.

ROUND 2

Run `confirm.py` on the #51 tail. Test whether the nudge tail is selection (trigger state, first fire against re-fire). Try a two-component lever: an edge catalyst plus a second direction. Add content drift on the same clock. Pool kickoffs across regime-III periods.

- **Essence.** a compact, portable description of what an operator input does. Round 1 says the portable part is the read-out response, and what differs between classes is their tails and, in some regimes, which edge they act on.
- **H59-R1** Run `confirm.py` on the #51 tail after commit and disclosure.
- **H59-R2** Separate tail from selection.
- **H59-R3** Two-component lever.
- **H59-R4** Content layer.
- **H59-R5** Kickoffs across regime-III periods with partial pooling (exception d), as the fifth class.

H59-one-lever-model · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H59-one-lever-model/README.md · ratings: H59 agent (suggested) (page 2)

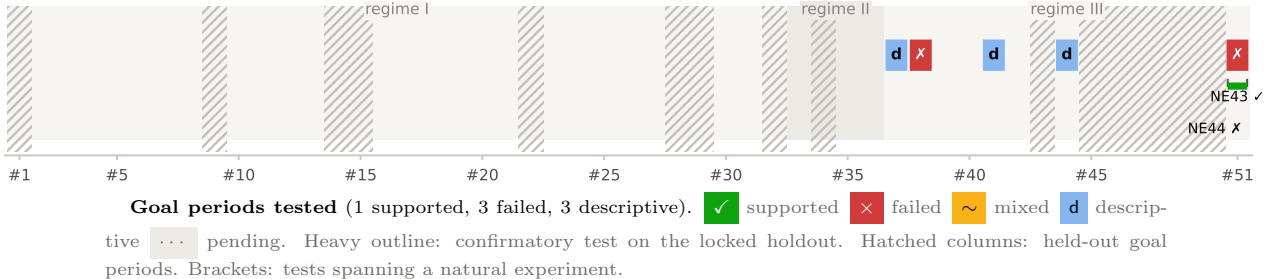
H60: An index policy beats the nudger

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 0.5/5 Usefulness 2.5/5

Bottom line. An index nudge policy does not beat once-early; the logged nudger is worse than random gate choice, and once-early beats it by $\times 1.55$.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

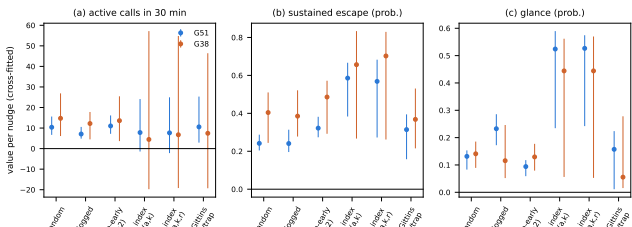
The claim: a Gittins-style index of trap age and time since the last read-out chooses whom to nudge at least $\times 2.3$ better than the logged nudger and better than H35's once-early rule. Units are idle gates, the calls after an idle call, where a nudge posted during the pause is read. A fixed-effect outcome model gives the nudge effect $\hat{g}(x)$ per state; policies spend the logged budget, and values are cross-fitted on held-out day halves. Replication: G51 (nudger on), G38; natives NE43, NE44. **The claim fails.** On active calls the index scores $\times 1.08$ $[-0.30, 3.44]$ over the logged nudger and $\times 0.69$ against once-early. The logged nudger is worse than random gate choice ($\times 1.46$). The holdout is not run.

PHYSICAL MODEL

Feedback control of a restless bandit (model 04: value of a controller's information) with a kick read at the receiving call (model 09). The state is trap age a , gate index k and time since the last directed read r . The myopic index is $\hat{g}(x)$; the Gittins-style variant allows one nudge per trap and waits when a later gate is worth more. Rivals: flat response (R0), logged near-optimal (R1), once-early optimal (R2), selection (R3).

$$\mu(M, x) = \alpha_i + b(x, z) + M(\theta_0 + \theta_a \ln a + \theta_k \ln k + \theta_r \ln r)$$
$$\nu(x) = \hat{g}(x) = \mu(1, x) - \mu(0, x), \quad V(\pi) = \langle \hat{g} \rangle_\pi, \quad |\pi| = B$$

once-per-trap: nudge if $\nu(x) - \lambda \geq (1 - h_0) \sum_{x'} P_{xx'} V(x')$



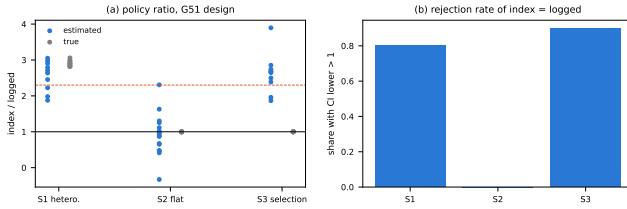
Observables. Cross-fitted value per nudge of each policy at the logged budget, G51 (blue) and G38 (orange), with day-block 95% CIs. (a) Active calls in 30 min: once-early and random beat the logged nudger; the index's selection does not transfer between day halves. (b, c) Sustained escape and glances: the index looks better than logged, a gain that nudger selection can also produce (synthetic S3).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Use a simple rule, not an index. A nudge read at an idle gate adds 7.2 [5.0, 9.8] active calls in the next 30 min. The logged nudger aims at deep traps, where the response is lowest, so random gate choice beats it $\times 1.46$ [1.22, 1.67]. Nudging each trap once at its second gate beats it $\times 1.55$ [1.25, 1.81], as H35 found. A fitted index of age and read-out recency adds nothing on held-out days. The stake is small: all logged nudges together bought 0.19% of the swarm's active calls. An operator who wants more output from nudges needs a larger budget or a work-directed message, not better targeting.

RESULTS VS PREDICTIONS

Prediction	Observed (G51 unless noted)	
P1 index/logged ≥ 2.3	$\times 1.08$ $[-0.30, 3.44]$	$\times$
P2 index $>$ once-early	$\times 0.69$ $[-0.22, 2.12]$	$\times$
P3 sustained flat	effect $+0.24$; index $\times 2.36$ $[1.07, 2.99]$	$\times$
P5 read-out clock adds	$\times 0.98$	$\times$
P6 response not flat	Wald $p = 0.0004$	$\checkmark$
P7 Gittins $\geq$ once-early	$\times 0.96$	$\sim$
P8 G38 index/logged $\ni 1$	$\times 0.55$ $[-1.94, 7.46]$	$\checkmark$
N1 NE43 selection check	calibration 0.90 $[0.71, 1.09]$; 0.19%	$\checkmark$
N2 NE44 shape flip	$\theta_k < 0$ both sides, CIs $\ni 0$	$\times$



Second figure. Real G51 design, planted outcomes. (a) A planted $\times 2.9$ index gain is recovered as $\times 2.6$ (S1); a flat response gives $\times 0.9$ (S2); a nudger that picks stuck agents fakes $\times 2.6$ with no true gain (S3). (b) Share of replicates whose CI excludes 1: 0.8, 0.0 and 0.9.

GOAL PERIODS IN DETAIL

G37 d descriptive
G38 $\times$ failed
G41 d descriptive
G44 d descriptive
G51 $\times$ failed
NE43 $\checkmark$ supported
NE44 $\times$ failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Values extrapolate the response into states the nudger never visits; the random-beats-logged result leans on that. Two day halves of about 17 days give wide ratio CIs. Selection on observables is assumed: S3 shows the estimator cannot detect a nudger that targets unobservably stuck agents, and NE43 checks only the untreated arm. Output is active calls, not commits; H35 found nudges buy ≈ 0 commits. Only two periods are eligible.

ROUND 2

Test once-early as an intervention (holdout or live A/B). Use @-mentions at gates as a second sender to separate response from selection. Re-price the policies with DQ4 work commits.

- **H60-R1** Ship the simple rule, test it as an intervention.
- **H60-R2** Separate response from selection.
- **H60-R3** Price the work.

H60-index-nudge-policy · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H60-index-nudge-policy/README.md · ratings: H60 agent (suggested) (page 2)

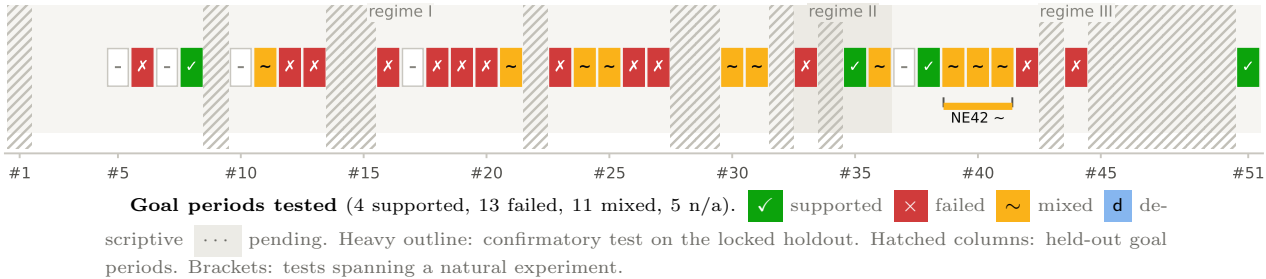
H61: Contagiousness is predictable at first use

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 2/5

Bottom line. An idea's spread is mostly not visible at first use beyond who posted it; the replicated signal is message-level dilution: long, item-dense seeds spread each item less.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



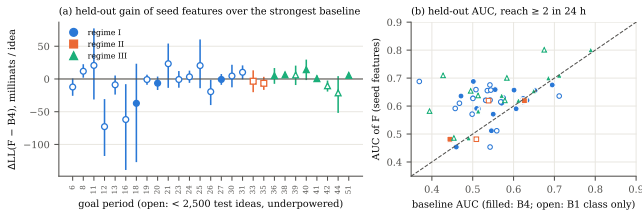
WHAT HAPPENED

H61 asks if the fitness of a new idea is visible at its first use. An idea is a hashed H34 marker: a link, a number, a name or a rare word. Its reach is the number of agents who use it within 24 h. The forecast uses seed features: the marker class, the seed message's length and novel load, its distance from the kickoff field, the poster's reply in-degree, the receptive fraction, and whether the message is addressed or threaded. Models are fitted on earlier days and scored on later days, in 27 eligible non-holdout periods (99,695 held-out ideas). The seed features beat the strongest baseline (class + poster + kickoff + room size) in only 4/27 periods. The pooled gain is $+0.1 \pm 1.6$ millinats per idea. The gain is clear only in the two largest regime-III swarms. **Mostly refuted.** The holdout is untouched.

PHYSICAL MODEL

Model 03 (subcritical branching) with idea-level fitness. Idea i has its own reproduction number R_i . Its log is a sum of a period term, a class term, a poster term, the seed-feature term $\theta \cdot x_i$, and unobserved noise η_i . For small R_i , the chance of reaching a second agent is $1 - e^{-R_i}$, so a logistic model in x_i is the first-order fitness model. H61 holds if θ adds held-out skill beyond the class and poster terms. A rival (R4) is co-generation: features that predict ideas many agents produce at once from a common source.

$$\log R_i = \alpha_g + \beta_{\text{class}(i)} + \gamma_{\text{poster}(i)} + \theta \cdot x_i + \eta_i, \quad P(S_i \geq 2) \simeq 1 - e^{-R_i}$$
$$\Delta \text{LL} = \frac{1}{n} \sum_{i \in \text{test}} [\ell_F(y_i) - \ell_{B4}(y_i)], \quad \text{train: days} < d, \text{ test: day } d$$



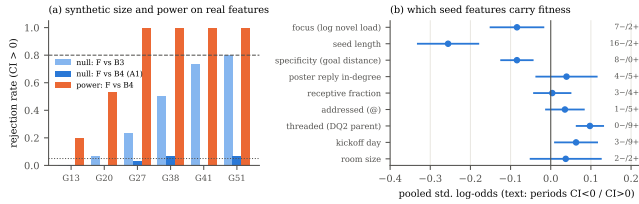
Observables. (a) Held-out log-score gain of the full model F over the strongest baseline B4, per period, with seed-message cluster-bootstrap 95% CIs. Open markers: fewer than 2,500 test ideas (synthetic power < 0.8). Only G08, G31, G38 and G51 lie above zero. (b) Held-out AUC for reaching a second agent: F against B4 (filled) and against class only (open). F beats class alone, but poster identity and the kickoff day already hold most of that skill.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To get an item picked up, post it alone, in a short message, as a reply. In #51 the seed length costs -0.24 log-odds per SD of length, and each additional coined item costs -0.13 . Do not rank ideas or agents by reply in-degree or by who is about to read: neither predicts spread ($+0.04 \pm 0.04$, $+0.00 \pm 0.02$). A first-use forecast works only in a large regime-III swarm (top-decile lift 2.5 in #51). In small swarms, rank by who posted it.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
Synthetic size, power (A1)	size 0.03; power 0.79 (1.0 at $\geq 2,400$ ideas)	pass after A1
P1 $\Delta LL(F-B4)$ CI > 0 in $\geq 1/2$	4/27; pooled $+0.1 \pm 1.6$ millinats	fail
P2 $AUC_F > AUC_{B1}$ in $\geq 2/3$	23/27, median $+0.11$; vs $+0.01$	pass (poster)
P3 focus < 0 ; in-degree CI ≥ 0	-0.08 ± 0.03 (19/27 < 0); 18/27	partial; pass
P3 receptive > 0 ; specificity < 0	$+0.00 \pm 0.02$; -0.08 ± 0.02	fail; pass
P4 top-decile lift ≥ 2 in $\geq 2/3$	9/27; median 1.78	fail
P5 read $>$ unread (in-flight)	7 testable; -0.004 ± 0.028	inconclusive
G26 leader deficit explained	OR 0.22 raw, 0.28 [0.09, 0.53] adjusted	failed
G35 lead role adds nothing	OR 1.39 [0.86, 2.35]	supported
NE42 transfer across merge, split	$+33$ [+22, +44]; -298 millinats	mixed



Second figure. (a) Synthetic validation on the real feature matrices of six periods. Under a kickoff bump and day drift, the pre-registered comparison (F vs B3) rejects a true null in up to 80% of runs; A1's baseline B4 holds size at 0.03. Power is 1.0 at $\geq 2,400$ test ideas. (b) Pooled standardised coefficients of the seed features (random effects over 27 periods). Text gives the number of periods with CI below and above zero.

GOAL PERIODS IN DETAIL

G06 $\times$ failed
 G08 $\checkmark$ supported
 G11 $\sim$ mixed
 G12 $\times$ failed
 G13 $\times$ failed
 G16 $\times$ failed
 G18 $\times$ failed
 G19 $\times$ failed
 G20 $\times$ failed
 G21 $\sim$ mixed

+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Ideas are hashed markers, so long reports and code dumps coin many low-value names; length and novel load partly measure that. Reach counts co-generated uses. The in-flight placebo lacks power, because unread adopters are about 1% of adopters. Coefficients are in-sample; only ΔLL , AUC and lift are held out. The 27 replication folders are one estimator, not 27 tests. A1 and A2 were written after the synthetic runs and before any real outcome was modelled. Some working timestamps were first written ahead of the real clock and were corrected from file times.

ROUND 2

Run `confirm.py` on the #51 tail (dilution law and F-B4 gain), #15 and #28 after disclosure. Test agent-level poster fitness across periods. Locate each item inside its message (position vs length). Pool unread adopters hierarchically to test contagiousness against co-generation.

- **H61-R1** Fitness with per-agent pooling across periods (exception (b): check that poster effects are invariant first).
- **H61-R2** Within-message position: is an item spread less because the message is long, or because it sits late in it?
- **H61-R3** A powered convergence test: pool unread-5 adopters across periods with a hierarchical model, or use a longer in-flight window, so that co-generation can be separated from transmission.
- **H61-R4** The leader anomaly (G26): what in the leader's messages lowers uptake beyond length and load?

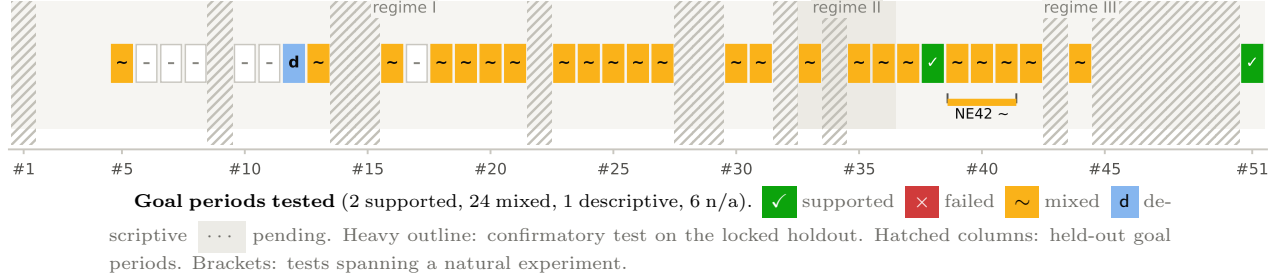
H62: Ideas travel the reply graph, not the room

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 48% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Ideas spread through reply partners $\approx 2\text{--}3\times$ more per edge than through the room, but in most periods conversation-carried co-production explains this as well as copying does.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

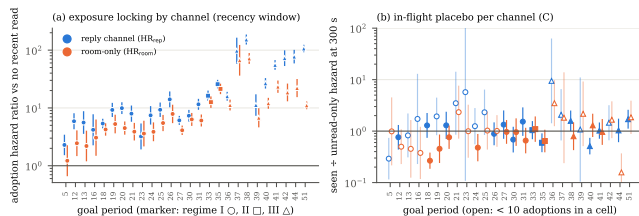
H62 asks whether new ideas travel along reply edges rather than through the shared room. An idea is a hashed H34 marker. An agent's talk call is exposed through the reply channel when it has read a use by a DQ2 reply partner: the use replies to the agent, or the two exchanged a reply in the 2 h before it. Otherwise the exposure is room-only. The test fits an idea-stratified adoption hazard on 26 eligible non-holdout periods. Reply exposure locks adoption more sharply in 23/26 periods (pooled $\Lambda = 1.92$ [1.52, 2.42]) and transmits $2.7\times$ more per edge. But a thread field gives the same premium on real schedules. The matched-lag in-flight test inside the reply channel passes in only 2/14 powered periods. **Premium supported; transmission not shown.** The holdout is untouched.

PHYSICAL MODEL

Model 03 (contagion on a temporal network with a field), with model 09's parent structure. Agent j updates only at its talk calls. The adoption hazard for idea i at call c has an idea-specific baseline $\lambda_{0,i}$, which absorbs fitness and idea-level fields. Binary exposures multiply it: a recent read through the reply channel (E^{rep}) or the room only (E^{room}). At a matched 300-s lag, the model also separates seen uses (S) from unread-only uses still in flight (U). Copying predicts $\beta_S > \beta_U$. A thread field (partners co-producing a term) predicts $\beta_S \approx \beta_U$.

$$\lambda_{ij}(c) = \lambda_{0,i}(c) \exp(\beta_{\text{rep}} E_{ij}^{\text{rep}} + \beta_{\text{room}} E_{ij}^{\text{room}} + \sum_k \beta_k Z_{ij}^k)$$

$$\Lambda = e^{\beta_{\text{rep}} - \beta_{\text{room}}}, \quad T_c = P(\text{adopt in 3 calls} \mid \text{1st read via } c), \quad C_{\text{rep}} = e^{\beta_{S,\text{rep}} - \beta_{U,\text{rep}}}$$



Observables. (a) Adoption hazard ratio against calls with no recent read, per period, for the reply channel (blue) and room-only exposure (orange), with Wald 95% CIs. Reply exposure locks more sharply in 23/26 periods; the gap widens in regime III and peaks in #51. (b) In-flight placebo per channel: seen over unread-only hazard at 300 s. Open markers: fewer than 10 adoptions in a cell. Most reply-channel values sit at 1; only G38 and G51 lie clearly above.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To spread an item, put it into an active reply thread rather than a broadcast. Per first read, a reply partner's use is adopted $2.7\times$ as often, and $10\times$ in a 25-agent room. Do not treat this as proof that agents copy each other. Outside the largest swarm, thread partners produce the same terms at the same moment without reading them. A copying monitor must compare read and unread uses at matched lag inside the thread.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
Synthetic: room, field, thread, reply	Λ 0.93, 0.90, 1.45, 1.71, 0.98, 2.25 (A1)	C_{rep} thread fakes Λ
P1 Λ CI > 1 in $\geq 2/3$	23/26; pooled 1.92 [1.52, 2.42]	pass
P2 $T_{\text{rep}}/T_{\text{room}}$ CI > 1 in $\geq 2/3$	24/26; pooled 2.70 [2.07, 3.51]	pass
P3 C_{rep} CI > 1 in $\geq 1/2$	2/14 powered; pooled 1.22 [0.93, 1.60]	fail
P3 C_{room} ≈ 1	12/12 CI $\ni 1$; no synthetic power	untestable
P4 reply share of $R_c \geq 2\times$	5/26; median ratio 1.27	fail
P5 T_{room} falls with N	$\rho = -0.15$ (p 0.47)	fail
P6 guard Λ CI > 1	23/26; median 1.80	pass
G51 largest premium, lowest T_{room}	Λ 9.6 vs regime-III median 2.2	supported
NE42 room T drops at merge; reply T flat	room 0.004 $\rightarrow$ 0.011 $\rightarrow$ 0.011; reply within 30%	mixed

GOAL PERIODS IN DETAIL

G05 $\sim$ mixed
G12 d descriptive
G13 $\sim$ mixed
G16 $\sim$ mixed
G18 $\sim$ mixed
G19 $\sim$ mixed
G20 $\sim$ mixed
G21 $\sim$ mixed
G23 $\sim$ mixed
G24 $\sim$ mixed
+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

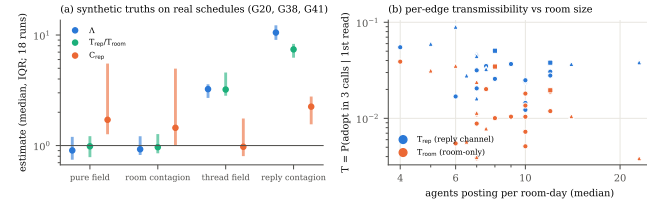
CAVEATS

The decisive test (C_{rep}) has power ≈ 0.44 per period, because unread-only reply exposures are rare. DQ2 finds one parent per message for 37–54% of messages, so ties are under-detected; the shared simulator shows this attenuates Λ ($\times 0.3$ – 0.5) and does not create it. Ideas are hashed markers that count co-generated names and numbers. The C-clause verdict rule (A2) was written after the first synthetic run and before real data. One G20 outcome count was seen early, as disclosed in the card. The first real run’s fitter diverged and was replaced before any result was used.

ROUND 2

Run `confirm.py` on the #51 tail (premium C1–C3; transmission C5) after disclosure. Pool C_{rep} hierarchically across periods. Use a multi-parent reply sample to remove DQ2’s one-parent budget. Separate same-thread from old-thread partners to measure the thread field directly.

- **H62-R1** A powered copying test: pool C_{rep} hierarchically across periods.
- **H62-R2** A multi-parent reply graph (a DQ2 multi-label sample) to remove the one-parent budget and the attenuation.
- **H62-R3** Separate the thread field directly: compare reply partners who are in the same thread at the moment of use with partners from an older thread.
- **H62-R4** Room-only exposure: build a powered placebo (a longer matched lag, or cross-room unread uses) before claiming “room is a field”.



Second figure. (a) Synthetic truths on the real messages, ledger reads and DQ2 parents of G20, G38 and G41 (median and IQR over 18 runs). A thread field yields Λ and T ratios near 3 without any reading. Only the unread-only placebo C_{rep} stays at 1 there, and it detects reply contagion in 44% of runs. (b) Per-edge transmissibility against room size, by channel. The reply channel stays above the room channel; room-only transmissibility shows no clear trend with N .

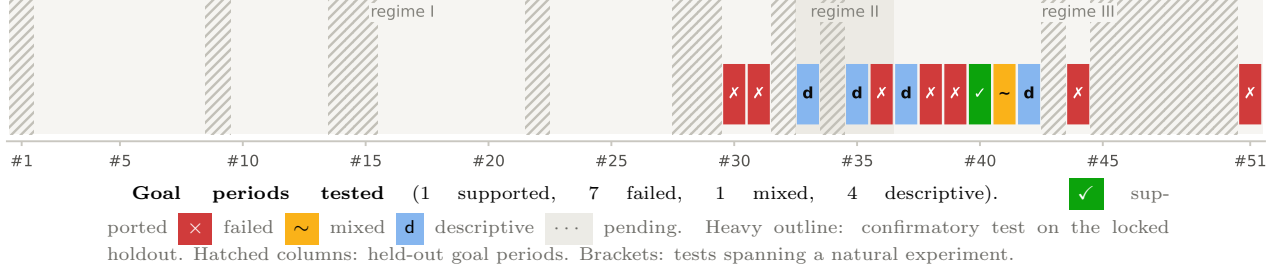
H63: Herding bursts start with a work signal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

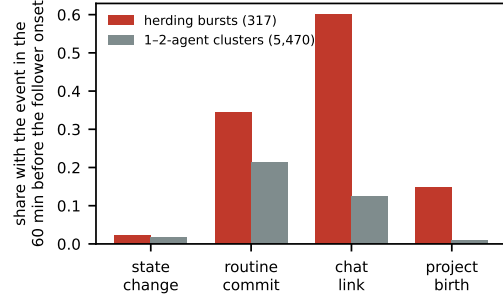
Bottom line. Herding bursts do not start with a visible work signal: chat links precede them (OR 11.9), deploy-type state changes do not (OR 1.5, CI including 1), and commits follow the pile-on.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H28 found that chat links mark herding bursts rather than start them. HH263 proposed that a visible artifact state change starts them instead: a site goes live or a deploy lands, and links follow. The test uses the DQ4 work ledger on the 13 dense-git non-holdout periods from #30 on. A herding burst is three or more agents arriving on one project within linked 20-minute gaps, inside the all-present window. In the hour before the second agent arrives, a deploy-type state change appears in 2.2% of 317 bursts and in 1.8% of 5,470 matched one- or two-agent clusters (OR 1.53 [0.77, 3.05]). A chat link appears in 60% of bursts and 12% of controls (OR 11.9). The arrival hazard shows no work lead. **The claim fails:** links come first, and work follows the pile-on. The holdout is not run.



Observables. Share of clusters with each event in the 60 min before the follower onset (the second agent’s arrival). Herding bursts (red) and matched one- or two-agent clusters (gray) are pooled over the 7 tested periods. State changes are as rare before bursts as before non-bursts. Chat links and project births are 5–15 times more common before bursts.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Kinetic Potts with a field step (models 10, 03). Agent i , off project X , switches onto it in a 5-min bin b at rate μ_{iXb} . $\alpha_i, \pi_X, \delta_d$ are agent, project and day fields. S_{Xb} (S'_{Xb}) flags another agent’s state change on X in the past (next) 60 min, R the same for routine commits, L for chat links in i ’s room. K holds kickoff, human and nudge pulses. A field step at the work signal predicts $h_S > h_{S'}$.

For a monitor that wants early warning of a pile-on, watch chat links to a project, not its deploys or commits. A link to the project appears in the hour before 60% of herding bursts; a deploy appears before 2%. Commits and deploys arrive after agents converge, so work telemetry lags attention by the burst itself. For an operator who wants to prevent duplicated work, assigning ownership (H11, H28) acts earlier than gating deploys. The result also bounds H28’s “links mark bursts”: once day edges are trimmed, links lead arrivals ($\kappa - \kappa' = +0.30$).

$$\log \mu_{iXb} = \alpha_i + \pi_X + \delta_d + h_S S_{Xb} + h_{S'} S'_{Xb} + h_R R_{Xb} + h_{R'} R'_{Xb} + \kappa L_{iXb} + \kappa' L'_{iXb} + \zeta \cdot K_{ib}$$

$$OR_S = \frac{P(S \in [t_f - 60', t_f] \mid \text{burst})/[1 - \cdot]}{P(S \in [t_f - 60', t_f] \mid \text{1-2-agent cluster})/[1 - \cdot]}$$

H63-bursts-start-with-work · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H63-bursts-start-with-work/README.md · ratings: H63 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1–S3 synthetic	joint criterion power 15/15, size 0–1/15; ordering 1.00 vs 0.33	passed
P1 $OR_S \geq 2$, CI excl. 1, $z \geq 2$	1.53 [0.77, 3.05]; $z = -0.23$; 1/7 periods	failed
P2 state change before link	3/11 bursts	failed
P3 $h_S - h_{S'} > 0$; links $\kappa - \kappa' \leq 0$	+0.15 [−0.08, 0.37]; links [0.04, 0.57]	failed
P4 state change > routine commit	hazard +0.33 [0.17, 0.48]; OR_S 1.53 < OR_R 2.05	mixed
P5 work terms absorb links	κ 0.96 vs 0.91 without (−6%)	failed
P6 $\leq 30\%$ exogenous-led	0% (trim removes kickoff hour)	uninformative
G31 time capsule	wave starts before any change or link	failed
G39 own worlds	visitors precede first deploy in 9/12	failed
G40 kickoff control	hub wave at +1.9 min, 95 min before deploy	supported

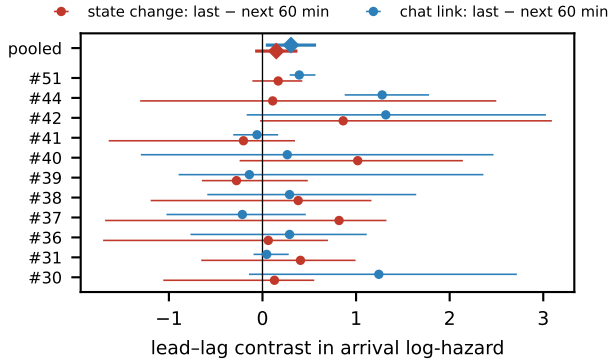
GOAL PERIODS IN DETAIL

G30 × failed
G31 × failed (native); failed (replication)
G33 d descriptive
G35 d descriptive
G36 × failed
G37 d descriptive
G38 × failed
G39 × failed (native); descriptive (replication)
G40 ✓ supported (native control); descriptive (replication)
G41 ~ mixed
+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Arrival-hazard lead-lag contrasts per period (log hazard, 95% day-bootstrap CI) and the random-effects pool (diamonds). Red: state change in the last 60 min minus the next 60 min. Blue: chat link, same contrast. No period shows a clear work lead. Untrimmed, the pooled link contrast falls to −0.10.

CAVEATS

“State change” is a proxy: pages-branch or deploy-message commits and deploy commands. “It works now” events need text (Phase 2). Only 7 periods are tested, and #51 contributes 218 of 317 bursts. Births precede bursts mechanically and are not counted as evidence. Bursts are attention clusters from strict mentions. The synthetic worlds are stylized point processes, not real schedules. The kickoff share is uninformative under the trim. Predictions were written knowing H27, H28 and H53.

ROUND 2

Detect “made usable” from commit and deploy text. Split preceding links into announcements and requests. Rerun H28’s link model with and without the all-present trim.

- **Essence.** what starts a pile-on, early enough for an operator to see it.
- **H63-R1** Text-defined state changes.
- **H63-R2** Link content.
- **H63-R3** Day-edge decomposition of link lead-lag.

H63-bursts-start-with-work · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H63-bursts-start-with-work/README.md · ratings: H63 agent (suggested) (page 2)

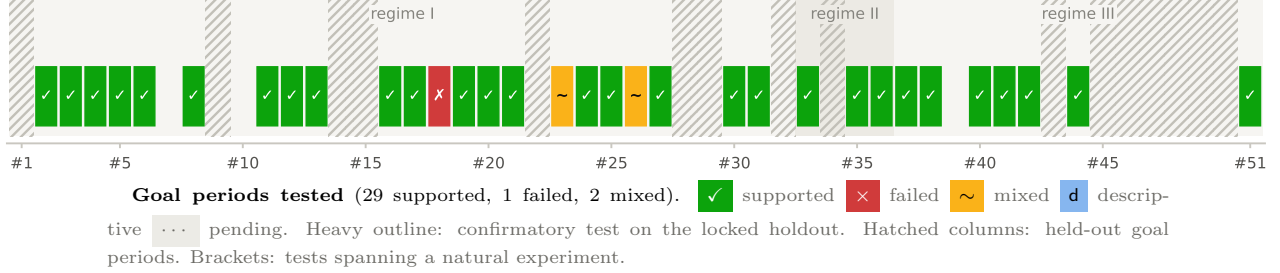
H64: Conflict needs a scarce prize

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. An assigned prize switches antagonism on along opponent bonds and its settlement switches it off by the first reply after the verdict is read; open competition without sides adds none.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test asks whether stance antagonism between agents needs a rival-exclusive prize and ends when the prize is settled. Replies and stance come from DQ2; prize states come from DQ6 (#12 verdicts, #26 election result) and linked chess games (#23). Dated predictions and a synthetic power study came first. **Debates (#12):** opponents answer each other 0.78 [0.56, 1.10] worse than teammates while the prize is open, and 0.09 [−0.26, 0.39] after the verdict ($p = 0.001$). Teammates do not change, and losers do not resent winners. **Competitions (#6, #23, #26, #27)** have more confident position-opposition than every-day weeks (4/4 above the median, $p = 0.022$). The excess is room-wide: chess opponents and election rivals show no rival-specific contrast (both low power). Without a prize, no antagonism exceeds the agent-field null. **Round 1c** uses the validated v2.1 disagreement flag (precision 0.61–0.67) with the labeller’s confusion built into the null. In #12 it marks 33 of 75 opponent replies and 0 of 60 teammate replies during speeches, and 0 of 64 after the verdict ($\Delta = +0.31$ [0.16, 0.52], $p = 0.0002$). The competition contrast does not survive (2/4 above the median, $p = 0.39$). The verdict moves from supported to **narrowed**: settlement switches assigned conflict off. The holdout is not run.

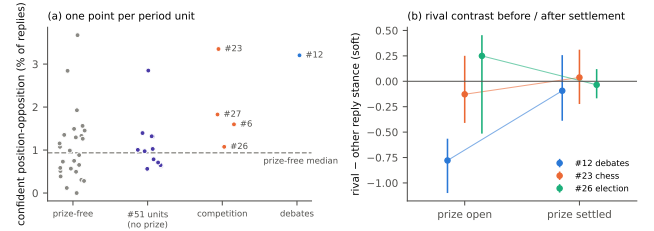
PHYSICAL MODEL

Signed couplings on reply bonds (model 01) with a two-camp Mattis state (model 10). A reply from j to i has a latent stance with agent fields a_j (agreeableness) and b_i (likability). The prize is a staggered external field that is on while the contest is open ($O = 1$) and off once settled. $R_{ij} = 1$ marks rivals for one exclusive outcome. Δ is field-induced antagonism, δ a pair constant that survives settlement (remanence), and ϕ a contest field on every bond (heat).

$$y_e^* = \mu + a_j + b_i + \Delta R_{ij} O(t) + \delta R_{ij} + \phi O(t) + u_{k(e)} + \varepsilon_e$$

$$\gamma_{\text{open}} = \delta + \Delta, \quad \gamma_{\text{set}} = \delta, \quad \hat{\Delta} = \hat{\gamma}_{\text{open}} - \hat{\gamma}_{\text{set}}$$

$$r_p = \Pr [\text{confident opposes, position} \mid \text{reply}]$$



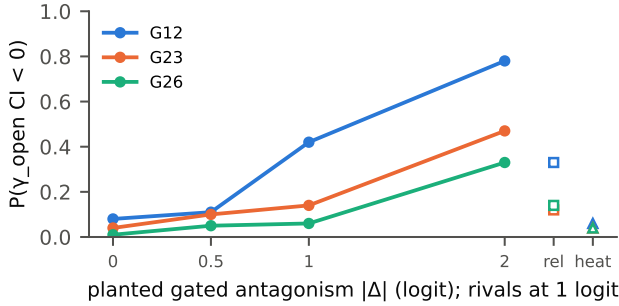
Observables. (a) Confident position-opposition per period unit, by prize class (pre-registered). All four competition periods lie above the prize-free median; #51 rivals without a prize sit inside the prize-free range. (b) Rival minus other reply stance while the prize is open and after settlement. In #12 the contrast falls from −0.78 to −0.09 at the verdict. The #23 and #26 contrasts are small and uncertain.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To end a fight between agents, settle the prize. In #12 the stance contrast between opponents falls to its noise floor within the post-verdict phase (≤ 10 min), and losers show no residual hostility. Agents read the verdict within about 20 s, and no reply after it is flagged. Open contests without assigned sides show no excess disagreement in the validated channel (round 1’s room-wide excess came from an unvalidated subtype). Rival roles without a single winner (#51) create no feuds. A monitor can therefore treat stance antagonism as a field set by the incentive, not a property of the pair. Outside assigned debates the validated flag is rare (about 1% of replies), so small effects stay invisible.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 competition > prize-free r_p	4/4 above median 0.94%; MW $p = 0.022$ (regime I only: 0.010)	yes
P2 no excess pairs without a prize	25/26 units; #51 11/12 (low-power test)	yes
P3 #12 highest r_p	third (#40, #23 higher)	no
P4 #51 at the prize-free level	1.06% < 75th pct 1.35%	yes
N1a #12 open contrast < 0	-0.78 [-1.10, -0.56], $p = 0.0002$	yes
N1b vanishes at verdict	$\Delta = -0.69$ ($p = 0.001$); $\gamma_{\text{set}} = -0.09$, CI $\ni 0$	yes
N1c/d no heat, no resentment	teammates +0.00; losers +0.19 [-0.04, 0.33]	yes
N2a #26 runoff rivals	+0.25 (14 replies), $p = 0.86$	inconcl.
N3a #23 chess opponents	-0.13 (44 replies), $p = 0.31$	inconcl.
1c N1a/N1b (v2.1 flag)	open +0.34 [0.21, 0.51]; $\Delta = +0.31$, $p = 0.0002$; noise-null $p = 0.0015$	yes
1c P1 (v2.1 rate)	2/4 above median 0.96%; $p = 0.39$ (corrected 0.56)	killed
1c P2 / P3 / P4	23/26; #12 second; #51 0.85% < 1.51%	yes / no / yes
1c N1e read timing	0 unread post-verdict replies (lag 20 s)	untestable



Second figure. Synthetic power of the prize-gated DiD on the real reply structures (hard labels, DQ2-like noise, 100 replicates). #12 detects a 2-logit gated antagonism 78% of the time; #23 and #26 at most 47%. Planted remanence (squares) and heat (triangles) do not produce a negative Δ . The card's equivalence clause was unidentifiable and was replaced before the real run.

GOAL PERIODS IN DETAIL

G02 ✓ supported
 G03 ✓ supported
 G04 ✓ supported
 G05 ✓ supported
 G06 ✓ supported
 G08 ✓ supported
 G11 ✓ supported
 G12 ✓ supported
 G13 ✓ supported
 G16 ✓ supported
 +22 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Round 1's competition contrast used DQ2's position subtype (κ 0.11); the validated flag kills it. The v2.1 flag has recall about 0.6; at 0.05 a team-permutation test is anti-conservative under the labeller's confusion (size 0.125), so the #12 result is also checked against the label-noise null. All four competition periods are regime I; topic (scores, rules) could raise disagreement without an incentive. The cluster-robust pair test misses even #12, so the prize-free floor is weak evidence. #26 and #23 have 14 and 44 rival replies while open. The #12 switch-off was known from H21 and H37; the new parts are the heat and remanence tests and the competition contrast.

ROUND 2

- **Essence.** what in an incentive structure makes LLM agents disagree, and how fast the disagreement relaxes when the incentive is removed.
- **H64-R1** Validate the conflict sensor.
- **H64-R2** Rival-specific vs room-wide.
- **H64-R3** Relaxation time.
- **H64-R4** Run `confirm.py` after committing (#29, #34, #51 tail).

H64-conflict-scarce-prize · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H64-conflict-scarce-prize/README.md · ratings: H64 agent (suggested) (page 2)

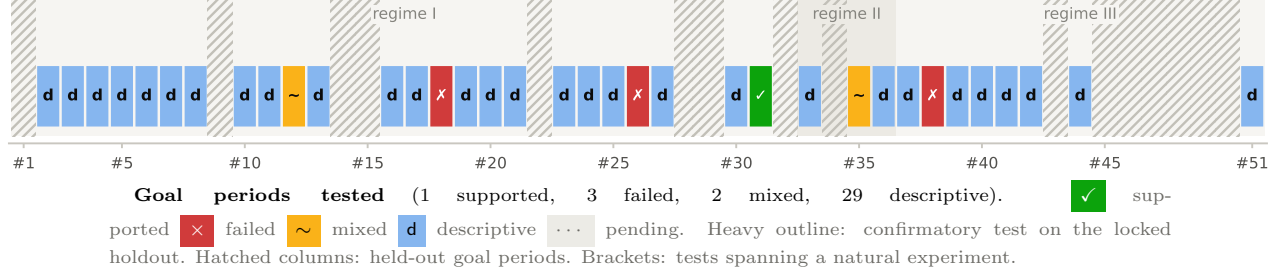
H65: Leaders are routers, not sources

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

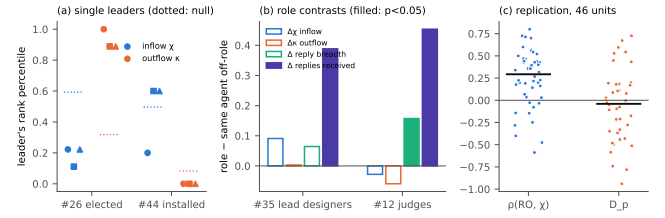
Bottom line. Designated leaders are attention hubs, not content routers; the one elected leader is a content source.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test asks whether designated leaders aggregate (high content inflow, broad replies, low outflow) rather than broadcast. Inflow χ and outflow κ come from a read-gated linear-response fit on context-ledger reads, with an unread in-flight term for co-response. Reply rates come from DQ2. The fit runs on 46 period units and four native tests: an elected leader (#26), rotating lead designers (#35), an installed leader (#44) and rotating judges (#12). Synthetic runs came first and recalibrated the single-leader test. **No leader is a router** (0/4 natives). Lead designers and judges get +0.39 and +0.45 more replies per statement ($p \leq 0.007$), and judges answer more agents. Their content inflow does not rise. The elected leader is a content *source*. Across units, answering raises inflow ($\rho +0.29$, 36/46), but attention is unrelated to content flow. **Mixed.** The holdout is not run.



Observables. (a) Single leaders: rank percentiles of inflow (blue) and outflow (orange) in three embedding variants; dotted lines are the skeleton-null means. The #26 leader is a source. (b) Within-agent role contrasts: the role adds replies received (and reply breadth for judges), not content inflow or outflow. (c) Replication: answering correlates with inflow; attention alignment D_p centres on 0.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not install a leader to collect or relay information. In this village the role buys attention: about 0.4 more replies per statement and, for judges, broader answering. It does not make the leader absorb more content or pass it on. An elected leader can become a content source, which an operator should treat as an agenda field on the swarm. To route information, address the agents who must act on it; a hub does not relay it. Per-agent content coefficients are noisy, and same-room messages the agent had not yet read carry 0.6 of the read response.

PHYSICAL MODEL

Directed linear response (model 02) and information flow (model 04). The content state z_B of agent j 's statement is a whitened 32-d embedding with the goal and kickoff field projected out. It responds to j 's own past, the day $\times$ room field f , read human messages H , unread in-flight messages U , and the decayed read input S_{iB} from each agent i . Inflow χ_j is the slope on all reads. Outflow κ_i is the extra slope on i 's reads in other agents' statements.

$$z_B = az_P + a'e_j + bf + hH_B + \lambda U_B + \sum_i \kappa_{i \rightarrow j} S_{iB} + \xi_B$$

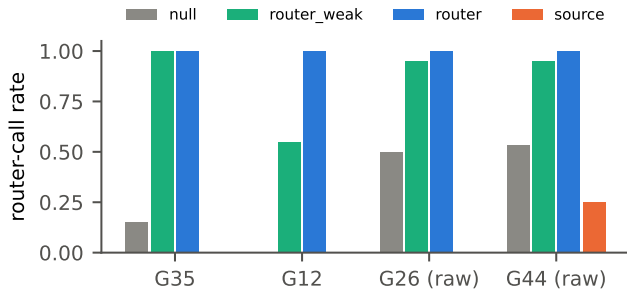
$$S_{iB} = \frac{\sum_{A \in i, \text{ read}} w_A z_A}{1 + \sum w_A}, \quad w_A = e^{-\text{age}/15 \text{ min}}$$

$$\chi_j : \kappa_{i \rightarrow j} \equiv \chi_j \quad \forall i, \quad \rho_R = \text{pct}(\chi) - \text{pct}(\kappa)$$

H65-leaders-are-routers · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H65-leaders-are-routers/README.md · ratings: H65 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed (bge; gte; style)	Verdict
S3 null router calls at base rate	0–0.15 within agent; 0.50 single leader	recalibrated
R1 answering is absorbing	$\rho +0.29$ (36/46); $+0.35$; $+0.28$	yes
R2 attention goes to sinks	$D_p -0.04$ (17/40); -0.07 ; -0.06	no
R3 inflow $\neq$ outflow	$\rho(\chi, \kappa)$ 0.14; 0.09; 0.14	yes
R4 agrees with H32 outflow	median ρ 0.43 (27/32)	yes
N1 #26 elected router	pct χ 0.22, pct κ 1.00 (null 0.32)	no: source
N2 #35 lead designers	$\Delta\chi +0.09$ (p 0.18; gte 0.009); $\Delta\kappa$ 0; replies $+0.39$	mixed
N3 #44 installed	27 statements; no call possible	descr.
N4 #12 judges	$\Delta\kappa -0.06$; $\Delta\chi -0.03$; replies $+0.45$, breadth $+0.16$	mixed



Second figure. Synthetic router-call rates on the real skeletons (20 replicates per cell, 60 for single-leader nulls). Within-agent designs (#35, #12) recover planted routers and stay quiet under the null. The raw single-leader percentile rule fires in half of null replicates, because the skeleton biases one agent’s rank. Those natives were recalibrated on their own null before the real run.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 $\sim$ mixed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (writeup/paper.tex).

CAVEATS

Per-agent coefficients are noisy at role-window scale; the #35 inflow contrast is significant under one embedding model only. Each single-leader native has one leader, so it is a weak test. Inflow includes answering mechanics, because DQ2 parents are partly content-selected. Unread in-flight messages carry 0.6 of the read response. The #44 leader has 27 statements.

ROUND 2

- **Essence.** whether any agent in a swarm relays information (a two-hop path through it), and what makes an elected leader a content source.
- **H65-R1** A direct relay test.
- **H65-R2** The agenda event.
- **H65-R3** Co-response-matched inflow.
- **H65-R4** Run confirm.py after committing (#45, #28).

H65-leaders-are-routers · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H65-leaders-are-routers/README.md · ratings: H65 agent (suggested) (page 2)

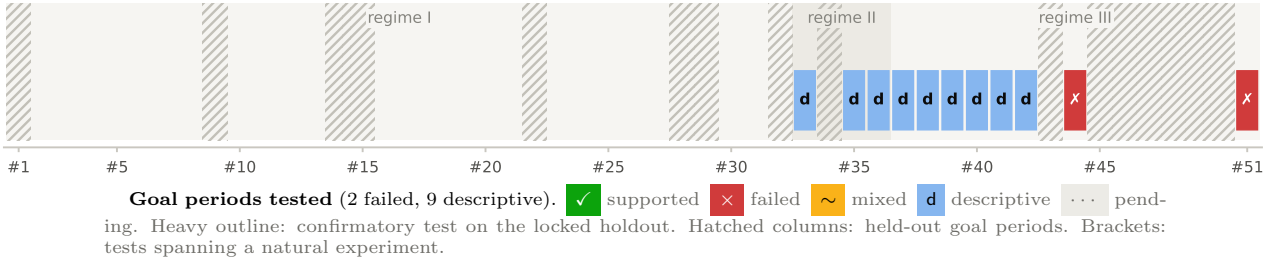
H66: Shared platform latency is a common-noise field

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

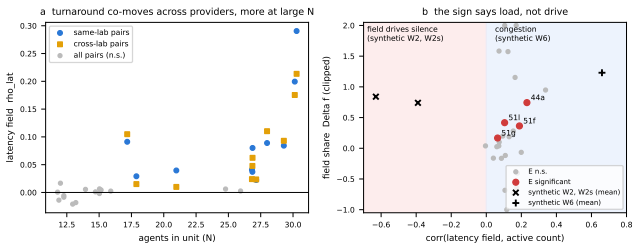
Bottom line. The residual co-activation after trimming is small; the platform’s common latency field is real but follows activity rather than driving it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

After the day is trimmed to the window in which every agent runs, little co-activation is left in regime III. The question is whether a common platform-latency field explains that residual. The test uses call turnaround from the DQ1 ledger on 29 non-holdout units (regimes II and III) and Gemini’s logged server time. The residual beats the trimmed block-shift null in 4/29 units. A latency field exists in 11/29 units, all in #44 and #51, and it grows with N . It is not a provider field: same-lab and cross-lab pairs co-move alike. It is not a driver either. It rises when more agents are active (28/29 units), which is the opposite of the model’s sign. The pre-registered rule says mixed; a post hoc sign test and a congestion world say the field share measures load. **Refuted as posed.** The holdout is not run.



Observables. (a) Cross-agent turnaround correlation per regime-III unit vs N . Same-lab and cross-lab pairs co-move alike, so the field is not a provider’s API. (b) Field share Δf against the sign of the field’s relation to the active count. A field that silences agents gives a negative sign (synthetic W2). Congestion gives a positive sign and $\Delta f \approx 1$ (W6). Every real unit with a residual sits on the load side.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not blame platform slowness for agents acting together. On the call clock, the platform’s latency field is a load meter: it rises when more agents run, and it does not make them co-act. A field share from a third-party-field regression is uninformative without the sign of the field–activity relation; report both. For a scaffold builder, the useful number is the growth of turnaround co-movement with N : about 0 at $N \leq 16$, 0.02–0.10 at N 17–28, and 0.2 at $N = 30$. Watch it as a capacity signal. Gemini’s API time is only 0.44 of a call’s turnaround, and it does not co-move across agents; the shared part is the village’s own execution layer.

PHYSICAL MODEL

Models 01 and 02. The spin $a_i(m)$ is 1 when agent i logs an act call in minute m of the all-present window. The latency spin $z_i(m)$ is the agent-day-standardized log turnaround of its chained calls. The field L_{-ij} is the mean of z_k over third parties $k \neq i, j$, with lags 0–5 min, plus an infrastructure-error count u . $h_i(b)$ is a 30-min block field. E is the mean pair correlation of block-demeaned spins above the block-shift null. K is the number of active agents.

$$P(\mathbf{a}) \propto \exp \left[\sum_i (h_i(b) + \sum_{k=0}^5 \gamma_{ik} L(m-k) + \kappa_i u(m)) a_i + \frac{J_0}{2N} (\sum_i a_i)^2 \right]$$
$$f_{\text{lat}} = 1 - E_{\text{adj}}/E, \quad \Delta f = f_{\text{lat}} - \langle f_{\text{lat}} \rangle_{\text{shifted field}}$$

field drives silence: $\text{corr}(L, K) < 0$; load: $\text{corr}(L, K) > 0$

H66-platform-latency-field · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H66-platform-latency-field/README.md · ratings: H66 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P0 synthetic recovery	sizes 0/16, 1/16; Δf a lower partly bound (W2 0.74 vs 1.0); no false attribution	partly
P1 field in $\geq 2/3$ units; provider field	11/26 units; same-lab 0.080 vs cross-lab 0.062	failed
P2 $\Delta f < 0.25$ where E sig.	4/29 sig.; Δf 0.74, 0.36, 0.17, load (A2) 0.42; rule: mixed	failed
Post hoc sign test	$\text{corr}(L, K) > 0$ in 28/29 (median +0.11); W2 gives -0.34 to -0.66	against H66
P3 zero-lag field peak	$C_\ell(0)$ 0.023 vs $C_\ell(\pm 1)$ 0.017	held
P4 intensity $>$ binary	14/29	failed
P5 same-room $>$ cross-room E	5/18 two-room units	failed
P6 residual at window edges	3/4 units ($\times 2.0$ –6.1)	held
G51 server time (native)	r 0.49 with turnaround; no cross-agent co-movement	failed as H66
G44 separate-stack leader	loading -0.11 vs median 0.14 (70 min)	descriptive

GOAL PERIODS IN DETAIL

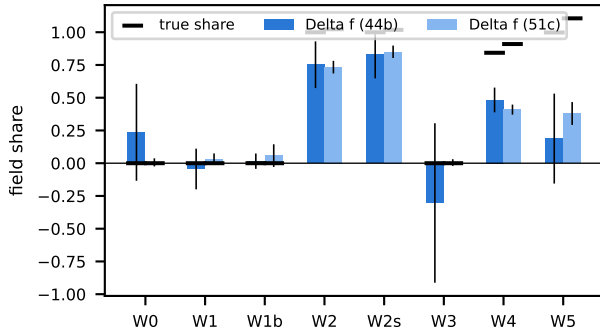
G33 d descriptive
G35 d descriptive
G36 d descriptive
G37 d descriptive
G38 d descriptive
G39 d descriptive
G40 d descriptive
G41 d descriptive
G42 d descriptive
G44 $\times$ failed

+1 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic swarms on real unit skeletons (44b, 51c; 8 seeds). Bars: Δf ; ticks: true field share. The estimator stays near 0 when the field does not act (W1, W3). It recovers 0.4–0.85 of a field that drives pauses (W2, W2s, W4).

CAVEATS

The sign test and the congestion world are post hoc (Amendment 2); they reverse a pre-registered “mixed”. Turnaround mixes generation, tool execution and overhead; only Gemini logs server time. Load may be congestion or shared heavy work; the data do not separate them. Only 4 units carry a residual, and their Δf intervals are wide. The separately served leader has 70 minutes. The holdout is not run.

ROUND 2

Fit a load curve (latency vs active count) per unit and its slope vs N . Use provider-specific error bursts as exogenous latency shocks. Run `confirm.py` after sign-off.

- **Essence.** separating exogenous platform fields from swarm-made fields in the residual co-activation.
- **H66-R1** Congestion as an order parameter.
- **H66-R2** An exogenous platform probe.
- **H66-R3** Separate execution from generation with the Anthropic and OpenAI token counts (output tokens per second) where usage is logged.
- **H66-R4** Holdout.

H66-platform-latency-field · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H66-platform-latency-field/README.md · ratings: H66 agent (suggested) (page 2)

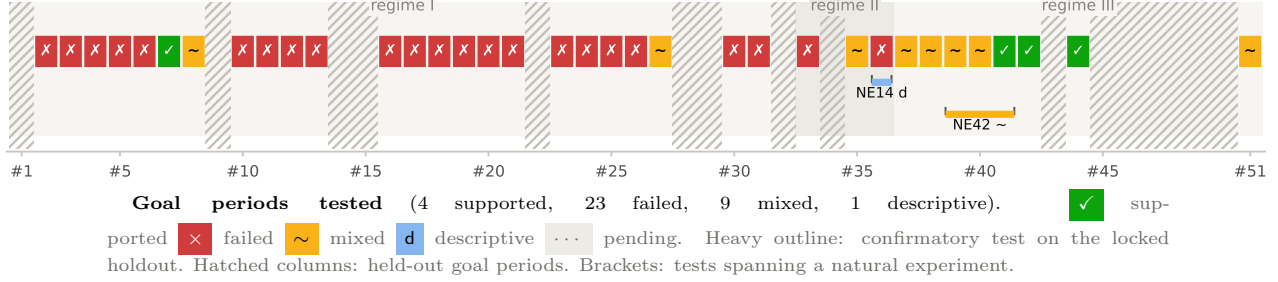
H67: A lagged criticality dial

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 3/5 Usefulness 2.5/5

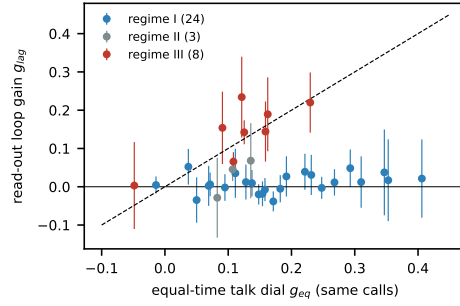
Bottom line. Read-out talk coupling is small and subcritical ($g_{lag} \approx 0.13$ in regime III, matching the Fano-implied 0.149 [0.123, 0.175]; ≈ 0 in regime I on either clock). H50's twice-larger value is an artifact of its estimator design.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H25's equal-time talk dial measures feedback within one minute. Coupling acts at the recipient's next model call, so HH262 asked whether a dial built on that read-out call shows larger gains that are still subcritical. Round 1 covered 66 non-reserved units in 35 periods, with 1.47M receiving calls. The read-out loop gain g_{lag} counts the extra talk messages that one message causes through reading. Every unit is subcritical (all upper bounds < 0.40). The gain is 0.13 in regime III and ≈ 0 elsewhere, and the **HH claim (lagged > equal-time) fails** (7/35 periods). Round 2 located H50's factor 2. In regime III it comes from H50's pair-RD design; in regime I it is call-class composition. H111's fluctuations give an independent total gain of 0.149 [0.123, 0.175], which matches g_{lag} . Regime I shows no gain on its chat clock either.



Observables. Round 1. Each point is one goal period (random-effects pool of its units, 95% CI). The x-axis is H25's equal-time talk dial on the same calls, the y-axis the read-out loop gain. Regime-I periods sit on $g_{lag} \approx 0$ while g_{eq} reads 0.05–0.40: a fast shared field, not read-out coupling. Regime-III periods lie near the diagonal at 0.07–0.23.

PHYSICAL MODEL

Linear response on the call clock (models 01, 09). Agent i 's call c is a talk call ($Y_{ic} = 1$) or not. R_{ic}^m counts peer messages read at c that were posted within the call's latency d_c before its start. P_{ic} counts messages posted within d_c after the start (in flight, unreadable). $a_{i,d,\kappa}$ is an agent $\times$ day $\times$ call-class field; $\bar{r}$ is the readers per message and $\bar{m}$ the messages per talk call. Round 2 adds the Fano-implied gain: H111's collective Fano ratio Φ inverted through its sum rule.

$$P(Y_{ic}=1) = a_{i,d,\kappa} + \beta_m R_{ic}^m + \beta_o R_{ic}^o + \beta_P P_{ic} + \gamma \cdot E_{ic} + \rho Y_{i,c-1}$$

$$J_1^* = \beta_m - \beta_P, \quad g_{lag} = \bar{m} \bar{r} J_1^*, \quad T/T_c = 1/g_{lag}$$

$$G_H = \bar{m} \bar{r} \sum_{h \leq H} \delta(h), \quad \delta(h) = \sum_{k < h} [\beta(R_{c-k}^m) - \beta(P_{c-k})]$$

$$\Phi_{obs}(10 \text{ min}) = \Phi_{pred}(g_{Fano}) \approx (1 - g_{Fano})^{-2}$$

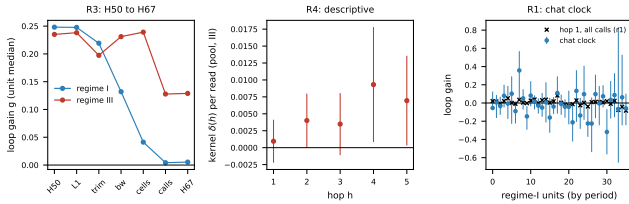
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: report g_{lag} , not the equal-time dial or a pair-RD jump, as distance to a talk cascade. One regime-III talk message causes about 0.13 further messages through reading (amplification $\times 1.15$, $T/T_c \approx 8$). The fluctuation bound caps the full gain, all hops included, at 0.175. Naming carries the coupling: a named message raises the next-call talk probability by 0.08, an unnamed one by 0.004. In regime I, talk is set by the chat-turn schedule, not by reading. No village configuration came near runaway, so this is a gauge, not a lever.

H67-lagged-criticality-dial · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H67-lagged-criticality-dial/README.md · ratings: H67 agent (suggested, round 2)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 subcritical every-where	max upper bound 0.39; all units < 0.5	supported
P2 $g_{lag} > g_{eq}$ in $\geq 2/3$	7/35 periods; median ratio 0.09	failed
P3 rank-consistent with H42	$\rho = 0.55$ ($p = 0.0007$)	supported
P5 named $\geq 5 \times$ un-named (III)	0.079 vs 0.004 ($\times 19$)	supported
P7 J_1^* falls with N	$\rho = -0.59$; g_{lag} flat	supported
<i>Round 2 (2026-10-05)</i>		
R3-P1 reproduce H50	L1/L0 0.94, ρ 0.87	supported
R3-P2 gap in cells + placebo (III)	gap in pair RD $\rightarrow$ call counts (share 1.0); placebo 0.016	failed
R3-P3 first stage $\Delta R \approx 1$	0.04	supported
R3-P4 cells remove $\geq 70\%$ (I)	69% (85–95% other readouts)	failed narrowly
R4 multi-hop G_5	synthetic failed; pool 0.24 $[-0.03, 0.51]$	descriptive
R4-P4 Fano check	g_{Fano} 0.149 $[0.123, 0.175]$ vs g_{lag} 0.127	hop closes 1
R1-P1 chat-clock (I) ≥ 0.05	gain -0.022 $[-0.052, 0.009]$	failed
R1-P2 explains excess	H111 r_F 1.35 unchanged	failed



Second figure. Round 2. Left: unit-median gain along the ladder from H50’s estimator to H67’s. Regime I collapses at the call-class cells; regime III drops only at the switch from pair RD to per-call counts. Middle: multi-hop kernel $\delta(h)$, regime-III pool (descriptive; the design failed its synthetic test). Right: regime-I units on the chat clock (points, 95% CI) and round 1’s hop-1 gain (crosses). Both sit at 0.

GOAL PERIODS IN DETAIL

G02 $\times$ failed (round 2, chat clock; round 1: failed)
G03 $\times$ failed (round 2, chat clock; round 1: failed)
G04 $\times$ failed (round 2, chat clock; round 1: failed)
G05 $\times$ failed (round 2, chat clock; round 1: supported)
G06 $\times$ failed (round 2, chat clock; round 1: failed)
G07 $\checkmark$ supported (round 2, chat clock; round 1: failed)
G08 $\sim$ mixed (round 2, chat clock; round 1: mixed)
G10 $\times$ failed (round 2, chat clock; round 1: failed)
G11 $\times$ failed (round 2, chat clock; round 1: failed)
G12 $\times$ failed (round 2, chat clock; round 1: failed)
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	2	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The multi-hop design is too noisy per unit (SD 0.4–1.5 in synthetic worlds) and cannot resolve hop 1 in real data. The full-gain bound rests on H111’s linear sum rule. Start-time error attenuates the chat-clock gain to about 0.23 of the truth, so regime I is bounded at < 0.04 , not zero. Post hoc: without conditioning on call mode, regime I shows 0.035 $[0.023, 0.046]$ above a call-skeleton null; 3/4 of the “read $\rightarrow$ chat-mode” effect is the chat-turn schedule. The unnamed part sits at the placebo’s resolution. Predictions were written knowing H25, H42, H50 and H111.

ROUND 2

Confirm $g_{lag} \approx g_{Fano}$ on the reserved regime-III units. Identify the pair-sampling term (0.18 vs 0.13). Test the chat-turn schedule as the regime-I fast field behind H111’s excess.

- **Essence.** a lag-aware, field-free loop gain that an operator can trust as “distance to runaway talk”, on the clock the scaffold imposes.
- **Round-3 redirects (newest).**
- **H67-R5** Reserved-data confirmation. Confirm $g_{lag} \approx g_{Fano}$ on the reserved regime-III units (re-freeze `confirm.py` with a `g_Fano` clause).
- **H67-R6** Pair-sampling term. Identify why the pair-weighted contrast (0.18) exceeds the per-call slope (0.13) in regime III.
- **H67-R7** A regime-I fast field. H111’s regime-I excess (r_F 1.35) is neither hop-1 nor chat-clock coupling.
- **H67-R8** Multi-hop with a donut. Rerun R4 with a 2-s donut and start-time error in the synthetic worlds, before any new reading of $\delta(h)$.

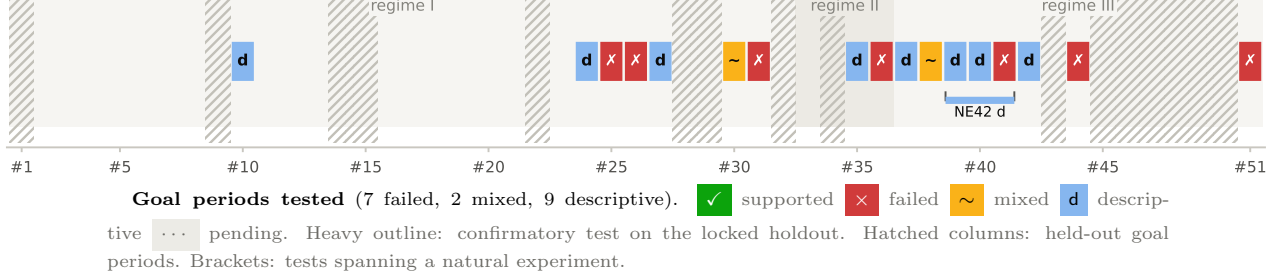
H68: Attention dilution is a mixture of two strategies

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 1/5

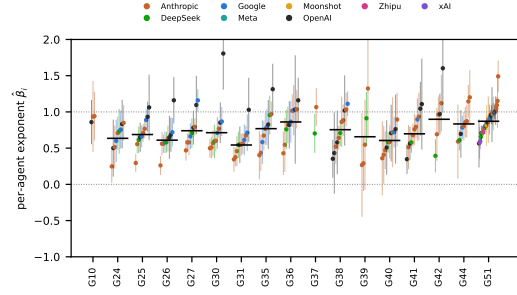
Bottom line. Every agent dilutes its attention with one narrow, unimodal exponent per period (between-agent SD 0.09–0.20), so $k^{-0.66}$ is a constant, not a mixture of two strategies.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH258 claims that the population dilution exponent ($k^{-0.66}$, H18) averages two kinds of agents: thin attenders ($\beta \approx 1$) and thread followers ($\beta \approx 0$), with lab-set weights. H68 fits one exponent per agent on H18's ledger pending sets in 17 periods (#10–#51, 170 agent-periods). A synthetic validation on the real skeletons came first. **The claim is refuted.** Agents scatter around the period mean with a between-agent SD of 0.09–0.20. The two-strategy world needs about 0.45. No period is bimodal (0/9 powered). In 7/9 powered periods the SD's upper 95% bound is below 0.35. No agent sits near $\beta = 0$ (lowest 0.25). Lab does not predict the exponent (p 0.37), and an agent's exponent barely persists across periods (ICC 0.23). Two natives: #51 fails, and NE42 has too few agents. The holdout script is written, not run.



Observables. Per-agent exponents $\hat{\beta}_i$ with 95% CIs, by period, colored by lab. The black bar is the random-effects mean. H68 predicts two clusters at 0 and 1 (dotted lines). Each period instead shows one cloud of width 0.1–0.2 around 0.55–0.87. No agent reaches 0, and the labs mix within every cloud.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 03 (contagion) with H18's uptake hazard. At talk turn τ of agent i , a pending sender j with n_j messages in a backlog of k is addressed with a cloglog hazard. $\theta_{i,d}$ is an agent-day propensity, γ a mention factor and β_i the agent's dilution exponent. The population layer is unimodal (U), the H68 two-delta mixture, or a free two-component mixture. Each is fitted on $\hat{\beta}_i$ with its own sampling variance s_i^2 .

$$P(r_{\tau j} = 1) = 1 - \exp[-\theta_{i,d} n_j k^{-\beta_i} e^{\gamma m_j}]$$

$$U: \hat{\beta}_i \sim \mathcal{N}(\mu, \tau^2 + s_i^2)$$

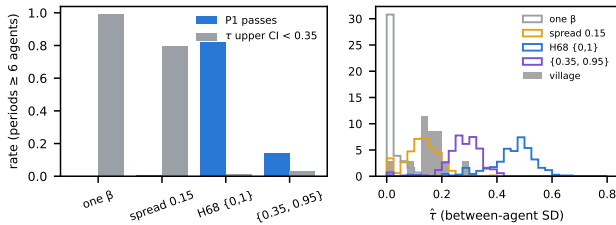
$$\text{H68: } \hat{\beta}_i \sim \pi \mathcal{N}(1, v_i) + (1 - \pi) \mathcal{N}(0, v_i), \quad v_i = \tau_w^2 + s_i^2$$

$$\text{LR} = 2[\ell_{M2} - \ell_U], \quad p \text{ by parametric bootstrap under } U$$

H68-dilution-mixture · generated 2026-10-07 17:50 by in-
fra/summaries/build_summaries.py · card hypotheses/H68-dilution-
mixture/README.md · ratings: H68 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 bimodal in $\geq 50\%$ powered	0/9; SD upper < 0.35 in 7/9; H68 fit 1–29 log-lik below U	failed
P2 lab $\eta^2 \geq 0.4$, $p < 0.05$; trait ≥ 0.5	η^2 0.16, p 0.37; ICC 0.23 [–0.10, 0.50]	failed
P3 mean of β_i = pooled β	15/15 within 0.035	passed
P4 $\rho(\beta, C) < -0.3$	+0.36 (n 170)	reversed
P5 mention vs reply rank ≥ 0.3	0.52 (n 156)	passed
#51: bimodal; timer wakes	lab; SD 0.15 [0.11, 0.21]; lab p 0.12; ρ 0.15	failed
NE42 trait merge	across the 5 agents; ρ 0.40 (p 0.50)	no power
Post hoc: H18’s per-message mention factor	#51 mean 0.87 $\rightarrow$ 0.63; all verdicts same	robust



Second figure. Synthetic validation on the real skeletons (17 periods, 4 worlds, 20 runs each). Left: in periods with at least 6 agents, the bimodality rule fires for the H68 world and never for one exponent. Right: the deconvolved spread $\hat{\tau}$. The village (gray) falls on the one-exponent and spread-0.15 worlds, far from H68’s 0.45.

GOAL PERIODS IN DETAIL

G10 d descriptive
G24 d descriptive
G25 $\times$ failed
G26 $\times$ failed
G27 d descriptive
G30 $\sim$ mixed
G31 $\times$ failed
G35 d descriptive
G36 $\times$ failed
G37 d descriptive

+8 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The response is H18’s mention proxy. Reply labels keep one parent per message, so only their agent ranking is used. Eligibility drops 53 of 223 agent-periods with few talks, and a quiet thread follower could hide there. The per-agent timer-wake check (#51, 24 agents) does not exclude reactive timing per agent. The trait estimator is noisy. The level of β depends on the mention-factor convention, but the spread does not. The holdout run needs a pending-set builder that keeps held-out days.

ROUND 2

- **H68-R1** The residual spread ($\tau \approx 0.15$) is agent $\times$ period.
- **H68-R2** Fix one mention-factor convention across H18 and H68 (per message) in the shared pending-set builder.
- **H68-R3** Use one β per period as the mean-field normalization in H03/H05; no composition correction is needed.

H68-dilution-mixture · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H68-dilution-mixture/README.md · ratings: H68 agent (suggested) (page 2)

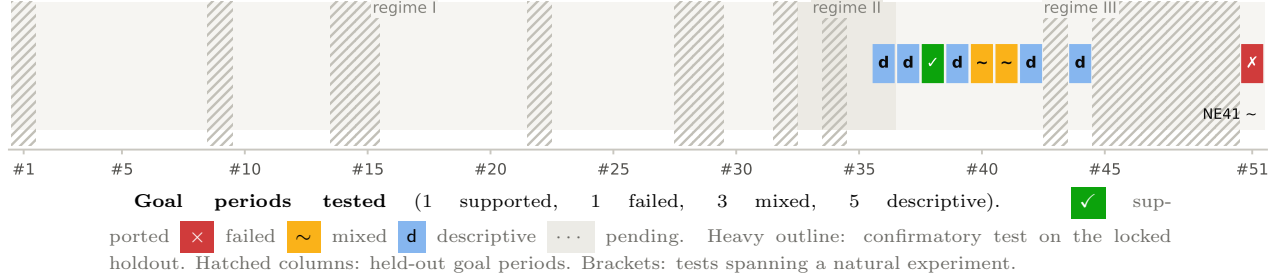
H69: Restatement loops are context fixed points

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 55% Faithfulness 2/5 Usefulness 2.5/5

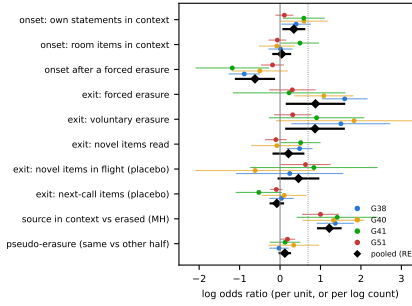
Bottom line. Restatement loops copy what the context still holds (source in context 3.4×), and an erasure ends them as a step, not a dose; own tool output does not trigger them, and novel input does not end them.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

HH259 claims that a restatement loop is a fixed point of an agent's context. It starts when the agent's own output fills the context above a threshold. It ends at an erasure or when novel input arrives. H69 maps 55k regime-III chat statements to their ledger calls and context segments (9 periods, 4 with ≥ 30 loop episodes). Synthetic validations on the real skeletons came first in both rounds. **The claim is half right.** The loop lives in the context window. A statement still in context is restated $3.4\times [2.6, 4.5]$ more often than an erased one at the same lag and call distance. A forced erasure (NE41) raises loop exit $2.4\times$ and lowers onset to $0.54\times$. Round 2 adds three results. Own tool output in tokens does not trigger loops. An erasure acts as a step, not a dose. Memory carries erased sources in #51 and #38 but not in pooled data. Novel input, nudges and human messages do not end loops. The holdout script is written, not run.



Observables. Round-1 effects as log odds ratios. Dots: G38, G40, G41, G51 with 95% CIs. Diamonds: random-effects pools. The source-in-context enrichment is $\ln 3.4$ in every period. The pseudo-erasure control (same vs other half of one segment) is about 0. Forced and voluntary erasures raise exit. Room items in context, novel reads and both placebos stay near 0. The dotted line marks $OR = 2$.

PHYSICAL MODEL

Models 02 and 08. The spin is the statement mode: restate ($r_t = 1$, a copy of an earlier own statement from an earlier call) or transform. At the producing call the context holds O_t own statements, U_t own tool tokens (prompt tokens minus calibrated room tokens and own chat, H45) and K_t room items since the last reset. Erasure (E_t) acts as a field toward exit; round 2 lets it scale with the removed own tokens D_t . The memory M_t survives the erasure. The decisive observables are Mantel-Haenszel odds ratios of a near-copy, in strata s of agent $\times$ lag $\times$ calls between: source in context vs erased, and erased source in memory vs not ($c \geq 0.5$ word 3-gram containment).

$$\text{logit } P(r_t=1 \mid r_{t-1}=0) = a_i + b_O \ln(1+O_t) + b_U \ln(1+U_t) + b_K \ln(1+K_t)$$

$$\text{logit } P(r_t=0 \mid r_{t-1}=1) = b_i + (c_E + c_D D_t) E_t$$

$$OR = \frac{\sum_s a_s d_s / n_s}{\sum_s b_s c_s / n_s}$$

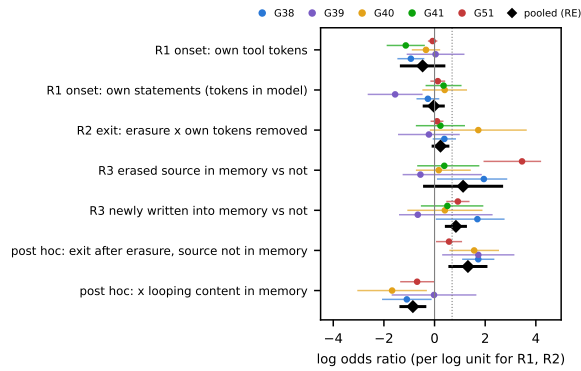
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A looping agent copies what its context window holds. To break a restatement loop, erase the context: one forced erasure more than doubles the exit odds and halves the onset odds, whatever the context size. Trimming tool output does not help: own tool tokens do not raise onset. Do not rely on messages: nudges read inside a loop give exit $0.81\times [0.49, 1.34]$. Check the agent's memory first. In #51, 73% of the loops that cross an erasure restate text the agent keeps in memory, and (post hoc) an erasure then helps less ($0.42\times$ the effect). The 41-call cap already acts as a loop breaker and costs 4–10% of output (H44).

H69-loops-context-fixed-points · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H69-loops-context-fixed-points/README.md · ratings: H69 agent (suggested, round 2)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed (pooled)	Verdict
P1 onset rises with s ; b_s 0.51 $[-0.13, 1.14]$; b_K +0.05; failed $b_K < 0$	b_O 0.32 $[0.08, 0.57]$	
P3 erasure exit OR ≥ 2	2.40 $[1.18, 4.88]$	passed
P4 in-context OR ≥ 2 ; 3.38 $[2.56, 4.46]$; 4/4; pseudo 1.11 pseudo ≈ 1		passed
P5 novel reads end loops	1.23 $[0.85, 1.78]$; in flight 1.58	failed
NE41: exit ≥ 2 ; onset	2.40; 0.54 $[0.34, 0.86]$	mixed
≤ 0.5		
G51: nudges end loops	0.81 $[0.49, 1.34]$	failed
R1 tool tokens: b_U CI	-0.47 $[-1.31, 0.37]$ (power 0.93)	passed
≥ 0		
R1 own count survives	b_O -0.03 $[-0.41, 0.35]$	failed
R2 step, not dose	c_D 0.24 $[-0.06, 0.53]$; 0.7 excluded	passed
R2 NE22 cap	2 loop statements	inconclusive
R3 memory OR ≥ 2 , CI	3.09 $[0.67, 14.1]$; G51 31.7, G38	failed
> 1	7.0	
R3 carrier vs salience	not identified (synthetic)	inconclusive



Second figure. Round-2 effects as log odds ratios (per log unit for R1 and R2). Dots: periods with 95% CIs; diamonds: random-effects pools. Own tool tokens and the own count give no onset effect. The erasure effect does not grow with the own tokens it removes. The memory odds ratio is large in G51 and G38 and near 1 in G39–G41. Post hoc: an erasure raises exit $3.7\times$ when the looping text is not in memory, and less when it is.

GOAL PERIODS IN DETAIL

G36 d descriptive
G37 d descriptive
G38 ✓ supported
G39 d descriptive
G40 ~ mixed
G41 ~ mixed
G42 d descriptive
G44 d descriptive
G51 × failed
NE41 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only four periods have ≥ 30 loop episodes, and the erasure exit effect varies across them (I^2 0.74). Statement geometry depends on the embedding model (enrichment gte 6.1, bge 3.5). Prompt tokens are dense only for Anthropic and Google; the token sample loses many OpenAI and DeepSeek statements, and there the round-1 own-count effect vanishes. No partial trim that removes own text exists in regime III, so the dose test spans full erasures only. Memory containment is lexical, and carriage cannot be told apart from salience. The memory moderation of erasure is post hoc.

ROUND 2

- **H69-R4** Test the post hoc memory moderation fresh: on the reserved targets in `confirm.py` (with sign-off), add "erasure $\times$ looping content in memory" as a frozen criterion.
- **H69-R5** Memory edits as an intervention: consolidations that drop the looping content from memory (compression) vs keep it; does the loop then end at the next erasure?
- **H69-R6** Lab composition of the own-count effect: refit round-1 onset by lab family (agent effects nested), since the token sample lost it.

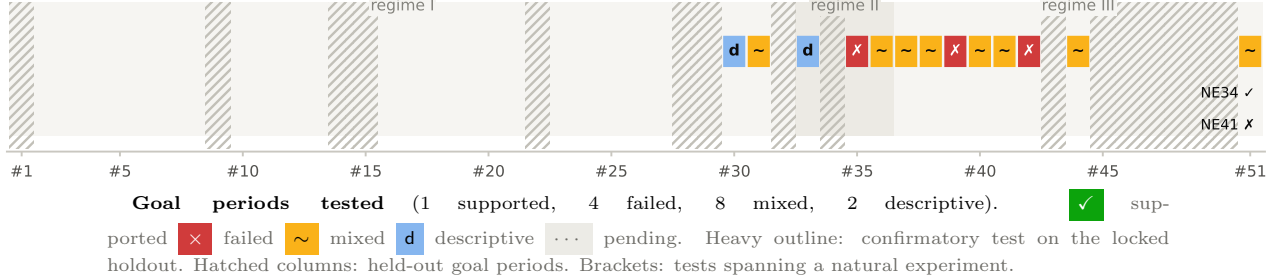
H70: The semantic information of the artifact store

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2/5

Bottom line. The artifact store tells an agent where to work after an erasure (0.14 bits, 89% return) but carries no measurable value per bit; the erased context window carries the value (41% of output, $\kappa_C \approx 5$ commits per 20 calls per bit).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH261 claims that the artifact store carries the swarm's day-scale semantic information, so losing it costs more than losing memory. H70 measures each channel at the two exogenous context erasures: the forced 41-call erasure (NE41, 21k events) and the night. It uses one shared estimator for HH307's κ table, on 13 non-holdout periods (#30–#51). **The artifact store holds *where* an agent works, not *how much* it gets done.** After a forced erasure, 89% of first commits return to the agent's own repo. The repo pointer carries 0.14 [0.12, 0.16] bits beyond agent identity. Re-reading the repo adds no output beyond its placebo association ($\Delta V_{\text{rel}} = -0.05 [-0.13, +0.01]$), so $\kappa_A \approx 0$. The context window holds the value: the erasure costs 41% of the next 20 calls' commits for 0.08 bits destroyed. A new goal, not a lost repo, resets allocation (return 10% vs 80%). The NE24 holdout test is written, not run.

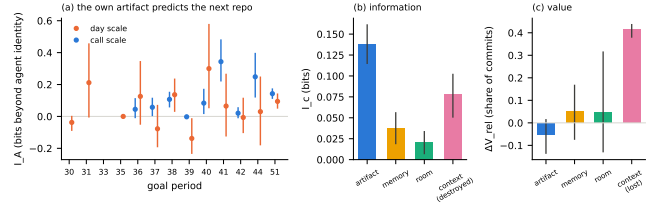
PHYSICAL MODEL

Model 04 (Kolchinsky–Wolpert, natural-scramble variant). The system is one agent; its state is its allocation X , the repo of its next work commit. A channel c (own artifact, memory note, room, context) stores a pointer S_c to a repo. A forced erasure or a night cuts the context and leaves the other channels. A matched placebo (pseudo-erasure, mid-day call) keeps the context. I_c is the information the channel carries about X ; ΔV_c is the output an open channel adds after the cut.

$$I_c = I_{\text{MM}}(X^+; S_c) - \langle I_{\text{MM}}(X^+; S_c^\pi) \rangle_{\pi \in \text{within agent}}$$

$$\log E[V] = \alpha_{\text{agent-period-arm}} + \beta_1 \text{open} + \beta_2 \text{open} \cdot \text{scramble}$$

$$\Delta V_{\text{rel}} = e^{\beta_2} - 1, \quad \kappa_C = \Delta V_c / I_c$$



Observables. (a) I_A per period: bits the last-commit repo carries about the next repo after an erasure (call scale, blue) or a night (orange), above the within-agent permutation floor. (b, c) The κ row, regime III pooled: information and value per channel. Only the context row has value. Its bits are the allocation information the erasure destroys. Its value is the output lost (41%).

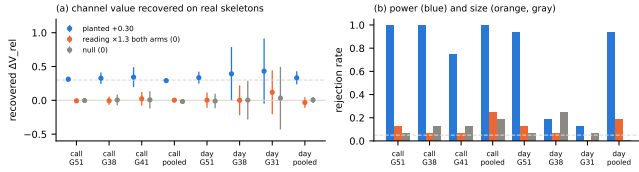
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a scaffold builder: the cost of a context wipe is lost working state, not lost orientation. Agents find their repo again (89%) whether or not they re-read it, and a re-read in the first five calls buys no measurable output. A memory note naming the repo buys none either. Measures that preserve working state (a longer cap, carrying the last tool outputs across the reset) are where the 41% per-wipe loss sits. Pointing the agent at its repo is not. To scramble allocation, change the goal; repo loss inside a goal was not observed outside the holdout. The shared estimator (`semantic_kappa.py`) is the reusable part for HH307.

H70-artifact-store-semantic-info · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H70-artifact-store-semantic-info/README.md · ratings: H70 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $I_A \geq 0.3$ bits	call 0.14 [0.12, 0.16] (7/9 periods > 0); day 0.08 (5/12)	failed
P2 $\Delta V_A > 0$ (call)	$\Delta V_{rel} -0.05 [-0.13, +0.01]$	failed (R2)
P3 artifact beats memory note	I 0.14 vs 0.04; $\Delta V_M +0.05$, CI incl. 0	mixed
P4 artifact beats room	I 0.14 vs 0.02; ΔV_R CI incl. 0	mixed
P5 context cost > 0 , $I_C < 0.3I_A$	cost 41% [38, 44]; $I_C = 0.56 I_A$	mixed
P6 $\kappa_A > \kappa_R > \kappa_M$	κ_C 5.2 [3.8, 8.4]; others CI incl. 0	failed
P7 return ≥ 0.8	F 0.89 (P 0.87); N 0.76 (PN 0.82)	mixed
NE41 κ table	value only in the context row	failed
NE34 goal resets allocation	new goal 0.10, within 0.80, supported #39→#40 0.78	



Second figure. Planted worlds on the real event skeleton (8 replicates). (a) A +30% channel value is recovered without bias. A “reading precedes writing” effect ($\times 1.3$ in both arms) reads as 0. An additive DiD read the same effect as -0.14 , which led to the Poisson estimator (A1). (b) Power is 1.0 at call scale and low per period at day scale. Size is about 0.1.

GOAL PERIODS IN DETAIL

G30 d descriptive
G31 $\sim$ mixed
G33 d descriptive
G35 $\times$ failed
G36 $\sim$ mixed
G37 $\sim$ mixed
G38 $\sim$ mixed
G39 $\times$ failed
G40 $\sim$ mixed
G41 $\sim$ mixed
+5 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The open flag is chosen by the agent, so ΔV rests on parallel trends between erasure and placebo arms. Size is about 0.1 ($B = 50$ in the synthetic), so single-period intervals are mildly anticonservative. I_A counts only bits beyond agent identity: where each agent has one repo (#35, #39) it is 0 by construction. Commits measure attempts, not value. The room row includes the consolidation backlog. No involuntary loss of an artifact occurs outside the holdout, so HH261’s “repo moved or broken” case waits for NE24.

ROUND 2

- **H70-R1** Value the artifact by what is re-read: join post-erasure reads to the files touched before the erasure (H44-R3) and test whether re-reading the exact pre-erasure files shortens the dip.
- **H70-R2** Add the history-search row (HH294: the 03-31/04-01 outage) and the kickoff row to the κ table with the shared estimator.
- **H70-R3** Run `confirm.py` on NE24 after disclosure: the one involuntary scramble of the whole artifact medium.
- **H70-R4** Allocation information by role: in #51 (private roles), split the artifact bits into identity, role and artifact parts.

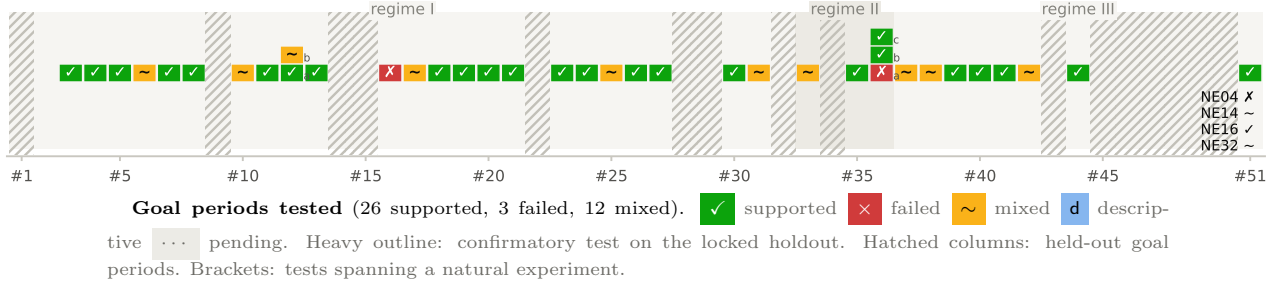
H71: Memory is a first-order homeostat

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 1.5/5 Usefulness 1.5/5

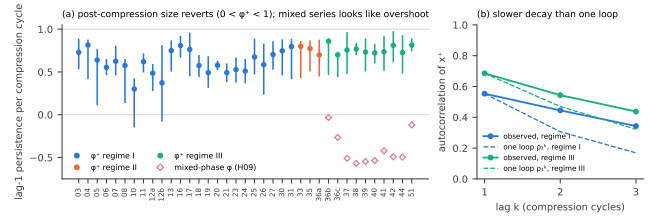
Bottom line. Memory size is regulated toward an agent set point on the consolidation clock, without overshoot; the loop is a fast near-deadbeat compression around a slowly drifting target, not a first-order homeostat.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH269 claims that memory size is a first-order homeostat with overshoot after erasures and a gain change at NE41. H71 fits the post-compression memory size from `memory_stats` (62k non-holdout cycles, 37 units, regimes I–III). Each regime-III consolidation writes an append snapshot, then a compression snapshot. **Memory is a homeostat, but not first order, and it does not overshoot.** Size reverts toward an agent set point in 37/37 units (ϕ^+ pooled 0.67 [0.63, 0.71] per cycle). H09's overshoot is the mixed sampling: the mixed series gives $\phi < 0$ in 10/10 regime-III units while $\phi^+ > 0$. Forced and voluntary consolidations give the same ϕ^+ . AR(2) terms are positive (median +0.19). NE14 raises the set point (+17–31%). The holdout script is not run.



Observables. (a) ϕ^+ per unit (agent bootstrap, half-panel jack-knife): all between 0 and 1. Open diamonds give H09's mixed-phase statistic for regime III. It is negative in every unit, which looks like overshoot. (b) Autocorrelation of x^+ at lags 1–3, averaged over units, against the one-loop prediction ρ_1^k (dashed). The slower decay shows a second, slow timescale.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a scaffold builder, memory size is a regulated variable with a per-agent set point (agent identity holds 91% of its variance, the lab 41%). Agents restore it within a few consolidations, so a memory-size alarm needs a per-agent baseline, not a global cap. Forced erasures do not destabilize it. The NE14 scaffold change raised the set point by about 20% and left the per-cycle gain almost unchanged, so a prompt change moves the target, not the loop. A monitor reading raw `memory_stats` will see a false overshoot ($\phi \approx -0.5$) from the two-snapshot write. Filter to compression rows first. The result is a design fact about the scaffold's memory loop. It does not say what the memory is worth (H15: about 0 at day scale).

PHYSICAL MODEL

Model 04 (a maintained store) with model 05's import and removal bookkeeping. The degree of freedom is x_n , the log size of an agent's memory after compression n . Growth adds Δ_n (appends), and compression maps the peak y_n back toward a set point μ_i . A single first-order loop gives an AR(1) with gain $1 - \phi$, where $\phi = b(1 + c)$. The rivals are a random walk ($\phi = 1$), a deadbeat template ($\phi = 0$) and an over-correcting controller ($\phi < 0$). A post hoc model adds a slowly drifting set point m_n .

$$y_n = x_{n-1} + \Delta_n, \quad \Delta_n = d_i + c(x_{n-1} - \mu_i) + \xi_n$$

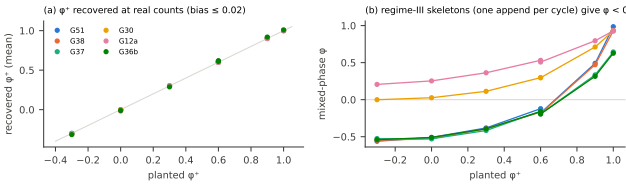
$$x_n = a_i + b y_n + \zeta_n, \quad \phi = b(1 + c)$$

$$x_n - \mu_i = \phi(x_{n-1} - \mu_i) + \varepsilon_n$$

post hoc: $x_n = m_n + e_n, \quad m_n = \rho_s m_{n-1} + \eta_n$

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $0 < \phi^+ < 0.9$, CI excl. 0, 1	35/37; pooled 0.67 [0.63, 0.71]; overshoot 0/37	supported
P2 AR(1) beats RW $\geq 70\%$	69% of 353 agent-periods; beats deadbeat 88%	narrowly failed
P3 mixed $\phi < 0$, $\phi^+ > 0$ (III)	10/10; one-loop synthetic matches 0/10	mixed
P4 no overshoot after forced	ϕ_F^+ 0.49–0.79; $ F - V < 0.15$ in 9/10	supported
P5 $ AR(2) < 0.1$	5/37; median +0.19	failed
P6 event clock	29/37 $ slope \leq 0.1$	supported
P7 agent-level gain	across-period r 0.52 (null p95 0.33)	supported
P8 $\phi^+ \approx b(1 + c)$	$b(1 + c)$ median 0.13 vs ϕ^+ 0.67	failed
NE14 / NE04	set point +17–31%, $\Delta\phi^+ +0.12$ / $\Delta\phi^+ +0.15$	mixed / failed
NE16 / NE32	no change / relaxation ≈ 0.9 per cycle	supported / mixed



Second figure. Synthetic one-loop dynamics on six real cycle skeletons. (a) The estimator recovers the planted ϕ^+ (bias ≤ 0.02 ; power 1.0 for an overshoot of -0.3). (b) With one append per cycle (regime III), the mixed-phase ϕ is negative for any planted $\phi^+ \leq 0.6$. This is H09's artifact. A post hoc two-timescale sawtooth matches the observed size in 7/8 units.

GOAL PERIODS IN DETAIL

G03 ✓ supported
G04 ✓ supported
G05 ✓ supported
G06 ~ mixed
G07 ✓ supported
G08 ✓ supported
G10 ~ mixed
G11 ✓ supported
G12a ✓ supported
G12b ~ mixed
+31 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	2	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The two-timescale model is post hoc: it explains P5 and P8 after they failed. Size is a capacity, not content; hashed-line turnover is not analysed. Regime-I cycles are session-scale and differ in structure from regime-III cycles, so pooling across regimes compares unlike cycles. NE04 has 7 agents and a goal change inside the window. NE16 has two days per side. The out-of-sample margin over a random walk is small (median MSE ratio 0.89) because the target drifts. The holdout is not used; #43, the memory goal, is the main confirmation target.

ROUND 2

- **H71-R1** Fit the two-timescale model directly (state-space Kalman fit per agent) and test the set point's drift against task covariates: does the target follow the agent's active repo count or its role (DQ4, DQ6)?
- **H71-R2** Set point vs viability: do agents whose memory is above its set point at a forced erasure recover output faster (H44 re-acquisition, H15 dip)?
- **H71-R3** Run `confirm.py` (#43 is the one goal that acts on the store) after disclosures.
- **H71-R4** Content turnover: the hashed-line Jaccard per cycle as a second order parameter (what is kept, not how much).

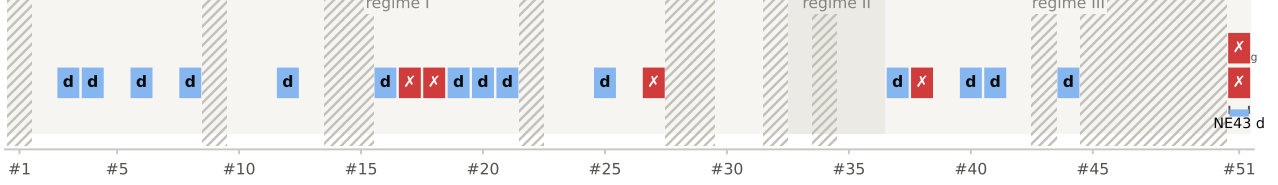
H72: Trap aging is input starvation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 70% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Trap aging is carried by the context's self-share, not by missing input; room chatter adds a weak hold that a mention cancels.

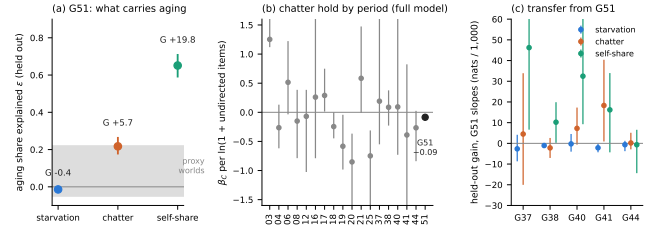
level: descriptive · pipeline checks 86% · updated 2026-10-05



Goal periods tested (6 failed, 15 descriptive). ✔ supported ✗ failed ⚡ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The claim: an idle agent's escape slows with trap age only because no new input has reached it. A wake is a call after an idle call; there the agent acts or idles again. Round 1 refuted the claim in-sample: the age slope survives every starvation clock (G51 $-0.48 \rightarrow -0.53$), and recent input *lowers* escape. Round 2 asks which input or context variable carries the aging, on the same 20,237 G51 wakes, scored on held-out days. Directed-input starvation explains none of it ($\epsilon = -0.01$). Undirected chatter explains 22%. The context's self-share (H16's own-call urn) explains 65% [59, 71], against $\leq 22\%$ in synthetic worlds where aging is intrinsic or frailty. Its coefficient is the urn's +1 (1.09), and its slope transfers from G51 to G38. Chatter adds a small, separate hold ($\beta_C = -0.085$) that a directed read cancels. The reserved data are not used.



Observables. (a) #51: the share of the aging clocks' held-out information that each mechanism makes redundant, with its own held-out gain (nats per 1,000 wakes). Grey band: the most a stand-in for wake index or per-trap frailty gives in synthetic worlds. (b) Chatter-hold slope per period, full model; only G51 is powered. (c) G51's slopes scored on the other regime-III periods: self-share helps in G37, G38 and G40; starvation never helps.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To judge how stuck an idle agent is, count its own re-pauses in the current context, not the minutes since someone wrote to it. A context full of its own pauses is the trap, and erasing it frees the agent (H16: 0.47 to 0.84 at the next wake). Quiet does not deepen the trap, and chatter does not break it. In #51, 48 undirected messages over five calls cut the next-wake escape odds by about a quarter, unless one message names the agent. The urn coefficient is portable: a monitor fitted on one period predicts another.

PHYSICAL MODEL

Model 09 (exogenous input events) and model 02 (kinetic escape) with an urn on the context. At wake g the agent acts (a run of ≥ 3 active calls) or idles. Clocks: trap age a and wake index k . Three mechanisms compete on one base (agent effects, pause, time of day, others' activity, reads at the wake, reset step, trap kind at start). S: minutes since the last directed read. C: undirected items read at the 5 calls before the wake. F: own idle calls as a share of own calls in the context segment. Traps that start with a consolidation are dropped.

$$\begin{aligned} \text{logit } P_g &= \alpha_i + z_g \cdot \delta + \beta_a \ln a_g + \beta_k \ln k_g \\ &\quad + \beta_S \ln s_g + \beta_C \ln(1 + U_g) + \beta_F \ln(1 - f_g) \\ \epsilon(M) &= 1 - \frac{\mathcal{L}_{B+M+A} - \mathcal{L}_{B+M}}{\mathcal{L}_{B+A} - \mathcal{L}_B} \quad (\text{held out}) \\ \text{rates race: } \beta_{\ln \lambda_u} &= -1; \quad \text{urn: } \beta_F = 1 \end{aligned}$$

H72-trap-aging-input-starvation · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H72-trap-aging-input-starvation/README.md · ratings: H72 round-2 agent (suggested v1; coordinator writes v2)

RESULTS VS PREDICTIONS

prediction (round 2)	observed (#51)	verdict
chatter dose holds (full model)	$\beta_C = -0.085$ $[-0.140, -0.045]$	holds
separable (< 30% change)	$-0.130 \rightarrow -0.085$ (34%)	fails
first wakes; agent + day FE	$-0.076; -0.090$	holds
a mention cancels the hold	$+0.14$ $[+0.06, +0.21]$	holds
rates race (slope -1)	-0.094 $[-0.148, -0.041]$	holds
replication (none above 0)	below 0: 4/18; above 0: G03, G17	fails
starvation explains no aging	$\varepsilon_S = -0.01$	holds
self-share explains most beyond proxy worlds (0.22)	$\varepsilon_F = 0.65$ vs $\varepsilon_C = 0.22$	holds
aging left over	$\varepsilon_{SCF} = 0.69$ $[0.64, 0.75]$	holds
self-share transfers to G38	$+10$ $[+0, +20]$ nats/1,000	holds

GOAL PERIODS IN DETAIL

G03 d descriptive
G04 d descriptive
G06 d descriptive
G08 d descriptive
G12 d descriptive
G16 d descriptive
G17 $\times$ failed
G18 $\times$ failed
G19 d descriptive
G20 d descriptive

+11 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

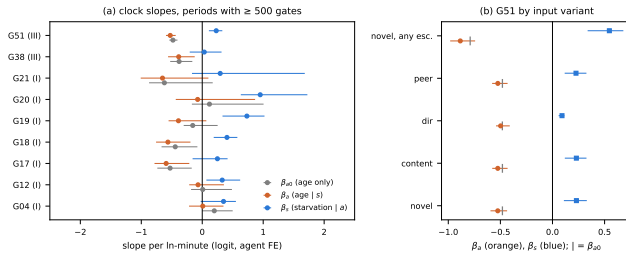
Only G51 is powered for the chatter hold; G03 and G17 have CIs above 0, and other periods are descriptive. A third of the naive dose slope is shared with self-share and starvation. The in-flight placebo is non-zero (+0.29) but leaves β_C unchanged; escaping calls may have longer model latency. Self-share and wake index correlate (-0.69); synthetic proxy worlds bound the overlap at 0.22. 31% of the clocks' information remains unexplained. The transfer fit was fixed post hoc (step halving).

ROUND 2

Confirm ε_F and the urn coefficient on reserved #51 days and #45–#50 with a re-frozen confirm script. Find what carries the last third of aging (memory, task state). Explain the in-flight term.

- **H72-R1** (done in round 2, 2026-10-05).
- **H72-R2** (done in round 2, 2026-10-05).
- **H72-R3** (open; needs re-freezing on round-2 statistics).

H72-trap-aging-input-starvation · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H72-trap-aging-input-starvation/README.md · ratings: H72 round-2 agent (suggested v1; coordinator writes v2) (page 2)



Second figure. Round 1. (a) Age slope without s (grey), with s (orange) and the starvation slope (blue), periods with ≥ 500 wakes, day-block 95% CIs: starvation absorbs no aging. (b) G51 under four input definitions and the glance outcome: the age slope never moves.

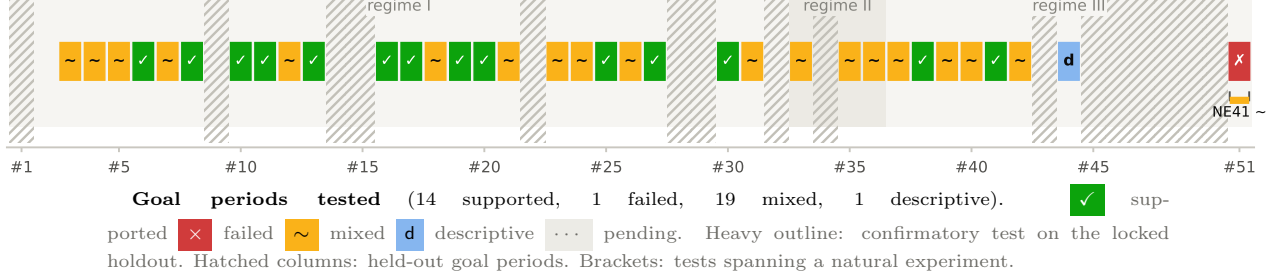
H73: Style = weights + context + register

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

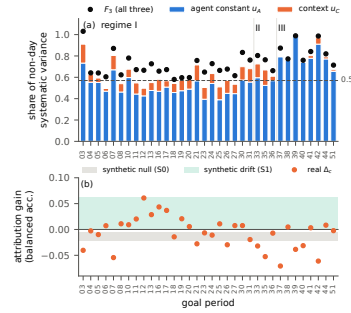
Bottom line. Style is an agent constant plus noise: context fill and assigned registers add small shares, and removing them does not improve attribution.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test splits an agent's chat style into an agent constant, a context-fill drift and an assigned-register shift. Style is H13's 17 type-controlled features. The fit runs per goal period on 34 non-holdout periods (126k messages). The three components explain most of the systematic variance that the day field leaves ($F_3 = 0.72$, IQR 0.65–0.80; 34/34 periods above 0.57). The agent constant carries almost all of it. Context fill adds a median share of 0.08, and 0.011 in regime III. Removing the context term does not raise attribution ($\Delta_c = -0.007$ [−0.017, +0.003]). At forced erasures the fitted drift predicts 1.7% of the style jump. Judging in #12 is a shared register. The #51 roles share none. The holdout script exists and is not run.



Observables. (a) Shares of the non-day systematic style variance per goal period. Blue is the agent constant, orange is the context term, and black dots are all three components (F_3). The agent constant is the largest share in every period. The context share falls to about 0.01 in regime III. (b) Attribution gain from the context model on blocks of five messages. Real gains (dots) sit in the synthetic null band, not in the planted-drift band.

PHYSICAL MODEL

Vector spins (model 11). The spin is a message's style vector x_m . The day field g_d and a time-of-day term τ are external fields. The agent constant q_i is the weights. The context term is a profile $c(s)$ over fill bins plus an agent drift b_i along $z = \log_2(1 + p)$, with p the calls since the last reset. ρ_r is an assigned register (judge, role). η is agent-day jitter. κ is the cell-means ceiling.

$$x_m = g_d + \tau(h_m) + q_i + c(s_m) + b_i z_m + \rho_{r_m} + \eta_{i,d} + \varepsilon_m$$

$$F_3 = \frac{\bar{R}^2(G+A+C+R) - \bar{R}^2(G)}{\kappa - \bar{R}^2(G)}, \quad u_X = \frac{\bar{R}^2(\text{all}) - \bar{R}^2(\text{all} \setminus X)}{\kappa - \bar{R}^2(G)}$$

$$\Delta_c = \text{acc}[q_j + c(s) + b_j z] - \text{acc}[q_j], \quad \beta = \frac{\sum \langle \Delta x, \Delta \hat{x}_{\text{ctx}} \rangle}{\sum \|\Delta \hat{x}_{\text{ctx}}\|^2}$$

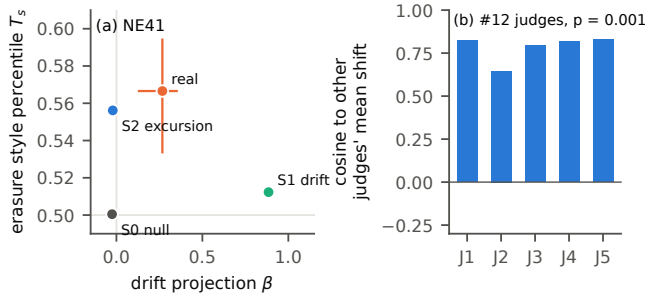
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For attribution, use the agent's mean style over at least five messages. Balanced accuracy is 0.63 against chance 0.10. Do not correct for context position: the correction gains −0.007. A style monitor can treat the agent constant as fixed within a period. It carries 0.55 of the non-day systematic variance, and agent-day jitter carries 0.18. Expect one message to be mostly noise: only 27% of its style variance is systematic. Known speech roles such as judging shift style in one shared direction ($\cos 0.78$). That shift does not confuse attribution. Context erasure moves style by a random amount with no fixed direction, so a monitor should not read a post-reset style change as a model change.

H73-style-three-components · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H73-style-three-components/README.md · ratings: H73 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $F_3 \geq 0.57$ in $\geq 2/3$	34/34; median 0.72 (agent-day ceiling 0.82)	pass
P2 agent largest; context $p < .05$	u_A largest 34/34 (0.55); u_C 23/26 (0.08)	pass
P3 context removal helps attribution	$\Delta_c -0.007 [-0.017, +0.003]$; 16/34 positive	fail
P4 dispersion grows with fill	1/5 dense periods; G51 negative	fail
P5 rivals (genre, own fill)	genre retention 1.04; received fill beats own fill 6/8 (regime III)	partial
N1 NE41 drift predicts the jump	β 0.27 [0.12, 0.36]; 1.7% of the jump; T_s 0.567 $\rightarrow$ 0.568	mixed
N2 #12 judges share a register	$\cos +0.78$ (p 0.001); u_R 0.056; attribution +0.01	mixed
N3 #51 roles are registers	Prankster 0.60; media $\cos -0.11$ (p 0.56)	fail
N4 leader writes like Kimi	rank 3 blind, 1 with agent drift (20 msgs)	descr.



Second figure. (a) Forced erasures (NE41): drift projection β against the style percentile T_s . The real point (orange, 95% CIs) sits near the synthetic directionless excursion (S2), far from a directed drift (S1). (b) #12: each judge's style shift points along the other judges' mean shift.

GOAL PERIODS IN DETAIL

G03 $\sim$ mixed
G04 $\sim$ mixed
G05 $\sim$ mixed
G06 $\checkmark$ supported
G07 $\sim$ mixed
G08 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\checkmark$ supported
G12 $\sim$ mixed
G13 $\checkmark$ supported

+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The 20 features mix formatting with topic-adjacent rates (digits, uppercase). F_3 rests on a cell ceiling that is noisy where cells are small. The synthetic bias reaches +0.07 near 0.55, and F_3 exceeds 1 in G03. In regimes I/II half of the context share is the chat versus computer-use level. Fill counts calls, not messages or tokens, and it may miss what drives the erasure excursion. The #51 role test is pre-registered on three-week eras. The onset-block result (Prankster 1.00) is post hoc. The natives are small: five judges, four media agents, 20 leader messages.

ROUND 2

Model the conversation state (received items, pending mentions, thread depth) in place of fill, with an in-flight placebo. Fit the persona shift as a transient with a decay time. Run the NE30 model swap on the holdout.

- **Essence.** which parts of an agent's style a monitor can trust as identity, and which parts the conversation writes.
- **H73-R1** Conversation-state model: replace fill with received items, pending @-mentions and thread depth (DQ2), and test accommodation with H57's in-flight placebo (posted-but-unread at matched lag).
- **H73-R2** Register as a transient: a daily event study of the #51 onset and NE38 (decay time of the persona shift), and the #12 judge shift by minute within a debate.
- **H73-R3** Excursion model: an Ornstein–Uhlenbeck process within a context segment on message position (H46-R2), with its direction random per segment; fit its variance growth and test it at NE41.
- **H73-R4** Model swaps: the NE30 test in the holdout (frozen as C6); in round 2 a larger leader sample (#45 is held out).

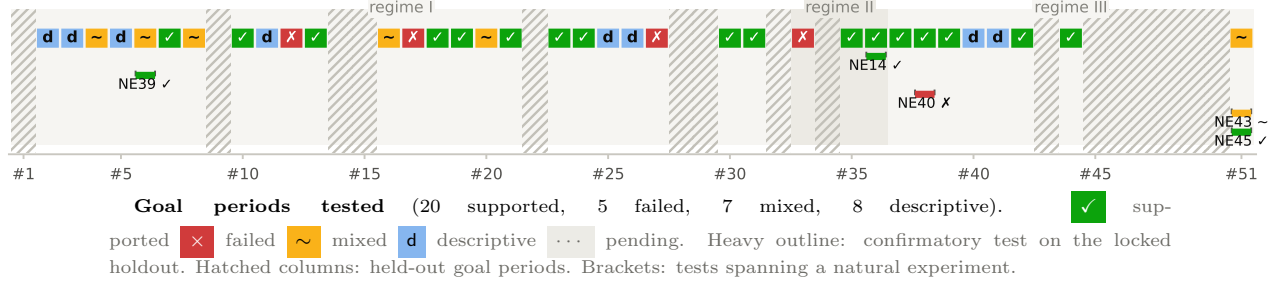
H74: An operator-grade change detector

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 2/5 Usefulness 3/5

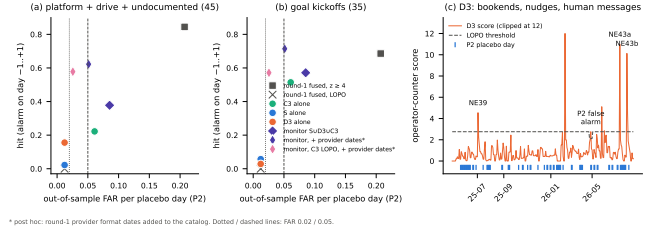
Bottom line. Calibrated out of sample, schema diff plus log-count operator counters plus topic shift is a usable monitor for drive, format and goal changes (FAR 0.05–0.085); scaffold changes stay invisible.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H74 builds a daily change detector for an agent swarm and scores it against 217 dated changes on 282 non-reserved days (April 2025 to September 2026). Round 1: a five-channel fused alarm was too noisy (18% false alarms on quiet days); a schema diff of the log records was precise. **Round 2** keeps three channels: the schema diff S , operator counters $D3$ (bookends, nudges, human messages) and $H36$'s frozen topic rule $C3$. Thresholds are calibrated on quiet days of the *other* goal periods, so every false-alarm rate (FAR) is out-of-sample. The monitor has FAR 0.085 [0.04, 0.17] per quiet day and hits goal kickoffs at 0.57 (p 0.003) and room changes at 6/6. $D3$ dates all three undocumented drive steps (NE39, NE43a, NE43b) at FAR 1/82. Scaffold tool changes are hit at chance. Verdict: **mixed**.



Observables. (a, b) Operating points: out-of-sample FAR per quiet day against hit rate (alarm on day $-1..+1$), for 45 platform, drive and undocumented events and for 35 goal kickoffs. Round 1's fused alarm (gray square) sits at FAR 0.21; with a calibrated threshold it fires once (gray cross). The monitor (violet) reaches FAR 0.085; pink and the small violet marker add round 1's provider-format dates to the catalog (post hoc). (c) The $D3$ counter channel over all days with its threshold (2.1–2.8): it fires at NE39 and both NE43 steps, and on one quiet day.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Models 02 and 11. A day d is one step of a driven system. A platform step changes record schemas for every agent at once; an operator step changes drive counters; a goal step is a field step on content. Counts are heavy-tailed, so each counter is logged, scored by a robust trailing z (10 days) and must persist for two days. Each channel k alarms above $\theta_k(g)$, the conformal $(1 - \alpha)$ quantile of its scores on the n placebo days outside goal period g (placebo: ≥ 2 active days from every catalogued event).

$$p_f(d) = \min\{|z_f(d-1)|, |z_f(d)|\}, \quad z_f = z[\log(1 + x_f)], \quad D3(d) = \max_f p_f(d)$$

$$\theta_k(g) = s_{k, ([1-\alpha)(n+1)]} \text{ over placebo days } \notin g, \quad \alpha = 0.02$$

$$\text{alarm}(d) = [z_S > \theta_S] \vee [D3 > \theta_D] \vee [R1 \geq 3 \vee Z_{\text{cont}} \geq 2]$$

H74-change-detector · generated 2026-10-07 17:50 by in-
fra/summaries/build_summaries.py · card hypotheses/H74-change-
detector/README.md · ratings: H74 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
r1 P1 tool changes AUC ≥ 0.70 , $p < 0.05$	AUC 0.68, p 0.059	mixed
r1 P4 fused FAR ≤ 0.10	0.18	failed
r2 P1a channel FAR ≤ 0.03 ; union ≤ 0.06	S 0.012, D3 0.012, C3 0.061; 0.085	partly
r2 P1b class hits in predicted bands	goal 0.57, drive 0.57, tool 0.35, held room 1.00, roster 0.26	
r2 P1c $p < 0.05$ drive; tool $p < 0.10$	goal, 0.003; 0.13; 0.55	partly
r2 P1e beats C3 alone by ≥ 0.15	pooled hit 0.22 $\rightarrow$ 0.38 at held +0.024 FAR	
r2 R1 kill rule (operator-grade)	FAR 0.085; tool p 0.55	mixed
r2 P3.1 NE39 alarms (heavy-tailed baseline)	quantile score no; amended D3 yes (4.5 > 2.76)	held (amended)
r2 P3.3 NE43a/b alarm	amended D3: both	held
r2 P2.1 NE40 by presence rule (recovery)	primary 2 days late; one-day and appearance rules on the day; 0–1/37 FAR	mixed
r2 P4 provider DiD	0/25 after Holm; minimum p 0.031	unpowered

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\sim$ mixed
G05 d descriptive
G06 $\sim$ mixed
G07 $\checkmark$ supported
G08 $\sim$ mixed
G10 $\checkmark$ supported
G11 d descriptive
G12 $\times$ failed
+30 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

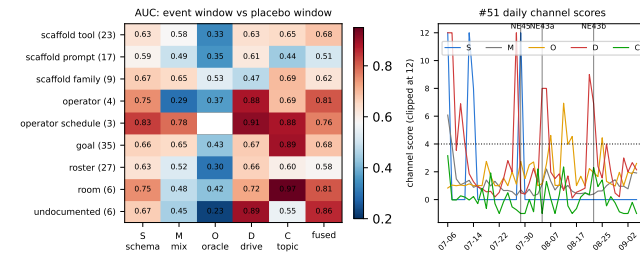
The pre-registered placebo pool held two provider format changes that round 1 found but did not catalog; they blind S (threshold 20). The catalog completion that fixes it is post hoc (FAR 0.051). The D3 channel and the accumulating presence rule are amendments after the synthetic study, before real data; synthetic power for count steps is 0.4–0.6. NE40 is a recovery, not a discovery. With dense events, random dates near other changes set a high chance level (0.33). The reserved data is not run.

ROUND 2

Re-freeze `confirm.py` on the round-2 monitor and run it on the reserved data after sign-off. Split the schema diff into platform and provider streams. Find an observable for scaffold changes (quantile-baselined mix: post hoc lead, p 0.016). Power the provider DiD with agent-day permutations.

- **Essence.** a cheap monitor that tells an operator, on the day, that the platform, the drive or the goal changed, with a known false-alarm rate.
- **Round-3 redirects (newest).**
- **H74-R5** (unchanged). Run `confirm.py` on the reserved data after sign-off.
- **H74-R6** Split S into platform and provider streams. Provider-envelope signatures change with the model vendor, not the village; alert on them separately.
- **H74-R7** Scaffold changes need a new observable. Neither schema, counters, topic nor action mix sees them beyond chance.
- **H74-R8** Powered provider DiD. Use agent-day block permutations instead of ~ 30 null windows, so that Holm over the tests can reject; test the infrastructure-error lead first.

H74-change-detector · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H74-change-detector/README.md · ratings: H74 agent (suggested) (page 2)



Second figure. Round 1. Left: AUC of the event window against placebo windows, by class and channel. Each channel sees its own class: schema and drive for platform and operator steps, topic for goals and rooms. Right: #51 daily channel scores; the schema channel fires on the search-tool rename (NE45), the drive channel on the bookend and nudger stops (NE43).

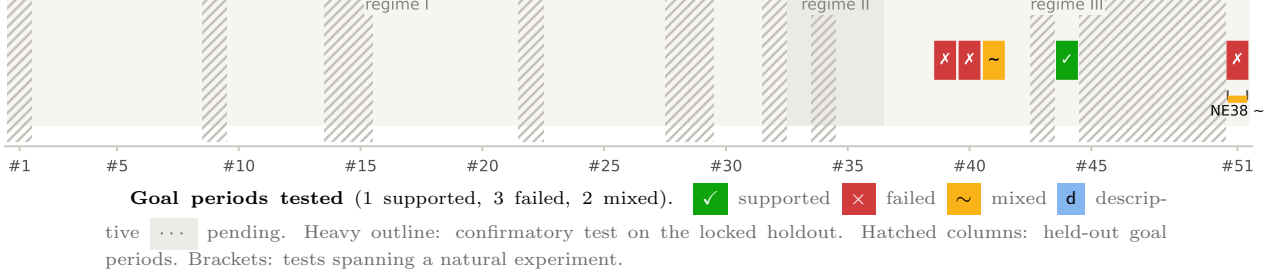
H75: A thermodynamic speed limit on re-allocation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

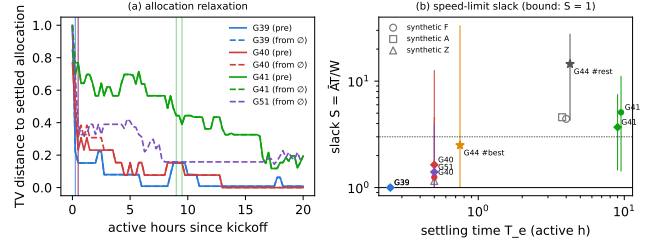
Bottom line. A kickoff that names its target re-allocates the swarm at the Wasserstein speed limit within half an active hour; a free-choice kickoff settles 5–40 times slower through churn, at the same switch rate.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

After a goal change, agents move their work to new repositories. I measured this move on DQ4 work commits in four regime-III kickoffs (G39, G40, G41, G51) and tested it against the Wasserstein speed limit $T \geq W/\bar{A}$. The synthetic check came first. In three kickoffs that name a target, the allocation freezes within 15–30 active min. Each moved agent switches once ($S = \bar{A}T/W = 1.00\text{--}1.40$), so the move sits on the bound. G41 (no shared target) takes 9.5h with $S = 5.1$ [1.4, 11.2]. In G44 the room with a named target settles $\times 5.7$ faster than the free-choice room at the same switch rate. The field-limited predictions (slack ≥ 3 , settling 2.5–10 h) failed in 3/4 periods. Eleven newcomers settle in a median 0.28 h. The cadence elasticity is not identifiable at $N \leq 25$. Holdout (NE20): not run.



Observables. (a) Distance of the repo allocation to its settled state after the kickoff. G39, G40 and G51 drop within 0.5 active h; G41 relaxes over 10 h. (b) Slack $S = \bar{A}T/W$ against the settling time T_e , with agent-bootstrap CIs. $S = 1$ is the speed limit. Grey: synthetic field-limited (F), activity-limited (A) and instant-freeze (Z) means. Stars: the two G44 rooms on the same days.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Useful as a design rule for kickoffs, with a small evidence base. If the kickoff names the artifact, the swarm re-points in 15–30 active min with no wasted switches. If agents must choose, re-pointing takes 4–10 h and 5–15 switches per net move. An operator who wants a fast, clean switch should name the repository or deliverable in the goal text. The speed-limit language adds little beyond this: the bound holds by counting, and the slack measures how specific the field is, not a thermodynamic cost. Whether faster call cadence speeds re-allocation is untested here; NE20 in the holdout can answer it.

PHYSICAL MODEL

Models 02 (kinetic jump process) and 05 (driven population). Each agent of a period is one copy of a Markov jump process on repos plus a null state; $p(t)$ is the share of agents on each repo. The kickoff is the field. W : total-variation distance between the allocations at $t = 0^-$ and at the settling time T . $\bar{A}$: switches per agent per active hour. S : slack. Σ_{ex} : excess entropy production (Kolchinsky et al. 2026).

$$\dot{p} = \nabla^\top j, \quad A = \sum_\rho j_\rho, \quad W = \frac{1}{2} \|p(T) - p(0)\|_1$$

$$\Sigma_{\text{ex}}(T) \geq 2W \tanh^{-1}(W/\bar{A}T) \Rightarrow T \geq \frac{W}{\bar{A}} \coth \frac{\Sigma_{\text{ex}}}{2W} \geq \frac{W}{\bar{A}}$$

$$S = \bar{A}T/W \quad (\geq 1; S = 1 : \text{every switch directed})$$

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 $\bar{A}T \geq W$	no violation (counting identity)	holds
P1 $S \geq 3$ in $\geq 2/3$	1.00, 1.25, 5.09, 1.40	fail (1/4)
P2 $R \geq 3$ in $\geq 2/3$	0.01, 0.00, 6.9, 0.31	fail (1/4)
P3 $T_e \in [2.5, 10]$ h	0.25, 0.5, 9.5, 0.5 h	fail; R2 wins
P4 pooled ε vs synthetic	-0.12 [-2.8, 2.6]	inconclusive
N1 G44 rooms	T ratio 5.7 vs activity rival 0.98	supported
N2 newcomers, NE38	median 0.28 h; $\varepsilon = +1.3 \pm 1.3$; NE38 2.0 h	mixed
tail slack S_{90}	1.15, 5.9, 8.9, 4.0	descriptive

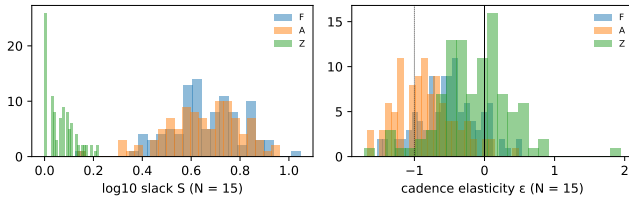
GOAL PERIODS IN DETAIL

G39 × failed
G40 × failed
G41 ~ mixed
G44 ✓ supported
G51 × failed
NE38 ~ mixed

FAITHFULNESS SCORECARD

A mapping 1 F identifiability 1
B assumptions 1 G ground truth 1
C adequacy 1 H comparative 1
D unfitted predictions 1 I transfer 1
E interventional 1

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).



Second figure. Synthetic agents observed only at their commits ($N = 15$, 100 replicates per scenario). Left: the slack separates an instant freeze (Z, $S \approx 1.15$) from churn-dominated settling. It does not separate field-limited (F) from activity-limited (A) settling, because both give $S \approx 4.5$. Right: the cadence elasticity is biased (-0.5 under F) and overlaps A.

CAVEATS

The bound holds by construction on the empirical ensemble, so P0 is a check, not a test. Commits are the only observation of the state: the settling time includes the first-commit delay, and the 15-min grid floors T_e at 0.25 h. G51 has no pre-kickoff allocation (the previous week is held out), so its W counts every first commit as a move. Each period gives one kickoff with 13–16 agents, and bootstrap CIs of S are wide outside G39. G44’s named-target room has 4 committing agents. The cadence elasticity is not identified.

ROUND 2

Score every regime-III kickoff for a named artifact and test S and T_e against it. Use the denser `artifact.mentions` channel to resolve sub-15-min settling. Run `confirm.py` (NE20 cadence DiD and transfer to #45–#47).

- **Essence.** an operator rule for how fast a swarm can be re-pointed, and whether the limit is the field, the agents’ switching, or their cadence.
- **H75-R1** Kickoff text vs slack across all regime-III kickoffs.
- **H75-R2** A denser allocation channel.
- **H75-R3** Excess EP of re-allocation.
- **H75-R4** Run `confirm.py` (NE20 DiD and transfer).

H75-reallocation-speed-limit · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H75-reallocation-speed-limit/README.md · ratings: H75 agent (suggested) (page 2)

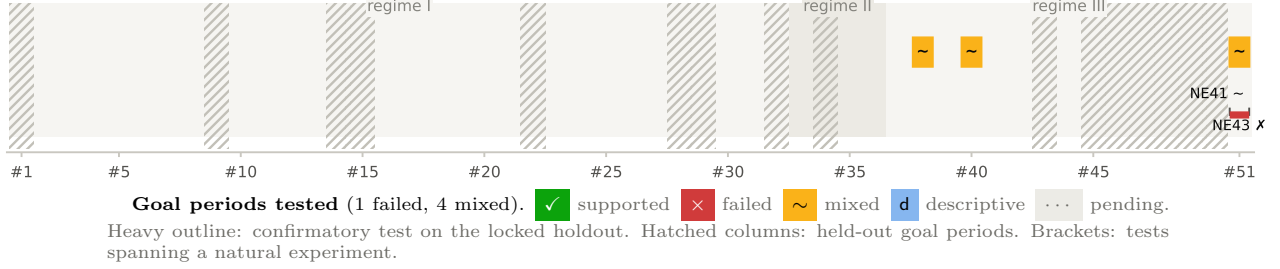
H76: Excess vs housekeeping irreversibility

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2/5

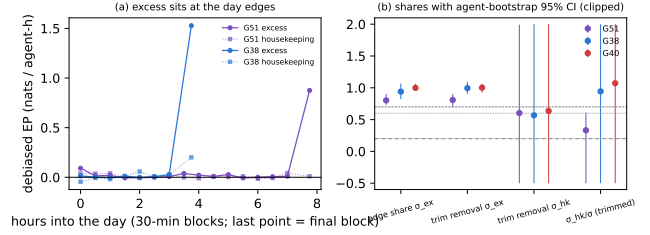
Bottom line. The swarm's excess behavioral irreversibility is the scheduler's day end, which trimming removes; kickoffs and the housekeeping work cycle are below the detection floor of a 12–27-agent ensemble with soft labels.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

I split the irreversibility of the swarm's behavior mix into excess (net change of the mix) and housekeeping (cycles with no net change). The split follows Kolchinsky et al. (2026) and uses ensemble fluxes of Jev v3 states over the agents present at each time, in G51, G38 and G40. The synthetic check came first. In every period, 80–100% of the excess falls in the day's first and last 30 min, against a 12–25% time share. The final block alone holds 72–100%. DQ8 trimming removes 81–100% of it. Kickoffs add no excess beyond an ordinary day start. The synthetic predicted this: a 5-h relaxation is too slow to show. Housekeeping does not separate from a reversible surrogate. A forced erasure gives a one-window excess pulse ($\times 5$ –12). Holdout: not run.



Observables. (a) Debiased excess (solid) and housekeeping (dotted) per 30-min block of the day, averaged over non-kickoff days, in nats per agent-hour. In G51 and G38 the excess sits in the final block; housekeeping is near zero everywhere. (b) Edge share of the excess, trimming removal of excess and of housekeeping, and the housekeeping share, with agent-bootstrap 95% CIs. Dotted line: the 0.6 edge criterion.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a measurement lesson, with one operator number. The behavior mix of an LLM swarm changes irreversibly at the scheduler's stop. Agents finish at staggered times, and that carries 72–100% of a day's excess. Any irreversibility or EP statistic on agent behavior must therefore trim to the all-present window first, or it measures the cron table (H38). A kickoff quench is invisible in 5-min behavior states: use content or repo allocation to see it (H54, H75). Housekeeping, the work cycle that could gauge real activity, is below the noise floor with soft labels at $N \leq 27$. For a monitor this gives a scheduler gauge, not a work gauge.

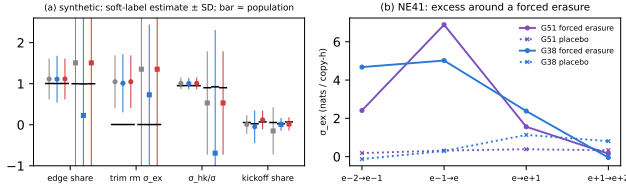
PHYSICAL MODEL

Models 02 (kinetic jump process) and 05 (driven population). The agents present in a 5-min window are exchangeable copies of a Markov chain on five behavior states: absent, work, explore, coord, wait. The fine version uses the 11 v3 states plus absent. j_{xy} is the soft transition flux per agent-step in a 30-min block. σ is the total EP, σ_{ex} the excess and σ_{hk} the housekeeping, all in nats per agent-step. The scheduler's start and stop and the kickoff are fields.

$$\begin{aligned} \sigma &= \sum_{x \neq y} j_{xy} \ln \frac{j_{xy}}{j_{yx}}, & \dot{p}_x &= \sum_y (j_{yx} - j_{xy}) \\ \sigma_{ex} &= \max_{\phi} \left[-\phi^T \dot{p} - \sum_{x \neq y} j_{xy} (e^{\phi_y - \phi_x} - 1) \right] \approx \dot{p}^T H^+ \dot{p} \\ \sigma_{hk} &= \sigma - \sigma_{ex} \geq 0, & \sigma_{ex} &= 0 \iff \dot{p} = 0 \end{aligned}$$

RESULTS VS PREDICTIONS

Prediction	Observed (G51, G38, G40)	Verdict
P1 kickoff above every day start	rank 0.06, 0.85, 0.33	fail (expected)
P1b decay $\tau \in [2.5, 10]$ h	CI spans 0.25–40 h	not identified
P2 edge share ≥ 0.6	0.80 [0.73, 0.90], 0.94, 1.00	yes (3/3)
P3a trim removes $\geq 70\%$ of σ_{ex}	0.81 [0.69, 0.90], 1.00, 1.00	yes (3/3)
P3b trim keeps σ_{hk}	removes 0.60, 0.57, 0.64 vs steps 0.28–0.30	fail (CI spans 0)
P4 $\sigma_{\text{hk}}/\sigma \geq 0.7$; vs H14	0.33, 0.94, 1.07 (CIs span 0); $\times 0.04$ –0.24 of H14	unresolved
N1 NE43 inside placebo range	edge $\times 0.34$ (range 0.73–2.51); recovers next week	fail
N2 NE41 pulse, σ_{hk} kept	$\sigma_{\text{ex}} \times 12$ [6, 27], $\times 5.2$ [3, 10]; $\sigma_{\text{hk}} \times 1.3$, $\times 0.38$	mixed



Second figure. (a) Synthetic five-state master equation at vil-lage sampling: soft-label estimates ($\pm$ SD) against the population value (bars). Edges and trimming are recovered at G51 scale; the kickoff share stays near 0 in every scenario. (b) NE41: excess per copy-hour around a forced erasure vs presence-matched placebo windows. The pulse is the consolidation gap, and it is gone one window later.

GOAL PERIODS IN DETAIL

G38 $\sim$ mixed
G40 $\sim$ mixed
G51 $\sim$ mixed
NE41 $\sim$ mixed
NE43 $\times$ failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The states are zero-shot soft labels (Jev v3). Soft labels attenuate housekeeping $\times 0.19$ in the synthetic, and “absent” also catches the consolidation gap. Block averaging gives lower bounds, and the block-flip floor is Monte Carlo (20 surrogates). The kickoff and housekeeping nulls are underpowered, not evidence of absence. Two estimator bugs were fixed after the first real-data pass (bootstrap floor, NE41 placebo; amendment A2). The NE43 dip coincides with the `#general/#focus` room split. No holdout run.

ROUND 2

Regress the end-block excess on the spread of agents’ last-call times. Redo housekeeping on hard fine action classes (H14). Give consolidation windows their own state and rerun NE41. Run `confirm.py` once the H14 holdout overlap is settled.

- **Essence.** a per-period gauge of the scheduler’s field, and a measured split of behavioral irreversibility into net change of the behavior mix and stationary cycling.
- **H76-R1** Excess on the stop schedule.
- **H76-R2** Housekeeping with hard states.
- **H76-R3** Kickoff excess at turn resolution.
- **H76-R4** Remove the consolidation gap.
- **H76-R5** Run `confirm.py` after the coordinator settles the H14 overlap.

H76-excess-housekeeping-split · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H76-excess-housekeeping-split/README.md · ratings: H76 agent (suggested) (page 2)

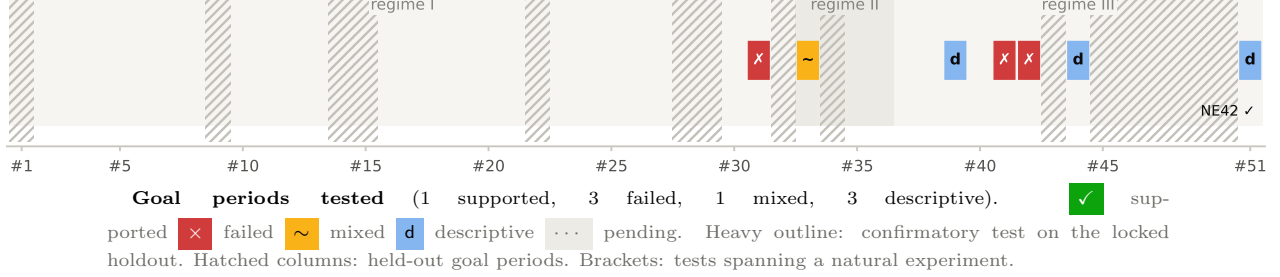
H77: Repos as replicators: the selection-resolution bound

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 1/5

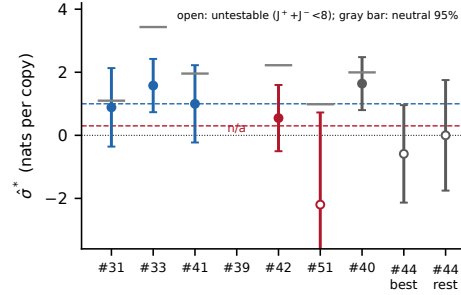
Bottom line. σ^* from commit logs is a bookkeeping convention inside a neutral band, and the resolution bound cannot be tested because winning repos are formed by the kickoff field.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H77 treats each repo as a replicator and measures the affinity σ^* of the top repo on its plateau. σ^* is the log ratio of recruitments to switch-outs. It then tests Kolchinsky's selection-resolution bound on rivals that die. Herding weeks should give $\sigma^* \geq 1$ nat and own-artifact weeks ≤ 0.3 . The data are DQ4 work commits in #31, #33, #41 (herding), #39, #42, #51 (own artifacts) and natives #44 and NE42. The herding weeks give 0.89, 1.58 and 1.00 nats, and #42 gives 0.55. The difference is 0.61 ± 1.23 , so HH325's kill condition is met. A neutral copying world on the same schedule gives the same values. The bound cannot be tested. The holdout is not run.



Observables. The affinity $\hat{\sigma}^*$ of the top repo per period, with a Poisson log-ratio 95% CI and host expiry $E = 100$. The dashed lines mark the predicted thresholds: ≥ 1 nat for herding (blue) and ≤ 0.3 for own artifacts (red). Gray bars show the 95th percentile of a neutral copying world on the same call schedule. No testable period exceeds its neutral bar. #40 (NE42 merge) is highest, but its hub is kickoff-named and every arrival into it is field-tagged.

PHYSICAL MODEL

Model 05 (Kolchinsky 2025, replicators in a flow reactor), with model 06 as the neutral null. A replicator is a repo j . Its copy number n_j counts the agents whose current work label is j . J^+ counts recruitments (copy steps). J^- counts switch-outs to another repo (the reverse step, uncopied). Dilution ϕ is a label that expires after E own calls without a commit. f is the rare-phase fitness: recruitments per host per 1,000 free calls while $n \leq 2$. s is a rival's selection coefficient. q is the order of uncopied.

$$\sigma^* = \ln \frac{J_T^+ + \frac{1}{2}}{J_T^- + \frac{1}{2}} \quad (\text{top repo } T, \text{ plateau})$$

$$s_k = 1 - f_k/f_T, \quad k \text{ extinct} \Rightarrow s_k \geq e^{-\sigma^*}$$

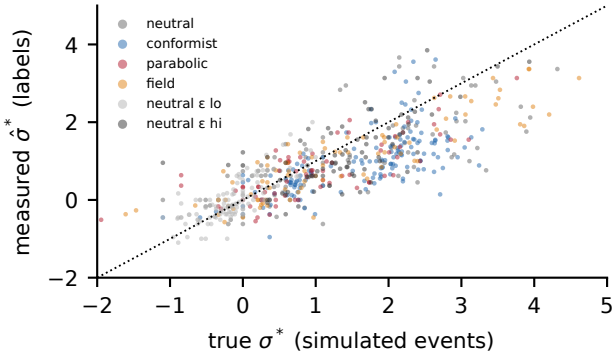
$$J_j^- \propto C_j^{\text{host}} n_j^{q-1}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator this result changes nothing. A recruitment/departure ratio from commit logs does not say whether a swarm selects among projects or only drifts. The ratio depends on how idle hosts are counted, and a neutral world on the same schedule reproduces it. The winning repo in herding weeks is the one the kickoff names, so the lever is the kickoff text (H54). One signature may be usable after confirmation. The per-host departure rate falls with repo size where repos are shared ($q < 1$). It rises with size in the private-role era ($q = 1.46 [1.22, 1.70]$).

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\sigma^* \geq 1$, herding	0.89, 1.58, 1.00 (1/3)	failed
P2 $\sigma^* \leq 0.3$, own	#42 0.55; #39, #51 untestable	failed
P3 contrast > 0.3 , $CI > 0$	0.61 ± 1.23	kill met
P4 bound on extinct rivals	$f_T = 0$ in herding weeks; #51 0/7	not ident.
P6 $ q - 1 < 0.3$	0.09–0.59 herding; 1.46, 1.90 own	failed
P7 above neutral 95%	0/3 herding weeks	failed
P8 synthetic bias ≤ 0.3	-1.2 to $+0.4$; test power 0	failed
NE42 σ^* (#40) highest	1.64 vs #41 1.00; hub field-made	supported
A0 frustrated herd	#31: 3 repos reached ≥ 3 hosts, died	supported



Second figure. Measured $\hat{\sigma}^*$ (labels, $E = 100$) against the true σ^* of the simulated events in six worlds on the real schedules. The estimator tracks the truth only roughly, and conformist worlds read low by about 1 nat. With $E = 300$ or $E = 50$ the bias reaches ± 2 nats.

GOAL PERIODS IN DETAIL

G31 × failed
G33 ~ mixed
G39 d descriptive
G41 × failed
G42 × failed
G44 d descriptive
G51 d descriptive
NE42 ✓ supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	0
B assumptions	0	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The theory defines σ by microscopic reversibility. Commit logs do not show it. Whether a host's loss counts as uncopied or dilution depends on the expiry E . Between $E = 50$ and 300, σ^* moves by 2.0–3.7 nats in 4/5 periods and changes sign in #31 and #41. $E = 100$ was fixed from the synthetic before the real run (A1). The H54 naming rule tags most herding arrivals, partly because agents name repos after the goal. Plateaus are short; #51 has none with flux.

ROUND 2

Observe reversibility directly from agent traces (move vs idle vs erasure); extend the bound to replicators with a formation term; test $q < 1$ vs $q > 1$ on the holdout.

- **H77-R1** Measure reversibility directly.
- **H77-R2** Bound with a field term.
- **H77-R3** q as the usable signature.

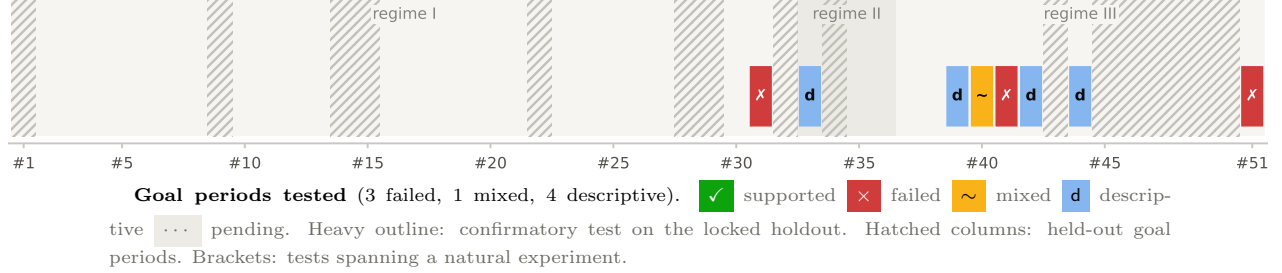
H78: Replicator growth order: herding vs division of labor

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1/5

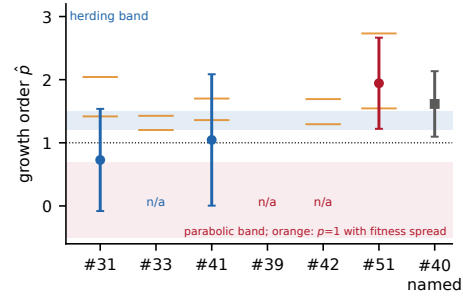
Bottom line. Repo growth order is not identifiable from commit logs: first-order copying plus fitness spread explains every $\hat{p}$, and herding sits on kickoff-named repos.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H78 treats each repo as a replicator and fits its growth order p on the swarm call clock. Herding weeks should give $p = 1.2\text{--}1.5$ and own-artifact weeks $p \leq 0.7$. The data are DQ4 work commits in three herding weeks (#31, #33, #41), three own-artifact points (#39, #42, #51) and two natives (#40, #44). Field-made arrivals are removed: births, kickoff-named repos (H54) and the H28 blind window. The herding weeks give $\hat{p} = 0.73 [-0.08, 1.54]$ and $1.05 [0.00, 2.09]$. #51 gives $1.94 [1.22, 2.67]$ from 271 recruitments, which is the wrong sign. First-order copying ($p = 1$) with a spread of repo fitness reproduces every value. Formation makes 76–100% of arrivals. The holdout is not run.



Observables. The growth order $\hat{p}$ per period (95% CI, formation-free recruitments, host expiry $E = 100$). The blue band is the herding prediction and the red band the parabolic prediction. Orange bars show the median $\hat{p}$ of simulated worlds on the same call schedule with $p = 1$ and repo fitness spread $\sigma_A = 0.5$ and 1.0 . Every testable estimate lies near a $p = 1$ world. The gray square is the #40 hub, a kickoff-named repo, which also grows superlinearly.

PHYSICAL MODEL

Model 05 (replicators) with model 06 as the null. A replicator is a repo j . Its copy number n_j is the count of agents whose current work label is j . A recruitment moves an agent's label into a repo that already has hosts. C_j^{free} counts the calls of agents who do not host j . κ_j is formation by a field (births, kickoff naming). A_j is a repo's fitness (attractiveness). The clock τ counts swarm calls (H40). $p = 1$ is neutral copying, $p > 1$ conformist, $p < 1$ parabolic.

$$\frac{dn_j^+}{d\tau} = \kappa_j + c A_j \frac{C_j^{\text{free}}}{C} n_j^p$$

$$R_{jb} \sim \text{Pois}(C_{jb}^{\text{free}} e^{\alpha + p \log n_{jb}}), \quad n_{jb} \geq 1$$

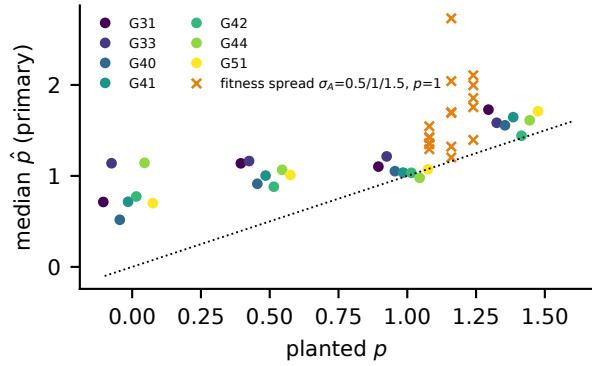
$$A_j \sim \text{LogN}(0, \sigma_A^2) \Rightarrow \hat{p} > p \text{ (pooled over repos)}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A monitor cannot read “winner-take-all dynamics” from how fast big repos recruit. Fitness differences between repos produce the same superlinear signal. Herding in these weeks is a repo that the kickoff names, and it takes every arrival. To prevent duplicated effort, an operator should control what the kickoff names. Watching the recruitment law does not help. The blind window is negligible in commit data (0.7% of recruitments). Most recruitments (66%) are returns to a repo the agent worked on before. This mostly restates H54 and H58 in replicator terms.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\hat{p} \in [1.2, 1.5]$, herding	#31 0.73, #41 1.05; #33 9 events	failed
P2 $\hat{p} \leq 0.7$, own	#51 1.94 [1.22, 2.67]; #39, #42 untestable	failed
P3 $\Delta p > 0$	-1.05	failed
P4 impostors shift < 0.2	#31 0.81, #33 0.33, #41 0.05	failed
P5 cross-lab within 0.3	#31 0.42, #41 -0.72, #51 2.43	failed
P7 recovery $p \in \{0.5, 1, 1.5\}$	$p = 1, 1.4$ recovered; 0.5 reads ≈ 1.0	failed
N #40 named hub $\hat{p} \leq 0.5$	1.62 [1.10, 2.13]; all 48 into the hub	mixed
A0 step from the field	calibrated in 2/6 (#41, #51); burst 3.9×	failed



Second figure. Synthetic recovery on the real schedules, 40 runs per world (10 for #51), measured through the same label builder. Planted $p = 1$ and $p = 1.4$ are recovered. A planted $p = 0.5$ and a field-only world ($p = 0$) read near 1. Worlds with $p = 1$ and repo fitness spread (orange) read 1.2–2.7.

GOAL PERIODS IN DETAIL

G31 × failed
 G33 d descriptive
 G39 d descriptive
 G40 ~ mixed
 G41 × failed
 G42 d descriptive
 G44 d descriptive
 G51 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The host expiry E sets which losses count. $E = 300$ inflated $\hat{p}$ through idle “ghost” hosts, so $E = 100$ was fixed before the real run (A1). The H54 naming rule tags 46–100% of recruitments. Agents name repos after the goal, so “named” is partly agent-made. The fitness worlds, the touch-based impostor class and the AR(1) step null were added after the first real run (A2). #33, #42 and #44 have under 15 formation-free recruitments. Commits are attempts.

ROUND 2

Within-repo dynamic-panel estimators validated on the fitness worlds; repo covariates as fitness; an explicit return term.

- **H78-R1** Separate order from fitness.
- **H78-R2** Model returns explicitly.
- **H78-R3** A better naming instrument.

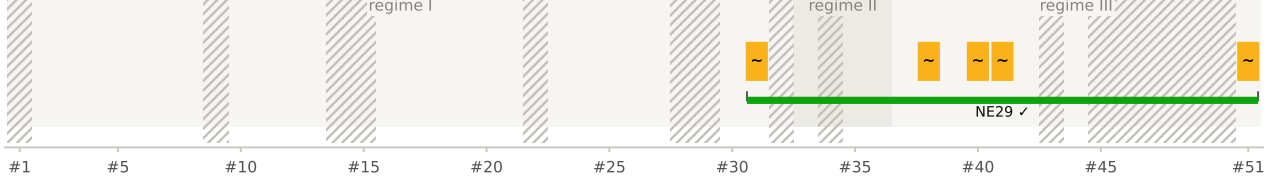
H79: An artifact-only autocatalytic set

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. Artifact catalysis in the village is first order: self-maintaining scripts and repos, no cross-catalytic cycles, and execution that follows writes rather than causes them.

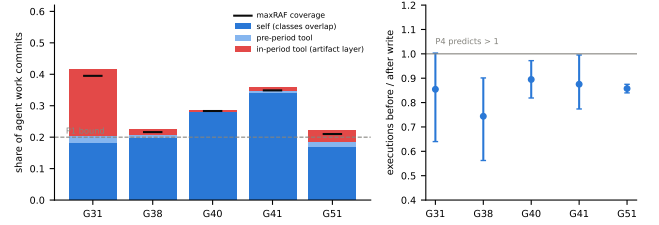
level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 supported, 5 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

H79 asks if the village's artifacts form an autocatalytic set when agents are not catalysts. A reaction is a write to repo X in a session where repo Y 's code runs first; cron streams catalyse their own repo. We built the network for G51, G31, G38 and G41, and for the natives G40 and NE29. Self-catalysis carries 85–99% of maxRAF commits. No period has a catalytic cycle of length ≥ 3 with ≥ 3 events per edge. G51 has fewer such cycles than the rewired null ($p = 0.01$). The maxRAF covers 21–40% of agent work commits, so P1 ($\leq 20\%$) fails. This coverage equals the execution share and the rewired null, and it falls to 0.3–14% without python heads. Executions follow writes more often than they precede them. Holdout script frozen, not run.



Observables. Left: share of agent work commits by catalyst class (classes overlap); bars mark maxRAF coverage; dashed line, P1's 20% bound. Self-catalysis dominates in every period. The in-period tool class, which would be the artifact layer, is $\leq 4\%$ except G31, where the rewired null gives the same value. Right: executions before / after a write, with day-bootstrap 95% CI. All five values are below 1, so P4 fails in reverse.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, the automata that agents build are single scripts that maintain their own repo. They do not form a self-sustaining tool ecosystem. When an agent leaves, nothing others depend on stops: the retiree's 120 repos catalysed 0 later commits. To audit what keeps running without agents, list the self-writing streams per repo. The G51 cron stream alone made 54,892 commits. A dependency-graph search for tool chains adds nothing to that list. The catalysis edge also points the wrong way in time, because agents mostly commit and then run code. So "this tool produced that work" cannot be read from session co-occurrence. This result is mostly a negative for the egregore reading.

PHYSICAL MODEL

Model 05 (replicators), RAF variant. Species: repos. Reaction $r = (X, Y)$: product X , catalyst Y , aggregated over session events. Food F : platforms, pre-period artifacts, operator-named artifacts. Agents are the uncatalysed background: their products are reactants, never catalysts. A maxRAF reaction needs $Y \in F$ or Y produced inside the set. Order parameters: coverage c , self share s , supported cycles n_3 , time ratio τ .

$$\mathcal{R}' \text{ RAF} \iff \forall r = (X, Y) \in \mathcal{R}' : Y \in F \cup \{X' : (X', Y') \in \mathcal{R}'\}$$

$$c = \frac{\sum_{e: r(e) \in \text{maxRAF}} n_e}{\sum_e n_e}, \quad s = \frac{\sum_{r=(X,X)} n_r}{\sum_{r \in \text{maxRAF}} n_r}$$

$$\text{irrRAF}_{|\cdot| \geq 2} = \text{simple cycle in } G = \{Y \rightarrow X : X, Y \notin F, X \neq Y\}$$

$$\tau = \frac{\#\{\text{commits with a repo run before the write}\}}{\#\{\text{commits with a repo run after the write}\}}$$

H79-artifact-autocatalytic-set · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H79-artifact-autocatalytic-set/README.md · ratings: H79 round-1 agent (suggested)

RESULTS VS PREDICTIONS

prediction	observed (5 periods)	verdict
P1 coverage ≤ 0.20	0.21–0.40 (G51 0.21 [0.19, 0.24]); failed* 0.003–0.14 without python	
P2 no ≥ 3 cycles	0 supported cycles; G51 below supported null (p_{low} 0.01)	supported
P3a self ≥ 0.5 of RAF	0.85–0.99	supported
P3b in-period tools ≤ 0.05	0.003–0.037; G31 0.21 (null 0.20)	4/5
P4 before/after > 1	0.74–0.90 (CI < 1 in 3/5)	failed
P5 CAF/RAF ≥ 0.9	G51 0.09; others 0.75–0.99	2/5
N1 G40 commons	0 cycles; self 0.98; N1c failed	mixed
N2 NE29 exit	0% of later work catalysed	supported
S1 synthetic	exact vs brute force; power 30/30, passed size 0/30	

*fails only under the head-only execution rule (post hoc variant).

GOAL PERIODS IN DETAIL

G31 $\sim$ mixed
G38 $\sim$ mixed
G40 $\sim$ mixed
G41 $\sim$ mixed
G51 $\sim$ mixed
NE29 $\checkmark$ supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

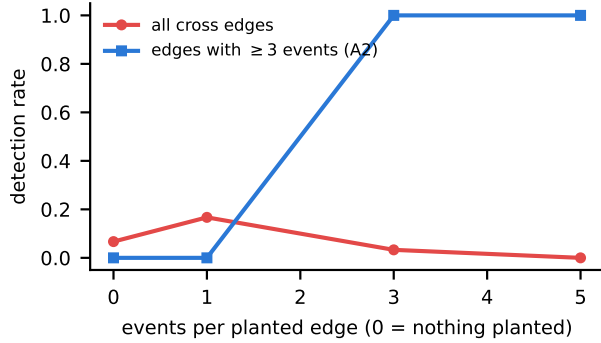
Execution is read from the bash head and the last logged working directory. python3 heredocs count as running the repo, and cwd precision is 0.89. The no-python variant moves coverage by a factor of 10 but leaves the structure unchanged. `first.t` is first mention, not creation, so some old repos enter as in-period artifacts. The rewired null was redesigned after a synthetic check (A2) and before the real run. Catalysis is session co-occurrence, not content dependence. NE29's retiree made few commits. Holdout not used.

ROUND 2

Content-dependent catalysis edges from file-level diffs; full command text for execution (H79-R1, R2).

- **Essence.** whether artifacts carry work forward without an agent in the loop. Round 1 says they do so only as single self-maintaining scripts; the open question is causal: does running an artifact make the next write happen, or only follow it?
- **H79-R1** Content-dependent catalysis.
- **H79-R2** Full command text for execution.
- **H79-R3** The automata as replicators (with H77/H78).
- **H79-R4** Run `confirm.py` on #46–#50 and the #51 tail (automation-dense, held out).

H79-artifact-autocatalytic-set · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H79-artifact-autocatalytic-set/README.md · ratings: H79 round-1 agent (suggested) (page 2)



Second figure. Synthetic network (3,000 events, neutral catalysis): rate of detecting a planted 3-cycle (cycle present and rewired $p < 0.05$) vs events per planted edge. Counting all cross edges gives false 3-cycles in 77% of empty networks and low power. Requiring ≥ 3 events per edge (amendment A2) gives size 0 and power 1.

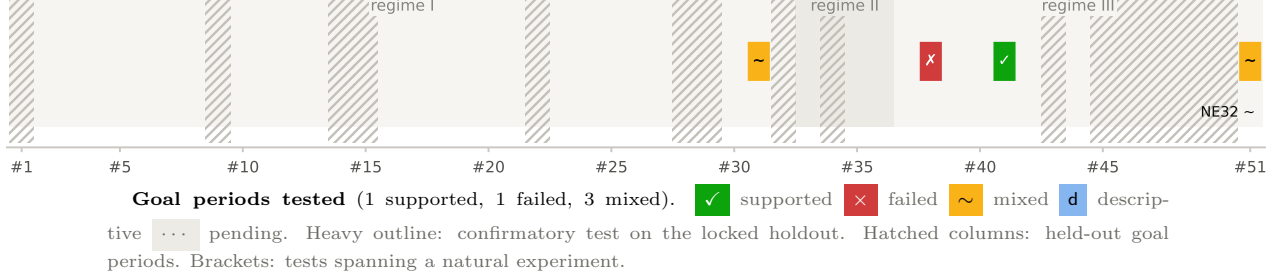
H80: Assembly index vs compression as an agent signature

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2.5/5 Usefulness 2/5

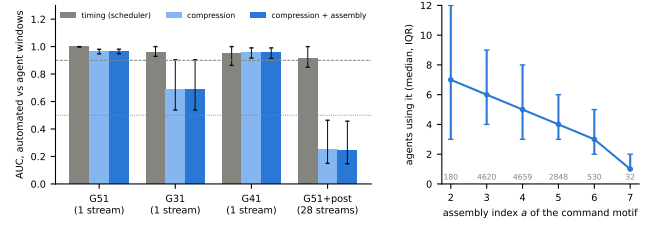
Bottom line. The assembly index is compression in disguise here, compression detects replay rather than automation, and the scheduler's timing carries the agent-vs-script label.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H80 tests assembly theory's index against compression as a signature of script work. The units are 16-commit windows labelled automated or agent (DQ4) in G51, G31 and G41. A fourth unit, G51+post, adds 28 scripts that commit after the village's last day. The index is a Re-Pair grammar bound, checked against an exact search. Adding the index to compression changes AUC by -0.007 to $+0.001$, so the kill (≥ 0.05) is far. Timing alone, the scheduler baseline, reaches $0.92-1.00$. Compression reaches 0.96 only for single cron streams that replay identical commits. Across the 28 scripts it falls to 0.25 . High-index, high-copy command motifs belong to single agents. Only 7.8% appear on a joiner's first day across ≥ 2 labs, so P5 fails. Holdout script frozen, not run.



Observables. Left: out-of-fold AUC with 95% cluster-bootstrap CI, for automated vs agent windows. Adding the assembly index (dark blue) never moves compression (light blue). Timing (gray) wins everywhere. With one cron stream compression looks strong. Across 28 scripts (repo-blocked) it inverts. Right: the median number of agents per command motif falls from 7 to 1 as a rises. Counts of motifs are printed at the bottom.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 04 used as a tool family. A window is a token string x of length $L = 16$; one token hashes one commit's size fields. The assembly index $a(x)$ is the fewest joins that build x with reuse. It equals the minimal grammar size minus the basis (Abrahão Cor. 8), and Re-Pair bounds it from above. The rivals are entropy rate, LZ76, LZ78, gzip and zstd (C), and the timing features (T). Motifs are 4–8-grams of command heads; n_i is the number of sessions holding motif i .

$$a(x) = \min_D \sum_{d \in D \cup \{x\}} (\text{tiles}(d \mid D_{<d}) - 1) \leq a_{\text{RP}} = |\mathcal{R}| + |x_{\text{RP}}| - 1$$

$$A = \sum_i e^{a_i} (n_i - 1) / N_T, \quad \text{NCD}(x, y) = \frac{C(xy) - \min(C_x, C_y)}{\max(C_x, C_y)}$$

$$\Delta \text{AUC}_A = \text{AUC}(C + a_{\text{RP}}) - \text{AUC}(C)$$

H80-assembly-vs-compression · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H80-assembly-vs-compression/README.md · ratings: H80 round-1 agent (suggested)

A monitor that must flag script commits under an agent's identity should use timing: off-hours activity, regular gaps, and no agent turn nearby. Timing alone gives $\text{AUC} \geq 0.91$ here. Compression catches only replay loops. A script that writes varied content looks like an agent, and the compression detector then flags 97% of a script-free period. The assembly index adds nothing to compression, so drop it. For a selection meter, high index $\times$ high copy here measures one agent repeating its own habit. That habit is present from the agent's first day, even in isolation. No operator decision changes because of assembly theory; the timing rule is the usable part.

RESULTS VS PREDICTIONS

prediction	observed	verdict
P1 C AUC ≥ 0.9	0.965 (G51, 1 stream); 0.25 (28 streams); G31 0.69	failed
P2 ΔAUC_A	$< +0.001, 0.000, -0.001, -0.007$	supported
P3 $\rho(a_{RP}, LZ78) \geq 0.9$	0.82 (G51), 0.98, 0.95	2/3
P4 T ≥ 0.85 ; C T 0.92–1.00; C adds ≤ 0.005 adds ≥ 0.02		half
P5 prior share $\geq 70\%$	7.8% (562 motifs); single-agent habits	failed
P6 no NCD fall	+0.005 (7 joiners)	supported
N1 NE32 isolated day 1	1.65 $\times$ incumbents' motif use	supported*
N2 G38 no au- tomation	C flags 97%; T 0%	failed
S1 synthetic	proxy $\geq$ exact, 97.6% exact; power 1.0	passed

*NE33 contrast arm uninformative (motif-set selection).

GOAL PERIODS IN DETAIL

G31 $\sim$ mixed
G38 $\times$ failed
G41 $\checkmark$ supported
G51 $\sim$ mixed
NE32 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

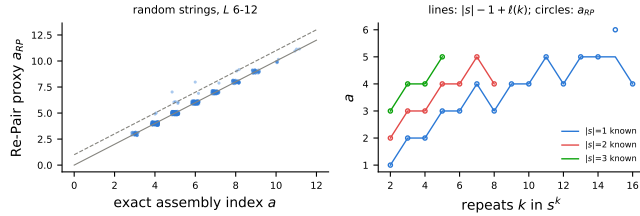
The label **automated** is defined by the time to the nearest agent turn, which favours timing by construction. In-period units have one automated stream each, so their AUCs compare one script against the agents. The powered unit (G51+post) uses scripts that run after the village stopped. Commit tokens hash size fields only; file paths could separate scripts better. Motif copy counts need ≥ 20 sessions inside G51, which excludes late joiners. Holdout not used.

ROUND 2

Copies per independent producer for ensemble assembly; file-path tokens; a timing-free label (H80-R1–R3).

- **Essence**. a log-only detector of "who did this, an agent or its script", and whether assembly theory's selection index measures anything in agent swarms beyond repetition.
- **H80-R1** Copies per independent producer.
- **H80-R2** Content tokens for commits.
- **H80-R3** A timing-free label.
- **H80-R4** Run `confirm.py` on #46–#50 and the #51 tail.

H80-assembly-vs-compression · generated 2026-10-07 17:50 by in-fra/summaries/build-summaries.py · card hypotheses/H80-assembly-vs-compression/README.md · ratings: H80 round-1 agent (suggested) (page 2)



Second figure. Left: the Re-Pair proxy against the exact assembly index on 500 random strings (L 6–12). It is never below the exact value and is exact in 97.6%. Right: s^k strings, where the exact index matches $|s| - 1 + \ell(k)$, with ℓ the shortest addition chain.

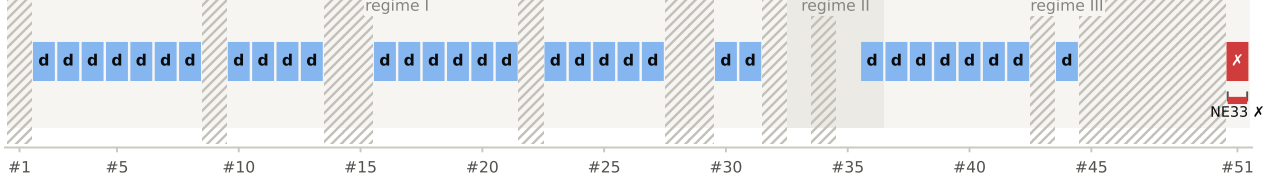
H81: Culture beyond composition: the emergent slow mode

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. In regime I the village content has a collective slow mode ($\tau_u \approx 25$ days) beyond its members and the measured goal and operator fields, carried across turnover; it does not follow the long-lived artifact record its members read, and an unmeasured calendar-time drift remains the open rival.

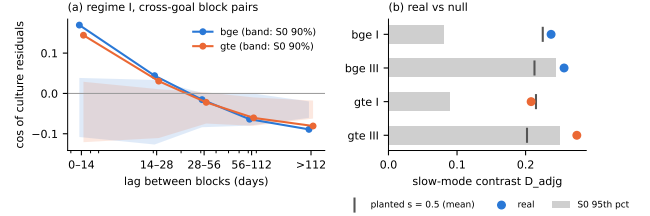
level: descriptive · pipeline checks 86% · updated 2026-10-04



Goal periods tested (2 failed, 32 descriptive). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The test asks whether village content carries a slow component that its members do not explain. The null is each member's personal vector (two-way fixed effects on its other goal periods) plus the measured goal fields. In regime I (24 goals) culture residuals of blocks ≤ 14 days apart align ($\bar{s} = 0.17$), decaying with $\tau_u \approx 23$ –28 days; the contrast is 3 times the synthetic null in both embedding models and survives turnover. Round 2 asked what carries it. It does not follow the long-lived artifact record its members read ($C_{R1} = +0.02 / -0.07$ against an exogenous-drift 95th percentile of 0.11/0.09, power 1.0). It survives removing operator directions beyond the centroid and is not concentrated in outside-world talk. H82's boundary trace fits the same τ . Calendar versus village time cannot be identified, so outside drift is narrowed, not beaten.



Observables. (a) Regime I: mean similarity of culture residuals between blocks of different goal periods against their lag. Bands are the 90% range of the synthetic composition null (S0) on the real panel. Similarity is high below two weeks and decays over about a month. (b) The slow-mode contrast D_{adjg} per model and regime (dots), the S0 95th percentile (grey bars) and the planted mean for a slow mode with half the period-field variance (ticks).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Expect a village to drift as a whole on a timescale of about a month, independent of who is in it. In regime I the slow-mode contrast matches a synthetic mode that carries half of the period-field variance ($s = 0.5$). That share is inferred from the match, not measured. A new kickoff does not reset the village: neighbouring goals share about 0.17 of their residual direction. Roster changes do not move the mode. Its carrier is still unknown, but it is not the long-lived artifact record (old sites, documents, repos), the operator's messages or outside-world talk, so curating those artifacts will not steer it. For monitoring, compare a village with its own last two to three weeks, not with its whole history.

PHYSICAL MODEL

Vector spins (model 11). The spin $v_{i,d}$ is agent i 's style-residualized content vector on day d (32-d). a_i is the personal vector (weights, family, style). h_G is the exogenous field of goal period G , removed by projection Π_G . w_b is a topic field with no memory. u_b is the collective slow mode, an Ornstein–Uhlenbeck vector with correlation time τ_u . The order parameter is the block culture residual u_b . The statistic D is the near-minus-far similarity of u_b across goal boundaries, adjusted for goal similarity (D_{adjg}) and member overlap (D_{adj}).

$$v_{i,d} = a_i + h_{G(d)} + w_{b(d)} + u_{b(d)} + e_{i,d}, \quad \langle u_b \cdot u_{b'} \rangle \propto e^{-|t_b - t_{b'}|/\tau_u}$$

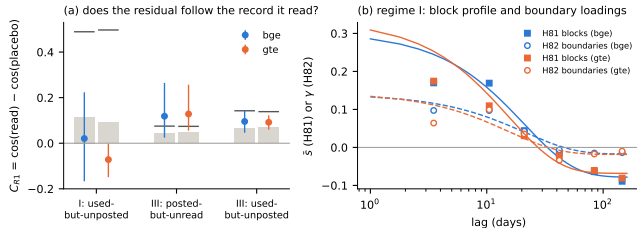
$$r_{i,b} = \Pi_G(\bar{v}_{i,b} - \hat{a}_i^{(-G)}), \quad u_b = \langle r_{i,b} \rangle_{i \in b}$$

$$D = \langle \cos(u_b, u_{b'}) \rangle_{\Delta t \leq 14 \text{ d}} - \langle \cos(u_b, u_{b'}) \rangle_{\Delta t \geq 42 \text{ d}}$$

H81-culture-beyond-composition · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H81-culture-beyond-composition/README.md · ratings: H81 agent (suggested, round 2)

RESULTS VS PREDICTIONS

Prediction	Observed (bge / gte)	Verdict
P2 slow mode $D_{\text{adj}} > S0$ q95	I: 0.237 / 0.208 (q95 0.081 / 0.089); $\tau_u = 28 \pm 5$ / 23 ± 4 d	pass (I)
P3 across turnover (D_{adj})	I: 0.166 / 0.170 (q95 0.10)	pass (I)
P4 roster jumps; N1 #51; N2 NE33	percentile 0.08 / 0.14; $L_{\text{slow}} < 0$; 0.14–0.89	fail
P5 without Gemini 2.5 Pro	$\times 1.06$; size-matched $\times 0.90$ / $\times 0.96$	pass
R1 follows read record (I)	$C_{R1} = +0.02$ / -0.07 (q95 0.11 / 0.09; power 1.0)	fail
R1 regime III (descr.)	$+0.12$ / $+0.13$ (q95 0.04 / 0.05)	pass
R2a operator beyond centroid	D_{adj} 0.244 / 0.229 after removal (q95 0.07 / 0.08)	survives
R2b outside-world topics	$L_{\text{out}} = +0.004$ / $+0.009$ (q95 0.018 / 0.014)	none
R2c calendar vs village clock	goal count lowest SSE ($\tau = 2.4$ / 1.7 goals); not identifiable	inconcl.
R4 one OU with H82	$\Delta\tau = -12$ / -11 d (band $[-33, +6]$); joint $\tau = 21$ / 16 d (weak)	pass



Second figure. Round 2. (a) Record-carrier contrast C_{R1} (read minus placebo record content) with goal-bootstrap 95% CIs; grey bars are the exogenous-drift (S_{ex}) 95th percentile, ticks the planted record-carriage mean ($\lambda = 1$). Regime I fails; regime III passes. (b) Regime I: H81 block similarity (squares) and H82 boundary loadings (circles) with the shared- τ OU fit.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 d descriptive

+24 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only measured exogenous directions are removed; a calendar-time drive reaching every agent without operator messages or outside-world talk would look the same, and the clock test cannot tell calendar from village time (synthetic accuracy ≤ 0.64). The regime-I artifact channel is thin (188 old-artifact chat mentions in 9 months). Regime-III record-following may partly be room topics (3–5 goals). The H82 one-mode band is wide. Power is 0.96–0.99 for a slow mode with half the period-field variance. τ_u is a descriptive fit.

ROUND 2

Test the rolling chat window (each block reads the previous block's tail) as the carrier. Use regime-I newcomers to separate calendar drift (immediate) from a village mode (needs reading). Redo regime-III record-following within rooms at matched lag.

- **Essence.** a village-level state variable that persists for weeks and survives turnover, and what carries it.
- **Round-3 redirects (newest).**
- Where round 2 went sideways: the regime-I record channel visible in the data (old-artifact links) is too sparse to carry the mode even in principle.
- **H81-R5** The rolling chat window as the carrier. Chat-mode prompts rebuild from recent chat, so each block reads the tail of the previous one.
- **H81-R6** Calendar-time drift through newcomers. A calendar drift reaches a newcomer on day 1; a village mode reaches it only after reading.
- **H81-R7** Regime-III record-following with room topics removed. Repeat C_{R1} within rooms (read vs in-flight at matched lag, H57-style) to separate record-reading from room-topic fields.

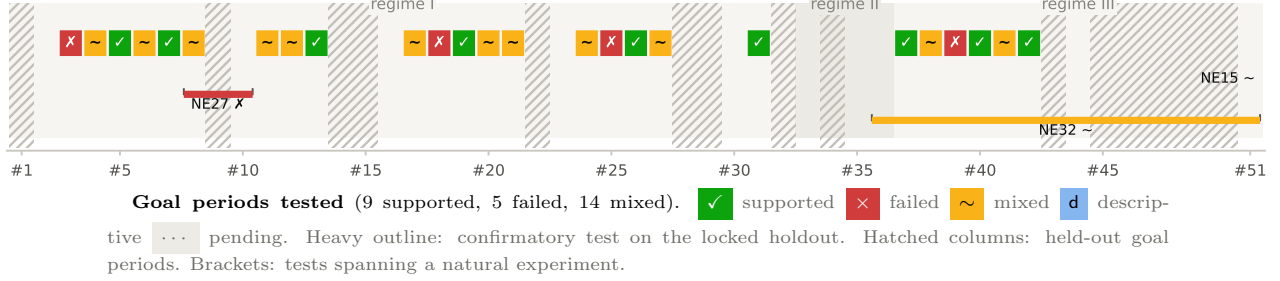
H82: Remanence: the endogenous part of the mean field

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. The previous goal leaves a persistent trace on day-1 content ($\Delta\gamma_1 \approx 0.15$, 25 boundaries), but it is each agent's own carry-over, not a decaying remanence of the village record.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

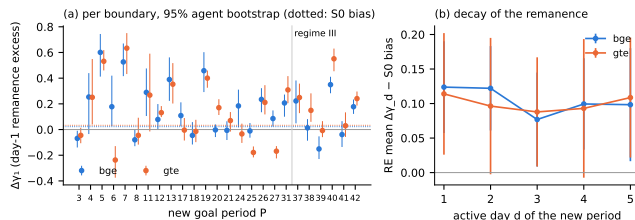
The test decomposes each agent's day-1 content in a new goal period into an exogenous field (new kickoff, goal text, previous kickoff, operator messages), the agent's prior from other periods, and the previous period's village centroid. It runs on 25 goal boundaries with both embedding models. The previous centroid loads by $\Delta\gamma_1 \approx 0.15$ beyond placebo centroids, 3 times the synthetic null. The trace does not decay over five days. The next period's centroid also loads, at about half the size. With the agent's own previous content as a regressor, the village centroid adds nothing ($-0.02 / -0.06$). Newcomers are too few to test. Read previous content predicts day-1 content better than unread content in one model. The holdout script exists and is not run.

PHYSICAL MODEL

Vector spins (model 11) with a two-part mean field. The spin $v_{i,d}$ is the agent-day content vector (32-d, style-residualized). X_{ex} are the exogenous directions. p_i is the agent's prior, fitted without periods P and $P-1$. $e_{P-1}^{(-i)}$ is the previous village centroid without agent i . Its loading γ is the endogenous field. The quench (H54) is the loading α on the new kickoff. Hysteresis is $\gamma > 0$ with $\gamma[P-1] > \gamma[P+1]$. A record-borne trace would also load in newcomers.

$$v_{i,d} = \sum_k \alpha_k X_{\text{ex},k} + \beta p_i^{(-P,-X)} + \gamma_d e_X^{(-i)} + \varepsilon_{i,d}$$

$$\Delta\gamma_d = \gamma_d[X=P-1] - \text{median}_{|Q-P|\geq 2} \gamma_d[X=Q], \quad A = \gamma_1[P-1] - \gamma_1[P+1]$$



Observables. (a) Day-1 remanence excess $\Delta\gamma_1$ per goal boundary with agent-bootstrap 95% intervals; dotted lines mark the synthetic null bias. Regime-I boundaries rest on 3–4 agents. (b) Random-effects mean of $\Delta\gamma_d$ minus the null bias over the first five active days. There is no decay.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A goal change re-points agents onto the new kickoff, but each veteran keeps its own previous themes for at least five days. That carry-over is about 0.15 in loading units, beyond the new fields and the agent's prior. For an operator: a new goal does not erase the old one, so expect about a week of mixed content. Retire or reset a specific agent's context if its old thread must stop. Nothing here shows that the village record carries the trace on its own. The trace follows members and partly what they read.

RESULTS VS PREDICTIONS

Prediction	Observed (bge / gte)	Verdict
P1 $\Delta\gamma_1 > S_0$ q95	0.158 / 0.153 (q95 0.055 / 0.063); 17/25, 16/25 positive	pass
P2 asymmetry $A > S_0$ q95	0.093 / 0.104 (q95 0.036 / 0.052); part is lag geometry	pass
P3 decay within days	$\Delta\gamma_d$ 0.15, 0.15, 0.11, 0.13, 0.13	fail
P4 newcomers carry it	5 boundaries, 1 newcomer each; power ≤ 0.18	inconcl.
N1 NE27 empty-memory join	no trace of #8 for veterans or newcomers	fail
N2 NE32 isolated newcomers	+0.036 [0.006, 0.081] / +0.010 [-0.020, 0.039]	mixed
N3 NE15 read vs unread	+0.071 [-0.007, 0.148] / +0.106 [0.039, 0.174]	mixed
PH after kickoff only	0.152 / 0.156	post hoc
PH own past as regressor	-0.024 / -0.060	post hoc

GOAL PERIODS IN DETAIL

G03 $\times$ failed
G04 $\sim$ mixed
G05 $\checkmark$ supported
G06 $\sim$ mixed
G07 $\checkmark$ supported
G08 $\sim$ mixed
G11 $\sim$ mixed
G12 $\sim$ mixed
G13 $\checkmark$ supported
G17 $\sim$ mixed

+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	0	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

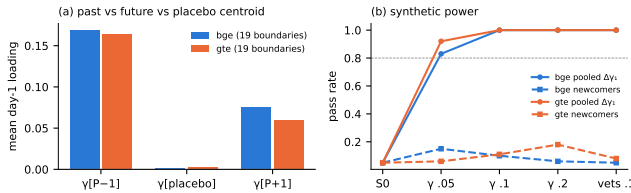
Each boundary is one regression on 3–15 agents. The placebo-corrected excess is biased upward under the null (+0.02 / +0.03), so decisions use the synthetic percentile. The newcomer clause, the HH’s central claim, is untestable at this sample size. The own-past regressor also contains the village state as the agent saw it, so the PH2 result says which regressor wins, not which mechanism acts. Part of the asymmetry is the lag geometry of H81’s slow mode.

ROUND 2

Model each agent’s carry-over with a decay over the whole period and compare it with H81’s τ_u . Pool all newcomer first weeks against same-day veterans, after a synthetic power check. Extend the read-vs-unread test to memory and history-search reads.

- **Essence.** whether a goal change resets the village or only re-points members who keep their own threads.
- **H82-R1** Individual carry-over model.
- **H82-R2** The record channel.
- **H82-R3** Newcomer power.

Second figure. (a) Mean day-1 loading on the previous, a placebo and the next period’s centroid (19 boundaries with an eligible next period). (b) Synthetic power on the real boundaries: the pooled test reaches 0.83–0.92 at $\gamma = 0.05$; the newcomer test stays below 0.2.



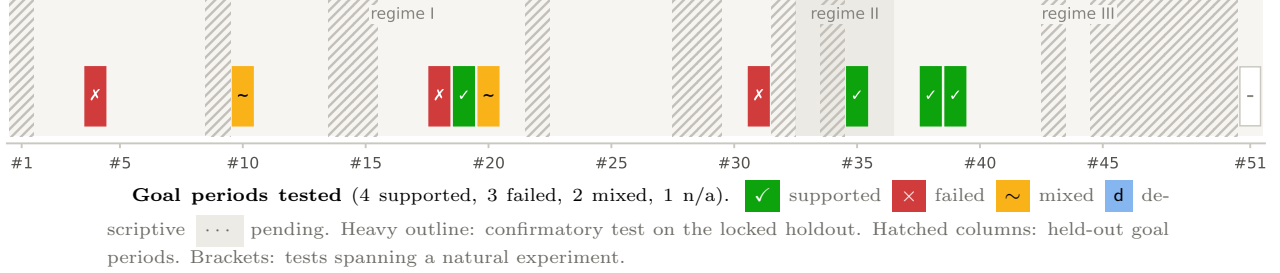
H83: Enculturation of newcomers

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Newcomers join the village's conversation at once and lose a first-week family signature, but no slower enculturation into a village culture is detectable at this power.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

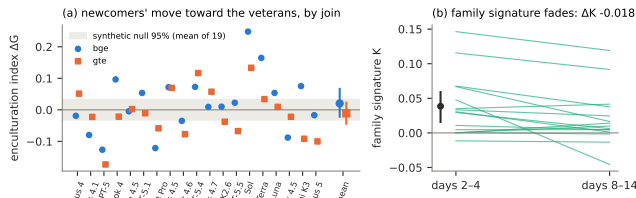
The test asks whether newcomers acquire a village culture. A newcomer's style-free chat content is compared with the veterans' same-day centroid in tenure days 2–4 and 8–14, after the goal and kickoff span is projected out. The veterans' change on the same days is subtracted. Over 19 joins (15 join days, three regimes) the enculturation index is $\Delta G = +0.020$ $[-0.025, +0.070]$ with bge and -0.011 $[-0.047, +0.026]$ with gte. Both sit below the calibrated null cut ($+0.032$). Newcomers start only 0.02 below the veterans. The synthetic power at half closure is 0.15–0.30, so the headline is inconclusive. A family signature at entry is real (13/15 joins) and fades by week two. In #38 newcomers align with their own room at once. NE32's isolation lasted 1.5–2 h, not a day. The holdout script is not run.

PHYSICAL MODEL

Vector spins (model 11). The spin is a statement's 32-d style-residualized unit vector $\tilde{u}_s$. Π_P projects out the goal and kickoff field. A newcomer's constant q_i relaxes toward the veterans' centroid $\bar{V}$ with closure λ as its veteran reads D_i grow. h_d is the day field. The observable $a_{i,d}$ is the mean statement cosine with the veterans' centroid V_d . ΔG is the newcomer's change minus the veterans'. K is the family signature against other newcomers' first-day baselines F_f .

$$\tilde{u}_s \propto \Pi_P [q_i(\tau) + h_d + \varepsilon_s], \quad q_i(\tau) = q_i^0 + \lambda(1 - e^{-D_i(\tau)/D^*})(\bar{V} - q_i^0)$$

$$\Delta G_i = (\bar{a}_{i,L} - \bar{a}_{i,E}) - \langle \bar{a}_{j,L} - \bar{a}_{j,E} \rangle_j, \quad a_{i,d} = \langle \tilde{u}_s \rangle \cdot \hat{V}_d$$



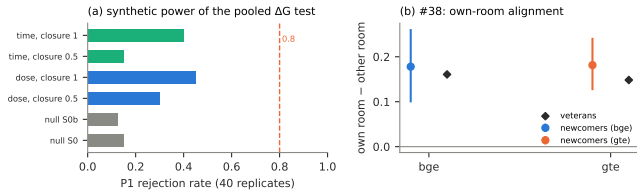
Observables. (a) Enculturation index per join (blue bge, orange gte) and the means with join-day bootstrap CIs. The gray band is the synthetic null 95% range of the mean. Joins scatter on both sides; the models disagree. (b) Family signature K per join at days 2–4 and 8–14. Newcomers start closer to their own family's newcomer baseline (black: mean, CI) and lose part of it.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An operator should not expect newcomers to need an induction period for content. They match the day's conversation from day 2: the gap to the veterans is -0.02 in cosine units and closes to 0 by days 8–14. Put a newcomer in the room whose work it should join; it aligns with that room's veterans at once (0.18, the veterans' own value). Expect a family flavor in its first week; it fades. Do not rely on reading volume: veteran items read do not predict the move ($\rho = +0.06$). Whether a deeper village culture exists is not settled here. The shared persistent content is about 3% of statement variance, too small to test with 19 joins.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 gap closes: $\Delta G > +0.032$	bge $+0.020$ $[-0.025, +0.070]$; gte -0.011 ; $\bar{G}_E -0.020$	inconclusive
P2 family signature fades	$K_E +0.038$ (13/15; gte $+0.031$); $\Delta K -0.018$ $[-0.033,$ $-0.004]$	pass (bge)
P3 dose response	$\rho +0.06$ (p 0.41)	fail (descr.)
P4 persistent part	$\Delta G^{mm} +0.040$ $[-0.030,$ $+0.110]$	fail (descr.)
P5 style approaches	$\Delta S +0.95$ $[-2.3, +4.5]$ (away)	fail
N1 NE32 isolated day	isolation 1.5–2 h, no statements; 733 veteran items read	n/a
N2 #38 own room	0.178 $[0.099, 0.262]$; veterans 0.161	supported
N3 NE27 kickoff-matched gap	-0.017 $[-0.103, +0.066]$; gte $+0.013$	mixed



Second figure. (a) Synthetic power of the pooled test on the real join schedule. Even full closure is detected in under half of the replicates. (b) #38: newcomers' own-room minus other-room alignment (CIs) equals the veterans' (diamonds).

GOAL PERIODS IN DETAIL

G04 × failed
G10 ~ mixed
G18 × failed
G19 ✓ supported
G20 ~ mixed
G31 × failed
G35 ✓ supported
G38 ✓ supported
G39 ✓ supported
G51 – n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The persistent shared component is small, so a null is inconclusive, not absent. bge and gte disagree in sign on ΔG ; statement-level geometry is model-dependent. Family baselines use 1–4 same-family newcomers per regime, and the fade is significant with bge only. Batch joins share windows (15 clusters). #51 roles are an agent-specific field; the own role text is projected out, but role topics remain. Windows can cross a goal boundary. The baselines without same-day joins and the NE32 premise check are post hoc.

ROUND 2

Test enculturation on discrete conventions (village-coined terms and file conventions) through each newcomer's first uses inside the logged light cone. Fit the decay time of the entry family signature per join.

- **Essence.** whether a village holds anything newcomers must learn, beyond the conversation they read.
- **H83-R1** Conventions, not centroids: follow village-coined terms and file conventions (H34 markers, HH291) through each newcomer's first uses, with ledger reads as the light cone (H41).
- **H83-R2** The entry family signature: find which content carries it (self-description, tooling, onboarding replies) and fit its decay time per join; run C3/C4 on the holdout joins.
- **H83-R3** Call-level reading: newcomers' first calls after a room move, read vs posted-but-unread at matched lag (H57), to separate reading from a shared field in the immediate convergence (#38 rooms).
- **H83-R4** Fix the NE32 catalog entry (isolation ≈ 1.5 –2 h) and drop NE32 as an isolation experiment.

H83-enculturation-of-newcomers · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H83-enculturation-of-newcomers/README.md · ratings: H83 agent (suggested) (page 2)

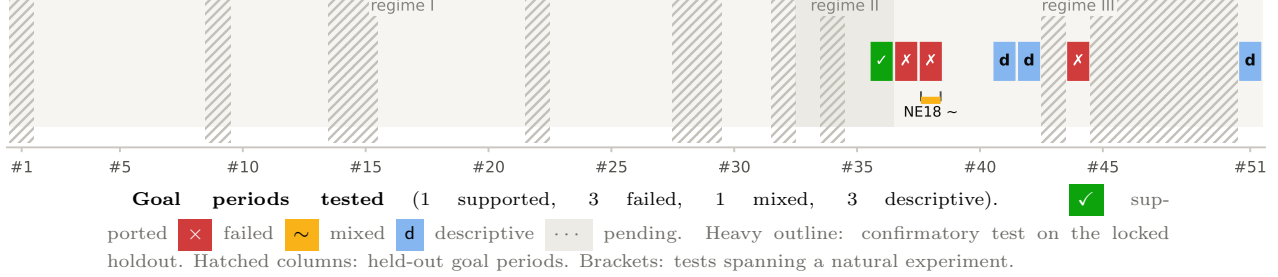
H84: History-search outage as a collective-memory scramble

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

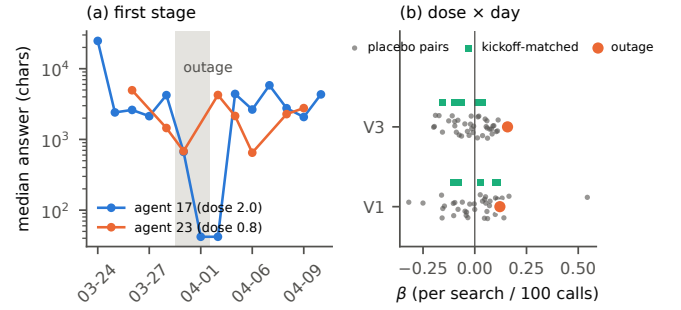
Bottom line. The history-search outage cost the scrambled agent no continuity and no output; the search channel carries about 0.01 bits of allocation information.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

On 2026-03-31 and 04-01 the history-search oracle returned near-empty answers. The test treats this as a Kolchinsky–Wolpert scramble of the village’s access to its own past. Each agent’s dose is its search rate in the five days before. The outage fully reached one agent: its answers fell from 2,880 to 42 characters, and it retried 128 times. Continuity did not dip. It rose with dose ($\beta = +0.12$ per search per 100 calls; placebo rank p for a dip 0.87), and so did references to earlier-goal artifacts. Commits per 20 calls did not move. A dip of 0.30 at the mean searcher dose is excluded (power 0.85–0.91); a dip of 0.20 is not. Search answers carry 0.012 bits about the next repo (7 periods). The holdout script exists and is not run.



Observables. (a) Median answer length per day for the two heaviest searchers (log scale). The outage (gray) collapses the top searcher’s answers to 42 characters; the second searcher’s answers halve. (b) The dose × outage slope (orange) against 38 placebo day pairs (gray) and five kickoff-matched pairs (green). Continuity (V1) and earlier-goal references (V3) move up, not down.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Semantic information, natural-scramble variant (model 04). The system X is one agent’s allocation: which repos its work commits go to. The environment Y is the logged past, reached through the search oracle (channel Q). The outage degrades Q . The dose d_a is searches per 100 calls before the outage. Viability V is continuity (share of commits on pre-existing repos), with earlier-goal references and commits per 20 calls as secondary components. O_t marks outage days and R_t recovery days. The permutation π shuffles the answer pointer S_Q within agent.

$$y_{a,t} = \alpha_a + \gamma_t + \beta d_a O_t + \beta_{\text{rec}} d_a R_t + \varepsilon_{a,t}, \quad \Delta V_Q = \beta \bar{d}_s$$

$$I_Q = \hat{I}_{\text{MM}}(X^+; S_Q) - \langle \hat{I}_{\text{MM}}(X^+; S_Q^\pi) \rangle_\pi, \quad \kappa_Q = \frac{\Delta V_Q}{I_Q}$$

Do not treat history search as load-bearing memory. Few agents use it: three of twelve searched more than once per 400 calls. When it broke, the one agent it reached retried and kept working from its own artifacts, with no loss of commits ($+0.03 \pm 0.51$ per 20 calls). An operator can let the search tool fail for a day without a measurable cost to project continuity. Invest in the artifact path instead: keep repos readable and named in the agent’s notes. Search answers name the agent’s own repo in 13% of #51 searches, and those searches preceded a return 99% of the time, but the agent chose the query.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 continuity dips $\propto$ dose	$\beta = +0.12$; rank p (dip) 0.87; above all 5 kickoff-matched	fail
P2 immediate recovery	searchers stay high on 04-02/03 ($\beta_{\text{rec}} = +0.24$)	fail
P3 duplicates up, old refs down	13 duplicates (unpowered); old refs +0.16, max of 38	fail
P4 first stage	1 of 3 searchers degraded (128 retries)	partial
P5 re-reads rise	+6.4 per 100 calls (perm p 0.048; rank 0.23)	partial
R $I_Q > 0$ in $\geq 2/3$	2/7; pooled 0.012 $[-0.011, 0.035]$ bits	fail
N2 NE18 rework helps	no first stage; $\beta = +0.055$, p 0.11	inconcl.
N3 failed answers cost	RR 1.25 $[0.52, 2.42]$ (#51, 83 failures)	not cons.

GOAL PERIODS IN DETAIL

G36	✓ supported
G37	× failed
G38	× failed
G41	d descriptive
G42	d descriptive
G44	× failed
G51	d descriptive
NE18	~ mixed (inconclusive: no first stage; power under 0.8)

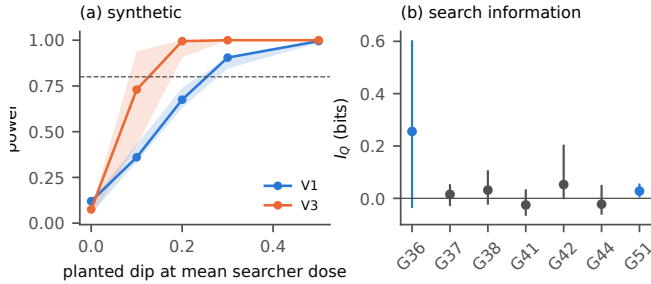
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

One agent was fully scrambled, for two days, so the bound applies to a heavy searcher, not to the village. The searchers' continuity is also high on the recovery days, so a dose $\times$ goal interaction (#37 $\rightarrow$ #38) could hide a small dip. Duplicate creation has 13 events and cannot speak. I_Q counts only repos named in answers. Failed searches in #51 are not exogenous. The single-agent residual check is post hoc. The NE18 test is oversized (0.12–0.22) and had no first stage.



Second figure. (a) Synthetic power of the placebo-rank test at real counts, against a planted dose-proportional dip (band: agent-day noise 0.3–1.0). Continuity reaches 0.8 only near a 0.30 dip. (b) Search-answer information about the next repo per period, with 95% bootstrap CIs; blue beats the within-agent permutation floor.

ROUND 2

Check NE45 (07-29, #51, 32 agents) for a search-failure burst; if present, rerun the dose design there. Run the NE25 confirmatory test after sign-off.

- **H84-R1** NE45 (2026-07-29, #51): check the search-tool schema change for a failure burst across 32 agents; if present, rerun the dose design there.
- **H84-R2** Call-scale event study in #51: runs of failed answers per agent as dose; commits and return in the next 40 calls; agent-day placebos.
- **H84-R3** Run `confirm.py` on NE25 after Vivian's sign-off (re-freeze if inputs change).
- **H84-R4** Search content beyond repos: discretize answers by the goal period they cite, to measure I_Q beyond repo naming.

H84-search-outage-memory-scramble · generated 2026-10-07 17:50 by `infra/summaries/build_summaries.py` · card hypotheses/H84-search-outage-memory-scramble/README.md · ratings: H84 agent (suggested) (page 2)

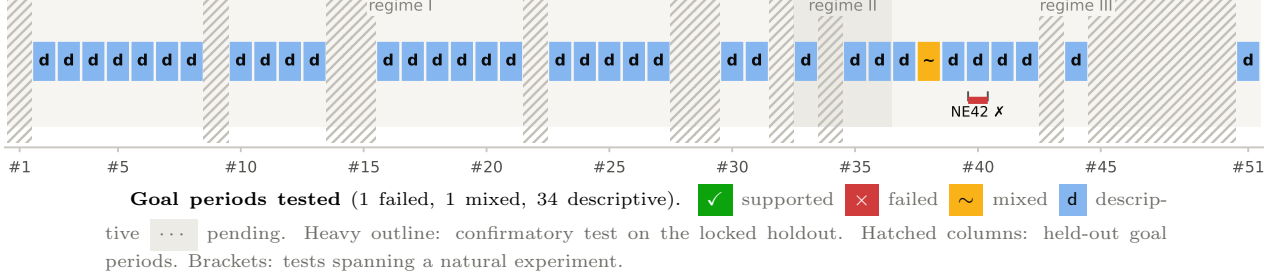
H85: Scaling of outputs with N

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Talk is a shared, nearly fixed channel, not a per-agent output: messages scale as $N^{0.33}$ while addressing per message rises as dilution predicts.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

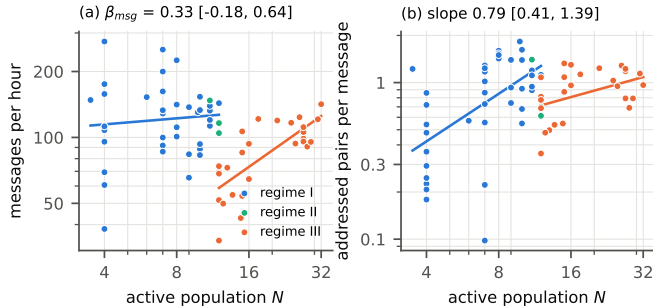
The test asks how the swarm's outputs scale with its active population N . Each of 71 non-holdout period units gives one point: output per scheduled hour against N , with regime intercepts and goal-cluster bootstrap CIs. Talk is sublinear: messages scale as $N^{0.33}$ $[-0.18, 0.64]$, and in regime I the room's talk volume is nearly fixed ($\beta = 0.09$). Addressing per message rises as $N^{0.79}$ $[0.41, 1.39]$, against 0.44 predicted from H18's dilution and the measured pending set ($k \propto N^{1.29}$). Reply parents per message do not rise (0.02), so the HH's reply kill fires. Work exponents are not identified. The holdout script exists and is not run.

PHYSICAL MODEL

Allometric scaling (model 14) with a per-call response kernel (model 02). Y_u is a unit's total output, T_u its scheduled hours, N_u its hour-weighted active population. α_r is a regime intercept and ζ_g a goal-period effect. Responses per talk call scale as $k^{1-\beta_d}$ with pending set $k \propto N^{\gamma_k}$ and H18's $\beta_d = 0.66$. DQ2 keeps one reply parent per message, which caps reply counts.

$$\ln(Y_u/T_u) = \alpha_{r(u)} + \beta_Y \ln N_u + \zeta_{g(u)} + e_u$$

$$\beta_{\text{ment}} - \beta_{\text{msg}} = (1 - \beta_d) \gamma_k, \quad \beta_{\text{reply}} - \beta_{\text{msg}} \approx (1 - \beta_d) \gamma_k \frac{\lambda(1-f)}{f}, \quad f = 1 - e^{-\lambda}$$



Observables. (a) Messages per scheduled hour against active population, one point per unit, colored by regime, with within-regime fits. Regime I is flat: more agents do not add talk. (b) Addressed agents per message against N . It rises in every regime, steepest in regime I.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not expect more talk from more agents. In regime I each agent kept its call clock, but its share of talking calls fell as $N^{-0.89}$, so the room's message rate stayed near 130 per hour (IQR 96–148) from 4 to 11 agents. In regime III talk grows as $N^{0.77}$. A message in a larger room names more agents: addressing per message grows as $N^{0.5}$ to $N^{0.8}$. Expect it, and route long rosters into rooms: within a day, the larger room talks less per agent (room exponent 0.36 in #38). Do not read reply-graph density as interaction load: reply parents per message stay flat in N (median 0.35 in regime I, 0.51 in III). Budget ± 0.5 on any output exponent fitted across goal periods.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 messages $\beta \in [0.85, 1.15]$	0.33 $[-0.18, 0.64]$; I 0.09, III 0.77	fail
P2 replies $\beta \in [1.15, 1.55]$	0.36 $[-0.46, 0.87]$; per message 0.02 vs 0.32	kill fires
P3 addressing/msg $= 0.34\gamma_k$	0.79 $[0.41, 1.39]$ vs 0.44 $[0.31, 0.52]$	pass
P4 $\gamma_k \in [0.7, 1.3]$	1.29 $[0.91, 1.52]$; rooms 1.02 $[0.88, 1.15]$	pass
P5 work $\beta = 1.0 \pm 0.1$	1.89 $[0.55, 7.76]$ (36 units)	unident.
P6 repos own vs shared	CIs span > 5	unident.
N1 NE42 merge	placebo $\times 2.5\text{--}7.5$; k ratio 1.34	fail
N2 #51 roster sweep	SD $\ln n$ 0.08; CIs span > 2	unident.
N3 #38 room contrast	msg 0.36 $[0.02, 0.82]$; ment 1.12	mixed

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 d descriptive
+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

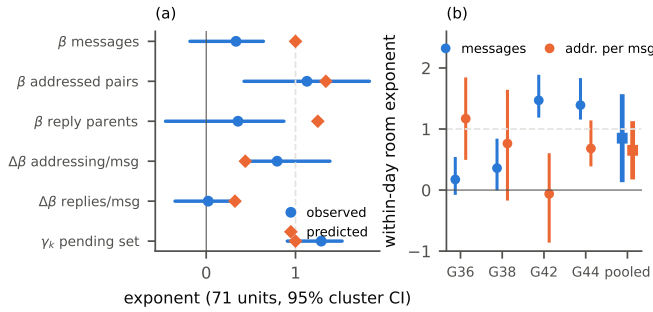
One decade of N , 35 goal clusters and regime intercepts give a level-exponent SD of 0.25 (synthetic), not the design 0.10. N grew with calendar time, so model generations ride along. Regime-I chat mode showed agents a capped recent-chat window, which may shape the talk budget. Addressed pairs count names, including co-addressing (post hoc: names per naming message rise as $N^{0.44}$ in regime I). Reply parents carry DQ2's one-parent budget. Room contrasts have no room fixed effect. NE42 and the #51 sweep could not identify a step.

ROUND 2

Fit a talk-budget model per call (talk probability against room traffic). Restrict addressing to senders with a pending message (the clean aggregate of H18). Measure work per incumbent agent across roster joins with agent fixed effects. Run the frozen holdout test after sign-off.

- **Essence.** whether the swarm produces more per agent as it grows (a collective benefit), and how much of its talk is a shared, capacity-limited channel.
- **H85-R1** Talk budget model.
- **H85-R2** Pending-sender addressing.
- **H85-R3** Work per agent across joins.
- **H85-R4** Uncapped replies.

Second figure. (a) Exponents across 71 units (blue, 95% goal-cluster CIs) against predictions (orange): the reply prediction is 1.25 after the one-parent budget. (b) Within-day room-size exponents in the four rooms-era periods where room sizes vary, with the random-effects mean (squares).



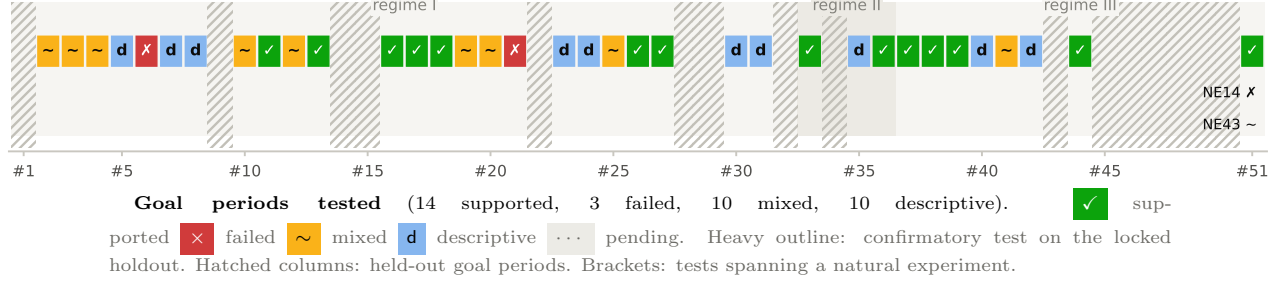
H86: Taylor's law as a shared-field gauge

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Taylor's c measures clock regularity, not the field; the pair-covariance $c_{\times}$ measures the field, which is small and mostly the scheduler.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

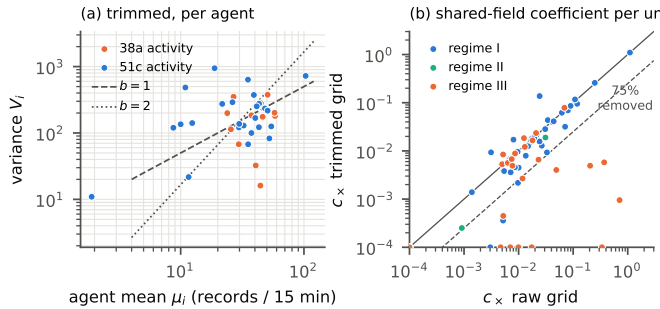
The test asks whether Taylor's law gives a one-number gauge of the shared field. Per period unit, each agent's activity is counted in 15-minute bins, on the raw day window and on DQ8's all-present (trimmed) window. The HH's coefficient c_T (variance = $a\mu + c_T\mu^2$ across agents) fails: Taylor's exponent is $b \approx 0.85$ and $c_T \leq 0$ in most of 70 units. The pair-covariance coefficient $c_{\times}$ works as a gauge. It is small (0.016 raw, 0.008 trimmed), and the shared share of excess variance is 0.08. Trimming removes 77% of it in regime III, matching H38's 70–80%. The gauge is now a per-unit column. The holdout script exists and is not run.

PHYSICAL MODEL

Fluctuation scaling (model 14) with a multiplicative field $h(t)$ (model 02). Agent i 's count in bin t is Poisson with rate $\mu_i \xi_t \zeta_{it}$. ξ_t is a field shared by all present agents (scheduler, kickoff, operator) with variance $c_{\times}$. ζ_{it} is the agent's private burstiness with variance s_i . $c_{\times}$ is read from pair covariances and needs no spread in μ . φ is the shared share of super-Poisson variance.

$$V_i = \mu_i + [c_{\times} + s_i(1 + c_{\times})]\mu_i^2, \quad \text{Cov}(y_i, y_j) = c_{\times} \mu_i \mu_j$$

$$c_{\times} = \frac{\sum_{i < j} \text{Cov}_{ij}}{\sum_{i < j} \mu_i \mu_j}, \quad \varphi = \frac{c_{\times} \sum_i \mu_i^2}{\sum_i (V_i - \mu_i)}, \quad r_{\times} = 1 - \frac{c_{\times}^{\text{trim}}}{c_{\times}^{\text{raw}}}$$



Observables. (a) Variance against mean per agent, trimmed activity, units 38a and 51c. The points do not follow $b = 2$; faster agents are more regular. (b) Shared-field coefficient per unit, raw against trimmed grid. Regime-III units fall toward the 75%-removed line; regime-I units sit on the diagonal.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Report `taylor_c_shared` on `activity_trim` (and `activity_raw`) from `per_period_estimates` next to any synchrony or influence claim. A trimmed $c_{\times}$ near 0.008 with $\varphi < 0.1$ means the swarm's activity fluctuations are about 90% private. Do not use Taylor's b or c_T as field measures in a clocked swarm: they track clock regularity. A raw $c_{\times}$ far above the trimmed one (e.g. 2.2 on 2026-03-31) flags a village-off gap inside the day window. Talk shares about 2.6 times more of its variance than activity (φ 0.21 vs 0.08).

RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction	Observed	Verdict
P1 raw $b \approx 2$, $c_T > 0$	b 0.85 [0.21, 1.21]; $c_T > 0$ in 31%	fail
P2 trimmed $b \leq 1.3$	0.82 (raw already < 1)	trivial
P3 III: $r_x \in [0.6, 0.9]$; $I < 0.4$	III 0.77 (13 units); I 0.06 (25)	pass (c_x)
P4 kill: III $\varphi_{\text{trim}} \geq 0.5$	φ 0.04; shift null $p < .05$ in 38%	not fired
P5 call clock lowers b	lower in 58% (need $\geq 67\%$)	fail
P6 commits; specificity	b 1.26; ρ 0.14 ($n = 6$)	b pass
P7 r_x vs H38 f_{scaffold}	ρ 0.19 (13 units)	fail
P8 talk $\varphi >$ activity	77% of 64 units; 0.21 vs 0.08	pass
N1 NE14 raw jump	one off-gap day; trimmed +0.001	fail
N2 NE43 operator stops	c_x unchanged; c_T no rise	mixed

A mapping 1 F identifiability 2
 B assumptions 1 G ground truth 1
 C adequacy 1 H comparative 1
 D unfitted predictions 1 I transfer 1
 E interventional 1
 Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

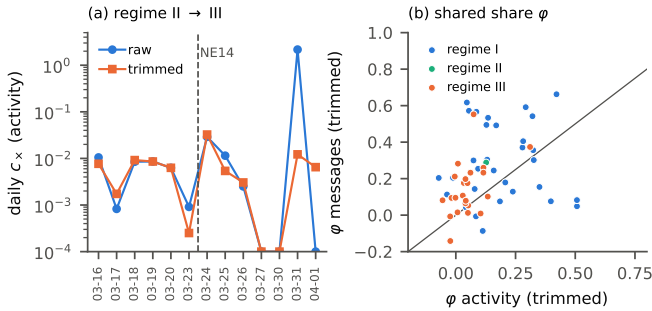
CAVEATS

Across-agent fits have 4–32 points and a narrow mean range in regime III, so unit-level b is unstable; the card uses medians. Day-bootstrap CIs under-cover (0.7–0.8 in synthetic runs). c_x counts equal-time coupling as field. Statistics depend on the 15-minute bin. The commit gauge exists in 13 units, so the kickoff-specificity test has 6 goal periods. The regularity reading of $b < 1$ is post hoc.

ROUND 2

Fit a clock-aware Taylor law (renewal model of each agent's calls) and report the residual shared term. Split c_x into equal-time and one-call-lagged parts on the call clock, which separates field from read-out coupling. Use raw minus trimmed c_x as a daily off-gap flag. Run the frozen holdout test after sign-off.

- **Essence.** one number per unit that says how much of the swarm's fluctuation is shared, after the scheduler is removed.
- **H86-R1** Clock-aware Taylor law.
- **H86-R2** Lagged vs equal-time c_x .
- **H86-R3** Off-gap detector.
- **H86-R4** Units 6a and 21b.



Second figure. (a) Daily c_x across NE14 (regime II to III). Raw and trimmed agree except on 2026-03-31, whose window holds a 513-minute all-silent gap. (b) Shared share φ on trimmed grids, messages against activity, one point per unit: talk shares more.

GOAL PERIODS IN DETAIL

G02 ~ mixed
 G03 ~ mixed
 G04 ~ mixed
 G05 d descriptive
 G06 x failed
 G07 d descriptive
 G08 d descriptive
 G10 ~ mixed
 G11 ✓ supported
 G12 ~ mixed

+27 more periods: see the card.

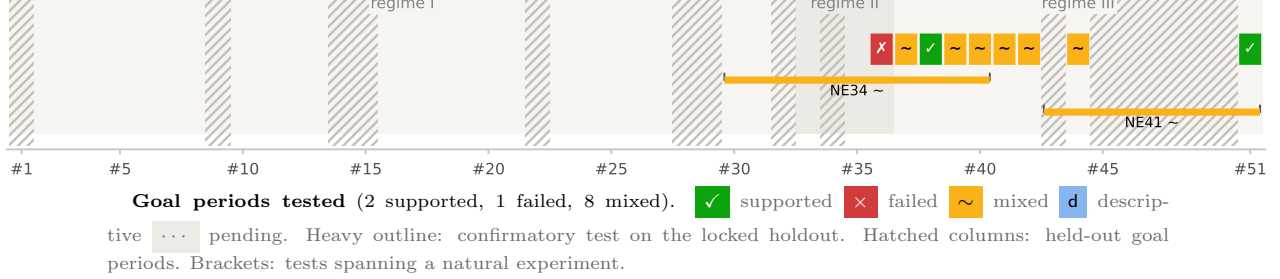
H87: κ table: commits per bit by channel

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2.5/5

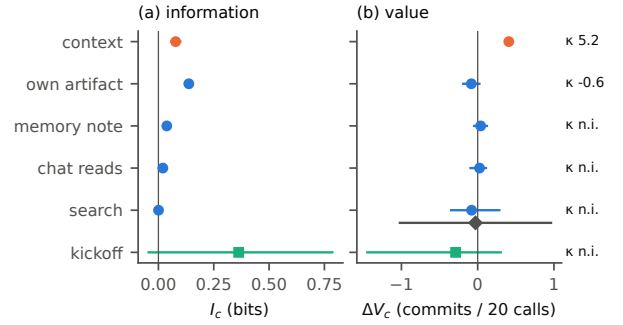
Bottom line. Of seven information channels, only the context window buys commits per bit ($\kappa_C = 5.2$); the artifact carries more bits but no measurable value.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test builds the Kolchinsky–Wolpert channel table: for each information channel, the allocation bits it carries (I_c) and the commits lost when it is naturally scrambled (ΔV_c). The ratio κ_c is commits per 20 calls per bit. The main design pools 18,760 non-holdout forced erasures against matched pseudo-erasures. Only the context window has value: $\kappa_C = 5.2$ [3.8, 7.9]. Its erasure costs 41% of output for 0.08 bits. The own artifact carries the most bits (0.14) but no value: $\kappa_A = -0.6$ [-1.6, +0.3]. Memory, chat, search and human messages carry ≤ 0.04 bits each, with value consistent with 0. The kickoff carries 0.36 bits and costs nothing. The predicted order (artifact first) fails. Receiving chat after an erasure raises output 22% without allocation bits. The holdout script exists and is not run.



Observables. The κ table at forced erasures. (a) Allocation information per channel, bits, 95% paired-bootstrap CIs. Kickoff (green square) is the day-scale goal change. (b) Commits per 20 calls lost when the channel is scrambled. Context (orange) is the erasure cost; the others are open $\times$ erasure differences. The gray diamond is the search outage (H84). Only context has a value whose CI excludes 0.

PHYSICAL MODEL

Semantic information, natural-scramble variant (model 04); plug-in estimators (model 12). The system X^+ is an agent's next allocation: the repo of its first work commit in 20 calls. Each channel c has a pointer S_c , the repo it names. It is open (o_c) when it names the agent's own artifact. The scramble s is a forced erasure. A pseudo-erasure at call 21 of a long segment is the placebo. V counts work commits in 20 calls; the Poisson fit also adjusts for $\log(1+V_{\text{pre}})$, commits in the 20 calls before. Strata g are agent $\times$ period. For the context row, $I_C = I_{s=0} - I_{s=1}$ and ΔV_C is the erasure cost; κ is defined only where the CI of I_c lies above 0.02 bits.

$$I_c = \hat{I}_{\text{MM}}(X^+; S_c)|_{s=1} - \langle \hat{I}_{\text{MM}}(X^+; S_c^\pi) \rangle_{\pi \in g}, \quad \kappa_c = \Delta V_c / I_c$$

$$\log E[V] = \phi_{g,s} + b_1 o_c + b_2 o_c s + \dots, \quad \Delta V_c = \bar{V}_{o,s}(1 - e^{-b_2})$$

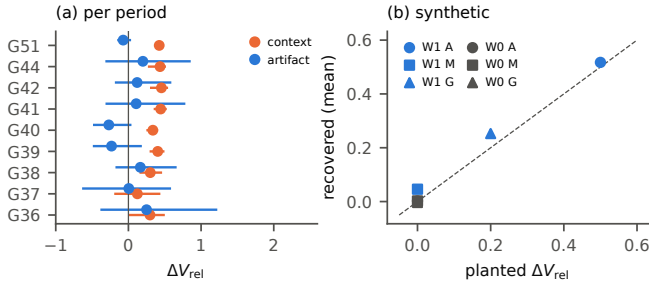
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Protect the context window, not the notes. A forced erasure costs 41% of the next 20 calls' commits, 5.2 commits per 20 calls per bit removed. Lengthening the 41-call cap is the lever with a measured payoff. Do not expect memory notes, larger memories, history search or repo-naming chat to buy it back: none shows value above 0. The own repo is the best pointer (0.14 bits; return 0.96 when re-read), but re-reading after an erasure gains nothing beyond what it gains at any call. Chat after an erasure does raise output (+22%), so a short status message to a freshly erased agent is a cheap candidate lever. It is untested as an intervention.

H87-kappa-channel-table · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H87-kappa-channel-table/README.md · ratings: H87 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\kappa_A > \kappa_C > \kappa_G > \kappa_K > \kappa_M$	$\kappa_C 5.2 \gg \kappa_A -0.6$ ($P = 1.00$); rest n.i., $\Delta V \approx 0$	fail
P2 $\kappa_C = 5.2 \pm 20\%$	5.24 [3.79, 7.88]; cost 41%	pass
P3 chat κ_G CI $\ni 0$	0.020 bits; $\Delta V_{\text{rel}} +0.04$ [-0.17, +0.25]	pass
P4 human row unpowered	no repo-naming item; dose +0.09 [-0.20, +0.40]	pass
P5 κ_Q CI $\ni 0$	0.012 bits; outage -0.03 ± 1.0 ; partial n.i.	partial
P6 $I_K \geq 0.2$, $\kappa_K \ni 0$	0.36 [-0.05, 0.79]; κ_K CI $\ni 0$	partial
P7 memory size null	+0.01 [-0.06, +0.09]	pass
P8 no V40 sign flip	none; artifact -0.19 [-0.25, -0.14]	pass
R context first, $\geq 2/3$	2/9 with I_C identified; cost > 0 in 7/9	fail



Second figure. (a) Per period: the erasure cost (orange) is 0.30–0.45 of output in 7/9 periods; the artifact’s open $\times$ erasure value (blue) straddles 0. (b) Synthetic at real counts: recovered vs planted ΔV_{rel} for three rows in a world with value (W1) and one without (W0).

GOAL PERIODS IN DETAIL

G36 $\times$ failed
 G37 $\sim$ mixed (inconclusive: κ_Q not identified)
 G38 $\checkmark$ supported
 G39 $\sim$ mixed
 G40 $\sim$ mixed
 G41 $\sim$ mixed
 G42 $\sim$ mixed
 G44 $\sim$ mixed
 G51 $\checkmark$ supported
 NE34 $\sim$ mixed

+1 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

κ is a ratio of two noisy numbers. It is defined for context and artifact pooled, and for context alone in only 2/9 periods. Allocation is coarse-grained to the repo, so chat and human messages score near 0 bits by construction. Open flags are chosen by the agent. Search (46 open erasures) and human rows are unpowered. The chat row reproduces H70’s room row. The per-period rule was clarified after the run; both counts are reported. $\eta = S/I$ is not identified: natural scrambles give one point, not the information–viability curve.

ROUND 2

Use a finer allocation variable (next behavior state or file) so chat can carry bits. Split the chat-after-erasure gain by item type with an in-flight placebo. Trace an information–viability curve with graded scrambles.

- **H87-R1** Finer X: the next behavior state (DQ3, 11 states) or the next file within the repo, so chat and human rows can carry measurable bits.
- **H87-R2** The chat-after-erasure value (+22%): split by item kind (mention of the agent, question, status), with H57’s in-flight placebo, and test whether it is a kick (activity) or information (content).
- **H87-R3** Graded scrambles for η : use erasure position within long segments and memory-size quantiles as a dose to trace an information/viability curve.
- **H87-R4** Move H70’s event builder and H87’s pointer builder into infra/shared/ (STANDARDS 8), so confirm.py runs without reading another hypothesis’s data folder.

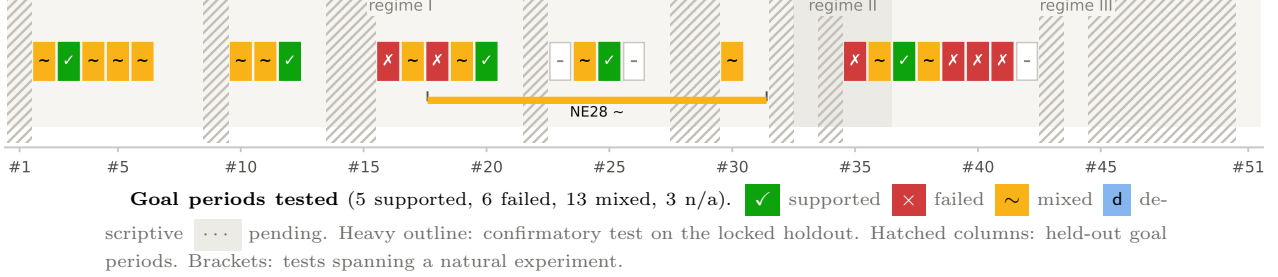
H88: Collective memory decays biexponentially

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2/5

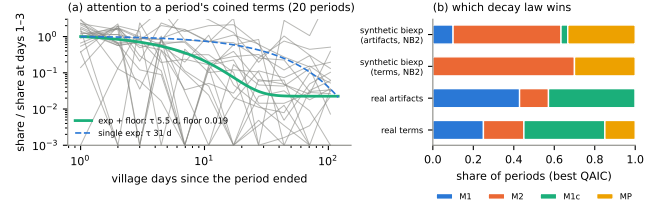
Bottom line. Attention to a finished period falls within a week to a small floor, carried alike by veterans and newcomers, not as a two-compartment biexponential.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test follows the village's attention to a finished goal period's items: its coined terms (H34 markers first seen there) and its artifacts. The observable is the veterans' daily share of uses for 120 village days after the period, with held-out days censored. Four decay laws are fitted per period. Candia's biexponential wins in 5 of 27 period-by-channel fits. An exponential with a floor wins in 11, the single exponential in 8. For terms, attention falls with $\tau = 5.5$ village days to a floor of 1.9% of the initial share (20 periods). The decay holds among veterans who stay, and with a closed roster. Newcomers are not slower than veterans (power 0.87–0.92). In 2 of 3 retirements, the retiree's items were never used again. The holdout script is not run.



Observables. (a) Veterans' share of uses of a period's terms, relative to days 1–3, for 20 periods (gray). The green curve is the median exponential-plus-floor fit; the dashed curve is a single exponential with the median τ . (b) Share of periods won by each decay law, real data against a synthetic biexponential truth at the real offsets.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A finished goal is gone from the village's attention within about a week: the share drops to 8% of its first-days value by days 21–40. If an operator wants old work reused, it must be re-injected (named in a kickoff or a file), not left to memory. Newcomers inherit a period's live topics as fast as veterans, so there is no "veterans' memory" to protect when rotating agents. Items used mostly by one agent tend to die when it retires (2 of 3 cases): before retiring an agent, hand its live repos and terms to another carrier. The persistent floor (1.9%) mixes the record with terms that models re-coin.

PHYSICAL MODEL

Collective memory (model 13, §4). In Candia's two-compartment model, communicative memory u decays at rate p and feeds cultural memory v at rate r ; v decays at rate q . Here k counts village days since the period ended. $y_{g,k}$ counts uses of the period's items by carrier group g (veterans or newcomers); $O_{g,k}$ counts the group's uses of all items of that kind. The share y/O is fitted with a single exponential, a biexponential, an exponential plus floor c , and a power law.

$$S(k) = Ne^{-(p+r)k} + \frac{Nr}{p+r-q}(e^{-qk} - e^{-(p+r)k}), \quad \tau_1 = \frac{1}{p+r}, \quad \tau_2 = \frac{1}{q}$$

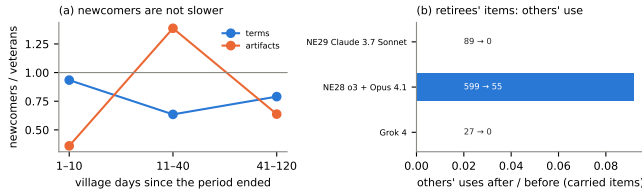
$$\mathbb{E}[y_{g,k}] = O_{g,k} f(k), \quad f \in \{Ae^{-k/\tau}, A_1e^{-k/\tau_1} + A_2e^{-k/\tau_2}, Ae^{-k/\tau} + c, Ak^{-\alpha}\}$$

$$R(\text{band}) = \sum_{P,k} y_{\text{new}} / \sum_{P,k} O_{\text{new}} (y_{\text{vet}} / O_{\text{vet}})$$

H88: collective memory-decay · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H88-collective-memory-decay/README.md · ratings: H88 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 biexponential modal	terms: floor 8, single 5, biexp 4, power 3 (of 20); artifacts: single 3, floor 3, biexp 1	fail
Kill 1: single exp $\geq 1/2$	5/20 terms, 3/7 artifacts	not met
P2 decay among present veterans	late/early share 0.079 (median); 13/18 CI < 1	pass
P3 newcomers only slow	R 0.93, 0.64, 0.79 by age; b $-0.17 [-0.80, +1.32]$	fail
P4 artifacts slower than terms	1/6 periods	fail
P5 newcomer uses after a read	59% in the logged cone	pass (descr.)
N1 closed roster biexp	0/3 (power law, power law, floor)	mixed
N2 NE27 newcomers slower	$R(\leq 15\text{ d})$ 0.98 [0.0, 2.7]	mixed
N3 carrier loss	Grok 4: 27 $\rightarrow$ 0; Sonnet 3.7: 89 $\rightarrow$ 0; pooled CI 0.01–2.2	mixed



Second figure. (a) Newcomers' uses over the uses expected at the veterans' share on the same day, by days since the period ended. No rise with age: newcomers carry the fast part too. (b) Other agents' later use of items carried by a retiree.

GOAL PERIODS IN DETAIL

G02 $\sim$ mixed
G03 $\checkmark$ supported
G04 $\sim$ mixed
G05 $\sim$ mixed
G06 $\sim$ mixed
G10 $\sim$ mixed
G11 $\sim$ mixed
G12 $\checkmark$ supported
G16 $\times$ failed
G17 $\sim$ mixed

+17 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Candia et al. 2019 is not in the literature folder; its form is quoted from memory. A 120-day horizon with censored stretches cannot separate a months-long τ_2 from a floor. Terms are hashed markers; 41% of newcomers' first uses come without a logged read, so re-coinage inflates the floor. Artifact mentions count working references. Overdispersion is large (ϕ 1–100). The NE27 shares use the pool of #2–#8 items. Reading the carrier-loss CI as uninformative is post hoc.

ROUND 2

Fit item-level survival with a cured fraction, with covariates: hardcoded in a repo, carried by one agent, named by a later kickoff. Test the floor across the history-search changes (NE04, NE18).

- **Essence.** which items outlive their period and what carries them (an artifact, a convention, one agent).
- **H88-R1** Item-level survival: time to last use per term and artifact, with covariates (hardcoded in a repo, carried by one agent, named by a later kickoff, in-cone share).
- **H88-R2** Carrier concentration: item survival vs the share of its uses by its top carrier (HH291's committed-carrier argument); the two clean retirements suggest single-carrier items die with their carrier.
- **H88-R3** Record access: test the floor across NE04 (history search added) and NE18 (verbatim 10-day search window) on items older than 10 village days.
- **H88-R4** Separate re-coinage from memory: family-matched control villages are not available, so use in-cone first uses only for the newcomer channel.

H88-collective-memory-decay · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H88-collective-memory-decay/README.md · ratings: H88 agent (suggested) (page 2)

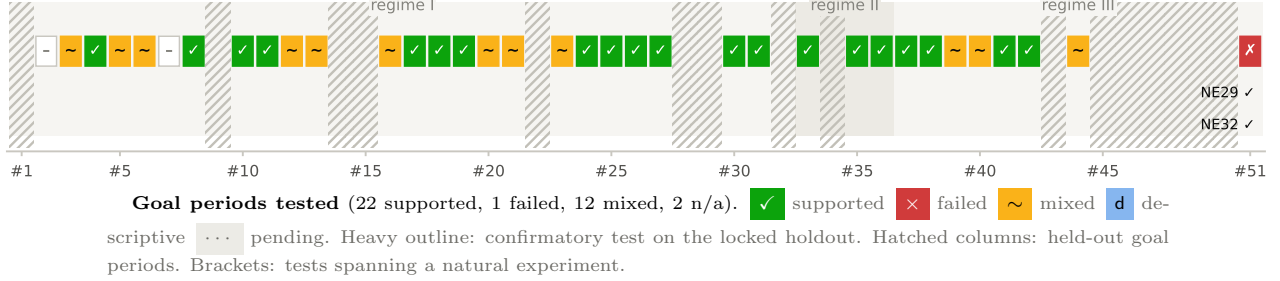
H89: Price equation for village culture

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

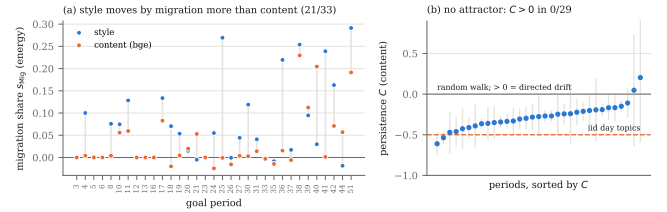
Bottom line. Content and conventions move by transmission and style partly by migration, but selection is undetectable and the remainder is day-topic swing, not a village attractor.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test splits the day-to-day change of the village's mean trait into selection, transmission and migration with a Price equation. Traits are content (DQ5 vectors, style removed), style (H13's 20 features) and convention use. Fitness counts in-cone adopters from the context ledger. Over 33 non-holdout periods (245 day transitions), stayers' own change carries the content change: median transmission share 0.98 (33/33). Selection carries 0.01 and never beats its permutation null. Style moves by migration more than content in 21 of 26 periods with turnover, but its migration share is only 0.04 (max 0.29). The kickoff direction carries 0.04, so the kill does not fire. The remainder swings day to day with no attractor (persistence $C > 0$ in 0/29). NE29 and NE32 pass. #51's eleven joins do not move its net style. The holdout script is not run.



Observables. (a) Migration share of the day-to-day change per goal period, for style (blue) and content (orange). Style sits above content in 21 of the 26 periods with any turnover. Seven periods have no turnover, so both shares are 0. (b) Persistence C of the non-kickoff content change, with 95% jackknife CIs. Independent daily topics give -0.5 . A village attractor needs $C > 0$, which no period reaches.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor of collective culture: within a goal period, the village's mean content follows what the stayers do each day, not who is present and not who is influential. Read a content shift as a day field acting on everyone. Influential agents do not pull the mean ($|s_{\text{sel}}| \approx 0.01$). Style is the roster-sensitive trait: it gets about ten times content's migration share (0.044 against 0.004). A style change on a day with a join or exit is therefore mostly composition. The split cannot tell copying from a common field at day resolution. Do not use it as evidence that agents copy each other. It is a bookkeeping identity with honest error bars. It is not a lever.

PHYSICAL MODEL

Cultural Price equation (model 13) with in-cone parentage. A trait $z_i(d)$ is agent i 's day- d mean. Generations are consecutive active days. Stayers S , leavers E and entrants I define the open population. Stayer j inherits $1 - \lambda_j$ from itself and λ_j from its in-cone sources. λ_j is the share of its first uses that are in-cone adoptions. Fitness w_i sums the parentage weights, $\bar{w} = 1$. Shares s_X are cross-fitted between message halves A and B (symmetrized). C is the cross-fitted cosine of consecutive non-kickoff changes.

$$\Delta \bar{z} = \text{Sel} + \text{Trans} + \text{Mig}, \quad \text{Sel} = \text{Cov}_S(w, z), \quad \text{Trans} = \text{E}_S(w \Delta z)$$

$$\text{Mig} = \frac{1}{|A_d|} \sum_I (z_j - \bar{z}_S) - \frac{1}{|A_d|} \sum_E (z_i - \bar{z}_S)$$

$$w_i = \sum_{j \in S} \alpha_{ji}, \quad \alpha_{jj} = 1 - \lambda_j$$

$$s_X = \sum_t \langle X_t^A, \Delta_t^B \rangle_{\text{sym}} / \sum_t \langle \Delta_t^A, \Delta_t^B \rangle$$

H89-price-equation-culture · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H89-price-equation-culture/README.md · ratings: H89 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1a style migration > content	21/33 (p 0.08); ties excluded 21/26	fail (narrow)
P1b migration largest for style	0/33; median s_{Mig} 0.044	fail
P2 content: transmission ≥ 0.5	33/33, both models; median 0.98	pass
P3 $ s_{\text{Sel}} \leq 0.2$	33/33 for each trait; median 0.01–0.03	pass
P4 selection beyond null	1–3 of 33 per trait (chance 1.7)	descr.
P5 kill (R CI $\ni 0$)	R 0.92, CI > 0 in 33/33; kickoff 0.04	not killed
P6 attractor $C > 0$	0/29; median -0.28	fail
N1 NE29 retirement	style 0.16/0.03 vs content 0.05/0.00	pass
N2 NE32 newcomers	entry style 0.08 vs 0.02; content cos +0.26	pass
N3 #51 growth	style roster share -0.14 [$-0.34, +0.05$]	fail

GOAL PERIODS IN DETAIL

G03 $\sim$ mixed
G04 $\checkmark$ supported
G05 $\sim$ mixed
G06 $\sim$ mixed
G08 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\checkmark$ supported
G12 $\sim$ mixed
G13 $\sim$ mixed
G16 $\sim$ mixed

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

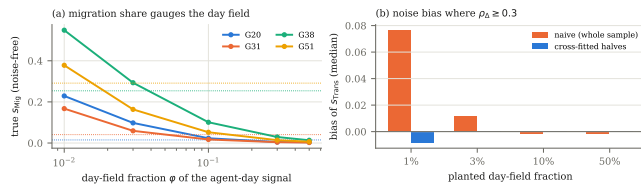
A day is a coarse generation. Presence turnover makes up most of the migration share; roster events are rare inside periods. The fitness spread is small ($\bar{\lambda} = 0.28$), so selection in a 20–32-dimensional trait is bounded by design, and no valid test has power for it. The synthetic copy scenario gives the same shares as a pure field. The unread placebo has a median of 8 adoptions per period. Cumulative shares are not causal: with persistent fitness, selection and transmission accumulate and cancel. N2 rests on one newcomer across the merge, and the gte model does not reproduce its content cosine.

ROUND 2

Measure heritability directly: regress each adopter's next-day trait on its in-cone parents beyond the day mean, with unread parents as the placebo. Run the split on single axes and across goal switches with roster change.

- **H89-R1** (separate copying from fields): a parent–offspring regression per transition: does j's day-d' trait move toward its in-cone parents' day-d traits beyond the day mean, with unread parents as the placebo?
- **H89-R2** (per-axis Price): run the partition on single interpretable axes (the goal.fields directions, H13 style features one at a time, single conventions) so selection is not diluted by 20–32 dimensions of day field.
- **H89-R3** (cross-boundary Price, exception (c)): run the split across goal switches with roster changes, where migration is large.
- **H89-R4** (a valid selection null): permute agent labels for the whole period (fitness trajectories kept), so the persistent-fitness inflation of the magnitude test goes away.

H89-price-equation-culture · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H89-price-equation-culture/README.md · ratings: H89 agent (suggested) (page 2)



Second figure. Synthetic at real counts. (a) The noise-free migration share falls steeply with the day-field fraction ϕ . Dotted lines are the observed style shares. They imply $\phi \approx 0.07$ for style and 0.32 for content. (b) Without cross-fitting, sampling noise inflates transmission: by +0.33 to +0.87 with no day field ($\phi = 0$), +0.05 to +0.23 at $\phi = 0.01$, < 0.01 at $\phi \geq 0.1$. The bars are period medians for $\phi \geq 0.01$. Cross-fitting removes the bias where $\rho_{\Delta} \geq 0.3$.

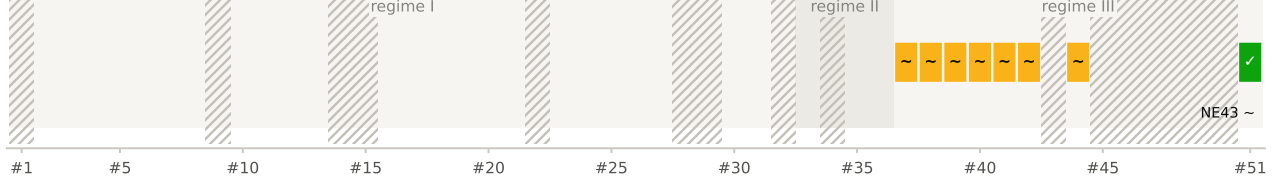
H90: Collective entropy production beyond the parts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. The swarm's only measurable collective arrow of time is agents talking in the minute after the agents who named them talked; it is coupling between named pairs, not a new collective variable, and only a 45-day period can resolve it.

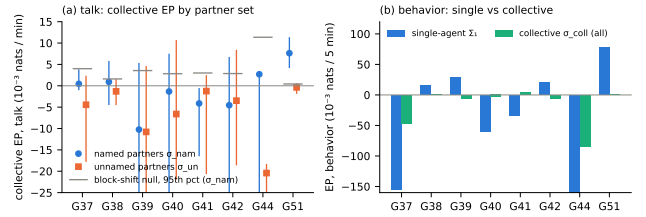
level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 supported, 8 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

I measured whether the swarm's joint dynamics are more irreversible than its agents' own dynamics. The estimator is the Aguilera-Ito-Kolchinsky max-ent bound on DQ8-trimmed windows, in three channels (5-min behavior states, 1-min talk, 1-min activity), on eight regime-III periods. Synthetic validation came first. In G51 (45 days, 21–32 agents) the talk channel carries a collective arrow of time: $\sigma_{\text{nam}} = 7.6 [4.3, 11.2] \times 10^{-3}$ nats/min on named partners, none on unnamed partners (block-shift $p = 0.01$). Single-agent talk is reversible, so this arrow is fully collective. Behavior is irreversible per agent (0.078 nats per 5 min) with a collective share of 0.003. The seven smaller periods are underpowered. Pairwise directions do not follow who names whom. Holdout (#51 tail): not run.



Observables. (a) Talk collective EP per period with the named-partner (blue) and unnamed-partner (orange) fields, day-bootstrap 95% CIs, and the block-shift null's 95th percentile (grey). Only G51 clears the null, on named partners only. (b) Behavior: single-agent bound Σ_1 (blue) and collective term (green). G51's single-agent arrow is clear; its collective term is 2×10^{-4} . G37 and G44 bars are clipped below the floor.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Two uses, both modest. First, a monitor that wants to detect agents coordinating should look at talk timing between agents that name each other: that is where the swarm's only joint arrow of time sits, at about 0.017 nats per agent-hour in #51. Second, a method warning: a block-shift or cross-day null does not separate coupling from a common drive with agent-specific lags; split partners by addressing instead. The detector needs about 40 days and 25 agents. On a 5-day, 15-agent week it has no power (power ≤ 0.05 at the #51 effect size). There is no egregore here: removing the naming partition removes the collective term.

PHYSICAL MODEL

Models 15 (EP inference) and 02 (kinetic spins with asymmetric couplings). Spins: an agent's behavior state (5 soft states) or its talk or activity in a minute. Single observables u_i : one agent's time-antisymmetric patterns. Coupling observables g_i : agent i 's state against a partner field h_i , the mean over agents who named i (named), never named either way (unnamed), or all others. Σ_g : the max-ent lower bound on entropy production for observables g ; θ : its multipliers.

$$\begin{aligned} \Sigma &\geq \Sigma_g = \max_{\theta} [\theta^\top \langle g \rangle - \ln \langle e^{-\theta^\top g} \rangle] \\ \Sigma_g &\approx 2 \langle g \rangle^\top K^{-1} \langle g \rangle \quad (\text{held out by day fold}) \\ g_i(t) &= s_i(t+1) h_i(t) - s_i(t) h_i(t+1) \\ h_i^{\text{nam}} &= \sum_j w_{ji} s_j / \sum_j w_{ji} \quad (w_{ji} : j \text{ names } i) \\ \sigma_{\text{coll}} &= \Sigma_{u \cup g} - \Sigma_u \geq 0, \quad \Delta_{\text{addr}} = \sigma_{\text{nam}} - \sigma_{\text{un}} \end{aligned}$$

H90-collective-entropy-production · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H90-collective-entropy-production/README.md · ratings: H90 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\rho_{\text{coll}} < 0.1$ (behavior, activity)	G51 behavior 0.003; activity $\Sigma \approx 10^{-4}$	supported
P2 talk $\sigma_{\text{coll}} > \text{shift null}$	G51 $p = 0.01$; 0/7 elsewhere	G51 only
P3 $\Delta_{\text{addr}} > 0$ (coupling test)	G51 $8.0 [5.1, 12.1] \times 10^{-3}$, $p = 0.01$; G38 $p = 0.08$	supported where powered
P4 rank vs H50 J_1	Spearman -0.29 ($n = 8$)	failed
P5 $\Sigma_1 \approx \sum_i \sigma_i$	gap ≤ 0.033 nats/step	supported
N1 G38 same room $>$ cross room	$+1.6$ ($p 0.03$) vs -0.7 ($p 0.66$) $\times 10^{-3}$	supported (weak)
N2 G51 pair θ follows naming	agreement 0.52 (null 0.55)	failed
N3 NE43 nudger off	σ_{coll} in placebo range; $\sigma_{\text{nam}} \times 0.80$	mixed

GOAL PERIODS IN DETAIL

G37	$\sim$ mixed (underpowered at the G51 effect size; "failed")
...	...
G38	$\sim$ mixed (replication underpowered; native N1 rooms: sup...)
G39	$\sim$ mixed (underpowered at the G51 effect size; "failed")
...	...
G40	$\sim$ mixed (no address contrast; one single-channel be-havi...)
G41	$\sim$ mixed (no address contrast; one single-channel be-havi...)
G42	$\sim$ mixed (underpowered at the G51 effect size; "failed")
...	...
G44	$\sim$ mixed (underpowered at the G51 effect size; "failed")
...	...
G51	$\checkmark$ supported (native N2 in this folder: failed)
NE43	$\sim$ mixed (σ_{coll} (all) inside the placebo range; σ_{nam} fa...)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

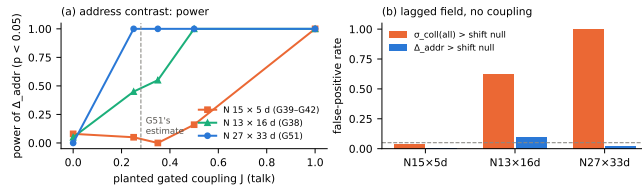
CAVEATS

One period carries the result. The 1-min talk bins put most read-out responses in the same bin as their cause, so the bound misses them and reads low. Naming uses any roster mention, not the leading-@ target. The block-shift null fails against lagged fields, so only the address contrast is a coupling claim. The estimator was fixed after a numerical failure on real behavior data (A2), then re-validated. The power reading of the small periods is post hoc (A3). No read vs unread split of the naming messages.

ROUND 2

Rebuild talk spins on each agent's call clock with the read-out call as origin, which should make 5-day periods testable. Split σ_{nam} by read vs posted-but-unread naming messages. Run `confirm.py` on the #51 tail after registering H90 in the holdout ledger.

- **Essence.** whether the swarm has collective order that no agent carries alone, measured as irreversibility, with the fields removed.
- **H90-R1** Call-clock talk spins.
- **H90-R2** Read vs posted-but-unread partner fields.
- **H90-R3** Leading-@ targets.
- **H90-R4** Hierarchical pooling across 5-day periods (exception (d)): a shrinkage estimate of Δ_{addr} per period, reported next to the per-period values.
- **H90-R5** Run `confirm.py` after registering H90 in the holdout ledger.



Second figure. Synthetic agents at the three real sizes with soft labels. (a) Power of the address contrast against planted gated coupling J ; the dashed line is G51's estimate ($J \approx 0.28$). (b) A common drive with agent-specific lags and no coupling: the collective term beats the block-shift null in 63–100% of large worlds, the address contrast in at most 10%.

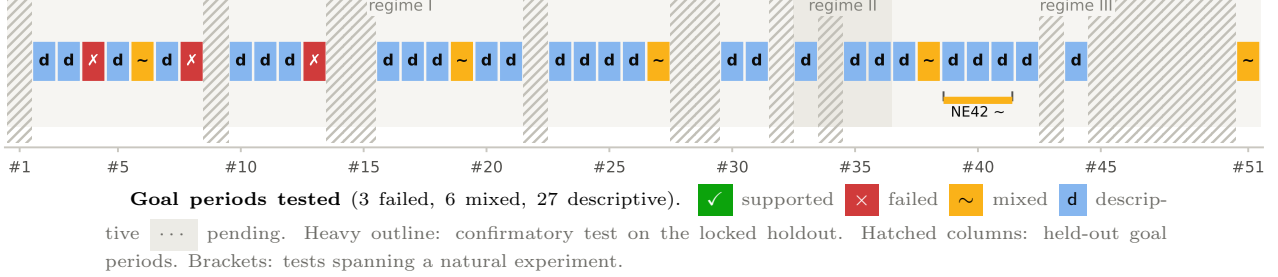
H91: Eigenvector rotation as a reorganization signal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1/5

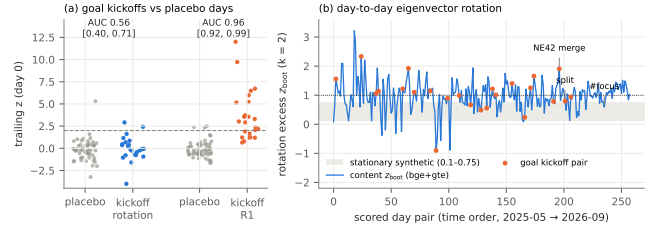
Bottom line. Collective-mode rotation does not mark goal changes; the content modes drift slowly every day, and only a room merge moves them beyond noise.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test asks whether the collective modes of the agent content matrix turn at reorganizations. For each pair of consecutive active days it measures how far the top two eigenvectors rotate, relative to the finite-sample noise of each day. It runs on 258 non-holdout day pairs in content (both embedding models) and 237 in talk. The rotation alarm does not flag goal kickoffs: AUC 0.56 [0.40, 0.71] on 23 kickoffs, against 0.96 for H36's centroid shift R1. It catches none of R1's 13 misses. Between events the content modes stay within the noise on 86% of quiet pairs. But every day rotates about one standard deviation more than a stationary swarm, with no spike at events. The NE42 room merge is the one event where talk and content both rotate ($p \leq 0.01$). The holdout script exists and is not run.



Observables. (a) Alarm scores on day 0 of 23–24 goal kickoffs (color) and on 51 placebo days (gray). The rotation alarm overlaps the placebos; R1 separates them. (b) Rotation excess z for every scored day pair in time order. It sits near 1 throughout, above the band of stationary synthetic swarms (gray). Kickoffs (orange) do not stand out. The NE42 merge is the largest content excess among room events.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Inverse Ising and vector spins (models 01, 11) with random-matrix theory. A content spin is an agent's 30-min window vector, centered on its own day. The day overlap matrix Q_d has eigenvectors v_k : its collective modes. v_1 is the uniform mode. The rotation δ_2 is the subspace distance between two days' top-2 modes. The null is finite- T eigenvector diffusion (the Dyson picture), estimated by block-bootstrapping each day. A is a trailing alarm in units of the recent spread.

$$Q_{ij}(d) = \frac{1}{W_d D} \sum_w \mathbf{z}_i(w) \cdot \mathbf{z}_j(w), \quad Q v_k = \lambda_k v_k$$

$$\delta_2(a, b) = \left[1 - \frac{1}{2} \|V_a^\top V_b\|_F^2 \right]^{1/2}, \quad z = \frac{\delta_2 - \langle \delta_2 \rangle_{\text{boot}}}{\sigma_{\text{boot}}}$$

$$\langle \sin^2 \theta_1 \rangle_{\text{Dyson}} \approx \frac{2}{T} \sum_{j>1} \frac{\lambda_1 \lambda_j}{(\lambda_1 - \lambda_j)^2}, \quad A(d) = \frac{z(d) - \text{med}_{10} z}{\text{sd}_{10} z}$$

H91-eigenvector-rotation-signal · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H91-eigenvector-rotation-signal/README.md · ratings: H91 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 synthetic size and power	size ≤ 0.05 ; power 0.2–0.4 at N 8–16, W 8; ≥ 0.85 at W 16	partial
P1 kickoff AUC ≥ 0.70 , not worse than R1	0.56 [0.40, 0.71]; R1 0.96; Δ -0.41 $[-0.56, -0.27]$	fail
P2 Dyson between events, $s_{\text{sig}} \leq 0.15$	content 0.14 (bge 0.20, gte 0.08); talk 0.17 (n 6)	content pass
P3 talk $z \geq 2$ at NE42	merge 1.97 (p 0.005); split 1.47 (p 0.025)	fail (threshold)
P4 complement, $\rho(A, R1) < 0.3$	0.21	pass
P5 model robustness	bge–gte ρ 0.33; style variant AUC +0.09	fail
P6 not composition	$\rho(z, \Delta N)$ 0.01–0.05	pass
N1 NE42 A-B-A	merge rotates talk and content; split weaker	mixed
N2 #51 baseline, #focus	content within noise (0.13); talk 0.40; #focus silent	mixed

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\times$ failed
G05 d descriptive
G06 $\sim$ mixed
G07 d descriptive
G08 $\times$ failed
G10 d descriptive
G11 d descriptive
G12 d descriptive
+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

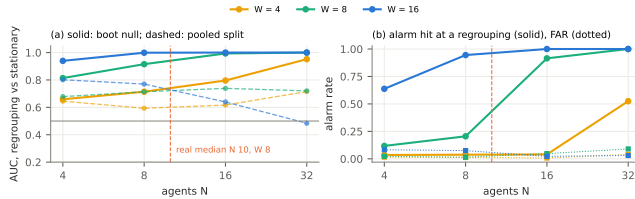
Most kickoff pairs are small ($N \approx 10$, 8 windows), where a strong regrouping is caught only 20–40% of the time. No kickoff fell in the powered range, so P1 cannot rule out rotation at strong regroupings. The null was changed once, after the synthetic study and before real data. Per-day rotation agrees only weakly between embedding models. The talk null is anti-conservative at $N \leq 8$. 27 of 35 periods have too few quiet pairs for a per-period verdict.

ROUND 2

Test rotation at operator room assignments and in 8-h, $N \geq 16$ periods (held-out #45–#50). Fit the daily drift as a rotational diffusion constant per period and place it on the phase diagram.

- **Essence.** whether the swarm’s co-movement structure has events at all, or only a slow daily drift that a monitor must treat as its baseline.
- **H91-R1** Test rotation where it is powered: operator room assignments (DQ6) and 8-h, $N \geq 16$ periods (#51 era, held-out #45–#50), not goal kickoffs.
- **H91-R2** Time the NE42 merge rotation with within-day halves, to see whether the modes turn at the merge hour.
- **H91-R3** Fit the daily drift ($z \approx 1$) as a rotational diffusion constant per period and use it as a phase-diagram coordinate.

H91-eigenvector-rotation-signal · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H91-eigenvector-rotation-signal/README.md · ratings: H91 agent (suggested) (page 2)



Second figure. Synthetic validation at village counts: a regrouping of half the agents (two rooms, loading 0.4, window persistence 0.5). (a) AUC against stationary days for the same-day bootstrap null (solid) and the pooled-split null (dashed). The split null loses power because it mixes the two days. (b) Trailing-alarm hit rate (solid) and false-alarm rate (dotted). The orange line marks the median real day.

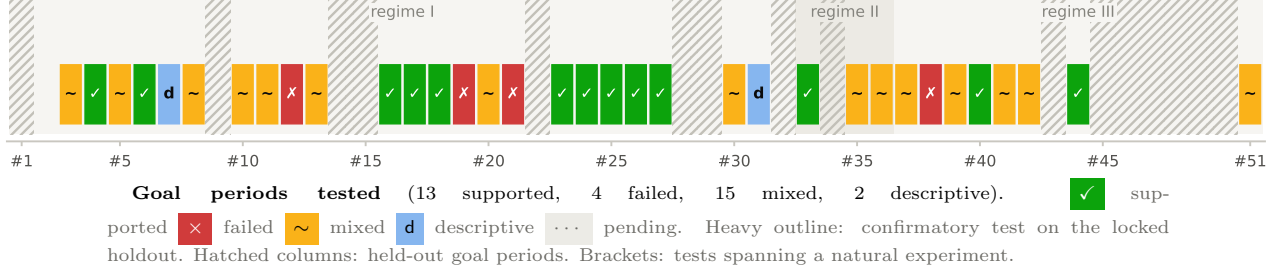
H92: RMT-cleaned matrices forecast tomorrow's alignment

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

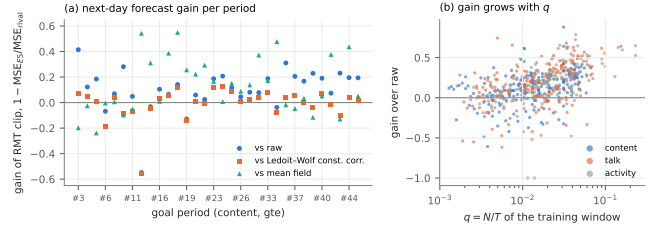
Bottom line. Random-matrix cleaning removes sampling noise from agent correlations as theory predicts, but the persistent structure is one uniform mode, so plain shrinkage does as well.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test asks which estimate of today's agent correlation matrix best forecasts tomorrow's. It compares the raw matrix, two Ledoit–Wolf shrinkages, a mean-field matrix and random-matrix clipping, which keeps only eigenmodes above a calibrated noise edge. Forecasts run inside 34 non-holdout goal periods (205 target days in content, both embedding models; talk and activity too). Clipping beats the raw matrix in 26/32 and 27/32 periods (mean gain 0.10 [0.03, 0.16]). The gain grows with $q = N/T$, as random-matrix theory predicts. Clipping does not beat shrinkage: it ties Ledoit–Wolf constant correlation within 2%. Talk pair structure is not forecastable beyond its mean. Cleaning erases the #38 talk room block. The holdout script exists and is not run.



Observables. (a) Gain of clipping over each rival, per goal period (content, gte). It is positive against raw (blue) in 27/32 periods and against mean field (green) in 23/32, and it scatters around zero against constant-correlation shrinkage (orange). (b) Gain over raw for every forecast against $q = N/T$ of the training window. The gain grows with q in content, talk and activity.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To track who will co-move with whom tomorrow, do not use the raw pair correlations. Clip them at the noise edge or shrink them toward the mean correlation: either removes about a tenth of the error, and more when the window is short. Constant-correlation shrinkage is the simpler choice and does as well. Expect little: the best forecast cuts the error of “no correlation” by only 30% in content. In talk the gain is 5% or less. Do not clip talk if room structure matters: in #38 clipping deleted the room block.

PHYSICAL MODEL

Inverse Ising (model 01) with random-matrix cleaning. A spin is an agent's day-centered 30-min content vector, or its talk or activity state per minute in the all-present window. S is the training correlation matrix over all earlier days of the unit, with N agents and T samples. λ_+ is the 95th percentile of the top eigenvalue when agents' series are shifted against each other. Clipping replaces every eigenvalue below λ_+ by their mean. The target is the next day's realized matrix Q .

$$\hat{C}_{\text{clip}} = \sum_{\lambda_k > \lambda_+} \lambda_k v_k v_k^\top + \bar{\lambda}_{\text{bulk}} \sum_{\lambda_k \leq \lambda_+} v_k v_k^\top, \quad \hat{C}_{\text{LW}} = (1 - \alpha)S + \alpha F$$

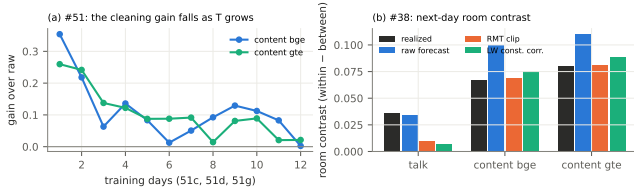
$$\text{MSE} = \langle (\hat{C}_{ij} - Q_{ij}(t+1))^2 \rangle_{i < j}, \quad r = 1 - \frac{\text{MSE}_{\text{clip}}}{\text{MSE}_{\text{rival}}}$$

$$\text{Var}(S_{ij}) \approx \frac{(1 - \rho_{ij}^2)^2}{T}, \quad \lambda_+^{\text{MP}} = (1 + \sqrt{N/T})^2$$

H92-rmt-cleaned-forecast · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H92-rmt-cleaned-forecast/README.md · ratings: H92 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 synthetic edge and ordering	spurious modes 3–17%; two-room advantage only at $N \geq 16$, $W \geq 8$	partial
P1 clip beats raw (content)	26/32, 27/32 periods; 0.10 [0.03, 0.16], 0.11 [0.04, 0.16]	pass
P2 clip beats best shrinkage	17/32 both models; vs const. corr. -0.020 , -0.008	fail (tie)
P3 channels	talk gain > 0 in 27/30, beats LW-CC 10/30; activity skill 0, raw -0.39	mixed
P4 gain grows with q	Spearman 0.45 – 0.65 , every channel	pass
P5 calibrated edge beats MP edge	$k_{MP} > k_{cal}$ in 20%; gain -0.015 , 0.000	fail
P6 longer training	helps raw (79–92%); hurts clip (-8%)	pass
N1 #51 gain falls with T	Spearman -0.69 , -0.90 ; beats LW-CC 4/7, 5/7 units	mixed
N2 #38 rooms kept	talk: kept on 2/10 days; content: 0.069 vs real 0.067	fail



Second figure. (a) #51: the gain of clipping over raw falls as the training window grows from 1 to 12 days, as the $1/T$ noise law predicts. (b) #38: next-day room contrast (within minus between rooms). In talk, clipping deletes it. In content, clipping keeps it and is closer to the realized value than raw.

GOAL PERIODS IN DETAIL

G03 ~ mixed
G04 ✓ supported
G05 ~ mixed
G06 ✓ supported
G07 d descriptive
G08 ~ mixed
G10 ~ mixed
G11 ~ mixed
G12 × failed
G13 ~ mixed

+24 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	2	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The target is itself a noisy one-day matrix, so all estimators share a large error floor; only differences are meaningful. The calibrated edge keeps a spurious mode in up to 17% of noise fits. Most periods have $N < 16$, where a real second mode is cut, so P2 is powered only in #42 and #51. Training matrices zero-fill absent agent-days. No natural experiment was tested.

ROUND 2

Add the rotationally invariant estimator and a one-mode-plus-rooms factor model. Use an exponentially weighted training window, since cleaned structure drifts. Report the cleaned top mode per period on the phase diagram.

- **Essence.** how many persistent collective modes the swarm has once sampling noise is removed; the tie with constant-correlation shrinkage says one.
- **H92-R1** Add the rotationally invariant estimator (Ledoit–Péché / Bun–Bouchaud–Potters) and a one-mode-plus-rooms factor model.
- **H92-R2** Forecast with an exponentially weighted training window: cleaned structure drifts, so recency should beat the expanding window.
- **H92-R3** Use the cleaned top mode as the per-period coupling coordinate on the phase diagram.

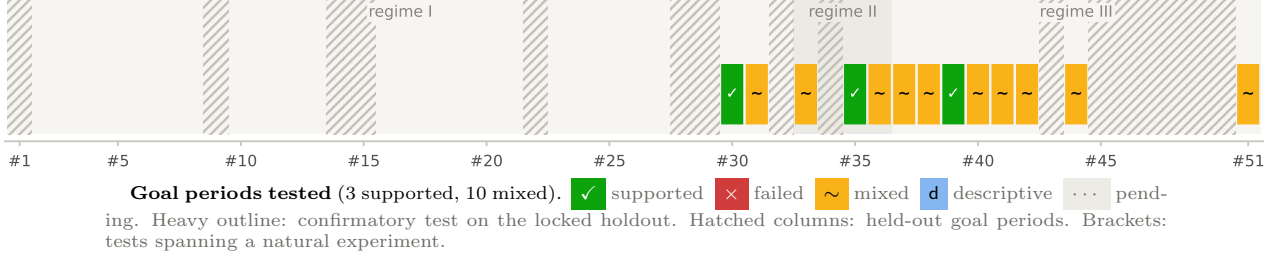
H93: Project choice as a Brock–Durlauf equilibrium

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

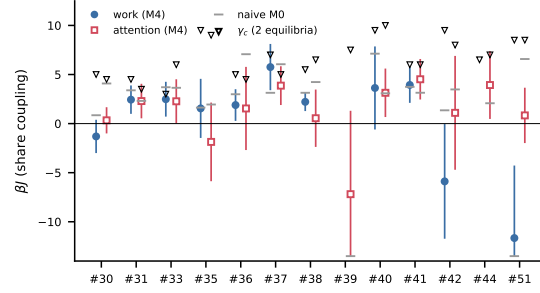
Bottom line. Project choice carries a share coefficient of +2 to +6 in shared weeks and −6 to −12 in own-role weeks, but repo bursts mimic it; at the fitted fields no period is bistable and no same-field replica splits.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H93 fits project choice as a Brock–Durlauf (BD) logit with a share coupling βJ , per period, on DQ4 work arrivals and attention labels in 13 periods (#30–#51). Multiple equilibria were predicted to split same-type weeks. The estimator M4 controls kickoff naming, habit and repo size; it passes the synthetic false-positive rule on all 13 schedules. $\beta J > 0$ (CI > 0) in 5 of 10 identified work periods, +1.9 to +5.8. It is negative in the own-role weeks (#51 – 11.7 [−19.0, −4.4]). A post hoc synthetic shows that fast repo bursts without coupling give the same values. Every period lies below its multiplicity boundary ($\beta J / \gamma_c \leq 0.86$). No same-field replica lands in a second state. The holdout is not run.



Observables. Share coupling βJ per period (95% CI; DL pool over testable units). Filled circles: work arrivals; open squares: attention switches. Gray dashes: the naive share-only estimate M0. Triangles: γ_c , where the period's fitted fields would first support a second equilibrium. Every point estimate lies below its triangle. The share coefficient is positive in shared-goal weeks and negative in the own-role weeks #42 and #51.

PHYSICAL MODEL

Model 10 (mean-field Potts) in its econ form, with model 05's repo fitness as the confound. An agent's state is its current repo or attention project. A choice event is a switch or a resumption. s_j is the share of the other active agents on option j . The fields are kickoff naming nm_j (H54), habit $prev_{ij}$ (the agent held j before) and log cumulative arrivals c_j (fitness proxy). βJ is the coupling in noise units. γ_c is the coupling at which the fitted fields first support two stable states.

$$P(j | e) \propto \exp(b_n nm_j + b_o prev_{ij} + b_c \ln(1+c_j) + \beta J s_j)$$

$$m_j = \langle \text{softmax}_j(h_i + \beta J m) \rangle_i, \quad \text{stable: } \text{Re } \lambda(\partial F / \partial m) < 1$$

$$\gamma_c = \min\{\beta J : \# \text{stable fixed points} \geq 2\}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Agents do pile onto projects that others already work on: from 0 to a third of the agents multiplies the pick odds by about 2–7 in shared weeks. That term cannot be told apart from a common burst of repo attention, so do not read it as imitation. The useful result is negative: at the fitted fields no week is bistable. An operator need not fear that a free week will lock into a herd by chance. Herding follows the fields: what the kickoff names, which team is assigned, and habit, which is the largest term (odds $\times 12\text{--}400$ of returning to one's own repo). To spread work, change the fields (assign owners, as in #51), not the chat.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 naive $M0 > 0$	8/10 work periods	supported
P2 fields+habit halve it	0/8 (ratio 0.58–2.8)	failed
P3 M4 CI $\ni 0$ in $\geq 1/2$	3/10; + in 5, – in 2	failed
P4 $P_{\text{multi}} < 0.5$ everywhere	51/54 fits; 3 small units fail	failed (letter)
P5 BD m^* ranks m	Spearman 0.77, 10 periods	supported
P6 work, attention same sign	7/11	failed
P8 BD vs neutral form tie	8/11 within 0.02 nats/event	supported
N G37 rooms, one field	$\Delta m = +0.22$ in band, $p = 0.50$	supported
N G44 free vs assigned	assigned arm +13.9: field absorbed	failed
N G51 12 same-goal units	no herded unit ($m \leq 0.21$); AF –11.7	mixed

GOAL PERIODS IN DETAIL

G30 ✓ supported
G31 ~ mixed
G33 ~ mixed
G35 ✓ supported
G36 ~ mixed
G37 ~ mixed
G38 ~ mixed
G39 ✓ supported
G40 ~ mixed
G41 ~ mixed

+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

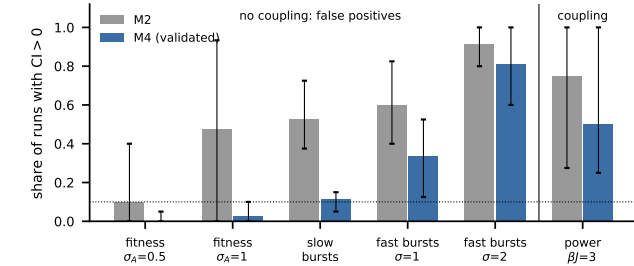
The burst worlds, the read split and the stability fix are post hoc (A2). H54's token rule marks 31–100% of chosen options as kickoff-named, so the kickoff field is measured poorly. An unmodelled field is absorbed into βJ ; #44's assigned team shows it. Work periods have 48–183 choice events outside #51, and M4 power at $\beta J = 3$ reaches 0.8 only in #37, #38 and #51. P_{multi} treats the fitted fields as fixed and the logit's IIA as true. Commits are attempts; attention labels are mentions.

ROUND 2

An in-flight design (hosts the chooser could not yet have seen) or exogenous share shocks (expiries, exits) to identify J ; assignment fields from DQ6 in the model.

- **H93-R1** Separate J from bursts.
- **H93-R2** Put assignment fields in the model.
- **H93-R3** Habit as an agent constant.

H93-brock-durlauf-project-choice · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H93-brock-durlauf-project-choice/README.md · ratings: H93 round-1 agent (suggested) (page 2)



Second figure. Synthetic identification on the real event schedules (40 runs per world; medians and ranges over skeletons). Static repo fitness spread fools the naive M2 but not M4. Fast repo-attention bursts (correlation time 20 arrivals) fool both. Power of M4 at $\beta J = 3$ is 0.25–1.00 by period.

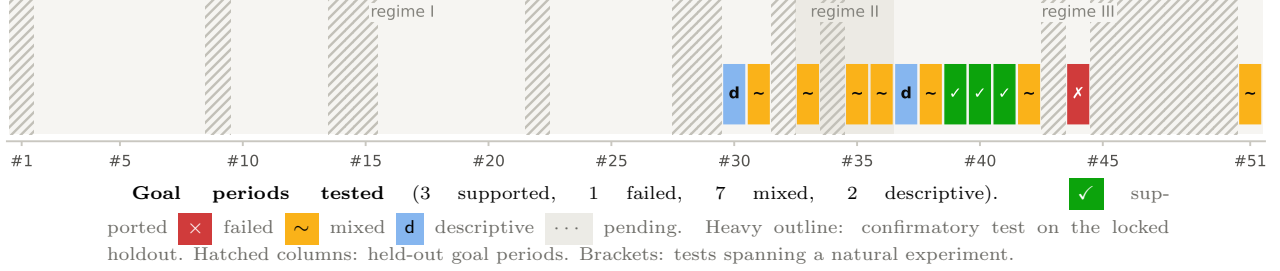
H94: Work allocation as a max-entropy equilibrium

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2.5/5 Usefulness 2/5

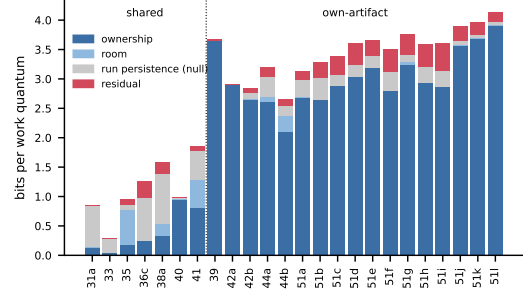
Bottom line. Work episodes are allocated at maximum entropy given agent activity, repo sizes, ownership and rooms; the ownership price is a goal-set field (2.2 nats in shared weeks, 7.8 in own-role units).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H94 asks whether the agent $\times$ repo allocation of DQ4 work is a maximum-entropy (Foley) equilibrium given its constraints. The data are 24 testable units in #31–#51, in 30-min work quanta grouped into episodes. A hierarchy of max-ent models adds agent activity, repo size, ownership and rooms. After these constraints, the information left above the run-persistence floor is ≤ 0.13 of $I(\text{agent}; \text{repo})$ in 23 of 24 units, and inside the floor's 95% band in 12. Ownership is a price set by the goal: median 2.2 nats in shared weeks, 7.8 in own-role units ($p = 0.0007$). Agents do not specialize beyond ownership. Episode concentration is at or above neutral copying in shared weeks; kickoff-named repos explain it in 2 of 6. The holdout is not run.



Observables. Bits of agent–repo information per work quantum, D_1 , split by what explains it: the ownership constraint, the room constraint, the run-persistence floor (what a max-ent draw with each agent-day's runs kept would show) and the residual. Shared-goal units on the left, own-artifact units on the right. Ownership removes 79–100% of 2.7–4.1 bits in own-role units. In #31a and #33, plain max-ent with margins already sits at the floor.

PHYSICAL MODEL

Models 10 and 05: a max-ent model of a categorical allocation, with repos as replicator abundances. A work quantum is one agent $\times$ 30-min window $\times$ repo with an agent commit; an episode is a run of quanta on one repo. The owner of a repo is its first committer in DQ4. λ_{own} and λ_{room} are Lagrange multipliers (prices, in nats). D_k is the KL divergence from the level- k max-ent table. κ places the repo-size law between Maxwell–Boltzmann (0, even) and Bose–Einstein (1, neutral Pólya copying).

$$P_k(i, j) \propto a_i b_j e^{\lambda_{\text{own}} \text{own}_{ij} + \lambda_{\text{room}} \text{room}_{ij}}$$

$$D_k = \text{KL}(\hat{P} \| P_k), \quad D_1 = I(\text{agent}; \text{repo}), \quad D_{k-1} - D_k = \text{KL}(P_k \| P_{k-1})$$

$$\kappa = \frac{H_{\text{MB}} - H_{\text{Obs}}}{H_{\text{MB}} - H_{\text{BE}}}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Work allocation in these swarms is statistically featureless once you know who works how much, which repos are big, who owns what and which room each agent sits in. There is no hidden plan to recover and no extra specialization. A monitor can therefore summarize allocation with two numbers per week: the ownership price λ_{own} and the concentration κ . The price is a lever: own-artifact goals raise it from about 2 to 8 nats, and a merge into one shared target (NE42) lowers it by ≥ 7 nats. To spread work, assign owners; to pool work, name one shared repo in the kickoff. Herding beyond neutral copying appears in 5 of 7 shared weeks, but kickoff naming accounts for it only twice.

H94-maxent-work-allocation · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H94-maxent-work-allocation/README.md · ratings: H94 round-1 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $D_1 - \text{floor} \geq 0.5$ bit everywhere	20/24; #31a 0.04, #33 -0.07	failed
P2 ownership ≥ 0.5 own, < 0.5 shared	17/17 own; 6/7 shared	supported
P3 residual $\leq 0.2 D_1$ (HH285)	23/24; in 95% band 12/24	supported
P4 $\lambda_{\text{own}} \text{ own} > \text{shared}$	7.8 vs 2.2 nats, $p = 0.0007$	supported
P5 $\kappa \geq 1$ shared, < 1 own	7/7 shared; 4/17 own	mixed
P6 named repos carry κ	2/6	failed
P7 singletons from ownership	better than M1 24/24; $\pm 25\%$ 9/24	failed
P8 specialization beyond ownership	breadth < 0.8 in 1/24	failed
N G40 NE42 merge	$\lambda_{\text{own}} -7$ nats, $\kappa 0.16 \rightarrow 3.6$	supported
N G44 planned vs free arm	planned arm owns more (10.2 vs 4.1)	failed

GOAL PERIODS IN DETAIL

G30 d descriptive
G31 $\sim$ mixed
G33 $\sim$ mixed
G35 $\sim$ mixed
G36 $\sim$ mixed
G37 d descriptive
G38 $\sim$ mixed
G39 $\checkmark$ supported
G40 $\checkmark$ supported
G41 $\checkmark$ supported

+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

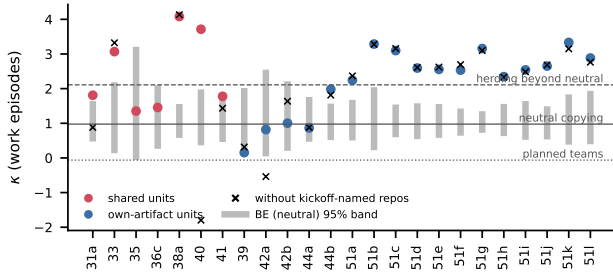
The persistence floor, the episode-level κ and the breadth test were adopted after the synthetic checks and before the real run (A1); the M3 solver and the agent-level κ are post hoc (A2). Max-ent with row, column and ownership constraints is close to saturated when most repos have one committer, so small residuals are partly by construction. In own-role units κ tracks agent activity, not herding. Owners before DQ4 and private repos are unseen. 12 units with < 100 quanta are descriptive.

ROUND 2

A reach (agents-per-repo) max-ent for herding free of activity; DQ6 rival pairs as a fourth constraint for #51; λ_{own} by day vs kickoff wording.

- **H94-R1** A herding index free of activity heterogeneity.
- **H94-R2** Close the #51 residual.
- **H94-R3** The ownership price as a phase-diagram axis.

H94-maxent-work-allocation · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H94-maxent-work-allocation/README.md · ratings: H94 round-1 agent (suggested) (page 2)



Second figure. Concentration κ of work episodes per repo (dots) against the 95% band of Bose–Einstein (neutral copying) draws at the same K and N (gray). Crosses: without kickoff-named repos. Lines: synthetic references for neutral copying, herding beyond neutral and planned teams.

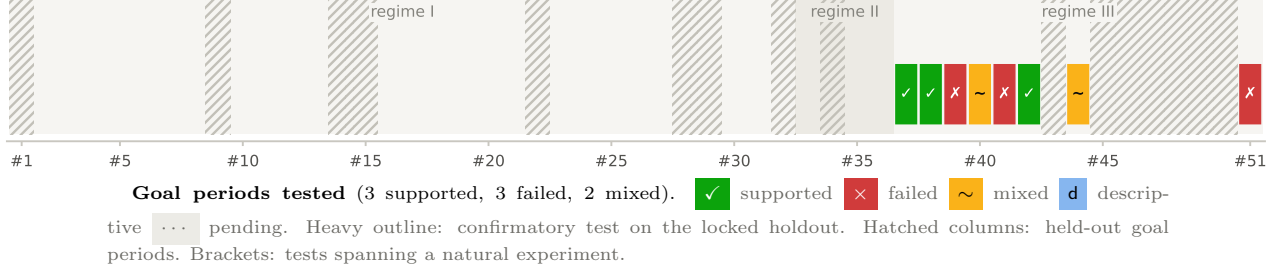
H95: Speed-limit slack as a kickoff-specificity gauge

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

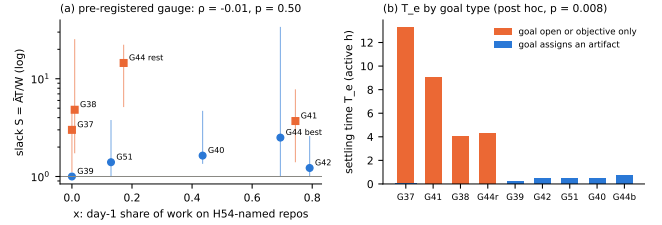
Bottom line. The slack separates assigned-artifact goals from open goals with no overlap, but the HH's specificity measure does not track either; the useful gauge is the goal type, found post hoc and frozen for the holdout.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

I tested whether H75's speed-limit slack $S = \bar{A}T/W$ falls with kickoff specificity across nine regime-III kickoff units (eight periods; G44's two rooms count separately). Specificity x was the HH's: the share of day-1 work on repos that H54 flags as kickoff-named. The synthetic check came first: S falls monotonically with the captured fraction of agents, but nine kickoffs give power 0.29. Result: $\rho(S, x) = -0.01$ ($p = 0.50$). The measure itself fails: a free research week scores $x = 0.74$ and "build your own world" scores 0. Post hoc, coding each goal by whether it assigns a concrete artifact splits the units perfectly: assigned $S = 1.0$ –2.5, open $S = 3.0$ –14.5 ($p = 0.008$). G44's two rooms replicate this on the same days. Holdout: not run.



Observables. (a) Slack against the pre-registered specificity x for the nine kickoff units, with agent-bootstrap 95% CIs (log scale). No trend. Colour is the post-hoc goal type: blue goals assign an artifact, orange goals leave the choice open. (b) Settling time by goal type: assigned goals settle in 0.25–0.75 active h, open goals in 4–13 h.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The operator rule is H75's, now on nine kickoffs: assign each agent (or the group) a concrete artifact in the goal text, and the swarm settles in under an active hour with about one switch per moved agent. Open or objective-only goals take 4–13 active hours and 3–15 switches per net move. The slack is a fine read-out of this after the fact. It is not a thermodynamic cost: the bound is a counting identity. The pre-registered gauge (word overlap between repo names and the goal text) is useless for steering and should not be used.

PHYSICAL MODEL

Models 15 (speed limit) and 11 (kickoff field). Each committing agent is a copy of a jump process on repos plus a null state; $p(t)$ is the share of agents on each repo after the kickoff. W : total-variation distance moved by the settling time T_e . $\bar{A}$: repo switches per agent per active hour. A captured fraction q of agents lands on its settled repo at its first commit; the rest churn between side repos first.

$$T_e \geq W/\bar{A} \quad (\text{counting identity}), \quad S = \bar{A}T_e/W \geq 1$$

$$S \approx \frac{q + (1-q)(1 + a\tau_s)}{q + (1-q)m} \xrightarrow{q \rightarrow 1} 1$$

$$x = \frac{\#\{\text{day-1 commits on named repos}\}}{\#\{\text{day-1 commits}\}}$$

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\rho(S, x) \leq -0.5$	$-0.01, p = 0.50$ ($n = 9$; power 0.29)	inconclusive; kill met
P2 $\rho(T_e, x) \leq -0.5$	-0.02	failed
P3 $\rho(S/S_0, x) \leq -0.3$	$+0.60$	failed
P4 naming, not concentration	partial -0.09 ; G39 has $x = 0$	failed
P5 text rival weaker	$\rho(S, S_{\text{text}}) = +0.63$	failed
N1 G44 rooms	$\Delta x = +0.52$; S ratio $\times 5.8$	supported
N2 G38 rooms	$\Delta x = 0.02$: no contrast	failed
Post hoc: goal type	assigned $S \leq 2.5$ (5/5), open $S \geq 3.0$ (4/4); $p = 0.008$	descriptive

GOAL PERIODS IN DETAIL

G37 ✓ supported
G38 ✓ supported
G39 × failed
G40 ~ mixed
G41 × failed
G42 ✓ supported
G44 ~ mixed (units: G44best mixed, G44rest supported)
G51 × failed

FAITHFULNESS SCORECARD

A mapping	0	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

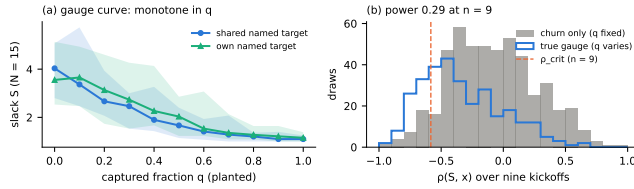
Nine kickoff units cannot give a powered rank test. The goal-type code was chosen after the run, from goal titles I had listed before it; it is frozen for the holdout. G44's #best room has 4 committing agents. G51's slack starts from the null state because its previous week is held out. Commits are the only observation of where agents work, so T_e includes the first-commit delay and has a 15-min floor. S and T_e are not independent of each other.

ROUND 2

Score each agent's own instruction (village plus private goal) for a named artifact and test S against that per-agent field. Merge the slack estimator with H75's in `infra/shared/`. Run `confirm.py` (C2: the four held-out kickoffs are coded open, so each should show $S \geq 2.5$ or $T_e \geq 2$ h).

- **Essence.** a one-number, log-computable gauge of how strongly a goal statement constrains where agents work.
- **H95-R1** A per-agent specificity.
- **H95-R2** More kickoffs.
- **H95-R3** Merge with H75.
- **H95-R4** Run `confirm.py` (C2 is the informative test).

H95-slack-specificity-gauge · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H95-slack-specificity-gauge/README.md · ratings: H95 agent (suggested) (page 2)



Second figure. H75's commit-observed agents with a captured fraction q . (a) The median slack falls monotonically with q for shared and per-agent targets (bands: 10–90%). (b) $\rho(S, x)$ over nine kickoffs: with churn rates varying 16× and q fixed (grey) the test has size 0.06; with q varying (blue) it reaches ρ_{crit} in only 29% of draws.

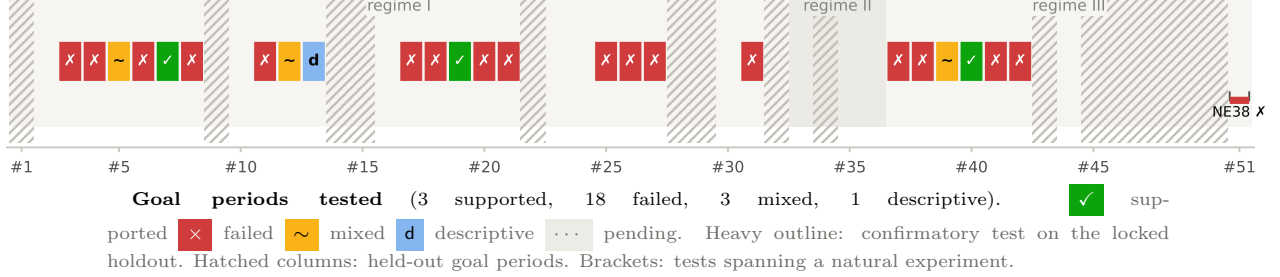
H96: Goal switches show hysteresis: a coercive field

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

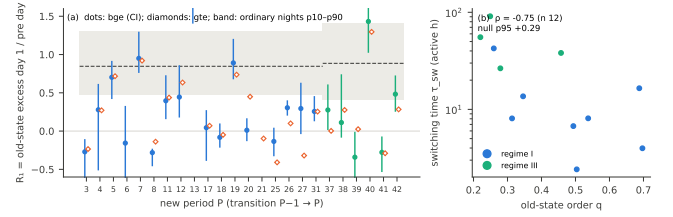
Bottom line. A goal switch quenches the swarm's content state within the first active hour, well beyond ordinary night-to-night drift; there is no sign of an order-dependent coercive field.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH123 claims that a goal switch shows hysteresis: the swarm keeps the old goal's state with a lag that grows with how ordered that state was, like a coercive field. H96 measures the old state's remanence after 24 non-holdout transitions (18 in regime I, 6 in regime III), with the new field projected out and placebo old states subtracted, in both embedding models. **A switch is a quench, not a hysteresis loop.** On day 1 the remanence is 0.27 (bge) / 0.18 (gte) of its pre-switch level, against 0.85–0.89 across an ordinary night (146 pseudo-switches). The order test is inconclusive: its synthetic power is 0.50, and the measured trend is negative ($\rho = -0.75$ / -0.52 , 12 transitions). One reassigned agent (NE38) loses its old role state faster than all 22 same-day placebos. The holdout script is written, not run.



Observables. (a) Persistence ratio R_1 per transition (bge with agent-bootstrap CI; gte diamonds). The grey band is the 10–90% range of ordinary nights. Most switches sit far below it. (b) Old-state order q against the switching time τ_{sw} for the 12 transitions with a measurable residual. HH123 predicts a rising trend; the data fall.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator who reassigns a swarm: there is nothing to clear first. The new kickoff removes 70–80% of the old state's content by the end of day 1, about 3× more than one night removes, and most of it within the first active hour. A single reassigned agent behaves the same way (NE38). A small residual of the old state can linger for about one active day (median τ_{sw} 7–15 active h), so expect stray references to the old goal on day 1, not a lag. The claim that ordered swarms resist a new goal is not supported. Its test is weak at this sample size, so it stays open, but the sign is wrong.

PHYSICAL MODEL

Model 11. Each agent is a unit spin $\mathbf{s}_i$ in the 32-d whitened, style-residualized content space. At the kickoff time t_0 the field steps from the old goal to the new one. The old state $\hat{e}_{old}$ is the leave-agent-out centroid of the old period's last days. $P_\perp$ removes the new field (kickoff, goal, room kickoffs). Placebo old states $\hat{e}_Q$ come from non-adjacent periods. Hysteresis predicts a decay time that grows with the old order q . The rivals are a quench and ordinary night-to-night drift.

$$M(t) = \langle \mathbf{s}_i(t) \cdot P_\perp \hat{e}_{old}^{(-i)} \rangle_i - \text{med}_Q \langle \mathbf{s}_i(t) \cdot P_\perp \hat{e}_Q^{(-i)} \rangle_i$$

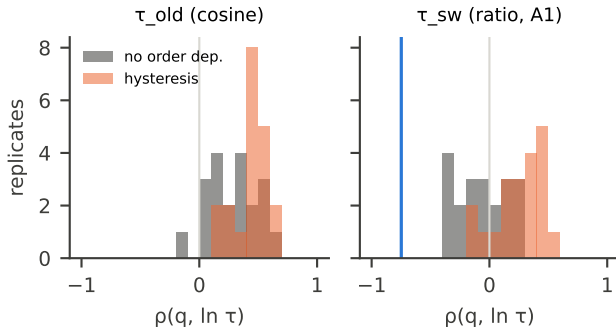
$$R_1 = M(\text{day } 1) / M(\text{pre}), \quad \Delta R_1 = R_1 - R_1^{\text{night}}$$

$$\ln \frac{M_{old}(h)}{M_{new}(h)} = L_0 - \frac{h}{\tau_{sw}}, \quad \text{HH123: } \ln \tau_{sw} = a + bq, \quad b > 0$$

H96-goal-switch-hysteresis · generated 2026-10-07 17:50 by in-
fra/summaries/build_summaries.py · card hypotheses/H96-goal-switch-
hysteresis/README.md · ratings: H96 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed (bge / gte)	Verdict
P1 quench: $\Delta R_1 < 0$, -0.62 $[-0.78, -0.45]$ / -0.72		supported
R_1 below nights in $\geq [-0.88, -0.56]$; 20/23 / 21/24		
2/3		
P2' $\rho(q, \ln \tau_{sw}) > +0.29$	-0.75 / -0.52 (n 12); power 0.50	inconclusive
P3 median $\tau_{old} < 8$ ac-	0.16 / 0.15 h (cosine scale)	supported
tive h		
P4 old kickoff fades first	10/15 / 10/19	holds (weak)
N1 G39 room memory	ratio 0.15 / 0.26; ordered room keeps less	mixed
N2 NE38 single agent	R_1 0.36 / 0.22, percentile 0/22	quench
per transition	18 quench, 3 lag, 2 mixed, 1 desc. – (bge)	



Second figure. Synthetic transitions on the real skeletons (Amendment 1). Left: with the cosine decay time τ_{old} , $\rho(q, \ln \tau)$ is positive even without hysteresis, because unit normalization slows the cosine decay of large components. Right: the ratio time τ_{sw} is centred under the null. Its power at a doubling over the IQR is 0.50. The blue line is the real value.

GOAL PERIODS IN DETAIL

G03 × failed
G04 × failed
G05 ~ mixed
G06 × failed
G07 ✓ supported
G08 × failed
G11 × failed
G12 ~ mixed
G13 d descriptive
G17 × failed

+15 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The order test has power 0.50, so its negative is not a refutation, and its 12 transitions are selected (those with a measurable residual). The cosine decay time is biased by the old state's amplitude; only the ratio time is usable for cross-period comparison. The new field is a text-embedding proxy and operator messages are not removed. Composition carry-over is reduced, not removed. bge and gte disagree on some single transitions (#19→#20). Regime II has too few placebo old states. G39's room contrast is confounded by three agents who changed rooms. The holdout is not used.

ROUND 2

Fit a two-stage model (fast quench fraction plus slow residual) and ask whether the residual lives in veterans' carried-over context or in the record (newcomers, H82). Pool with H82's boundary rows to raise power.

- **H96-R1** Raise power for the order test: pool transitions with H82's boundary rows (same transitions, different estimator) as robustness, and add the held-out transitions in the confirm run (20 targets).
- **H96-R2** Two-stage model: fit a fast quench fraction plus a slow residual (two exponentials, or a ratio model with a quench offset) and test whether the residual is veterans' context carry-over (regime III) or record-borne (newcomers; H82 N1/N2).
- **H96-R3** Explain anti-coercivity: are ordered old states the ones whose new kickoff is most specific (H54's quench depth)?
- **H96-R4** Run `confirm.py` after Vivian's sign-off and the ledger disclosure (H82, H54, H10 plan the same targets).

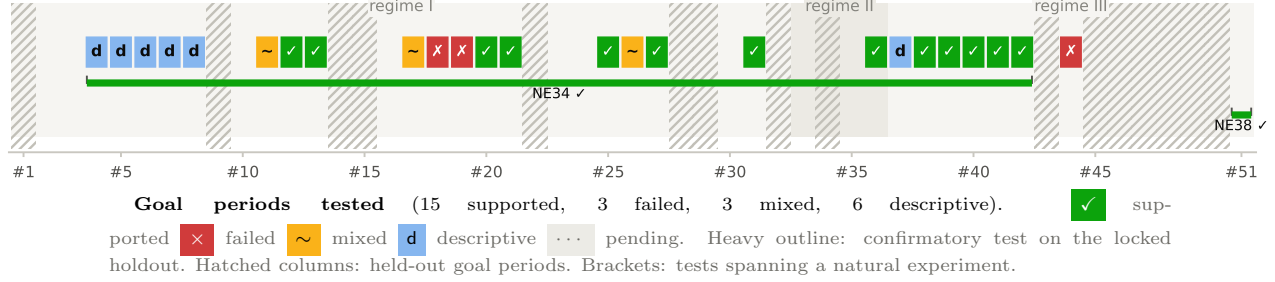
H97: A restoring-force law for the kickoff quench

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 1.5/5

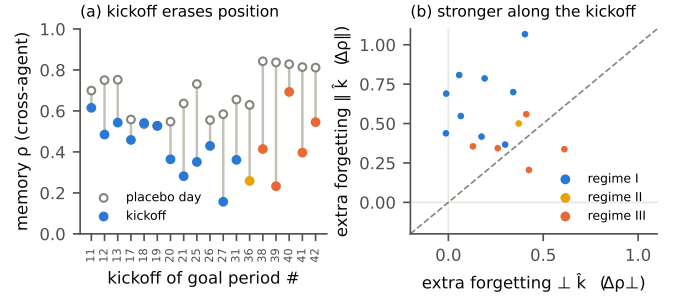
Bottom line. A goal kickoff is a restoring force that erases about a quarter of each agent's relative position beyond an ordinary day, stiffer along the goal and overshooting on day 1; the agent's rate is not shown to be a trait.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH246 says each agent relaxes into the kickoff target at its own rate, $\Delta v_i = \chi_i(\hat{k} - v_i)$. I measured how much of each agent's pre-kickoff position survives day 1, against ordinary days of the same agents, on 18 kickoffs (regimes I–III, 6–15 agents). The synthetic check came first and replaced a biased slope by a scale-free correlation. The kickoff erases position: memory falls from 0.68 to 0.42, $\Delta\rho = +0.26 \pm 0.04$, positive in 16/18 and in all 7 input variants. The law's shape fails. Forgetting is 2–3× larger along $\hat{k}$ than across it, and day 1 overshoots the settled level ($a = +0.12 \pm 0.02$). The per-agent χ_i does not repeat across kickoffs (split-half $r = 0.30$, $p = 0.14$; power 0.30). A one-agent reassignment (NE38) acts as a kickoff on that agent only. Holdout: not run.



Observables. (a) Memory of the pre-kickoff position at each kickoff (filled) and on ordinary days of the same agents (open). The kickoff is below the placebo in 16 of 18. (b) Extra forgetting along the kickoff direction against the transverse one. Points above the diagonal mean a stiffer well along $\hat{k}$; transverse forgetting is still positive (median 0.25), so the kickoff is more than a push along $\hat{k}$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Model 11 with overdamped relaxation (model 02). Agent state v_i : mean of unit regime-whitened statement vectors ($d = 32$) on a day. Field: the kickoff, direction $\hat{k}$ from shared `goal_fields`. Memory ρ : cross-agent correlation of positions before and after a boundary, with split-half true variances. ρ_0 : the same on ordinary day boundaries. χ_i : one agent's forgetting of its offset from the others. a : day-1 displacement beyond the law, relative to the days 2–5 level.

$$v_i^{(1)} = t + \beta_i(v_i^{(0)} - t) + \eta_i, \quad \chi_i = 1 - \beta_i$$

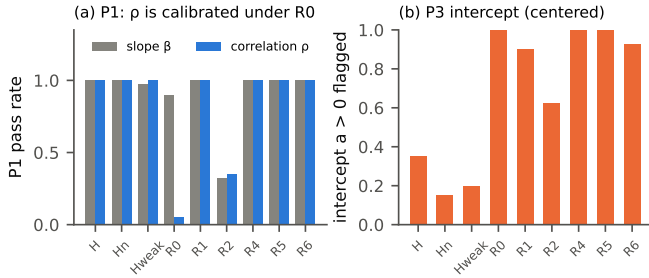
$$\rho = \frac{\text{cov}_i(v^{(1)}, v^{(0)})}{\sqrt{\text{var}_{\text{true}}(v^{(0)}) \text{var}_{\text{true}}(v^{(1)})}}, \quad \Delta\rho = \rho_0 - \rho_{\text{kick}}$$

$$a = \langle \Delta_i \rangle - (1 - \beta_{\parallel}) \langle T - u_i \rangle \quad (= 0 \text{ under the law})$$

Useful as a correction, small as a lever. A kickoff does more than add a common push: it scrambles where each agent was, along the goal and, less, across it. Expect day-1 content to overshoot and then settle back (H54's remanence); do not read the day-1 level as the steady state. The per-agent rate is not a stable trait in these data, so an operator cannot yet rank agents by how fast they adopt a goal. To retarget one agent without disturbing the others, change only its goal text (NE38). For anyone measuring memory in embeddings: a slope between unit-normalized segments fakes forgetting when the common topic grows; use the disattenuated correlation.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\Delta\rho > 0$	$+0.26 \pm 0.04$; 16/18; 7/7 variants	supported
P2 isotropy $\beta_{\parallel} - \beta_{\perp}$	-0.17 ± 0.16 ; style -0.37 ; $\Delta\rho_{\parallel}$ 0.63 vs $\Delta\rho_{\perp}$ 0.25	passes; fragile
P3 intercept $a = 0$	$+0.12 \pm 0.02$ (gte $+0.17$)	failed (R5/R6)
P4 χ_i agent constant	r 0.30 (p 0.14); gte 0.48 (p 0.03); centered -0.06	inconclusive
P6 no lab effect	$p = 0.49$	supported
P8 weekend gap	ρ_0 0.73 vs 0.69	supported
N1 NE38 one agent	χ 0.57/0.70 vs others' 90th pct 0.47/0.40	supported
N2 G44 rooms	$+0.00 / -0.16$	failed
N3 G26 leader's text	below 3/3 placebos; no move toward it	failed (con-founded)



Second figure. Synthetic swarms at the real structure of 23 kickoffs, 40 replicates per scenario. (a) A slope-based P1 passes under pure translation (R0) in 90%; the correlation passes in 5%. (b) The centered intercept flags overshoot, common-mode jumps and constant speed (R0, R4–R6) in 90–100%, and the true law in 15–35%.

GOAL PERIODS IN DETAIL

G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G11 $\sim$ mixed
G12 $\checkmark$ supported
G13 $\checkmark$ supported
G17 $\sim$ mixed
G18 $\times$ failed
+17 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The target direction is a text embedding of the kickoff. Day 1 includes quoting the kickoff text. Each kickoff has 6–15 agents, so per-kickoff intervals are wide; the claim rests on the meta-analysis. The isotropy verdict depends on the estimator and the style variant. The previous period's last day can carry wrap-up content that ordinary days lack. Agent constancy has power 0.30, so its miss is not a negative. The G26 test cannot separate the leader's announcement from the day-1 transient.

ROUND 2

Resolve the relaxation in hours after the kickoff, gate it on each agent's read-out call (context ledger), and pool χ_i hierarchically. Run `confirm.py` on the 13 held-out kickoffs.

- **H97-R1** Time-resolved relaxation.
- **H97-R2** Read-out-resolved forgetting.
- **H97-R3** Agent constancy with power.

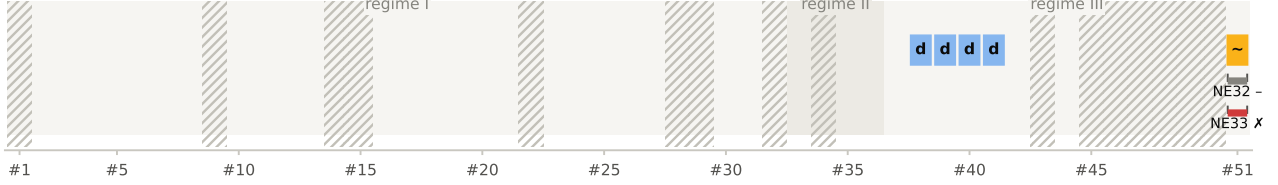
H98: #51 is a random-field system

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1/5

Bottom line. #51's static structure is a random field with a weak pull, but its dynamics are not independent agents in a field: rivals co-move mostly through conversation chosen by niche, and some days move together.

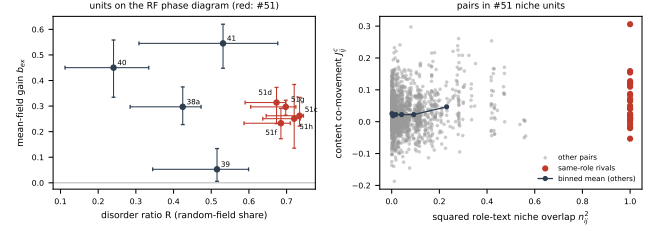
level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 failed, 1 mixed, 4 descriptive, 1 n/a). ✓ supported ✗ failed ◻ mixed ◻ descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

H22 refuted a spin glass in #51 and described a random-field system instead. H98 fits that picture as a mean-field random-field $O(n)$ model on 9 non-holdout regime-III units: 5 private-role #51 units and 4 shared-goal weeks (#38–#41). After style removal, 67–73% of the static content configuration in #51 is agent-specific (disorder ratio R), against 24–53% in shared-goal weeks. A weak positive pull acts on top ($b_{ex} = 0.23\text{--}0.31$). Same-role rivals co-move because their role texts overlap: the niche slope fitted on non-rivals predicts the rival excess (bge). But reads and replies absorb 70% of that slope, and day-to-day overlaps are synchronized in 2 of 5 units. **The claim fails by its pre-registered rule;** the static random-field part stands. The holdout is not run.



Observables. Left: each unit on the random-field phase diagram (disorder ratio R , agent bootstrap CI; mean-field gain b_{ex} , day-bootstrap CI). #51 units (red) sit at higher R than every shared-goal week. Right: content co-movement against squared role-text overlap for all pairs in the 7 #51 niche units. Rivals (red) have identical role texts ($n^2 = 1$). The binned mean of non-rivals rises with n^2 .

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Mean-field random-field $O(n)$ model (models 01, 11). The spin s_{id} is agent i 's whitened, style-residualized content on day d ($n = 32$). The quenched field h_i holds role, own projects and priors; $\hat{r}_i$ is the whitened role text. J pulls toward the room mean; g_d is the day field. Within a day, window fluctuations follow a shared drive η_w filtered by each agent's niche projector P_i , so co-movement grows with the squared niche overlap n_{ij}^2 .

$$s_{id} \propto h_i + J m_{-i,d} + g_d + z_{id} + \epsilon_{id}, \quad R = \frac{\Delta^2}{\Delta^2 + M^2}$$

$$x_{iw} = (\alpha P_i + \beta) \eta_w + \xi_{iw} \Rightarrow \mathbb{E} J_{ij}^c \propto \alpha^2 n_{ij}^2 + 2\alpha\beta + n\beta^2$$

$$\hat{T}_{SR} = \beta_n (\overline{n^2}_{SR} - \overline{n^2}_U), \quad G = T_{SR} - \hat{T}_{SR}$$

H98-random-field-51 · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H98-random-field-51/README.md · ratings: H98 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 $R(\#51) > \text{every contrast unit}$	0.67–0.73 vs 0.24–0.53 (gte same; supported not without style removal)	supported
P2 $b_{ex} > 0$, all < 0.5	5/5 #51 CI > 0 (0.23–0.31); #41 0.55	mixed
P3 no collective switching (W)	BC < 0.555 in 5/5; W 3.05 (p failed (2/5) 0.007), 1.61 (p 0.048) in 51d, 51g	failed
P4 niche slope predicts rival excess	β_n 0.064 [0.009, 0.118]; G -0.034 bge only [−0.095, 0.028]; gte β_n 0.010 (n.s.)	bge only
P5 reads+replies keep $\geq 70\%$ of β_n	0.084 $\rightarrow$ 0.025 (30%)	failed
P6 stance ferromagnetic	mean 0.51–0.64; β_s 0.019 [−0.066, 0.103]	supported
P7 role text in the field	R_{role}^2 0.05–0.14, $p \leq 0.001$ (7/7)	supported
NE32 isolated newcomers	0–1 statements while isolated	untestable
#focus split	pull 0.42 $\rightarrow$ 0.01 (one mover), unchanged (other)	mixed
NE33 joiners on day 1	own-role percentile 0.57 vs 0.83	failed

GOAL PERIODS IN DETAIL

G38 d descriptive
G39 d descriptive
G40 d descriptive
G41 d descriptive
G51 $\sim$ mixed
NE33 $\times$ failed
NE32 – n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

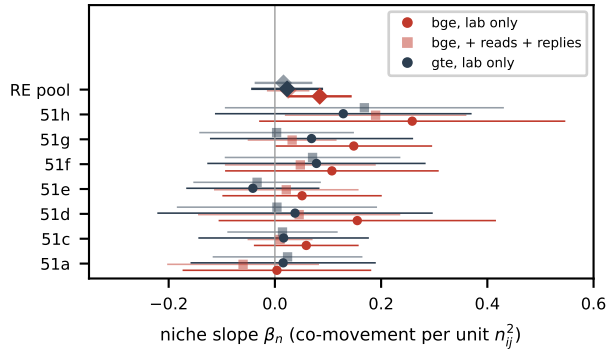
The niche result holds in bge only; #51 is the least model-robust period (DQ5). $\hat{T}_{SR}$ extrapolates from non-rival overlaps $n^2 \leq 0.56$ to rivals' $n^2 = 1$. DQ2 reply parents are partly content-selected, so the reply control may over-adjust; reads alone remove about 30%. R depends on the reference centre (Amendment A0) and on style residualization. b_{ex} includes any common window drive. The sources of the W excess (a join day, a slow drift) are post hoc readings. Two of three natives failed or were untestable; the room split has two agents.

ROUND 2

Model reading as niche-gated coupling $J_{ij} = J_0 f(n_{ij}^2)$ and test co-movement given reads. Add a day-level amplitude field to the overlap model. Fit newcomers' settling time into their role field.

- **Essence.** how much of a mixed-motive swarm's structure is set by assigned roles (field), and how much by who reads whom (coupling).
- **H98-R1** Niche-selected coupling.
- **H98-R2** Day-level amplitude field.
- **H98-R3** Settling time.

Second figure. Niche slope β_n per #51 unit and random-effects pool (diamonds), with same-lab control only (circles) and with ledger reads and DQ2 replies added (squares). In bge the pooled slope falls from 0.084 to 0.025. In gte (dark) it is null from the start.



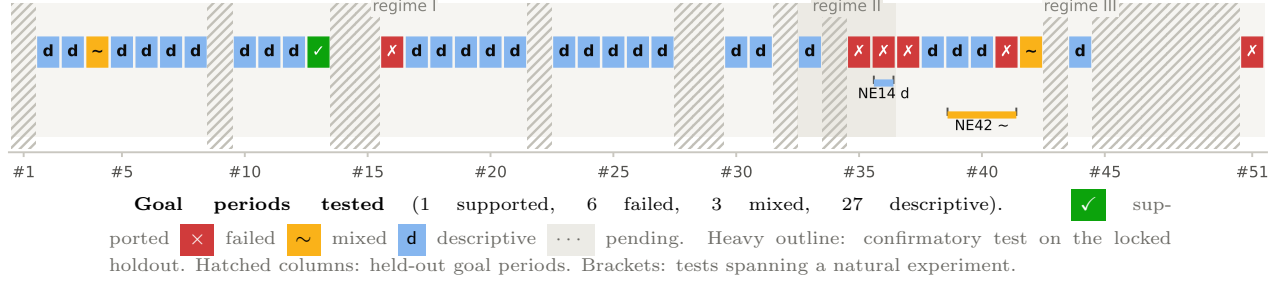
H99: Fluctuations predict relaxation (mean-field Glauber)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 3/5 Usefulness 2/5

Bottom line. The regime-III talk memory that mean-field Glauber misses is coupling delayed by one read-out call. It survives scheduler removal, respects the room and address gates, and matches H67's read-out gain. The tenfold nudge excess was an alignment artefact.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Mean-field Glauber dynamics ties the gain g read from equal-time fluctuations to how much slower the collective mode relaxes than one agent. Round 1 (70 units) found that in regime III the collective talk mode keeps more lag-1 memory than this predicts ($\Delta\rho_1$, post hoc). The open rival was a slow common drive, the scheduler. Round 2 re-indexed talk on each agent's call clock, removed the scheduler field, and added partition contrasts. **The excess is read-out coupling, not the scheduler.** It survives every removal (+0.052 median; #51 +0.042 [0.014, 0.070]), lives inside rooms ($\Pi \approx 15$), and equals what H67's read-out gain produces ($g_{1,\text{fit}}/g_{\text{lag}} = 0.96$). The $\times 9.5$ nudge result is withdrawn. The holdout is not run.

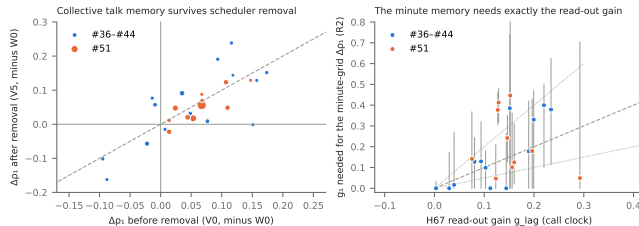
PHYSICAL MODEL

Models 02, 01 and 09. Talk spin $s_i(t)$ per minute on the call-based all-present window, or Y_{ic} per receiving call c . Read-out coupling acts at the recipient's next call: messages posted before t_c are read; messages posted during the call are in flight. A slow common drive acts on every agent at once, whether or not it reads. Each unit's null is the W0 world: independent talk at its real calls.

$$\Delta\rho_1 = \rho_c(1) - \rho_\perp(1)^{1-g_X}, \quad \Delta\rho_1^c = \Delta\rho_1 - \langle\Delta\rho_1\rangle_{W0}$$

$$P(Y_{ic}=1) = a_{i,d,\kappa} + \sum_h \beta_h R_{ic}^h + \beta_0 P_{ic}^0 + \beta_X X_{ic}^1 + \rho Y_{i,c-1}$$

$$g_1 = \bar{m} \bar{r} (\beta_1 - \beta_X), \quad \Omega = \sum_k G(k) / [G(p) \sum_k S(k) / S(p)]$$



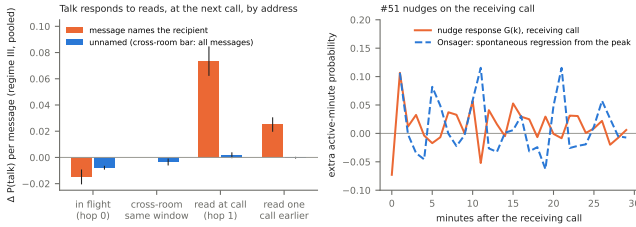
Observables. Left: one point per regime-III unit (size $\propto$ trimmed calls). The W0-corrected collective memory before (V0) and after (V5) removing the call-window, day-edge, exogenous and activity-field structure. Points stay near the diagonal, so the removals change little. Right: the hop-1 gain that a call-skeleton simulator needs to reproduce each unit's minute memory, against H67's read-out gain. Bars are CIs; dotted lines mark $\times 2$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: in regime III, read the collective talk memory as read-out coupling of size $g \approx 0.11\text{--}0.14$ (push amplified $\times 1.15$), not as a scheduler artefact. Report $\Delta\rho_1^c$ next to g_{lag} ; they agree. For an operator: naming the recipient is the lever. A named message raises the reader's talk chance by 7.3 points at its next call and by 2.5 points one call later; an unnamed one by 0.2. A nudge's response is about what the target's own spontaneous wake predicts; it is not tenfold.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
R1 synthetic (real skeletons)	J_1^* only in coupling worlds; g_1 recovered +10%; room partition fails only under a room-local drive	3/5 pass, 2 marginal
Q-S1 survives removal (V5)	median +0.052; +0.042 [0.014, 0.070]	#51 supported
Q-S2 no tracking of c_x	Spearman -0.10 (p 0.64)	supported
Q-P1 room gate (minute grid)	same 0.0165, cross 0.0011 [-0.003, 0.005]	supported
Q-P2 address gate (call clock)	named $\times 10$; placebos negative	mixed
Q-R1a self memory > 0	-0.048 (refractory)	failed
Q-R1b not instantaneous	in flight -0.015 vs read $+0.073$ (named)	supported
Q-R1c kernel $> \text{one call}$	named 0.025 (ratio 0.34); unnamed 0	mixed
Q-R2 $g_{1,\text{fit}} \approx g_{\text{lag}}$	ratio 0.96 (median, 24 units)	supported
Q-R4a nudge $\Omega \leq 2$	1.29; A_{30} 0.44 [-0.41, 1.23] (post hoc)	supported
Q-R4b human $\Omega \in [0.5, 2]$	talk per call 2.53 [1.63, 3.44] (post hoc)	mixed



Second figure. Left: regime-III pooled talk response per message by window on the call clock. Messages a call cannot read (in flight, other rooms) carry a small negative association. Read messages carry a positive one, almost all from messages that name the recipient, which keep acting one call later. Right: #51 nudge activity response aligned on the target's receiving call, with the matched spontaneous regression (Onsager) scaled to its peak.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\sim$ mixed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 d descriptive

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

$\Delta\rho_1$ was chosen after round-1 data (A2). The round-2 tests were pre-registered, with estimator fixes found in the synthetic (B1). The room partition cannot exclude a room-local drive; the call-clock gates do. The in-flight placebo is negative in regime III, so the in-flight-gated gain (0.23) overstates; the room-gated one (0.11) is used. R4 normalization changed after a first run (B2, post hoc), and the nudge activity CI is wide. Regime I stays untestable on this design.

ROUND 2

Holdout on #43 (V5 $\Delta\rho_1^c$, room partition, $g_{1,\text{fit}}/g_{\text{lag}}$) via a re-frozen confirm script. Fit the full per-call kernel of named messages against H40's decay. Identify the negative in-flight field.

- **Essence.** a dynamical check that tells a coupling from a field without inferring J , on the clock at which agents actually update.
- **Round-3 redirects (newest).**
- **H99-R6** Holdout. Run V5 $\Delta\rho_1^c$, the room partition and $g_{1,\text{fit}}/g_{\text{lag}}$ on #43 (regime III) through a re-frozen confirm.py, after Vivian's sign-off.
- **H99-R7** Named hop-2 kernel. Fit the full per-call kernel of named messages (hops 1–5) and test it against H40's per-call decay ϕ .
- **H99-R8** Negative in-flight field. Every unreadable message around a call lowers its talk probability, in every room (β -0.005 to -0.008).
- **H99-R3** (regime-I power) and H99-R5 (content) remain open.

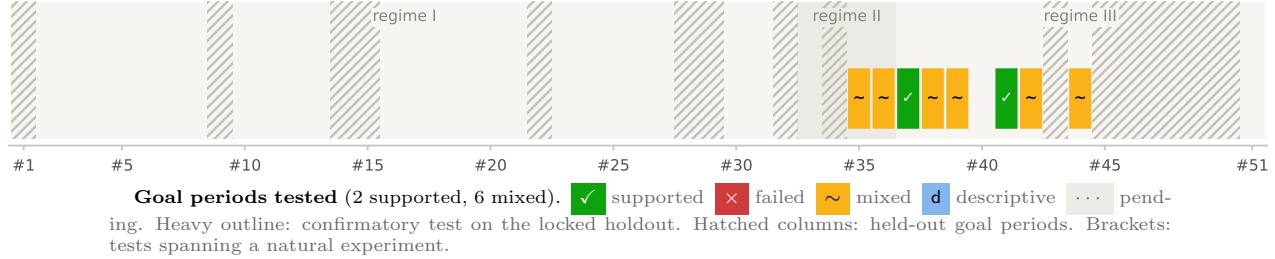
H100: #best/#rest: explicit vs spontaneous symmetry breaking

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

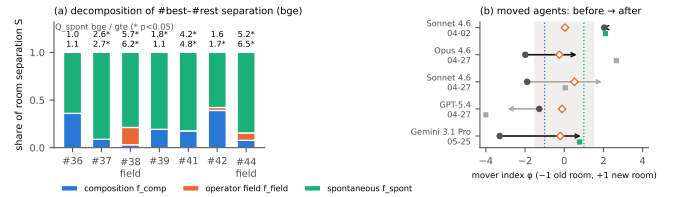
Bottom line. The #best/#rest content split is mostly endogenous and regenerates at each goal; the operator’s room instruction triggers a larger split but not along its own text, and composition explains little.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH66 says the #best/#rest content difference has two parts: a response to the operator’s room field (explicit symmetry breaking) and a spontaneous divergence that stays with the room when members swap. H100 decomposes the room separation in 8 non-holdout two-room periods (#35–#44) into composition, field and spontaneous shares. It also follows 5 operator moves. **The difference is endogenous.** After the members’ own constants and the measured field directions are removed, the rooms still differ beyond random groupings in 5/7 regime-III periods (f_{spont} 0.58–0.91). Composition is small. The kickoff-text direction carries a chance share, although the two fielded weeks separate most. The divergence does not survive the next goal. One mover adopts its new room within a day, and one keeps its old position all week. The holdout (NE19) is not run.



Observables. (a) Shares of the room separation per regime-III period (bge). Numbers give Q_{spont} for bge and gte (* $p < 0.05$). Composition (blue) and the operator-field direction (orange) are small; the endogenous part (green) dominates. (b) Mover index before → after each operator move. Grey arrows: the post-move native contrast is not significant. Diamonds: composition prediction. Squares: gte.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, rooms with identical instructions still diverge (Q_{spont} 1.8–4.2 in 3/5 such weeks). Different instructions about double the separation (Q 5–7), but the content does not line up with the instruction text, so a text-similarity monitor of “is each room on its brief” will miss it. Room identity does not carry over a goal change: regroup freely, nothing persists. A moved agent may align with its new room on day 1 or keep its old project for a week. Check its work, not its room. This is a modest design fact. It says where room differences come from, not how to steer them.

PHYSICAL MODEL

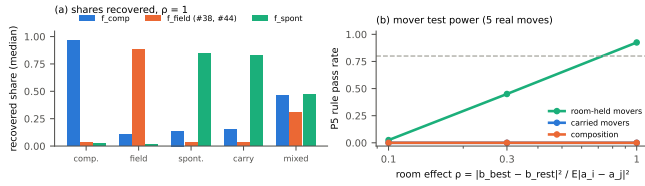
Model 11 (soft-spin $O(n)$, $n = 32$ whitened dimensions) with the room as a Potts domain index (model 10). The spin $\mathbf{x}_{i,d}$ is agent i ’s style-residualized content on day d . $\mathbf{g}_d$ is the day field (goal, schedule), $\mathbf{a}_i$ the agent constant, $\mathbf{h}_r$ the operator room field (room kickoff, operator messages) with susceptibility χ , and $\mathbf{m}_r$ the spontaneous room magnetization. S is the split-half room separation and Q its excess over random partitions. The mover index φ is +1 for a native of the new room.

$$\begin{aligned}\mathbf{x}_{i,d} &= \mathbf{g}_d + \mathbf{a}_i + \chi \mathbf{h}_{r(i,d)} + \mathbf{m}_{r(i,d)} + \boldsymbol{\epsilon}_{i,d} \\ S &= \Delta^{(1)} \cdot \Delta^{(2)}, \quad Q = S / \langle S \rangle_{\text{relabel}} \\ f_{\text{comp}} &= \Delta_{\hat{a}} \cdot \Delta / S, \quad f_{\text{field}} = \sum_k (\Delta^{(1)} \cdot \mathbf{u}_k) (\Delta^{(2)} \cdot \mathbf{u}_k) / S \\ \varphi_k &= [\text{sim}(k, \text{new}) - \text{sim}(k, \text{old})] / C_{\text{native}}\end{aligned}$$

H100-room-symmetry-breaking · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H100-room-symmetry-breaking/README.md · ratings: H100 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: size, shares, mover power	Q size 0.04; shares separate mostly worlds; power 0.93 at $\rho = 1$, 0.45 at 0.3	
P1 $Q > 1$ in $\geq 6/8$	bge 5/8, gte 6/8	mixed
P2 $f_{comp} \leq 0.3$	0.03–0.18 where rooms separate; supported constants invariant (r 0.49 vs 0.14)	
P3 field share ≥ 0.15 in #38, #44	0.18 (null 0.16), 0.08 (null 0.03); failed gte 0.00, 0.04	
P4 spontaneous, identical kickoffs	$Q_{spont} p < 0.05$ in 3/5; #41 4.23 supported (p 0.001)	
P5 movers adopt new room	3/5 bge, 2/5 gte; GPT-5.4 carries mixed ($\varphi -3.0$)	
P6 no remanence	$R(\#39, \#41) z -1.6$; all pairs supported $ z < 2$	
P7 swap-carry null	$R_{room} -0.23, R_{agent} -0.21$ (n.s.) supported	
Post hoc: fielded weeks largest	Q 6.85, 5.09 vs median 1.68 (rank lead p 0.048)	



Second figure. Synthetic validation on the real agent-day skeleton (40 worlds per setting). (a) Recovered shares at $\rho = 1$ in the composition, field, spontaneous, carried-mover and mixed worlds. (b) Pass rate of the mover rule against room-effect size; carried and composition worlds never pass.

GOAL PERIODS IN DETAIL

G35 ~ mixed
G36 ~ mixed
G37 ✓ supported
G38 ~ mixed
G39 ~ mixed
G41 ✓ supported
G42 ~ mixed
G44 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Five movers, all at goal starts; at 04-27 the native contrast is marginal (p 0.04–0.10) and the mover test has power 0.45 at $\rho = 0.3$. The field is measured as the whitened kickoff-text direction, so a failed P3 does not say the instruction had no effect. The endogenous share includes room drives the members create, such as shared repos and the #35 forks; coupling through reading is not separated (convergence impostor open). Periods are 3–17 days. The ranking of fielded weeks and the per-mover narratives are post hoc.

ROUND 2

Confirm on NE19 (#45). Measure the field as DQ4 work per room instead of kickoff text. Test movers' drift toward read vs not-yet-read new-room messages at matched age.

- **Essence.** whether a room is an order parameter of its own (a domain that holds content independent of its members and its instructions), and what an operator's room assignment does to content.
- **H100-R1** Confirm on NE19 (confirm.py C1) and the held-out two-room weeks.
- **H100-R2** Read-out mechanism for movers.
- **H100-R3** The field as work, not text.
- **H100-R4** GPT-5.4's carry and Sonnet 4.6's pre-alignment.
- **H100-R5** Coordination with H81–H83 (composition, remanence, enculturation): the same agent constants and the same "no remanence" reading at village level.

H100-room-symmetry-breaking · generated 2026-10-07 17:49 by infra/summaries/build-summaries.py · card hypotheses/H100-room-symmetry-breaking/README.md · ratings: H100 agent (suggested) (page 2)

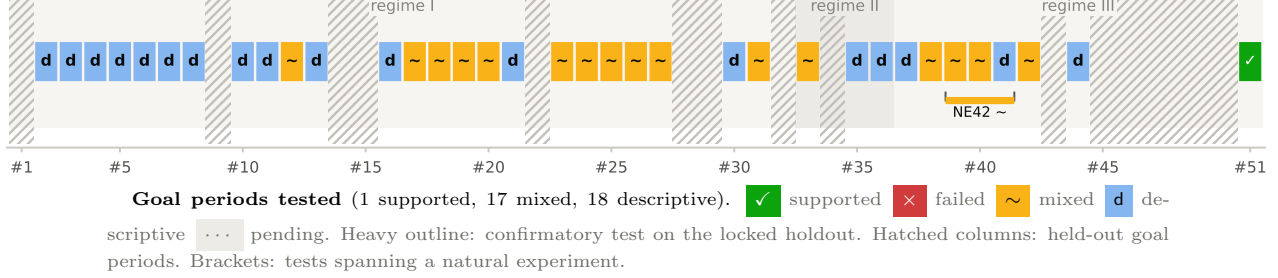
H101: Pairwise Ising vs full multi-information in co-usage

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

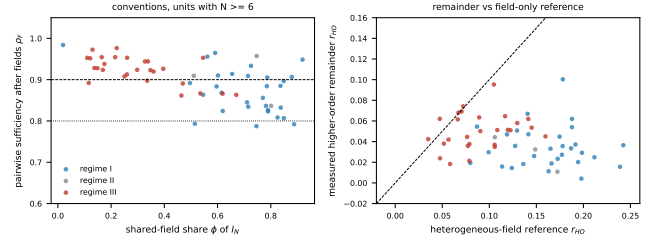
Bottom line. Convention co-usage is pairs plus fields (field-removed pairwise sufficiency 0.91, remainder 3–5% of I_N and field-sized); no separable group term is resolved, and the raw Schneidman ratio is uninformative at village sampling.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH322 asked whether co-usage of conventions has structure beyond pairs: a “group mind” would leave multi-information that a pairwise maximum-entropy model cannot capture after fields are removed. The test covers 71 non-holdout units (282 days): per day, which active agents used which H34 marker, on random 6-agent subsets. The raw Schneidman ratio is $I_2/I_N = 0.95$, but the synthetic shows this is near 1 even with a planted group term. After removing agent rates and a shared item-popularity field, pairs capture a median $\rho_F = 0.91$ of the remaining multi-information. A higher-order remainder of 3–5% of I_N is significant in 83% of units with $N \geq 6$, and it stays inside what agent-specific responses to a shared field produce in 92%. Known teams and rooms appear as signed pairwise couplings. The holdout is not run.



Observables. Units with $N \geq 6$. Left: pairwise sufficiency after field removal against the field share. Regime-I units (blue) carry most multi-information in the uniform field ($\phi \approx 0.7$); regime-III units (red) carry most in pairs ($\phi \approx 0.26$) and sit at $\rho_F \approx 0.93$. Right: the measured higher-order remainder against the remainder of a heterogeneous-field model fitted to the same subsets; almost every unit lies below the diagonal.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: co-usage of names, numbers and coinages across an agent swarm is described by who-pairs-with-whom plus shared hot topics. Rooms and teams show up as pairwise blocks (opponents avoid each other’s terms: $J = -0.60$ vs $+0.15$ within teams). Nothing here needs a group-level interaction, so a pairwise model of co-usage is a sufficient compressed description for steering or auditing. The cynical reading is that this instrument cannot see a group mind of moderate size: a heterogeneous shared field leaves the same remainder. Raw $I_2/I_N \approx 1$ is not evidence of anything at these rates.

PHYSICAL MODEL

Models 01 and 12. Sample: one marker used in agent chat on one day; spin $s_i = 1$ if agent i used it. Fields h_i (agent-day rate), $V(K)$ (shared item popularity, $K = \text{users}$), couplings J_{ij} . Models are fitted exactly on the $2^6 - 1$ patterns with $K \geq 1$; entropies are bias-corrected by a parametric bootstrap from the fitted K-pairwise model.

$$P(\mathbf{s}) \propto \exp\left(\sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + V(K)\right), \quad K = \sum_i s_i \geq 1$$

$$I_N = S_1 - S_N, \quad \rho_{\text{raw}} = \frac{S_1 - S_2}{S_1 - S_N}, \quad \phi = \frac{S_1 - S_K}{S_1 - S_N}$$

$$\rho_F = \frac{S_K - S_{2K}}{S_K - S_N}, \quad r_{HO} = \frac{S_{2K} - S_N}{S_1 - S_N}$$

H101-pairwise-vs-multi-information · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H101-pairwise-vs-multi-information/README.md · ratings: H101 agent (suggested)

RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction (dated)	Observed	Verdict
Y1–Y5 synthetic	bias correction recovers ρ_F 0.93–1.00; planted group term never gives $\rho_F < 0.8$; raw ratio uninformative	2 pass, 1 fail
R1 $\rho_F \geq 0.9$ in $\geq 70\%$	median 0.91; 56% of resolved units	mixed
R2 field share $\phi \geq 0.5$	0.56 overall; I 0.71, III 0.26	supported overall
R3 raw $I_2/I_N \geq 0.9$	0.95	uninformative
R4 remainder in $\leq 20\%$	83% of $N \geq 6$ units (pairwise world $\leq 33\%$)	failed
R6 ϕ larger in regime III	0.26 vs 0.71 (smaller)	failed
R7/R8 (A1) remainder common, field-sized	83%; $\leq$ field reference in 92%	supported
R5 projects	15 resolved; 5 candidates, days pooled	mixed
Natives G12, NE42, G51	teams, rooms = signed pair blocks (p 0.0005); Roudi size trend	mixed / mixed / supported

A mapping 1 F identifiability 1
 B assumptions 1 G ground truth 2
 C adequacy 1 H comparative 1
 D unfitted predictions 1 I transfer 0
 E interventional 1
 Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

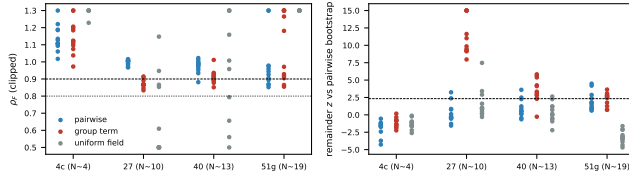
CAVEATS

A group interaction of the planted size and a heterogeneous shared field give the same remainder, so the egreore question is open, not closed. Markers within one message are not independent, so item-level resampling understates uncertainty; ratios carry no CI in the estimates table. Subsets of 6 agents marginalize hidden agents, which creates real subset-level higher order. Units with 4 agents (most of 2025) have no power. Projects pool days, so their day field is not removed.

ROUND 2

Fit a latent-factor max-ent rival (pairs plus agent-specific field loadings); resample by message; confirm C1–C4 on #22, #28, #43.

- **Essence.** whether co-usage needs any interaction beyond pairs once agent-specific responses to shared fields are modelled.
- **H101-R1** Latent-factor max-ent.
- **H101-R2** Message-level resampling.
- **H101-R3** Projects with day fields.
- **H101-R4** Debate pooling.



Second figure. Synthetic validation at the real day counts, item counts and agent rates of four units. Left: field-removed sufficiency in pairwise (blue), planted group (red) and uniform-field (grey) worlds. Right: the remainder z against the pairwise bootstrap. The group term is detected at $N \geq 10$, never at $N = 4$, and never pushes ρ_F below 0.8.

GOAL PERIODS IN DETAIL

G02 d descriptive
 G03 d descriptive
 G04 d descriptive
 G05 d descriptive
 G06 d descriptive
 G07 d descriptive
 G08 d descriptive
 G10 d descriptive
 G11 d descriptive
 G12 ~ mixed

+26 more periods: see the card.

H101-pairwise-vs-multi-information · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H101-pairwise-vs-multi-information/README.md · ratings: H101 agent (suggested) (page 2)

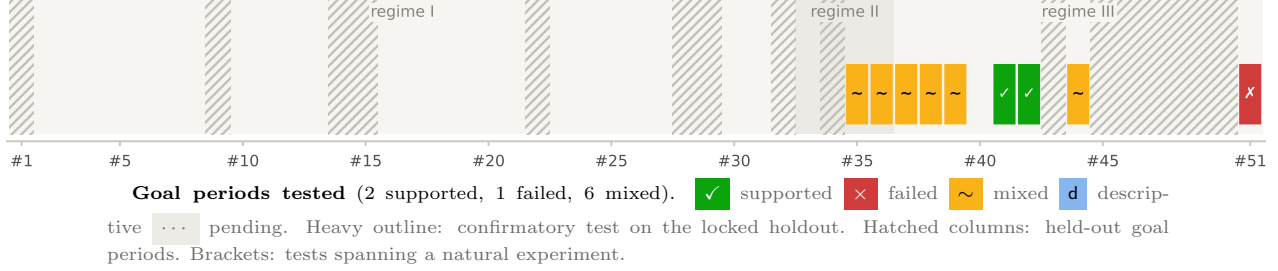
H102: Domain walls between rooms, set by bridging agents

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

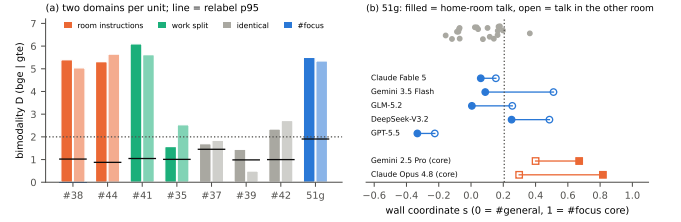
Bottom line. Rooms that do different work form two content domains separated by a sharp, empty wall; hoppers switch domains with the room they speak in and carry nothing home, so bridging agents do not widen the wall.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH332 says that rooms doing different things form two content domains. Agents who hop between rooms would sit inside the wall, and the wall would widen with their cross-room read-outs. H102 projects style-residualized content on the inter-room axis in 9 non-holdout units (#35–#44 and the #focus weeks of #51). It follows the five #focus hoppers and one #36 hopper. **The domains are real; the wall is not.** Where rooms do different work, stayers split into two groups (D 5.3–6.1; relabel 95th percentile ≈ 1). Hoppers stay inside their home domain. The one frequent hopper sits at $s = 0.06$ against a stayers' 95th percentile of 0.21. Hopping read-outs have no effect ($\kappa_R -0.014 [-0.042, 0.024]$). A hopper's talk follows the room it is speaking in. HH332's kill rule fires. The holdout is not run.



Observables. (a) Bimodality D per unit (bge solid, gte light); the black line is the relabel 95th percentile with the axis refitted. Instruction weeks, #41 and the #focus room separate most. (b) #51g wall coordinate. Grey: #general stayers. Blue: hoppers' home-room talk (filled) and their talk inside #focus (open). Orange: the two #focus residents' own-room and #general talk. No agent's home-room talk sits in the wall.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 11 (kinetic soft-spin, $n = 32$) read along one axis like an Ising order parameter (model 01). Agent content $\mathbf{x}_i$ relaxes toward the content it reads, with weights π_{ij} from the context ledger, and toward its domain field $\mathbf{h}_D$ (room work or instructions). The wall coordinate s is 0 at the home-domain centroid and 1 at the other domain. p_i is the share of read items from the other domain. Coupling predicts $s \approx \kappa p$; a room-local field predicts that s jumps with the room of the statement.

$$\mathbf{x}_i \leftarrow (1 - \lambda)\mathbf{x}_i + \lambda[w \sum_j \pi_{ij}\mathbf{x}_j + (1 - w)(\mathbf{h}_{D(i)} + \mathbf{a}_i)] + \boldsymbol{\eta}$$

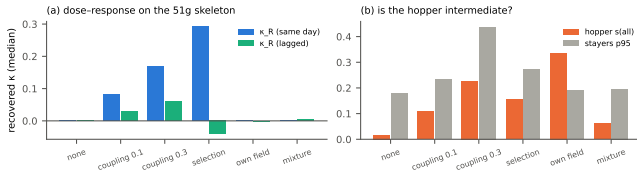
$$s_i = \frac{(\mathbf{x}_i - \mathbf{c}_{\text{home}}) \cdot \mathbf{u}}{(\mathbf{c}_{\text{other}} - \mathbf{c}_{\text{home}}) \cdot \mathbf{u}}, \quad s_i \approx \kappa p_i$$

$$D = \frac{\bar{z}_B - \bar{z}_A}{\sqrt{(\sigma_A^2 + \sigma_B^2)/2}}$$

H102-room-domain-walls · generated 2026-10-07 17:49 by in-
fra/summaries/build_summaries.py · card hypotheses/H102-room-domain-
walls/README.md · ratings: H102 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: D size, κ_R power	D size 0; κ_R power ≥ 0.93 , size 0.05–0.18	mostly
P1 $D \geq 2$ in #38, #44, #35, #41	5.40, 5.31, 1.57 (gte 2.53), 6.10; mostly identical median 1.70	
P2 hoppers intermediate	Table 5 0.06 vs stayers p95 0.21; failed none above	
P3 $\kappa_R > 0$, $> \kappa_U$, lagged	−0.014 [−0.042, 0.024]; lag failed −0.004	
P4 wall width grows with reads	Spearman 0.30 (p 0.43); weak #38/#44 I 0.07/0.17	
P5 axis = kickoff direction	cos 0.02 (p 0.95), 0.27 (p 0.14) failed	
P6 #36 hopper shift ≥ 0.3	≥ 0.25 (gte 0.20); back to −0.13 after	mixed
HH332 kill rule	hoppers not intermediate	failed



Second figure. Synthetic worlds on the real #51g skeleton (40 each). (a) Coupling gives a same-day and a lagged κ_R . Selection gives a same-day κ_R only. An own-field world gives none. (b) Only the own-field world makes hoppers intermediate without a dose.

GOAL PERIODS IN DETAIL

G35 $\sim$ mixed
 G36 $\sim$ mixed
 G37 $\sim$ mixed
 G38 $\sim$ mixed
 G39 $\sim$ mixed
 G41 $\checkmark$ supported
 G42 $\checkmark$ supported
 G44 $\sim$ mixed
 G51 $\times$ failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

One hopper has ≥ 5 hop-days, and hopping read-outs total about 50 items. The test rules out $\kappa \geq 0.1$ per e-fold of reads, not H48's small per-read pull. The #focus domain has two residents. #best had three stayers in #35–#37, which makes D and the interior occupancy noisy there. The day bootstrap for κ_R is liberal (size up to 0.18). Statement-level geometry depends on the embedding model; hopper positions agree to 0.05–0.25 between bge and gte. The “situational” reading is post hoc.

ROUND 2

Measure on-entry and on-exit switching times per statement with ledger call times. Use NE42's merged week as the all-hopper limit.

- **Essence.** what sets the width of the boundary between two content domains in a swarm, and whether bridging agents widen it.
- **H102-R1** Situational vs carried state at the statement level.
- **H102-R2** The #35 room episodes (03-19/20).
- **H102-R3** Larger dose: NE42's merged week (#40).
- **H102-R4** Field as work: replace the kickoff-text direction with DQ4 repo labels per room (as H100-R3).

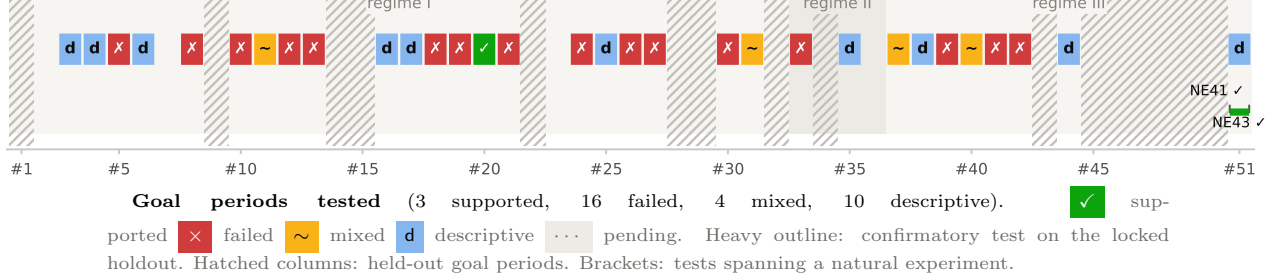
H103: Nights demagnetize: remanence decays per night

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1/5

Bottom line. The content magnetization decays with active work, not per night; nights add no demagnetizing step and slightly re-orient agents toward the current goal.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH331 claims that nights demagnetize the swarm: the remanent content field (a finished kickoff, the previous goal's state) falls in steps at night boundaries by a fixed factor per night. H103 compares four clocks (nights, active hours, wall time, context resets) on 30-min agent windows in 31 non-holdout periods, plus a two-time night step in 41 period units, in both embedding models. **Nights do not demagnetize.** Where the kickoff remanence decays (22/31 periods), active hours fit as well as or better than nights in 17/22 (bge) / 16/22 (gte). At fixed active lag an agent's content is as similar across a night as within a day ($\beta_N = -0.001 [-0.023, +0.021]$). Across the night the kickoff alignment rises a little. Context resets and the end of the operator's bookends change nothing. The holdout script is written, not run.

PHYSICAL MODEL

Model 11 with model 15's question of where in time relaxation happens. The spin is an agent's 30-min content vector $\mathbf{s}_{iw}$ (32-d, whitened, style-residualized). The remanent magnetization is the decoy-corrected alignment with the period's kickoff $\hat{k}_P$. It relaxes as $A_\infty + \Delta A f(t)$ on one of four clocks, with a common time-of-day profile $s(\text{slot})$ that removes the scheduler field. HH331 is the night clock $f = \lambda^N$, AC demagnetization by a fixed factor per night.

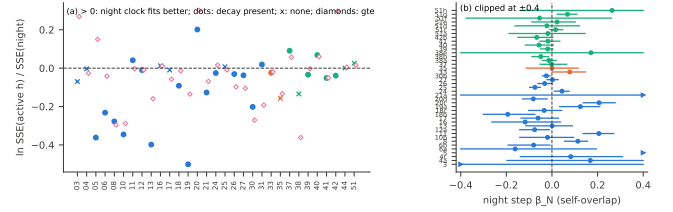
$$A_w = \langle \cos(\mathbf{s}_{iw}, \hat{k}_P) - \overline{\cos(\mathbf{s}_{iw}, \hat{k}_q)} \rangle_i$$

$$A_w = A_\infty + \Delta A f(t_w) + s(\text{slot}_w)$$

$$f \in \{\lambda^N, e^{-H/\tau_H}, e^{-W/\tau_W}, e^{-R/\rho}\}, \text{ nested } \lambda^N e^{-H/\tau_H}$$

$$\cos(\mathbf{s}_{iw}, \mathbf{s}_{iw'}) = \alpha_i + \gamma_{\text{slots}} + g(\Delta H)$$

$$+ \beta_N N + \beta_G N \Delta G/24 + \beta_{\text{gap}} \text{gap}$$



Observables. (a) Log ratio of the active-hour to the night-clock residual per period; values below 0 mean active hours fit better. Filled dots: decay present. Most sit below 0 in both models. (b) The two-time night step β_N per period unit (bge, agent-cluster bootstrap CI, clipped at ± 0.4). It scatters around 0 with no sign preference.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a swarm operator: a night does not reset the swarm's content. What the swarm remembers of a kickoff or of the last goal fades with work done, not with nights or wall time. Shutting the village down overnight neither helps nor hurts that memory, and forcing context erasures inside the day does not move content either. Each morning the agents lean slightly back toward the current goal, which suits goals kept in the prompt. To clear a stale theme, change the goal; waiting overnight does nothing. The result mostly rules out a mechanism; its direct operator value is small.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed (bge / gte)	Verdict
P1 night clock best in $\geq 2/3$	night best 5/22 / 3/22; $SSE_H \leq$ failed (kill) SSE_N 17/22 / 16/22	
P1 RE $\ln \lambda < 0$	+0.06 [-0.11, +0.23] / -0.19 failed [-0.38, +0.00]	
P2 previous-centroid level > 0 in $\leq 1/3$	11/23 / 10/23; mostly too weak to clock	failed
P3 RE $\beta_N < 0$	-0.001 [-0.023, +0.021] / -0.002 failed (41 units)	
P3 $\beta_G \approx 0$	-0.017 [-0.030, -0.005] per extra 24 h	failed (small)
P3 $\beta_{gap} \approx 0$	+0.099 [+0.055, +0.142]	failed (sign +)
N1 NE43 bookends	$\Delta\beta_N$ +0.023 [-0.017, +0.062]	supported
N2 NE41 resets	β_R +0.0003 [-0.0011, +0.0018] per reset	supported
N3 midday break	0.37 across vs 0.35 same-day vs descriptive 0.52 cross-night	

GOAL PERIODS IN DETAIL

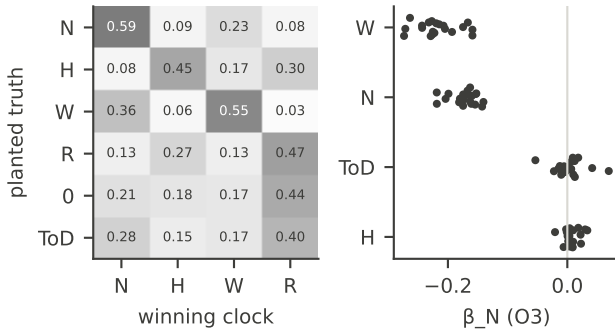
G03 d descriptive
G04 d descriptive
G05 $\times$ failed
G06 d descriptive
G08 $\times$ failed
G10 $\times$ failed
G11 $\sim$ mixed
G12 $\times$ failed
G13 $\times$ failed
G16 d descriptive

+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic validation on the real window skeletons. Left: the winning clock under each planted truth (rows). Night and active-hour truths separate (accuracy about 0.82); a time-of-day field never reads as decay. Right: the repaired two-time design gives $\beta_N \approx 0$ under active-hour and time-of-day truths and detects a night jump and a wall clock.

CAVEATS

Active hours and reset counts are nearly collinear, so the decay clock is "activity" rather than either one. The night factor is not robust in sign between models (λ 1.06 vs 0.83). The positive night step matches a morning re-orientation, but a time-of-day field gives the same sign in the synthetic, so it is not a clean measurement of re-magnetization. Per-unit two-time CIs are anti-conservative (10–25% under null truths), and regime-I units with 3-h days identify β_N poorly. Only one village-off midday break is usable. The kickoff field is a text-embedding proxy. The holdout is not used.

ROUND 2

Add the agent's own call count (H40's clock) to the two-time regression; model the daily sawtooth as activity decay plus a morning pulse, before and after the goal and kickoff entered the prompt (NE08, NE13).

- **H103-R1** Put the agent's own call count (the H40 clock) into the O3 regression next to active hours and resets; test whether calls absorb the active-hour decay.
- **H103-R2** Model the daily sawtooth explicitly: an activity-clock decay plus a morning re-orientation pulse toward the prompt-resident goal; compare eras before and after NE08/NE13 (goal and kickoff in the prompt).
- **H103-R3** Rerun O2 with H82's remanence series once it exists, to see whether its veteran/newcomer split changes the clock.
- **H103-R4** Run `confirm.py` after Vivian's sign-off and the ledger disclosure.

H103-nights-demagnetize · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H103-nights-demagnetize/README.md · ratings: H103 agent (suggested) (page 2)

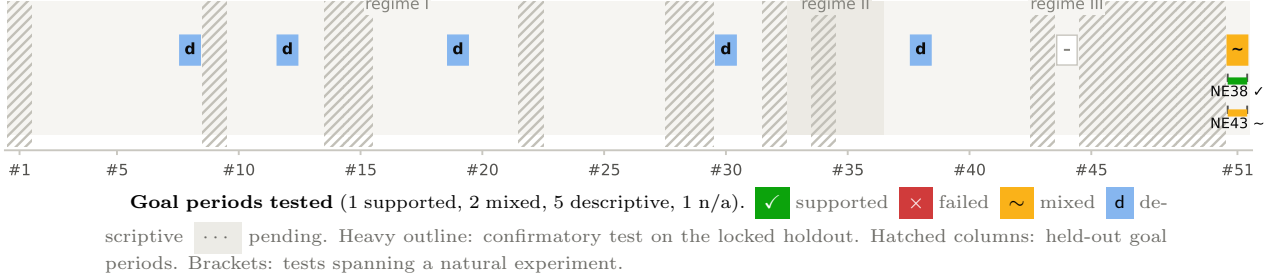
H104: Barkhausen avalanches when a field is stepped

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

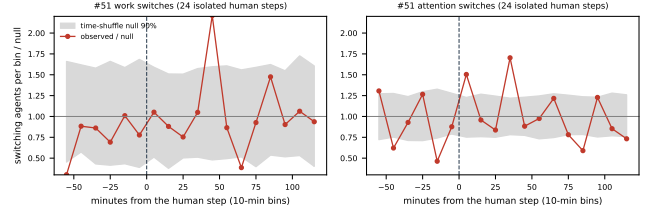
Bottom line. Human messages do not release switch avalanches in the AI Village: the step response is within 10% of the null where it can be measured, and the crackling tail cannot be identified with fewer than 25 steps.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH330 proposed Barkhausen crackling: when a human steps the external field mid-period, agents switch projects in avalanches with a heavy-tailed size law ($\tau \approx 3/2$), and between steps they switch at random. The test uses work switches (DQ4) and attention switches (project states) on 8 eligible non-holdout period-channels. Only #51 has power (24 isolated human steps). There, switching in the hour after a step is $\times 1.09$ [0.84, 1.32] (work) and $\times 1.07$ [0.97, 1.18] (attention) the time-shuffle null. A planted $\tau = 3/2$ avalanche gives $\times 1.87$ and $\times 1.25$. Big bursts are as frequent between steps as after them. Between steps, switching is near-Poisson given each agent's rate. Kickoffs do move agents. **The claim fails**; the tail exponent is not identifiable at village counts. The holdout is not run.



Observables. Distinct agents switching per 10-min bin around the 24 isolated human steps in #51, divided by the time-shuffle null (each agent-day's switches rotated within its at-risk span). Gray: the null's 90% band. The curve stays inside the band in both channels. Single bins outside it (+45 min work, +5 and +35 min attention) are about what 18 bins give by chance.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, a mid-period message is a weak lever on what a swarm works on. It moves the addressed agent (NE38) but does not set off a cascade: the upper bound on the response is 18–32% more switching in the next hour. To redirect a whole swarm, use a kickoff-scale change: on kickoff days 48–100% of agents switch in the first 2 h, against 17–86% on other days. For a monitor, switching between interventions is close to Poisson given each agent's own rate ($D \approx 1.1$ – 1.2), so an unusual cluster of switches is a real event. Crackling exponents need far more steps than the village has. Do not fit them to an LLM swarm's logs at this scale.

PHYSICAL MODEL

Zero-temperature mean-field random-field Ising/Potts model driven by field steps (models 01, 10, 14). Each agent is a Potts spin σ_i (its current project) with a quenched field h_i and a coupling J to the swarm. A human session at t_k steps the field by ΔH_k . Spins that cross threshold flip and can trigger others: an avalanche of A_k flips. The burst size S_k adds background flips B_k , whose distribution the time shuffle supplies.

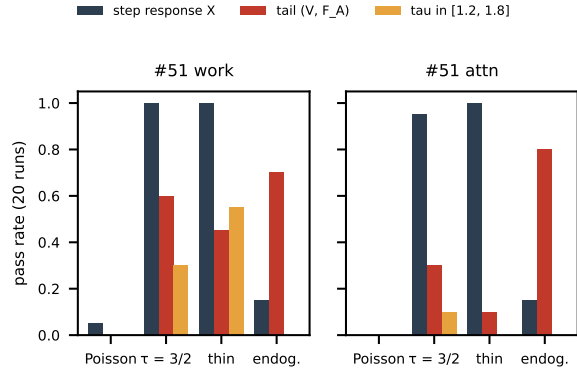
$$P(A) \propto A^{-\tau} \mathcal{D}(A/A^*), \quad \tau = \frac{3}{2} \text{ (mean-field RFIM)}, \quad S_k = A_k + B_k$$

$$X = \frac{\langle S_k \rangle}{\langle S_k^{\text{null}} \rangle}, \quad F_A = \frac{\text{Var } S - \text{Var } S^{\text{null}}}{\langle S \rangle - \langle S^{\text{null}} \rangle}, \quad D = \frac{\text{Var } S_{\text{quiet}}}{\text{Var } S_{\text{null}}^{\text{quiet}}}$$

H104-barkhausen-avalanches · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H104-barkhausen-avalanches/README.md · ratings: H104 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 steps release switches ($X > 1$)	pooled attention 0.91 [0.72, 1.14], work 1.04 [0.86, 1.25]; #51 1.07, 1.09 (CIs incl. 1)	failed
P2 heavy tail, $\tau \in [1.2, 1.8]$	$V \in 1.02, 1.19$; F_A CIs incl. 1; no avalanche component	inconclusive (A1)
P3 Poisson between steps; BR > 1	#51 D 1.12, 1.18; BR CIs incl. 1	D yes; BR failed
P4 burst grows with step size	$\rho -0.30, -0.10$ (n.s.)	failed
P5 kickoffs give the largest bursts	$K > 1$ in 7/8 (1.16–3.97)	supported
P7 no reverse causation	pre-step X 0.82, 0.93	supported
NE38 local step stays local	target switches; others inside band	supported
NE43 nudger stop	ratio 1.22 [0.72, 2.03], 0.96 [0.86, 1.07]; inside placebo	mixed
G44 room-local sponse	no response to localize	n/a



Second figure. Synthetic validation at #51 counts (20 runs per world). The step response X separates a response from none. The tail criterion ($V > 1$ and F_A CI > 1) and the τ window pass as often for a thin Poisson response, or for endogenous bursts, as for $\tau = 3/2$ avalanches. The tail is not identifiable at 24 steps.

GOAL PERIODS IN DETAIL

G08 d descriptive
 G12 d descriptive
 G19 d descriptive
 G30 d descriptive
 G38 d descriptive
 G51 ~ mixed
 NE38 ✓ supported
 NE43 ~ mixed
 G44 – n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only #51 has power; the other periods have 3–4 isolated steps. Attention switches have 15-min resolution. The at-risk trim removes each agent's first 30 min, so morning steps rarely qualify. Work switches need a commit, so idle agents cannot switch, and the read-out split is uninformative. The step size (message count) is a crude field magnitude. Single variants show at most a 10–20% response (switch counts: attention $\times 1.13$, p 0.013). Predictions were written knowing H28, H53 and H63.

ROUND 2

Event-study the kickoffs as the real field steps. Restrict human steps to messages that name a project. Move the spin to 30-min content states, which flip far more often than projects and may give tail power.

- **Essence.** which inputs move a swarm's allocation of effort, and by how much.
- **H104-R1** Kickoff-scale steps.
- **H104-R2** Directed vs broadcast human steps.
- **H104-R3** Avalanches in content, not projects.

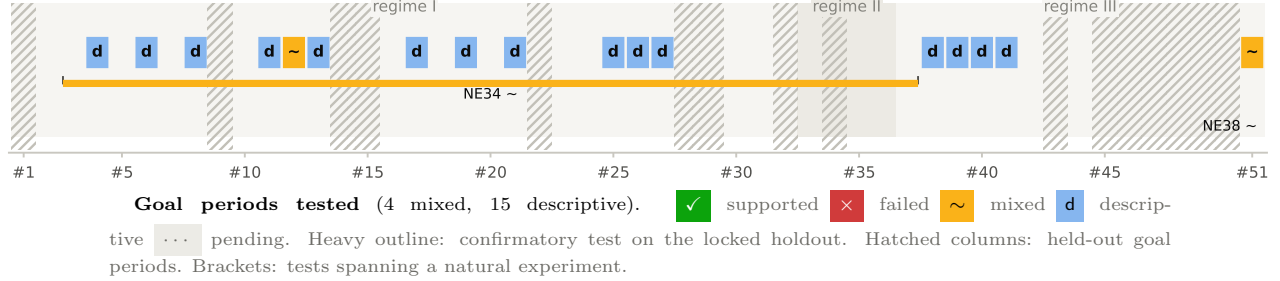
H105: Goals act on a two-state order parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1/5

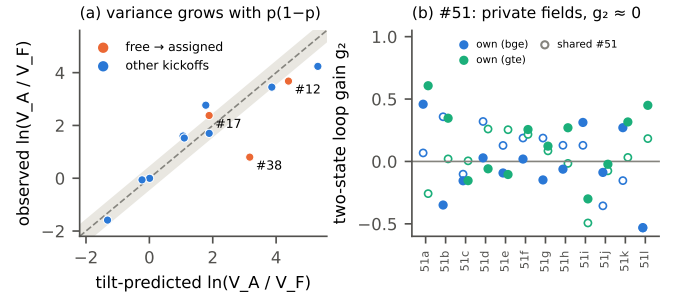
Bottom line. Goals act on a two-state occupancy whose variance grows as $p(1-p)$, but the free-week tilt forecast is not identifiable at village sampling, so the Legendre question stays open.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H10's Legendre test failed on continuous alignment. HH130 proposed a fairer test: treat each agent-window as on-goal or off-goal and ask whether a uniform field (tilt) on the free week predicts the assigned week's occupancy variance and who moves. I ran it on three free → assigned pairs, 16 kickoffs and three natives, in two embedding models. The synthetic came first and showed the tilt test is not identifiable at village sampling: with a true tilt it passes in 30% of pairs, and in 40% under a coupling change. The two-state picture holds descriptively. Occupancy rises from the decoy floor (0.02–0.05) to 0.23–0.44, its variance grows as $p(1-p)$ (Spearman 0.86 over 16 kickoffs), and transverse occupancy stays flat. Who moves: inconclusive in 3/3 pairs. Holdout: not run.



Observables. (a) Observed against tilt-predicted growth of the occupancy variance, free (or previous) week to assigned week, for 16 kickoffs; the band is $\pm \ln 1.5$. The ranks agree ($\rho_s = 0.86$), but only 7/16 fall in the band; #38a sits far below. (b) #51: loop gain of each agent's spin along its own private goal (filled) and along the shared #51 direction (open), per unit. Private fields do not synchronize, except the role-onset unit 51a.

PHYSICAL MODEL

Model 01, forward mean field (Curie–Weiss with agent fields), in 0/1 spins. Spin $\sigma_{i,t}$: agent i 's statements in 30-min window t are mostly on-goal (projection on $\hat{g}$ above the 95th percentile of decoy statements). Field h_i : each agent's on-goal tendency. Coupling J_2 : synchronized switching. Occupancy p_i : mean spin. The tilt adds one λ to every h_i and keeps J_2 . g_2 : two-state loop gain.

$$P(\sigma) \propto \exp\left(\sum_i h_i \sigma_i + \frac{J_2}{2N} n^2\right), \quad n = \sum_i \sigma_i$$

$$\text{Var}(n) = \frac{S}{1 - J_2 \bar{q}}, \quad S = \sum_i p_i(1 - p_i), \quad g_2 = J_2 \bar{q}$$

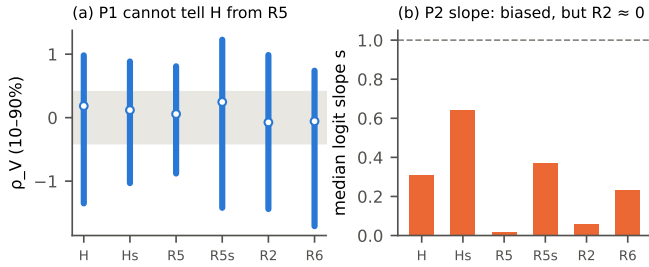
$$\text{tilt: } h_i \rightarrow h_i + \lambda \Rightarrow \rho_V = \ln \frac{V_A^{\text{obs}}}{V_A(\lambda, J_2^F)} \approx 0$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a methods result. The two-state view is the right bookkeeping for a goal: it moves only the on-goal occupancy, from about 0.03 to 0.2–0.45, and leaves other directions alone. But the forecast HH130 wanted, steerability from a free week, cannot be made from village logs with this instrument: free weeks have almost no on-goal windows, so their coupling is unmeasurable, and 2–4 statements per window misclassify the spin. An operator learns nothing new beyond H10 and H54. A monitor builder learns that occupancy measured this way needs a calibrated classifier before any fluctuation–response forecast.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 tilt variance	$\rho_V -0.71, +0.53, -2.45$; gte flips #12a	not identifiable
P2 uniform logit shift	$s -0.60, +0.38, +2.53$; calibrated 0/3	inconclusive
P3 J_2 unchanged	$\Delta g_2 +0.64, +0.26, +0.03$	not identifiable
P4 transverse flat	$p_{\perp} \approx 0.02$ in F and A	supported
P6 kickoffs	7/16 in band; $\rho_s 0.86$ ($p 0.0003$)	half / half
N1 G51 private goals	$g_2 -0.08$ (bge), $+0.19$ (gte)	mixed
N2 G12 debate schedule	removes 50% of excess (gte); no excess (bge)	mixed
N3 NE38 one spin	$0 \rightarrow 0.96$; others $+0.01 / +0.05$	mixed



Second figure. Latent Curie-Weiss spins at the real window structure of the three pairs, observed through statements as in the real pipeline (30 replicates per scenario). (a) The tilt (H) and a coupling change (R5) give the same ρ_V spread. (b) The logit slope is biased low under the tilt (0.31) but near 0 under a common target (R2), so P2 keeps some resolution.

GOAL PERIODS IN DETAIL

G04 d descriptive
G06 d descriptive
G08 d descriptive
G11 d descriptive
G12 $\sim$ mixed
G13 d descriptive
G17 d descriptive
G19 d descriptive
G21 d descriptive
G25 d descriptive
+9 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

"On-goal" is a text-embedding threshold; the 90% and 98% variants shift p but not the conclusions. Spins built from 2-4 statements are often wrong, which drives the non-identifiability. The two models disagree on the sign of ρ_V in #12a and on both natives' second clauses. Free weeks are not field-free. The variance-growth agreement across kickoffs follows from any binary occupancy and does not test the tilt. The on-goal work variant was not run.

ROUND 2

Build a calibrated on-goal classifier with a measured confusion matrix, prove identifiability on the synthetic, and pool free-week couplings hierarchically before retesting the tilt.

- **H105-R1** Fix the instrument, not the model.
- **H105-R2** Pool the free-week coupling hierarchically (exception (d)) across the free weeks of one regime to measure J_2 at all, then predict each assigned week.
- **H105-R3** Separate drive from coupling in #12 with the DQ6 debate schedule as an explicit time-dependent field and a read-out (ledger) lag test for the residual.

H105-two-state-goal-order · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H105-two-state-goal-order/README.md · ratings: H105 agent (suggested) (page 2)

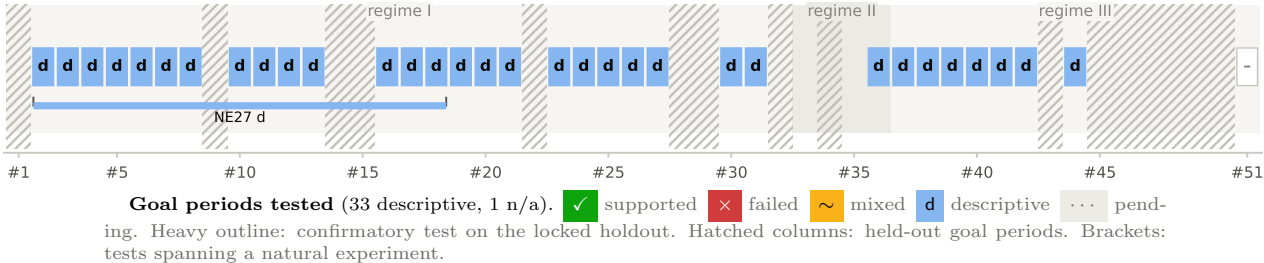
H106: A finite magnet's magnetization wanders as 1/N: the slow mode's decay time should grow with village size

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 0.5/5

Bottom line. The $1/N$ finite-magnet signature of H81's slow mode is not identifiable in one village (power 0.00, oracle SD 0.45; real $\alpha_k +0.40 / +1.22$ with CIs of ± 2), so this test does not weigh on the outside-drift rival; post hoc, the mode is confined to the $N < 8$ era.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

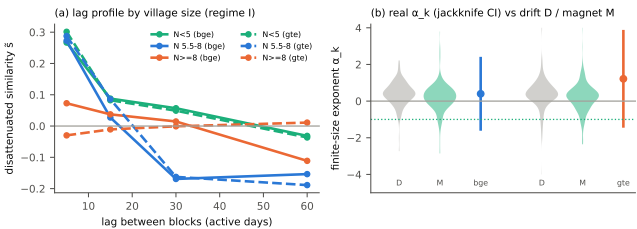
The test asks whether H81's regime-I culture slow mode has the inertia of a finite ordered magnet, whose direction wanders at a rate $\propto 1/N$, or the rate of an outside drift, which does not depend on N . Culture vectors come from random 4-agent subsets in each goal×week block (24 goals, N 4–12), with split-half disattenuation. A synthetic on the real panel, run before any real statistic, shows that the slope cannot be measured: power 0.00 against $\alpha_k = -1$, and a noiseless oracle still has SD 0.45. The real slopes are $\alpha_k = +0.40$ [−1.62, 2.41] (bge) and $+1.22$ [−1.45, 3.88] (gte). The NE27 jump (N 4→7) and the #51 joins are unpowered. Post hoc, the slow mode is strong before mid-November 2025 and weak or gone after it.

PHYSICAL MODEL

Vector spins (model 11) with finite-size scaling (model 14). $u(t)$ is the block culture residual of H81: members' content after two-way fixed-effect personal vectors and goal directions are removed. It is an Ornstein–Uhlenbeck vector with fixed variance and rate $k(t)$. A finite magnet has $\alpha_k = -1$; an outside drift has $\alpha_k = 0$. N is the daily active population; time is the active-day index a . R_b is the split-half reliability of block b .

$$du = -k(t) u dt + \sqrt{2k(t)\sigma_u^2} dW, \quad k(t) = k_6 (N(t)/6)^{\alpha_k}$$

$$\tilde{s}_{bb'} = \frac{\langle \cos(u_b^{(4)}, u_{b'}^{(4)}) \rangle}{\sqrt{\hat{R}_b \hat{R}_{b'}}} = A \exp\left(-k_6 \int_{a_b}^{a_{b'}} (N/6)^{\alpha_k} da\right) + s_\infty$$



Observables. (a) Disattenuated similarity of 4-agent culture vectors between blocks of different goals against lag, for pairs inside each village-size class. Only the $N \geq 8$ class lacks the near-lag similarity. (b) Real α_k with delete-one-goal jackknife 95% CIs (dots) against the synthetic drift (D, $\alpha = 0$) and magnet (M, $\alpha = -1$) distributions on the real panel. The two worlds overlap almost completely.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Nothing changes for an operator. One village cannot show whether its culture has inertia that grows with headcount. Regime I holds one slow-mode trajectory of about 13 decay times, and N rises in calendar order. A $1/N$ law in a timescale needs several villages of different size. The useful by-product is post hoc: the slow mode lives in the small-village months (near-lag similarity 0.25–0.28 at $N < 8$, 0.00–0.11 at $N \geq 8$). If it is confirmed, do not expect a village of 10 to carry a month-long culture the way a village of 4–7 did. For H81, split the slow-mode tests at 2025-11-17.

RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction	Observed (bge / gte)	Verdict
P1 α_k CI excludes 0, includes -1	$+0.40 [-1.62, 2.41]$ / $+1.22 [-1.45, 3.88]$; power 0.00	inconcl.
Evidence (A3)	LR magnet/drift 0.90 / 0.65; D percentile 0.49 / 0.92	uninformative
P2 NE27 $\Delta \ln k \approx -0.55$	$+0.41$ / $+0.38$ (CIs $\pm 2-4$); power 0.06 / 0.09	inconcl.
P3 full roster more negative	V3 $+0.32$ / $+1.11$ vs V1 $+0.40$ / $+1.22$	weak yes
P4 variants keep sign	V2, V4, V5 flip (synthetic SD 1.2-2.7)	no
N2 #51 joins $\Delta \phi > 0$	NE32 -0.09 / -0.38 ; NE33 $+0.44$ / -0.07 ; power ≤ 0.21	inconcl.
Regime III (descr.)	$+0.33$ / -0.54 , CIs $\pm 7-10$	no info
PH1 era drop (post hoc)	$d_p = -0.14$ / -0.28 ; percentile in D 0.11 / 0.00, in M 0.09 / 0.01	gte only

A mapping 1 F identifiability 0
 B assumptions 1 G ground truth 0
 C adequacy 0 H comparative 0
 D unfitted predictions 0 I transfer 0
 E interventional 0
 Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

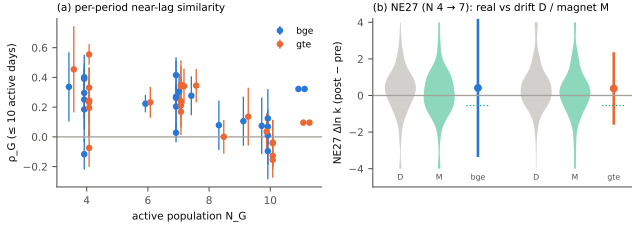
CAVEATS

N is collinear with calendar time: three levels (4, 6-8, 8-12) in date order, with NE03, NE04 and NE08 inside the span. The synthetic assumes one stationary OU mode; the real amplitude changes by era, which neither world contains. The estimator is shifted by $+0.4$ and attenuated to 0.2 of the true slope, though its CI covers the truth in 87-99% of worlds. Per-period ρ_G rests on 1-11 pairs. The era result is post hoc, clear in gte and marginal in bge.

ROUND 2

Confirm the era drop on the held-out goals (#9, #14, #15 vs #22, #28, #29; coordinate with H81). Test inertia cross-sectionally on parallel room trajectories of different size (#38, #44; H108), or across villages.

- **Essence.** whether the slow mode has inertia that scales with the number of carriers (endogenous) or not (outside drift).
- **H106-R1** Era split (feeds H81 round 2).
- **H106-R2** Many trajectories instead of one.
- **H106-R3** Transfer (Q7).



Second figure. (a) Per-period near-lag similarity ρ_G (blocks of other goals within 10 active days) against the active population; bars are ± 1.96 SE. (b) NE27 rate step $\Delta \ln k$ with jackknife CIs against the drift and magnet worlds; dotted ticks mark the magnet value -0.55 .

GOAL PERIODS IN DETAIL

G02 d descriptive
 G03 d descriptive
 G04 d descriptive
 G05 d descriptive
 G06 d descriptive
 G07 d descriptive
 G08 d descriptive
 G10 d descriptive
 G11 d descriptive
 G12 d descriptive

+24 more periods: see the card.

H106-slow-mode-finite-size · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H106-slow-mode-finite-size/README.md · ratings: H106 agent (suggested) (page 2)

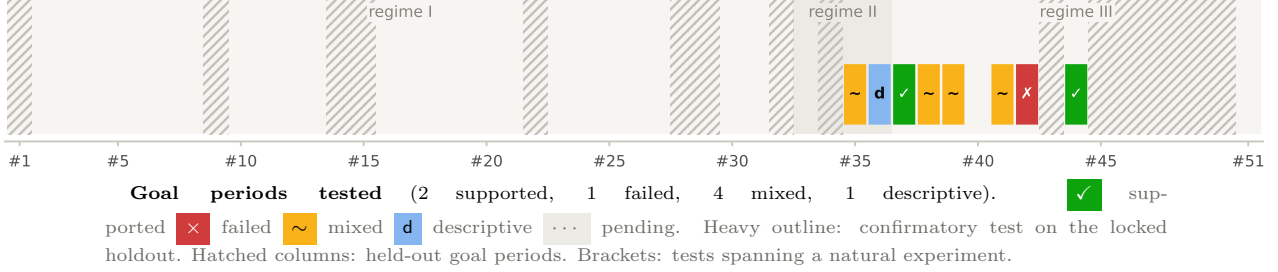
H107: Does the room split grow from zero (an instability) or appear at once (a hidden field)?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1/5

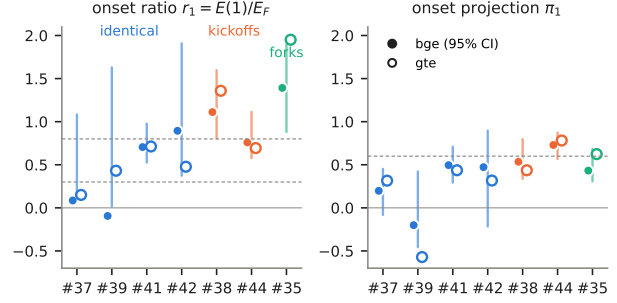
Bottom line. The room content split has no single onset shape: it grows from zero in two identical-kickoff weeks and is nearly full on day 1 in two others, its day-1 direction is half kept, and inherited repos set it only on one week's first day.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH337 asks whether H100's spontaneous #best/#rest content split is a true symmetry breaking or a hidden field. True symmetry breaking starts at zero on day 1 and grows, with a direction chosen by early fluctuations. A hidden field, such as the work each room inherits, gives a full split on day 1 along a direction the members' previous repos predict. H107 measures the day-1 split against the final one in 8 non-holdout two-room periods, with the inherited repo field as an explicit covariate. **Neither picture covers all four identical-kickoff weeks.** In #37 and #39 the split starts near zero and grows. In #41 and #42 it is at 70–90% of final size on day 1, but the day-1 direction is only half kept. Inherited repos set the day-1 direction in #42 only.



Observables. Onset ratio r_1 (left) and onset projection π_1 (right) per period; filled = bge with agent-bootstrap 95% CI, open = gte. Dashed lines: the SSB (0.3) and kill (0.8) thresholds; the positive-control threshold 0.6. Identical kickoffs (blue) split two ways: #37 and #39 start near zero, #41 and #42 near full size. Room kickoffs (orange) and forks (green) give full-size day-1 splits, but only #44 keeps its direction.

PHYSICAL MODEL

Model 11 (soft-spin $O(n)$, $n = 32$ whitened dimensions), with the inherited work as a model-10 Potts field. $\mathbf{x}_{i,b}$ is agent i 's style-residualized content in time bin b , centred on the bin and minus its leave-period-out constant. Δ is the room order parameter. $E(b)$ is its squared size in excess of random joint relabels. The inherited field $\mathbf{h}_r$ weights each repo's content vector $\mathbf{v}_\sigma$ by the members' pre-period commit shares $w_{i\sigma}$.

$$\Delta(b) = \bar{\mathbf{x}}_{\text{best}}(b) - \bar{\mathbf{x}}_{\text{rest}}(b), \quad C(b, b') = \Delta(b) \cdot \Delta(b') - \langle \cdot \rangle_\pi$$

$$r_1 = E(1)/E_F, \quad c_1 = C(1, F)/\sqrt{E(1)E_F}, \quad \pi_1 = C(1, F)/E_F$$

$$\mathbf{h}_r = \langle \sum_\sigma w_{i\sigma} \mathbf{v}_\sigma \rangle_{i \in r}, \quad f_{\text{repo}} = (\Delta^{(1)} \cdot \hat{\mathbf{u}})(\Delta^{(2)} \cdot \hat{\mathbf{u}})/S$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, this mostly says what not to infer. The room difference you see on day 1 of a week with identical instructions is a poor guide to the week: in some weeks it is absent and grows, and where it exists its direction is only about half kept ($c_1 \approx 0.5$). The work the members bring in rarely sets it, which weakens the “hidden field” explanation of H100's residual without proving true symmetry breaking. Day resolution cannot separate a field present at $t = 0$ from an instability that grows within two hours. This is a measurement of onset shape, not a lever.

H107-room-split-onset · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H107-room-split-onset/README.md · ratings: H107 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: size, step, SSB, repo power	E_F size 0.02; kill 75%, $\pi_1 \geq 0.6$ 100% (step); SSB 76% (slow); repo size 0.03–0.05, power 1.0	passed
P1 SSB clause $\geq 3/4$; kill $\geq 2/4$; median $\pi_1 \geq 0.5$	2/4 (#37, #39); 0/4; 0.33 (gte 0.32)	not met
P2 first half-day ≤ 0.3 in $\geq 3/4$	2/4 (#41 0.82, #42 2.38)	not met
P3 repo clause $\geq 2/4$; $\kappa_w > 0$	period 0/4, day 1 1/4 (#42: 0.34, p 0.002); $\kappa_w > 0$ 4/4	not met; met
P4 positive control $\pi_1 \geq 0.6$	#44 0.73; #38 0.53 (r_1 1.11)	failed (half)
P5 forks $\pi_1 \geq 0.6$	0.43 (gte 0.62)	failed
P6 #39 inherits #38 work	repo 0.00; carried content (gte 0.22)	mixed
P7 #41 recalls #39	z -4.5 (anti-aligned); #40 repos n.s.	failed (reversed)
HH337	2 supported, 1 failed, 1 mixed; P4 failed	mixed

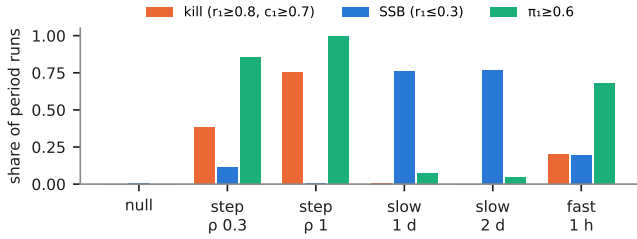
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Four identical-kickoff periods with 3–5 #best agents; r_1 CIs span 1.0–1.9 in #37, #39 and #42. A growth time below two active hours cannot be told from a field at $t = 0$. The positive control is half met: #38 has a full-size day-1 split whose direction is half kept, a day-1 transient (post hoc check at days 4–5: π_1 0.56). Repo vectors need same-regime statements naming the repo before the period, so #37's field rests on four days. The #41 anti-alignment and the reshuffle reading of #39 are post hoc.



Second figure. Synthetic validation on the real statement skeletons (40 worlds, 7 periods): share of period runs in which the HH kill, the SSB clause and $\pi_1 \geq 0.6$ fire. Step worlds trigger the kill and π_1 ; slow growth triggers the SSB clause; an instability with a growth time of about one hour looks like a step at day resolution.

ROUND 2

Hour-resolution onset on day 1 with hierarchical pooling across periods; the reshuffle onset on NE19 (#45, confirm C4); what the day-1 transient follows (operator messages vs the first repos touched).

- **Essence.** whether a room's content order is born from a field the room inherits or from an instability, and how fast it orients.
- **H107-R1** Hour resolution on day 1.
- **H107-R2** What sets the onset shape.
- **H107-R3** Contemporaneous work field (H100-R3): the room's period-P repo shares as a Potts field; does the final split lie along what each room builds?
- **H107-R4** Day-1 transient mechanism.

GOAL PERIODS IN DETAIL

G35 ~ mixed
G36 d descriptive
G37 ✓ supported
G38 ~ mixed
G39 ~ mixed
G41 ~ mixed
G42 × failed
G44 ✓ supported

H107-room-split-onset · generated 2026-10-07 17:49 by in-fra/summaries/build_summaries.py · card hypotheses/H107-room-split-onset/README.md · ratings: H107 agent (suggested) (page 2)

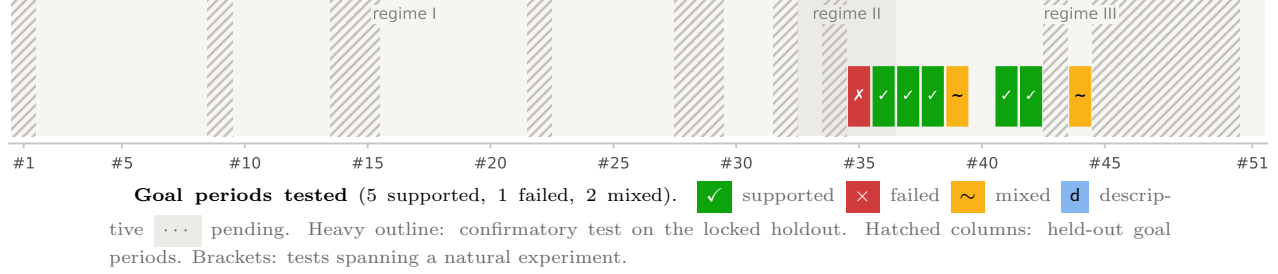
H108: Goldstone wandering: the spontaneous room direction should drift, while a fielded one stays pinned

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1/5

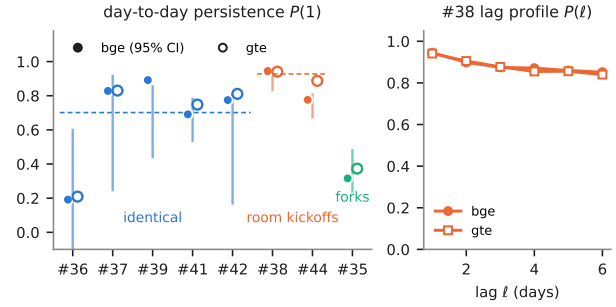
Bottom line. The room content direction persists from day to day inside a week and persists most under room-specific instructions (pooled cosine 0.93 vs 0.70), but the contrast rests on one long week and cannot yet be told apart from slower drift of a larger split.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH338 says that a spontaneously broken continuous symmetry has a soft direction. The room content difference of identical-kickoff weeks should then wander from day to day (a Goldstone mode), while room-specific instructions pin it. H108 measures the day-to-day persistence of the #best/#rest content direction in 8 non-holdout periods, corrected for sampling noise by joint relabels. **The pre-registered contrast passes, but it is fragile.** Directions persist inside a week (cosine 0.69–0.95 in 6/8 periods). The fielded weeks are more persistent (pooled 0.93 vs 0.70; ratio of decorrelation rates 4.7). The contrast rests on the one long fielded week, #38, and it is confounded with split size. The fork week does not pin.



PHYSICAL MODEL

Model 11 (soft-spin $O(n)$, $n = 32$). The room order parameter $\mathbf{m}(d) = \Delta(d)$ is the daily #best – #rest content difference (style-residualized, day-centred, agent constants removed). Without a field its direction $\hat{\mathbf{m}}$ diffuses on the sphere with $D \propto 1/(N_{\text{eff}}|\mathbf{m}|^2)$; a field $\mathbf{h}$ adds a restoring torque, so the angle is an Ornstein–Uhlenbeck process with a plateau. C and E are excess cross-products over joint relabels; $P(\ell)$ estimates the lag- ℓ cosine.

$$d\hat{\mathbf{m}} = -\Gamma\chi(\mathbf{h} \cdot \hat{\mathbf{m}}_{\perp}) dt + \sqrt{2D} d\boldsymbol{\xi}_{\perp}, \quad D \simeq \sigma^2/(2N_{\text{eff}}|\mathbf{m}|^2)$$

$$P(\ell) = \frac{\sum_d C(d, d+\ell)}{\sqrt{\sum_d E(d) \sum_d E(d+\ell)}}, \quad D_{\theta} = -\ln P(1)$$

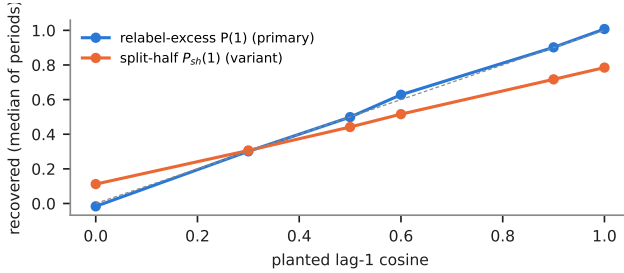
$$R_D = D_{\theta}^{\text{identical}}/D_{\theta}^{\text{fielded}}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, the useful fact is modest: inside a week the content difference between rooms keeps most of its direction (cosine about 0.7–0.9 a day), and one set of different room instructions held it nearly fixed for three weeks. Whether that is the instruction pinning the direction or simply a bigger split drifting more slowly (Goldstone predicts both) is not decided by seven periods. A work assignment alone (the forks) did not hold the content direction. Treat this as a description of room dynamics, not a lever.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: bias, recovery, contrast power	pinned $P(1)$ 0.97–1.08; planted cosines ± 0.05 ; contrast 100%/0%/0%	passed
P1 $R_D \geq 2$, CI > 1, both models	4.7 [2.1, 5.9]; gte 5.2 [2.1, 6.4]	supported
robustness (post hoc)	without #38: 1.4 (gte 3.0); field vs size not separable	fragile
P2 D_θ rises with $1/(N \mathbf{m} ^2)$	Spearman 0.10 (bge, 5 pts), 0.40 (gte)	weak
P3 persistence beats relabel $\geq 5/8$	6/8 (gte 5/8)	met
P4 #38 plateau	slope $-0.019/\text{day}$, $P(6)$ 0.85	met
P5 forks pin	$P(1)$ 0.32; ω 0.68 vs 0.27	failed
HH338	identical rotation $\geq 2\times$ fielded	supported, fragile



Second figure. Synthetic validation on the real statement skeletons (40 worlds per setting): the relabel-excess estimator recovers the planted day-to-day cosine from 0 to 1; the within-day split-half correction named in the HH is biased low under pinning (it keeps each agent’s day deviation as signal), so it is reported but not scored.

GOAL PERIODS IN DETAIL

G35 $\times$ failed
G36 $\checkmark$ supported
G37 $\checkmark$ supported
G38 $\checkmark$ supported
G39 $\sim$ mixed
G41 $\checkmark$ supported
G42 $\checkmark$ supported
G44 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Two fielded periods, one of them 17 days long; #44 alone gives R_D 1.4 in bge. Goldstone diffusion slows with split size, and the fielded rooms split most; a log–log fit over seven periods leaves the field coefficient at -0.17 ± 1.33 . Identical periods have 2–4 day pairs, and agent-bootstrap CIs sit below the point estimate where duplicated agents shift the null. Day 1 is a transient (H107) and enters the pairs. The leave-one-out and size checks are post hoc.

ROUND 2

Compare fielded and identical weeks at matched split size; drop the kickoff day; test transverse kicks (does one room’s operator message rotate the direction more without a field?).

- **Essence.** whether a room’s content order is pinned by its instructions or diffuses freely, at matched order-parameter size.
- **H108-R1** Match on split size.
- **H108-R2** Exclude the kickoff-day transient.
- **H108-R3** Transverse kicks.
- **H108-R4** Why the forks do not pin.

H108-goldstone-room-wandering · generated 2026-10-07 17:49 by infra/summaries/build-summaries.py · card hypotheses/H108-goldstone-room-wandering/README.md · ratings: H108 agent (suggested) (page 2)

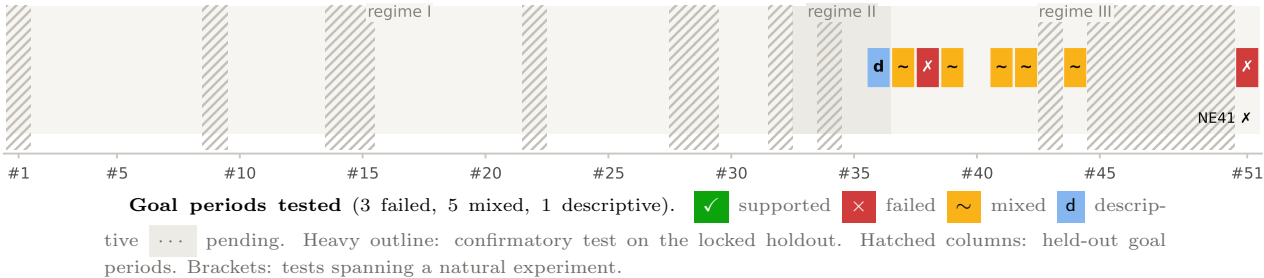
H109: The forced erasure is a demagnetizing pulse: where is the room's order stored?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. A forced context erasure does not demagnetize an agent's room alignment ($\delta_F = -0.08 [-0.23, 0.07]$, power 1.00 at a 30% drop); room order lives in the agent's work, prompt and memory, not in its context window.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



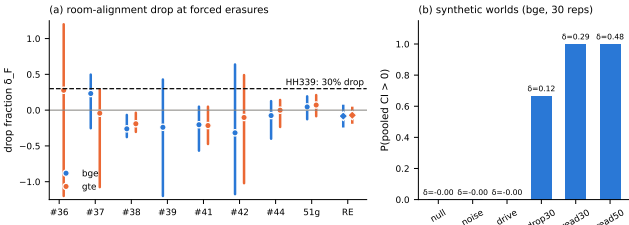
WHAT HAPPENED

HH339 says that a forced context erasure (NE41, the 41-call cap) is a demagnetizing pulse. If a room's content order is held by coupling through the context window, an agent's alignment with its room's direction should drop right after an erasure and recover as it re-reads the room. H109 projects every statement on the leave-agent-out spontaneous room axis (H100's Δ_{spont}) in 8 non-holdout regime-III two-room periods and compares 3,477 forced erasures with gap-matched within-segment boundaries of the same agent and day. **There is no pulse.** Pooled $\delta_F = -0.08 [-0.23, 0.07]$ (gte -0.07), where a 30% drop is detected with power 1.00. HH339's kill fires. The holdout is not run.

PHYSICAL MODEL

Model 11 (soft-spin $O(32)$ in a two-domain room system) with the erasure as a natural scramble of one channel (model 04). A statement is $\tilde{\mathbf{x}}_t = \mathbf{a}_i + \sigma_r m(t) \mathbf{u}_P + \boldsymbol{\eta}_t$. Context-held order: m drops by δ after an erasure and recovers with the room items R read. Self-made field (own work, prompt, memory): $\delta = 0$. The kickoff, held in the prompt since NE13, must not move in either case.

$$a_t = \sigma_{r(t)} \langle \tilde{\mathbf{x}}_t - \hat{\mathbf{a}}_i, \hat{\mathbf{u}}_i \rangle, \quad m(R) = m_\infty [1 - \delta e^{-R/R_0}]$$
$$\delta_F = -\frac{\sum_s n_{F,s} (\bar{D}_{F,s} - \bar{D}_{W,s})}{\sum_s n_{F,s} \bar{A}_{\text{pre},F}}, \quad D = \bar{a}_{\text{post}} - \bar{a}_{\text{pre}}$$



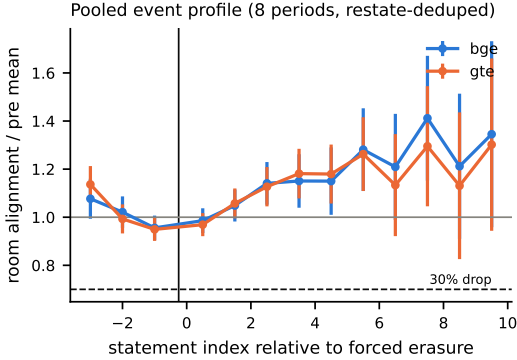
Observables. (a) Drop fraction δ_F per period (bge blue, gte orange; 95% agent-day bootstrap CIs, clipped at ± 1.2) and the random-effects mean (diamonds). The dashed line is HH339's 30% drop. Periods whose pre-erasure alignment is not identified are omitted. (b) Real-skeleton synthetic worlds: probability that the pooled CI excludes 0, with each world's noise-free estimand. The null, erasure-noise and common-drive worlds never reject; the 30% world always does.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: resetting an agent's context does not erase its room's content identity. Alignment with the room stays within -7% to $+23\%$ of its pre-erasure level (pooled 95% CI). Use erasures to break loops or clear clutter (H44, H69) without worrying about room differentiation. To change what a room talks about, change its work or instructions. Reading the room has only a small pull ($\kappa_R \approx 0.02$ per e-fold of items read).

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 size ≤ 0.07 ; power ≥ 0.8	size 0.00; power 1.00 pooled, 2/8 mostly periods	mostly
P1 $\delta_F \geq 0.30$ (HH339); kill	$-0.08 [-0.23, 0.07]$; gte $-0.07 [-0.18, 0.04]$	killed
P2 $ \delta_K < 0.15$	$0.08 [-0.12, 0.28]$; gte 0.00	holds
P3 no drop: κ_R CI $\ni 0$	$0.019 [0.003, 0.034]$; gte 0.017	failed
P4 drop at $R = 0$	no drop; $R = 0$: -0.10	moot
P5 $\delta_V \approx \delta_F$	-0.04 vs -0.08	holds
P6 dedupe adds a drop	$+0.005 / -0.03$	≈ 0
#38 (powered)	$-0.26 [-0.38, -0.07]$: a rise	failed
51g (powered)	$0.05 [-0.13, 0.19]$	failed



Second figure. Pooled event profile: room alignment of the last three statements before and the first ten after a forced erasure, divided by the pre-erasure mean (95% CIs). No dip at the erasure; the later climb is post hoc and mixes segment lengths (later positions come only from long segments).

GOAL PERIODS IN DETAIL

G36 d descriptive
G37 ~ mixed
G38 x failed
G39 ~ mixed
G41 ~ mixed
G42 ~ mixed
G44 ~ mixed
G51 x failed
NE41 x failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Scoped statements are sparse (median 2 per reset segment), so effect sizes are read on the estimand scale (Amendment 1); HH339's literal "30% for 3 statements" world has an estimand of only 0.12. Per-period power is ≥ 0.8 only in #38 and 51g; six per-period negatives are inconclusive. The room axis is a period-level ruler; a faster, other-direction room component would be missed. A drop that recovers before the first post-erasure statement is invisible. The post-erasure rise ($-0.12 [-0.21, -0.03]$ direct pooled) is post hoc.

ROUND 2

Test whether room alignment falls with context fill inside a segment (does context dilute room order?), and measure the drop on the call clock in the first 1–3 calls after an erasure. Confirm on #45–#47 and #50.

- **H109-R1** What holds the order?
- **H109-R2** The post-erasure rise.
- **H109-R3** A call-clock version.
- **H109-R4** Confirm on #45–#47, #50 after Vivian's sign-off and the ledger disclosure.

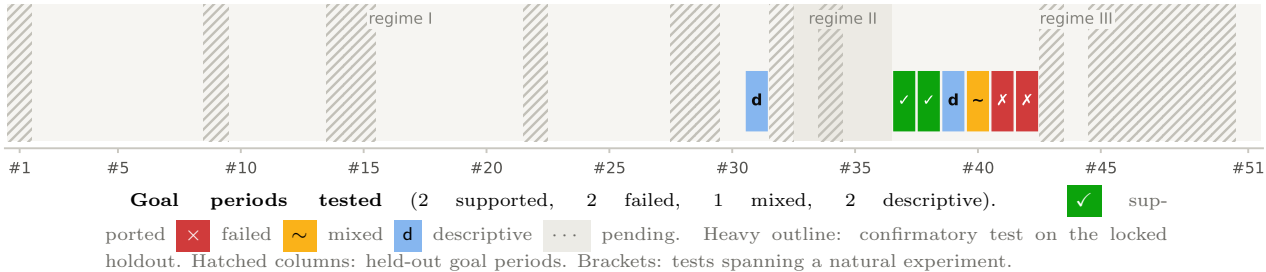
H110: Exchange bias: an agent's own artifact shifts its goal-switch loop

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. An agent's own live repo does not pin its old goal state: at most a small day-1 residual (0.30 [−0.01, 0.57] of the pre level) that is not tied to continued commits and is gone by day 2.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



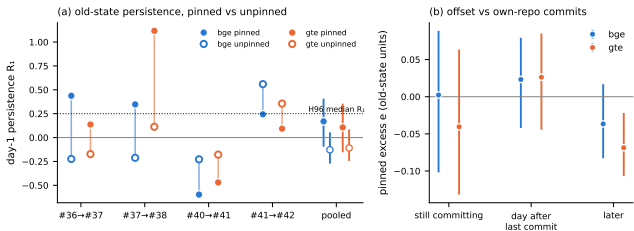
WHAT HAPPENED

HH340 says that an agent's own repo acts like the pinned layer in exchange bias. Agents still committing to their own repo at a goal boundary should keep the old goal state longer (decay $\geq 2\times$ slower), and the offset should end within a day of their last commit to that repo. H110 re-implements H96's field-orthogonal old-state remanence per agent and splits it by DQ4 pinning at 7 non-holdout transitions (4 with both groups). **A small day-1 asymmetry, no exchange bias.** Pinned agents keep $R_1 = 0.17$ vs -0.13 (difference 0.30 [−0.01, 0.57]; gte 0.22), but the offset is not tied to continued commits, is gone by day 2, and reverses under an any-repo definition. Neither HH340's rule nor its kill fires. The holdout (incl. NE24) is not run.

PHYSICAL MODEL

Model 01 (hysteresis under a stepped field) measured with the O(32) content vector of model 11. A free spin's old-state component decays as $m_i(h) = m_i(0)e^{-\lambda h}$ (H96: a quench, $R_1 \approx 0.25$). A spin bonded to a pinned layer feels an extra field $J_{\text{ex}}\pi_i(h)$ while it still commits to its own repo ($\pi = 1$), which shifts its loop: slower decay, and a residual offset that ends with the last commit.

$$\frac{dm_i}{dh} = -\lambda(m_i - m_i^\infty), \quad m_i^\infty = J_{\text{ex}}\pi_i(h)/\lambda$$
$$R_1^g = \frac{\sum_{i \in g} m_i(\text{day 1})}{\sum_{i \in g} m_i(\text{pre})}, \quad \rho_\lambda = \frac{\ln \max(R_1^U, 0.02)}{\ln \max(R_1^P, 0.02)}$$



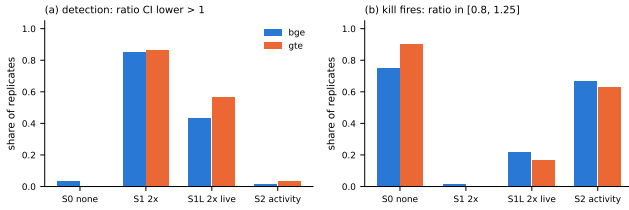
Observables. (a) Day-1 persistence R_1 of pinned (filled) and unpinned (open) agents at the four transitions with both groups, and pooled with agent-bootstrap 95% CIs (bge blue, gte orange). Two transitions favour pinned agents, one quenches both, one reverses. (b) Pinned excess over same-boundary unpinned agents, split by own-repo commits: while still committing, the day after the last commit, and later. No offset while committing.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: do not expect agents with a live own repo to resist a goal change. By day 2 they have switched like everyone else (difference 0.11 of the pre level, CI includes 0). The kickoff alone moves them; archiving their old repo is not needed for the content switch. The own repo stores where an agent works (H70), not what it is thinking about.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 size; HH power; kill discrimination	size ≤ 0.03 ; HH power 0.47–0.60; kill fires 0.75–0.90 vs 0–0.02	amended (A1)
P1 $\rho_\lambda \geq 2$, CI > 1 ; kill [0.8, 1.25]	2.2 / 1.7, CI lower at floor; dR 0.30 [−0.01, 0.57]	neither
P2 offset ends with last commit	$e(\text{live})$ 0.00 [−0.10, 0.09]	failed
P3 own-state $\rho_\lambda > 1$	1.15 [0.39, 3.33]	failed
P4 share pinned vs R_1 (not blind)	ρ 0.57 / 0.25, n 7	weak
G40 continuing > stop-ping	−0.01 [−0.21, 0.18]	null
G39 movers by #38 commits	1 kept committing	n/a
Any-repo pinning	dR −0.23 (reversed)	against



Second figure. Synthetic validation on the real skeleton (bge 60, gte 30 replicates per world). (a) The detection rule (decay-ratio CI above 1) has size ≤ 0.03 with no pinning or an activity confound, and power 0.85–0.87 at a 2× slower decay; less if the slowdown lasts only while agents commit. (b) The kill fires in 75–90% of no-pinning worlds and 0–2% of 2× worlds.

GOAL PERIODS IN DETAIL

G31 d descriptive
G37 ✓ supported
G38 ✓ supported
G39 d descriptive
G40 ~ mixed
G41 × failed
G42 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only 4 transitions have both groups (26 pinned, 28 unpinned agent-transitions). The decay ratio is unresolved when the unpinned group is fully quenched ($R_1 \leq 0$); the persistence difference is the readable number. The switcher variant is significant in bge only (0.37 [0.05, 0.66]). Ownership uses non-holdout commits only. The transition-level correlation is not blind (H96 had reported #39→#40).

ROUND 2

Test whether the day-1 asymmetry is a mid-task effect (open vs closed task at the last pre-kickoff commit), pool with H96's transitions, and run the NE24 confirm.

- **H110-R1** Pre-switch task state.
- **H110-R2** Pool with H96's transitions (same estimator, regime I has sparse commits) and add the held-out transitions in the confirm run for power.
- **H110-R3** A commit-level offset.
- **H110-R4** NE24 confirm after Vivian's sign-off.

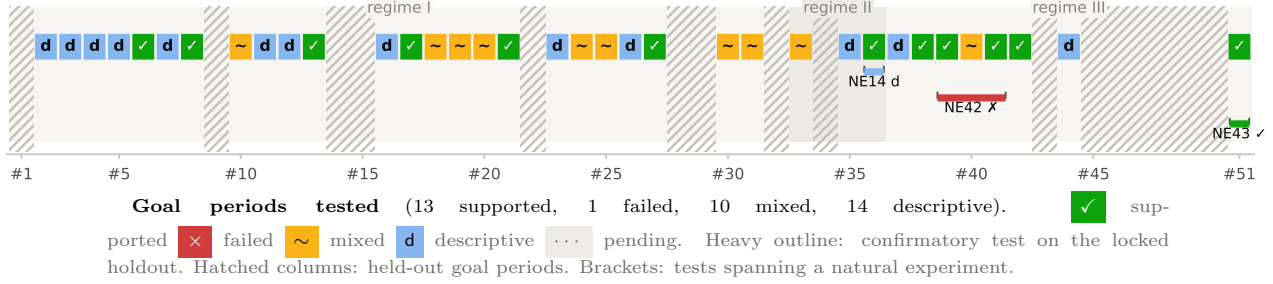
H111: A fluctuation–response sum rule for talk: Fano factor = $1/(1 - g)^2$

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2/5

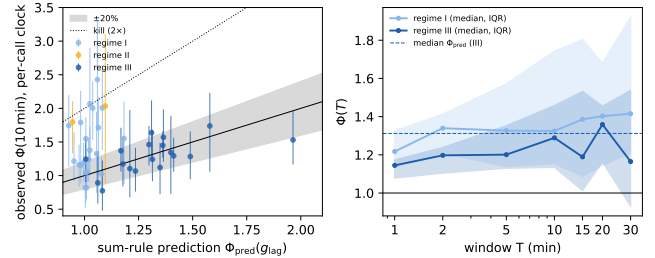
Bottom line. In regime III, talk obeys a fluctuation–response sum rule with no free parameter: the collective Fano ratio equals $1/(1 - g_{\text{lag}})^2$ to within 9% (pooled over 18 units), so the read-out loop accounts for all collective talk variance.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH342 proposed a parameter-free check of the read-out loop gain. For a linear point process with branching ratio g , collective count variance at long windows exceeds the independent-agent value by $1/(1 - g)^2$. H111 compares this sum rule, with H67's g_{lag} , against the collective Fano ratio Φ of 10-min talk counts in 66 non-holdout units (35 periods). In regime III the rule closes: the pooled ratio $r_F = \Phi/\Phi_{\text{pred}}$ is 1.01 [0.94, 1.09] over 18 units, with no between-unit spread, and $\Phi - 1$ tracks g_{lag} (ρ 0.60). No unit reaches the kill ($r_F \geq 2$). In regime I, where $g_{\text{lag}} \approx 0$, talk co-varies $1.34 \times [1.18, 1.52]$ more than the loop allows. The holdout is not run.



Observables. Left: observed $\Phi(10 \text{ min})$ (per-call clock, all-present window, hour-block demeaned) against the parameter-free prediction from H67's g_{lag} , per unit with 95% CIs. Regime-III units (dark) lie on the line; regime-I units (light) sit above it at $\Phi_{\text{pred}} \approx 1$. Right: $\Phi(T)$ medians and IQRs. Regime III is flat from 5 to 30 min near the median prediction, so no slow field remains.

PHYSICAL MODEL

Linear multivariate Hawkes process on the call clock (model 09) with a fluctuation decomposition (model 14). Each agent has private talk noise of any form (clocked calls, own bursts). Reading couples agents through a cross-response matrix K with eigenvalue g on each room's collective mode. n_i is agent i 's talk count in a window, Φ the ratio of collective to summed private variance, N the present agents, n_b the room sizes. Φ divides out every private noise level.

$$C(T)/T \rightarrow (I - K)^{-1} \Sigma (I - K)^{-\top}, \quad \Phi(T) = \frac{\text{Var} \sum_i n_i}{\sum_i \text{Var} n_i}$$

$$\Phi_{\text{pred}}(g) = \frac{N(1 - g)^{-2}}{\sum_b [(1 - g)^{-2} + \frac{n_b - 1}{(1 + g/(n_b - 1))^2}]} \approx \frac{1}{(1 - g)^2}$$

$$r_F = \Phi(10 \text{ min}) / \Phi_{\text{pred}}(g_{\text{lag}})$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, H67's read-out gain is the whole story of collective talk clustering in the current scaffold: a push is amplified by $1/(1 - g) \approx 1.15$ and the swarm's talk variance by $1/(1 - g)^2 \approx 1.32$, with nothing left over for a hidden field at 10–30 min. A monitor can use Φ on 10-min talk counts as a cheap, label-free check of the coupling gauge: Φ above $1/(1 - g_{\text{lag}})^2$ by more than ~ 0.3 flags a field or a coupling the read-out estimator misses. In regime-I-style scheduled chat, expect that excess ($\Phi \approx 1.3$ with $g_{\text{lag}} \approx 0$).

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 regime III $r_F \in [0.8, 1.2]$ in $\geq 2/3$	$r_F \in 14/18$; pooled 1.01 [0.94, 1.09]	supported
K kill: $r_F \geq 2$ in $> 1/2$	0/18	not met
P2 untrimmed trimmed	$> 11/18$; gap +0.03 (raw +0.22)	failed
P3 excess rises with T	1/17	failed
P4 regime I $\Phi > 1.2$ in $\geq 2/3$	13/21; pooled r_F 1.34	failed narrowly
P5 $\rho(\Phi - 1, g_{\text{lag}}) \geq 0.3$	all 0.09; regime III 0.60	failed / III yes
P6 within $>$ cross room	10/12; cross > 0 in 0/12	mixed
P7 call clock lowers Φ	13/18	supported
NE43 bookends: trim flat, raw falls	-0.01, +0.08; +0.23 $\rightarrow$ -0.05	supported
NE42 Φ drops in #40	-0.04 vs predicted 0.41	failed

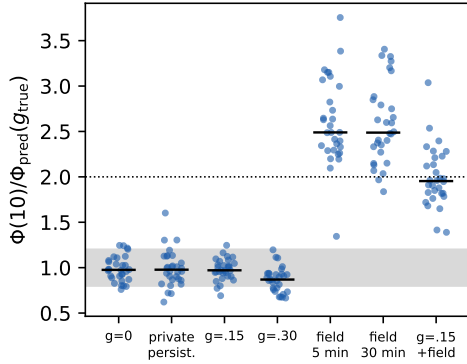
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	2	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Per-unit CIs are ± 0.3 – 0.5 , so the HH's $\pm 20\%$ per unit is not resolvable; the pooled regime-III ratio carries the claim. A fast field ($\tau \leq 5$ min) and a missed coupling look alike in $\Phi(T)$, so the regime-I excess is unattributed. The per-call clock was adopted after the synthetic run showed a call-grid field (Amendment A1, before real data). The block-shift identity $F/F_{\text{shift}} = \Phi$ fails within 10% in half the units. Predictions were written knowing H67's g_{lag} .



Second figure. Synthetic validation on the real call grids of units 27, 41 and 51e (10 replicates per world). Coupling alone is recovered within $\pm 3\%$ at $g = 0.15$ and reads 5–20% low at $g = 0.30$. Private persistence leaves Φ at 1. Shared rate fields of 5 or 30 min give $\Phi/\Phi_{\text{pred}} = 2.3$ – 2.9 .

ROUND 2

Recompute Φ_{pred} in regime I with a next-chat-call gain; split the regime-I excess with a window-level in-flight placebo; run the frozen confirm after H67's.

- **Essence.** a closure test that says, per period, how much collective talk variance the measured coupling explains and how much is field.
- **H111-R1** Regime-I clock.
- **H111-R2** Fast-field separation.
- **H111-R3** NE42 #40.
- **H111-R4** Holdout.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 ✓ supported
G07 d descriptive
G08 ✓ supported
G10 ~ mixed
G11 d descriptive
G12 d descriptive
+28 more periods: see the card.

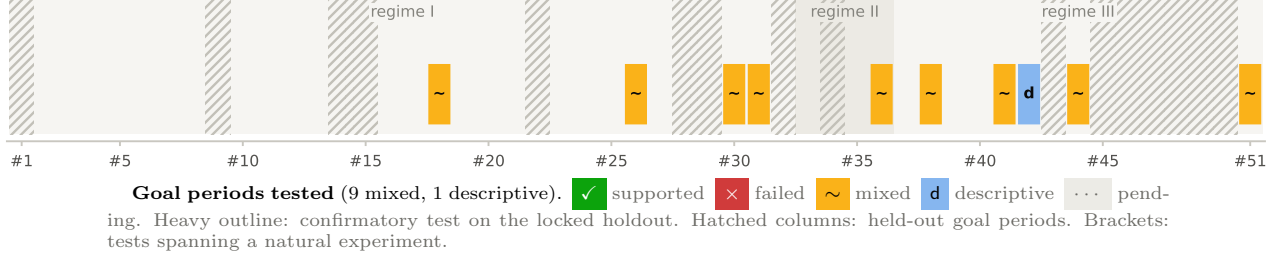
H112: Crossing claims anti-coordinate: parallel updates make 2-cycles

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 1/5

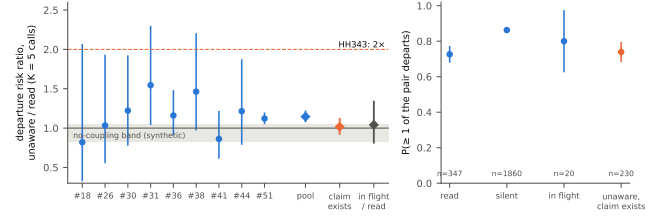
Bottom line. Co-switch departures do not depend on whether the movers had read each other once chat engagement is matched (RR 1.02 [0.92, 1.13]); HH343's 2× crossing effect and 2-cycle are absent.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H112 tests a Little-model prediction: when two agents move onto the same project without having read each other (a parallel update), they should both leave again (a period-2 cycle); when one moved after reading the other (a sequential update), the move should stick. Units are co-switch pairs: two agents' call-level switch-ins onto one project within 15 min, classed by the context ledger as read, in flight or silent. Over 9 testable non-holdout periods (2,227 pairs), unaware pairs depart within 5 calls 1.15 [1.08, 1.22]× as often as read pairs, not the predicted $\geq 2\times$. With a partner claim present in both classes the ratio is 1.02 [0.92, 1.13]. There is no mutual leaving and no departure at the read. The holdout is not run.



Observables. Left: departure risk ratio, unaware over read pairs, per testable period (95% CI), the pool, the pool with a partner claim present in both classes (orange; post hoc A2) and the literal crossing contrast (in flight over read, 20 pairs). Gray band: the no-coupling range on the real schedules. Dashed: HH343's 2×. Right: departure rates by class; silent pairs (no claim) depart most.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Two agents that land on the same project at the same time do not bounce off each other. Their fate does not depend on who read whom once chat about the project is matched. Most simultaneous moves are silent: nobody wrote about the project, and the "move" is often a glance. An operator need not stagger agents or force them to read before claiming work to avoid thrashing: at village cadence true crossings are rare (1% of co-switches) because a message is read in about 20 s.

PHYSICAL MODEL

Model 02's update-rule table with model 10's categorical states. Agent i holds project σ_i . At a call it updates with probability p_u (plus p_r after reading news). It draws a project from a logit in the fields h and the number $\hat{n}_{ij}$ of other agents it *believes* are on j . Beliefs change only at the ledger read-out of a claim, so the read-out delay sets the update order. For $J < 0$ a parallel pair is frustrated once both read: Little's 2-cycle.

$$P(\sigma_i \rightarrow j) \propto \exp(h_j + J \hat{n}_{ij}(t)), \quad \hat{n}_{ij} \text{ updated at read-out only}$$

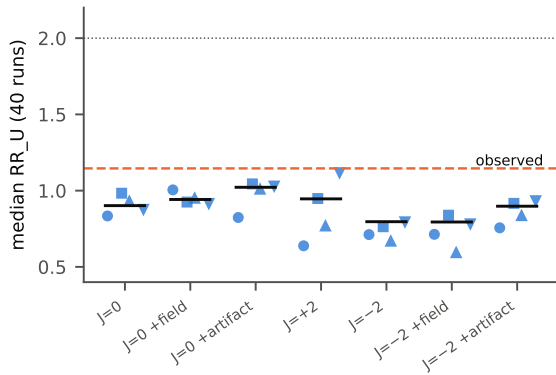
$$RR_U = \frac{P(\geq 1 \text{ departs in } K=5 \text{ calls} \mid \text{unaware})}{P(\geq 1 \text{ departs} \mid \text{read})} \quad (\text{MH over lag } \times \text{ named})$$

$$M = \frac{P(\text{both depart})}{P(i \text{ departs}) P(j \text{ departs})}$$

H112-crossing-claims-two-cycles · generated 2026-10-07 17:49 by in-fra/summaries/build_summaries.py · card hypotheses/H112-crossing-claims-two-cycles/README.md · ratings: H112 round-1 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $RR_U \geq 2$, CI > 1	1.15 [1.08, 1.22], 9 periods	failed
P1' in flight vs read ≥ 2	1.04 [0.80, 1.35], 20 pairs	failed (descr.)
P2 mutual leaving after unaware	M 1.07 vs 1.19 (read)	failed (descr.)
P3 sequential sticks	S 0.70, but biased without coupling	not a test
P4 own-role $\geq 1.5\times$ shared	ratio 0.88 [0.73, 1.06]	failed
P5 $RR_U > 1$ in $\geq 2/3$	7/9 periods	supported
N G51 departure at the read	0.79 [0.37, 1.64]	failed
N G44 free vs assigned arm	#best: 0 read pairs	descriptive
A2 claim present (post hoc)	1.02 [0.92, 1.13], 577 pairs	—



Second figure. Synthetic worlds on the real call schedules of #18, #31, #38 and #51 (40 runs; markers per schedule, bars the pool). No world reaches $RR_U = 2$. Read-gated avoidance ($J = -2$) gives $RR_U \approx 0.8$: the first mover reads the second mover's claim and leaves. The observed 1.15 (dashed) lies above every world: it is engagement selection, not coupling.

GOAL PERIODS IN DETAIL

G18 ~ mixed
G26 ~ mixed
G30 ~ mixed
G31 ~ mixed
G36 ~ mixed
G38 ~ mixed
G41 ~ mixed
G42 d descriptive
G44 ~ mixed
G51 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Departures use touch labels, so 73–86% of pairs have a departure within 5 calls; that caps any risk ratio near 1.35. The engagement control and the risk differences are post hoc (A2). In-flight pairs number 26 in all, so the literal crossing test is unpowered. The work channel (119 pairs) gives 0.99 [0.84, 1.18]. Regime I/II call starts are modelled, not logged.

ROUND 2

Joins from DQ4 commits or a two-touch rule (base rate below 50%); first- vs second-mover departures in read pairs of #42 and #51, the sign that read-gated avoidance predicts.

- **H112-R1** Measure joins, not touches.
- **H112-R2** Crossings by design.
- **H112-R3** Read-gated avoidance's own sign.

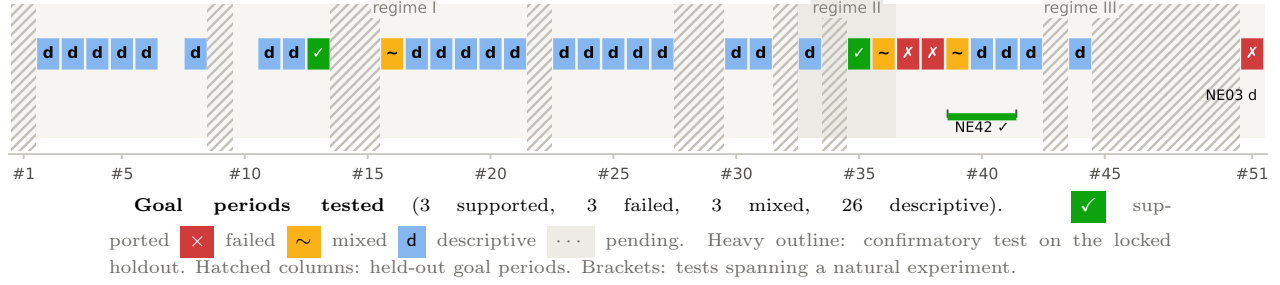
H113: Read-out channel capacity: information per call grows as $k^{(1-\beta)} \approx k^{0.34}$

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Content uptake per read-out call grows as $k^{0.2-0.3}$ (wake batches 0.31 [0.25, 0.37]), excluding both a one-message capacity and H18's corrected $k^{0.50}$; HH345's 0.34 is excluded by the talk-call pool.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

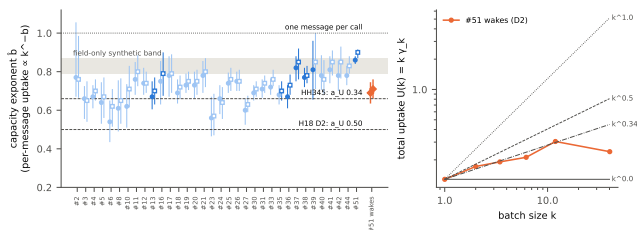
H113 measures how much content an agent takes up from a batch of k peer messages read before it speaks. Each row is a talk call: the reader's next statement, orthogonalized to its previous statement and the goal, is regressed on the summed batch vectors with a gain $\gamma_1 k^{-b}$. HH345 predicted total uptake $\propto k^{0.34}$; the coordinator later corrected H18's value to $k^{0.50}$. On #51 timer-wake batches, where others set k , $a_U = 1 - b = 0.31$ [0.25, 0.37]. Pooled over the 6 periods identified against a topic field, talk calls give 0.23 [0.16, 0.31]: the pre-set kill (0.34 excluded) is met, and 0.50 is excluded everywhere. The holdout is not run.

PHYSICAL MODEL

Models 04 and 12: a linear read-out channel into the destination's next statement, with an in-flight placebo source. x_m : DQ5 style_resid32 vector of batch message m ; y : the reader's statement; P removes the reader's previous statement and the goal directions. $\gamma(k)$ is the per-message uptake slope, $U(k) = k\gamma(k)$ the total uptake per call. $b = 0$: no bottleneck; $b = 0.66$: HH345; $b = 1$: one message per call.

$$Py = \gamma_1 k^{-b} \sum_{m \in B} Px_m + \gamma_F \sum_{m \in F} Px_m + \epsilon, \quad U(k) = k\gamma(k) \propto k^{a_U}, \quad a_U = 1 - b$$

$$I(k) = -\frac{D}{2} \log_2(1 - R_{cv}^2(k)), \quad a_I \approx 1 - 2b + r \quad (\text{low SNR})$$



Observables. Left: capacity exponent $\hat{b}$ per testable period (95% CI; circles bge, squares gte; dark = identified against a field by the

placebo rule) and on #51 timer-wake batches (orange). Dashed lines mark HH345 ($a_U = 0.34$) and the corrected H18 value (0.50); dotted, one message per call. Gray band: what a topic field alone produces. Right: total uptake $U(k)$ on the wake batches, against reference slopes; it grows about as $k^{0.3}$ up to $k \approx 12$ and then flattens.

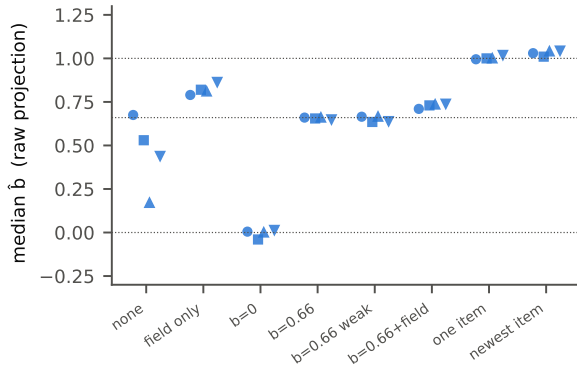
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Reading more messages before speaking helps, but with steep diminishing returns: ten messages buy about twice the content of one. An operator who wants an agent to act on a specific message should send it alone or make it the newest; burying it in a batch of ten dilutes it about fivefold. The bottleneck is per call, so faster call cadence, not longer batches, raises throughput.

H113-readout-channel-capacity · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H113-readout-channel-capacity/README.md · ratings: H113 round-1 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1a pooled $a_U \ni 0.34$ vs corrected 0.50 (N1)	0.23 [0.16, 0.31] bge; 0.22 gte excluded in every identified period	failed (kill) failed
P2 $b > 0$; P3 $b < 1$	6/6 and 6/6 (bge)	supported
P4 read beats in flight	+0.014 [0.006, 0.022]; γ_F/γ_1 0.31	supported
P5 $a_I \approx 1 - 2b + r$	7/11 within 0.2	failed (narrow)
P6 two models agree	34/34 within 0.2	supported
P7 newest item favoured	+0.034 [0.025, 0.043]	supported
N1 D2 wakes = talk calls	0.69 vs 0.86 ($\Delta - 0.17$)	failed
N2 NE03 fetch limit	#10a underpowered	descriptive
N3 NE42 merge invariance	$ \Delta \hat{b} \leq 0.09$, $k \times 1.5$	supported (weak)



Second figure. Synthetic worlds on the real read-out skeletons of #18, #31, #38 and #51 (10–20 runs): the estimator recovers $b = 0, 0.66$ and 1 within 0.05. A topic field with no reading gives $\hat{b} \approx 0.8$, so only periods whose read messages beat in-flight messages at matched age are scored.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G08 d descriptive
G11 d descriptive
G12 d descriptive
G13 ✓ supported
G16 ~ mixed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

A time-local field mimics the exponent; 27 of 34 testable periods fail the placebo rule and are not scored. In-flight messages co-respond to the same batch, so the placebo is conservative. Talk-call batch sizes are partly reactive; the wake design avoids that. The linear channel cannot tell averaging from picking one item. Synthetic runs were cut to 10 per world on three skeletons by machine load.

ROUND 2

Exogenous- k designs in every regime-III period; a selection-vs-averaging mixture on per-item alignment; capacity as an agent constant.

- **H113-R1** A field-robust exponent.
- **H113-R2** Selection vs averaging.
- **H113-R3** Capacity as an agent constant.

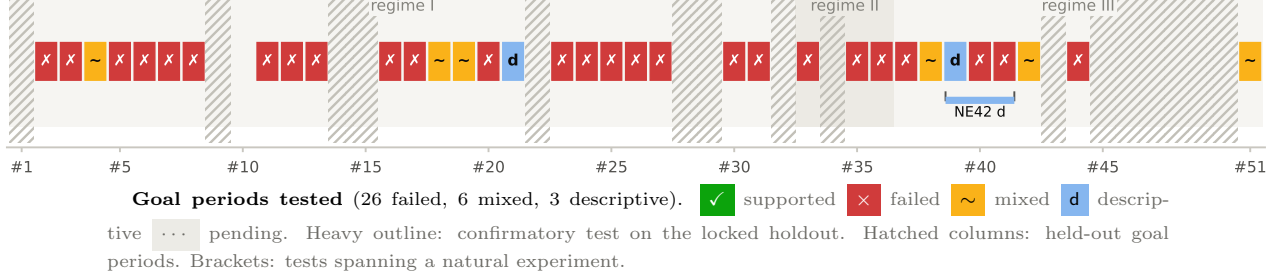
H114: A Griffiths phase: rare strong pairs give a subcritical swarm heavy-tailed talk

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1/5

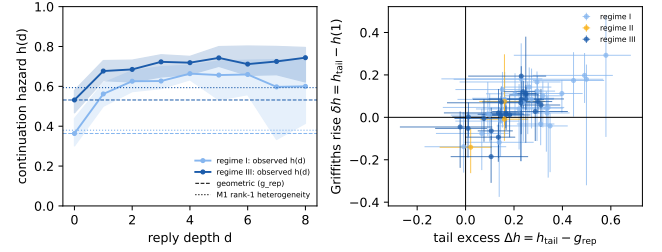
Bottom line. Reply chains are heavier than uniform subcritical branching because of first-order thread momentum, not because rare strong agent pairs form locally supercritical regions.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH346 proposed a Griffiths phase: a swarm subcritical on average whose rare strongly coupled pairs make reply chains heavy-tailed. H114 measures reply depth on DQ2 parent links in 66 non-holdout units (54 usable). Chains are heavier than a uniform subcritical process: the tail continuation hazard exceeds the reply branching ratio in 52/54 units (CI > 0 in 33/54). The excess sits at depth 1: a reply's parent is itself a reply more often than the mean allows (+0.16, 54/54 units). The hazard then rises little (significant in 10/54). Strong pairs (pair gain > 0.5) exist in 18/66 units, are half of them readers answering three hub authors, and carry the tail in 1/16. **The Griffiths mechanism fails**; thread momentum explains the tail. The holdout is not run.



Observables. Left: continuation hazard $h(d)$ by reply depth (median and IQR over units), against the geometric law at g_{rep} (dashed) and the rank-1 agent model M1 (dotted). The hazard jumps at depth 1 and stays nearly flat. Right: per-unit tail excess Δh against the Griffiths rise δh . Most units have $\Delta h > 0$ with δh near 0.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

A reply forest (models 09, 03, 01). Each agent message has at most one agent parent. Its depth D counts links to the root. A uniform subcritical process gives a geometric depth law at the reply branching ratio g_{rep} . A Griffiths phase mixes continuation rates: strong pairs (i, j) with pair gain g_{ij} near 1 make the hazard $h(d)$ rise with depth. Thread momentum makes h jump at $d = 1$ and stay flat. M1 and M2 are first-order author chains without and with pair-specific transitions.

$$S(d) = P(D \geq d), \quad h(d) = \frac{S(d+1)}{S(d)}, \quad g_{\text{rep}} = S(1)$$

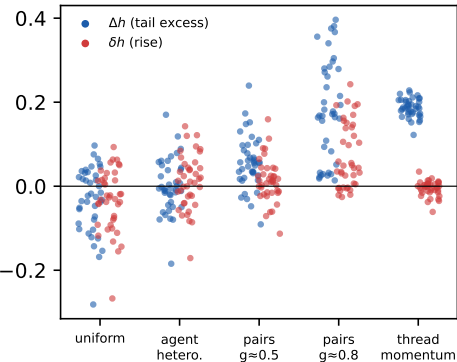
$$h_{\text{tail}} = \frac{\sum_{d \geq 4} S(d+1)}{\sum_{d \geq 4} S(d)}, \quad \Delta h = h_{\text{tail}} - g_{\text{rep}}, \quad \delta h = h_{\text{tail}} - h(1)$$

$$g_{ij} = L_{ij}/R_{ij}, \quad \text{Mk} : s_d = p \odot (P_k s_{d-1})$$

H114-griffiths-phase-pairs · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H114-griffiths-phase-pairs/README.md · ratings: H114 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 $\Delta h > 0$ in $\geq 2/3$	CI 33/54; point 52/54	failed narrowly
P2 rise $\delta h > 0$ in $\geq 1/2$	10/54 (median +0.05)	failed
P3 strong pair in $\geq 1/2$ of units	18/66; 37 pairs; 4 mutual	failed
P4 strong cut carries the tail	1/16	failed
P5 M2 beats M1	39/54, M2-M1 +0.003	null in size
P6 tail above M2 (momentum)	31/54	supported
P7 $\rho(\Delta h, \text{strong share}) \geq 0.4$	-0.14; exponent unidentifiable	failed
P8 naming $\geq 1.2\times$; no family	0.73 $\times$; OR 0.31	mixed
G51 assigned pairs strong / higher g	0/176 / MW p 0.0003	failed / yes



Second figure. Synthetic forests planted on real message skeletons (4 units, 10 replicates). Uniform and agent-heterogeneity worlds give $\Delta h \leq 0$. Strong pairs at $g \approx 0.8$ give $\Delta h > 0$ and a rise $\delta h > 0$; at $g \approx 0.5$ they are invisible. Thread momentum gives $\Delta h \approx 0.19$ with no rise.

GOAL PERIODS IN DETAIL

G02 $\times$ failed
G03 $\times$ failed
G04 $\sim$ mixed
G05 $\times$ failed
G06 $\times$ failed
G07 $\times$ failed
G08 $\times$ failed
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		
Level: hypothesis . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

CAVEATS

DQ2’s parent rule favors named partners and its reply probability keeps a thread-membership component, so label-driven momentum is possible (R4); a label-free reply proxy is needed. Pairs with $g \approx 0.5$ are not detectable at these counts (Amendment A1). One parent per message projects a richer reply graph. P1’s CI count (61%) misses its 2/3 bar. Predictions were written knowing H62’s thread-field result.

ROUND 2

Rebuild chains from ledger timing without DQ2 labels; split momentum into read vs in-flight parents; model hub authors as fields.

- **Essence.** whether swarm talk has a disordered, locally near-critical structure an operator could target, or only conversation-level memory.
- **H114-R1** Label-free momentum.
- **H114-R2** Momentum as a field or a coupling.
- **H114-R3** Hubs.
- **H114-R4** Holdout.

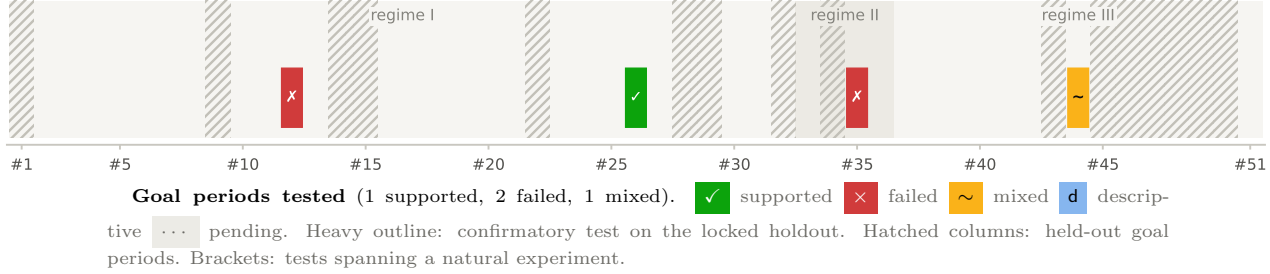
H115: Blind role recovery in the debate week: the judge is a sink of antisymmetric coupling (#12)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. The judge is not recoverable blind as a talk sink (rank 1 in 3/10 debates; a sink ≥ 0.6 is excluded), and debate talk couples across teams, not within them ($J_{ST} - J_{OT} = -0.16 \pm 0.08$, $p = 0.01$).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

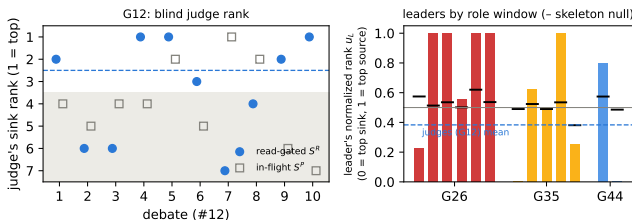
HH353 asked whether the per-call talk kinetic Ising, fitted blind to timing, ranks each debate's judge as the top sink of antisymmetric coupling. In the ten #12 debates (7 agents, 3,644 calls) the ranking was frozen and hashed before any role label was read. The judge ranks first in 3 of 10 debates (median rank 2.5, mean normalized rank 0.38; uniform $p = 0.15$, field-only skeleton $p = 0.24$). The predicted team blocks come out reversed: debaters talk after reading an opponent, not a team-mate. As replication, the #26 elected leader is the top talk source in 4 of 6 windows; #35 lead designers and the #44 leader show nothing. The holdout is not run.

PHYSICAL MODEL

Model 02. Spin $s_i(c) = \pm 1$: call c of agent i talks or not, on the ledger call clock. Input $x_{ij}(c) = 1$ if j posted in i 's room since i 's previous call (read-gated); x_{ij}^P : posted during the call (in-flight placebo). Fields h : agent $\times$ call mode and agent $\times$ debate phase. Couplings $J_{ij} = \alpha_i + \beta_j$ (in, out); sink score $S_i = \text{in} - \text{out}$. Pair-type model: J by same team, opposite team, judge and bench.

$$P(s_i(c) = +1) = \frac{e^{H_i(c)}}{2 \cosh H_i(c)}, \quad H_i = h_{i,\kappa} + h_{i,\phi} + J_{ii}s_i(c^-) + \sum_j J_{ij}x_{ij} + \sum_j J_{ij}^P x_{ij}^P$$

$$S_i = \sum_j (J_{ij} - J_{ji}), \quad u = \frac{\text{rank}(S) - 1}{n - 1} \quad (0 = \text{top sink})$$



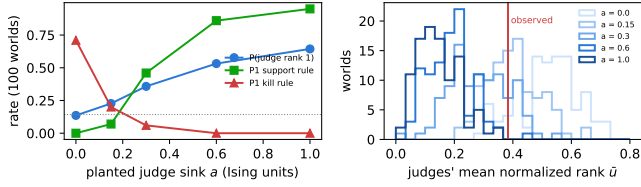
Observables. Left: the judge's blind sink rank in each debate under read-gated (blue) and in-flight (grey) couplings; the shaded band is the HH kill region. Right: the designated leader's normalized rank per role window (bars) against its field-only skeleton null (black ticks); 1 is the top source. The #26 leader is a source; judges lean slightly toward the sink end.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: one-way roles do not show up as one-way talk timing at debate scale. A judge whose talk follows debaters' talk by ≥ 0.6 Ising units per read debater is excluded, so blind role detection from who-talks-after-whom fails for 7 agents and 300 calls per episode. Opposition does show up: opponents answer each other (cross-team J 0.20 vs within-team 0.04). A cheap opposing-camp detector is therefore cross-block talk coupling, not within-block coupling. An elected leader can be the top talk source, but a single-agent rank needs a skeleton null (its null mean is 0.38 to 0.62, not 0.5).

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 judge top sink median 2.5, rank 1 in 3/10; $\bar{u} = 0.38$ (median ≤ 2 , rank 1 in ≥ 4) $p = 0.15 / 0.24$		not supported
P1b within-agent contrast > 0	+0.39 (perm. $p = 0.12$)	not supported
P1c read beats in-flight	mean rank 3.3 vs 4.0	direction only
P2 $J_{ST} - J_{OT} > 0$	-0.16 ± 0.08 ; perm. $p(\leq) = 0.01$	failed (reversed)
P3 rulings as steps	$J_{DJ} - J_{DJ}^P = +0.02 \pm 0.11$; $ \Delta h $ in band	failed
R1 leaders in half ($\geq 2/3$)	9/13 pooled; G26 5/6, G35 3/5, G44 1/2	G26 only
Synthetic (F)	size ≤ 0.01 ; 80% power at sink $a = 0.6$; team power 0.87	passed



Second figure. Synthetic validation on the real #12 skeleton and call order (100 worlds per point). Left: judge rank-1 rate, support rule and kill rule against the planted sink a . Right: the judges' mean normalized rank in planted worlds; the observed value (red) sits at the median of $a = 0.15$ worlds and excludes $a = 0.6$.

GOAL PERIODS IN DETAIL

G12 $\times$ failed
G26 $\checkmark$ supported
G35 $\times$ failed
G44 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Regime-I talk may not run on the all-call clock (H67); the chat-mode clock variant moves the judge to median rank 4 but has little power. Per-debate fits use 170–630 calls, so only large role signals are resolvable. The pooled judge in- vs out-coupling ($J_{JD} = 0.28$, $J_{DJ} = 0.07$) points the predicted way but uses labels and is post hoc. The leader-as-source replication rests on one unit (Fisher $p \approx 0.08$). The verdict step's sign reversal was found after the pre-registered $|\Delta h|$ rule failed.

ROUND 2

Blind team recovery from cross-block coupling (antiferromagnetic ground state); split reads into named and unnamed messages; confirm the focal-agent source rank on NE37 (#48).

- **H115-R1** Blind team recovery from cross-team coupling.
- **H115-R2** Addressed reads.
- **H115-R3** Leader as source, more units.
- **H115-R4** Regime-I clock.

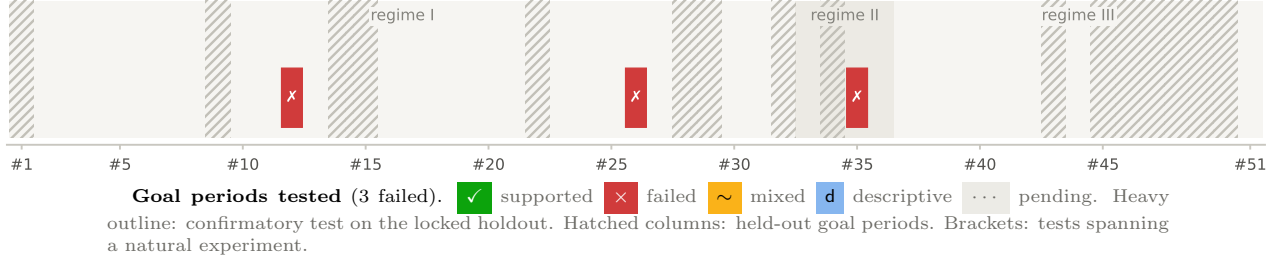
H116: The election as a coupling step: influence should flow toward the winner after the result (#26)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. Winning the election did not raise the per-call talk response to the winner ($\Delta J_{out} = +0.09 \pm 0.33$; a step ≥ 0.75 excluded), and designated judges and lead designers carry no out-coupling step; a post hoc step at the re-election is unexplained.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

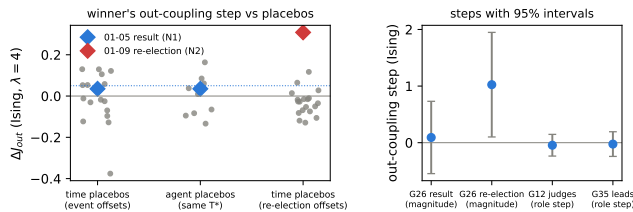
HH354 asked whether winning the #26 election steps up the winner's out-coupling: after the result at 19:35 UTC on 2026-01-05, do the other nine agents talk more often after reading agent 17? In 83-min windows on both sides the step is $+0.09 \pm 0.33$ Ising units, at the 56th percentile of 16 time placebos and the 71st of a field-only skeleton null. The in-coupling stays flat and EP is below resolution. Debate judges (#12) and daily lead designers (#35) carry no out-coupling step either. The pre-registered negative control failed: the 01-09 re-election shows a step of $+0.31 \pm 0.21$, above every placebo. The holdout is not run.

PHYSICAL MODEL

Model 02 on the call clock. Spin $s_j(c) = \pm 1$: call c of agent j talks or not. Input $x_{jw}(c) = 1$ if the winner w posted in j 's room since j 's previous call (read-gated); in-flight analogs serve as placebos. Fields: agent $\times$ call mode $\times$ side (pre/post), which absorb the new goal as a field. Couplings: J_{out} (readers' response to w), J_{in} (w 's response to others), each with a post-result step Δ .

$$P(s_j(c) = +1) = \frac{e^{H_j}}{2 \cosh H_j}, \quad H_j = h_{j,\kappa,\text{side}} + J_{jj}s_j(c^-) + (J_{out} + \Delta J_{out} \text{ post})x_{jw} + (J_{oth} + \Delta J_{oth} \text{ post})X_{j,-w}$$

$$\text{detect} \iff \Delta J_{out} \geq 0.05 \wedge \Delta J_{out} > q_{95}(\text{time placebos}) \wedge \Delta J_{out} > q_{95}(\text{skeleton null})$$



Observables. Left: the winner's out-coupling step ($\lambda = 4$) at the election result (blue) and at the re-election (red) against time

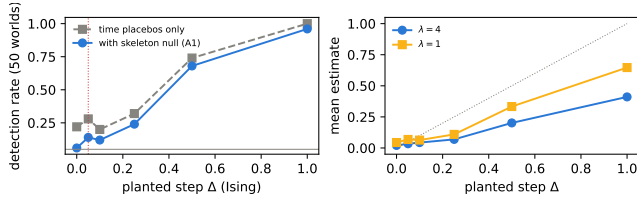
placebos at matched offsets and agent placebos. Right: unpenalized steps with 95% intervals for the two #26 events and the pooled role steps of #12 judges and #35 lead designers.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: electing or appointing a leader does not, by itself, make the other agents answer that agent more per call. A step of ≥ 0.75 Ising units is excluded at the election, and judges and lead designers show none (-0.04 ± 0.10 , -0.02 ± 0.11). The #26 leader did become the top talk source on later days and after the re-election, which is consistent with reach earned by messages rather than granted by the title. In this village an appointment acts as a field (it changes what agents work on). One event cannot resolve the 0.05 step HH354 asked about.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
N1 $\Delta J_{out} \geq 0.05$, beyond placebos	+0.035 (λ 4); pct 0.56 time, 0.71 skeleton; +0.09 $\pm$ 0.33	not detected
N1b ΔJ_{in} in band	-0.004 in [-0.15, +0.18]	consistent
N1c winner agent placebos	above pct 0.56 of 9	failed
N2 no step at re-election	+0.31 $\pm$ 0.21, above all placebos	failed
N3 read step > in-flight	-0.02 $\pm$ 0.28	failed
N4 EP rises	+0.002 nats/call; bound ≤ 0	failed
R role steps (#12, #35)	-0.04 $\pm$ 0.10 (p 0.76); -0.02 $\pm$ 0.11 (p 0.83)	failed
Synthetic (F)	size 0.06 with both nulls; power 0.14 at 0.05, 0.96 at 1.0	passed



Second figure. Synthetic validation on the real #24–#27 skeletons (50 worlds). Left: detection rate against the planted step; time placebos alone are anti-conservative (size 0.22), the skeleton null restores 0.06. Right: ridge shrinkage of the estimate ($\lambda = 4$ halves it), so magnitudes come from a $\lambda = 0.01$ fit.

GOAL PERIODS IN DETAIL

G12 × failed
G26 × failed
G35 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

One event gives a step SE of about 0.2 (shrunk) to 0.33 (unpenalized), so HH354's 0.05 is untestable here. The re-election step rests on 44 / 54 exposed reader-calls, untrimmed windows (A0) and post hoc placebo checks. Regime-I talk may not run on the all-call clock (H67). The multipartite EP bound is ≤ 0 at 10 agents and 1,500 calls, so N4 has no power. Judges' and leads' role steps are ridge-shrunk.

ROUND 2

Model the leader's out-coupling as a function of its directive messages; pool all designations hierarchically for an SE near 0.05; pre-register the day-scale step from 01-05 to the term days; confirm on NE37 (#47 → #48).

- **H116-R1** Activity, not title.
- **H116-R2** Pool designations.
- **H116-R3** Day-scale step.
- **H116-R4** EP instrument.

H116-election-coupling-step · generated 2026-10-07 17:49 by infra/summaries/build-summaries.py · card hypotheses/H116-election-coupling-step/README.md · ratings: H116 agent (suggested) (page 2)

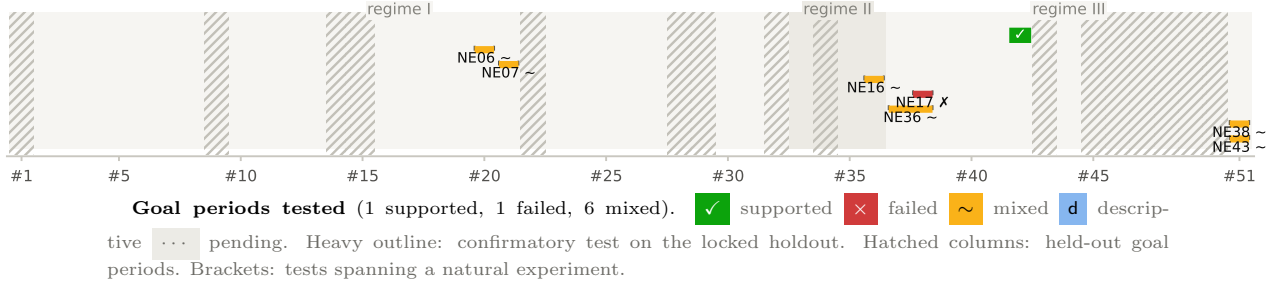
H117: J should not move when the rules do: the mid-goal reset in #32 (NE35)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

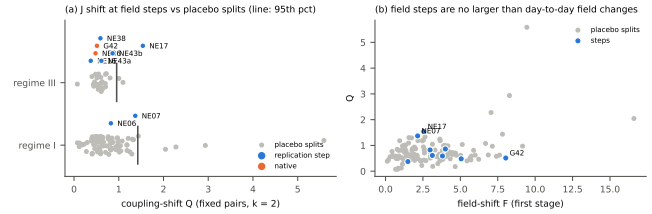
Bottom line. The fitted talk-spin coupling matrix stays at its day-to-day noise level across field steps, including the one goal switch that clearly moved the fields; most documented rule changes are not field steps in the talk data.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH355 says that a pure field change (a rule reset, a tool change, a new goal) moves each agent's field h_i but not the couplings J_{ij} . A fitted J that jumps at such a step is a field artifact. The target, NE35 in #32, is held out. H117 fits a kinetic Ising model to minute talk spins on both sides of 9 non-holdout steps (2 days each side). It compares the change in J with 109 placebo splits at ordinary day boundaries. J stays at its placebo level, but most steps are not field steps in the data. The coupling shift Q exceeds the placebo 95th percentile at 1 of 9 steps (NE17, 7 pairs, $p = 0.037$, no field shift). The field shift F beats day-to-day changes at one step only: the #41→#42 goal switch. There $Q = 0.51$ ($p = 0.74$). The holdout is not run.



Observables. (a) Coupling shift Q at the steps (blue: replication; orange: natives) against placebo splits at ordinary day boundaries (grey; line: 95th percentile), by regime. Only NE17 lies beyond it. (b) Q against the field shift F . Steps sit inside the placebo cloud on both axes; G42 (goal switch) has the largest F and a low Q .

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: a goal or rule change does not rewire who responds to whom in talk timing. At #38-type sizes, a change of ± 1.2 logistic units on half the pairs would be seen with probability 0.85, and none is seen. Talk couplings carry over a reset. Only room moves changed them, and those changes were small (switched pairs $Q = 0.93$ against fixed pairs 0.61). For an analyst: a fitted J on talk spins with block fields is not a field artifact at this resolution. Most documented rule changes, however, do not move talk fields more than an ordinary night does.

PHYSICAL MODEL

Model 02, kinetic Ising with parallel updates. A talk spin $s_i(t) = \pm 1$ marks whether agent i posted chat in minute t . Each agent's next spin is a logistic function of its own spin, the 5-min boxcar m_j of every other agent, and a per-agent field for each (day, 30-min block). The fields absorb the schedule and any field step. Under HH355, h_i moves across a step while J does not. A second estimator works on the call clock. It compares messages read just before a call (J^R) with messages posted during it and not yet read (J^U).

$$\text{logit } P(s_i^{t+1} = 1) = h_i + \delta_i(b) + K_i s_i^t + K'_i m_i^t + \sum_j J_{ij} m_j^t$$

$$Q = \langle (J_{ij}^{\text{after}} - J_{ij}^{\text{before}})^2 / (\sigma_a^2 + \sigma_b^2) \rangle_{\text{fixed pairs}}, \quad J^s = \frac{1}{2} (J_{ij} + J_{ji})$$

$$F = \langle (\bar{h}_a - \bar{h}_b)^2 / (\sigma_a^2 + \sigma_b^2) \rangle_i$$

H117-coupling-invariant-rule-reset · generated 2026-10-07 17:49 by in-
fra/summaries/build_summaries.py · card hypotheses/H117-coupling-invariant-
rule-reset/README.md · ratings: H117 agent (suggested)

RESULTS VS PREDICTIONS

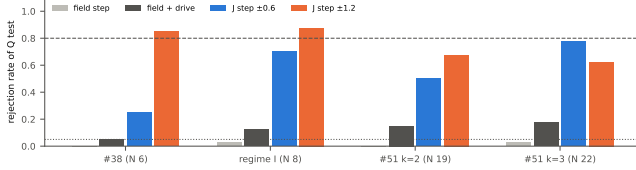
Prediction (locked)	Observed	Verdict
P0 size ≤ 0.10 ; power ≥ 0.8	size 0.00–0.03; power 0.85–0.88 at ± 1.2 , 0.25–0.78 at ± 0.6	partly
P1 Q $p > 0.05$ at $\geq 6/7$ steps	6/7 (NE17 p 0.037)	supported
P2 field step F at $\geq 4/7$	0/7 (p 0.37–0.85)	failed
P3 D_F within placebo	6/7	supported
P4 read-gated Q_R ; $J^R > J^U$	5/5; $J^R - J^U > 0$ on 9/10 sides	supported
P5 goal switches move J more	medians 0.50 vs 0.82	failed
G42 goal switch	Q 0.51 (p 0.74), F p 0.037	supported
NE38 one agent's role	its $\Delta \bar{h}$ rank 1/19, row Q rank 5/19	supported
HH kill ($\geq 2/7$ steps)	1/7	not fired

G42 ✓ supported
 NE06 ~ mixed
 NE07 ~ mixed
 NE16 ~ mixed
 NE17 × failed
 NE36 ~ mixed
 NE38 ~ mixed
 NE43 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic worlds on real step skeletons (80 null and 40 alternative worlds each). The Q test rejects a pure field step at ≤ 0.03 and a field step with a stronger common drive at 0.05–0.18. It detects rewiring of ± 1.2 on half the pairs at 0.63–0.88; at ± 0.6 only regime I reaches 0.7.

GOAL PERIODS IN DETAIL

CAVEATS

4–22 eligible agents per step, because agents with fewer than 10 talk minutes per side drop out (including both movers). The regime-III placebo pool has 26 splits, so $p \geq 0.037$. NE17's failure rests on 7 pairs. Q cannot see a global rescaling of all couplings. The naive estimator was not more fragile in the synthetic, so the field-leak rival is untested rather than rejected. The absolute J^R and J^U are confounded by call length; only their difference is read. NE35 sits on the regime I→II boundary.

ROUND 2

Run the frozen NE35/NE44 confirm after sign-off. Add log latency to the call model. Pool goal switches with fixed rooms hierarchically, using more days per side.

- **Essence.** is a fitted coupling a property of the read path or of the task? Settle it with steps that move fields far beyond nightly changes (goal switches with fixed rooms: #41 → #42, #39 → #40 within classes) and with more days per side.

H117-coupling-invariant-rule-reset · generated 2026-10-07 17:49 by in-fra/summaries/build_summaries.py · card hypotheses/H117-coupling-invariant-rule-reset/README.md · ratings: H117 agent (suggested) (page 2)

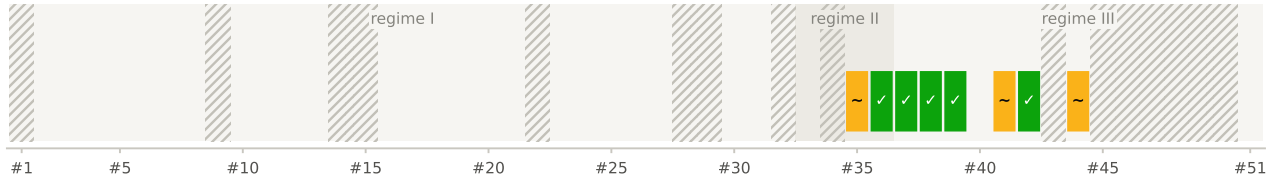
H118: The forked RPG is two replicas of one dynamics: damage spreading between rooms (NE15, #35)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. After the NE15 fork the rooms' content shared no same-day state beyond day 1 and was far from identical replicas from the first hour, while their code diverged slowly with a frozen core and correlated damage; the content damage time and any rules plateau are not identified at these counts.

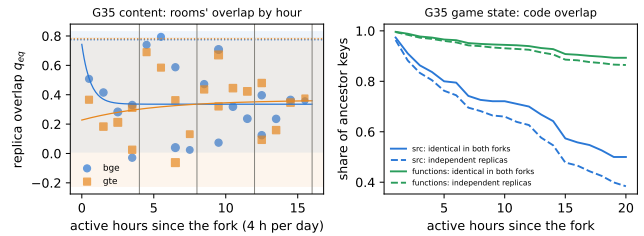
level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (5 supported, 3 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

HH358 treats the NE15 fork as a replica experiment. On 2026-03-16 the village split into #best (3 agents) and #rest (9), and each room took its own copy of the RPG built in #34: the same rules and starting state, different players. In kinetic Ising, two replicas of one dynamics either stay together (frozen or field-ordered phase) or separate at a rate set by the noise (damage spreading). H118 measures the overlap of the two rooms' content per active hour and the overlap of the two forks' code per hour. **Content and code behave oppositely.** The rooms' content is already far from identical replicas in the first hour and shares no same-day state after day 1. The code separates slowly on the edit clock, keeps a frozen core of untouched files, and its damage is correlated.



Observables. Left: hourly replica overlap of the #35 rooms (bge filled, gte squares; size = statement weight), with the fitted relaxation, the identical-replica level from joint relabels (dotted) and the identical-kickoff band (shaded). The overlap is low from the first hour and flat over five days. Right: the share of ancestor src files and functions identical in both forks (solid) against independent replicas (dashed). Code damage grows over 20 active hours and stays correlated.

PHYSICAL MODEL

Model 02 (replicas of a kinetic spin dynamics) with model-11 content spins. Content: $\mathbf{c}_r(b)$ is room r 's agent-weighted centroid of style-residualized statement vectors in active hour b , centred on a leave-own-period-out regime reference so that the shared static component stays in. Split halves give unbiased squared norms S_r . Code: each ancestor key (file, function) is a spin that keeps or changes its inherited value; two independent replicas are identical at a key with probability c_{ACB} .

$$q(b, b') = \frac{\mathbf{c}_A(b) \cdot \mathbf{c}_B(b')}{\sqrt{\langle S_A \rangle \langle S_B \rangle}}, \quad M(d) = q_{eq}(d) - q_{lag}(d), \quad q_{eq}(t) = q_\infty + (q_0 - q_\infty)e^{-t/\tau_D}$$

$$X = q_{code} - c_{ACB}, \quad F = P(\text{unchanged in both forks})$$

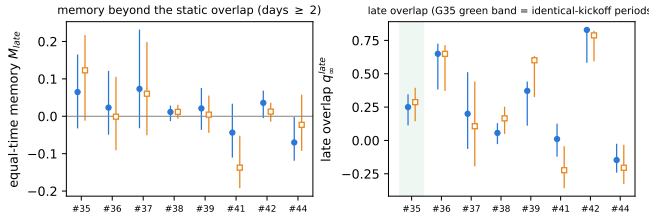
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator who forks a team onto copies of one artifact: the copies' conversations stop sharing a state within the first hour, so expect no residual coordination from the common start. The artifacts themselves diverge slowly and at the same hot spots: about half of the inherited code stays untouched in both copies, and identity exceeds independent copies by 0.12. A shared artifact acts as a field on where work lands, not on what the teams say. Two numbers to plan with: content overlap at the first hour 0.51 (vs 0.80 for random teams), code identity after five days 0.50.

H118-forked-rpg-replicas · generated 2026-10-07 17:49 by infra/summaries/build_summaries.py · card hypotheses/H118-forked-rpg-replicas/README.md · ratings: H118 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: memory size, τ_D , plateau, drive	memory test fixed to M_{late} (size 0.07–0.13); τ_D slow-world power 0.58; plateau PI power 0.16–0.29; drive caught 100%	1/4 as written (Amendment 1)
P1 G35 fast: $\tau_D \leq 4$ h, no memory	τ_D 0.6 h, CI 0.25–80 (gte 5.2 h, no decay); M_{late} 0.07 [−0.03, 0.17]	memory met; τ_D inconclusive
P2 no plateau; HH kill	q_{∞}^{late} 0.25 [0.11, 0.35], band rank 2/5 (36a 0.62); kill silent	consistent, unpowered
P3 code slow, frozen, $X > 0$	day 1 0.84 (pred. ≥ 0.85); end 0.50; F 0.50; X 0.116, p 6e-12	3/4
P4 replication: no memory $\geq 5/7$	6/7 in each model; $\tau_D \leq 4$ h	3/7 memory met
P5 control (G38, G44 below band median)	G38, G44 0.06, −0.15 (gte 0.17, −0.21)	supported
HH358	content fast-looking but unidentified; code slow and correlated	mixed



Second figure. Left: equal-time memory M_{late} (days ≥ 2) per two-room period with agent $\times$ bin bootstrap 95% intervals (bge filled, gte open). It covers 0 in #35 and 6/7 kickoff restarts; #44 (and #41 in gte) is negative: same-day anti-alignment. Right: late overlap q_{∞}^{late} . The two different-kickoff periods (#38, #44) are the lowest, as the control predicts; #35 sits inside the identical-kickoff band.

GOAL PERIODS IN DETAIL

G35 $\sim$ mixed
G36 $\checkmark$ supported
G37 $\checkmark$ supported
G38 $\checkmark$ supported
G39 $\checkmark$ supported
G41 $\sim$ mixed
G42 $\checkmark$ supported
G44 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

One fork event with 3 #best agents: agent-bootstrap intervals are wide, and in regime II no agent constants exist, so composition is part of the day-1 separation. The real-skeleton synthetic shows that τ_D and the plateau rule are not identifiable at these counts, so “no plateau” is not a negative result. Memory in this estimator sees only same-day shared state; slow memory shared across days is invisible to it. Code identity is keyed (moved code counts as changed), and the game-state predictions were written after reading H07’s day-level results.

ROUND 2

Hierarchical τ_D across the eight kickoffs; agent constants for regime-II agents from #33 and #36a; test whether #34 touch counts (H07’s C1) predict the shared hot spots behind X ; link same-day anti-alignment to room-specific operator messages. Confirmation on #45–#47 and #50 (frozen, not run).

- **Essence.** whether a shared artifact acts as a field on where forked teams work (code hot spots) while their conversations decorrelate at once.
- **H118-R1** Hierarchical damage time: pool the hourly relaxation across the eight kickoffs with period-specific τ_D and shrinkage (CLAUDE.md exception (d)); one period cannot identify it.
- **H118-R2** Composition-free overlap: agent constants for regime-II agents from #33 and #36a remove R-comp from G35’s day-1 separation.
- **H118-R3** Where damage lands: test whether H07’s #34 touch counts predict the shared hot spots behind the identity excess X (H07’s C1).
- **H118-R4** Same-day anti-alignment (G41, G44): link negative memory to room-specific operator messages per day.

H118-forked-rpg-replicas · generated 2026-10-07 17:49 by infra/summaries/build-summaries.py · card hypotheses/H118-forked-rpg-replicas/README.md · ratings: H118 agent (suggested) (page 2)

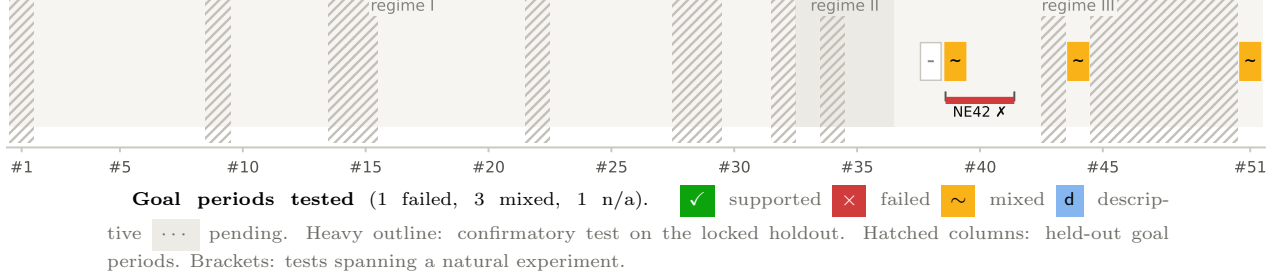
H119: The room merge switches a known adjacency on and off: recover it from the dynamics (NE42)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

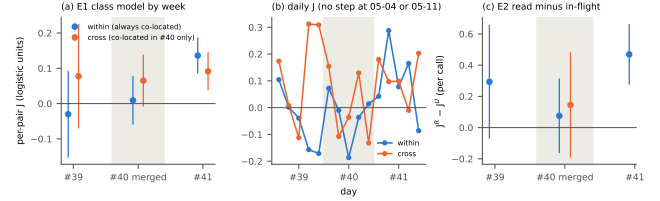
Bottom line. The NE42 merge did not switch on a coupling between the two groups; per-pair talk coupling vanished for all pairs in the merged room and returned within the old rooms after the split, which is attention dilution, not adjacency gating.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH359 says that NE42's merge, one week in which #best and #rest shared one room, switches the read path between the two groups on and off. A kinetic Ising coupling fitted from talk timing should follow it. It should be near 0 across the wall, positive in the merged week, back near 0 after the split, and carried by read-outs. H119 fits per-pair class couplings on minute talk spins for 11 agents over #39, #40 and #41. It adds a call-clock contrast of read against in-flight messages and four other room switches. **The cross coupling does not rise:** the merge contrast is $M = -0.02 [-0.13, 0.09]$ per pair, against a synthetic power of 0.95 at 0.3. Within-room pairs lose their coupling in the merged week as well, and regain it after the split (read minus in-flight 0.47 [0.27, 0.66]). HH359's kill rule fires. The holdout is not run.



Observables. (a) Per-pair class couplings by week (95% CI); grey band: the merged week. Cross pairs (orange) never rise, and within pairs (blue) are near 0 until #41. (b) Daily fits show no step on 05-04 or 05-11. (c) Read minus in-flight on the call clock: no class couples in the merged week; within pairs couple after the split.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: merging two groups into one room did not make them couple. For a week it removed measurable talk coupling for everyone: read-gated coupling of old room-mates went from 0.29 to 0.08 to 0.47 across #39, #40 and #41. To link two groups, keep rooms small and move or post across instead of merging. A read path is necessary for coupling but not sufficient: in a 14-agent room the per-pair coupling falls below what five 4-hour days resolve.

PHYSICAL MODEL

Model 02, kinetic Ising on talk spins with a block-structured J . Each pair class c (within, cross) shares one per-pair coupling J_c , and recipient fields per (day, 30-min block) absorb the schedule and the goal. Adjacency gating predicts $J_x \approx 0$ when the groups sit in different rooms and $J_x \approx J_w$ when they share one. Attention dilution predicts that per-pair coupling falls as the room grows. On the call clock, a read-gated coupling raises talk after messages read just before a call (J^R), not after messages posted during it and not yet read (J^U).

$$\text{logit } P(s_i^{t+1} = 1) = h_i + \delta_i(b) + K_i s_i^t + K'_i m_i^t + \sum_c J_c \sum_{j \in c(i)} m_j^t$$

$$M = J_x^{(40)} - \frac{1}{2}(J_x^{(39)} + J_x^{(41)}), \quad R_m = J_x^{(41)} - J_x^{(39)}$$

$$C_{RU} = J^R - J^U \quad (\text{read } [t_c - L, t_c] \text{ vs. unread } [t_c, t_c + L])$$

H119-room-merge-adjacency · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H119-room-merge-adjacency/README.md · ratings: H119 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic power and size	M power 0.95; C_{RU} power 0.73, supported size 0.00	
P1 $M > 0$ (merge rise)	-0.019 $[-0.127, 0.088]$; relabel p	failed
	0.43	
P2 no remanence	R_m 0.014 $[-0.14, 0.17]$	trivial
P3 no partition memory	0.056 $[-0.010, 0.122]$	trivial
P5 daily step at merge	merged 0.002 vs 0.126 (p 0.95)	failed
P6 read-gated cross in #40	C_{RU} 0.15 $[-0.19, 0.48]$	failed
P7 switch contrast 0	$S > 2/4$ significant (#focus); 4/4 sign	partly
HH kill	M CI includes 0	fires

GOAL PERIODS IN DETAIL

G39 $\sim$ mixed
G44 $\sim$ mixed
G51 $\sim$ mixed
NE42 $\times$ failed
G38 $-$ n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

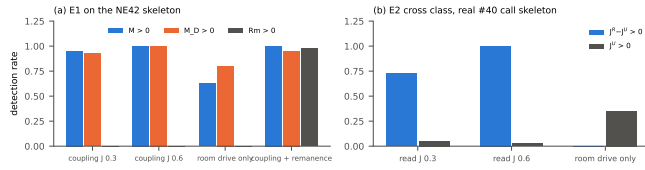
11 eligible agents. GPT-5 (the mirror arm) and most movers drop out by the 10-talk-minute rule. The minute estimator resolves per-pair couplings of about 0.2 and above. The absolute J^R and J^U are confounded by call length; only their difference is read. The minute estimator's switch contrast is not specific (room-drive false positives 0.58–0.95), and the read-gated attribution holds at one of five switches. #40's shared objective coincides with the merge, but coupling fell.

ROUND 2

Fit coupling against room size across all one- and two-room weeks (a dilution exponent). Pre-register the dilution null for the held-out #48–#49 merge.

- **Essence.** does a read path make coupling, or does attention per partner? Fit J against room size across all two-room and one-room weeks (dilution exponent, H18) instead of an on/off switch.

H119-room-merge-adjacency · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H119-room-merge-adjacency/README.md · ratings: H119 agent (suggested) (page 2)



Second figure. Synthetic worlds on the real NE42 skeleton (40 each). (a) The minute estimator detects a gated coupling (M) and remanence, but a shared room drive alone also gives $M > 0$ in 0.63 of worlds. (b) The call-clock read-minus-in-flight contrast detects coupling and stays silent under a room drive, although J^U alone is positive.

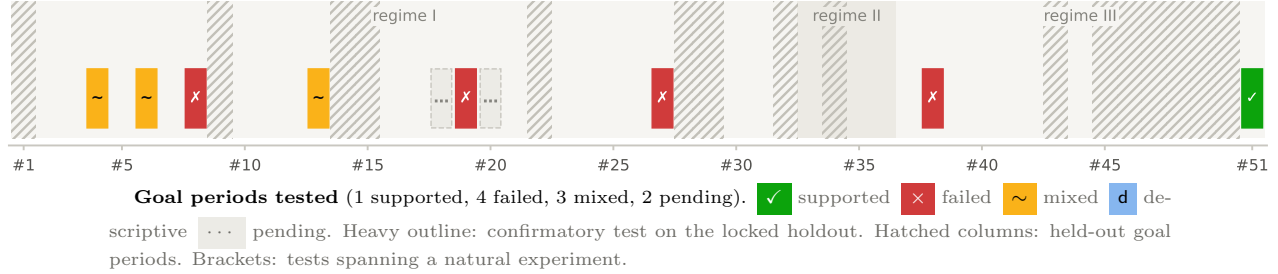
H120: Is a goal period a stationary NESS? Within-period drift of J and EP (#38, #51 main)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 3/5 Usefulness 2.5/5

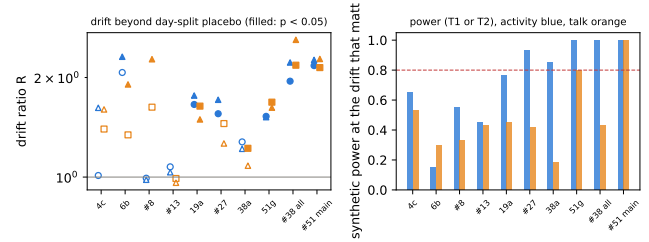
Bottom line. Inside long goal periods the kinetic Ising parameters of activity and talk drift beyond day-split placebo noise (#38: R 1.95 and 2.18, Holm ≤ 0.008 ; #51: 2.17 and 2.14), with no step at NE17, while an 8-day unit shows no drift; entropy-production drift is not identifiable.

level: descriptive · pipeline checks 83% · updated 2026-10-04



WHAT HAPPENED

Every hypothesis in this project fits its model inside one goal period and treats the fit as one point on a phase diagram. That assumes the period is a stationary nonequilibrium steady state. HH360 tests the assumption on the longest periods: the 17-day charity drive (#38), predicted stationary, and the #51 main body (45 days), predicted to drift. H120 fits a kinetic Ising model to 1-min activity and talk spins of the core agents and asks whether the first and second halves (or a linear trend) differ more than random day halves do. **#38 is not stationary, and the HH's kill fires.** Its halves differ about twice as much as random day halves in both channels. The drift is gradual, not a step at NE17. The 8-day unit 38a shows no drift.



Observables. Left: drift ratio R for the contiguous split (circles, squares) and the linear trend (triangles) per window; activity blue, talk orange; filled = permutation $p < 0.05$. $R \approx 1$ is a stationary window. #38, #51 main, 51g, #27 and 19a drift; 38a, #13, 4c do not. Right: synthetic power at the drift that matters (a re-randomization of J at its own scale over the window); the dashed line marks 0.8. Regime-I units with 4–6 agents are underpowered.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 02 (kinetic Ising, logistic Glauber updates on a 1-min grid inside the DQ8 all-present window). Core agent i updates with field h_i , couplings J_{ij} (incl. persistence J_{ii}), weekday fields $\phi_{i,w}$ and a session-length field ψ_i , ridge 1 on J in every window. Drift test: an efficient score statistic W for $\theta_i(d) = \theta_i + \tau_d \delta_i$ on (h_i, J_i) , from per-day gradients g_d and Hessians H_d at the pooled fit, calibrated by permuting days. Entropy production: the corrected held-out Newton bound on antisymmetric lagged pair and single-agent observables.

For anyone fitting models to a swarm's logs, including this project: a goal period is not one steady state. Kinetic Ising parameters move by about 0.2–0.4 placebo SD per day, so a 17-day period fit averages over a walking state. Two rules follow. Fit in windows of about one week (the 8-day unit 38a passes; the 13-day unit 51g does not). Report long-period parameters as trajectories or weekly points, not as one constant. Step changes do not explain the drift: NE17 is an ordinary boundary.

$$\text{logit } P(s'_i = 1 \mid \mathbf{s}) = h_i + \sum_j J_{ij} s_j + \phi_{i,w(d)} + \psi_i L_d$$

$$U_i = \sum_d \tau_d P g_{i,d}, \quad W = \sum_i U_i^\top V_i^{-1} U_i, \quad R = W_{\text{obs}} / (W)_{\text{random day splits}}$$

H120-period-ness-stationarity · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H120-period-ness-stationarity/README.md · ratings: H120 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: size, power, step rank	size ≤ 0.12 per test; activity power 1.00 (#38, #51), 0.85 (38a); EP power ≤ 0.30 ; step top-10% 100%	passed (EP unpow-ered)
P1 #38 stationary (HH)	R 1.95 / 2.18; trend 2.22 / 2.60; Holm ≤ 0.008 ; robust to weekday strata and day-1 drop	failed (kill)
P2 38a stationary	R 1.28 / 1.22; Holm ≥ 0.16 ; power 0.85	supported
P3 #51 main drifts	R 2.17 / 2.14, p 0.001	supported
P4 boundaries top 20%	NE17 0.36 / 0.21; #51 08-05 0.13 (activity)	failed / supported
P5 replication: no drift $\geq 6/8$	4/8 (3 underpowered); 8, 19a, 27, 51g drift	failed
P6 EP stable in #38, drifts in #51	no EP drift detected anywhere testable	inconclusive
HH360	#38 drifts; #51 drifts; no NE17 step	failed

GOAL PERIODS IN DETAIL

G04 $\sim$ mixed
G06 $\sim$ mixed
G08 $\times$ failed
G13 $\sim$ mixed
G19 $\times$ failed
G27 $\times$ failed
G38 $\times$ failed
G51 $\checkmark$ supported
G18 $\dots$ pending
G20 $\dots$ pending

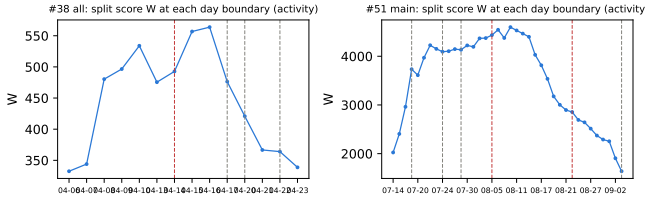
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The drift statistic includes the fields h_i as well as J ; post hoc, J alone also drifts in #38 and #51, and #38 drifts before its first roster join. Non-core agents (joiners) act as an unmodelled field. Talk is underpowered in most windows, and EP drift is unpowered everywhere; the 51g EP rejection rests on negative bounds and is not read as an arrow of time. First-order Markov dynamics on a 1-min grid is assumed. Regime-I units with 4-6 agents cannot tell stationarity from drift.



Second figure. Split score W at every day boundary of #38 (left) and the #51 main body (right), activity channel. Dashed lines: step changes (red: the pre-registered ones). A step would make one boundary stand out; instead W rises smoothly toward the middle, the signature of a trend. Only #51's 08-05 boundary (#focus room, bookends stop) ranks in the top 13%.

ROUND 2

Fit $J(t)$ with a random-walk prior and report a drift rate per day; estimate the longest stationary window per regime; regress the weekly drift on roster churn, operator volume and project churn; run the same test on H91's content modes. Confirmation on the #51 tail and unit 1a (frozen, not run).

- **Essence.** how fast a swarm's dynamical parameters walk inside a period, and what window a stationary fit can use.
- **H120-R1** Time-varying parameters: fit $J(t)$ with a random-walk prior (Kalman or GP over days), report the drift rate per day with an interval, and place periods on the phase diagram as trajectories.
- **H120-R2** Stationary window length: estimate the longest window whose split test passes, per regime (38a passes at 8 days; 51g fails at 13).
- **H120-R3** Drivers of the walk: regress the weekly one-step δJ on roster share outside the core, operator message volume and project churn (DQ4).
- **H120-R4** Content analog: the same score test on H91's content modes, to tie the two "about 1 SD a day" drifts.

H120-period-ness-stationarity · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H120-period-ness-stationarity/README.md · ratings: H120 agent (suggested) (page 2)

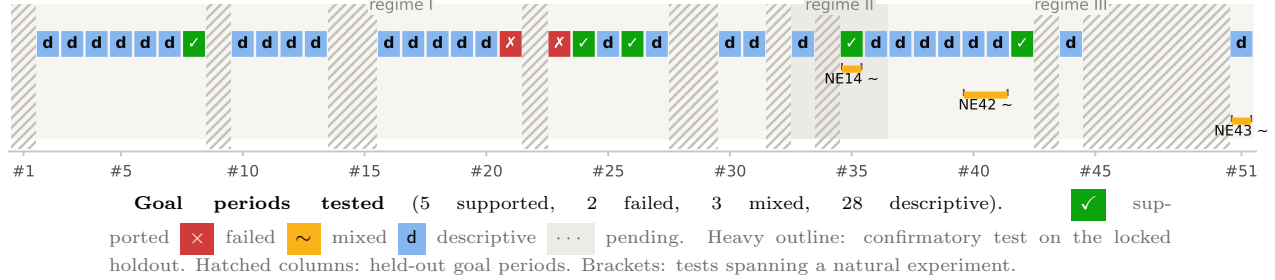
H121: The daily boot is a reproducible quench: hundreds of replicas of relaxation to the steady state

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1/5

Bottom line. The daily boot is a one-call talk spike plus, in regime III, a slow day-scale talk excess; the HH's single-exponential relation $\tau = \tau_0/(1 - g_{lag})$ fails in regime III ($K_{boot} = 22$ [6.9, 71]).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Every active day, each agent restarts from off. HH361 treats this daily boot as one quench, repeated on every day. It predicts that the collective talk share relaxes as one exponential on the call clock, with $\tau = \tau_0/(1 - g_{lag})$ from H67's read-out gain. The test uses 3,135 boot replicas in 67 non-holdout units (34 periods). In regimes I and II the first call of the day talks with probability 0.49 against a steady state of 0.12, and the excess is gone by the next call ($\tau_{boot} \approx 1$ call, as predicted). In regime III the pooled slowing ratio is $K_{boot} = 22$ [6.9, 71], so **the kill fires**. There the curve is a weak first-call spike plus a slow talk excess that lasts hours. Collapse across days and the late-booter contrast are untestable. The holdout is not run.

PHYSICAL MODEL

Mean-field Glauber on the call clock (model 02). Agent i 's k -th receiving call of a day has talk spin Y_{ik} ; $k = 0$ is its boot. $m(k)$ is the mean over agent-days. ρ_{self} is an agent's lag-1 talk autocorrelation across consecutive steady-state calls. τ_0 is its relaxation time, floored at one call (one heat-bath update). g_{lag} is H67's read-out loop gain, used unfitted. The rival is a slow start-up field.

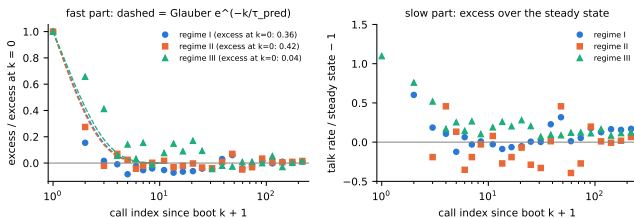
$$m(k) = m_\infty + A e^{-k/\tau_{boot}}, \quad \tau_{pred} = \frac{\max(1, -1/\ln \rho_{self})}{1 - g_{lag}},$$

post hoc: $e(k) = m(k) - \bar{m}_{20-60}, \quad \tau_{fast} = -1/\ln[e(1)/e(0)]$

Observables. Left: talk excess over the call-20-60 plateau, divided by its first-call value, pooled per regime. Dashed lines are the Glauber prediction $e^{-k/\tau_{pred}}$. Regimes I-II drop within one call. Regime III decays over about two calls, then keeps a tail. Right: talk rate over the steady state. Regime III stays 10-25% above it for hundreds of calls (regime I/II first-call values, +3 and +7, are off scale).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, the daily restart costs one call of extra chatter in chat-mode scaffolds. In the computer-use scaffold it adds a 10-20% talk excess that fades over hours. Neither depends measurably on coupling. The coupling moves τ_{pred} by at most $\times 1.4$ ($g_{lag} \leq 0.29$), and the single-agent memory sits at the one-call floor in 58 of 67 units. So the boot is not a usable probe of g_{lag} . Treat it as a scheduler field and trim it: drop the first call of each agent-day (regimes I-II), and in regime III treat the first hours as non-stationary. This is a measurement hygiene rule, not a lever.



RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1-S4 synthetic (real call grids)	bias $\leq 19\%$; ρ_{self} bias -0.03 ; collapse stat. replaced (A1)	2/4 pass
P1 regime-III $K_{boot} \in [0.5, 2]$ (kill)	$22 [6.9, 71]$; 37% of units in band	failed
P1' $K_{boot} > 2$ in $\geq 2/3$ I 35%, II 0%, III 63% of units		failed
P2 one exponential (CV)	95-100% pass; test unpowered	uninformative
P3 collapse across days (kill)	10/175 and 0/93 days resolved	untestable
P4 late booters $\tau_{late} \leq 5 / 0 / 19$ late agent-days τ_{sync}		untestable
P5 regime III vs I differs	ratio 48 (post hoc fast part 4.4)	supported
P6 phase slope	power 0.05	not re-reported
Post hoc fast part	K_{fast} 0.51 [0.39, 0.63] (I), 2.0 [0.8, 23] (III)	post hoc
Natives NE14, NE43, NE42	registered fits at bounds	mixed $\times 3$

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

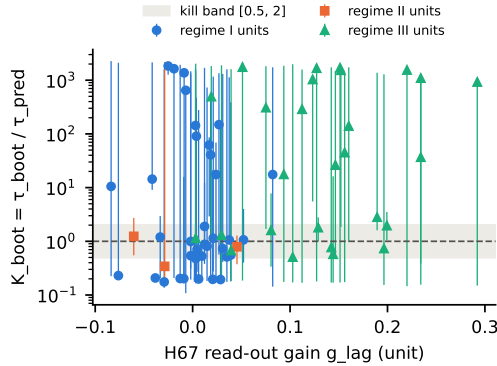
CAVEATS

τ_0 is set by the one-call floor, not by measured memory, in 58 of 67 units, so $K \approx 1$ in regimes I-II is nearly trivial. The fast/slow split is post hoc. The slow excess may be a diurnal talk profile rather than a relaxation. Per-day fits are unresolved: 4-30 boots per day are too few, so collapse is untested. The synthetic did not include a two-timescale world. H99 round 2 withdrew the $\times 9.5$ kick excess that motivated the HH. The registered bootstrap CIs are mildly anti-conservative (coverage 0.78-0.84).

ROUND 2

Pre-register a two-component boot fit with a synthetic two-timescale world. Use mid-day restarts and late booters as clock-shifted replicas to separate the slow excess from a diurnal profile. Test K_{fast} on the holdout (C2).

- **H121-R1** Pre-register the two-component boot.
- **H121-R2** Separate the slow excess from a diurnal profile.
- **H121-R3** Make the coupling term testable.
- **H121-R4** Kickoff boots.



Second figure. Registered slowing ratio per unit against H67's g_{lag} . The grey band is the kill band. Regime-I and II units cluster near $K = 1$ or below (spike-only fits), but some lock onto slow drifts ($K \sim 10^3$). Regime-III units spread over four decades: the single exponential picks the spike or the tail.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 ✓ supported
G10 d descriptive
G11 d descriptive
G12 d descriptive
+28 more periods: see the card.

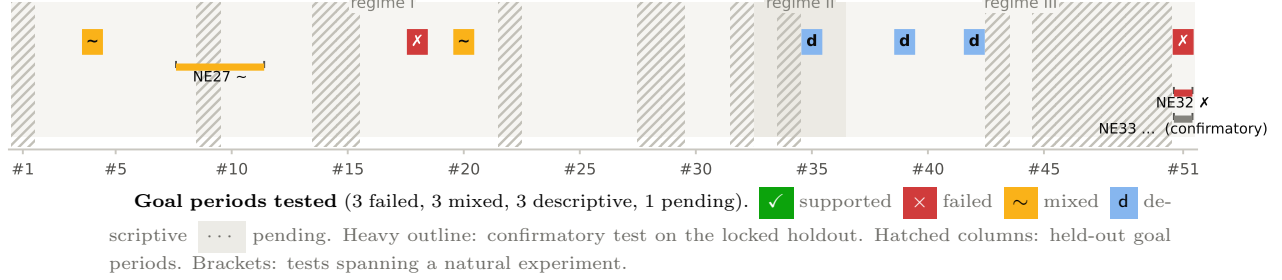
H122: A batch join is a spin addition: do incumbents respond as the fitted couplings say? (NE27, NE33)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1/5

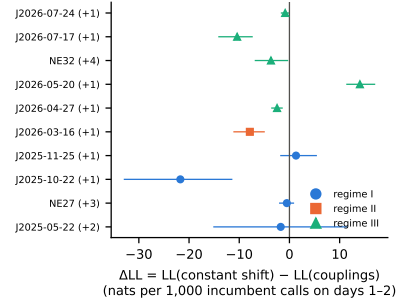
Bottom line. Fitted newcomer couplings do not predict incumbents' first two days after a join better than a constant field shift (6/10 events favour the shift; NE32 $\Delta LL = -3.7$ [-6.9, -0.2]); the HH fails.

level: hypothesis · pipeline checks 84% · updated 2026-10-04



WHAT HAPPENED

When agents join, a kinetic Ising model adds J_{iNs_N} to every incumbent's field. HH363 predicts that newcomer couplings fitted on days 3–7 after a join, applied back to days 1–2, predict the incumbents' per-call talk better than a constant field shift. The test uses 10 non-holdout join events (4 in regime I, 1 in II, 5 in III), with a log-loss cross-fit on held-out days. **The HH fails.** The constant shift wins in 6 of 10 events, including NE32, the stand-in for NE33 ($\Delta LL = -3.7$ [-6.9, -0.2] nats per 1,000 calls; the kill fires). Couplings win in 1. Without the #51 kickoff day in NE32's baseline, the result weakens to -1.6 [-3.8, +0.7] (post hoc). NE33 itself is held out and blind.



Observables. Each row is one join event (number of newcomers in brackets), with ΔLL and its 95% block-bootstrap CI. Values left of zero mean the constant shift predicts days 1–2 better than the fitted couplings. Six events sit left with CIs excluding zero, one sits right (Gemini 3.5 Flash, 05-20), and three include zero (all small regime-I joins).

PHYSICAL MODEL

Kinetic Ising on the call clock (model 02). Incumbent i 's receiving call c has talk spin Y_{ic} . Its field sums incumbent baselines $a_{i,\kappa}$ and the messages read at the call (R , ledger rule). The H18 dilution factor scales read couplings. Newcomers add a read coupling J_N (MJ) or a uniform shift δ (MC). The in-flight placebo P^N counts newcomer messages posted during the call: unreadable, at matched lag.

$$\text{logit } P(Y_{ic}=1) = a_{i,\kappa} + \left(\frac{\bar{N}_{post}}{\bar{N}_{pre}} \right)^{-0.66} (\beta_m R_{ic}^m + \beta_o R_{ic}^o + \beta_P P_{ic}) + \gamma \cdot E_{ic} + \rho Y_{i,c-1} + \phi(k)$$

$$\begin{aligned} \text{MC: } & + \delta & \text{MJ: } & + J_N R_{ic}^N & \text{MJ}_{pl}: & + J_P P_{ic}^N \\ \Delta LL = & \langle \ell_{MC} \rangle_{d1-2} - \langle \ell_{MJ} \rangle_{d1-2} & (\delta, J \text{ fit on days 3-7}) \end{aligned}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator adding agents: the incumbents' first-day response is a uniform step whose sign depends on what else happens that week. A kickoff, a goal change or the end of a kickoff transient matters more than whom the newcomers talk to. Per newcomer message read, an incumbent's talk probability moves by about 0.009 (regime III median). A message that names the incumbent moves it more (named exceeds unnamed in 8/10 events). Practical rule: do not forecast a join's effect from couplings. If you want incumbents to engage a newcomer, have the newcomer name them. Do not schedule a join on a kickoff day if you want to measure it.

H122-batch-join-spin-addition · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H122-batch-join-spin-addition/README.md · ratings: H122 agent (suggested)

RESULTS VS PREDICTIONS

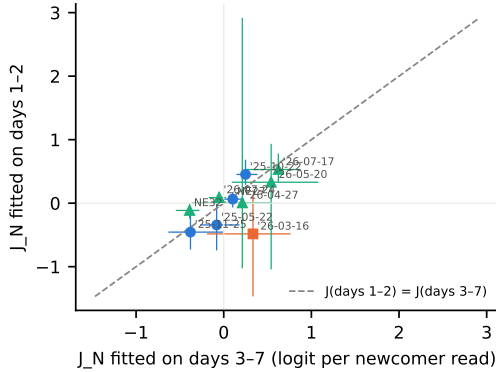
Prediction (dated)	Observed	Verdict
S1–S4 synthetic (NE32)	power 1.00, kill power 1.00, size 0.04; NE27 power 0.47–0.64	pass (NE32)
P1 NE32 MJ beats MC (kill)	−3.68 [−6.92, −0.25]; post hoc w/o kickoff −1.6 [−3.8, 0.7]	failed
P2 NE27: $J_N \approx 0$, step = kickoff	J_N 0.10 [−0.03, 0.23]; step inside 12-kickoff range	supported
P3 fast step	1.82, 0.85, 0.29 (3 resolved)	mixed
$J(1-2)/J(3-7) \in [0.5, 2]$		
P4 read beats in-flight (where MJ wins)	+0.03 [−0.13, 0.20] (1 event)	weak
P5 dilution helps	4/10 events	failed
P6 newcomers couple like incumbents	NE32 −0.36 [−1.02, 0.32]	supported
P7 MJ wins in $\geq 1/2$ of MJ regime III	1/5, MC 4/5	failed
NE33 (confirmation not run; frozen only)	confirm.py	pending

GOAL PERIODS IN DETAIL

NE33	...	pending (holdout)
G04	~	mixed
G18	×	failed
G20	~	mixed
G35	d	descriptive
G39	d	descriptive
G42	d	descriptive
G51	×	failed
NE27	~	mixed
NE32	×	failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		
Level: hypothesis . Scale 0 not done or failed, 1 partial, 2 passed (writeup/paper.tex).			



Second figure. Newcomer read coupling fitted on days 3–7 against the same coupling fitted on days 1–2, per event (95% CIs). Points near the dashed diagonal couple the same from day 1. Couplings exist in some events but are small and change sign across events, so they cannot carry the day-1–2 response.

CAVEATS

Kickoff fields sit inside several windows (NE27 day 1, two single joins on day 1, NE32's baseline). The constant-shift model absorbs them, so this design cannot separate a join field from a kickoff field. Single joins give 22–447 day-1–2 calls with a newcomer read. NE27 is underpowered and its kill is anti-conservative (size 0.12). The logistic model is linear in clipped read counts. The dilution exponent is H18's, not refitted.

ROUND 2

Run join-free pseudo-joins in regime III to size the calendar drift of δ . Rebuild baselines without kickoff days. Fit a saturating read response. Run NE33 on the holdout with the frozen script.

- **H122-R1** Join-free placebo in regime III.
- **H122-R2** Kickoff-free PRE windows.
- **H122-R3** Saturating read response.
- **H122-R4** NE33 on the holdout with the frozen confirm.py.

H122-batch-join-spin-addition · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H122-batch-join-spin-addition/README.md · ratings: H122 agent (suggested) (page 2)

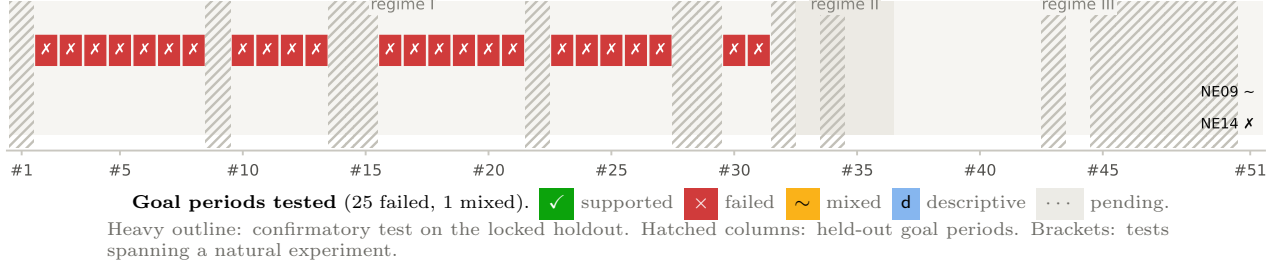
H123: Regime I's turn order is a sweep: equilibrium-looking statistics with nonzero entropy production

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1.5/5

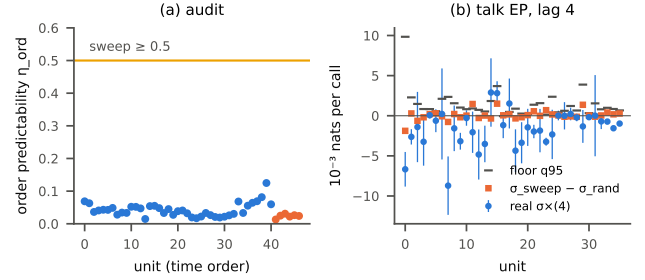
Bottom line. Regime I's turn order is self-clocked asynchronous, not a sweep, and the order alone makes no entropy production; the swarm's regime-I cross-agent irreversibility is at the floor.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

I tested whether regime I's update order is a fixed sweep, which keeps equilibrium snapshots but breaks detailed balance. The export has no turn-pointer history, so I audited the call log first: who calls next, on 41 non-holdout regime-I units. No unit is a sweep. All 41 are self-clocked asynchronous: calls of different agents overlap (concurrency median 0.997), order predictability is $\eta_{\text{ord}} = 0.04$ (a sweep needs ≥ 0.5), and return gaps are burstier than random. Being named lifts the next call only $\times 1.13$. A symmetric-coupling kinetic Ising driven by the real order makes no entropy production ($\sigma_{\text{sweep}} - \sigma_{\text{rand}}$ median 2×10^{-5} nats per call). Measured cross-agent talk EP sits at the floor in 29 of 31 units. Snapshots fit model 01 as well as in regime III. Holdout: not run.



Observables. (a) Order predictability of the next caller per unit (blue regime I, orange the NE14 units of regimes II/III) against the sweep threshold. Every unit sits below 0.13. (b) Talk EP at lag 4 per unit: real $\sigma_{\times}$ with 95% intervals, the block-flip floor (dash) and the scheduler's own prediction $\sigma_{\text{sweep}} - \sigma_{\text{rand}}$ (orange). The scheduler's term is zero everywhere; two units clear the floor, at the chance rate.

PHYSICAL MODEL

Models 02 (kinetic Ising, update-rule table) and 15 (EP inference). One step is one model call; $a(t)$ is the calling agent. Spin $s_i = \pm 1$: whether agent i 's latest call produced chat (talk), or whether it is in computer use (mode). At its call, agent a resamples s_a by heat bath with field h_a and couplings J_{aj} ; others hold. Σ : entropy production; $\sigma_{\times}(L)$: its cross-agent part at lag L calls; $J_s = (J + J^{\top})/2$.

$$P(s'_a = +1 | \mathbf{s}) = [1 + e^{-2(h_a + \sum_{j \neq a} J_{aj} s_j)}]^{-1}, \quad a = a(t)$$

$$g_{ij}^L(t) = s_i(t+L) s_j(t) - s_i(t) s_j(t+L)$$

$$\sigma_{\times}(L) = \Sigma_{u \cup g^L} - \Sigma_u \quad (\text{held-out Newton bound})$$

$$\text{sweep: } \sigma_{\times}(1) = 0 < \sigma_{\times}(L \geq 2); \quad \text{random: } \sigma_{\times} = 0 \quad (J = J^{\top})$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A practical simplification, not a lever. Regime I's scheduler does not create an arrow of time by itself: each agent runs on its own call clock, and the event order is close to random. Kinetic Ising and EP fits on the call clock can therefore treat the update order as random sequential with agent-specific rates; no sweep correction is needed. A scaffold builder who wants detailed balance in a multi-agent loop gets it from independent per-agent clocks; a fixed round robin would add a scheduler arrow that is detectable at lag ≥ 2 calls. The HH's premise rested on a pointer whose history the export does not keep.

H123-regime1-turn-sweep · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H123-regime1-turn-sweep/README.md · ratings: H123 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 (HH) sweep in $\geq 2/3$ units	0/41 sweep; 41/41 self-clocked	failed
P1b chat-mode calls a sweep	0/41 (36 mixed, 5 random)	failed (HH)
P1c naming lift ≤ 1.25	36/41; max 1.32; none ≥ 1.5	supported
P2 (HH) snapshots fit model 01 as in regime III	ρ_2 0.87 vs 0.88 ($N = 4$ subsets)	supported (descr.)
P3 (HH) $\sigma_{\times} \approx \sigma_{\text{sweep}} > 0$	$\sigma_{\text{sweep}} - \sigma_{\text{rand}}$ median 2×10^{-5} ; $\sigma_{\times}$ at floor 29/31	failed
P4 sweep signature	absent; above-floor units positive at lag 1	supported
N1 NE14 class invariant	6/6 self-clocked; concurrency 1.00 both sides	supported
N2 NE09 coupling step	RMS J_s +33%, intervals overlap	inconclusive
N3 G27 chat-mode order	η_{ord} 0.005 vs 0.004 shuffled	not a sweep

GOAL PERIODS IN DETAIL

G02 $\times$ failed
G03 $\times$ failed
G04 $\times$ failed
G05 $\times$ failed
G06 $\times$ failed
G07 $\times$ failed
G08 $\times$ failed
G10 $\times$ failed
G11 $\times$ failed
G12 $\times$ failed

+16 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	2	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

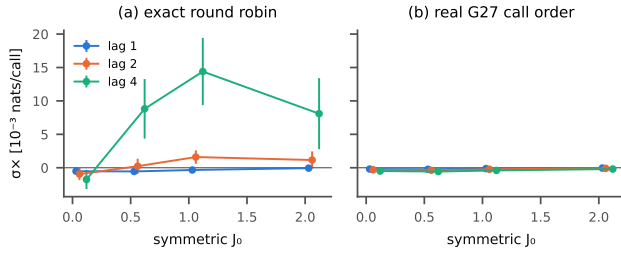
The turn pointer has no logged history; the audit reconstructs the order from call starts, whose times are partly calibrated (DQ1 ledger). Spins are coarse (talk or mode of the latest call). Fitted regime-I couplings are weak (RMS $J_s \approx 0.09$), so even a perfect sweep would make little EP here; the null result on EP is partly a consequence. The block-flip floor alone has a 10–25% false-positive rate (A1 added the σ_{rand} floor). The snapshot clause is trivially high at $N = 4$. Family invariance was not checked.

ROUND 2

Run an NE09 event study with placebo boundaries (G24 is the one lead). Test whether the $\times 1.13$ naming lift lives in chat-mode wake-ups and why it disappears after NE14.

- **Essence.** whether the scaffold’s scheduling, rather than the agents, creates any of the swarm’s irreversibility; round 1 says the order does not, so the open part is the small state-dependent wake-up and the one possible coupling step.
- **H123-R1** NE09 event study.
- **H123-R2** Use the event clock without a sweep correction.
- **H123-R3** Named-next lift.
- **H123-R4** Family invariance.

Second figure. Synthetic symmetric- J kinetic Ising at G27’s step counts (20 worlds per point, mean $\pm$ SD). (a) An exact round robin: lag 1 stays at zero, lags 2 and 4 rise with J_0 , the sweep signature. (b) The real G27 call order: no EP at any $J_0 \leq 2$, the same as a random order.



H123-regime1-turn-sweep · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H123-regime1-turn-sweep/README.md · ratings: H123 agent (suggested) (page 2)

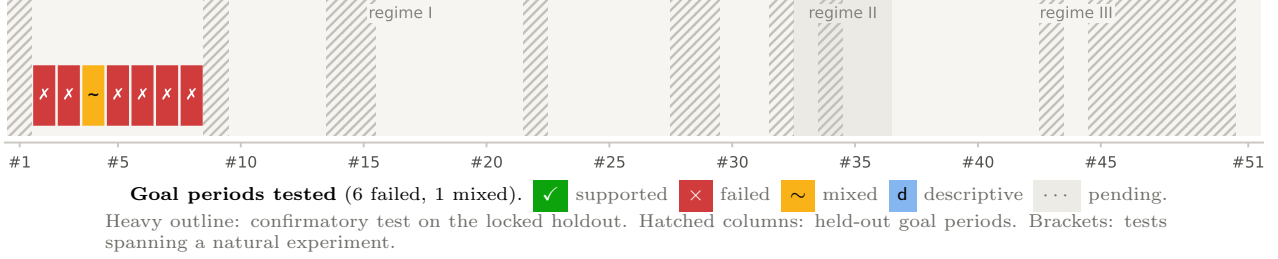
H124: Four agents for weeks: an exact kinetic Ising benchmark for the mean-field approximations (#4, #6)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

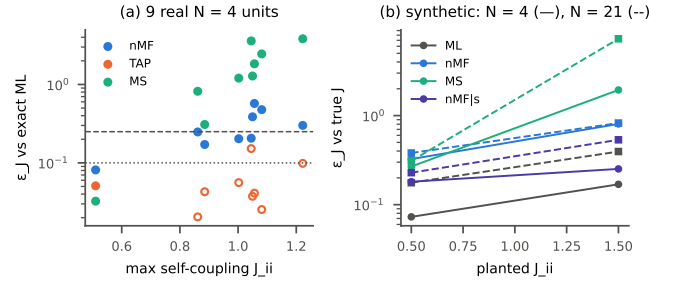
Bottom line. Mean-field inversions of the kinetic Ising match exact likelihood only when agents' own states do not persist; on the village's call clock they mostly do, so use exact ML.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

With four agents the kinetic Ising likelihood is exact and cheap, so I compared exact maximum likelihood (ML) with three mean-field inversions (naive MF, TAP, Mézard–Sakellariou MS) on nine closed-roster regime-I units, on the per-call talk clock and a 1-min grid. Synthetic worlds on the real #4/#6 and #51 schedules came first. The HH's claim fails: TAP or MS is within 10% of ML in 1 of 9 units. That unit (4c) is the only one with weak own-state persistence (self-coupling $J_{ii} \leq 0.6$). Elsewhere (J_{ii} 0.86–1.22) TAP has no solution on 25–75% of rows, MS overshoots the couplings up to $\times 3.8$, and naive MF shrinks them (median error 0.25). The synthetic shows the same law at $N = 4$ and $N = 21$. Off-diagonal couplings are at noise level in 7/9 units. Holdout: not run.



Observables. (a) Off-diagonal coupling error of each inversion against exact ML, per real unit, against the unit's largest self-coupling (open circles: TAP solved on only some rows; dotted 10%, dashed 25%). Only the weak-persistence unit is inside 10% for all three. (b) Synthetic error against the true J (medians; $N = 4$ on the 4c schedule at $\sigma_J = 0.3$, $N = 21$ on #51's, pooled over σ_J): MS and naive MF break as J_{ii} grows; exact ML and the own-spin stratified nMF (A1) hold.

PHYSICAL MODEL

Model 02, kinetic Ising. At each call of agent i , its talk spin $s'_i = \pm 1$ (did the call produce chat) depends on the four latest talk spins, its own included. J_{ij} : coupling of i to j 's state; J_{ii} : persistence of i 's own state; h_i : field. From i 's rows: $m'_i = \langle s'_i \rangle$, C the covariance of $\mathbf{s}$, D_i the covariance of s'_i with $\mathbf{s}$. The inversions differ only in the gain a_i ; TAP has a real solution only when $\sum_j B_{ij}^2 (1 - m_j^2) \leq \frac{4}{27} (1 - m_i^2)$, $B_i = D_i C^{-1}$.

$$P(s'_i = +1 | \mathbf{s}) = [1 + e^{-2(h_i + \sum_j J_{ij} s_j)}]^{-1} \quad (\text{exact ML: logistic})$$

$$J_i = a_i^{-1} D_i C^{-1}, \quad a_i^{\text{nMF}} = 1 - m_i^2$$

$$a_i^{\text{TAP}} = (1 - m_i^2) [1 - (1 - m_i^2) \sum_j J_{ij}^2 (1 - m_j^2)]$$

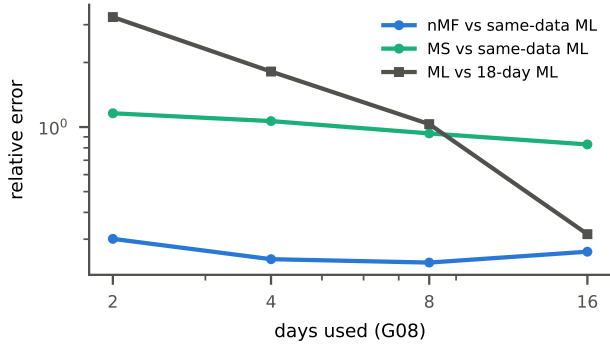
$$a_i^{\text{MS}} = \langle 1 - \tanh^2(g_i + x \sqrt{\Delta_i}) \rangle_x, \quad \Delta_i = J_i C J_i^T$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A method rule for this project, with a number. On the call clock, agents' own states persist ($J_{ii} \approx 0.9$ –1.2), and then TAP and MS are not usable and naive MF underestimates couplings by 5–55%. Exact per-row logistic ML costs seconds at $N = 21$, so large- N kinetic Ising fits here should use it (or pseudo-likelihood) and report J_{ii} . Mean-field inversions are acceptable on channels without persistence (1-min talk: MS within 10% in 6/9 units). For an operator nothing changes; this protects the coupling estimates other hypotheses report.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 (HH) TAP or MS within 10% (G04/G06)	1/4 units (4c: TAP 0.05, MS 0.03)	failed
Kill: none within 25%	2/4 primary units (threshold); 4/9 all	at threshold
P2 naive MF > 10%	8/9; shrink 0.45–0.95	supported
P3 MS $\leq$ TAP < nMF	only 4c; nMF best full-cover in 8/9	failed
P4 ranking carries to $N = 21$	synthetic, best method agrees 15/16 cells	supported
P5 off-diagonal J near null	7/9 inside shift null; ML noise 0.6–1.1	supported
P6 1-min activity nMF > 25%	7/8 units	supported
N1 coupling lower in #6 than #4	no difference ($\bar{J}$ 0.004–0.05)	failed
N2 ML noise below bias by 8 days	bias flat; crossover 16–18 days	mixed



Second figure. Native N2 on G08 (18 days, 20 subsamples per length). The inversions’ bias against same-data ML is flat in the number of days, while ML’s distance to the full 18-day fit falls from 3.3 to 0.32. Below about 16 days, sampling noise dominates any inversion’s bias.

GOAL PERIODS IN DETAIL

G02 $\times$ failed
G03 $\times$ failed
G04 $\sim$ mixed
G05 $\times$ failed
G06 $\times$ failed
G07 $\times$ failed
G08 $\times$ failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The benchmark compares inversions with ML on the same data, and the off-diagonal ML couplings are themselves noise in 7/9 units, so coupling errors partly measure noise handling. The EP error ε_σ is not interpretable (reference EP at noise level). Plefka2[t] (Aguilera 2021) was not implemented. The own-spin stratified variants (A1) were added after the synthetic, before real data. Units with 2–3 days have degenerate intervals.

ROUND 2

Repeat the benchmark where couplings are real (regime-III named talk, H50) and add Plefka2[t] once its notes are in the literature folder.

- **Essence.** which coupling estimator to trust at $N = 21$; round 1 says exact (or pseudo-) likelihood on the call clock, because own-state persistence breaks the mean-field inversions.
- **H124-R1** Large-N rule.
- **H124-R2** Plefka2[t].
- **H124-R3** Real couplings.
- **H124-R4** Markov order.

H124-small-n-meanfield-benchmark · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H124-small-n-meanfield-benchmark/README.md · ratings: H124 agent (suggested) (page 2)

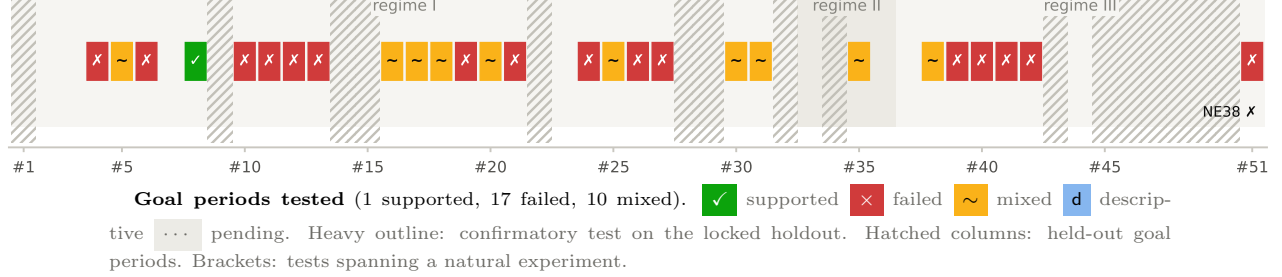
H125: The kickoff response is a damped oscillator: look for the day-2 undershoot

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 1.5/5

Bottom line. The kickoff response is overdamped: content overshoots the goal on day 1 and settles within two active days without swinging below it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH366 says the kickoff response is an underdamped oscillator: content overshoots the settled goal level on day 1 and swings below it on days 2–3. A fading kickoff field gives the overshoot without the swing. I measured each agent’s excess alignment with the kickoff on 27 kickoffs (regimes I–III, 4–21 agents) and compared days 2–3 with days 4–5, against 80 ordinary-day origins. The synthetic check came first: the test has size 0 and power 1 at damping $\zeta \leq 0.5$. Day 1 overshoots ($E_1 = +0.055 \pm 0.012$, 22/27). Days 2–3 do not undershoot: $U = -0.006 [-0.020, +0.008]$ (bge), $-0.002 [-0.017, +0.013]$ (gte), inside the placebo spread. The response is overdamped. Holdout: not run.

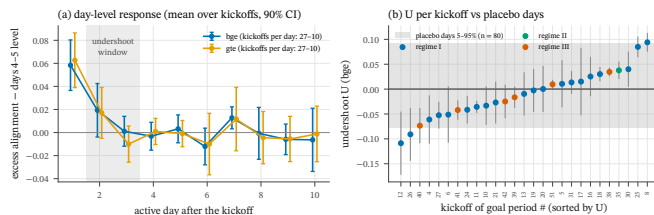
PHYSICAL MODEL

Model 11 with the overdamped relaxation of model 02 as the rival. Spin: agent i ’s statement vectors in the regime-whitened 32-d basis. Field: the kickoff, direction $\hat{k}$ from shared `goal_fields`. Order parameter: the excess alignment $a = \langle z, \hat{k} \rangle$ minus the mean over decoy kickoffs, averaged linearly (no renormalized windows). Clock: active hours h since the kickoff (H103). M_{osc} : damped oscillator; M_{fade} : overdamped spin in a fading field (two exponentials). Both carry a time-of-day profile.

$$M_{osc} : A(h) = A_{\infty} + Be^{-\lambda h} \cos(\omega h + \phi), \quad \zeta = \lambda / \sqrt{\lambda^2 + \omega^2}$$

$$M_{fade} : A(h) = A_{\infty} + c_1 e^{-h/\tau_1} + c_2 e^{-h/\tau_2}$$

$$U = \langle A_{i,4-5} - A_{i,2-3} \rangle_i, \quad \zeta_{pk} = \delta / \sqrt{\pi^2 + \delta^2}, \quad \delta = \ln(E_1/U)$$



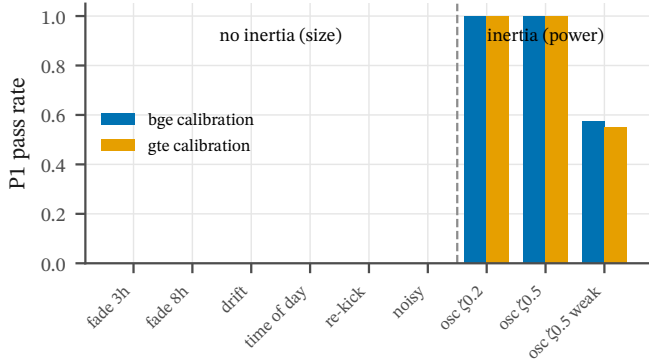
Observables. (a) Mean excess alignment relative to the days 4–5 level, by active day, over 27 kickoffs (both embedding models, 90% CI). Day 1 overshoots; day 2 keeps $+0.02$; day 3 and later sit at the settled level. The gray band marks where M_{osc} predicts an undershoot. (b) U per kickoff (bge) against ordinary-day origins (gray: 5–95%). Kickoffs do not exceed the placebo spread.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Read the steady state two active days after a kickoff. Day-1 content is about 0.055 above the settled alignment and day 2 about 0.02 above; there is no later dip to wait out. A drop in goal alignment on day 2 or 3 is the kickoff’s salience fading, not a loss of the goal. An operator who wants to know where a swarm settled can stop watching after day 3.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 undershoot $U > 0$, beyond placebo	bge $-0.006 [-0.020, +0.008]$, p 0.79; gte -0.002 , p 0.69	failed (kill)
P2 M_{osc} wins $\geq 2/3$	18/27, 19/27; null size 0.12–0.20	no weight
P3 $\zeta_{\text{pk}} < 1$	$\zeta_{\text{pk}} \geq 1$ in both models	failed
S1 overshoot $E_1 > 0$	$+0.055 \pm 0.012$ (22/27); gte $+0.062$	supported
S2 $U(\text{III}) > U(\text{I})$	-0.019 vs -0.010 , p 0.65	not supported
N1 G51 own roles ($N = 21$)	$U_{\text{own}} -0.009 / -0.026$	failed
N2 NE38 one agent	no overshoot, U at others' median	failed
N3 G38 ring-down	M_{osc} wins; size 0.19–0.26; ζ 0.56 vs 0.04	weak



Second figure. Synthetic swarms on the real skeleton of 27 kick-offs (40 replicates per world, both calibrations). P1 passes in no world without inertia (fading field, drift, time-of-day profile, day-4 re-kick, doubled day noise) and in every replicate with $\zeta = 0.2$ or 0.5 at the observed overshoot.

GOAL PERIODS IN DETAIL

G04 $\times$ failed
G05 $\sim$ mixed
G06 $\times$ failed
G08 $\checkmark$ supported
G10 $\times$ failed
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
G16 $\sim$ mixed
G17 $\sim$ mixed
+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The target is a text embedding of the kickoff, and day 1 contains restatements of it, which raise E_1 but not U . Only 10 long periods supply placebo origins. Damping between 0.5 and 1 gives an undershoot below $0.2 \times E_1$ and is not excluded. Model comparison by SSE is liberal for single kickoffs, so G38's preference for M_{osc} is weak. Two calendar days include village-off gaps that lengthen the active-hour clock.

ROUND 2

Fit the decay of the read-out spike (H127) on the call and hour clocks; remove kickoff restatements before measuring E_1 ; run `confirm.py` on the held-out kickoffs.

- **H125-R1** The overshoot at call resolution.
- **H125-R2** Restatement vs re-orientation.
- **H125-R3** G38's non-monotone 10 days.

H125-kickoff-damped-oscillator · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H125-kickoff-damped-oscillator/README.md · ratings: H125 agent (suggested) (page 2)

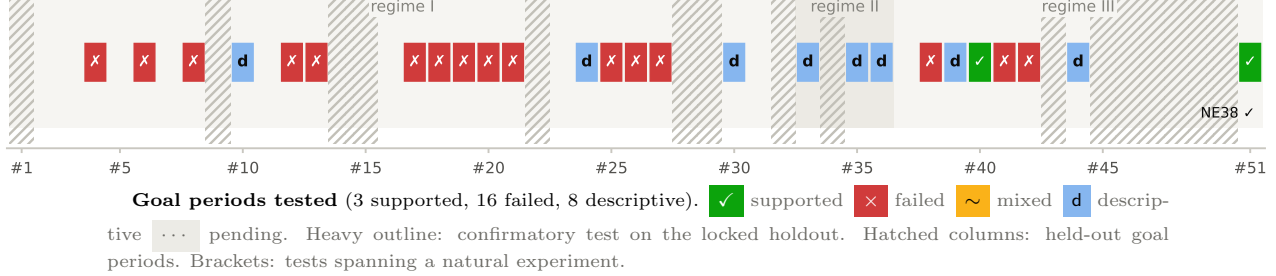
H126: Goal occupancy is a telegraph process: dwell times predict the occupancy

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. An assigned goal acts on both telegraph rates: it raises the on-rate and lowers the off-rate ($\Delta \ln k_{\text{off}} < 0$ in 15/17 kickoffs and in the #12 debate schedule), so it is a symmetric field, not an attempt-rate field. Dwell shape is not identifiable here.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH367 models each agent's on-goal state as a random telegraph signal. In that model dwells are exponential, two mean dwells predict the occupancy $p = k_{\text{on}} / (k_{\text{on}} + k_{\text{off}})$, and a kickoff raises only k_{on} . H126 fits a two-state hidden Markov model to H105's decoy-threshold on-goal labels on each agent's ledger call clock: 39 non-holdout units, 17 kickoffs, the #12 debate schedule, #51 private goals and Opus 5's reassignment. The decisive result is that a kickoff lowers k_{off} as well as raising k_{on} : $\Delta \ln k_{\text{off}}$ is below 0 in 15 of 17 kickoffs. The dwell-shape and occupancy clauses are not identifiable at village sampling (synthetic, before data).

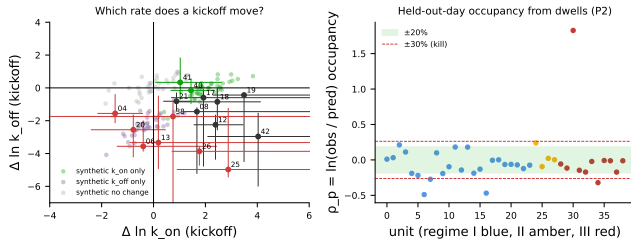
PHYSICAL MODEL

Telegraph spin read through a noisy channel (models 10 and 02). The latent $s_{i,c} \in \{\text{off}, \text{on}\}$ of agent i switches per ledger call with $a_i = k_{\text{on}}$ and $b_i = k_{\text{off}}$. Statements are observed at irregular call gaps Δ and labelled on-goal with sensitivity q_1 and false-positive rate q_0 . Rival: a 4-state aggregated chain whose dwells are mixtures of two geometrics. Kickoff designs fit segment rates before and after the boundary with shared emissions.

$$P^\Delta = \Pi + \lambda^\Delta (I - \Pi), \quad \lambda = 1 - a - b, \quad \Pi_{\cdot, \text{on}} = \frac{a}{a+b}$$

$$P(b_k=1 | s) = q_0 + (q_1 - q_0) \mathbb{I}[s = \text{on}], \quad \tau_{\text{on}} = 1/b, \quad \tau_{\text{off}} = 1/a$$

one-rate field: $\Delta \ln a > 0$, $\Delta \ln b = 0$; Glauber: $\Delta \ln a = -\Delta \ln b$



Observables. Left: kickoff rate changes with 90% agent-bootstrap CIs (green: k_{on} only; red: k_{off} dominant; black: both move). Faint points are synthetic k_{on} -only, k_{off} -only and no-change worlds on the real skeletons. Most real kickoffs sit below zero on the k_{off} axis. Right: held-out-day occupancy ratio ρ_p per unit; the $\pm 30\%$ kill fires in 5/39 units. The synthetic shows this check passes under every rival.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, a goal works by holding agents on goal as much as by recruiting them. A kickoff multiplies the per-call chance of switching onto the goal by about 5 and divides the chance of leaving it by about 6 (medians over 17 kickoffs). Latent on-goal stints last a median of 78 calls. Expect occupancy to respond to anything that interrupts agents (lower k_{off} is what a goal buys), not only to how often it is announced.

RESULTS VS PREDICTIONS

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

Prediction (dated)	Observed	Verdict
P1 exponential dwells	shape test power 0.05–0.10 (synthetic); “heavy” 6/39	not identifiable
P2 p from dwells within 20%	28/39; kill ($> 30\%$) 5/39; passes in all synthetic worlds	non-discriminating
P3 kickoff raises k_{on} only	2/17; k_{off} falls in 15/17	failed
P4 occupancy carried by k_{on}	$\rho(\ln k_{\text{on}}, \text{logit } p)$ 0.37; $\rho(\ln k_{\text{off}})$ −0.50	failed
P5 call clock beats wall (III)	9/11 units	supported
N1 #12 debate field on k_{on} only	+1.27 / −1.37 [−2.28, −0.35]	failed
N2 #51 per-agent rates persist	median ρ 0.61 (8 pairs)	supported
N3 NE38 k_{on} only	+3.2 / −5.1 [−12.9, +0.2]	weak pass
Overall (A1 rule)	P3 fails	failed

CAVEATS

The on-goal label is a text-embedding proxy (bge decoy threshold); gte doubles the heavy-tail counts. In each kickoff’s pre-segment, occupancy along the next goal sits at the decoy floor ($p \approx 0.04$), so its k_{off} rests on few latent stints, although the synthetic gave 0/30 false k_{off} calls under a true k_{on} -only change. Sparse regime-III units give unstable short dwells. A kickoff smoke test printed one result (K38) after Amendment A1 and before the full run.

ROUND 2

Calibrate the classifier externally (Jev on-goal sample) to restore shape power; fit semi-Markov dwell hazards on #51 own-goal sequences; fit time-dependent rates through the kickoff day; run the frozen confirm (held-out kickoffs, C1: k_{off} falls in $\geq 2/3$).

- **H126-R1** A calibrated classifier.
- **H126-R2** A semi-Markov fit for #51.
- **H126-R3** Kickoff time course.

GOAL PERIODS IN DETAIL

G04 × failed
G06 × failed
G08 × failed
G10 d descriptive
G12 × failed
G13 × failed
G17 × failed
G18 × failed
G19 × failed
G20 × failed
+17 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

H126-telegraph-goal-occupancy · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H126-telegraph-goal-occupancy/README.md · ratings: H126 agent (suggested) (page 2)

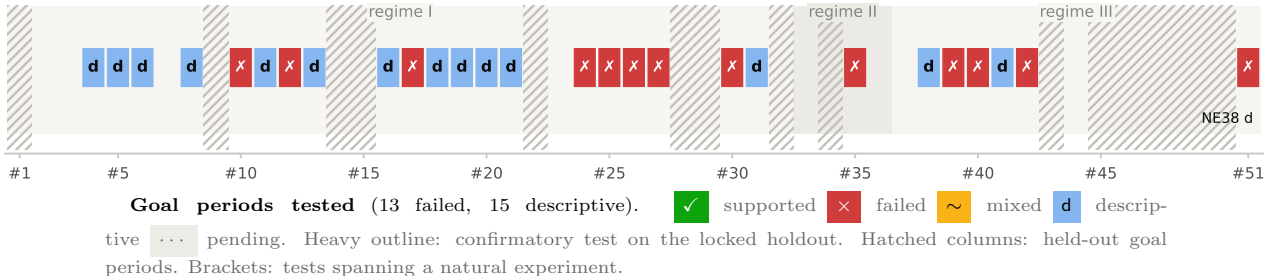
H127: Content trails a moving goal like an overdamped particle: agents that call more catch up faster

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

Bottom line. An agent adopts a new goal within its read-out call; there is no catch-up time for the call rate to set, and what follows is a decay of the kickoff's salience.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH368 says content trails a new goal like an overdamped particle, catching up at a rate set by each agent's model calls: the half-alignment time should scale as $1/(\text{call rate})$. I fitted each agent's day-1 approach to its settled alignment after 27 kickoffs (152 rising agents) on hour and call clocks, timed from the kickoff and from the agent's read-out call (context ledger). The synthetic check came first and replaced three estimators that read-out gating fools. There is no approach to time. 74% of rising agents are already at their day-1 plateau in the read-out call itself; alignment then decays from about $2.4\times$ to $1\times$ the plateau over about 64 calls. The call-rate slope is $+0.59 [-0.45, +1.75]$: inconclusive. Holdout: not run.

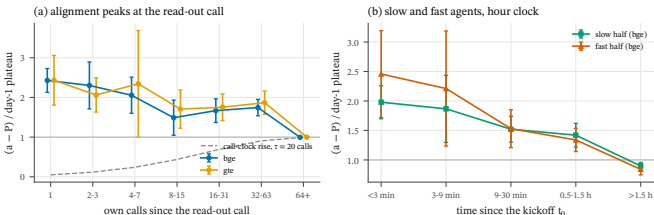
PHYSICAL MODEL

Model 11 with Glauber kinetics (model 02): the call is the attempt. Spin: agent i 's statement vectors in the regime-whitened 32-d basis. Field: the kickoff, direction $\hat{k}$; a is the excess alignment over decoy kickoffs (H125). Per agent: pre level P_i , settled level S_i (days 2–5), and a one-parameter rise on clock c (hours or own calls, from t_0 or from the read-out call). Rival R_L : the field arrives as a step at the read-out call (H08 gating) and nothing relaxes.

$$a_i(t) = P_i + A_i(1 - e^{-c_i(t)/\tau}), \quad T_{1/2} = \tau \ln 2$$

call clock: $\log T_{1/2,i}^H = \text{const} - \log r_i$ ($s = -1$); hour clock: $s = 0$

$$\Delta \text{SSE}_{\text{ro}} = \frac{\text{SSE}_H - \text{SSE}_C}{\text{SSE}_0 - \min(\text{SSE}_H, \text{SSE}_C)} \quad (\text{from the read-out call})$$



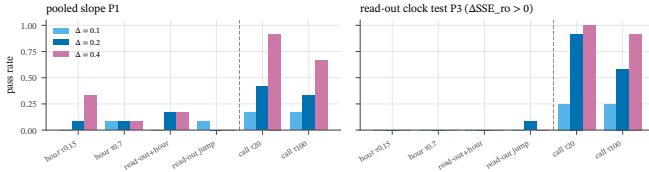
Observables. (a) Day-1 excess alignment of rising agents, normalized to each agent's late day-1 plateau (calls ≥ 64 after read-out, equal to 1 by construction), against own calls since the read-out call; both embedding models, 90% CI. Dashed: a call-clock rise with $\tau = 20$ calls. The data start at the top and decay. (b) The same against time since the kickoff, for the slower and faster halves of agents by call rate (bge).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Expect a new goal to show up in an agent's output at its first read-out call, about 75 s after the kickoff, not after a catch-up period. Speeding up an agent's calls will not make it adopt a goal faster; it is already adopted within one call. What does relax over the next hour is the kickoff's salience, from $2.4\times$ to $1\times$ the plateau. Do not judge adoption from the first statements, which overstate it.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 pooled slope near -1	+0.59 [-0.45, +1.75]; gte +0.42; power $\approx$ 0.35	inconclusive
P2 collapse CR < 1	0.65 / 0.19; ties at the floor	non-diagnostic
P3 $\Delta\text{SSE}_{\text{ro}} > 0$	-0.0009 [-0.0019, +0.0001]	not supported
P4 relaxation beyond read-out	immediate share 0.74 (relaxation worlds $\leq$ 0.20)	failed: R_L
S1 slope steeper in II-III	I +0.34, II-III +1.00	not supported
S4 rising share $\geq$ 0.5	0.60 / 0.57	supported
N1 G51 slope (21 agents)	all half times at the floor	failed
N2 NE38 prediction	Opus 5 $T_{1/2}$ 0.20 h; predictions at the floor	uninformative



Second figure. Synthetic swarms on the real skeleton (27 kickoffs, real statement, call and read-out times; 12 replicates per world and rise amplitude). The pooled slope reaches power 0.67–0.92 only at a rise of 0.4. The read-out-origin clock test separates hour and call clocks at a rise $\geq$ 0.2 and stays silent under a read-out jump.

GOAL PERIODS IN DETAIL

G04 d descriptive
G05 d descriptive
G06 d descriptive
G08 d descriptive
G10 $\times$ failed
G11 d descriptive
G12 $\times$ failed
G13 d descriptive
G16 d descriptive
G17 $\times$ failed

+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The first statements after a read-out likely restate the kickoff, so part of the spike is copying, not re-orientation. Half times below the 0.36-s grid floor are ties, which make per-kickoff slopes 0. The median rise (0.15–0.19) is below the amplitude where the slope test has power, so the slope miss is not a negative. Five kickoffs lack a previous non-holdout day. The rise curve is a post hoc description.

ROUND 2

Fit the clock of the spike's decay from the read-out call (it is large enough to test); measure resolved rises at single-agent role changes (NE38 and #51's later reassignments); run `confirm.py`.

- **H127-R1** The clock of the spike's decay.
- **H127-R2** Resolved rises.
- **H127-R3** Restatement.

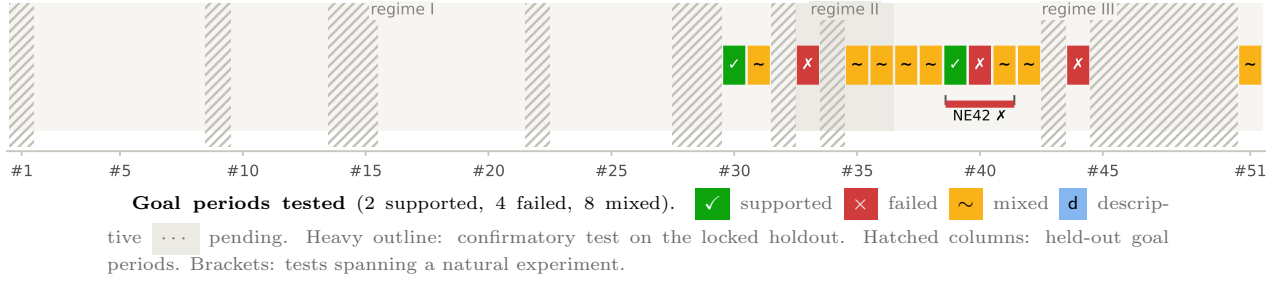
H128: Free kickoffs coarsen, named kickoffs freeze: a kinetic Potts quench

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. After a goal kickoff, the spread of agents over work projects does not separate free from named kickoffs (freeze time $p = 0.32$, excess area $p = 0.73$, power ≥ 0.96), and free weeks lose projects by finishing, not merging (merge share 0.00–0.38): HH369's coarsening-vs-freeze split is killed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

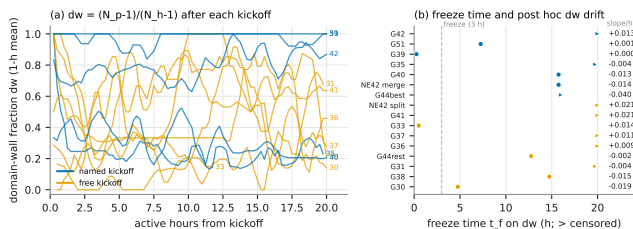
HH369 says that a goal kickoff is a zero-temperature Potts quench on the agents' work projects. A free kickoff should coarsen: projects merge and their number falls as $t^{-\alpha}$ with $\alpha \approx 0.3$ –0.5. A kickoff that names its target should freeze at once. H128 replays every agent's work-ledger host label after 12 non-holdout kickoffs (7 free, 5 named; regimes I–III) and two natives (#44 rooms, NE42). **Free and named kickoffs give the same curves:** freeze time $p = 0.32$, excess area $p = 0.73$, where a true coarsening-vs-freeze world gives $p < 0.05$ with power 0.96–1.00. Free weeks lose projects by finishing, not merging (merge share 0.00–0.38 in 7/7). HH369's kill is met. The holdout is not run.

PHYSICAL MODEL

Model 10, kinetic Potts quench with births. At each of its work windows an agent picks repo j with $P(j) \propto e^{\beta J s_j + h_j + b[j=\sigma_i]}$ plus a new-repo option. Free kickoff: $h = 0$, so domains merge by coupling (mean-field voter: $N_p \propto 1/t$). Named kickoff: $h_T \gg 1$ on a shared target or one target per agent, so the count jumps to its final value in one round. Rival R1: projects end by finishing (expiry), with no merging.

$$d_w(t) = \frac{N_p(t) - 1}{N_h(t) - 1}, \quad t_f = \min\{t : |d_w(s) - d_w^\infty| \leq \delta \ \forall s \geq t\}, \quad \delta = \max\left(\frac{1}{N_h - 1}, 0.15 d_w^\infty\right)$$

$$A_u = \int_{t_{pk}}^{t_{pk} + 8h} \frac{d_w(t) - d_w^\infty}{d_w^{pk} - d_w^\infty} dt, \quad c_m = \frac{\# \text{ merge deaths}}{\# \text{ project deaths}}$$



Observables. (a) Domain-wall fraction d_w after each kickoff (1-h running mean; named blue, free orange). Per-agent named weeks (#39, #51) sit at $d_w \approx 1$; free weeks stay scattered, fragment (#41) or drift slowly. (b) Freeze time t_f per unit (triangles: censored at the horizon), with the post hoc drift of d_w per active hour. The dashed line is the 3-h freeze criterion. The kickoff code does not order either quantity.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: naming the target in a kickoff does not make the swarm converge on fewer projects. Own-artifact goals freeze at one project per agent from the first hour. A free goal leaves agents scattered or splitting, and projects end when work stops, not by merging (merge share ≤ 0.38). To get agents onto one repo, give a shared target to a swarm that is already scattered (NE42: d_w 1 to 0.2 over about 16 active hours). Expect hours, not minutes.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
F: rule valid on skeletons	d_w 12/13; HH's N_p 0/16 (A1)	d_w only
P1 free: $t^{-\alpha}$, $\alpha \in 1/7$ (#30, $\hat{\alpha}$ 0.44 [0.32, 0.60])		failed
P2 named: freeze ≤ 3 h	1/5 (#39, frozen at $d_w = 1$)	failed
P3 free slower (kill test)	t_f $p = 0.32$; A_u $p = 0.73$	killed
P4 merging, not finishing	c_m 0.00–0.38 in 7/7 free	failed
P5 power beats exponential	3/7	failed
N1 #44 rooms	free room flat at 0.92; named drifts	failed
N2 NE42 A-B-A	named merge coarsens (0.26); failed free splits	
N3 #37 replicas	one room < 2 hosts	descr.

GOAL PERIODS IN DETAIL

G30 ✓ supported

G31 $\sim$ mixedG33 $\times$ failedG35 $\sim$ mixedG36 $\sim$ mixedG37 $\sim$ mixedG38 $\sim$ mixed

G39 ✓ supported

G40 $\times$ failedG41 $\sim$ mixed

+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The observable was changed before real data (A1): N_p failed the synthetic rule on every skeleton. Power law vs exponential is not identified at village sizes (a true coarsening selects the power law in $\leq 48\%$ of runs), so P1 and P5 were weak by design; the kill contrast P3 is the powered test. Host labels lapse after 100 own calls, so “finishing” mixes shipped and abandoned projects. The NE42 coarsening reading and the drift signs are post hoc (A2). #44 #best has 4 hosts (below the testability floor).

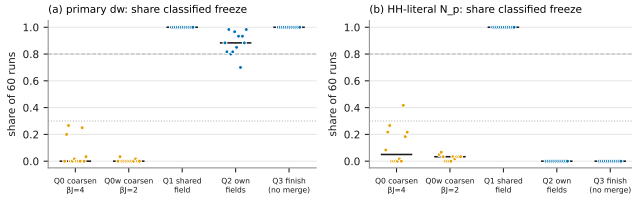
ROUND 2

Measure field strength per agent (share of first post-kickoff commits on the settled repo), split finishing into shipped vs dropped with deploy commits, and test “shared target into a scattered start” on held-out #45. Confirm C1–C3 on #46–#49.

- **H128-R1** A per-agent field strength, not a kickoff code.
- **H128-R2** Separate finishing from abandonment.
- **H128-R3** Shared target into a scattered start.

H128-coarsening-vs-freeze · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H128-coarsening-vs-freeze/README.md · ratings: H128 agent (suggested) (page 2)

Second figure. Synthetic validation on the real skeletons (60 runs per world and period). (a) The primary d_w classifies coarsening worlds as slow and field or finishing worlds as frozen. (b) The HH’s literal project count N_p calls per-agent freezes and pure finishing “slow” in every run: it tracks how many agents hold a label, not the number of domains (Amendment A1).



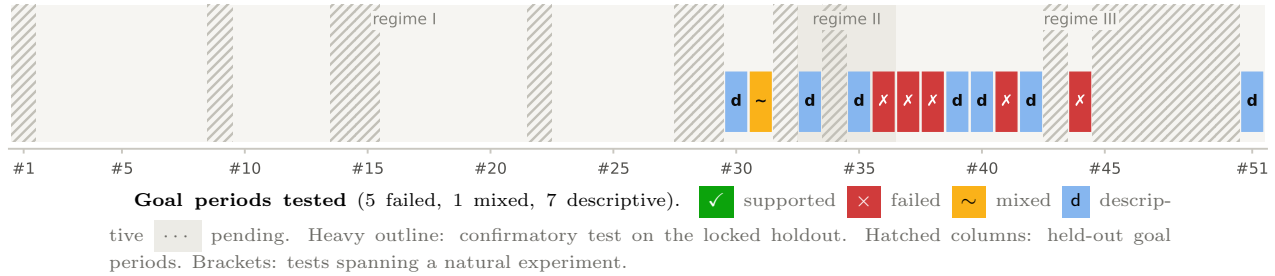
H129: Project hopping carries cycle currents: the sticky Potts walker breaks detailed balance

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1/5

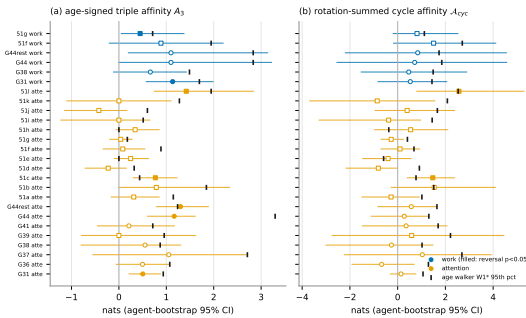
Bottom line. Agents' project hops carry no circulation beyond age drift (cycle affinity above a calibrated age walker in 0/9 shared-goal unit-channels, power 0.55–0.97); net flow to newer projects is small but consistent (30/37 positive) and dwell hazards age rather than stay geometric (32/39).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH370 says that agents hopping between projects behave like a sticky Potts walker whose flows are driven around cycles ($A \rightarrow B \rightarrow C \rightarrow A$) by projects being born and dying, so detailed balance breaks with a circulation from older to newer projects, and dwell times are geometric. H129 extracts every agent’s sequence of distinct projects (work host labels and attention labels) in 13 non-holdout periods and 12 #51 units and measures triple affinities against a per-agent time-reversal null and against an age walker calibrated to each unit. **There is no circulation beyond age drift:** the rotation-summed cycle affinity beats the age walker in 0/9 shared-goal unit-channels (power 0.55–0.97 for a 2-nat cycle). HH370’s literal test passes in only 3/9. Dwell hazards age (32/39). The holdout is not run.



Observables. (a) Age-signed triple affinity A_3 and (b) rotation-summed cycle affinity $\mathcal{A}_{\text{cyc}}$ per testable unit-channel (work blue, attention orange; agent-bootstrap 95% CIs; filled markers: per-agent reversal null $p < 0.05$; circles shared-goal, squares own-role). Black ticks: 95th percentile of the age walker calibrated to the unit’s net age flux and return share. No shared-goal unit exceeds its tick in (b).

PHYSICAL MODEL

Models 10, 02 and 15. Sticky walker: per own call an agent leaves with probability r ; at a hop it picks j with $P(j) \propto e^{\alpha_j + b[\text{held}] + \lambda z_j(t)}$, z_j the standardized age rank among available projects. Any potential (fixed attractiveness, age bias λ) makes the three rotations of a triangle cancel; a cyclic force κ does not. Age drift is excess entropy production; a cycle is housekeeping.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: agents do not rotate through projects. They drift slightly toward the newest ones (net flow 0.03–0.21 of hops) and become stickier the longer they stay (hazard falls by about 20–60% per e-fold of dwell). To move an agent off a project, act early in its stay; an agent several hours into a project rarely leaves on its own. Expect few returns-in-a-ring; plan handoffs as one-way moves to new work.

$$A_3 = \ln \frac{N(omn) + \frac{1}{2}}{N(nmo) + \frac{1}{2}}, \quad \mathcal{A}_{\text{cyc}} = \sum_{\text{rot}} \ln \frac{N(\text{rot}) + \frac{1}{2}}{N(\overline{\text{rot}}) + \frac{1}{2}} \approx 2 \ln \frac{P_{om} P_{mn} P_{no}}{P_{on} P_{nm} P_{mo}}$$

$$m_2 = \frac{n_{\uparrow} - n_{\downarrow}}{n_{\uparrow} + n_{\downarrow}}, \quad \ln h(d) \simeq a + \gamma \ln d$$

H129-project-cycle-currents · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H129-project-cycle-currents/README.md · ratings: H129 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
F: valid circulation test	$\mathcal{A}_{cyc} + W1^* 5/6$ skeletons; C_2 not identified	partly
P1 $A_3 > 0$ vs reversal (kill)	3/9 shared unit-channels	killed
P2 $\mathcal{A}_{cyc}$ beyond age 0/9 walker		failed
P3 Hodge curl	0/5; not identified (A1)	n.i.
P4 $m_2 > 0$, $p < 0.05$, $\geq 2/3$	8/34; sign + in 30/37 (post hoc)	failed
P5a geometric dwell	6/39; aging in 32/39	failed
P5b rate falls with habit	$\rho = -0.29$ (p 0.53)	weak
P6 own-role smaller	reversed (2 vs 3 units)	failed
N2 #38 drift, no cycle	m_2 0.17 (p 0.03); $\mathcal{A}_{cyc}$ inside	supported

GOAL PERIODS IN DETAIL

G30 d descriptive
G31 $\sim$ mixed
G33 d descriptive
G35 d descriptive
G36 $\times$ failed
G37 $\times$ failed
G38 $\times$ failed
G39 d descriptive
G40 d descriptive
G41 $\times$ failed

+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

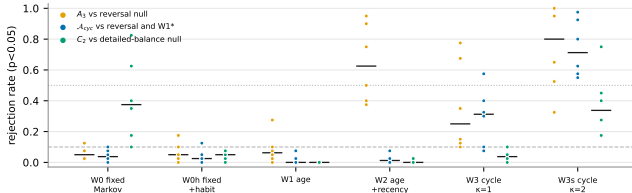
Work hops are few outside #51 (2–120 per period); most per-period verdicts rest on 30–170 triples, and weak cycles ($\kappa = 1$: power 0.07–0.57) are not excluded. Attention labels are artifact mentions, not work. Age is the first non-holdout appearance; projects started before the record look young. The m_2 sign test and #51 pooling are post hoc (A2); two own-role #51 attention units pass both nulls, about the chance rate for 14 tests.

ROUND 2

Model dwell as an aging renewal process and relate its exponent to habit and to H72's gate aging; replicate the two #51 hits across units; bound the excess EP of the age flow. Confirm C0–C3 on #45–#47.

- **H129-R1** Aging dwell as a model.
- **H129-R2** Pool cycle tests across #51's private-goal units on the work channel once more units pass the triple floor (only 2/12 now); use the 12 attention units as a replica test of the two hits (51c, 51l).
- **H129-R3** Excess bound.

Second figure. Synthetic validation on six real skeletons (40 datasets per world). A_3 against the reversal null rejects in an age walker with recency and no cycle (W2), so it measures drift. $\mathcal{A}_{cyc}$ against both nulls stays below 0.10 in every no-cycle world and detects a 2-nat cycle (W3s) in 0.55–0.97 of datasets. The detailed-balance null for the Hodge curl rejects in the reversible fixed world W0 (finite-time transients), so C_2 is not identified.



H129-project-cycle-currents · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H129-project-cycle-currents/README.md · ratings: H129 agent (suggested) (page 2)

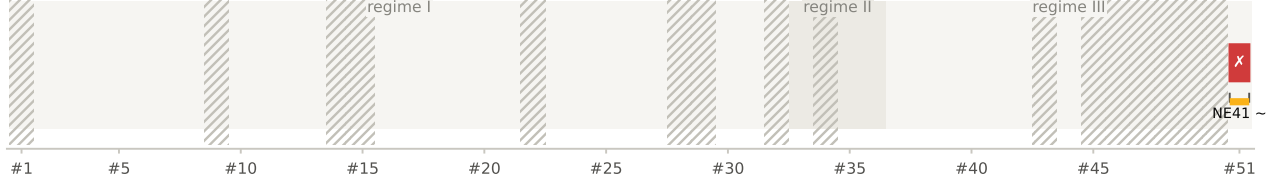
H130: #51 agents are Ornstein–Uhlenbeck particles in private wells: reads kick them and the kick decays at the well’s rate

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. In #51 a read pulls the next statement toward the sender (J_K 0.044 [0.038, 0.050], 11/12 units, both models), but the pull fades about 15× faster (≈ 7 calls) than the agent’s own content memory (γ_{auto} 0.0094/call, homogeneous over 12 units).

level: descriptive · pipeline checks 86% · updated 2026-10-04

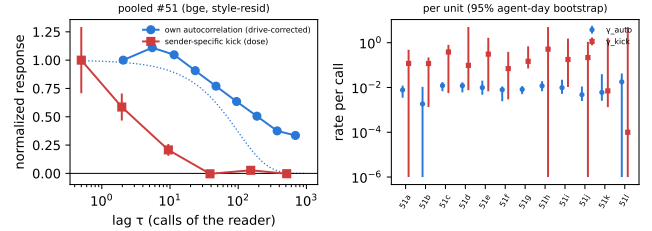


Goal periods tested (1 failed, 1 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending.

Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

HH371 proposed the simplest kinetic version of H98’s random field: each #51 agent is an Ornstein–Uhlenbeck particle in its own well, every read is a small kick toward the sender, and the kick decays at the well’s own rate. H130 measures both rates on the per-call clock in all 12 non-holdout #51 units (45 days, 40,958 statements, 930k reads), with two embedding models. A read does kick: the read-minus-in-flight jump is $J_K = 0.044$ [0.038, 0.050], positive in every unit with ≥ 3 days. The well relaxes at $\gamma_{\text{auto}} = 0.0094$ per call (≈ 100 calls, 1 h), the same in every unit. The kick fades $\approx 15\times$ faster (ρ_γ 14.8, 90% CI [7.2, 30.4]), so the one-rate kill fires. The holdout is not run.



Observables. Left: pooled #51 (bge). Own content autocorrelation, corrected for the common drive (blue), against the sender-specific kick per read from the dose regression (red), both normalized, against lag in the reader’s calls. The dotted line is $e^{-\gamma_{\text{auto}}\tau}$. The kick is gone after ≈ 10 calls; the memory lasts ≈ 100 . Right: per-unit rates with 95% agent-day bootstrap CIs.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Vector OU dynamics on the call clock (model 11) with a measured response tested against fluctuations (model 02). x_i is agent i ’s 32-d whitened, style-residualized content minus its well centre h_i (leave-day-out mean). γ is the well stiffness per call, κ the kick per read toward the sender’s message z_m , ξ the agent’s own noise, D_r a room drive. A linear system has one rate for response and memory; a kick held only in the context window would decay faster.

$$x_i(n+1) = (1 - \gamma) x_i(n) + \kappa \sum_{m \text{ read at } n} (z_m - h_i) + \xi_i(n)$$

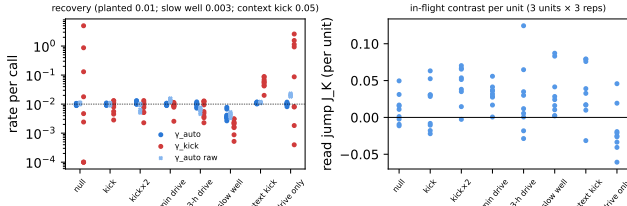
$$C(\tau) = \langle x_B \cdot x_{B'} \rangle - C_\times(|t_{B'} - t_B|) \rightarrow \gamma_{\text{auto}}$$

$$y_{Bj} = x_B \cdot \widehat{h_j} - h_i = \sum_b \beta_b R_b^{(j)} + \dots \rightarrow \gamma_{\text{kick}}, \quad \rho_\gamma = \gamma_{\text{kick}} / \gamma_{\text{auto}}$$

H130-ou-private-wells-51 · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H130-ou-private-wells-51/README.md · ratings: H130 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1/K2 read kick > in-flight	0.044 [0.038, 0.050]; 7/7 units	≥ 3 supported
P2/K1 $\rho_\gamma \in [0.5, 2]$	14.8 [7.2, 30.4]; gte 20.2	failed (kill)
P3 co-movement rises with reads	β_R 0.018 [0.012, 0.024]	supported
P4 pair decay at γ	$\gamma_\times / \gamma_{\text{auto}} \approx 6-8$	failed
P5 one well rate, 30-600 calls	I^2 0; $1/\gamma \approx 106$ calls	supported
P6 call clock fits better	5/12 units	not supported
N1 rival kick ratio $\in [0.5, 2]$	2.4 [-0.7, 5.3]	inconclusive
N2 NE41: memory, R_C 1.03 [0.95, 1.10]; R_K undefined	kick survive erasure	mixed
N3 joiners relax at γ	flat at 0.39 of own well on day 1	failed



Second figure. Synthetic validation on the real #51 skeleton (real statements, calls, reads, resets; 3 units $\times$ 3 replicates). The drive-corrected γ_{auto} recovers the planted 0.01 under 5-min to 3-h room drives; the raw estimator does not. The sender-specific γ_{kick} recovers 0.01 and separates a context-held kick (0.05). The registered message-direction profile read the drive's rate and was replaced before real data (A1).

GOAL PERIODS IN DETAIL

G51 $\times$ failed
NE41 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		
Level: descriptive . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

CAVEATS

The estimators for both rates were changed after the synthetic and before real data (A1). γ_{kick} rests on the first 2–3 lag bins, so its CI spans $\times 4$, but the lower 90% bound of ρ_γ is 7. In one-room #51 every agent reads every message, so beyond the first read-out the kick is identified only through sender-specific doses. The erasure test of the kick is undefined (the kick has decayed by lag 4). One private-role period; no transfer test.

ROUND 2

Resolve the kick kernel at 1-call steps and split replies from plain reads; fit joiners' well formation over days; run the read jump and the two rates on #38–#44; run the frozen confirm on the #51 tail after sign-off.

- **Essence.** what clock the swarm's influence runs on, separate from what clock its memory runs on.
- **H130-R1** Fast-kick kernel.
- **H130-R2** Kick vs reply.
- **H130-R3** Well formation.
- **H130-R4** Transfer.

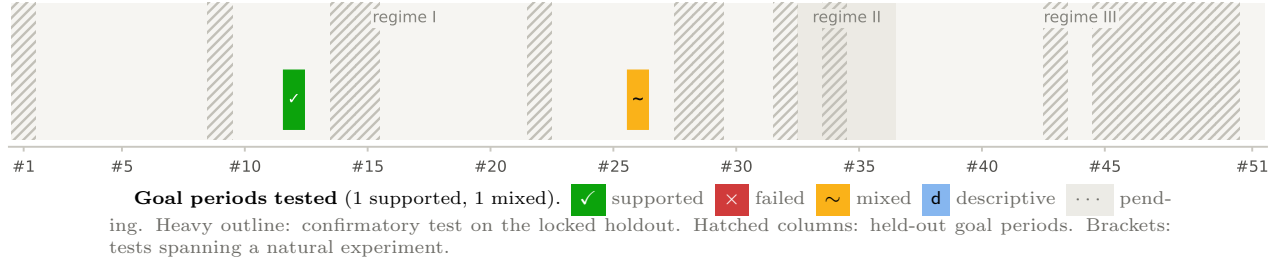
H131: Assigned antagonism switches off in one read-out after the prize goes: a field quench with no remanence

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. In the #12 debates, rival antagonism (40% of rival replies flagged while open; 0% between teammates) is fully off by each rival's first reply after it reads the verdict (0/21). The read-out vs wall-clock question is untestable here.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH374 asks whether assigned antagonism switches off at each former rival's first call after it reads the verdict, as a weakly coupled kinetic Ising spin relaxes in one update when its field goes. H131 tests this in the ten judged debates of #12 and the #26 runoff with the validated stance v2.1 conflict flag and the DQ1 read-out ledger. In #12, 34 of 85 rival replies carry the flag while the prize is open; teammates carry 0 of 87. After each speaker reads the verdict, rivals drop to 0 of 57, and each rival's first post-read reply is already at 0 of 21. The HH's kill test cannot run: no reply falls between a verdict and its speaker's read, so read and wall-clock alignment coincide.

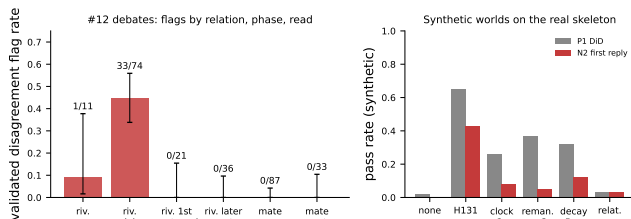
PHYSICAL MODEL

Kinetic Ising on reply bonds (model 02) with a staggered two-camp field (model 10, $q = 2$). A reply by j to i from j 's call c carries a latent conflict z . $R_{ij} = 1$ for drafted rivals, $O_j(c) = 1$ while j has not read the verdict, h is the prize field, and u_j, v_i are agent fields. The flag y is z seen through the labeller: recall $\rho \approx 0.6$, false-positive rate $\varepsilon \approx 0.005$. Remanence lets h decay over κ post-read talk calls; H131 says $\kappa \rightarrow 0$.

$$\logit P(z = 1) = \mu + u_j + v_i + h R_{ij} O_j(c) e^{-(k-1)/\kappa}$$

$$P(y = 1) = \rho P(z = 1) + \varepsilon [1 - P(z = 1)]$$

$$\Delta = (\bar{y}_{R=1} - \bar{y}_{R=0})_{\text{open}} - (\bar{y}_{R=1} - \bar{y}_{R=0})_{\text{read}}$$



Observables. Left: #12 flag rates (Wilson 95% CIs) for rival and teammate replies by phase and read status; "1st post-read" is each speaker's first rival reply after its ledger read of the verdict. Right: pass rates of the read-gated DiD (P1) and the first-reply test (N2) in planted worlds on the real reply skeleton (100 replicates each). N2 passes in 43% of read-gated worlds and 5–12% of slow-relaxation worlds.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, assigned antagonism is a field: settle the contest and make sure each party reads the settlement, and its conflict ends by its first reply after that. In #12 that took a median of 93 s after the verdict, with no cool-down and no resentment. The design cannot yet say whether the read itself or simply the passage of time ends the conflict, because in scheduled chat every agent read the verdict within seconds.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1 read = clock on skeleton	0/262 replies differ	confirmed
S2 P1 power ≥ 0.8	0.65	failed (A1)
P1 #12 read-gated DiD $\Delta > 0$	+0.32 [0.16, 0.52], p 0.0002; γ_{set} +0.03	pass
P1 #26	-0.01, p 0.72 (power 0.11)	inconclusive
N1 kill: in-flight replies	0 rival in #12 and #26	untestable
N2 first post-read reply at baseline	0/21 vs open 0.45 (p 0.0002)	pass
N2 soft residual	0.031 vs 0.45 (7%)	small
N3 switch-on (descriptive)	0.44 vs 0.39	non-discriminating
Overall (A1 rule)	P1 + N2 pass; N1 untestable	narrowed

GOAL PERIODS IN DETAIL

G12 ✓ supported
G26 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	2
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		
Level: descriptive . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

CAVEATS

The verdict also ends the debate protocol, so “nothing left to argue” explains the switch-off as well as a removed field; the open-phase teammate contrast (0/87) is the only evidence that the antagonism is relation-specific. All flags come from one period and ten debates. The flag’s precision is 0.61–0.67. Zero flags among 57 post-read rival replies bounds the latent rate at $\lesssim 0.08$. The DiD sign was corrected and the first-reply cell redefined before outcomes (Amendment A1).

ROUND 2

Find an in-flight partition (regime II/III settlements, held-out #29/#34 via the frozen confirm); label whether post-verdict rival replies still discuss the motion; report a posterior for the latent post-read rate.

- **H131-R1** Find an in-flight partition.
- **H131-R2** Separate the field from the topic.
- **H131-R3** Latent-scale bound.

H131-antagonism-off-one-readout · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H131-antagonism-off-one-readout/README.md · ratings: H131 agent (suggested) (page 2)

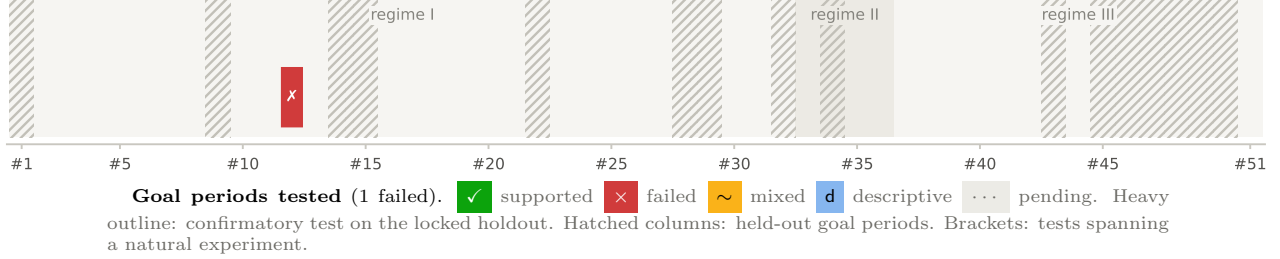
H132: Debate is a two-sublattice limit cycle: topic ping-pong at lag one

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1/5

Bottom line. In #12's debates the team-centred issue projection persists across consecutive turns of opposite teams ($\rho_1 +0.16$, shuffle p 0.002), with no lag-one alternation beyond the turn-shuffled null ($\Lambda +0.05$, p 0.08; power 1.00 at $k_\times = 0.6$).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH375 proposed that a debate is a two-sublattice limit cycle: each team rebuts the other's last move, so the issue content alternates sign at lag one and the cycle runs one way. H132 tests this in #12's ten Asian-Parliamentary debates (306 turns, drafted Government and Opposition teams). Each turn is projected on the motion's support-oppose axis, built from the motion text without labels, and each team's static position is removed. The content does not alternate. Consecutive cross-team turns are *positively* correlated ($\rho_1 = +0.16$, shuffle $p = 0.002$), and the alternation $\Lambda = \rho_2 - \rho_1 = +0.05$ sits inside the turn-shuffled null ($p = 0.08$). The kill fires; the synthetic power for $k_\times = 0.6$ is 1.00. The holdout is not run.

PHYSICAL MODEL

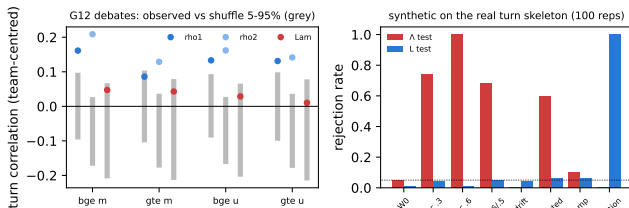
Alternating-update two-sublattice linear dynamics (model 02) with antiferromagnetic block coupling (model 01). y_t is turn t 's whitened projection on the issue axis $\hat{a}_d$; $\epsilon_t = \pm 1$ marks Government or Opposition; μ_d is the assigned (staggered) field; u_t is the dynamic issue state with antiferromagnetic response $k_\times$; z_t removes each team's mean. Speakers alternate by format, so only z_t can carry a cycle.

$$\begin{aligned} y_t &= \epsilon_t \mu_d + a_{i(t)} + u_t + e_t, & u_{t+1} &= -k_\times u_t + \eta_{t+1} \\ \rho_1 &\approx -k_\times r, & \rho_2 &\approx k_\times^2 r, & \Lambda &= \rho_2 - \rho_1 \\ A_\gamma &= \rho_{G \rightarrow O}(1) - \rho_{O \rightarrow G}(1), & L &= \langle a_t g_{t+1} - g_t a_{t+1} \rangle \end{aligned}$$

Observables. Left: observed ρ_1 , ρ_2 and Λ (dots) against the 5–95% band of 5,000 turn shuffles (grey) in four input variants. A cycle needs ρ_1 below its band; it sits above it. Right: rejection rates on the real turn skeleton (100 replicates per world): the Λ test has size 0.05 and power 0.74–1.00 for cycles; drift never triggers it; a rotation is caught only by L .

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator running adversarial or debate formats: assigned sides show up as a static tilt in content, and turn-to-turn content persists across the team line. Expect a rebuttal to restate the point it answers (lag-one correlation $\approx +0.16$ on the issue axis, $+0.15$ in full content), not to swing the topic back and forth. Use stance labels, not embeddings, to see opposition between moves.



RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1/K $\Lambda > 0$, $\rho_1 < 0$	$\Lambda +0.047$ (p 0.08); $\rho_1 +0.161$	failed (kill)
P2 $A_\gamma \neq 0$	$+0.048$ (p 0.71)	failed
P3 no within-team alternation	same-team $\rho +0.30$	supported
P4 cycle area $L \neq 0$	-0.071 (p 0.39)	failed
P5 topic echo $\rho_1^{\text{vec}} > 0$	$+0.145$ (p 0.0002)	supported
N1 verdict switch-off	$\Lambda_{\text{post}} -0.18$, no cycle to stop	uninformative
N2 read gate	$\Delta\rho_1 -0.006$ (p 0.47)	failed
N3 re-drafted pairs	-0.047 (18 pairs, p 0.29)	failed

G12 $\times$ failed

GOAL PERIODS IN DETAIL

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

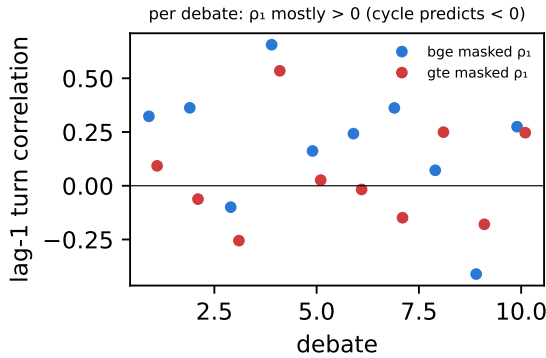
Embeddings encode negation poorly, so a rebuttal that restates a claim reads as agreement on the issue axis; the antiferromagnet H21 and H37 found lives in reply stance. Cycles weaker than $k_\times \approx 0.4$ at H21-sized noise are not excluded (power 0.74 at 0.3). A staggered-field ramp gives false Λ in 10% of synthetic worlds. The negative L in all variants is post hoc. Ten debates, one period, regime I.

ROUND 2

Re-run the turn statistics on stance v2.1 and DQ2 reply stance to the previous turn; fit a common-drift plus team-persistence model; try other alternating formats (#23, #26); run the frozen #34 confirm after sign-off.

- **Essence**. whether assigned opposition creates a dynamic coupling (answer the other side) or only a static field (hold your side).
- **H132-R1** Stance sequence.
- **H132-R2** Drift model.
- **H132-R3** Other alternating formats.

H132-debate-limit-cycle · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H132-debate-limit-cycle/README.md · ratings: H132 agent (suggested) (page 2)



Second figure. Lag-1 turn correlation per debate in the two masked embedding models. A limit cycle predicts negative values in most debates; $\rho_1 > 0$ in 8/10 debates in bge and 6/10 in gte.

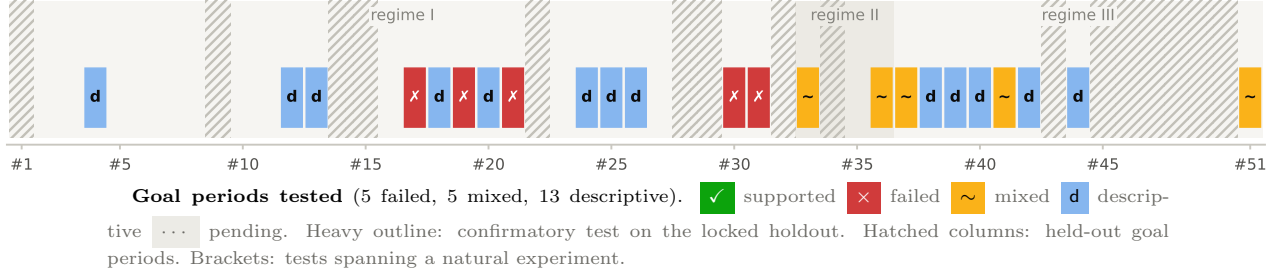
H133: Read-out Glauber Potts: an agent switches project only at a call, driven by the named messages it just read

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 1/5 Usefulness 1/5

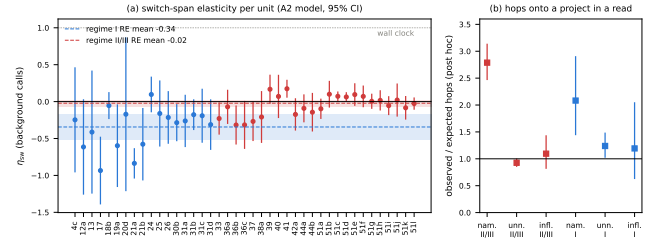
Bottom line. In regime II/III an agent's background project-switch hazard runs on its own call clock (switch-span elasticity $-0.02 [-0.07, +0.02]$, 24 units). Named-read coupling is untestable at the planted effect; post hoc, hops onto a project named to the agent are $2.8\times$ expected, unnamed $0.93\times$, in flight $1.1\times$.

level: hypothesis · pipeline checks 86% · updated 2026-10-07



WHAT HAPPENED

HH376 says that an agent changes project only at its own model call, by a Glauber update whose odds for project b rise with the read messages that name the agent and link b ; unnamed links add nothing, and without reads the switch rate is constant per call, not per hour. H133 labels every call with a project (shared `project_calls`: 2.6M calls, 61,939 non-reserved hops) and fits a conditional logit with project $\times$ active-hour effects in 41 non-reserved units. **The coupling test is untestable as planned:** named reads about a non-current project are so rare that a planted $\gamma_{\text{nam}} = 1$ gives under 2 such hops per unit (power ≈ 0 ; Amendment A1). **The clock test holds in regime II/III:** background switches do not scale with wall time ($\eta_{\text{sw}} = -0.02 [-0.07, +0.02]$, 24 units). The reserved periods are not run.



Observables. (a) Background switch-span elasticity η_{sw} per unit (A2 model, agent-cluster 95% CIs; blue regime I, red regime II/III; bands: random-effects means). No unit approaches the wall clock at 1. Regime II/III sits at 0; regime I is negative. (b) Post hoc, descriptive: hops onto a project in a read message, observed over the count expected without read terms, summed over units (Poisson 95%). Named reads: 2.79; unnamed: 0.93; named in flight (not yet readable): 1.10.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: an agent's spontaneous project switching is a per-call rate, so slowing its call rate slows its drift between projects, and long pauses do not add switches. Descriptively, a message that names an agent and links a project goes with about 2.8 times the expected switches onto that project. A link the agent reads without being named goes with none. To move an agent, name it in the message with the link. This rule is not yet a tested result.

$$u_{\text{stay}} = \alpha_i + \phi \ln(1 + d)$$

$$u_b = \kappa_{b,h} + \sum_k \gamma_k \ln(1 + N_b^k) + \rho[\text{held}] + \beta_s s_b$$

$$\text{cloglog } P(\text{hop} \mid \text{no read}) = \alpha_i + \eta_{\text{sw}} \ln \text{span}_c + \dots$$

H133-readout-glauber-potts · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H133-readout-glauber-potts/README.md · ratings: H133 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
F: O1–O3 testable	power ≈ 0 per unit at $\gamma = 1$	A1
P1 $\gamma_{\text{nam}} > 0$	9/24 units (13 estimable); RE +2.3	untestable
P2 $\gamma_{\text{un}} = 0, \Delta\gamma > 0$	RE $\gamma_{\text{un}} + 0.09 [-0.16, 0.34]$	untestable
P3 named $>$ in flight	γ_{if} in 3 units	untestable
P4 $\eta_{\text{sw}} \ni 0, \not\ni 1$ (II/III)	$-0.02 [-0.07, +0.02]$	holds
P5 regime I $\eta_{\text{sw}} > 0$	$-0.34 [-0.51, -0.17]$	failed
P6 read-hop lag ≤ 1 : ≥ 0.55		not met
N2 #51 timer wakes $\eta \ni 0$	$+0.46 [+0.09, +0.84]$	failed
N4 G31 $\eta > 0$	$-0.22 [-0.34, -0.09]$	failed
Kill B (wall clock)	CI excludes 1 in 41/41	not fired

GOAL PERIODS IN DETAIL

G04 d descriptive
G12 d descriptive
G13 d descriptive
G17 $\times$ failed
G18 d descriptive
G19 $\times$ failed
G20 d descriptive
G21 $\times$ failed
G24 d descriptive
G25 d descriptive
+13 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

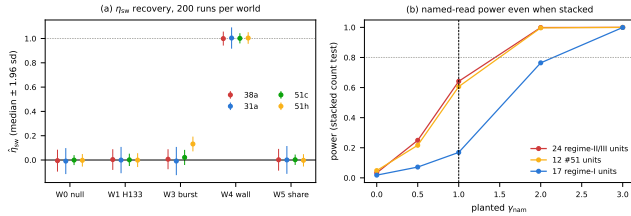
The read ratios are post hoc and descriptive: they were added after the synthetic run made the planned test untestable, and units with few named hops are noisy. A named message may answer the reader's own earlier mention of the project (not yet controlled). Call hops are fine-grained: 26% return to the previous project within 5 calls. $\eta_{\text{sw}} < 0$ in regime I and > 0 at timer wakes are unexplained; a burst field can also give $\eta \approx +0.1$.

ROUND 2

Pre-register the named-read ratio with power at $\gamma \approx 2$ and hierarchical shrinkage; control for the reader's own last touch of the project; split spans by gap kind to explain the regime-I and timer-wake elasticities. Confirm C1 (call clock) on #45–#47 and the #51 tail.

- **Essence.** whether naming an agent in a message that links a project moves that agent onto the project, beyond the attention burst that produces the message. Round 1 saw a $2.8\times$ lift only post hoc, so round 2 must test it with power at the observed scale.
- **H133-R1** Test the named-read ratio with power at $\gamma \approx 2$.
- **H133-R2** Add a sender-side impostor.
- **H133-R3** Explain $\eta \ni 0$ in regime I and $\eta \not\ni 0$ at timer wakes.
- **H133-R4** Fit the deferred variants.

H133-readout-glauber-potts · generated 2026-10-07 17:50 by in-fra/summaries/build-summaries.py · card hypotheses/H133-readout-glauber-potts/README.md · ratings: H133 agent (suggested) (page 2)



Second figure. Synthetic validation on four real skeletons. (a) η_{sw} is recovered without bias in null, H133, share and wall-clock worlds (200 runs each); an attention burst biases it to $+0.13$ on 51h. (b) Even stacking all 24 regime-II/III units, a count test of named-read hops has power 0.64 at the planted $\gamma_{\text{nam}} = 1$. It reaches 1.0 only at $\gamma_{\text{nam}} = 2$, so the per-unit coupling test was declared untestable before real data.

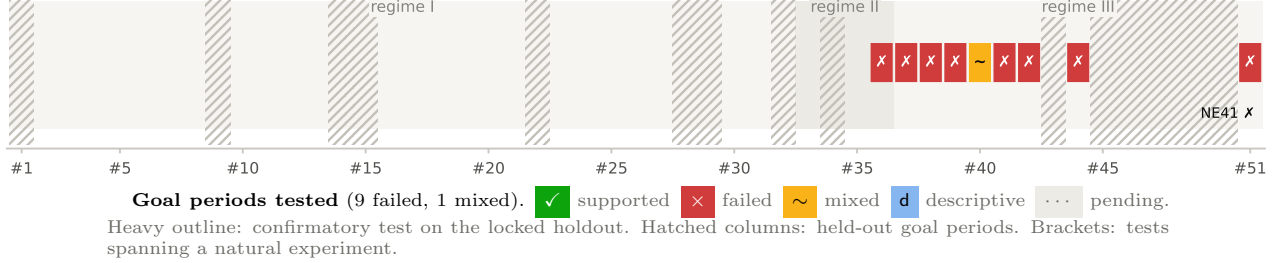
H134: A Pólya-urn self-field holds an agent on its project: the leave odds fall as the project's share of the agent's own context rises

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 0.5/5 Usefulness 1/5

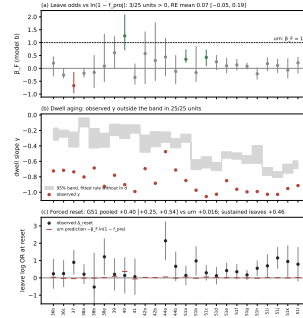
Bottom line. Project stickiness ages strongly, but the context's project self-share does not carry it; a context reset releases the agent as a step, not as the urn's dose.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

The claim (HH377): an agent stays on a project because its own context is full of that project. The field toward the current project is $-\ln(1-f)$, with f the share of the agent's own calls in the context segment that touched the project (a Pólya urn, as H16 and H72 found for idle traps). The urn predicts a leave slope of 1 on $\ln(1-f)$, and its per-call rule should reproduce the observed aging of project dwell without fitting it. On 25 regime-III units (989,450 calls, 58,043 project visits) the slope is $+0.07 [-0.05, +0.19]$, and the rule misses the dwell aging in 25/25 units. The context still matters as a step: a forced reset raises leaving by $+0.40$ log-odds in G51, 24 times the urn's prediction. The reserved data are not used.



Observables. (a) Urn slope β_F per unit with agent-day block-bootstrap 95% CIs; the dashed line is the urn's value 1. (b) Observed dwell slope γ (red) against the 95% band of the fitted per-call rule without dwell (grey, posterior-predictive on the observed risk calls). (c) Leave log odds ratio at the first call after a forced reset, matched on agent, dwell and self-share decile, with the urn's predicted step (red).

PHYSICAL MODEL

Model 10 (kinetic Potts walker) with a self-field, model 02 (escape kinetics), model 08 (self-copying). A visit is a run of calls with one carried project label (shared per-call labels, expiry after 100 own calls). At each own call the agent leaves (touches a new project) or stays. Base terms: agent effects, time in the day, previous call kind, reset at the call, named reads about other projects (K). Rivals: the artifact store holds the project (no reset step), intrinsic aging, per-visit frailty.

$$\text{logit } P(\text{leave}_c) = \alpha_i + z_c \cdot \delta + \beta_F \ln(1 - f_c) + \beta_d \ln d_c + \gamma_R K_c$$

$$f_c = \frac{n_a}{n_{\text{own}} + \frac{1}{2}} \quad (0 \text{ after a reset})$$

$$\text{urn: } \beta_F = 1, \beta_d = 0; \quad \Delta_{\text{reset}} = -\beta_F \ln(1 - f_{\text{pre}})$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A context reset frees an agent from its project, but not in proportion to how much of the context the project fills. In #51 a forced reset raises the leave odds by a factor of about 1.5 at the next call, at fixed dwell. To move an agent off a stuck project, reset its context; do not expect a context crowded with other work to loosen it. Agents get stickier the longer they stay (leave odds fall by about 45% per e-fold of dwell), and the self-share does not explain it.

H134-polya-urn-project-stickiness · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H134-polya-urn-project-stickiness/README.md · ratings: H134 round-1 agent (suggested v1; coordinator writes v2)

RESULTS VS PREDICTIONS

prediction	observed	verdict
P1 $\beta_F > 0$ in $\geq 2/3$ units	3/25	fails
P2 β_F CI $\ni 1$; RE mean $\ni 1$	6/25; RE +0.07 [-0.05, +0.19]	fails
P3 $\varepsilon(F)$ above proxy band	+0.001 vs band 0.055	unpowered
P4 dwell tail follows (Kill A)	γ outside band 25/25	fails
P5 aging left over ($\beta_d < 0$)	G51 -0.58 [-0.62, -0.55]	holds
N1 reset step $\approx$ urn's placebo (20 calls ear-lier)	+0.40 [+0.25, +0.54] vs +0.016 [-0.19, +0.14]	fails
N2 same slope by goal type	difference +0.84 [+0.31, +1.37]	fails

GOAL PERIODS IN DETAIL

G36 × failed
G37 × failed
G38 × failed
G39 × failed
G40 ~ mixed
G41 × failed
G42 × failed
G44 × failed
G51 × failed
NE41 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Hops come from mentions: 27% are flickers ($A \rightarrow B \rightarrow A$ within 5 calls); without them the reset step is +0.46 (post hoc). Context segments last at most 41 calls, so on this skeleton even a true urn makes little dwell aging; $\varepsilon(F)$ and a null reset step were unpow-ered (synthetic power 0.45–0.65 and 0.52), and both were declared before the data. The card's forward tail test failed its size check (0.14–0.27) and was replaced by a posterior-predictive test (size 0.02–0.03). The richer self-share that counts the room's messages about the project gives larger slopes in G51 (+0.5 to +1.4); it is reported, not tested.

ROUND 2

Model the reset as a step clock (own calls since the last forced reset) and measure its decay. Separate intrinsic aging from per-visit frailty with repeat visits to one project. Explain why the self-share slope is positive in own-role weeks and negative in shared-goal weeks. Re-run with a minimum-dwell hop.

- **Essence.** where an agent's project commitment is stored, and which operator action releases it; round 1 says the con-text acts as a reset step, not as a self-share dose.
- **H134-R1** Reset as a step clock.
- **H134-R2** Aging vs frailty.
- **H134-R3** Goal-type slope.
- **H134-R4** Sustained hops.

H134-polya-urn-project-stickiness · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H134-polya-urn-project-stickiness/README.md · ratings: H134 round-1 agent (suggested v1; coordinator writes v2) (page 2)

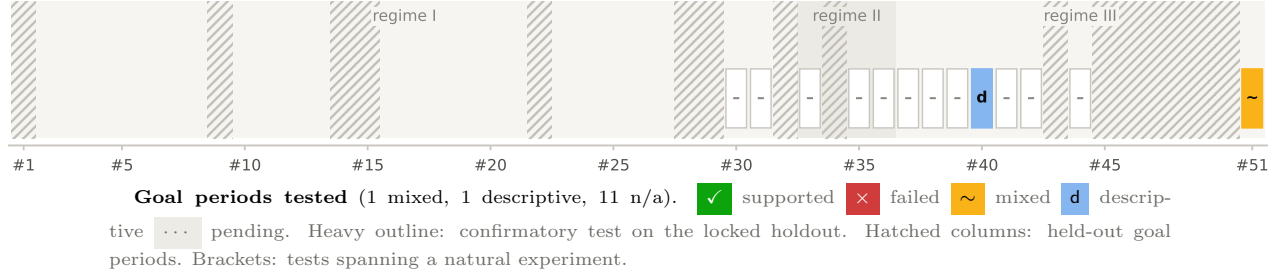
H135: Max-ent allocation as the steady state of a detailed-balance Potts walker: hop-rate ratios from max-ent occupancies

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 40% Faithfulness 0.5/5 Usefulness 1/5

Bottom line. In five own-role #51 attention units, hop destinations are not heat-bath on the max-ent occupancies ($\psi = 0.44$ [0.41, 0.48], habit $\rho = 1.9$ [1.4, 2.4], 5/5); zero net flux holds as written but is unpowered, so detailed balance is neither shown nor refuted.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

HH378 says that H94's max-ent allocation is the steady state of a reversible kinetic Potts walker, so the hop-rate ratio between two projects equals their max-ent occupancy ratio. H135 rebuilds every agent's project hops (work host labels; attention labels) and refits H94's max-ent occupancies π per period unit on non-reserved data, then tests zero net flux between projects that live through the unit and the heat-bath destination law. **The round is inconclusive by the card's rule.** The strict co-alive rule leaves 0/62 unit-channels testable, and the zero-flux test has no power against the rival walkers. On the five #51 attention units testable under the card's 80% variant, **destinations are not heat-bath:** $\psi = 0.44$ [0.41, 0.48] instead of 1, with a habit bonus $\rho = 1.9$ nats.

PHYSICAL MODEL

Models 10 and 15. Agent i walks on the available projects with field $h_i(j) = \ln \pi_i(j)$, π_i the agent-conditional of H94's max-ent fit (margins, ownership, rooms). At each own call it updates with an aging hazard; the heat-bath update draws the new project from π_i . Detailed balance then gives zero net flux on every pair and destinations $\propto \pi_i$. Rivals: age drift, herding sink, cycle force, habit.

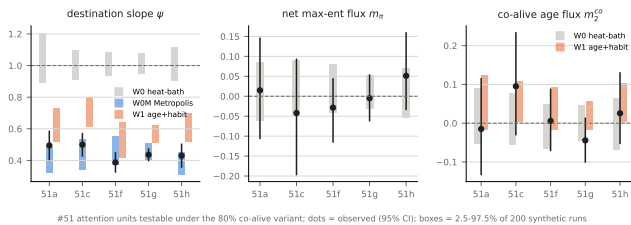
$$\frac{k_{ab}}{k_{ba}} = \frac{n_{ab}}{n_{ba}} \frac{T_b}{T_a} \stackrel{\text{DB}}{=} \frac{\pi_b}{\pi_a}, \quad m_\pi = \frac{\sum_{\text{pairs}} (n_{ab} - n_{ba}) \operatorname{sgn} \ln(\pi_b/\pi_a)}{\sum (n_{ab} + n_{ba})}$$

$$P(b \mid \text{hop from } a) \propto \exp[\psi \ln \pi_i(b) + \rho [b \text{ held}]], \quad \text{heat-bath: } \psi = 1, \rho = 0$$

Observables. Observed statistics per testable unit (#51 attention, 80% co-alive variant; dots with 95% CIs) against 200 synthetic runs on each unit's own skeleton (boxes, 2.5–97.5%). Left: the destination slope ψ sits far below the heat-bath band (gray) and inside the Metropolis band (blue) in 4/5 units. Middle and right: the net max-ent flux and the co-alive age flux sit inside the heat-bath band, but the synthetic shows this test cannot reject the rivals.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: the max-ent prices (ownership, rooms) set where agents spend their time, but they do not set where a moving agent goes next. A hopping agent responds to a project's max-ent weight with less than half the heat-bath exponent and prefers projects it has held before by about $e^{1.9} \approx 7\times$. To redirect agents, name the target explicitly; raising its general attractiveness moves hop destinations only weakly.



H135-detailed-balance-potts-walker · generated 2026-10-07 17:50 by
infra/summaries/build_summaries.py · card hypotheses/H135-detailed-balance-
potts-walker/README.md · ratings: H135 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
Precondition (strict co-alive)	0/62 unit-channels; 5 under the 80% variant	untestable
S1 O1 is an identity	passes in up to 1.00 of cycle-world runs	holds
P1 rate-ratio slope (HH)	1.11–1.60, $r > 0$ in 5/5	consistent only
P2 zero net flux m_π	5/5 inside band; RE -0.002 $[-0.04, 0.04]$	unpowered
P3 zero co-alive age flux	4/5 inside	unpowered
P4 heat-bath destinations	0/5 (ψ 0.39–0.50; ρ 1.2–2.6)	failed
N1 #38, N2 #44	too few co-alive hops	untestable
N3 #40 hub flux days 1–2	0.00 work, 0.04 attention	unpowered

GOAL PERIODS IN DETAIL

G40 d descriptive
G51 $\sim$ mixed
G30 – n/a
G31 – n/a
G33 – n/a
G35 – n/a
G36 – n/a
G37 – n/a
G38 – n/a
G39 – n/a
+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

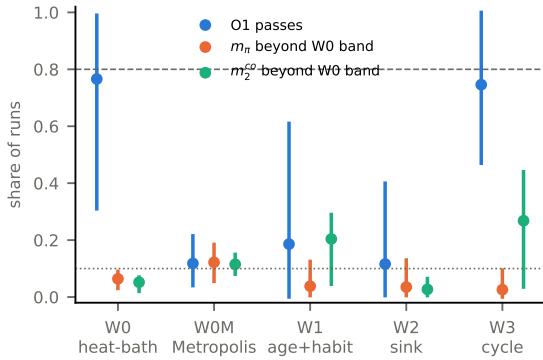
Only own-role #51 attention units are testable, and only under the 80% co-alive variant; attention labels are artifact mentions, not work. The origin $\times$ destination LR test was recalibrated after the synthetic (its χ^2 form rejects 0.76–1.00 of heat-bath runs). $\psi < 1$ may partly be attenuation from noise in π_i (cross-fit not run). The N3 hub counts were seen before the band. Owners come from non-reserved commits (37 repos differ from H94).

ROUND 2

Build a flux test with power: a time-varying or nonreciprocal field (flux after an owner arrives; H137's one-way pairs). Fit the destination law on all hops in every unit and compare ψ across own-role and shared weeks. Cross-fit π on odd and even days.

- **H135-R1** A flux test with power.
- **H135-R2** Destination law.
- **H135-R3** Cross-fit π (odd/even days) to check that $\psi \neq 1$ is not attenuation from noise in π_i .

H135-detailed-balance-potts-walker · generated 2026-10-07 17:50 by `infra/summaries/build_summaries.py` · card hypotheses/H135-detailed-balance-potts-walker/README.md · ratings: H135 agent (suggested) (page 2)



Second figure. Synthetic validation on the five testable skeletons (200 runs per world; dots = mean, bars = range over skeletons). O1 passes as often in the cycle world as in the reversible one, so it is an occupancy identity. The flux statistics leave the heat-bath band in at most 0.29 of age-drift and sink runs: a fixed age preference and symmetric herding are themselves reversible on living projects. Only a cycle force produces flux.

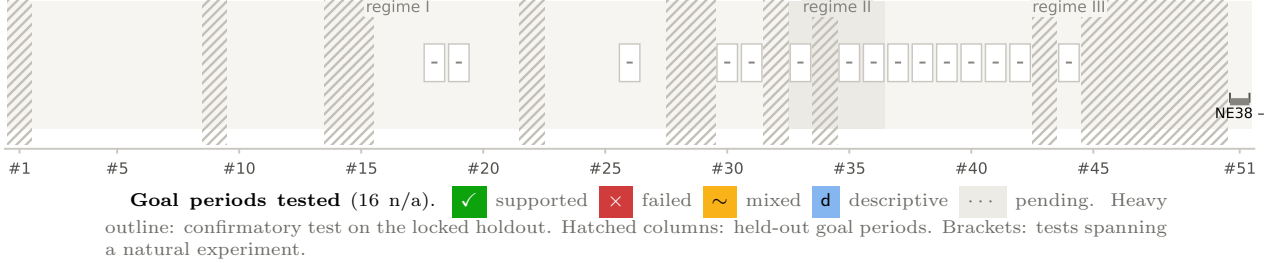
H136: Zero-temperature Potts freeze at the read-out call: each agent commits to a named target at its first read of the kickoff, not at the kickoff time

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 35% Faithfulness 0.5/5 Usefulness 1/5

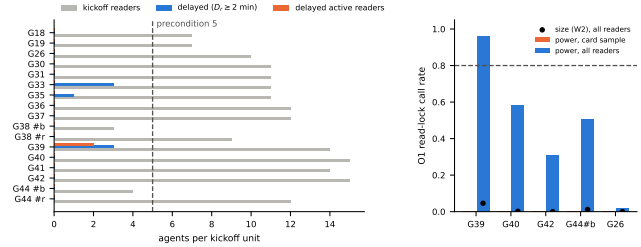
Bottom line. In 17 day-start kickoff units, 171 of 178 kickoff readers (96%) read the kickoff within 2 active min, at day boot, with at most 2 delayed active readers per unit, so the read-locked vs clock freeze test is untestable on exploration data.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

HH379 asks whether each agent commits to a kickoff's named target at its own first call that reads the kickoff, plus about one call, as a zero-temperature Potts spin aligns at its first update in a strong field. The rival is a common delay after the kickoff post (R-clock). The test needs agents who were active before the kickoff but read it late. H136 counted them first, before computing any freeze time. In 17 non-reserved kickoff units (G18–G44, regimes I–III), no unit has more than 2 such delayed active readers; 5 are needed. The kill is untestable on exploration data, and the outcome analysis stopped there.



Observables. Left: per kickoff unit, the kickoff readers, the readers with $D_r \geq 2$ active min, and the delayed active readers (also active in the 30 active min before t_k); the dashed line is the precondition of 5. Right: synthetic O1 power (W1 read-locked) on five real skeletons, 500 runs per world, for the card's sample (all zero) and for all readers, with the size under W2 (clock).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Zero-temperature kinetic Potts quench with read-out updates (model 10; asynchronous updates, model 02). A field h on the named target x appears at the kickoff post t_k . Agent i updates only at its own calls, and the field enters at its kickoff read-out call $t_{r,i}$ (DQ1 ledger). With h above every other field, the agent freezes on x K_i calls after the read, with K_i independent of the read delay $D_{r,i}$. R-clock instead freezes at $t_k + D_i$, truncated at the read.

$$t_{f,i} = t_{r,i} + K_i \text{ calls}, \quad K_i \sim \text{Geom}, \quad K_i \perp D_{r,i} = t_{r,i} - t_k$$

$$\text{R-clock: } t_{f,i} = \max(t_k + D_i, t_{r,i}), \quad D_i \perp D_{r,i}$$

$$\text{O1: } D_{f,i} = a + b D_{r,i} \quad (\text{H136: } b = 1; \text{ R-clock: } b \approx 0)$$

H136-freeze-at-first-kickoff-read · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H136-freeze-at-first-kickoff-read/README.md · ratings: H136 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S0 ≥ 2 units with ≥ 5 delayed active readers	0/17; max 2 (G39)	failed
S1 power < 0.8 below 10 readers	card sample 0.00 in 5/5 skeletons	holds
P1 read alignment $b \ni 1$	not computed	untestable
P2 $K \leq 2$ for $\geq 1/2$	not computed (stop)	not run
P3 $\rho(K, D_r) \ni 0$	not computed; O3 power ≤ 0.08	untestable
P4 (A1) read beats peer-link anchor	not computed; read = post anchor	not run
N1 G44 rooms	0 + 0 delayed active readers	untestable
N2 G40 hub link	0 of 15	untestable
N3 NE38	no readable message	excluded
Overall	kill untestable	inconclusive

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

All 17 kickoffs are posted 0.6–1.6 min before the day window opens, so the kickoff read, the first call of the day and the post are the same instant to within one call: no reader has an own call between t_k and t_r . The active-before window therefore reaches back to the previous evening, and in 5 units that day is reserved. Without the active-before condition the count is still at most 3. The synthetic all-readers power of 0.96 in G39 rests on one reader who read the next day, which is the late-starter impostor; it is not a result. Amendment A1 merged the post and day-start anchors in O4 before any outcome.

ROUND 2

GOAL PERIODS IN DETAIL

G18 – n/a (untestable)
 G19 – n/a (untestable)
 G26 – n/a (untestable)
 G30 – n/a (untestable)
 G31 – n/a (untestable)
 G33 – n/a (untestable)
 G35 – n/a (untestable)
 G36 – n/a (untestable)
 G37 – n/a (untestable)
 G38 – n/a (untestable)
+6 more periods: see the card.

Find fields that arrive mid-day through a readable message (operator reassignments with a chat source, room kickoffs inside a window). Check the reserved kickoffs' post times against the window start, a calendar fact, before spending a reserved use. Pre-register any all-readers widening with the late-starter world as its null.

- **Essence.** whether a field acts on each agent at its own read-out call; this needs a field that arrives while agents are already calling, not at day boot.
- **H136-R1** Mid-day fields.
- **H136-R2** Reserved kickoffs: check the post time first.
- **H136-R3** All-readers widening (post hoc idea).

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

H136-freeze-at-first-kickoff-read · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H136-freeze-at-first-kickoff-read/README.md · ratings: H136 agent (suggested) (page 2)

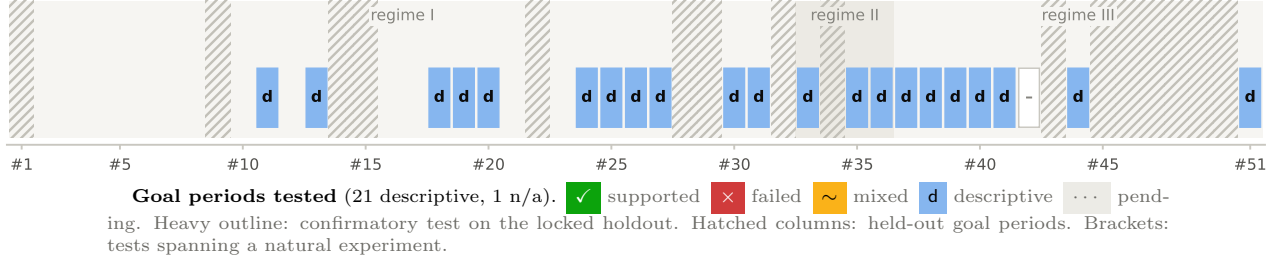
H137: Nonreciprocal Potts: project hops break detailed balance only along named pairs

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 0.5/5 Usefulness 1/5

Bottom line. Untestable at village counts (pooled power 0.18 at $J = 1$). Descriptively, follow hops show no naming direction once hop propensity is controlled ($\theta_{\text{name}} = +0.01 [-0.07, +0.08]$, 41 units); the card's uncontrolled -0.39 is a propensity artifact.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

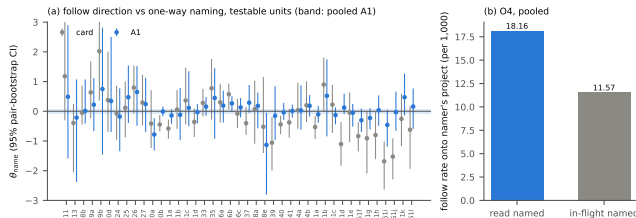
HH380 says that a one-way name steers project choice: when agent j names agent i and i does not name j back, i should join j 's project more often than j joins i 's, so project hops break detailed balance only along one-way named pairs. H137 builds follow hops from the shared per-call project labels in 71 non-reserved units, classifies every agent pair by naming (one-way, mutual, none) and fits the follow direction against the naming direction with popularity controls. **The test cannot decide at village counts:** on real skeletons a coupling of $J = 1$ moves one-way follow counts by 1–3%, and the pooled power is 0.18. Descriptively, once each agent's hop propensity is controlled, follow direction does not track naming ($\theta_{\text{name}} = +0.01 [-0.07, +0.08]$, 41 testable units).

PHYSICAL MODEL

Models 10, 02 and 15. Kinetic Potts walker: at its call c agent i hops to project b with utility $u_b = \kappa_b + \sum_j J \ln(1 + R_{i \leftarrow j}(c)) [\sigma_j(c) = b] + \text{habit} + \text{share}$, where $R_{i \leftarrow j}$ counts j 's messages naming i read in i 's last 10 calls. Couplings exist only from named senders, so $J_{i \leftarrow j} \neq J_{j \leftarrow i}$ on one-way pairs, and the pair current runs from namer to named. The pair entropy production is Schnakenberg's edge term.

$\text{logit } P(a \text{ joins } b) = \theta_{\text{name}} z_{ab} + \delta \Delta \ln \text{pop} + \epsilon \Delta \ln \text{indeg} + \epsilon_{\text{act}} \Delta \ln \text{act}$

$$\sigma_{ij} = (F_{i \leftarrow j} - F_{j \leftarrow i}) \ln \frac{F_{i \leftarrow j} + \frac{1}{2}}{F_{j \leftarrow i} + \frac{1}{2}}, \quad A = \ln \frac{F_{\text{named} \leftarrow \text{namer}} + \frac{1}{2}}{F_{\text{namer} \leftarrow \text{named}} + \frac{1}{2}}$$



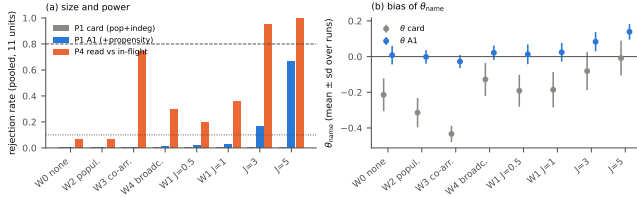
Observables. (a) Follow-direction coefficient θ_{name} per testable unit (95% pair-bootstrap CIs). Gray: the card's estimator (popularity and in-degree controls). Blue: Amendment A1 (adds each member's hop propensity). The blue band is the pooled A1 estimate over 62 units. The card's estimator sits below zero; A1 sits on zero. (b) Pooled rate of moving onto the namer's project at a call that read a named message, against a call with a named message still in flight (matched lag).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: naming an agent does not recruit it to your project. Over 1,130 one-way named pairs, named agents joined their namers' projects no more often than the reverse, once each agent's hop rate is controlled. Agents who name others also hop more, so raw counts make the namer look like the follower. To move an agent, H137 gives no lever; a direct request at the hop call is the next thing to test.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
S0 power ≥ 0.8 at $J = 1$	0.18 (P1, 62 units); 0.00 (P3)	failed
P1 $\theta_{\text{name}} > 0$	A1 +0.01 [−0.07, +0.08]; card −0.39 [−0.55, −0.23]	untestable
P2 symmetric otherwise	33/41 units inside N1b (A2)	descriptive
P3 $\sigma_{\text{one}} - \sigma_{\text{none}} > 0$	excess over N1b +0.9 [−1.4, +3.3]	untestable
P4 read > in-flight	+6.6 [+4.6, +8.7] $\times 10^{-3}$	descriptive
P5 one-way < 1/3	median 0.35; 19/41 units	not supported
N1 G51 $\theta > 0$	−0.07 [−0.21, +0.07]	not met
N2 G38 same room	+0.15 [−0.20, +0.40]; 1 pair	not met
Popularity kill	raw $A = -0.38$ [−0.44, −0.31]	no fire



Second figure. Synthetic validation on 11 real skeletons (G38, G31, 51c, 51h; 60–300 runs per world). (a) Rejection rates. The card’s P1 never rejects, because its θ_{name} is biased negative. A1 has size ≤ 0.013 but power 0.02 at $J = 1$ and 0.67 only at $J = 5$. P4 has power 0.36 at $J = 1$, but co-arrival (W3) fakes it in 74% of runs. (b) Bias: with no following (W0) the card’s θ_{name} is -0.21 ± 0.09 , because namers hop more; A1 removes the bias.

GOAL PERIODS IN DETAIL

G11 d descriptive
G13 d descriptive
G18 d descriptive
G19 d descriptive
G20 d descriptive
G24 d descriptive
G25 d descriptive
G26 d descriptive
G27 d descriptive
G30 d descriptive

+12 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Every result is descriptive: S0 failed, so no statistic is a powered test, and both HH kills need power to fire. Amendments A1–A3 (propensity control, propensity-adjusted direction null, null-centred contrast) were written before real data. At the full pool A1 keeps a +0.04 bias. All pair-EP classes exceed their nulls, so pair direction carries structure that naming does not explain. The $J \approx 5$ exclusion and the class-excess comparison are post hoc. Attention labels only; no work-channel follows.

ROUND 2

Move the coupling test to the hop call (conditional logit over destinations with the namer’s project as an alternative), add a same-hour co-arrival exclusion before reading P4, and build a pair-persistence null for σ . Confirm only what then passes on #45–#47.

- **H137-R1** Coupling at the hop call.
- **H137-R2** Pair-persistence null for σ .
- **H137-R3** P4 without co-arrival.

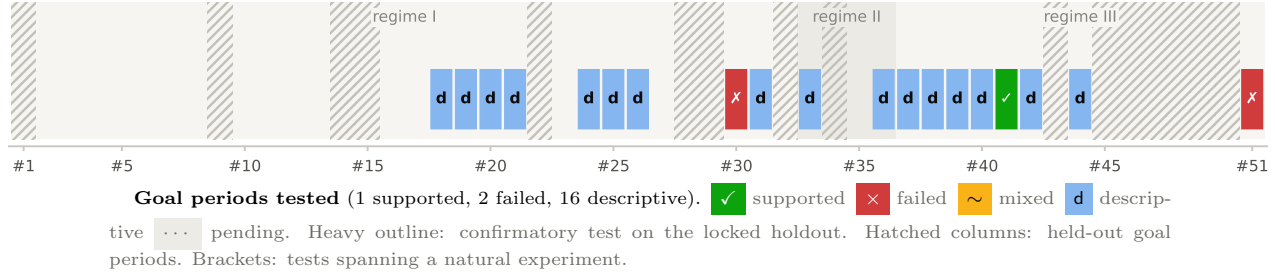
H138: Glauber escape from a project grows with the number of open options

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 1/5 Usefulness 1/5

Bottom line. Within goal periods the per-call leave hazard does not grow with the number of open projects (pooled elasticity $-0.66 [-1.64, 0.32]$ work, $-0.28 [-0.42, -0.15]$ attention), so the Glauber law $\varepsilon_q = 1$ is rejected; the HH381 kill itself is unpowered.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

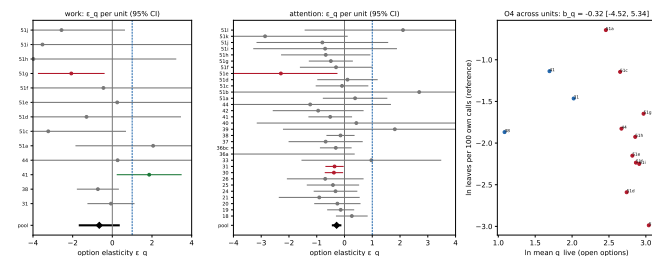
HH381 says that an agent working on one project leaves in the way a Glauber spin leaves its Potts state: at each of its own calls, the chance of leaving grows in proportion to the number of other open options, $h \propto (q-1)e^{-\beta\Delta}$. H138 builds one row per agent, 30-min window and project visit (work: DQ4 host labels; attention: strict project mentions) and fits the leave hazard per own call against the number of projects other agents held in the last 2 active hours. **Leaving does not rise with the open options:** the pooled option elasticity is $-0.66 [-1.64, 0.32]$ in the work channel (13 units) and $-0.28 [-0.42, -0.15]$ in the attention channel (31 units). The literal law $\varepsilon_q = 1$ is rejected. The card's kill ($\varepsilon_q = 0$ vs 0.5) is unpowered in the work channel (0.43), so the verdict by its rule is inconclusive.

PHYSICAL MODEL

Models 10 (kinetic Potts) and 02 (single-spin kinetics). Heat-bath walker on the agent's call clock: from project a it moves to b with probability $\nu e^{u_b} / (e^{u_a} + \sum_{b'} e^{u_{b'}})$. For rare leaves $h \approx \nu e^{-u_a} Z_{\text{alt}}$; with equal attractions $Z_{\text{alt}} = (q-1)e^u$, so $\varepsilon_q = \partial \ln h / \partial \ln(q-1) = 1$ at a fixed stay field. Rival R-finish: the agent leaves when its task ends, $\varepsilon_q \approx 0$.

$$\text{cloglog } P(\text{leave in } w) = \ln c_{iw} + \theta_i + s(t) + \varepsilon_q \ln q_{\text{live}} + \gamma_d \ln(1+d) + \beta_a \ln(1+a_a) + \beta_c m + \beta_{\text{own}} o$$

$$q_{\text{live}}(i, w) = \#\{b \neq a : \text{ held by } j \neq i \text{ in } w - 4..w - 1\}, \quad Z_{\text{alt}} = \sum_{b \in C} e^{u_b}$$



Observables. (a) Work and (b) attention option elasticity $\hat{\varepsilon}_q$ per testable unit (cloglog hazard per own call, agent effects, active-time spline; agent-cluster 95% CIs; green CI above 0, red below 0). Black: DerSimonian–Laird pool. Dashed line: the Glauber value 1. (c) Cross-unit reference leave rate against the mean number of open options (blue shared-goal, red own-role units); the slope is not identified (power 0.12).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: opening more parallel projects does not make agents drop their current one faster. Expect about 0.06–0.3 leaves per 100 own calls in the work channel whatever the number of open projects; in own-role weeks agents leave slightly less when many projects are live. Plan churn from task length, not from the size of the menu. Whether naming a target project pulls agents is open (a small attention signal, below the label-noise level).

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
S1–S3 synthetic	unbiased; size 0.055; power 0.43 at 0.5	passed
P1 $\varepsilon_q > 0$ (work)	1/13 units; −0.66 [−1.64, 0.32]	pool against
A3 bound $\varepsilon_q < 1$	upper limit 0.32	$\varepsilon_q = 1$ rejected
P2 $\varepsilon_Z \in [0.5, 1.5]$	0.22 [−0.12, 0.55]; Z wins 8/13	not passed
P3 lag > lead	−0.51 [−1.47, 0.45]	failed
P4 $R_{\text{own}} \in [0.5, 2]$	0/13	failed
P5 cross-unit b_q	−0.32 [−4.5, 5.3]	descriptive
N1 G38 terciles	ratio 2.89 vs $\times 1.5$	against
N2 G44 rooms	HR 1.97 vs 3.32 (15 leaves)	descriptive
N3 G51	−1.83 [−3.19, −0.46]	failed
Kill (work)	CI includes 0; power 0.43	untestable
Attention	−0.28 [−0.42, −0.15]; power 1.00	against

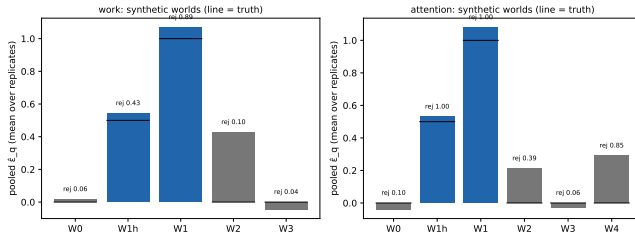
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Work leaves are few outside #51 (25–88 per unit), so no single work unit has power 0.8. The attention negative appears only after the active-time spline (agent-only fit +0.18 [−0.02, 0.38]; post hoc). The #51 negative rests on Wald CIs; the within-day permutation gives $p \geq 0.095$ in each #51 unit. Labels are 30-min windows, coarser than read-out delays. Reserved periods were not run.



Second figure. Synthetic validation on the real skeleton of every testable unit (200 replicates per world): pooled $\hat{\varepsilon}_q$ by world (line: truth) with its rejection rate of $\varepsilon_q = 0$. The estimator is unbiased; renewal (W0) and shared-drive (W3) worlds keep size 0.04–0.06. Work power is 0.89 at $\varepsilon_q = 1$ but 0.43 at 0.5. Attention label noise (W4) fakes $\hat{\varepsilon}_q \approx 0.3$.

ROUND 2

Refit on the call clock (one row per own call) to raise power and make the lead placebo work; explain the negative elasticity with a same-time cross-agent q shuffle; test kickoff-named against free options with H54's targets; reconcile the rising work dwell hazard with H129's aging.

- **H138-R1** Move to the call clock.
- **H138-R2** Explain the negative elasticity.
- **H138-R3** Named vs free options.
- **H138-R4** Reconcile dwell.
- **H138-R5** Confirmation.

GOAL PERIODS IN DETAIL

G18 d descriptive
G19 d descriptive
G20 d descriptive
G21 d descriptive
G24 d descriptive
G25 d descriptive
G26 d descriptive
G30 $\times$ failed
G31 d descriptive
G33 d descriptive
+9 more periods: see the card.

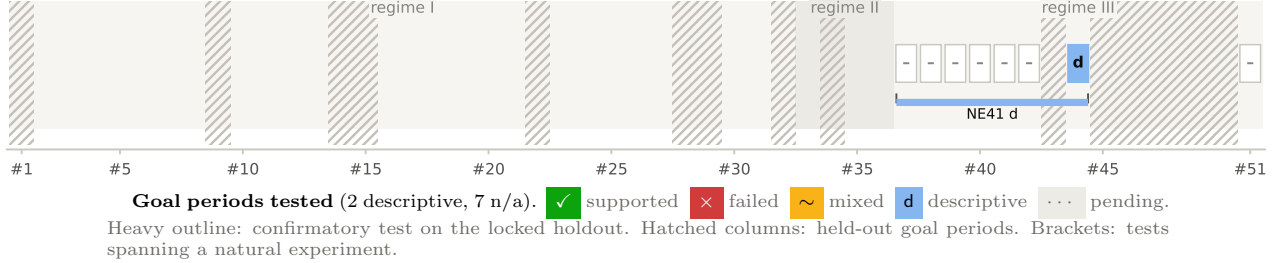
H139: A fast read kick on a slow private well predicts how content variance splits between the two

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 0.5/5 Usefulness 1/5

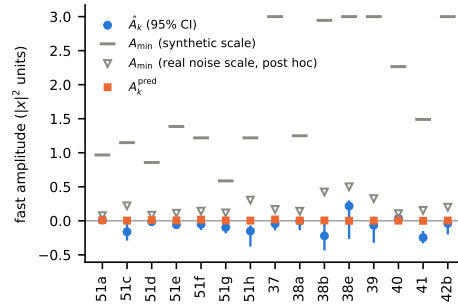
Bottom line. In 15 regime-III units the read-driven fast content variance predicted from H130's kick (median A_k^{pred} 0.0044, $\approx 2.5\%$ of the dynamic variance) lies 7 to $>1,000\times$ below the smallest resolvable fast amplitude, so HH382's variance-split test is untestable at this data size.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

H130 found that in #51 a read kicks the reader's content ($J_K = 0.044$) and the kick fades within ≈ 7 calls, while the agent's own well relaxes in ≈ 100 calls. HH382 asked whether that fast kick fixes, with no new fit, how much of an agent's content variance is fast. H139 measures the own content autocovariance at 1-call resolution in 15 regime-III units (#37–#42, #51) and compares its fast amplitude with the shot-noise prediction. A synthetic check on the real skeletons decided first whether the prediction is resolvable. It is not: the predicted fast amplitude is 7 to $>1,000$ times below the smallest resolvable one in every unit. The verdict is inconclusive (below resolution).



Observables. Fitted fast amplitude $\hat{A}_k$ (blue, 95% agent-day bootstrap CI) per testable unit, the predicted A_k^{pred} (orange) and the smallest resolvable amplitude $A_{\min}$ (gray bar: synthetic; open triangle: rescaled to the real noise level, post hoc). The prediction sits far below both resolution marks in every unit.

PHYSICAL MODEL

Two-rate Langevin relaxation (model 16) on the per-call clock. Content $x = s + k$: a slow part s held by the agent's private well (rate γ_s) and a fast part k made of read kicks of size J that decay at γ_k . For independent kick directions, Campbell's theorem fixes the fast amplitude from the read rate $\bar{r}$ (DQ1 ledger), J and γ_k (H130). The own autocovariance, corrected for the common drive, then has two exponentials and a plateau.

$$C(\tau) = A_k(1 - \gamma_k)^\tau + A_s(1 - \gamma_s)^\tau + B, \quad \tau \geq 1 \text{ call}$$

$$A_k^{\text{pred}} = \bar{r} J^2 / [1 - (1 - \gamma_k)^2], \quad Q_k = \hat{A}_k / A_k^{\text{pred}}$$

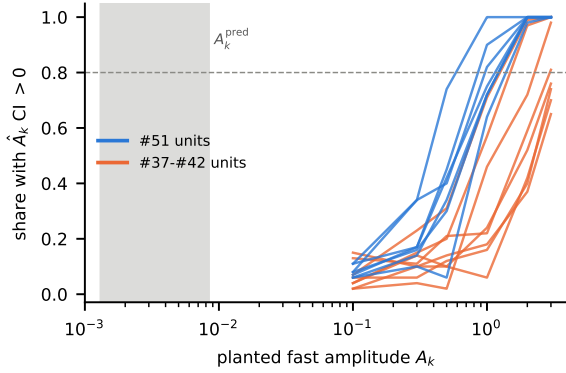
$$S1: A_{\min} > 2A_k^{\text{pred}} \Rightarrow \text{P1 untestable}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, the read channel is not visible in an agent's overall content fluctuations: at about one read per call, H130's kick predicts a fast share of $\approx 2.5\%$ of the dynamic content variance (median), well inside the noise of $\sim 10^3$ – 10^4 statement pairs per unit. To see what reading does to an agent, measure the response along the sender's direction (H130's dose design), not the agent's total variance.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1 $A_{\min} > 2A_k^{\text{pred}}$	15/15 units; ratio 92→2,000 (6.8–9,993 real scale)	as predicted
S2 Q_k unbiased $\pm 30\%$	not resolvable at $\times 5, \times 20$	failed
S3 R_{fast} separates worlds	37–59% of replicates	failed
P1 $Q_k \in [0.5, 2]$ (kill)	$\hat{A}_k - 0.058$ $[-0.099, -0.017]$ vs 0.0044	untestable
P2 fast part $\propto$ reads	not computed	untestable
P3 slow share ≥ 0.8	13/15 units	non-diagnostic
P4 slow share is an agent constant	$\rho - 0.20, p 0.81$	failed
P5 two rates beat one	9/15 (null 34–58%)	non-diagnostic
N1 NE41 fast part erased	$R_{\text{slow}} 1.06$ $[0.98, 1.14]$	untestable



Second figure. Synthetic check on the real skeletons of 15 units (real statements, calls, ledger reads, resets; 50–100 replicates per world). Share of replicates in which $\hat{A}_k$'s 95% CI excludes 0, against the planted fast amplitude. Blue: #51 units; orange: #37–#42 units. The gray band is the range of A_k^{pred} . Detection reaches 80% only at $A_k \approx 0.6$ –3.

GOAL PERIODS IN DETAIL

G44 d descriptive (no unit with ≥ 3 days).
 NE41 d descriptive. N1 is untestable (S1 fired in every unit...
 G37 – n/a (inconclusive: below resolution). S1 fired before...
 G38 – n/a (inconclusive: below resolution). S1 fired before...
 G39 – n/a (inconclusive: below resolution). S1 fired before...
 G40 – n/a (inconclusive: below resolution). S1 fired before...
 G41 – n/a (inconclusive: below resolution). S1 fired before...
 G42 – n/a (inconclusive: below resolution). S1 fired before...
 G51 – n/a (inconclusive: below resolution). S1 fired before...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The synthetic noise level was higher than the real one; rescaling to the real bootstrap SE (post hoc) still leaves $A_{\min}/A_k^{\text{pred}} \geq 6.8$. The drive correction subtracts conversational co-movement at short wall lags, and the lag-1 covariance sits below the lag-2 covariance in 14/15 units, so the constrained $\hat{A}_k$ is negative (post hoc). γ_{kick} was not re-estimated in the shared weeks. Agent-level slow shares are unstable.

ROUND 2

Measure the read echo along sender directions (H130's dose design), which has the power the variance split lacks. Explain the lag-1 dip (reply vs own-task alternation, or producing-call errors). Correct for the drive with unread partners only, and calibrate the synthetic noise to the real variance before new power claims.

- **Essence.** how much of an agent's moment-to-moment content is the echo of what it read.
- **H139-R1** Measure the echo where it is large.
- **H139-R2** The lag-1 dip.
- **H139-R3** Drive correction without conversation.
- **H139-R4** Calibrate the synthetic noise scale to the real $-x-^2$ and dynamic variance before any new power claim.

H139-two-rate-variance-split · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H139-two-rate-variance-split/README.md · ratings: H139 agent (suggested) (page 2)

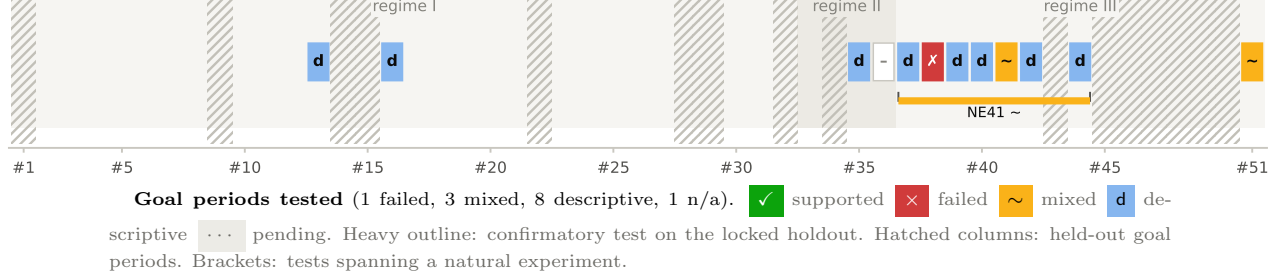
H140: DeGroot at the read-out call: the self-weight of a content step equals the agent's own share of its context

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. The self-weight does not follow the context self-share (pooled $\hat{w}_1 = -0.01$ $[-0.12, 0.10]$, 25 regime-III units, power 1.00 at 0.5); the read weight grows as $k^{0.22}$ $[0.13, 0.31]$.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

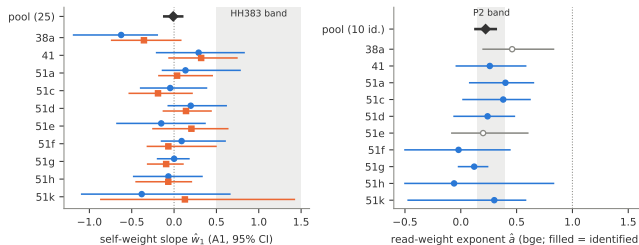
H140 asks whether an agent's next statement is a DeGroot average of its own recent content and the batch it just read, with the self-weight equal to its share of its own context window. Each row is a regime-III talk call. The statement vector is regressed on the agent's last two statements (instrumented by older ones), the read batch, the unreadable in-flight messages and the agent's well centre. HH383 predicted a self-weight slope $w_1 \approx 1$ in the self-share. Over 25 units the pooled slope is -0.01 $[-0.12, 0.10]$, with synthetic power 1.00 at $w_1 = 0.5$: the pre-set kill fires. The read weight is sublinear, $a = 0.22$ $[0.13, 0.31]$, as H113 found.

PHYSICAL MODEL

Models 11, 02 and 04: a linear vector-spin update on the call clock (Friedkin-Johnsen DeGroot). y : the reader's statement (DQ5 style_resid32, goal and kickoff directions removed); p : its last two statements in the context segment; $s = \sum_{m \in B} x_m$: the read batch of size k ; s^F : messages posted but not yet readable; h : the leave-day-out well centre. $s_{\text{self}} = \text{ctx_pos} / (\text{ctx_pos} + k_{\text{ctx}})$ is the agent's own share of its context. Amendment A1 (before data) adds wall-clock and batch-size nuisance terms to w_{self} .

$$y = w_{\text{self}} p + \gamma_1 k^{-b} s + \gamma_F s^F + w_{\text{anchor}} h + \epsilon, \quad W_{\text{read}}(k) = \gamma_1 k^a, \quad a = 1 - b$$

$$w_{\text{self}} = w_0 + w_1 s_{\text{self}} + w_2 (1 - \gamma_{\text{auto}})^{\Delta n} + w_3 e^{-\Delta t / 30 \text{ min}} + w_5 \ln(1 + k) + w_6 \ln(1 + k_F)$$



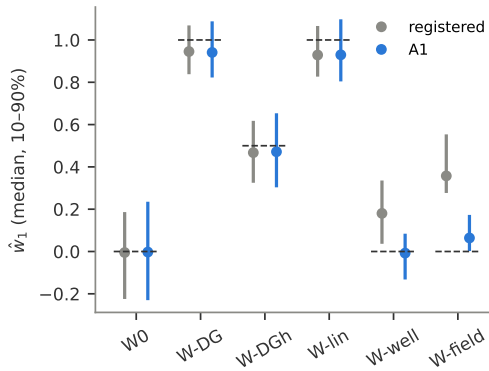
Observables. Left: self-weight slope $\hat{w}_1$ per testable unit (≥ 300 talk calls; circles bge, squares gte; 95% CI) and the DerSimonian-Laird pool over all 25 units (diamond). The gray band is HH383's prediction. Right: read-weight exponent $\hat{a}$ (bge); filled points are units where read messages beat in-flight messages at matched age (identified). The gray band is the registered P2 range; dotted line, no bottleneck.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Changing what fills an agent's context does not change how much it sticks to its own recent content. An operator who wants an agent to listen more will not get it by trimming the agent's own turns from the context. Even a forced erasure leaves about 95% of the weight on the erased content. The lever that does work is the batch: ten messages read at once carry only about $10^{0.22} \approx 1.7$ times the pull of one.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
S1 synthetic recovery, W-well size	registered: size 0.14–1.00; A1: 0.00–0.07	amended
S2 power at $w_1 = 0.5$	1.00 (#51 pool)	passed
P1 $\hat{w}_1 \in [0.5, 1.5]$, CI > 0	$-0.01 [-0.12, 0.10]$; gte -0.02	failed (kill)
P2 $\hat{a} \in [0.15, 0.40]$	0.22 [0.13, 0.31], 10 units	supported
P3 call-gap term $w_2 > 0$	0.35 $[-0.33, 1.02]$	failed
P4 read beats in flight	γ_F/γ_1 0.02; $+0.056 [0.042, 0.071]$	supported
P5 closure $\Sigma \approx 1$	1.04–1.05 in 6/6 bins	void
N1 NE41 erasure	$\Delta\hat{w}/\Delta s$ 0.29 (gte 0.55)	mixed
N2 #51 timer wakes	$\hat{w}_1$ 0.04; $\hat{a}$ 0.19 [0.07, 0.32]	supported (weak)



Second figure. Synthetic worlds on the real skeletons of 51c, 51g, #38 and #41 (100 runs each): median $\hat{w}_1$ and 10–90% range for the registered estimator (gray) and the A1 estimator (blue); dashed lines mark the truth. The registered estimator reads an Ornstein–Uhlenbeck well (W-well) or a time-local field (W-field) as a positive slope; A1 removes most of that leak before any real data were used.

GOAL PERIODS IN DETAIL

G13 d descriptive
G16 d descriptive
G35 d descriptive
G37 d descriptive
G38 $\times$ failed
G39 d descriptive
G40 d descriptive
G41 $\sim$ mixed
G42 d descriptive
G44 d descriptive
+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

A time-local field still leaks a small positive slope (0.04–0.14) into A1 on synthetic data, so the real $\hat{w}_1 \approx 0$ is, if anything, an upper bound. The closure test is void: any stationary world gives $\Sigma \approx 1$. The NE41 native is not blind (H46, H130, H44 were known) and its registered ratio splits by embedding model. Only 10 of 25 units have ≥ 300 scored calls. Chat-mode periods have no self-share.

ROUND 2

Model the self-weight as an H130 well plus a wall-clock term; a blind NE41 test of erased-content weight on reserved resets; a batch-size self-weight law (the A1 term $\ln(1+k)$ has weight -0.026).

No round-2 redirects yet.

H140-degroot-readout-self-weight · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H140-degroot-readout-self-weight/README.md · ratings: H140 round-1 agent (suggested) (page 2)

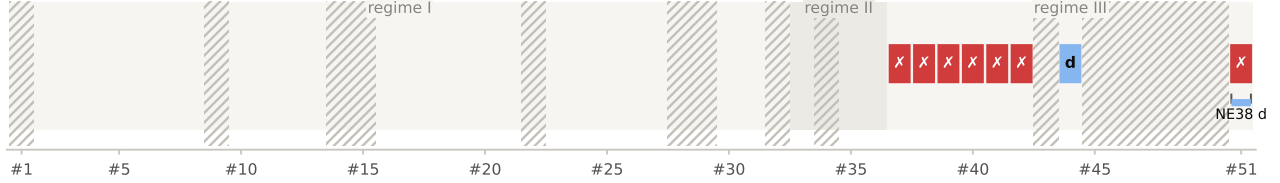
H141: Content moves on an easy plane set by the goal and room texts: slow along it, fast across it

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. In 25 regime-III units, content decays along the goal/kickoff/room text plane at a rate within $\times 3$ of its rate across it (pooled ρ_A 0.68 [0.50, 0.94], bge; gte 0.78 [0.60, 1.02]); the plane is no slower than random planes (1/15 units) and has no extra day-scale memory (contrast -0.02 $[-0.10, 0.06]$).

level: hypothesis · pipeline checks 71% · updated 2026-10-07



Goal periods tested (7 failed, 2 descriptive). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

HH384 proposed that the operator's texts set an easy plane for agent content: along the span of the goal, kickoff and room-kickoff texts, content should relax at least $10\times$ more slowly than across it. H141 builds that plane from texts only and measures memory along and across it in 25 regime-III units of #37–#44 and #51 (#51 with each agent's own role text), in two embedding models. Content is not slower along the plane. The pooled anisotropy ratio is $\rho_A = 0.68$ (90% CI [0.50, 0.94]; gte 0.78 [0.60, 1.02]), so the kill fires. The plane is no slower than random planes, and it has no extra day-to-day memory. The reserved periods are not run.

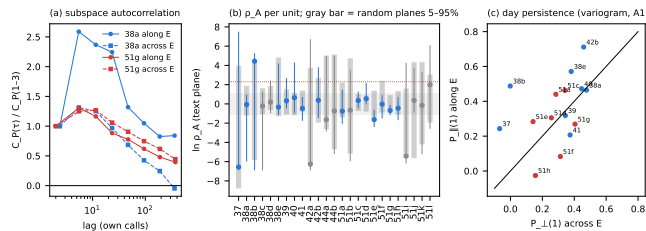
PHYSICAL MODEL

An anisotropic Ornstein–Uhlenbeck well on the call clock (model 16, anisotropy language of model 11). x is an agent's 32-d whitened, style-residualized content minus its leave-day-out well. Γ relaxes x at $\gamma_{\parallel}$ inside the text plane E and at $\gamma_{\perp}$ across it. With isotropic noise, the variance ratio equals the rate ratio. Rivals: isotropy, a hard goal axis (H97), content's own principal modes, and a common drift.

$$x(n+1) = (1 - \gamma_{\parallel} P_E - \gamma_{\perp} (1 - P_E)) x(n) + \xi(n), \quad \rho_A = \gamma_{\perp} / \gamma_{\parallel}$$

$$C_P(\tau) = \langle P_{x_B} \cdot P_{x_{B'}} \rangle - C_P^{\times}(|t_{B'} - t_B|) = A_P e^{-\gamma_P \tau} + B_P$$

$$P_P(1) = 1 - G_P(1)/G_P(\geq 2), \quad G_P(\ell) = \frac{1}{2} \langle |P(\bar{z}_d - \bar{z}_{d+\ell})|^2 \rangle - \text{noise}$$



Observables. (a) Drive-corrected subspace autocorrelation along (solid) and across (dashed) the text plane in 38a and 51g, normalized at lag 1–3 calls. (b) $\ln \rho_A$ per unit with 90% agent-day bootstrap CIs (blue: testable units); gray bars are the 5–95% band of 200 random planes; the dotted line is HH384's $\rho_A = 10$. (c) Day-to-day persistence from the noise-corrected day variogram, along against across E ; points sit on the diagonal.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, a goal or kickoff text does not make agents hold its themes longer than other themes. Content along the text directions turns over at about the same rate as other content (rate ratio 0.5–1.0, 90% CI), within a working day and from day to day. Do not rely on the goal text to keep a theme alive; restate it. The texts do point into content's high-variance space (post hoc: variance per dimension $\approx 1.2\times$ random), so they shape where content sits, not how long it stays.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 pooled $\rho_A \geq 10$	0.68 [0.50, 0.94]; gte 0.78	failed
Kill: CI inside $[\frac{1}{3}, 3]$	inside, all 3 variants	fires
Reverse kill $\rho_A < \frac{1}{3}$	0.68	no
P2 above random 95th pct	1/15 units	failed
P3 $V'_A \approx \rho_A$	7/15; both ≈ 1	mixed
P4 kick lag-1/lag-0 $\leq \frac{1}{4}$	0.42 [0.17, 0.75]	inconclusive
P5 day memory $P_{ } > 0/15$; -0.02 [-0.10, 0.06]		failed
$P_{\perp}$		
N1 #38 room axes	text axis not slow 3/3; $\hat{m}$ not slow 0/3	failed
N2 #51 own planes	0.67 [0.45, 0.99]; 0/7	failed
N3 NE38 rotation	order does not flip	descriptive

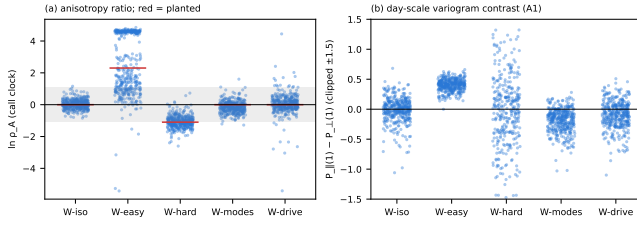
GOAL PERIODS IN DETAIL

G37	× failed
G38	× failed
G39	× failed
G40	× failed
G41	× failed
G42	× failed
G44	d descriptive
G51	× failed
NE38	d descriptive

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic validation on the real skeletons of 51c, 51g, 38a and 41 (100 replicates per world). (a) The call-clock $\ln \rho_A$ is unbiased in the isotropic, hard and drift worlds; in the easy world ($\rho_A = 10$) the slow rate often hits the fit edge, so the day scale carries the slowness test. (b) The amended day variogram separates the easy world from all others; the registered split-half estimator did not (it also flagged the fast plane), so it was replaced before real data.

CAVEATS

The registered day-scale estimator was replaced before real data (A1), and P3 and P4 were operationalized then. The call clock cannot resolve along-plane memory beyond a day ($\gamma \lesssim 0.001/\text{call}$), so P1 alone is underpowered; the kill and the day variogram carry the verdict. Short units (1–2 days) enter only the pool. The kick split cannot see an along-plane kick. NE38 is one agent with 3–10 day pairs.

ROUND 2

Fit the full relaxation matrix per unit to find any slow mode; test the texts as a static field (the post-hoc variance excess); rerun the #38 room axis with the day variogram.

- **H141-R1** Where is content slow?
- **H141-R2** Soft but not slow.
- **H141-R3** Room axis at the day scale.

H141-easy-plane-anisotropy · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H141-easy-plane-anisotropy/README.md · ratings: H141 agent (suggested) (page 2)

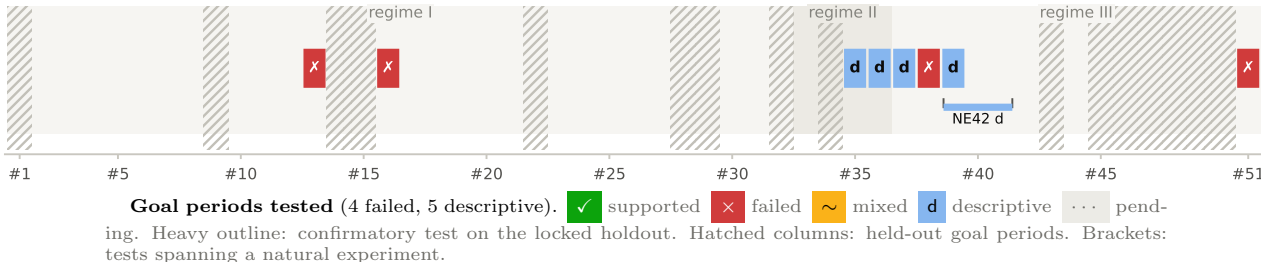
H142: The content response to aligned reads saturates as a Langevin-function torque

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 50% Faithfulness 1/5 Usefulness 1/5

Bottom line. The step toward a direction does not grow with the number of aligned reads at fixed batch size and topic; the Langevin form does not beat a line out of fold, and only the newest item pulls.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



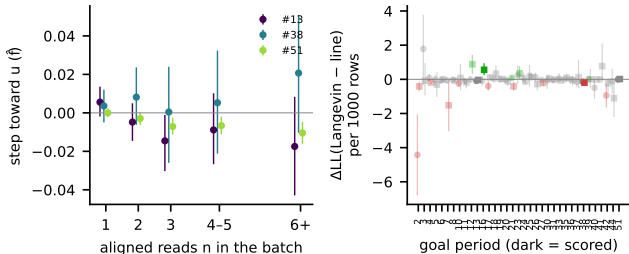
WHAT HAPPENED

H142 asks whether the content step toward a direction saturates with the number of aligned reads in one batch, as a classical moment saturates in a field (the Langevin function). Rows are (talk call, direction) pairs; directions are day-fold clusters of reader statements. Synthetic checks on real skeletons showed that the card’s form mistakes H113’s linear channel for saturation (60–100% of runs), so Amendment A1 lets the amplitude vary with batch size. On the powered, field-identified periods (bge #13, #38, #51; gte adds #16) the step does not grow with aligned reads at all: the pooled out-of-fold $\Delta LL(\text{Langevin} - \text{line})$ is $-9.0 [-17.2, -2.2]$ nats (bge), and $\hat{f}(1) = 0.000 [-0.002, 0.002]$ on #51. The pre-set kill fires. Only the newest batch item pulls.

PHYSICAL MODEL

Model 11 (vector spins, mean-field equation of state). The reader’s projected statement is a moment; each aligned read adds to a field along a direction u , one of $K = 8$ day-fold cluster centroids. n : aligned reads (cosine ≥ 0.3) in the ledger batch of size k . α_c : call effect; δ : direction $\times$ room $\times$ hour effect (hot topic). Rivals: a line in n (H113 at fixed k), a power law, the newest item.

$$y_{c,u} = \alpha_c + \delta_{u,\text{room},h} + a_{b(k)} h(n_{c,u}) + g(n_{c,u}^F) + \beta_{\text{new}} \text{newest}_{c,u} + \beta_{\text{name}} n_{c,u}^{\text{name}} + \epsilon$$
$$h(n) \in \{n, n^p, L(cn)\}, \quad L(x) = \coth x - 1/x, \quad n_{\text{sat}} = 3/c, \quad \Delta_{\text{curv}} = \ln \frac{f(2)}{f(1)} - \ln \frac{f(6+)}{f(3)}$$



Observables. Left: the model-free step curve $\hat{f}(n)$ (95% agent-day cluster bootstrap) on the three scored bge periods; it is flat at zero, not a saturating rise. Right: out-of-fold $\Delta LL(\text{Langevin} - \text{line})$ per 1000 rows for every period (circles bge, squares gte; dark = scored; green CI > 0 , red < 0 , gray includes 0).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator the answer is simple. Repeating the same point in one batch does not move an agent further toward it: the second to sixth aligned message adds nothing measurable once the batch size and the topic of the hour are held fixed. What moves the next statement is the newest message ($+0.010 [0.007, 0.014]$ per direction on #51). Put the key point last; do not send it several times in one batch. The Langevin picture is not needed to say this; the curve has no rise to saturate.

RESULTS VS PREDICTIONS

Prediction	Observed (bge; gte)	Verdict
P1 pooled $\Delta LL(L - \text{line}) > 0$	-9.0 [-17.2, -2.2]; +6.1 [-4.1, 15.7]	failed (kill)
P2 $\Delta_{\text{curv}} > 0$	undefined: no positive $\hat{f}$ bins	failed
P3 $n_{\text{sat}} \in [1.5, 6]$	37 [3.4, 407]; 12 [1.8, 77]	failed
P4 read beats in flight	-0.0025 [-0.0078, 0.0029]	failed
N1 #51 wakes	-5.3 [-13.3, 1.0]; -7.4 [-18.7, 0.1]	failed
N2 NE42 n_{sat} fixed	n_{sat} unidentified, sides unpowered	descriptive
S1 power at #51	1.00, size 0.02 (after A1)	supported
S2 Δ_{curv} size	≤ 0.01	supported
S3 $< 1/2$ powered	3/6; 4/7	failed

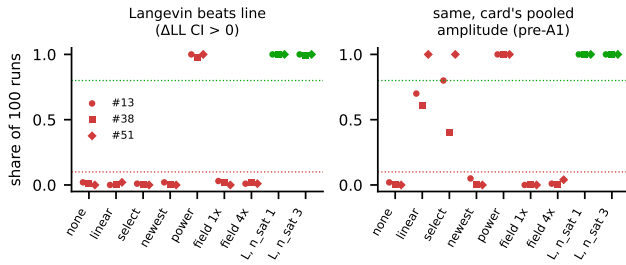
GOAL PERIODS IN DETAIL

G13 × failed
G16 × failed
G35 d descriptive
G36 d descriptive
G37 d descriptive
G38 × failed
G39 d descriptive
G51 × failed
NE42 d descriptive

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic worlds on the real skeletons of #13, #38, #51 (100 runs each). Left: with batch-size amplitudes (A1), the Langevin worlds pass (power 0.99–1.00) and the linear, selection, newest-item and field worlds stay ≤ 0.03 . The power-law world also beats the line; $\Delta LL(L - \text{power})$ separates it (0.00). Right: the card's original single amplitude passes the linear and selection worlds in 40–100% of runs.

CAVEATS

Directions are clusters of reader statements, a coarse ruler; an aligned read is one with cosine ≥ 0.3 to a centroid. The direction $\times$ room $\times$ hour effects and the window-field projection could absorb real uptake that follows the hour's topic; post hoc variants without them keep the flat curve. The result is a null at the H113-calibrated effect size (power ≥ 0.87), not a bound on tiny effects. #16 (gte only) shows a positive ΔLL with no positive bin. Reserved periods were not run.

ROUND 2

Decompose H113's batch uptake into newest, aligned-cluster and off-cluster parts on the same calls; fit the newest-item pull against age and rank.

- **H142-R1** Newest-item channel.
- **H142-R2** Named aligned items.
- **H142-R3** Where does H113's uptake go?

H142-langevin-torque-saturation · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H142-langevin-torque-saturation/README.md · ratings: H142 round-1 agent (suggested) (page 2)